

backend/app/__init__.py

```
# ===== 文件: backend/app/__init__.py =====
1  """FastAPI backend package for the OMELET Lab system."""
2
3
4  # ===== 文件: backend/app/auth.py =====
5  from datetime import timedelta
6  import logging
7
8
9  from fastapi import APIRouter, Cookie, Depends, Header, HTTPException, Request, Response, status
10 from sqlalchemy import select, update
11 from sqlalchemy.exc import IntegrityError
12 from sqlalchemy.orm import Session
13
14 from .config import get_settings
15 from .database import get_session
16 from .email_verification import (
17     create_register_email_code,
18     ensure_email_not_registered,
19     normalize_email,
20     try_normalize_email,
21     verify_register_email_code,
22 )
23 from .models import User, UserSession
24 from .schemas import (
25     AccessTokenResponse,
26     AuthResponse,
27     EmailCodeRequest,
28     LoginRequest,
29     MessageResponse,
30     RegisterRequest,
31     TurnstilePassRequest,
32     TurnstilePassResponse,
33     UserResponse,
34 )
35 from .security import (
36     create_access_token,
37     create_refresh_token,
38     create_turnstile_pass_token,
39     clear_refresh_cookie,
40     hash_password,
41     hash_token,
42     REFRESH_COOKIE_NAME,
43     set_refresh_cookie,
44     utc_now,
45     verify_password,
46 )
47 from .turnstile import verify_turnstile_access, verify_turnstile_token
48
49 router = APIRouter(prefix="/api/auth", tags=["auth"])
50 logger = logging.getLogger(__name__)
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46
47
48 def get_user_by_username(session: Session, username: str) -> User | None:
49     return session.scalar(select(User).where(User.username == username))
50
51
52 def get_user_by_login_identifier(session: Session, identifier: str) -> User | None:
53     normalized_email = try_normalize_email(identifier)
54     if normalized_email is not None:
55         user = session.scalar(select(User).where(User.email == normalized_email))
56         if user is not None:
57             return user
58
59     return get_user_by_username(session, identifier.strip())
60
61
62 def get_active_user_or_401(session: Session, identifier: str, password: str) -> User:
63     user = get_user_by_login_identifier(session, identifier)
64     if not user or user.status != "active" or not verify_password(password, user.password_hash):
65         raise HTTPException(
66             status_code=status.HTTP_401_UNAUTHORIZED,
67             detail="用户名或密码错误",
68         )
69     return user
70
71
72 def create_user_session(session: Session, user: User) -> tuple[str, str]:
73     settings = get_settings()
74     refresh_token = create_refresh_token()
75     refresh_token_hash = hash_token(refresh_token)
76     user_session = UserSession(
77         user_id=user.id,
78         refresh_token_hash=refresh_token_hash,
79         expires_at=utc_now() + timedelta(days=settings.refresh_token_expire_days),
80     )
81     session.add(user_session)
82     access_token = create_access_token(user.id, user.username)
83     return access_token, refresh_token
84
85
86 def revoke_all_user_sessions(session: Session, user_id: int, revoked_at) -> None:
87     session.execute(
88         update(UserSession)
89         .where(UserSession.user_id == user_id, UserSession.revoked_at.is_(None))
90         .values(revoked_at=revoked_at),
91     )
92
93
94 def _raise_refresh_error(message: str) -> None:
95     """通过异常响应明确清除浏览器中的刷新 Cookie。"""
```

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```
96     cookie_response = Response()
97     clear_refresh_cookie(cookie_response)
98     raise HTTPException(
99         status_code=status.HTTP_401_UNAUTHORIZED,
100         detail=message,
101         headers={"set-cookie": cookie_response.headers["set-cookie"]},
102     )
103
104
105 def parse_bearer_token(authorization: str | None) -> str:
106     if not authorization:
107         raise HTTPException(status_code=status.HTTP_401_UNAUTHORIZED, detail="缺少登录凭证")
108
109     scheme, _, token = authorization.partition(" ")
110     if scheme.lower() != "bearer" or not token:
111         raise HTTPException(status_code=status.HTTP_401_UNAUTHORIZED, detail="登录凭证格式无效")
112     return token
113
114
115 @router.post("/turnstile-pass", response_model=TurnstilePassResponse)
116 def create_turnstile_pass(payload: TurnstilePassRequest):
117     verify_turnstile_token(payload.turnstile_token, payload.action)
118     pass_token, expires_at = create_turnstile_pass_token()
119     expires_in_seconds = max(0, int((expires_at - utc_now()).total_seconds()))
120     return TurnstilePassResponse(
121         turnstile_pass_token=pass_token,
122         expires_at=expires_at,
123         expires_in_seconds=expires_in_seconds,
124     )
125
126
127 @router.post("/register/email-code", response_model=MessageResponse)
128 def send_register_email_code(
129     payload: EmailCodeRequest,
130     request: Request,
131     session: Session = Depends(get_session),
132 ):
133     verify_turnstile_access(payload.turnstile_token, payload.turnstile_pass_token, "register")
134
135     email = normalize_email(payload.email)
136     client_ip = request.client.host if request.client else None
137     try:
138         create_register_email_code(session, email, client_ip)
139         session.commit()
140     except HTTPException:
141         session.rollback()
142         raise
143
144     return MessageResponse(message="验证码已发送")
145
```

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```
146
147 def get_current_user(
148     authorization: str | None = Header(default=None),
149     session: Session = Depends(get_session),
150 ) -> User:
151     from .security import decode_access_token
152
153     token = parse_bearer_token(authorization)
154     user_id = decode_access_token(token)
155     if user_id is None:
156         raise HTTPException(status_code=status.HTTP_401_UNAUTHORIZED, detail="登录凭证无效或已过期")
157
158     user = session.get(User, user_id)
159     if not user or user.status != "active":
160         raise HTTPException(status_code=status.HTTP_401_UNAUTHORIZED, detail="用户不可用")
161     return user
162
163
164 @router.post("/register", response_model=AuthResponse)
165 def register(payload: RegisterRequest, response: Response, session: Session = Depends(get_session)):
166     settings = get_settings()
167     email: str | None = None
168     email_code_record = None
169
170     if settings.email_verification_required:
171         if payload.email is None:
172             raise HTTPException(status_code=status.HTTP_400_BAD_REQUEST, detail="请填写邮箱")
173         email = normalize_email(payload.email)
174         ensure_email_not_registered(session, email)
175         email_code_record = verify_register_email_code(session, email, payload.email_code)
176     elif payload.email:
177         email = normalize_email(payload.email)
178         ensure_email_not_registered(session, email)
179
180     user = User(
181         username=payload.username,
182         email=email,
183         password_hash=hash_password(payload.password),
184         role="user",
185         status="active",
186     )
187     session.add(user)
188     try:
189         session.flush()
190     except IntegrityError as exc:
191         session.rollback()
192         raise HTTPException(status_code=status.HTTP_409_CONFLICT, detail="用户名或邮箱已存在") from exc
193
194     user.last_login_at = utc_now()
195     if email_code_record is not None:
```

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196         email_code_record.consumed_at = utc_now()
197
198     access_token, refresh_token = create_user_session(session, user)
199     session.commit()
200     session.refresh(user)
201     set_refresh_cookie(response, refresh_token)
202
203     return AuthResponse(
204         user=UserResponse.model_validate(user),
205         access_token=access_token,
206     )
207
208
209 @router.post("/login", response_model=AuthResponse)
210 def login(payload: LoginRequest, response: Response, session: Session = Depends(get_session)):
211     verify_turnstile_access(payload.turnstile_token, payload.turnstile_pass_token, "login")
212
213     user = get_active_user_or_401(session, payload.username, payload.password)
214     user.last_login_at = utc_now()
215     access_token, refresh_token = create_user_session(session, user)
216     session.commit()
217     session.refresh(user)
218     set_refresh_cookie(response, refresh_token)
219
220     return AuthResponse(
221         user=UserResponse.model_validate(user),
222         access_token=access_token,
223     )
224
225
226 @router.get("/me", response_model=UserResponse)
227 def me(user: User = Depends(get_current_user)):
228     return UserResponse.model_validate(user)
229
230
231 @router.post("/refresh", response_model=AccessTokenResponse)
232 def refresh(
233     response: Response,
234     refresh_token: str | None = Cookie(default=None, alias=REFRESH_COOKIE_NAME),
235     session: Session = Depends(get_session),
236 ):
237     if not refresh_token:
238         _raise_refresh_error("缺少刷新凭证")
239
240     token_hash = hash_token(refresh_token)
241     now = utc_now()
242     user_session = session.scalar(
243         select(UserSession)
244         .where(UserSession.refresh_token_hash == token_hash)
245         .with_for_update(),
```

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```
246     )
247     if not user_session:
248         _raise_refresh_error("刷新凭证无效或已过期")
249
250     if user_session.revoked_at is not None:
251         revoke_all_user_sessions(session, user_session.user_id, now)
252         session.commit()
253         logger.warning(
254             "refresh token reuse detected",
255             extra={"user_id": user_session.user_id, "session_id": user_session.id},
256         )
257         _raise_refresh_error("检测到刷新凭证重复使用, 请重新登录")
258
259     if user_session.expires_at <= now or user_session.user.status != "active":
260         user_session.revoked_at = now
261         session.commit()
262         _raise_refresh_error("刷新凭证无效或已过期")
263
264     user_session.revoked_at = now
265     access_token, next_refresh_token = create_user_session(session, user_session.user)
266     session.commit()
267     set_refresh_cookie(response, next_refresh_token)
268     return AccessTokenResponse(
269         access_token=access_token,
270     )
271
272
273 @router.post("/logout", response_model=MessageResponse)
274 def logout(
275     response: Response,
276     refresh_token: str | None = Cookie(default=None, alias=REFRESH_COOKIE_NAME),
277     session: Session = Depends(get_session),
278 ):
279     if not refresh_token:
280         clear_refresh_cookie(response)
281         return MessageResponse(message="已退出登录")
282
283     token_hash = hash_token(refresh_token)
284     user_session = session.scalar(
285         select(UserSession).where(UserSession.refresh_token_hash == token_hash),
286     )
287     if user_session and user_session.revoked_at is None:
288         user_session.revoked_at = utc_now()
289         session.commit()
290
291     clear_refresh_cookie(response)
292     return MessageResponse(message="已退出登录")
293
```

===== 文件: backend/app/config.py =====

backend/app/config.py

```
1 from functools import lru_cache
2 from pathlib import Path
3 from urllib.parse import quote_plus
4
5 from pydantic import Field, model_validator
6 from pydantic_settings import BaseSettings, SettingsConfigDict
7
8
9 BASE_DIR = Path(__file__).resolve().parents[1]
10 JWT_SECRET_PLACEHOLDER = "replace_with_a_random_secret"
11 MIN_JWT_SECRET_BYTES = 32
12
13
14 def validate_jwt_secret_key(value: str | None, app_env: str) -> str:
15     """校验 JWT 密钥，避免应用在空密钥或弱密钥下运行。"""
16     environment = (app_env or "development").strip().lower()
17     secret = (value or "").strip()
18
19     if not secret:
20         if environment == "production":
21             raise ValueError("生产环境必须配置 JWT_SECRET_KEY")
22         raise ValueError(f"{environment} 环境必须配置 JWT_SECRET_KEY")
23     if secret == JWT_SECRET_PLACEHOLDER:
24         raise ValueError("JWT_SECRET_KEY 仍是配置模板占位值，请替换为随机密钥")
25     if len(secret.encode("utf-8")) < MIN_JWT_SECRET_BYTES:
26         raise ValueError(f"JWT_SECRET_KEY 至少需要 {MIN_JWT_SECRET_BYTES} 字节")
27     return secret
28
29
30 class Settings(BaseSettings):
31     app_env: str = Field("development", validation_alias="APP_ENV")
32     max_dataset_file_size_mb: int = Field(100, ge=1, validation_alias="MAX_DATASET_FILE_SIZE_MB")
33     max_matrix_rows: int = Field(50000, ge=1, validation_alias="MAX_MATRIX_ROWS")
34     max_matrix_columns: int = Field(5000, ge=1, validation_alias="MAX_MATRIX_COLUMNS")
35     max_mat_variables: int = Field(100, ge=1, validation_alias="MAX_MAT_VARIABLES")
36     user_storage_quota_mb: int = Field(2048, ge=1, validation_alias="USER_STORAGE_QUOTA_MB")
37     dataset_parse_timeout_seconds: float = Field(30.0, gt=0, validation_alias="DATASET_PARSE_TIMEOUT_SECONDS")
38     db_host: str = Field("127.0.0.1", validation_alias="DB_HOST")
39     db_port: int = Field(3306, validation_alias="DB_PORT")
40     db_name: str = Field("soft_web", validation_alias="DB_NAME")
41     db_user: str = Field("root", validation_alias="DB_USER")
42     db_password: str = Field("", validation_alias="DB_PASSWORD")
43     db_charset: str = Field("utf8mb4", validation_alias="DB_CHARSET")
44     db_pool_size: int = Field(5, validation_alias="DB_POOL_SIZE")
45     db_max_overflow: int = Field(10, validation_alias="DB_MAX_OVERFLOW")
46     db_pool_recycle: int = Field(1800, validation_alias="DB_POOL_RECYCLE")
47     db_echo_sql: bool = Field(False, validation_alias="DB_ECHO_SQL")
48
49     jwt_secret_key: str | None = Field(None, validation_alias="JWT_SECRET_KEY")
50     jwt_algorithm: str = Field("HS256", validation_alias="JWT_ALGORITHM")
```

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```
51     access_token_expire_minutes: int = Field(30, validation_alias="ACCESS_TOKEN_EXPIRE_MINUTES")
52     refresh_token_expire_days: int = Field(7, validation_alias="REFRESH_TOKEN_EXPIRE_DAYS")
53     dataset_storage_dir: Path = Field(
54         BASE_DIR / "storage" / "datasets",
55         validation_alias="DATASET_STORAGE_DIR",
56     )
57     result_storage_dir: Path = Field(
58         BASE_DIR / "storage" / "results",
59         validation_alias="RESULT_STORAGE_DIR",
60     )
61     task_executor_enabled: bool = Field(True, validation_alias="TASK_EXECUTOR_ENABLED")
62     task_poll_interval_seconds: float = Field(1.0, validation_alias="TASK_POLL_INTERVAL_SECONDS")
63     task_max_runtime_seconds: float = Field(3600.0, gt=0, validation_alias="TASK_MAX_RUNTIME_SECONDS")
64     task_heartbeat_interval_seconds: float = Field(10.0, gt=0, validation_alias="TASK_HEARTBEAT_INTERVAL_SECONDS")
65     task_heartbeat_timeout_seconds: float = Field(60.0, gt=0, validation_alias="TASK_HEARTBEAT_TIMEOUT_SECONDS")
66     turnstile_secret_key: str = Field("", validation_alias="TURNSTILE_SECRET_KEY")
67     turnstile_verify_url: str = Field(
68         "https://challenges.cloudflare.com/turnstile/v0/siteverify",
69         validation_alias="TURNSTILE_VERIFY_URL",
70     )
71     turnstile_timeout_seconds: float = Field(5.0, validation_alias="TURNSTILE_TIMEOUT_SECONDS")
72     turnstile_pass_expire_minutes: int = Field(5, validation_alias="TURNSTILE_PASS_EXPIRE_MINUTES")
73     email_verification_required: bool = Field(True, validation_alias="EMAIL_VERIFICATION_REQUIRED")
74     email_code_expire_minutes: int = Field(10, validation_alias="EMAIL_CODE_EXPIRE_MINUTES")
75     email_code_resend_cooldown_seconds: int = Field(60, validation_alias="EMAIL_CODE_RESEND_COOLDOWN_SECONDS")
76     smtp_host: str = Field("smtp.qq.com", validation_alias="SMTP_HOST")
77     smtp_port: int = Field(465, validation_alias="SMTP_PORT")
78     smtp_username: str = Field("", validation_alias="SMTP_USERNAME")
79     smtp_password: str = Field("", validation_alias="SMTP_PASSWORD")
80     smtp_from_email: str = Field("", validation_alias="SMTP_FROM_EMAIL")
81     smtp_from_name: str = Field("新材料数据分析系统", validation_alias="SMTP_FROM_NAME")
82     smtp_use_ssl: bool = Field(True, validation_alias="SMTP_USE_SSL")
83     smtp_timeout_seconds: float = Field(10.0, validation_alias="SMTP_TIMEOUT_SECONDS")
84
85     cors_origins: str = Field(
86         "http://localhost:5173,http://127.0.0.1:5173,http://localhost:5174,http://127.0.0.1:5174",
87         validation_alias="CORS_ORIGINS",
88     )
89
90     model_config = SettingsConfigDict(
91         env_file=str(BASE_DIR / "config" / "database.env"),
92         env_file_encoding="utf-8",
93         extra="ignore",
94     )
95
96     @model_validator(mode="after")
97     def validate_security_settings(self):
98         self.jwt_secret_key = validate_jwt_secret_key(self.jwt_secret_key, self.app_env)
99         return self
100
```


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```
101     @property
102     def database_url(self) -> str:
103         password = quote_plus(self.db_password)
104         return (
105             f"mysql+pymysql://{self.db_user}:{password}"
106             f"@{self.db_host}:{self.db_port}/{self.db_name}?charset={self.db_charset}"
107         )
108
109     @property
110     def allowed_origins(self) -> list[str]:
111         return [origin.strip() for origin in self.cors_origins.split(",") if origin.strip()]
112
113
114 @lru_cache
115 def get_settings() -> Settings:
116     return Settings()
117
```

```
# ===== 文件: backend/app/database.py =====
1 from collections.abc import Generator
2
3 from sqlalchemy import create_engine, event
4 from sqlalchemy.orm import DeclarativeBase, Session, sessionmaker
5 from sqlalchemy.pool import StaticPool
6
7 from .config import get_settings
8
9
10 class Base(DeclarativeBase):
11     pass
12
13
14 def make_engine(database_url: str | None = None):
15     settings = get_settings()
16     url = database_url or settings.database_url
17     kwargs = {"future": True, "echo": settings.db_echo_sql}
18
19     if url.startswith("sqlite"):
20         kwargs.update(
21             {
22                 "connect_args": {"check_same_thread": False},
23                 "poolclass": StaticPool,
24             },
25         )
26     else:
27         kwargs.update(
28             {
29                 "pool_pre_ping": True,
30                 "pool_size": settings.db_pool_size,
31                 "max_overflow": settings.db_max_overflow,
```

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```
32         "pool_recycle": settings.db_pool_recycle,
33     },
34 )
35
36 created_engine = create_engine(url, **kwargs)
37
38 if created_engine.dialect.name == "mysql":
39     @event.listens_for(created_engine, "connect")
40     def _set_mysql_utc(dbapi_connection, connection_record) -> None:
41         """让数据库默认时间函数和直接 SQL 写入都使用 UTC。"""
42         cursor = dbapi_connection.cursor()
43         try:
44             cursor.execute("SET time_zone = '+00:00'")
45         finally:
46             cursor.close()
47
48     return created_engine
49
50
51 def make_session_factory(engine):
52     return sessionmaker(bind=engine, autoflush=False, autocommit=False, future=True)
53
54
55 engine = make_engine()
56 SessionLocal = make_session_factory(engine)
57
58
59 def get_session() -> Generator[Session, None, None]:
60     with SessionLocal() as session:
61         yield session
62
63
64 # ===== 文件: backend/app/datasets.py =====
65 1 import asyncio
66 2 import hashlib
67 3 import io
68 4 import json
69 5 import logging
70 6 import multiprocessing
71 7 import os
72 8 import pickle
73 9 import re
74 10 import shutil
75 11 import tempfile
76 12 import zipfile
77 13 from dataclasses import dataclass
78 14 from datetime import datetime, timezone
79 15 from pathlib import Path
80 16
81 17 import numpy as np
```

backend/app/datasets.py

```
18 import scipy.io as sio
19 from fastapi import APIRouter, Depends, File, Form, HTTPException, Query, UploadFile
20 from fastapi.responses import StreamingResponse
21 from sqlalchemy import asc, case, desc, func, select
22 from sqlalchemy.exc import IntegrityError
23 from sqlalchemy.orm import Session
24
25 from .auth import get_current_user
26 from .config import get_settings
27 from .database import get_session
28 from .models import AnalysisTask, Dataset, DatasetQuality, DatasetRevision, OperationLog, User
29 from .storage_cleanup import enqueue_cleanups, process_pending_cleanup_jobs
30 from .time_utils import format_utc_iso, utc_now
31 from .schemas import (
32     DatasetBulkRequest,
33     DatasetCatalogItemResponse,
34     DatasetCatalogPageResponse,
35     DatasetParseResponse,
36     DatasetRenameRequest,
37     DatasetRevisionResponse,
38     MessageResponse,
39 )
40
41 router = APIRouter(prefix="/api/datasets", tags=["datasets"])
42 LABEL_VARIABLE_NAMES = ("y", "label", "labels")
43 SAFE_FILENAME_PATTERN = re.compile(r"^[0-9A-Za-z._-]+")
44 UPLOAD_CHUNK_SIZE = 1024 * 1024
45 logger = logging.getLogger(__name__)
46
47
48 class MatValidationError(ValueError):
49     def __init__(self, message: str, status_code: int = 400):
50         super().__init__(message)
51         self.status_code = status_code
52
53
54 @dataclass(frozen=True)
55 class MatLimits:
56     max_file_size_bytes: int
57     max_matrix_rows: int
58     max_matrix_columns: int
59     max_variables: int
60     timeout_seconds: float
61
62
63 @dataclass
64 class StagedUpload:
65     path: Path
66     filename: str
67     size: int
```

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```
68     file_hash: str
69
70     def cleanup(self) -> None:
71         # 临时文件不属于数据库事务；无论解析或提交在哪一步失败，都必须单独清理。
72         try:
73             self.path.unlink(missing_ok=True)
74         except OSError:
75             logger.warning("清理临时数据集文件失败：%s", self.path.name)
76
77
78     def _variable_shape(value) -> list[int]:
79         return [int(dim) for dim in getattr(value, "shape", [])]
80
81
82     def _format_label(value) -> str:
83         if isinstance(value, np.generic):
84             value = value.item()
85         if isinstance(value, bytes):
86             value = value.decode(errors="ignore")
87         if isinstance(value, float) and value.is_integer():
88             return str(int(value))
89         return str(value)
90
91
92     def _flatten_values(value) -> np.ndarray:
93         values = np.asarray(value).squeeze().reshape(-1)
94         if np.issubdtype(values.dtype, np.floating):
95             values = values[~np.isnan(values)]
96         return values
97
98
99     def _find_label_variable(mat: dict):
100         for key in LABEL_VARIABLE_NAMES:
101             if key in mat:
102                 return key, mat[key]
103
104         for name, value in mat.items():
105             if name.startswith("__"):
106                 continue
107             if name.lower() in LABEL_VARIABLE_NAMES:
108                 return name, value
109
110         return None, None
111
112
113     def _build_label_distribution(label_values) -> list[dict]:
114         labels = _flatten_values(label_values)
115         if labels.size == 0:
116             return []
117
```

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```
118     unique_labels, counts = np.unique(labels, return_counts=True)
119     total = int(counts.sum())
120
121     return [
122         {
123             "label": _format_label(label),
124             "count": int(count),
125             "percent": round(int(count) * 100 / total) if total else 0,
126         }
127         for label, count in zip(unique_labels, counts)
128     ]
129
130
131 def _format_value_range(values: np.ndarray) -> str:
132     if values.size == 0:
133         return ""
134
135     if np.issubdtype(values.dtype, np.number):
136         numeric_values = values.astype(float)
137         if numeric_values.size == 0:
138             return ""
139         return f"{{format_label(float(numeric_values.min()))}} - {{format_label(float(numeric_values.max()))}}"
140
141     unique_values = np.unique(values)
142     if unique_values.size == 1:
143         return _format_label(unique_values[0])
144     return f"{{format_label(unique_values[0])}} - {{format_label(unique_values[-1])}}"
145
146
147 def _build_cluster_stats(matrix) -> list[dict]:
148     values = np.asarray(matrix)
149     if values.ndim < 2:
150         return []
151
152     stats = []
153     for index in range(values.shape[1]):
154         column_values = _flatten_values(values[:, index])
155         stats.append(
156             {
157                 "name": f"base_{{index + 1}}",
158                 "clusterCount": int(np.unique(column_values).size),
159                 "range": _format_value_range(column_values),
160             },
161         )
162
163     return stats
164
165
166 def _safe_filename(filename: str) -> str:
167     source = Path(filename).name
```

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```
168     suffix = Path(source).suffix.lower()
169     stem = Path(source).stem or "dataset"
170     safe_stem = SAFE_FILENAME_PATTERN.sub("_", stem).strip(".") or "dataset"
171     return f"{safe_stem[:80]}{suffix[:16]}"
172
173
174 def _dataset_name(filename: str) -> str:
175     return (Path(filename).stem or "未命名数据集")[:128]
176
177
178 def _mat_limits() -> MatLimits:
179     settings = get_settings()
180     return MatLimits(
181         max_file_size_bytes=int(settings.max_dataset_file_size_mb) * 1024 * 1024,
182         max_matrix_rows=int(settings.max_matrix_rows),
183         max_matrix_columns=int(settings.max_matrix_columns),
184         max_variables=int(settings.max_mat_variables),
185         timeout_seconds=float(settings.dataset_parse_timeout_seconds),
186     )
187
188
189 def _visible_mat_variables(mat: dict) -> dict:
190     return {name: value for name, value in mat.items() if not name.startswith("__")}
191
192
193 def _validate_loaded_mat(mat: dict, limits: MatLimits) -> None:
194     """在子进程内完成结构校验，避免畸形数组进入主进程和数据库事务。"""
195     variables = _visible_mat_variables(mat)
196     if not variables:
197         raise MatValidationError(".mat 文件不包含可用变量")
198     if len(variables) > limits.max_variables:
199         raise MatValidationError(
200             f".mat 文件变量数量超过限制（最多 {limits.max_variables} 个）",
201             status_code=413,
202         )
203
204     for name, value in variables.items():
205         if not isinstance(value, np.ndarray):
206             raise MatValidationError(f"变量 "{name}" 结构异常，仅支持 NumPy 数组")
207         if value.dtype.hasobject or value.dtype.kind in {"O", "V"}:
208             raise MatValidationError(f"变量 "{name}" 包含不支持的对象类型")
209         if value.ndim not in {1, 2}:
210             raise MatValidationError(f"变量 "{name}" 结构异常，仅支持一维或二维数组")
211         if any(int(dimension) <= 0 for dimension in value.shape):
212             raise MatValidationError(f"变量 "{name}" 包含空维度")
213
214         if value.ndim == 1 and value.shape[0] > limits.max_matrix_rows:
215             raise MatValidationError(
216                 f"变量 "{name}" 长度超过限制（最多 {limits.max_matrix_rows} 行）",
217                 status_code=413,
```

```
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218         )
219         if value.ndim == 2:
220             rows, columns = (int(value.shape[0]), int(value.shape[1]))
221             if rows > limits.max_matrix_rows:
222                 raise MatValidationError(
223                     f"矩阵行数超过限制（最多 {limits.max_matrix_rows} 行）",
224                     status_code=413,
225                 )
226             if columns > limits.max_matrix_columns:
227                 raise MatValidationError(
228                     f"矩阵列数超过限制（最多 {limits.max_matrix_columns} 列）",
229                     status_code=413,
230                 )
231
232         if np.issubdtype(value.dtype, np.number):
233             try:
234                 finite = np.isfinite(value)
235             except TypeError as exc:
236                 raise MatValidationError(f"变量“{name}”数值结构异常") from exc
237             if not bool(np.all(finite)):
238                 raise MatValidationError(f"变量“{name}”包含 NaN 或 Inf")
239
240         variable_info = {
241             name: {"shape": _variable_shape(value), "dtype": str(value.dtype)}
242             for name, value in variables.items()
243         }
244         main_variable, _ = _find_main_variable(mat, variable_info)
245         if not main_variable:
246             raise MatValidationError("未检测到有效的二维基础聚类矩阵")
247         main_matrix = variables[main_variable]
248         if main_matrix.ndim != 2 or not np.issubdtype(main_matrix.dtype, np.number):
249             raise MatValidationError("基础聚类矩阵必须是二维数值数组")
250
251
252 def _mat_parse_worker(
253     source_path: str,
254     result_path: str,
255     filename: str,
256     max_matrix_rows: int,
257     max_matrix_columns: int,
258     max_variables: int,
259     result_queue,
260 ) -> None:
261     limits = MatLimits(
262         max_file_size_bytes=0,
263         max_matrix_rows=max_matrix_rows,
264         max_matrix_columns=max_matrix_columns,
265         max_variables=max_variables,
266         timeout_seconds=0,
267     )
```

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```

268     try:
269         mat = sio.loadmat(source_path)
270         _validate_loaded_mat(mat, limits)
271         parsed = _parse_loaded_mat(filename, mat)
272         if result_path:
273             with Path(result_path).open("wb") as output:
274                 pickle.dump(_visible_mat_variables(mat), output, protocol=pickle.HIGHEST_PROTOCOL)
275             result_queue.put({"ok": True, "parsed": parsed})
276     except MatValidationError as exc:
277         result_queue.put(
278             {
279                 "ok": False,
280                 "statusCode": exc.status_code,
281                 "message": str(exc),
282             },
283         )
284     except Exception:
285         result_queue.put(
286             {
287                 "ok": False,
288                 "statusCode": 400,
289                 "message": "无法解析 .mat 文件",
290             },
291         )
292
293
294 def _controlled_parse_mat_file(
295     source_path: Path,
296     filename: str,
297     *,
298     include_mat: bool = False,
299     limits: MatLimits | None = None,
300 ) -> tuple[dict, dict | None]:
301     """以受控 spawn 子进程解析 MAT 文件，并由父进程负责超时和资源回收。"""
302     limits = limits or _mat_limits()
303     try:
304         file_size = source_path.stat().st_size
305     except OSError as exc:
306         raise HTTPException(400, "上传文件临时副本不可用") from exc
307     if file_size <= 0:
308         raise HTTPException(400, "上传文件为空")
309     if file_size > limits.max_file_size_bytes:
310         raise HTTPException(
311             413,
312             f"文件大小超过限制 (最大 {limits.max_file_size_bytes // (1024 * 1024)} MB)",
313         )
314
315     result_path: Path | None = None
316     if include_mat:
317         result_file = tempfile.NamedTemporaryFile(prefix="soft-web-mat-", suffix=".pickle", delete=False)

```



```
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```

```
318         result_path = Path(result_file.name)
319         result_file.close()
320
321     context = multiprocessing.get_context("spawn")
322     result_queue = context.Queue(maxsize=1)
323     process = context.Process(
324         target=_mat_parse_worker,
325         args=(
326             str(source_path),
327             str(result_path) if result_path else "",
328             _safe_filename(filename),
329             limits.max_matrix_rows,
330             limits.max_matrix_columns,
331             limits.max_variables,
332             result_queue,
333         ),
334     )
335
336     try:
337         process.start()
338         process.join(limits.timeout_seconds)
339         if process.is_alive():
340             process.terminate()
341             process.join(2)
342             if process.is_alive():
343                 process.kill()
344                 process.join()
345             raise HTTPException(408, "数据集解析超时, 请检查文件复杂度")
346
347     try:
348         result = result_queue.get(timeout=2)
349     except Exception as exc:
350         raise HTTPException(400, "无法解析 .mat 文件") from exc
351
352     if not result.get("ok"):
353         status_code = int(result.get("statusCode") or 400)
354         message = str(result.get("message") or "无法解析 .mat 文件")
355         raise HTTPException(status_code, message)
356
357     parsed = result["parsed"]
358     parsed_mat = None
359     if result_path is not None:
360         with result_path.open("rb") as source:
361             parsed_mat = pickle.load(source)
362     return parsed, parsed_mat
363 finally:
364     if result_path is not None:
365         try:
366             result_path.unlink(missing_ok=True)
367         except OSError:
```

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```
368         logger.warning("清理矩阵临时交换文件失败：%s", result_path.name)
369     result_queue.close()
370     result_queue.join_thread()
371     if process.pid is not None:
372         process.close()
373
374
375 def _find_main_variable(mat: dict, variables: dict) -> tuple[str | None, list[int]]:
376     # 默认优先使用 E; 没有 E 时, 使用面积最大的二维矩阵作为基础聚类矩阵。
377     main_var = None
378     main_shape = [0, 0]
379     if "E" in mat and len(_variable_shape(mat["E"])) >= 2:
380         main_var = "E"
381         main_shape = _variable_shape(mat["E"])
382
383     for name, info in variables.items():
384         if main_var == "E":
385             break
386         shape = info["shape"]
387         if len(shape) >= 2 and shape[0] * shape[1] > main_shape[0] * main_shape[1]:
388             main_var = name
389             main_shape = shape
390
391     return main_var, main_shape
392
393
394 def _parse_loaded_mat(filename: str, mat: dict) -> dict:
395     variables = {}
396     for name, value in _visible_mat_variables(mat).items():
397         variables[name] = {
398             "shape": _variable_shape(value),
399             "dtype": str(value.dtype),
400         }
401
402     main_var, main_shape = _find_main_variable(mat, variables)
403     label_var, label_values = _find_label_variable(mat)
404     label_distribution = _build_label_distribution(label_values) if label_values is not None else []
405     has_labels = len(label_distribution) > 0
406     sample_count = int(main_shape[0]) if len(main_shape) >= 1 else 0
407     base_count = int(main_shape[1]) if len(main_shape) >= 2 else 0
408     main_matrix = mat[main_var] if main_var else None
409
410     return {
411         "fileName": filename,
412         "variables": variables,
413         "mainVariable": main_var,
414         "labelVariable": label_var,
415         "sampleCount": sample_count,
416         "baseCount": base_count,
417         "classCount": len(label_distribution),
```

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```
418         "hasLabels": has_labels,
419         "matrixShape": f"{main_var}: {sample_count} x {base_count}" if main_var else "",
420         "labelShape": f"{label_var}: {sample_count}" if label_var else "",
421         "labelDistribution": label_distribution,
422         "clusterStats": _build_cluster_stats(main_matrix) if main_matrix is not None else [],
423     }
424
425
426 def _parse_mat_content(filename: str, content: bytes) -> dict:
427     if not filename.lower().endswith(".mat"):
428         raise HTTPException(400, f"不支持的文件格式: {filename}, 目前仅支持 .mat")
429     limits = _mat_limits()
430     if len(content) > limits.max_file_size_bytes:
431         raise HTTPException(
432             413,
433             f"文件大小超过限制 (最大 {limits.max_file_size_bytes // (1024 * 1024)} MB)",
434         )
435     staged_file = tempfile.NamedTemporaryFile(prefix="soft-web-mat-", suffix=".mat", delete=False)
436     staged_path = Path(staged_file.name)
437     try:
438         staged_file.write(content)
439         staged_file.close()
440         parsed, _ = _controlled_parse_mat_file(staged_path, filename, limits=limits)
441         return parsed
442     finally:
443         try:
444             staged_file.close()
445         except OSError:
446             pass
447         try:
448             staged_path.unlink(missing_ok=True)
449         except OSError:
450             logger.warning("清理矩阵临时文件失败: %s", staged_path.name)
451
452
453 def _persist_parse_log(
454     session: Session,
455     user: User,
456     *,
457     filename: str,
458     success: bool,
459     detail: dict,
460 ) -> None:
461     """解析接口单独提交审计日志, 避免日志事务影响文件解析结果。"""
462     session.add(
463         OperationLog(
464             user_id=user.id,
465             action="dataset_parse" if success else "dataset_parse_failed",
466             level="info" if success else "error",
467             message=(f"数据集文件解析成功: " if success else "数据集文件解析失败: ") + filename,
```

```
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```

```
468         detail_json=json.dumps(detail, ensure_ascii=False),
469     ),
470 )
471 try:
472     session.commit()
473 except Exception:
474     session.rollback()
475     logger.exception("写入数据集解析操作日志失败")
476
477
478 def _build_quality_summary(parsed: dict) -> tuple[str, list[str]]:
479     issues: list[str] = []
480     sample_count = int(parsed.get("sampleCount") or 0)
481     base_count = int(parsed.get("baseCount") or 0)
482     main_variable = parsed.get("mainVariable")
483
484     if not main_variable or sample_count <= 0 or base_count <= 0:
485         return "error", ["未检测到有效的基础聚类矩阵"]
486
487     if not parsed.get("hasLabels"):
488         issues.append("未提供真实标签，部分评估指标不可用")
489
490     label_shape = parsed.get("variables", {}).get(parsed.get("labelVariable"), {}).get("shape", [])
491     if label_shape:
492         label_count = int(np.prod(label_shape))
493         if label_count != sample_count:
494             issues.append("标签数量与样本数量不一致")
495
496     return ("warning" if issues else "ready"), issues
497
498
499 def _decode_issues(value: str | None) -> list[str]:
500     if not value:
501         return []
502     try:
503         decoded = json.loads(value)
504     except json.JSONDecodeError:
505         return []
506     return [str(item) for item in decoded] if isinstance(decoded, list) else []
507
508
509 def _upsert_quality(session: Session, dataset: Dataset, parsed: dict) -> DatasetQuality:
510     status, issues = _build_quality_summary(parsed)
511     quality = session.scalar(select(DatasetQuality).where(DatasetQuality.dataset_id == dataset.id))
512     if quality is None:
513         quality = DatasetQuality(dataset_id=dataset.id, status=status, issues_json=json.dumps(issues,
514 ensure_ascii=False))
515         session.add(quality)
516     else:
517         quality.status = status
```

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```
517     quality.issues_json = json.dumps(issues, ensure_ascii=False)
518     return quality
519
520
521 def _quality_for_dataset(session: Session, dataset: Dataset, parsed: dict | None = None) -> DatasetQuality:
522     quality = session.scalar(select(DatasetQuality).where(DatasetQuality.dataset_id == dataset.id))
523     if quality is not None:
524         return quality
525
526     parsed = parsed or _parse_stored_dataset_file(dataset)
527     quality = _upsert_quality(session, dataset, parsed)
528     session.flush()
529     return quality
530
531
532 def _next_version(session: Session, dataset_id: int) -> int:
533     current = session.scalar(
534         select(func.max(DatasetRevision.version)).where(DatasetRevision.dataset_id == dataset_id),
535     )
536     return int(current or 0) + 1
537
538
539 def _record_revision(
540     session: Session,
541     dataset: Dataset,
542     parsed: dict,
543     action: str,
544     quality: DatasetQuality | None = None,
545 ) -> DatasetRevision:
546     quality = quality or _quality_for_dataset(session, dataset, parsed)
547     revision = DatasetRevision(
548         dataset_id=dataset.id,
549         version=_next_version(session, dataset.id),
550         action=action,
551         name=dataset.name,
552         original_filename=dataset.original_filename,
553         storage_path=dataset.storage_path,
554         file_hash=dataset.file_hash,
555         sample_count=dataset.sample_count,
556         base_cluster_count=dataset.base_cluster_count,
557         has_ground_truth=dataset.has_ground_truth,
558         cluster_count=dataset.cluster_count,
559         quality_status=quality.status,
560         quality_issues_json=quality.issues_json,
561     )
562     session.add(revision)
563     return revision
564
565
566 def _parse_stored_dataset_file(dataset: Dataset) -> dict:
```

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```
567     path = Path(dataset.storage_path)
568     if not path.exists():
569         return {}
570
571     try:
572         parsed, _ = _controlled_parse_mat_file(path, dataset.original_filename)
573         return parsed
574     except HTTPException:
575         return {}
576
577
578 def _stored_file_size(dataset: Dataset) -> int:
579     path = Path(dataset.storage_path)
580     if not path.exists():
581         return 0
582     return path.stat().st_size
583
584
585 def _task_summaries(
586     session: Session,
587     user_id: int,
588     dataset_ids: list[int],
589 ) -> dict[int, tuple[int, datetime | None]]:
590     if not dataset_ids:
591         return {}
592
593     rows = session.execute(
594         select(
595             AnalysisTask.dataset_id,
596             func.count(AnalysisTask.id),
597             func.max(
598                 case(
599                     (AnalysisTask.status == "succeeded", AnalysisTask.finished_at),
600                     else_=None,
601                 ),
602             ),
603         )
604         .where(
605             AnalysisTask.user_id == user_id,
606             AnalysisTask.dataset_id.in_(dataset_ids),
607         )
608         .group_by(AnalysisTask.dataset_id)
609     ).all()
610     return {int(dataset_id): (int(count), last_finished_at) for dataset_id, count, last_finished_at in rows}
611
612
613 def _catalog_item(
614     session: Session,
615     dataset: Dataset,
616     parsed: dict | None = None,
```

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```
617     quality: DatasetQuality | None = None,
618     task_summary: tuple[int, datetime | None] | None = None,
619 ) -> DatasetCatalogItemResponse:
620     parsed = parsed or _parse_stored_dataset_file(dataset)
621     sample_count = int(parsed.get("sampleCount") or dataset.sample_count)
622     base_count = int(parsed.get("baseCount") or dataset.base_cluster_count)
623     has_labels = bool(parsed.get("hasLabels", dataset.has_ground_truth))
624     class_count = int(parsed.get("classCount") or dataset.cluster_count or 0) if has_labels else 0
625     quality = quality or _quality_for_dataset(session, dataset, parsed)
626     current_version = session.scalar(
627         select(func.max(DatasetRevision.version)).where(DatasetRevision.dataset_id == dataset.id),
628     )
629     task_count, last_analysis_at = task_summary or _task_summaries(
630         session,
631         dataset.user_id,
632         [dataset.id],
633     ).get(dataset.id, (0, None))
634
635     return DatasetCatalogItemResponse(
636         id=dataset.id,
637         name=dataset.name,
638         createdAt=format_utc_iso(dataset.created_at) or "",
639         fileSizeBytes=_stored_file_size(dataset),
640         sampleCount=sample_count,
641         baseCount=base_count,
642         classCount=class_count,
643         hasLabels=has_labels,
644         taskCount=task_count,
645         lastAnalysisAt=format_utc_iso(last_analysis_at),
646         version=int(current_version or 1),
647         qualityStatus=quality.status,
648         qualityIssues=_decode_issues(quality.issues_json),
649         matrixShape=parsed.get("matrixShape") or f"E: {sample_count} x {base_count}",
650         labelShape=parsed.get("labelShape") or (f"y: {sample_count}" if has_labels else ""),
651         labelDistribution=parsed.get("labelDistribution") or [],
652         clusterStats=parsed.get("clusterStats") or [],
653     )
654
655
656 async def _stage_upload(file: UploadFile) -> StagedUpload:
657     """先分块写入临时文件并校验大小，解析和正式落盘阶段不再持有完整文件字节。"""
658     filename = file.filename or ""
659     if not filename:
660         raise HTTPException(400, "未选择文件")
661     if Path(filename).suffix.lower() != ".mat":
662         raise HTTPException(400, f"不支持的文件格式: {filename}，目前仅支持 .mat")
663
664     limits = _mat_limits()
665     staged_file = tempfile.NamedTemporaryFile(prefix="soft-web-upload-", suffix=".mat", delete=False)
666     staged_path = Path(staged_file.name)
```

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```
667     digest = hashlib.sha256()
668     size = 0
669     try:
670         while True:
671             chunk = await file.read(UPLOAD_CHUNK_SIZE)
672             if not chunk:
673                 break
674             size += len(chunk)
675             if size > limits.max_file_size_bytes:
676                 raise HTTPException(
677                     413,
678                     f"文件大小超过限制（最大 {limits.max_file_size_bytes // (1024 * 1024)} MB）",
679                 )
680             staged_file.write(chunk)
681             digest.update(chunk)
682     staged_file.close()
683     if size == 0:
684         raise HTTPException(400, "上传文件为空")
685     return StagedUpload(
686         path=staged_path,
687         filename=filename,
688         size=size,
689         file_hash=digest.hexdigest(),
690     )
691 except Exception:
692     try:
693         staged_file.close()
694     except OSError:
695         pass
696     try:
697         staged_path.unlink(missing_ok=True)
698     except OSError:
699         logger.warning("清理上传临时文件失败: %s", staged_path.name)
700     raise
701
702
703 def _storage_directory(user_id: int) -> Path:
704     settings = get_settings()
705     return Path(settings.dataset_storage_dir) / str(user_id)
706
707
708 def _storage_usage_bytes(user_id: int) -> int:
709     storage_dir = _storage_directory(user_id)
710     if not storage_dir.exists():
711         return 0
712     total = 0
713     for path in storage_dir.rglob("*"):
714         try:
715             if path.is_file():
716                 total += path.stat().st_size
```


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```
717         except OSError:
718             continue
719     return total
720
721
722 def _ensure_user_storage_quota(user_id: int, additional_bytes: int) -> None:
723     settings = get_settings()
724     quota_bytes = int(settings.user_storage_quota_mb) * 1024 * 1024
725     used_bytes = _storage_usage_bytes(user_id)
726     if used_bytes + additional_bytes > quota_bytes:
727         used_mb = round(used_bytes / (1024 * 1024), 2)
728         quota_mb = int(settings.user_storage_quota_mb)
729         raise HTTPException(
730             413,
731             f"用户存储空间不足: 已使用 {used_mb} MB, 配额为 {quota_mb} MB",
732         )
733
734
735 def _file_sha256(path: Path) -> str:
736     digest = hashlib.sha256()
737     with path.open("rb") as source:
738         while True:
739             chunk = source.read(UPLOAD_CHUNK_SIZE)
740             if not chunk:
741                 break
742             digest.update(chunk)
743     return digest.hexdigest()
744
745
746 def _save_staged_upload(user_id: int, staged: StagedUpload, filename: str | None = None) -> tuple[str, Path]:
747     """通过同目录临时文件完成原子落盘，避免数据库记录指向未写完的文件。"""
748     filename = filename or staged.filename
749     storage_dir = _storage_directory(user_id)
750     storage_dir.mkdir(parents=True, exist_ok=True)
751
752     timestamp = datetime.now(timezone.utc).strftime("%Y%m%d%H%M%S%f")
753     storage_path = storage_dir / f"{timestamp}_{staged.file_hash[:12]}_{_safe_filename(filename)}"
754     temporary_storage = tempfile.NamedTemporaryFile(
755         prefix="soft-web-storage-",
756         suffix=".tmp",
757         dir=storage_dir,
758         delete=False,
759     )
760     temporary_path = Path(temporary_storage.name)
761     try:
762         temporary_storage.close()
763         with staged.path.open("rb") as source, temporary_path.open("wb") as destination:
764             shutil.copyfileobj(source, destination, length=UPLOAD_CHUNK_SIZE)
765         os.replace(temporary_path, storage_path)
766     except Exception:
```

```
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```

```
767         try:
768             temporary_path.unlink(missing_ok=True)
769         except OSError:
770             logger.warning("清理存储临时文件失败: %s", temporary_path.name)
771         raise
772
773     return staged.file_hash, storage_path
774
775
776 def _raw_label_values(value) -> np.ndarray:
777     return np.asarray(value).squeeze().reshape(-1)
778
779
780 def _merge_label_values(current_value, incoming_value) -> np.ndarray:
781     current_labels = _raw_label_values(current_value)
782     incoming_labels = _raw_label_values(incoming_value)
783     combined_labels = np.concatenate([current_labels, incoming_labels], axis=0)
784
785     current_array = np.asarray(current_value)
786     if current_array.ndim == 2 and current_array.shape[1] == 1:
787         return combined_labels.reshape(-1, 1)
788     if current_array.ndim == 2 and current_array.shape[0] == 1:
789         return combined_labels.reshape(1, -1)
790     return combined_labels
791
792
793 def _build_appended_mat_file(
794     current_path: Path,
795     current_filename: str,
796     incoming_path: Path,
797     incoming_filename: str,
798     output_path: Path,
799     limits: MatLimits,
800 ) -> dict:
801     """在独立输出文件中合并矩阵；原版本只有在数据库提交成功后才进入清理队列。"""
802     current_parsed, current_mat = _controlled_parse_mat_file(
803         current_path,
804         current_filename,
805         include_mat=True,
806         limits=limits,
807     )
808     incoming_parsed, incoming_mat = _controlled_parse_mat_file(
809         incoming_path,
810         incoming_filename,
811         include_mat=True,
812         limits=limits,
813     )
814     if current_mat is None or incoming_mat is None:
815         raise HTTPException(400, "无法读取追加数据的矩阵内容")
816
```

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```
817     current_main_var = current_parsed.get("mainVariable")
818     incoming_main_var = incoming_parsed.get("mainVariable")
819     if not current_main_var:
820         raise HTTPException(400, "当前数据集缺少有效的基础聚类矩阵，无法追加")
821     if not incoming_main_var:
822         raise HTTPException(400, "追加文件缺少有效的基础聚类矩阵")
823
824     current_matrix = np.asarray(current_mat[current_main_var])
825     incoming_matrix = np.asarray(incoming_mat[incoming_main_var])
826     if current_matrix.ndim < 2 or incoming_matrix.ndim < 2:
827         raise HTTPException(400, "基础聚类矩阵必须是二维结构，无法追加")
828     if current_matrix.shape[1:] != incoming_matrix.shape[1:]:
829         raise HTTPException(400, "追加文件的基础聚类矩阵列数必须与当前数据集一致")
830
831     current_has_labels = bool(current_parsed.get("hasLabels"))
832     incoming_has_labels = bool(incoming_parsed.get("hasLabels"))
833     if current_has_labels != incoming_has_labels:
834         raise HTTPException(400, "追加文件的标签状态必须与当前数据集一致")
835
836     combined_mat = _visible_mat_variables(current_mat)
837     combined_mat[current_main_var] = np.concatenate([current_matrix, incoming_matrix], axis=0)
838
839     if current_has_labels:
840         current_label_var = current_parsed.get("labelVariable")
841         incoming_label_var = incoming_parsed.get("labelVariable")
842         if not current_label_var or not incoming_label_var:
843             raise HTTPException(400, "追加文件的标签变量无法识别")
844
845         current_labels = _raw_label_values(current_mat[current_label_var])
846         incoming_labels = _raw_label_values(incoming_mat[incoming_label_var])
847         if current_labels.size != int(current_parsed.get("sampleCount") or 0):
848             raise HTTPException(400, "当前数据集的标签数量与样本数量不一致，无法追加")
849         if incoming_labels.size != int(incoming_parsed.get("sampleCount") or 0):
850             raise HTTPException(400, "追加文件的标签数量与样本数量不一致")
851
852         combined_mat[current_label_var] = _merge_label_values(
853             current_mat[current_label_var],
854             incoming_mat[incoming_label_var],
855         )
856
857     _validate_loaded_mat(combined_mat, limits)
858     sio.savemat(str(output_path), combined_mat)
859     output_size = output_path.stat().st_size
860     if output_size > limits.max_file_size_bytes:
861         raise HTTPException(
862             413,
863             f"追加后的文件大小超过限制（最大 {limits.max_file_size_bytes // (1024 * 1024)} MB）",
864         )
865     parsed, _ = _controlled_parse_mat_file(output_path, current_filename, limits=limits)
866     return parsed
```

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```
867
868
869 def _get_user_dataset(session: Session, dataset_id: int, user: User) -> Dataset:
870     dataset = session.scalar(
871         select(Dataset).where(Dataset.id == dataset_id, Dataset.user_id == user.id),
872     )
873     if dataset is None:
874         raise HTTPException(404, "数据集不存在")
875     return dataset
876
877
878 def _find_name_conflict(
879     session: Session,
880     user: User,
881     name: str,
882     exclude_dataset_id: int | None = None,
883 ) -> Dataset | None:
884     stmt = select(Dataset).where(
885         Dataset.user_id == user.id,
886         func.lower(Dataset.name) == name.strip().lower(),
887     )
888     if exclude_dataset_id is not None:
889         stmt = stmt.where(Dataset.id != exclude_dataset_id)
890     return session.scalar(stmt)
891
892
893 def _ensure_quality_records(session: Session, datasets: list[Dataset]) -> None:
894     updated = False
895     for dataset in datasets:
896         quality = session.scalar(select(DatasetQuality.id).where(DatasetQuality.dataset_id == dataset.id))
897         if quality is None:
898             _quality_for_dataset(session, dataset)
899             updated = True
900     if updated:
901         session.commit()
902
903
904 def _delete_datasets(session: Session, datasets: list[Dataset]) -> set[tuple[str, Path]]:
905     """删除前检查正式任务引用，并把所有版本文件登记到事务性清理队列。"""
906     dataset_ids = [dataset.id for dataset in datasets]
907     task_count = session.scalar(
908         select(func.count(AnalysisTask.id)).where(AnalysisTask.dataset_id.in_(dataset_ids)),
909     )
910     if task_count:
911         raise HTTPException(409, f"数据集仍被 {task_count} 个分析任务引用，请先删除关联任务后再删除数据集")
912
913     revision_paths = session.scalars(
914         select(DatasetRevision.storage_path).where(DatasetRevision.dataset_id.in_(dataset_ids)),
915     ).all()
916     cleanup_paths = {("dataset", Path(path)) for path in revision_paths}
```

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```
917     cleanup_paths.update(("dataset", Path(dataset.storage_path)) for dataset in datasets)
918
919     # 文件清理任务和数据库删除必须处于同一事务; 事务回滚时文件仍保持可用。
920     enqueue_cleanups(session, cleanup_paths)
921
922     for dataset in datasets:
923         session.delete(dataset)
924     try:
925         session.commit()
926     except IntegrityError as exc:
927         session.rollback()
928         raise HTTPException(409, "数据集已被分析任务引用, 请先删除关联任务后再删除数据集") from exc
929     return cleanup_paths
930
931
932 @router.get("", response_model=DatasetCatalogPageResponse)
933 def list_datasets(
934     q: str = "",
935     data_type: str | None = Query(default=None, alias="dataType"),
936     label_status: str = Query(default="all", alias="labelStatus"),
937     usage: str = "all",
938     quality_status: str = Query(default="all", alias="qualityStatus"),
939     sort_by: str = Query(default="createdAt", alias="sortBy"),
940     sort_order: str = Query(default="desc", alias="sortOrder"),
941     page: int = Query(default=1, ge=1),
942     page_size: int = Query(default=20, ge=1, le=100, alias="pageSize"),
943     user: User = Depends(get_current_user),
944     session: Session = Depends(get_session),
945 ):
946     all_user_datasets = session.scalars(select(Dataset).where(Dataset.user_id == user.id)).all()
947     _ensure_quality_records(session, all_user_datasets)
948
949     stmt = select(Dataset).where(Dataset.user_id == user.id)
950     if q.strip():
951         stmt = stmt.where(Dataset.name.ilike(f"%{q.strip()}%"))
952     if data_type == "混合":
953         stmt = stmt.where(False)
954     if label_status == "labeled":
955         stmt = stmt.where(Dataset.has_ground_truth.is_(True))
956     elif label_status == "unlabeled":
957         stmt = stmt.where(Dataset.has_ground_truth.is_(False))
958     if quality_status in {"ready", "warning", "error"}:
959         stmt = stmt.join(DatasetQuality).where(DatasetQuality.status == quality_status)
960     if usage in {"used", "unused"}:
961         referenced_dataset_ids = select(AnalysisTask.dataset_id).where(AnalysisTask.user_id == user.id)
962         if usage == "used":
963             stmt = stmt.where(Dataset.id.in_(referenced_dataset_ids))
964         else:
965             stmt = stmt.where(Dataset.id.not_in(referenced_dataset_ids))
966
```

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```
967     total = int(session.scalar(select(func.count()).select_from(stmt.subquery())) or 0)
968     task_count_subquery = (
969         select(AnalysisTask.dataset_id, func.count(AnalysisTask.id).label("task_count"))
970         .where(AnalysisTask.user_id == user.id)
971         .group_by(AnalysisTask.dataset_id)
972         .subquery()
973     )
974     sort_columns = {
975         "createdAt": Dataset.created_at,
976         "name": Dataset.name,
977         "sampleCount": Dataset.sample_count,
978         "baseCount": Dataset.base_cluster_count,
979         "taskCount": func.coalesce(task_count_subquery.c.task_count, 0),
980     }
981     sort_column = sort_columns.get(sort_by, Dataset.created_at)
982     ordering = asc(sort_column) if sort_order == "asc" else desc(sort_column)
983     page_stmt = (
984         stmt.outerjoin(task_count_subquery, task_count_subquery.c.dataset_id == Dataset.id)
985         .order_by(ordering, desc(Dataset.id))
986         .offset((page - 1) * page_size)
987         .limit(page_size)
988     )
989     datasets = session.scalars(page_stmt).all()
990     qualities = {
991         quality.dataset_id: quality
992         for quality in session.scalars(
993             select(DatasetQuality).where(DatasetQuality.dataset_id.in_([dataset.id for dataset in datasets])),
994         ).all()
995     }
996     task_summaries = _task_summaries(session, user.id, [dataset.id for dataset in datasets])
997     total_pages = max(1, (total + page_size - 1) // page_size) if total else 0
998
999     return DatasetCatalogPageResponse(
1000         items=[
1001             _catalog_item(
1002                 session,
1003                 dataset,
1004                 quality=qualities.get(dataset.id),
1005                 task_summary=task_summaries.get(dataset.id),
1006             )
1007             for dataset in datasets
1008         ],
1009         total=total,
1010         page=page,
1011         pageSize=page_size,
1012         totalPages=total_pages,
1013     )
1014
1015
1016 @router.post("/", response_model=DatasetCatalogItemResponse)
```

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```
1017 async def upload_dataset(
1018     file: UploadFile = File(...),
1019     allow_duplicate: bool = Form(default=False, alias="allowDuplicate"),
1020     user: User = Depends(get_current_user),
1021     session: Session = Depends(get_session),
1022 ):
1023     """上传、解析、落盘和数据库写入构成一个业务事务，失败时同时清理两侧状态。"""
1024     staged: StagedUpload | None = None
1025     storage_path: Path | None = None
1026     try:
1027         staged = await _stage_upload(file)
1028         dataset_name = _dataset_name(staged.filename)
1029         if not allow_duplicate and _find_name_conflict(session, user, dataset_name):
1030             raise HTTPException(409, f"已存在同名数据集 “{dataset_name}”，请使用追加数据或确认保留独立副本")
1031
1032         _ensure_user_storage_quota(user.id, staged.size)
1033         parsed, _ = await asyncio.to_thread(_controlled_parse_mat_file, staged.path, staged.filename)
1034         file_hash, storage_path = await asyncio.to_thread(_save_staged_upload, user.id, staged)
1035
1036         dataset = Dataset(
1037             user_id=user.id,
1038             name=dataset_name,
1039             original_filename=staged.filename,
1040             storage_path=str(storage_path),
1041             file_hash=file_hash,
1042             sample_count=parsed["sampleCount"],
1043             base_cluster_count=parsed["baseCount"],
1044             has_ground_truth=parsed["hasLabels"],
1045             cluster_count=parsed["classCount"] if parsed["hasLabels"] else None,
1046             status="ready",
1047         )
1048         session.add(dataset)
1049         session.flush()
1050         quality = _upsert_quality(session, dataset, parsed)
1051         _record_revision(session, dataset, parsed, "uploaded", quality)
1052         session.flush()
1053         session.refresh(dataset)
1054         response = _catalog_item(session, dataset, parsed, quality)
1055         session.commit()
1056     except Exception:
1057         session.rollback()
1058         if storage_path is not None:
1059             storage_path.unlink(missing_ok=True)
1060         raise
1061     finally:
1062         if staged is not None:
1063             staged.cleanup()
1064
1065     return response
1066
```

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```
1067
1068 @router.put("/{dataset_id}", response_model=DatasetCatalogItemResponse)
1069 async def replace_dataset_file(
1070     dataset_id: int,
1071     file: UploadFile = File(...),
1072     allow_duplicate: bool = Form(default=False, alias="allowDuplicate"),
1073     user: User = Depends(get_current_user),
1074     session: Session = Depends(get_session),
1075 ):
1076     """保存旧版本快照后切换文件引用，提交失败时继续保留当前可用版本。"""
1077     dataset = _get_user_dataset(session, dataset_id, user)
1078     staged: StagedUpload | None = None
1079     storage_path: Path | None = None
1080     try:
1081         staged = await _stage_upload(file)
1082         next_name = _dataset_name(staged.filename)
1083         if not allow_duplicate and _find_name_conflict(session, user, next_name, exclude_dataset_id=dataset.id):
1084             raise HTTPException(409, f"已存在同名数据集 “{next_name}”，请确认是否保留独立副本")
1085
1086         _ensure_user_storage_quota(user.id, staged.size)
1087         current_path = Path(dataset.storage_path)
1088         current_parsed, _ = await asyncio.to_thread(
1089             _controlled_parse_mat_file,
1090             current_path,
1091             dataset.original_filename,
1092         )
1093         current_quality = _quality_for_dataset(session, dataset, current_parsed)
1094         if _next_version(session, dataset.id) == 1:
1095             _record_revision(session, dataset, current_parsed, "uploaded", current_quality)
1096
1097         parsed, _ = await asyncio.to_thread(_controlled_parse_mat_file, staged.path, staged.filename)
1098         file_hash, storage_path = await asyncio.to_thread(_save_staged_upload, user.id, staged)
1099
1100         dataset.name = next_name
1101         dataset.original_filename = staged.filename
1102         dataset.storage_path = str(storage_path)
1103         dataset.file_hash = file_hash
1104         dataset.sample_count = parsed["sampleCount"]
1105         dataset.base_cluster_count = parsed["baseCount"]
1106         dataset.has_ground_truth = parsed["hasLabels"]
1107         dataset.cluster_count = parsed["classCount"] if parsed["hasLabels"] else None
1108         dataset.status = "ready"
1109         dataset.created_at = utc_now().replace(microsecond=0)
1110
1111         session.flush()
1112         quality = _upsert_quality(session, dataset, parsed)
1113         _record_revision(session, dataset, parsed, "replaced", quality)
1114         session.flush()
1115         session.refresh(dataset)
1116         response = _catalog_item(session, dataset, parsed, quality)
```


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```
1117         session.commit()
1118     except Exception:
1119         session.rollback()
1120         if storage_path is not None:
1121             storage_path.unlink(missing_ok=True)
1122         raise
1123     finally:
1124         if staged is not None:
1125             staged.cleanup()
1126
1127     return response
1128
1129
1130 @router.post("/{dataset_id}/append", response_model=DatasetCatalogItemResponse)
1131 async def append_dataset_file(
1132     dataset_id: int,
1133     file: UploadFile = File(...),
1134     user: User = Depends(get_current_user),
1135     session: Session = Depends(get_session),
1136 ):
1137     """校验新旧矩阵结构后生成完整新版本，不直接修改正在使用的原文件。"""
1138     dataset = _get_user_dataset(session, dataset_id, user)
1139     staged: StagedUpload | None = None
1140     output_staged: StagedUpload | None = None
1141     output_path: Path | None = None
1142     storage_path: Path | None = None
1143     try:
1144         staged = await _stage_upload(file)
1145         current_path = Path(dataset.storage_path)
1146         if not current_path.exists():
1147             raise HTTPException(409, "当前数据集文件不存在，无法追加")
1148
1149         current_parsed, _ = await asyncio.to_thread(
1150             _controlled_parse_mat_file,
1151             current_path,
1152             dataset.original_filename,
1153         )
1154         current_quality = _quality_for_dataset(session, dataset, current_parsed)
1155         if _next_version(session, dataset.id) == 1:
1156             _record_revision(session, dataset, current_parsed, "uploaded", current_quality)
1157
1158         output_file = tempfile.NamedTemporaryFile(prefix="soft-web-append-", suffix=".mat", delete=False)
1159         output_path = Path(output_file.name)
1160         output_file.close()
1161         parsed = await asyncio.to_thread(
1162             _build_appended_mat_file,
1163             current_path,
1164             dataset.original_filename,
1165             staged.path,
1166             staged.filename,
```

```
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```

```
1167         output_path,
1168         _mat_limits(),
1169     )
1170     output_staged = StagedUpload(
1171         path=output_path,
1172         filename=dataset.original_filename,
1173         size=output_path.stat().st_size,
1174         file_hash=await asyncio.to_thread(_file_sha256, output_path),
1175     )
1176     _ensure_user_storage_quota(user.id, output_staged.size)
1177     file_hash, storage_path = await asyncio.to_thread(
1178         _save_staged_upload,
1179         user.id,
1180         output_staged,
1181         dataset.original_filename,
1182     )
1183
1184     dataset.storage_path = str(storage_path)
1185     dataset.file_hash = file_hash
1186     dataset.sample_count = parsed["sampleCount"]
1187     dataset.base_cluster_count = parsed["baseCount"]
1188     dataset.has_ground_truth = parsed["hasLabels"]
1189     dataset.cluster_count = parsed["classCount"] if parsed["hasLabels"] else None
1190     dataset.status = "ready"
1191
1192     session.flush()
1193     quality = _upsert_quality(session, dataset, parsed)
1194     _record_revision(session, dataset, parsed, "appended", quality)
1195     session.flush()
1196     session.refresh(dataset)
1197     response = _catalog_item(session, dataset, parsed, quality)
1198     session.commit()
1199 except Exception:
1200     session.rollback()
1201     if storage_path is not None:
1202         storage_path.unlink(missing_ok=True)
1203     raise
1204 finally:
1205     if staged is not None:
1206         staged.cleanup()
1207     if output_staged is not None:
1208         output_staged.cleanup()
1209     elif output_path is not None:
1210         output_path.unlink(missing_ok=True)
1211
1212     return response
1213
1214
1215 @router.patch("/{dataset_id}", response_model=DatasetCatalogItemResponse)
1216 def rename_dataset(
```

```
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```

```
1217     dataset_id: int,
1218     payload: DatasetRenameRequest,
1219     user: User = Depends(get_current_user),
1220     session: Session = Depends(get_session),
1221 ):
1222     dataset = _get_user_dataset(session, dataset_id, user)
1223     next_name = payload.name.strip()
1224     if not payload.allowDuplicate and _find_name_conflict(session, user, next_name, exclude_dataset_id=dataset.id):
1225         raise HTTPException(409, f"已存在同名数据集 “{next_name}” , 请更换名称或确认保留重名副本")
1226
1227     dataset.name = next_name
1228     parsed = _parse_stored_dataset_file(dataset)
1229     quality = _quality_for_dataset(session, dataset, parsed)
1230     _record_revision(session, dataset, parsed, "renamed", quality)
1231     session.commit()
1232     session.refresh(dataset)
1233     return _catalog_item(session, dataset, parsed, quality)
1234
1235
1236 @router.post("/bulk-delete", response_model=MessageResponse)
1237 def bulk_delete_datasets(
1238     payload: DatasetBulkRequest,
1239     user: User = Depends(get_current_user),
1240     session: Session = Depends(get_session),
1241 ):
1242     datasets = session.scalars(
1243         select(Dataset).where(Dataset.user_id == user.id, Dataset.id.in_(payload.datasetIds)),
1244     ).all()
1245     if len(datasets) != len(set(payload.datasetIds)):
1246         raise HTTPException(404, "部分数据集不存在或无权访问")
1247
1248     cleanup_paths = _delete_datasets(session, datasets)
1249     process_pending_cleanup_jobs(limit=max(20, len(cleanup_paths)), bind=session.get_bind())
1250     return MessageResponse(message=f"已删除 {len(datasets)} 个数据集, 相关文件已进入清理流程")
1251
1252
1253 @router.post("/export")
1254 def export_datasets(
1255     payload: DatasetBulkRequest,
1256     user: User = Depends(get_current_user),
1257     session: Session = Depends(get_session),
1258 ):
1259     datasets = session.scalars(
1260         select(Dataset).where(Dataset.user_id == user.id, Dataset.id.in_(payload.datasetIds)),
1261     ).all()
1262     if len(datasets) != len(set(payload.datasetIds)):
1263         raise HTTPException(404, "部分数据集不存在或无权访问")
1264
1265     archive = io.BytesIO()
1266     with zipfile.ZipFile(archive, "w", zipfile.ZIP_DEFLATED) as zip_file:
```

backend/app/datasets.py

```
1267         for dataset in datasets:
1268             storage_path = Path(dataset.storage_path)
1269             if storage_path.exists():
1270                 zip_file.write(storage_path, arcname=f"{dataset.name}_{dataset.id}{storage_path.suffix}")
1271         archive.seek(0)
1272         return StreamingResponse(
1273             archive,
1274             media_type="application/zip",
1275             headers={"Content-Disposition": "attachment; filename=datasets-export.zip"},
1276         )
1277
1278
1279 def _revision_response(revision: DatasetRevision) -> DatasetRevisionResponse:
1280     return DatasetRevisionResponse(
1281         id=revision.id,
1282         version=revision.version,
1283         action=revision.action,
1284         name=revision.name,
1285         originalFilename=revision.original_filename,
1286         createdAt=format_utc_iso(revision.created_at) or "",
1287         sampleCount=revision.sample_count,
1288         baseCount=revision.base_cluster_count,
1289         classCount=revision.cluster_count or 0,
1290         hasLabels=revision.has_ground_truth,
1291         qualityStatus=revision.quality_status,
1292         qualityIssues=_decode_issues(revision.quality_issues_json),
1293     )
1294
1295
1296 @router.get("/{dataset_id}/versions", response_model=list[DatasetRevisionResponse])
1297 def list_dataset_versions(
1298     dataset_id: int,
1299     user: User = Depends(get_current_user),
1300     session: Session = Depends(get_session),
1301 ):
1302     _get_user_dataset(session, dataset_id, user)
1303     revisions = session.scalars(
1304         select(DatasetRevision)
1305         .where(DatasetRevision.dataset_id == dataset_id)
1306         .order_by(DatasetRevision.version.desc()),
1307     ).all()
1308     return [_revision_response(revision) for revision in revisions]
1309
1310
1311 @router.post("/{dataset_id}/quality", response_model=DatasetCatalogItemResponse)
1312 def recheck_dataset_quality(
1313     dataset_id: int,
1314     user: User = Depends(get_current_user),
1315     session: Session = Depends(get_session),
1316 ):
```

backend/app/datasets.py

```
1317     dataset = _get_user_dataset(session, dataset_id, user)
1318     parsed = _parse_stored_dataset_file(dataset)
1319     quality = _upsert_quality(session, dataset, parsed)
1320     session.commit()
1321     session.refresh(dataset)
1322     return _catalog_item(session, dataset, parsed, quality)
1323
1324
1325 @router.delete("/{dataset_id}", response_model=MessageResponse)
1326 def delete_dataset(
1327     dataset_id: int,
1328     user: User = Depends(get_current_user),
1329     session: Session = Depends(get_session),
1330 ):
1331     dataset = _get_user_dataset(session, dataset_id, user)
1332     cleanup_paths = _delete_datasets(session, [dataset])
1333     process_pending_cleanup_jobs(limit=max(20, len(cleanup_paths)), bind=session.get_bind())
1334     return MessageResponse(message="数据集已删除，相关文件已进入清理流程")
1335
1336
1337 @router.post("/parse", response_model=DatasetParseResponse)
1338 async def parse_dataset_file(
1339     file: UploadFile = File(...),
1340     user: User = Depends(get_current_user),
1341     session: Session = Depends(get_session),
1342 ):
1343     source_filename = file.filename or ""
1344     safe_filename = _safe_filename(source_filename) if source_filename else "unknown"
1345     staged: StagedUpload | None = None
1346
1347     try:
1348         staged = await _stage_upload(file)
1349         parsed, _ = await asyncio.to_thread(_controlled_parse_mat_file, staged.path, staged.filename)
1350     except HTTPException as exc:
1351         session.rollback()
1352         _persist_parse_log(
1353             session,
1354             user,
1355             filename=safe_filename,
1356             success=False,
1357             detail={"statusCode": exc.status_code, "errorType": type(exc).__name__},
1358         )
1359     logger.error("数据集文件解析失败: %s (状态码: %s)", safe_filename, exc.status_code)
1360     raise
1361 except Exception as exc:
1362     session.rollback()
1363     _persist_parse_log(
1364         session,
1365         user,
1366         filename=safe_filename,
```

backend/app/datasets.py

```
1367         success=False,
1368         detail={"statusCode": 500, "errorType": type(exc).__name__},
1369     )
1370     logger.exception("数据集文件解析出现未预期异常: %s", safe_filename)
1371     raise HTTPException(500, "数据集解析失败") from exc
1372 else:
1373     _persist_parse_log(
1374         session,
1375         user,
1376         filename=safe_filename,
1377         success=True,
1378         detail={
1379             "sampleCount": parsed.get("sampleCount", 0),
1380             "baseCount": parsed.get("baseCount", 0),
1381             "classCount": parsed.get("classCount", 0),
1382             "hasLabels": bool(parsed.get("hasLabels")),
1383         },
1384     )
1385     return parsed
1386 finally:
1387     if staged is not None:
1388         staged.cleanup()
1389
```

```
# ===== 文件: backend/app/email_verification.py =====
1 from datetime import timedelta
2 from email.message import EmailMessage
3 from email.utils import formataddr
4 import hashlib
5 from html import escape
6 import hmac
7 import logging
8 import re
9 import secrets
10 import smtplib
11 import ssl
12
13 from fastapi import HTTPException, status
14 from sqlalchemy import select
15 from sqlalchemy.orm import Session
16
17 from .config import get_settings
18 from .models import EmailVerificationCode, User
19 from .product import SOFTWARE_SHORT_NAME
20 from .security import utc_now
21
22
23 logger = logging.getLogger(__name__)
24
25 EMAIL_CODE_PURPOSE_REGISTER = "register"
```

backend/app/email_verification.py

```
26 EMAIL_PATTERN = re.compile(r"^[^\s@]+@[^\s@]+\.[^\s@]+$")
27 EMAIL_CODE_PATTERN = re.compile(r"^\d{6}$")
28 DEFAULT_EMAIL_BRAND_NAME = SOFTWARE_SHORT_NAME
29 EMAIL_BRAND_COLOR = "#2f8ff0"
30
31
32 def normalize_email(value: str | None) -> str:
33     email = (value or "").strip().lower()
34     if len(email) > 255 or not EMAIL_PATTERN.fullmatch(email):
35         raise HTTPException(status_code=status.HTTP_400_BAD_REQUEST, detail="邮箱格式不正确")
36     return email
37
38
39 def try_normalize_email(value: str | None) -> str | None:
40     email = (value or "").strip().lower()
41     if len(email) > 255 or not EMAIL_PATTERN.fullmatch(email):
42         return None
43     return email
44
45
46 def generate_email_code() -> str:
47     return f"{secrets.randbelow(1_000_000):06d}"
48
49
50 def hash_email_code(email: str, code: str) -> str:
51     settings = get_settings()
52     payload = f"{email}:{code}".encode("utf-8")
53     return hmac.new(settings.jwt_secret_key.encode("utf-8"), payload, hashlib.sha256).hexdigest()
54
55
56 def ensure_email_not_registered(session: Session, email: str) -> None:
57     exists = session.scalar(select(User.id).where(User.email == email))
58     if exists is not None:
59         raise HTTPException(status_code=status.HTTP_409_CONFLICT, detail="该邮箱已注册")
60
61
62 def get_email_brand_name() -> str:
63     brand_name = get_settings().smtp_from_name.strip()
64     return brand_name or DEFAULT_EMAIL_BRAND_NAME
65
66
67 def build_email_code_plain_text(email: str, code: str) -> str:
68     settings = get_settings()
69     brand_name = get_email_brand_name()
70     return "\n".join(
71         [
72             f"{email}, 您好: ",
73             "您的验证码是: ",
74             code,
75             f"验证码将在 {settings.email_code_expire_minutes} 分钟后失效。",
```

```
backend/app/email_verification.py
```

```
76         "如果不是您本人操作，请忽略此邮件。",
77         f"This email was sent by {brand_name}. Please do not reply directly.",
78     ],
79 )
80
81
82 def build_email_code_html(email: str, code: str) -> str:
83     settings = get_settings()
84     brand_name = get_email_brand_name()
85     email_html = escape(email, quote=True)
86     code_html = escape(code, quote=True)
87     brand_html = escape(brand_name, quote=True)
88     expire_minutes = settings.email_code_expire_minutes
89
90     return f"""<!doctype html>
91 <html lang="zh-CN">
92 <head>
93 <meta charset="utf-8">
94 <meta name="viewport" content="width=device-width, initial-scale=1">
95 <title>邮箱验证码</title>
96 </head>
97 <body style="margin:0;padding:0;background:#f1f5f9;">
98 <table role="presentation" width="100%" cellpadding="0" cellspacing="0" border="0"
99 style="width:100%;border-collapse:collapse;background:#f1f5f9;">
100 <tr>
101 <td align="center" style="padding:48px 16px;">
102 <table role="presentation" width="640" cellpadding="0" cellspacing="0" border="0"
103 style="width:100%;max-width:640px;border-collapse:separate;border-spacing:0;background:#ffffff;border-radius:16px;
104 overflow:hidden;">
105 <tr>
106 <td bgcolor="{EMAIL_BRAND_COLOR}" style="padding:34px
107 48px;background:{EMAIL_BRAND_COLOR};color:#ffffff;font-family:Lato,'Helvetica Neue',Arial,'Microsoft
108 YaHei',sans-serif;font-size:28px;line-height:1.3;font-weight:800;">
109 邮箱验证码
110 </td>
111 </tr>
112 <tr>
113 <td style="padding:44px 48px 48px;background:#ffffff;color:#1f2937;font-family:Lato,'Helvetica
114 Neue',Arial,'Microsoft YaHei',sans-serif;">
115 <p style="margin:0 0 28px;font-size:20px;line-height:1.7;color:#1f2937;">{email_html}，您好：</p>
116 <p style="margin:0 0 26px;font-size:20px;line-height:1.7;color:#1f2937;">您的验证码是：</p>
117 <table role="presentation" width="100%" cellpadding="0" cellspacing="0" border="0"
118 style="width:100%;border-collapse:collapse;">
119 <tr>
120 <td align="center" style="padding:22px 0 34px;color:#111827;font-family:Lato,'Helvetica
121 Neue',Arial,'Microsoft YaHei',sans-serif;font-size:48px;line-height:1;font-weight:800;letter-spacing:12px;white-sp
122 ace:nowrap;mso-line-height-rule:exactly;">
123 {code_html}
124 </td>
125 </tr>
126 </table>
127 </td>
128 </tr>
129 </table>
130 </body>
131 </html>
132 """
```


backend/app/email_verification.py

```
117         </table>
118         <p style="margin:0 0 28px;font-size:20px;line-height:1.7;color:#1f2937;">验证码将在 <strong
style="font-weight:800;">{expire_minutes}</strong> 分钟后失效。</p>
119         <p style="margin:0;font-size:20px;line-height:1.7;color:#1f2937;">如果不是您本人操作，请忽略此邮件。</p>
120     </td>
121 </tr>
122 <tr>
123     <td bgcolor="#f8fafc" style="padding:24px 48px
30px;background:#f8fafc;color:#6b7280;font-family:Lato,'Helvetica Neue',Arial,'Microsoft
YaHei',sans-serif;font-size:16px;line-height:1.55;">
124         This email was sent by {brand_html}. Please do not reply directly.
125     </td>
126 </tr>
127 </table>
128 </td>
129 </tr>
130 </table>
131 </body>
132 </html>"""
133
134
135 def build_register_email_message(email: str, code: str) -> EmailMessage:
136     settings = get_settings()
137     username = settings.smtp_username.strip()
138     from_email = (settings.smtp_from_email or username).strip()
139     brand_name = get_email_brand_name()
140
141     message = EmailMessage()
142     message["From"] = formataddr((brand_name, from_email))
143     message["To"] = email
144     message["Subject"] = f"[{brand_name}] 邮箱验证码"
145     message.set_content(build_email_code_plain_text(email, code))
146     message.add_alternative(build_email_code_html(email, code), subtype="html")
147     return message
148
149
150 def deliver_email_message(message: EmailMessage) -> None:
151     settings = get_settings()
152     username = settings.smtp_username.strip()
153     password = settings.smtp_password.strip()
154     from_email = (settings.smtp_from_email or username).strip()
155
156     if not settings.smtp_host.strip() or not username or not password or not from_email:
157         raise HTTPException(status_code=status.HTTP_503_SERVICE_UNAVAILABLE, detail="邮件服务未配置")
158
159     context = ssl.create_default_context()
160     try:
161         if settings.smtp_use_ssl:
162             with smtplib.SMTP_SSL(
163                 settings.smtp_host,
```

```
backend/app/email_verification.py
```

```
164         settings.smtp_port,
165         timeout=settings.smtp_timeout_seconds,
166         context=context,
167     ) as smtp:
168         smtp.login(username, password)
169         smtp.send_message(message)
170     else:
171         with smtplib.SMTP(
172             settings.smtp_host,
173             settings.smtp_port,
174             timeout=settings.smtp_timeout_seconds,
175         ) as smtp:
176             smtp.starttls(context=context)
177             smtp.login(username, password)
178             smtp.send_message(message)
179     except (OSError, smtplib.SMTPException) as exc:
180         logger.warning("Register email verification code send failed: %s", exc)
181         raise HTTPException(status_code=status.HTTP_503_SERVICE_UNAVAILABLE, detail="验证码邮件发送失败") from exc
182
183
184 def send_email_code(email: str, code: str) -> None:
185     deliver_email_message(build_register_email_message(email, code))
186
187
188 def create_register_email_code(session: Session, email: str, client_ip: str | None = None) -> None:
189     settings = get_settings()
190     now = utc_now()
191
192     ensure_email_not_registered(session, email)
193
194     recent_code = session.scalar(
195         select(EmailVerificationCode)
196         .where(
197             EmailVerificationCode.email == email,
198             EmailVerificationCode.purpose == EMAIL_CODE_PURPOSE_REGISTER,
199             EmailVerificationCode.consumed_at.is_(None),
200             EmailVerificationCode.created_at >= now - timedelta(seconds=settings.email_code_resend_cooldown_seconds),
201         )
202         .order_by(EmailVerificationCode.created_at.desc(), EmailVerificationCode.id.desc()),
203     )
204     if recent_code is not None:
205         raise HTTPException(status_code=status.HTTP_429_TOO_MANY_REQUESTS, detail="验证码发送过于频繁，请稍后再试")
206
207     for existing_code in session.scalars(
208         select(EmailVerificationCode).where(
209             EmailVerificationCode.email == email,
210             EmailVerificationCode.purpose == EMAIL_CODE_PURPOSE_REGISTER,
211             EmailVerificationCode.consumed_at.is_(None),
212         ),
213     ):
214         existing_code.consumed_at = now
```

```
backend/app/email_verification.py
```

```
214         existing_code.consumed_at = now
215
216     code = generate_email_code()
217     session.add(
218         EmailVerificationCode(
219             email=email,
220             purpose=EMAIL_CODE_PURPOSE_REGISTER,
221             code_hash=hash_email_code(email, code),
222             client_ip=client_ip,
223             expires_at=now + timedelta(minutes=settings.email_code_expire_minutes),
224             created_at=now,
225         ),
226     )
227     send_email_code(email, code)
228
229
230 def verify_register_email_code(session: Session, email: str, code: str | None) -> EmailVerificationCode:
231     normalized_code = (code or "").strip()
232     if not EMAIL_CODE_PATTERN.fullmatch(normalized_code):
233         raise HTTPException(status_code=status.HTTP_400_BAD_REQUEST, detail="邮箱验证码不正确或已过期")
234
235     now = utc_now()
236     record = session.scalar(
237         select(EmailVerificationCode)
238         .where(
239             EmailVerificationCode.email == email,
240             EmailVerificationCode.purpose == EMAIL_CODE_PURPOSE_REGISTER,
241             EmailVerificationCode.consumed_at.is_(None),
242         )
243         .order_by(EmailVerificationCode.created_at.desc(), EmailVerificationCode.id.desc()),
244     )
245     if record is None or record.expires_at <= now:
246         raise HTTPException(status_code=status.HTTP_400_BAD_REQUEST, detail="邮箱验证码不正确或已过期")
247
248     if not hmac.compare_digest(record.code_hash, hash_email_code(email, normalized_code)):
249         raise HTTPException(status_code=status.HTTP_400_BAD_REQUEST, detail="邮箱验证码不正确或已过期")
250
251     return record
252
253
254 # ===== 文件: backend/app/migrations.py =====
255 1 """Alembic 版本核验。
256 2
257 3 生产启动只允许读取数据库迁移状态，不在应用进程内执行升级或 ALTER TABLE。
258 4 部署流程应先执行 ``python -m alembic upgrade head``，再启动 FastAPI。
259 5 """
260 6
261 7 from pathlib import Path
262 8
263 9 from alembic.config import Config
```

backend/app/migrations.py

```
10 from alembic.migration import MigrationContext
11 from alembic.script import ScriptDirectory
12 from sqlalchemy import Engine, inspect
13
14
15 BACKEND_ROOT = Path(__file__).resolve().parents[1]
16 ALEMBIC_CONFIG_PATH = BACKEND_ROOT / "alembic.ini"
17
18
19 class DatabaseMigrationError(RuntimeError):
20     """数据库未升级到应用要求版本时抛出的启动错误。"""
21
22
23 def _alembic_config() -> Config:
24     """使用绝对脚本路径，避免从项目根目录启动时找不到 revisions。"""
25     config = Config(str(ALEMBIC_CONFIG_PATH))
26     config.set_main_option("script_location", str(BACKEND_ROOT / "alembic").replace("%", "%%"))
27     return config
28
29
30 def get_alembic_heads() -> tuple[str, ...]:
31     """读取代码仓库中的迁移 head，不访问数据库。"""
32     return tuple(ScriptDirectory.from_config(_alembic_config()).get_heads())
33
34
35 def ensure_database_is_current(engine: Engine) -> None:
36     """确认数据库存在且已升级到代码仓库的全部 head。
37
38     该函数只执行 ``SELECT`` 和 Alembic 版本读取，不会创建版本表、写入版本号或修改业务表。
39     """
40     expected_heads = set(get_alembic_heads())
41     if not expected_heads:
42         raise DatabaseMigrationError("代码仓库没有可用的 Alembic head，无法校验数据库版本")
43
44     with engine.connect() as connection:
45         if "alembic_version" not in inspect(connection).get_table_names():
46             raise DatabaseMigrationError(
47                 "数据库缺少 alembic_version，启动已拒绝；请先执行：python -m alembic upgrade head",
48             )
49
50         current_heads = set(MigrationContext.configure(connection).get_current_heads())
51
52     if not current_heads:
53         raise DatabaseMigrationError(
54             "数据库 alembic_version 没有版本记录，启动已拒绝；请先执行：python -m alembic upgrade head",
55         )
56     if current_heads != expected_heads:
57         current_label = ", ".join(sorted(current_heads)) or "无"
58         expected_label = ", ".join(sorted(expected_heads))
59         raise DatabaseMigrationError(
```

backend/app/migrations.py

```
60         f"数据库迁移版本落后或不兼容：当前为 [{current_label}], 代码要求 [{expected_label}]; "
```

61 "请先执行: python -m alembic upgrade head",

62)

63

===== 文件: backend/app/models.py =====

1 """业务数据模型及删除策略。

2

3 数据集被任务引用时禁止物理删除；任务删除后结果和导出记录由级联关系清理，磁盘文件另由清理队列处理。

4 """

5

6 from datetime import datetime

7

8 from sqlalchemy import BigInteger, Boolean, Float, ForeignKey, Index, Integer, String, Text, UniqueConstraint, func

9 from sqlalchemy.dialects import mysql

10 from sqlalchemy.orm import Mapped, mapped_column, relationship

11

12 from .database import Base

13 from .time_utils import UTCDateTime, utc_now

14

15

16 # 既有 MySQL 库的用户和数据集主键为 BIGINT UNSIGNED; SQLite 测试保留 INTEGER 才能正常自增。

17 IDENTIFIER_TYPE = (

18 BigInteger()

19 .with_variant(mysql.BIGINT(unsigned=True), "mysql")

20 .with_variant(Integer(), "sqlite")

21)

22

23

24 class User(Base):

25 __tablename__ = "users"

26

27 id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, primary_key=True, autoincrement=True)

28 username: Mapped[str] = mapped_column(String(64), unique=True, index=True)

29 email: Mapped[str | None] = mapped_column(String(255), unique=True, index=True, nullable=True)

30 password_hash: Mapped[str] = mapped_column(String(255))

31 role: Mapped[str] = mapped_column(String(32), default="user")

32 status: Mapped[str] = mapped_column(String(32), default="active", index=True)

33 last_login_at: Mapped[datetime | None] = mapped_column(UTCDateTime(), nullable=True)

34 created_at: Mapped[datetime] = mapped_column(UTCDateTime(), default=utc_now, server_default=func.now())

35 updated_at: Mapped[datetime] = mapped_column(

36 UTCDateTime(),

37 default=utc_now,

38 server_default=func.now(),

39 onupdate=utc_now,

40)

41

42 sessions: Mapped[list["UserSession"]] = relationship(back_populates="user")

43 datasets: Mapped[list["Dataset"]] = relationship(back_populates="user")

44

backend/app/models.py

```
45
46 class EmailVerificationCode(Base):
47     """邮箱验证码只保存哈希值，避免数据库泄漏时暴露仍在有效期内的明文验证码。"""
48
49     __tablename__ = "email_verification_codes"
50
51     id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, primary_key=True, autoincrement=True)
52     email: Mapped[str] = mapped_column(String(255), index=True)
53     purpose: Mapped[str] = mapped_column(String(32), default="register", index=True)
54     code_hash: Mapped[str] = mapped_column(String(255))
55     client_ip: Mapped[str | None] = mapped_column(String(64), nullable=True)
56     expires_at: Mapped[datetime] = mapped_column(UTCDateTime(), index=True)
57     consumed_at: Mapped[datetime | None] = mapped_column(UTCDateTime(), nullable=True)
58     created_at: Mapped[datetime] = mapped_column(
59         UTCDateTime(), default=utc_now, server_default=func.now(), index=True,
60     )
61
62
63 class UserSession(Base):
64     __tablename__ = "user_sessions"
65
66     id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, primary_key=True, autoincrement=True)
67     user_id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, ForeignKey("users.id", ondelete="CASCADE"), index=True)
68     refresh_token_hash: Mapped[str] = mapped_column(String(255), unique=True)
69     expires_at: Mapped[datetime] = mapped_column(UTCDateTime(), index=True)
70     revoked_at: Mapped[datetime | None] = mapped_column(UTCDateTime(), nullable=True)
71     created_at: Mapped[datetime] = mapped_column(UTCDateTime(), default=utc_now, server_default=func.now())
72
73     user: Mapped[User] = relationship(back_populates="sessions")
74
75
76 class Dataset(Base):
77     """用户上传数据集的当前版本，历史文件由 DatasetRevision 单独保留。"""
78     __tablename__ = "datasets"
79     __table_args__ = (
80         Index("idx_datasets_user_created_at", "user_id", "created_at"),
81     )
82
83     id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, primary_key=True, autoincrement=True)
84     user_id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, ForeignKey("users.id", ondelete="CASCADE"), index=True)
85     name: Mapped[str] = mapped_column(String(128))
86     original_filename: Mapped[str] = mapped_column(String(255))
87     storage_path: Mapped[str] = mapped_column(String(500))
88     file_hash: Mapped[str] = mapped_column(String(128))
89     sample_count: Mapped[int] = mapped_column(Integer)
90     base_cluster_count: Mapped[int] = mapped_column(Integer)
91     has_ground_truth: Mapped[bool] = mapped_column(Boolean, default=False)
92     cluster_count: Mapped[int | None] = mapped_column(Integer, nullable=True)
93     status: Mapped[str] = mapped_column(String(32), default="ready", index=True)
94     created_at: Mapped[datetime] = mapped_column(
```

```
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```

```
95         UTCDateTime(), default=utc_now, server_default=func.now(), index=True,
96     )
97
98     user: Mapped[User] = relationship(back_populates="datasets")
99
100
101 class DatasetRevision(Base):
102     __tablename__ = "dataset_revisions"
103     __table_args__ = (
104         UniqueConstraint("dataset_id", "version", name="uk_dataset_revisions_dataset_version"),
105     )
106
107     id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, primary_key=True, autoincrement=True)
108     dataset_id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, ForeignKey("datasets.id", ondelete="CASCADE"),
109                                             index=True)
110     version: Mapped[int] = mapped_column(Integer)
111     action: Mapped[str] = mapped_column(String(32))
112     name: Mapped[str] = mapped_column(String(128))
113     original_filename: Mapped[str] = mapped_column(String(255))
114     storage_path: Mapped[str] = mapped_column(String(500))
115     file_hash: Mapped[str] = mapped_column(String(128))
116     sample_count: Mapped[int] = mapped_column(Integer)
117     base_cluster_count: Mapped[int] = mapped_column(Integer)
118     has_ground_truth: Mapped[bool] = mapped_column(Boolean, default=False)
119     cluster_count: Mapped[int | None] = mapped_column(Integer, nullable=True)
120     quality_status: Mapped[str] = mapped_column(String(16), default="ready")
121     quality_issues_json: Mapped[str | None] = mapped_column(Text, nullable=True)
122     created_at: Mapped[datetime] = mapped_column(
123         UTCDateTime(), default=utc_now, server_default=func.now(), index=True,
124     )
125
126 class DatasetQuality(Base):
127     __tablename__ = "dataset_qualities"
128
129     id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, primary_key=True, autoincrement=True)
130     dataset_id: Mapped[int] = mapped_column(
131         IDENTIFIER_TYPE,
132         ForeignKey("datasets.id", ondelete="CASCADE"),
133         unique=True,
134         index=True,
135     )
136     status: Mapped[str] = mapped_column(String(16), default="ready", index=True)
137     issues_json: Mapped[str | None] = mapped_column(Text, nullable=True)
138     checked_at: Mapped[datetime] = mapped_column(
139         UTCDateTime(), default=utc_now, server_default=func.now(), onupdate=utc_now,
140     )
141
142
143 class AnalysisTask(Base):
```

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```

144     """正式分析任务；承接 OMELET / OMELET-SV 的创建、调度、进度和结果关联。
145
146     worker_id 与 heartbeat_at 共同表示当前执行租约，避免多个调度实例重复领取同一任务。
147     """
148
149     __tablename__ = "analysis_tasks"
150     __table_args__ = (
151         Index("idx_analysis_tasks_user_dataset", "user_id", "dataset_id"),
152         Index("idx_analysis_tasks_dataset_status_finished", "dataset_id", "status", "finished_at"),
153         Index("idx_analysis_tasks_worker_id", "worker_id"),
154         Index("idx_analysis_tasks_heartbeat_at", "heartbeat_at"),
155         Index("idx_analysis_tasks_status_queued", "status", "queued_at", "id"),
156         Index("idx_analysis_tasks_status_heartbeat", "status", "heartbeat_at"),
157         Index("idx_analysis_tasks_user_status_queued", "user_id", "status", "queued_at"),
158     )
159
160     id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, primary_key=True, autoincrement=True)
161     user_id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, ForeignKey("users.id", ondelete="CASCADE"), index=True)
162     dataset_id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, ForeignKey("datasets.id", ondelete="RESTRICT"),
163     index=True)
164     name: Mapped[str] = mapped_column(String(128), default="")
165     mode: Mapped[str] = mapped_column(String(32), default="OMELET-SV")
166     status: Mapped[str] = mapped_column(String(32), default="draft", index=True)
167     progress: Mapped[float] = mapped_column(Float, default=0.0)
168     current_run: Mapped[int] = mapped_column(Integer, default=0)
169     current_iter: Mapped[int] = mapped_column(Integer, default=0)
170     max_iter: Mapped[int] = mapped_column(Integer, default=20)
171     params_json: Mapped[str] = mapped_column(Text, default="{}")
172     error_message: Mapped[str | None] = mapped_column(Text, nullable=True)
173     failure_reason: Mapped[str | None] = mapped_column(String(64), nullable=True)
174     current_stage: Mapped[str | None] = mapped_column(String(64), nullable=True)
175     worker_id: Mapped[str | None] = mapped_column(String(128), nullable=True)
176     heartbeat_at: Mapped[datetime | None] = mapped_column(UTCDateTime(), nullable=True)
177     retry_count: Mapped[int] = mapped_column(Integer, default=0, server_default="0")
178     queued_at: Mapped[datetime | None] = mapped_column(UTCDateTime(), nullable=True, index=True)
179     started_at: Mapped[datetime | None] = mapped_column(UTCDateTime(), nullable=True)
180     finished_at: Mapped[datetime | None] = mapped_column(UTCDateTime(), nullable=True)
181     created_at: Mapped[datetime] = mapped_column(
182         UTCDateTime(), default=utc_now, server_default=func.now(), index=True,
183     )
184     updated_at: Mapped[datetime] = mapped_column(
185         UTCDateTime(),
186         default=utc_now,
187         server_default=func.now(),
188         onupdate=utc_now,
189     )
190
191 class TaskResult(Base):
192     """任务唯一结果索引，矩阵和标签等大文件只在数据库中保存受控路径。"""

```


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```

193     __tablename__ = "task_results"
194     __table_args__ = (
195         UniqueConstraint("task_id", name="uk_task_results_task_id"),
196     )
197
198     id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, primary_key=True, autoincrement=True)
199     schema_version: Mapped[int | None] = mapped_column(Integer, nullable=True)
200     task_id: Mapped[int] = mapped_column(
201         IDENTIFIER_TYPE,
202         ForeignKey("analysis_tasks.id", ondelete="CASCADE"),
203         index=True,
204     )
205     metrics_json: Mapped[str | None] = mapped_column(Text, nullable=True)
206     kernel_weights_json: Mapped[str | None] = mapped_column(Text, nullable=True)
207     convergence_json: Mapped[str | None] = mapped_column(Text, nullable=True)
208     preview_json: Mapped[str | None] = mapped_column(Text, nullable=True)
209     labels_path: Mapped[str | None] = mapped_column(String(500), nullable=True)
210     ca_matrix_path: Mapped[str | None] = mapped_column(String(500), nullable=True)
211     s_matrix_path: Mapped[str | None] = mapped_column(String(500), nullable=True)
212     z_matrix_path: Mapped[str | None] = mapped_column(String(500), nullable=True)
213     runtime_seconds: Mapped[float | None] = mapped_column(Float, nullable=True)
214     created_at: Mapped[datetime] = mapped_column(UTCDateTime(), default=utc_now, server_default=func.now())
215
216
217 class TaskExport(Base):
218     """用户生成的交付档案记录，实际 ZIP 文件由存储清理队列维护生命周期。"""
219     __tablename__ = "task_exports"
220     __table_args__ = (
221         Index("idx_task_exports_task_id_id", "task_id", "id"),
222     )
223
224     id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, primary_key=True, autoincrement=True)
225     task_id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, ForeignKey("analysis_tasks.id", ondelete="CASCADE"),
226         index=True)
227     export_type: Mapped[str] = mapped_column(String(32))
228     name: Mapped[str | None] = mapped_column(String(128), nullable=True)
229     items_json: Mapped[str | None] = mapped_column(Text, nullable=True)
230     status: Mapped[str] = mapped_column(String(32), default="ready")
231     filename: Mapped[str] = mapped_column(String(255))
232     storage_path: Mapped[str] = mapped_column(String(500))
233     file_size: Mapped[int | None] = mapped_column(Integer, nullable=True)
234     created_at: Mapped[datetime] = mapped_column(UTCDateTime(), default=utc_now, server_default=func.now())
235
236 class TaskTemplate(Base):
237     """用户保存的任务参数模板，便于重复创建同类分析。"""
238
239     __tablename__ = "task_templates"
240
241     id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, primary_key=True, autoincrement=True)

```

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```
242     user_id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, ForeignKey("users.id", ondelete="CASCADE"), index=True)
243     name: Mapped[str] = mapped_column(String(128))
244     mode: Mapped[str] = mapped_column(String(32), default="OMELET-SV")
245     params_json: Mapped[str] = mapped_column(Text, default="{}")
246     created_at: Mapped[datetime] = mapped_column(
247         UTCDateTime(), default=utc_now, server_default=func.now(), index=True,
248     )
249
250
251 class OperationLog(Base):
252     __tablename__ = "operation_logs"
253
254     id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, primary_key=True, autoincrement=True)
255     user_id: Mapped[int | None] = mapped_column(IDENTIFIER_TYPE, ForeignKey("users.id", ondelete="SET NULL"),
256         nullable=True)
257     task_id: Mapped[int | None] = mapped_column(
258         IDENTIFIER_TYPE,
259         ForeignKey("analysis_tasks.id", ondelete="SET NULL"),
260         nullable=True,
261         index=True,
262     )
263     action: Mapped[str] = mapped_column(String(64), index=True)
264     level: Mapped[str] = mapped_column(String(16), default="info")
265     message: Mapped[str] = mapped_column(String(500))
266     detail_json: Mapped[str | None] = mapped_column(Text, nullable=True)
267     created_at: Mapped[datetime] = mapped_column(
268         UTCDateTime(), default=utc_now, server_default=func.now(), index=True,
269     )
270
271 class StorageCleanupJob(Base):
272     """数据库提交后异步删除磁盘文件的幂等队列任务。"""
273
274     __tablename__ = "storage_cleanup_jobs"
275     __table_args__ = (
276         Index("idx_storage_cleanup_jobs_status_retry", "status", "next_attempt_at", "id"),
277         Index("idx_storage_cleanup_jobs_created_at", "created_at"),
278         Index("idx_storage_cleanup_jobs_status", "status"),
279         Index("idx_storage_cleanup_jobs_next_attempt_at", "next_attempt_at"),
280         UniqueConstraint("storage_path", name="uk_storage_cleanup_jobs_storage_path"),
281     )
282
283     id: Mapped[int] = mapped_column(IDENTIFIER_TYPE, primary_key=True, autoincrement=True)
284     storage_path: Mapped[str] = mapped_column(String(500), nullable=False)
285     storage_kind: Mapped[str] = mapped_column(String(32), nullable=False)
286     status: Mapped[str] = mapped_column(String(16), default="pending")
287     attempts: Mapped[int] = mapped_column(Integer, default=0, server_default="0")
288     last_error: Mapped[str | None] = mapped_column(Text, nullable=True)
289     next_attempt_at: Mapped[datetime] = mapped_column(
290         UTCDateTime(), default=utc_now, server_default=func.now(),
```

```
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```

```
291     )
292     created_at: Mapped[datetime] = mapped_column(
293         UTCDateTime(), default=utc_now, server_default=func.now(),
294     )
295     completed_at: Mapped[datetime | None] = mapped_column(UTCDateTime(), nullable=True)
296
```

```

# ===== 文件: backend/app/product.py =====
1  """软件身份信息的唯一来源。
2
3  软著材料、API 元数据和通知邮件都引用这里的名称与版本，避免不同模块各自维护品牌文本。
4  """
5
6  SOFTWARE_NAME = "新材料数据分析---拓扑感知多核集成聚类分析系统"
7  SOFTWARE_SHORT_NAME = "新材料数据分析系统"
8  SOFTWARE_VERSION = "V1.0"
9
```

```

# ===== 文件: backend/app/schemas.py =====
1  from datetime import datetime
2  from typing import Literal
3
4  from pydantic import BaseModel, Field, field_serializer, model_validator
5
6  from .time_utils import format_utc_iso
7
8  TurnstileAction = Literal["login", "register"]
9
10
11 class RegisterRequest(BaseModel):
12     username: str = Field(min_length=3, max_length=64)
13     email: str | None = Field(default=None, max_length=255)
14     password: str = Field(min_length=6, max_length=128)
15     confirm_password: str = Field(min_length=6, max_length=128)
16     email_code: str | None = Field(default=None, min_length=6, max_length=6)
17     turnstile_token: str | None = Field(default=None, max_length=2048)
18
19     @model_validator(mode="after")
20     def passwords_match(self):
21         if self.password != self.confirm_password:
22             raise ValueError("两次密码不一致")
23         return self
24
25
26 class EmailCodeRequest(BaseModel):
27     email: str = Field(min_length=5, max_length=255)
28     turnstile_token: str | None = Field(default=None, max_length=2048)
29     turnstile_pass_token: str | None = Field(default=None, max_length=2048)
30
31
```

backend/app/schemas.py

```
32 class LoginRequest(BaseModel):
33     username: str = Field(min_length=3, max_length=255)
34     password: str = Field(min_length=6, max_length=128)
35     turnstile_token: str | None = Field(default=None, max_length=2048)
36     turnstile_pass_token: str | None = Field(default=None, max_length=2048)
37
38
39 class TurnstilePassRequest(BaseModel):
40     turnstile_token: str = Field(min_length=1, max_length=2048)
41     action: TurnstileAction
42
43
44 class TurnstilePassResponse(BaseModel):
45     turnstile_pass_token: str
46     expires_at: datetime
47     expires_in_seconds: int
48
49     @field_serializer("expires_at", when_used="json")
50     def serialize_expires_at(self, value: datetime) -> str:
51         return format_utc_iso(value) or ""
52
53
54 class UserResponse(BaseModel):
55     id: int
56     username: str
57     email: str | None = None
58     role: str
59     status: str
60     last_login_at: datetime | None = None
61
62     model_config = {"from_attributes": True}
63
64     @field_serializer("last_login_at", when_used="json")
65     def serialize_last_login_at(self, value: datetime | None) -> str | None:
66         return format_utc_iso(value)
67
68
69 class AuthResponse(BaseModel):
70     user: UserResponse
71     access_token: str
72     token_type: str = "bearer"
73
74
75 class AccessTokenResponse(BaseModel):
76     access_token: str
77     token_type: str = "bearer"
78
79
80 class MessageResponse(BaseModel):
81     message: str
```

backend/app/schemas.py

```
82
83
84 class DatasetRenameRequest(BaseModel):
85     name: str = Field(min_length=1, max_length=128)
86     allowDuplicate: bool = False
87
88
89 class DatasetBulkRequest(BaseModel):
90     datasetIds: list[int] = Field(min_length=1, max_length=100)
91
92
93 class LabelDistributionItem(BaseModel):
94     label: str
95     count: int
96     percent: int
97
98
99 class ClusterStatItem(BaseModel):
100     name: str
101     clusterCount: int
102     range: str
103
104
105 class DatasetVariableResponse(BaseModel):
106     shape: list[int]
107     dtype: str
108
109
110 class DatasetParseResponse(BaseModel):
111     fileName: str
112     variables: dict[str, DatasetVariableResponse]
113     mainVariable: str | None = None
114     labelVariable: str | None = None
115     sampleCount: int
116     baseCount: int
117     classCount: int
118     hasLabels: bool
119     matrixShape: str
120     labelShape: str
121     labelDistribution: list[LabelDistributionItem] = Field(default_factory=list)
122     clusterStats: list[ClusterStatItem] = Field(default_factory=list)
123
124
125 class DatasetCatalogItemResponse(BaseModel):
126     id: int
127     name: str
128     createdAt: str
129     fileSizeBytes: int
130     sampleCount: int
131     baseCount: int
```

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```
132     classCount: int
133     hasLabels: bool
134     dataType: str = "数值"
135     taskCount: int = 0
136     lastAnalysisAt: str | None = None
137     version: int = 1
138     qualityStatus: str = "ready"
139     qualityIssues: list[str] = Field(default_factory=list)
140     matrixShape: str
141     labelShape: str
142     labelDistribution: list[LabelDistributionItem] = Field(default_factory=list)
143     clusterStats: list[ClusterStatItem] = Field(default_factory=list)
144
145
146 class DatasetCatalogPageResponse(BaseModel):
147     items: list[DatasetCatalogItemResponse] = Field(default_factory=list)
148     total: int
149     page: int
150     pageSize: int
151     totalPages: int
152
153
154 class DatasetRevisionResponse(BaseModel):
155     id: int
156     version: int
157     action: str
158     name: str
159     originalFilename: str
160     createdAt: str
161     sampleCount: int
162     baseCount: int
163     classCount: int
164     hasLabels: bool
165     qualityStatus: str
166     qualityIssues: list[str] = Field(default_factory=list)
167
168
169 class AnalysisTaskParams(BaseModel):
170     nBase: int = Field(default=20, ge=1, le=500)
171     sigma: float = Field(default=1.0, ge=0)
172     lambdaValue: float = Field(default=5.0, ge=0, alias="lambda")
173     gamma: float = Field(default=5.0, ge=0)
174     anchor: int = Field(default=10, ge=1, le=1000)
175     runs: int = Field(default=10, ge=1, le=100)
176     maxIter: int = Field(default=10, ge=1, le=200)
177     randomSeed: int = Field(default=1, ge=0, le=2147483647)
178
179     model_config = {"populate_by_name": True}
180
181
```

backend/app/schemas.py

```
182 class AnalysisTaskCreateRequest(BaseModel):
183     datasetId: int
184     name: str | None = Field(default=None, max_length=128)
185     mode: str = Field(default="OMELET-SV")
186     params: AnalysisTaskParams | None = None
187     startImmediately: bool = False
188     templateId: int | None = None
189
190
191 class AnalysisTaskUpdateRequest(BaseModel):
192     name: str | None = Field(default=None, max_length=128)
193     mode: str | None = None
194     params: AnalysisTaskParams | None = None
195
196
197 class AnalysisTaskBulkRequest(BaseModel):
198     taskIds: list[int] = Field(min_length=1, max_length=100)
199     action: str
200
201
202 class AnalysisTaskMetricsSummary(BaseModel):
203     acc: float | None = None
204     nmi: float | None = None
205     ari: float | None = None
206     f1: float | None = None
207
208
209 class AnalysisTaskResponse(BaseModel):
210     id: int
211     name: str
212     mode: str
213     status: str
214     progress: float
215     currentRun: int = 0
216     totalRuns: int = 1
217     queuePosition: int | None = None
218     currentIter: int
219     maxIter: int
220     currentStage: str | None = None
221     datasetId: int
222     datasetName: str
223     params: dict
224     errorMessage: str | None = None
225     failureReason: str | None = None
226     metricsSummary: AnalysisTaskMetricsSummary | None = None
227     runtimeSeconds: float | None = None
228     createdAt: str
229     startedAt: str | None = None
230     finishedAt: str | None = None
231     updatedAt: str
```

backend/app/schemas.py

```
232
233
234 class AnalysisTaskPageResponse(BaseModel):
235     items: list[AnalysisTaskResponse] = Field(default_factory=list)
236     total: int
237     page: int
238     pageSize: int
239     totalPages: int
240
241
242 class AnalysisTaskStatsResponse(BaseModel):
243     total: int = 0
244     draft: int = 0
245     queued: int = 0
246     running: int = 0
247     succeeded: int = 0
248     failed: int = 0
249     cancelled: int = 0
250     todayCompleted: int = 0
251     averageRuntimeSeconds: float | None = None
252     failureRate: float = 0.0
253
254
255 class AnalysisTaskLogItem(BaseModel):
256     id: int
257     action: str
258     level: str
259     message: str
260     createdAt: str
261     detail: dict | None = None
262
263
264 class AnalysisTaskLogPageResponse(BaseModel):
265     items: list[AnalysisTaskLogItem] = Field(default_factory=list)
266     total: int
267
268
269 class TaskResultEnvelope(BaseModel):
270     state: str
271     task: AnalysisTaskResponse | None = None
272     result: dict | None = None
273
274
275 class TaskExportCreateRequest(BaseModel):
276     items: list[str] = Field(min_length=1, max_length=12)
277     name: str | None = Field(default=None, max_length=128)
278
279
280 class TaskExportResponse(BaseModel):
281     id: int
```


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```
282     name: str
283     items: list[str] = Field(default_factory=list)
284     itemCount: int
285     status: str
286     filename: str
287     fileSize: int
288     createdAt: str
289     downloadUrl: str
290
291
292 class TaskExportListResponse(BaseModel):
293     items: list[TaskExportResponse] = Field(default_factory=list)
294
295
296 class TaskTemplateCreateRequest(BaseModel):
297     name: str = Field(min_length=1, max_length=128)
298     mode: str = Field(default="OMELET-SV")
299     params: AnalysisTaskParams | None = None
300
301
302 class TaskTemplateResponse(BaseModel):
303     id: int
304     name: str
305     mode: str
306     params: dict
307     createdAt: str
308
309
310 class TaskTemplateListResponse(BaseModel):
311     items: list[TaskTemplateResponse] = Field(default_factory=list)
312
313
314 # ===== 文件: backend/app/security.py =====
315 1 from datetime import datetime, timedelta
316 2 import hashlib
317 3 import secrets
318 4
319 5 import bcrypt
320 6 from fastapi import Response
321 7 from jose import JWTError, jwt
322 8
323 9 from .config import get_settings
324 10 from .time_utils import utc_now
325 11
326 12
327 13 REFRESH_COOKIE_NAME = "soft_web_refresh"
328 14 REFRESH_COOKIE_PATH = "/api/auth"
329 15 REFRESH_COOKIE_SAMESITE = "lax"
330 16
331 17
```

backend/app/security.py

```
18 def hash_password(password: str) -> str:
19     return bcrypt.hashpw(password.encode("utf-8"), bcrypt.gensalt()).decode("utf-8")
20
21
22 def verify_password(password: str, password_hash: str) -> bool:
23     try:
24         return bcrypt.checkpw(password.encode("utf-8"), password_hash.encode("utf-8"))
25     except ValueError:
26         return False
27
28
29 def hash_token(token: str) -> str:
30     return hashlib.sha256(token.encode("utf-8")).hexdigest()
31
32
33 def create_refresh_token() -> str:
34     return secrets.token_urlsafe(48)
35
36
37 def set_refresh_cookie(response: Response, refresh_token: str) -> None:
38     settings = get_settings()
39     environment = settings.app_env.strip().lower()
40     response.set_cookie(
41         key=REFRESH_COOKIE_NAME,
42         value=refresh_token,
43         max_age=settings.refresh_token_expire_days * 24 * 60 * 60,
44         httponly=True,
45         secure=environment not in {"development", "test"},
46         samesite=REFRESH_COOKIE_SAMESITE,
47         path=REFRESH_COOKIE_PATH,
48     )
49
50
51 def clear_refresh_cookie(response: Response) -> None:
52     response.delete_cookie(
53         key=REFRESH_COOKIE_NAME,
54         path=REFRESH_COOKIE_PATH,
55         samesite=REFRESH_COOKIE_SAMESITE,
56     )
57
58
59 def create_access_token(user_id: int, username: str) -> str:
60     settings = get_settings()
61     expires_at = utc_now() + timedelta(minutes=settings.access_token_expire_minutes)
62     payload = {
63         "sub": str(user_id),
64         "username": username,
65         "exp": expires_at,
66         "type": "access",
67     }
```

```
backend/app/security.py
```

```
68     return jwt.encode(payload, settings.jwt_secret_key, algorithm=settings.jwt_algorithm)
69
70
71 def create_turnstile_pass_token() -> tuple[str, datetime]:
72     settings = get_settings()
73     expires_at = utc_now() + timedelta(minutes=settings.turnstile_pass_expire_minutes)
74     payload = {
75         "sub": "turnstile",
76         "scope": "auth",
77         "exp": expires_at,
78         "type": "turnstile_pass",
79     }
80     return jwt.encode(payload, settings.jwt_secret_key, algorithm=settings.jwt_algorithm), expires_at
81
82
83 def decode_turnstile_pass_token(token: str) -> bool:
84     settings = get_settings()
85     try:
86         payload = jwt.decode(token, settings.jwt_secret_key, algorithms=[settings.jwt_algorithm])
87     except JWTError:
88         return False
89
90     return payload.get("type") == "turnstile_pass" and payload.get("scope") == "auth"
91
92
93 def decode_access_token(token: str) -> int | None:
94     settings = get_settings()
95     try:
96         payload = jwt.decode(token, settings.jwt_secret_key, algorithms=[settings.jwt_algorithm])
97     except JWTError:
98         return None
99
100     if payload.get("type") != "access":
101         return None
102
103     subject = payload.get("sub")
104     if subject is None:
105         return None
106
107     try:
108         return int(subject)
109     except ValueError:
110         return None
111
```

```
# ===== 文件: backend/app/storage_cleanup.py =====
```

```
1 """数据库提交后的文件清理队列。
2
3 数据库事务不能和本地文件系统组成同一个原子事务，因此删除接口只在数据库
4 事务中登记清理任务。文件操作放到提交之后执行，失败时保留可重试的队列记录，
```

backend/app/storage_cleanup.py

```
5 避免出现数据库仍指向已删除文件的情况。
6  """
7
8  from __future__ import annotations
9
10 import logging
11 import shutil
12 from dataclasses import dataclass
13 from datetime import timedelta
14 from pathlib import Path
15
16 from sqlalchemy import select
17 from sqlalchemy.orm import Session, sessionmaker
18
19 from .config import get_settings
20 from .database import SessionLocal
21 from .models import StorageCleanupJob
22 from .time_utils import utc_now
23
24
25 logger = logging.getLogger(__name__)
26
27 CLEANUP_KINDS = {"dataset", "result", "export"}
28 CLEANUP_LEASE_SECONDS = 300
29 MAX_RETRY_DELAY_SECONDS = 3600
30
31
32 @dataclass(frozen=True)
33 class CleanupSummary:
34     attempted: int = 0
35     succeeded: int = 0
36     failed: int = 0
37
38
39 def _cleanup_root(kind: str) -> Path:
40     if kind == "dataset":
41         return Path(get_settings().dataset_storage_dir).resolve()
42     if kind in {"result", "export"}:
43         return Path(get_settings().result_storage_dir).resolve()
44     raise ValueError(f"不支持的文件清理类型: {kind}")
45
46
47 def normalize_cleanup_path(path: str | Path, kind: str) -> Path:
48     """校验并规范化待清理路径，禁止队列删除存储根目录之外的文件。"""
49     if kind not in CLEANUP_KINDS:
50         raise ValueError(f"不支持的文件清理类型: {kind}")
51     normalized = Path(path).resolve()
52     root = _cleanup_root(kind)
53     if normalized == root:
54         raise ValueError("禁止把存储根目录加入文件清理队列")
```

backend/app/storage_cleanup.py

```
55     try:
56         normalized.relative_to(root)
57     except ValueError as exc:
58         raise ValueError("文件清理路径不在受控存储目录内") from exc
59     return normalized
60
61
62 def enqueue_cleanup(session: Session, path: str | Path, kind: str) -> StorageCleanupJob:
63     """在当前业务事务中登记一次幂等的文件清理任务。"""
64     normalized = normalize_cleanup_path(path, kind)
65     normalized_path = str(normalized)
66     job = session.scalar(
67         select(StorageCleanupJob)
68         .where(StorageCleanupJob.storage_path == normalized_path)
69         .with_for_update(),
70     )
71     if job is not None:
72         if job.status == "failed":
73             job.status = "pending"
74             job.last_error = None
75             job.next_attempt_at = utc_now()
76     return job
77
78     job = StorageCleanupJob(
79         storage_path=normalized_path,
80         storage_kind=kind,
81         status="pending",
82         attempts=0,
83         next_attempt_at=utc_now(),
84     )
85     session.add(job)
86     session.flush()
87     return job
88
89
90 def enqueue_cleanups(session: Session, paths: set[tuple[str, Path]]) -> list[StorageCleanupJob]:
91     jobs: list[StorageCleanupJob] = []
92     for kind, path in sorted(paths, key=lambda item: (item[0], str(item[1]))):
93         jobs.append(enqueue_cleanup(session, path, kind))
94     return jobs
95
96
97 def _delete_path(path: Path) -> None:
98     if path.is_symlink() or path.is_file():
99         path.unlink(missing_ok=True)
100     elif path.is_dir():
101         shutil.rmtree(path)
102
103
104 def _retry_delay(attempts: int) -> int:
```

```
backend/app/storage_cleanup.py
```

```
105     return min(MAX_RETRY_DELAY_SECONDS, 2 ** min(max(attempts, 1), 10))
106
107
108 def _claim_job(session_factory) -> tuple[int, str, str, int] | None:
109     now = utc_now()
110     with session_factory() as session:
111         job = session.scalar(
112             select(StorageCleanupJob)
113             .where(
114                 StorageCleanupJob.status.in_(("pending", "failed", "processing")),
115                 StorageCleanupJob.next_attempt_at <= now,
116             )
117             .order_by(StorageCleanupJob.next_attempt_at, StorageCleanupJob.id)
118             .limit(1)
119             .with_for_update(skip_locked=True),
120         )
121     if job is None:
122         return None
123     job.status = "processing"
124     job.attempts = int(job.attempts or 0) + 1
125     job.next_attempt_at = now + timedelta(seconds=CLEANUP_LEASE_SECONDS)
126     session.commit()
127     return job.id, job.storage_path, job.storage_kind, job.attempts
128
129
130 def process_pending_cleanup_jobs(*, limit: int = 20, session_factory=None, bind=None) -> CleanupSummary:
131     """处理有限数量的清理任务；数据库异常交给调用方记录并在下一轮重试。"""
132     factory = session_factory or (sessionmaker(bind=bind, autoflush=False, future=True) if bind is not None else
133     SessionLocal)
134     attempted = succeeded = failed = 0
135     for _ in range(max(0, int(limit))):
136         claimed = _claim_job(factory)
137         if claimed is None:
138             break
139         job_id, raw_path, kind, attempts = claimed
140         attempted += 1
141         path = Path(raw_path)
142         try:
143             normalize_cleanup_path(path, kind)
144             _delete_path(path)
145         except Exception as exc: # pragma: no cover - filesystem errors are platform-specific
146             failed += 1
147             with factory() as session:
148                 job = session.get(StorageCleanupJob, job_id)
149                 if job is not None:
150                     job.status = "failed"
151                     job.last_error = str(exc)[:2000]
152                     job.next_attempt_at = utc_now() + timedelta(seconds=_retry_delay(attempts))
153                     session.commit()
154             logger.warning("文件清理失败，将在稍后重试：%s (%s)", path, exc)
```

```
backend/app/storage_cleanup.py
```

```
154         else:
155             succeeded += 1
156             with factory() as session:
157                 job = session.get(StorageCleanupJob, job_id)
158                 if job is not None:
159                     job.status = "succeeded"
160                     job.last_error = None
161                     job.completed_at = utc_now()
162                     session.commit()
163     return CleanupSummary(attempted=attempted, succeeded=succeeded, failed=failed)
164
```

```
# ===== 文件: backend/app/task_executor.py =====
1 """数据库驱动的单任务执行调度器。
2
3 执行器只负责领取、监控和持久化；算法计算在独立子进程中完成，避免阻塞 Web 服务进程。
4 """
5
6 from __future__ import annotations
7
8 import json
9 import os
10 import queue
11 import shutil
12 import signal
13 import socket
14 import subprocess
15 import sys
16 import threading
17 import time
18 import uuid
19 from datetime import datetime, timedelta
20 from pathlib import Path
21 from typing import Any
22
23 from sqlalchemy import or_, select, update
24
25 from .config import get_settings
26 from .database import SessionLocal
27 from .models import AnalysisTask, Dataset, OperationLog, TaskResult
28 from .storage_cleanup import process_pending_cleanup_jobs
29 from .time_utils import utc_now
30
31
32 EVENT_PREFIX = "OMELET_EVENT "
33 TERMINAL_STATUSES = {"succeeded", "failed", "cancelled"}
34
35
36 def _now() -> datetime:
37     return utc_now().replace(microsecond=0)
```

backend/app/task_executor.py

```
38
39
40 def _load_json(value: str | None, default: Any) -> Any:
41     if not value:
42         return default
43     try:
44         return json.loads(value)
45     except json.JSONDecodeError:
46         return default
47
48
49 def _dump_json(value: Any) -> str:
50     return json.dumps(value, ensure_ascii=False)
51
52
53 def _is_in_directory(path: Path, directory: Path) -> bool:
54     try:
55         path.resolve().relative_to(directory.resolve())
56         return True
57     except ValueError:
58         return False
59
60
61 def _terminate_process_tree(process: subprocess.Popen[str], wait_timeout: float = 5.0) -> None:
62     """结束算法进程及其子进程，避免只杀掉外层 Python 进程。"""
63     if process.poll() is not None:
64         return
65
66     if os.name == "nt":
67         try:
68             subprocess.run(
69                 ["taskkill", "/PID", str(process.pid), "/T", "/F"],
70                 check=False,
71                 stdout=subprocess.DEVNULL,
72                 stderr=subprocess.DEVNULL,
73                 timeout=max(5.0, wait_timeout),
74             )
75         except (OSError, subprocess.SubprocessError):
76             # 精简 Windows 环境可能没有 taskkill，保留 Popen 的兜底终止。
77             try:
78                 process.terminate()
79             except OSError:
80                 pass
81     else:
82         try:
83             os.killpg(os.getpgid(process.pid), signal.SIGTERM)
84         except (OSError, ProcessLookupError):
85             try:
86                 process.terminate()
87             except OSError:
```


backend/app/task_executor.py

```
88         pass
89
90     try:
91         process.wait(timeout=wait_timeout)
92         return
93     except subprocess.TimeoutExpired:
94         pass
95
96     if os.name == "nt":
97         try:
98             subprocess.run(
99                 ["taskkill", "/PID", str(process.pid), "/T", "/F"],
100                 check=False,
101                 stdout=subprocess.DEVNULL,
102                 stderr=subprocess.DEVNULL,
103                 timeout=max(5.0, wait_timeout),
104             )
105         except (OSError, subprocess.SubprocessError):
106             try:
107                 process.kill()
108             except OSError:
109                 pass
110     else:
111         try:
112             os.killpg(os.getpgid(process.pid), signal.SIGKILL)
113         except (OSError, ProcessLookupError):
114             try:
115                 process.kill()
116             except OSError:
117                 pass
118     try:
119         process.wait(timeout=wait_timeout)
120     except subprocess.TimeoutExpired:
121         pass
122
123
124 class TaskExecutionManager:
125     """使用数据库租约的单并发任务调度器。
126
127     一个部署实例只能启用一个调度器；多个 API 实例应将 TASK_EXECUTOR_ENABLED 设为 false。
128     数据库租约仍会阻止误启动的第二个实例重复领取同一任务。
129     """
130
131     def __init__(self) -> None:
132         self.worker_id = f"{socket.gethostname()}:{os.getpid()}:{uuid.uuid4().hex[:16]}"
133         self._lock = threading.RLock()
134         self._wake_event = threading.Event()
135         self._stop_event = threading.Event()
136         self._thread: threading.Thread | None = None
137         self._process: subprocess.Popen[str] | None = None
```

backend/app/task_executor.py

```
138         self._active_task_id: int | None = None
139
140     def start(self) -> None:
141         settings = get_settings()
142         if not settings.task_executor_enabled:
143             return
144         with self._lock:
145             if self._thread is not None and self._thread.is_alive():
146                 return
147             self._stop_event.clear()
148             self._recover_interrupted_tasks()
149             self._thread = threading.Thread(target=self._dispatch_loop, name="omelet-task-executor", daemon=True)
150             self._thread.start()
151
152     def stop(self) -> None:
153         self._stop_event.set()
154         self._wake_event.set()
155         with self._lock:
156             process = self._process
157             task_id = self._active_task_id
158             if process is not None and process.poll() is None:
159                 _terminate_process_tree(process)
160             if task_id is not None:
161                 self._requeue_active_task(task_id)
162             thread = self._thread
163             if thread is not None and thread.is_alive():
164                 thread.join(timeout=8)
165
166     def notify(self) -> None:
167         self._wake_event.set()
168
169     def cancel_active_process(self, task_id: int) -> None:
170         with self._lock:
171             if task_id != self._active_task_id or self._process is None:
172                 return
173             process = self._process
174             if process.poll() is None:
175                 _terminate_process_tree(process)
176
177     def _dispatch_loop(self) -> None:
178         poll_interval = max(0.2, float(get_settings().task_poll_interval_seconds))
179         last_cleanup_at = 0.0
180         while not self._stop_event.is_set():
181             if time.monotonic() - last_cleanup_at >= 5.0:
182                 try:
183                     process_pending_cleanup_jobs(limit=10)
184                 except Exception:
185                     # 清理器故障不应阻断任务调度，下一轮继续尝试。
186                     pass
187             last_cleanup_at = time.monotonic()
```

backend/app/task_executor.py

```
188         try:
189             claimed = self._claim_next_task()
190         except Exception:
191             # 数据库短暂不可用时保留调度线程，下一轮继续尝试。
192             self._wake_event.wait(timeout=poll_interval)
193             self._wake_event.clear()
194             continue
195         if claimed is None:
196             self._wake_event.wait(timeout=poll_interval)
197             self._wake_event.clear()
198             continue
199         try:
200             self._execute_task(claimed)
201         except Exception as exc:
202             # 兜底保护调度线程；正常失败路径会在 _execute_task 内写入更具体的原因。
203             task_id = int(claimed["taskId"])
204             self.cancel_active_process(task_id)
205             try:
206                 self._mark_failed(task_id, "executor_error", f"执行器异常: {exc}")
207             except Exception:
208                 pass
209
210     def _recover_interrupted_tasks(self) -> None:
211         """只回收心跳超时的运行任务，仍由其他 worker 持有租约的任务保持不动。"""
212         settings = get_settings()
213         stale_before = _now() - timedelta(seconds=float(settings.task_heartbeat_timeout_seconds))
214         with SessionLocal() as session:
215             running = session.scalars(
216                 select(AnalysisTask)
217                 .where(
218                     AnalysisTask.status == "running",
219                     or_(
220                         AnalysisTask.heartbeat_at.is_(None),
221                         AnalysisTask.heartbeat_at <= stale_before,
222                     ),
223                 )
224                 .order_by(AnalysisTask.id)
225                 .with_for_update(skip_locked=True),
226             ).all()
227         for task in running:
228             task.status = "queued"
229             task.retry_count = int(task.retry_count or 0) + 1
230             task.progress = 0.0
231             task.current_run = 0
232             task.current_iter = 0
233             task.current_stage = "select_base"
234             task.started_at = None
235             task.finished_at = None
236             task.queued_at = _now()
237             task.worker_id = None
```

backend/app/task_executor.py

```
238         task.heartbeat_at = None
239     session.add(
240         OperationLog(
241             user_id=task.user_id,
242             task_id=task.id,
243             action="task_recovered",
244             level="warning",
245             message="服务重启后任务心跳已超时，任务重新排队，将从头执行",
246             detail_json=_dump_json({"retryCount": task.retry_count}),
247         ),
248     )
249     if running:
250         session.commit()
251
252 def _claim_next_task(self) -> dict[str, Any] | None:
253     """锁定候选行后再按 queued 条件更新，确保多实例只能有一个领取成功。"""
254     with SessionLocal() as session:
255         task = session.scalar(
256             select(AnalysisTask)
257             .where(AnalysisTask.status == "queued")
258             .order_by(AnalysisTask.queued_at, AnalysisTask.id)
259             .limit(1)
260             .with_for_update(skip_locked=True),
261         )
262     if task is None:
263         return None
264
265     dataset = session.scalar(select(Dataset).where(Dataset.id == task.dataset_id))
266     if dataset is None or not Path(dataset.storage_path).is_file():
267         failed = session.execute(
268             update(AnalysisTask)
269             .where(AnalysisTask.id == task.id, AnalysisTask.status == "queued")
270             .values(
271                 status="failed",
272                 failure_reason="dataset_missing",
273                 error_message="任务数据集文件不存在，无法执行分析",
274                 finished_at=_now(),
275                 worker_id=None,
276                 heartbeat_at=None,
277             ),
278         )
279     if failed.rowcount == 1:
280         session.add(
281             OperationLog(
282                 user_id=task.user_id,
283                 task_id=task.id,
284                 action="task_failed",
285                 level="error",
286                 message="任务数据集文件不存在，无法执行分析",
287             ),
```

```
backend/app/task_executor.py
```

```
288         )
289         session.commit()
290         return None
291
292     claimed_at = _now()
293     claimed = session.execute(
294         update(AnalysisTask)
295         .where(AnalysisTask.id == task.id, AnalysisTask.status == "queued")
296         .values(
297             status="running",
298             progress=0.0,
299             current_run=1,
300             current_iter=0,
301             current_stage="select_base",
302             started_at=claimed_at,
303             finished_at=None,
304             error_message=None,
305             failure_reason=None,
306             worker_id=self.worker_id,
307             heartbeat_at=claimed_at,
308         ),
309     )
310     if claimed.rowcount != 1:
311         session.rollback()
312         return None
313
314     payload = {
315         "taskId": task.id,
316         "userId": task.user_id,
317         "workerId": self.worker_id,
318         "mode": task.mode,
319         "params": _load_json(task.params_json, {}),
320         "datasetPath": dataset.storage_path,
321     }
322     session.add(
323         OperationLog(
324             user_id=task.user_id,
325             task_id=task.id,
326             action="task_started",
327             message="OMELET 子进程已启动",
328             detail_json=_dump_json(
329                 {"mode": task.mode, "datasetId": task.dataset_id, "workerId": self.worker_id},
330             ),
331         ),
332     )
333     session.commit()
334     return payload
335
336 def _execute_task(self, payload: dict[str, Any]) -> None:
337     """启动受控子进程，同时维护心跳、运行时上限和标准输出事件通道。"""
```

backend/app/task_executor.py

```
338         task_id = int(payload["taskId"])
339         user_id = int(payload["userId"])
340         settings = get_settings()
341         result_root = Path(settings.result_storage_dir).resolve()
342         temporary_dir = result_root / ".tmp" / f"{user_id}-{task_id}-{uuid.uuid4().hex}"
343         temporary_dir.mkdir(parents=True, exist_ok=True)
344         job_path = temporary_dir / "job.json"
345         job_path.write_text(
346             json.dumps(**payload, "outputDir": str(temporary_dir)), ensure_ascii=False,
347             encoding="utf-8",
348         )
349
350         command = [sys.executable, "-m", "app.task_worker", "--job", str(job_path)]
351         environment = os.environ.copy()
352         environment.setdefault("PYTHONUTF8", "1")
353         popen_kwargs: dict[str, Any] = {
354             "cwd": str(Path(__file__).resolve().parents[1]),
355             "stdout": subprocess.PIPE,
356             "stderr": subprocess.STDOUT,
357             "text": True,
358             "encoding": "utf-8",
359             "errors": "replace",
360             "bufsize": 1,
361             "env": environment,
362         }
363         if os.name == "nt":
364             popen_kwargs["creationflags"] = getattr(subprocess, "CREATE_NEW_PROCESS_GROUP", 0)
365         else:
366             popen_kwargs["start_new_session"] = True
367
368         try:
369             process = subprocess.Popen(command, **popen_kwargs)
370         except OSError as exc:
371             self._mark_failed(task_id, "executor_start_failed", str(exc))
372             shutil.rmtree(temporary_dir, ignore_errors=True)
373             return
374
375         with self._lock:
376             self._process = process
377             self._active_task_id = task_id
378
379         heartbeat_stop = threading.Event()
380         heartbeat_lost = threading.Event()
381
382         def heartbeat_loop() -> None:
383             interval = max(0.5, float(settings.task_heartbeat_interval_seconds))
384             while not heartbeat_stop.wait(interval):
385                 try:
386                     heartbeat_alive = self._touch_heartbeat(task_id)
387                 except Exception:
```

```
backend/app/task_executor.py
```

```
388         heartbeat_alive = False
389     if not heartbeat_alive:
390         heartbeat_lost.set()
391     with self._lock:
392         active_process = self._process
393     if active_process is not None and active_process.poll() is None:
394         _terminate_process_tree(active_process)
395     return
396
397     heartbeat_thread = threading.Thread(
398         target=heartbeat_loop,
399         name=f"omelet-task-heartbeat-{task_id}",
400         daemon=True,
401     )
402     heartbeat_thread.start()
403
404     output_queue: queue.Queue[str | None] = queue.Queue()
405     output_closed = threading.Event()
406
407     def read_output() -> None:
408         try:
409             assert process.stdout is not None
410             for line in process.stdout:
411                 output_queue.put(line)
412         finally:
413             output_closed.set()
414             output_queue.put(None)
415
416     reader_thread = threading.Thread(target=read_output, name=f"omelet-task-output-{task_id}", daemon=True)
417     reader_thread.start()
418
419     completed_manifest: Path | None = None
420     last_error = ""
421     timed_out = False
422     started_monotonic = time.monotonic()
423
424     try:
425         while True:
426             if (
427                 not timed_out
428                 and process.poll() is None
429                 and time.monotonic() - started_monotonic >= float(settings.task_max_runtime_seconds)
430             ):
431                 timed_out = True
432                 _terminate_process_tree(process)
433
434             try:
435                 line = output_queue.get(timeout=0.2)
436             except queue.Empty:
437                 if process.poll() is not None and output_closed.is_set() and output_queue.empty():
```

```
backend/app/task_executor.py
```

```
438         break
439     continue
440
441     if line is None:
442         if process.poll() is not None and output_queue.empty():
443             break
444         continue
445
446     line = line.strip()
447     if not line:
448         continue
449     if line.startswith(EVENT_PREFIX):
450         try:
451             event = json.loads(line[len(EVENT_PREFIX) :])
452         except json.JSONDecodeError:
453             last_error = "任务子进程发送了无法解析的进度事件"
454             continue
455         event_type = event.get("type")
456         if event_type == "completed":
457             completed_manifest = Path(str(event.get("manifestPath", "")))
458         elif event_type == "error":
459             last_error = str(event.get("message")) or "算法执行失败"
460         else:
461             try:
462                 self._persist_progress(task_id, event)
463             except Exception as exc:
464                 last_error = f"进度写入失败: {exc}"
465     else:
466         last_error = line[-500:]
467
468     try:
469         return_code = process.wait(timeout=2)
470     except subprocess.TimeoutExpired:
471         _terminate_process_tree(process)
472     try:
473         return_code = process.wait(timeout=5)
474     except subprocess.TimeoutExpired:
475         # 进程句柄异常时也要继续落库和清理临时目录，不能让调度线程退出。
476         return_code = int(process.poll() if process.poll() is not None else -9)
477 finally:
478     heartbeat_stop.set()
479     heartbeat_thread.join(timeout=2)
480     with self._lock:
481         if self._process is process:
482             self._process = None
483             self._active_task_id = None
484
485 if self._task_is_cancelled(task_id) or self._stop_event.is_set() or heartbeat_lost.is_set():
486     shutil.rmtree(temporary_dir, ignore_errors=True)
487     return
```


backend/app/task_executor.py

```
488         if timed_out:
489             self._mark_failed(
490                 task_id,
491                 "timeout",
492                 f"任务运行超过最大时长 {float(settings.task_max_runtime_seconds):g} 秒，已终止完整进程树",
493             )
494             shutil.rmtree(temporary_dir, ignore_errors=True)
495             return
496     if return_code != 0 or completed_manifest is None:
497         self._mark_failed(
498             task_id,
499             "execution_error",
500             last_error or f"OMELET 子进程异常退出，退出码 {return_code}",
501         )
502         shutil.rmtree(temporary_dir, ignore_errors=True)
503         return
504     if not _is_in_directory(completed_manifest, temporary_dir) or not completed_manifest.is_file():
505         self._mark_failed(task_id, "invalid_result", "任务结果清单不存在或位于受控目录之外")
506         shutil.rmtree(temporary_dir, ignore_errors=True)
507         return
508     try:
509         manifest = json.loads(completed_manifest.read_text(encoding="utf-8"))
510         self._persist_completed_result(task_id, user_id, temporary_dir, manifest)
511     except Exception as exc:
512         self._mark_failed(task_id, "persist_error", f"结果持久化失败: {exc}")
513         shutil.rmtree(temporary_dir, ignore_errors=True)
514
515     def _touch_heartbeat(self, task_id: int) -> bool:
516         with SessionLocal() as session:
517             result = session.execute(
518                 update(AnalysisTask)
519                 .where(
520                     AnalysisTask.id == task_id,
521                     AnalysisTask.status == "running",
522                     AnalysisTask.worker_id == self.worker_id,
523                 )
524                 .values(heartbeat_at=_now()),
525             )
526             if result.rowcount != 1:
527                 session.rollback()
528                 return False
529             session.commit()
530             return True
531
532     def _persist_progress(self, task_id: int, event: dict[str, Any]) -> None:
533         """仅当前 worker 可以推进任务进度，失去租约后的迟到事件直接忽略。"""
534         with SessionLocal() as session:
535             task = session.scalar(
536                 select(AnalysisTask).where(
537                     AnalysisTask.id == task_id,
```

```
backend/app/task_executor.py

538             AnalysisTask.status == "running",
539             AnalysisTask.worker_id == self.worker_id,
540         ),
541     )
542     if task is None:
543         return
544     task.heartbeat_at = _now()
545     progress = event.get("progress")
546     if progress is not None:
547         task.progress = max(float(task.progress or 0), min(99.0, float(progress)))
548     task.current_stage = str(event.get("stage") or task.current_stage or "select_base")
549     if event.get("run") is not None:
550         next_run = max(1, int(event["run"]))
551         if next_run != int(task.current_run or 0):
552             task.current_iter = 0
553             task.current_run = next_run
554     if event.get("iteration") is not None:
555         task.current_iter = int(event["iteration"])
556     if event.get("type") == "run_completed":
557         session.add(
558             OperationLog(
559                 user_id=task.user_id,
560                 task_id=task.id,
561                 action="task_run_completed",
562                 message=f"第 {task.current_run} 轮实验完成",
563                 detail_json=_dump_json(event.get("metrics") or {}),
564             ),
565         )
566     session.commit()

def _persist_completed_result(
    self,
    task_id: int,
    user_id: int,
    temporary_dir: Path,
    manifest: dict[str, Any],
) -> None:
    """在锁定任务行后完成目录切换和结果写入，首个成功结果一经提交便不可覆盖。"""
    if manifest.get("schemaVersion") != 1:
        raise ValueError("不支持的任务结果版本")
    artifacts = manifest.get("artifacts")
    if not isinstance(artifacts, dict):
        raise ValueError("任务结果缺少产物清单")
    required_keys = {"labels", "ca", "s", "z"}
    if not required_keys.issubset(artifacts):
        raise ValueError("任务结果缺少关键矩阵或标签产物")
    for relative_path in artifacts.values():
        artifact_path = temporary_dir / str(relative_path)
        if not _is_in_directory(artifact_path, temporary_dir) or not artifact_path.is_file():
            raise ValueError("任务结果包含不安全或不存在的产物路径")
```

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```
588
589     result_root = Path(get_settings().result_storage_dir).resolve()
590     final_dir: Path | None = None
591     moved = False
592     try:
593         with SessionLocal() as session:
594             task = session.scalar(
595                 select(AnalysisTask)
596                     .where(
597                         AnalysisTask.id == task_id,
598                         AnalysisTask.status == "running",
599                         AnalysisTask.worker_id == self.worker_id,
600                     )
601                 .with_for_update(),
602             )
603         if task is None:
604             shutil.rmtree(temporary_dir, ignore_errors=True)
605             return
606
607         # 首次执行保持既有目录布局; 重试使用同级独立目录, 避免旧目录清理延迟时误删新结果。
608         retry_count = int(task.retry_count or 0)
609         final_dir = result_root / str(user_id) / str(task_id)
610         if retry_count > 0:
611             final_dir = result_root / str(user_id) / f"{task_id}-attempt-{retry_count}"
612         final_dir.parent.mkdir(parents=True, exist_ok=True)
613
614         existing = session.scalar(
615             select(TaskResult).where(TaskResult.task_id == task_id).with_for_update(),
616         )
617         if existing is not None:
618             # 重复完成事件只清理本次临时目录, 绝不覆盖已提交结果。
619             shutil.rmtree(temporary_dir, ignore_errors=True)
620             return
621         if final_dir.exists():
622             raise ValueError("任务结果目录已存在, 拒绝覆盖已有结果")
623
624         shutil.move(str(temporary_dir), str(final_dir))
625         moved = True
626         result = TaskResult(
627             task_id=task_id,
628             schema_version=1,
629             metrics_json=_dump_json(manifest["metrics"]),
630             kernel_weights_json=_dump_json(manifest["kernelWeights"]),
631             convergence_json=_dump_json(manifest["convergence"]),
632             preview_json=_dump_json(manifest["preview"]),
633             labels_path=str(final_dir / str(artifacts["labels"])),
634             ca_matrix_path=str(final_dir / str(artifacts["ca"])),
635             s_matrix_path=str(final_dir / str(artifacts["s"])),
636             z_matrix_path=str(final_dir / str(artifacts["z"])),
637             runtime_seconds=float(manifest["runtimeSeconds"]),
```

backend/app/task_executor.py

```
638         )
639         session.add(result)
640         task.status = "succeeded"
641         task.progress = 100.0
642         task.current_iter = int(task.max_iter or 0)
643         task.current_run = int(_load_json(task.params_json, {})).get("runs") or 1)
644         task.current_stage = "persist"
645         task.finished_at = _now()
646         task.error_message = None
647         task.failure_reason = None
648         task.worker_id = None
649         task.heartbeat_at = None
650         session.add(
651             OperationLog(
652                 user_id=task.user_id,
653                 task_id=task.id,
654                 action="task_succeeded",
655                 message="OMELET 分析完成, 结果已持久化",
656                 detail_json=_dump_json(
657                     {
658                         "runtimeSeconds": result.runtime_seconds,
659                         "metrics": manifest["metrics"].get("aggregate", {}),
660                     },
661                 ),
662             ),
663         )
664         session.commit()
665     except Exception:
666         if moved and final_dir is not None:
667             shutil.rmtree(final_dir, ignore_errors=True)
668         raise
669
670 def _task_is_cancelled(self, task_id: int) -> bool:
671     with SessionLocal() as session:
672         task = session.get(AnalysisTask, task_id)
673         return task is not None and task.status == "cancelled"
674
675 def _mark_failed(self, task_id: int, reason: str, message: str) -> None:
676     with SessionLocal() as session:
677         task = session.scalar(
678             select(AnalysisTask)
679             .where(
680                 AnalysisTask.id == task_id,
681                 AnalysisTask.worker_id == self.worker_id,
682             )
683             .with_for_update(),
684         )
685         if task is None or task.status in TERMINAL_STATUSES:
686             return
687         task.status = "failed"
```

```
backend/app/task_executor.py
```

```
688         task.failure_reason = reason
689         task.error_message = message[:1000]
690         task.finished_at = _now()
691         task.worker_id = None
692         task.heartbeat_at = None
693         session.add(
694             OperationLog(
695                 user_id=task.user_id,
696                 task_id=task.id,
697                 action="task_failed",
698                 level="error",
699                 message=task.error_message[:500],
700                 detail_json=_dump_json({"failureReason": reason}),
701             ),
702         )
703         session.commit()
704
705     def _requeue_active_task(self, task_id: int) -> None:
706         with SessionLocal() as session:
707             task = session.scalar(
708                 select(AnalysisTask)
709                 .where(
710                     AnalysisTask.id == task_id,
711                     AnalysisTask.status == "running",
712                     AnalysisTask.worker_id == self.worker_id,
713                 )
714                 .with_for_update(),
715             )
716             if task is None:
717                 return
718             task.status = "queued"
719             task.retry_count = int(task.retry_count or 0) + 1
720             task.progress = 0.0
721             task.current_run = 0
722             task.current_iter = 0
723             task.current_stage = "select_base"
724             task.started_at = None
725             task.finished_at = None
726             task.queued_at = _now()
727             task.worker_id = None
728             task.heartbeat_at = None
729             session.add(
730                 OperationLog(
731                     user_id=task.user_id,
732                     task_id=task.id,
733                     action="task_requeued",
734                     level="warning",
735                     message="服务停止，任务已重新排队",
736                     detail_json=_dump_json({"retryCount": task.retry_count}),
737                 ),
```

```
backend/app/task_executor.py
738         )
739         session.commit()
740
741
742 task_execution_manager = TaskExecutionManager()
743
744
745 # ===== 文件: backend/app/task_worker.py =====
746 1 """单个分析任务的子进程入口。
747 2
748 3 该模块把平台任务参数转换为算法调用，并把数值结果整理为可校验、可下载的标准产物。
749 4 """
750 5
751 6 from __future__ import annotations
752 7
753 8 import argparse
754 9 import csv
755 10 import json
756 11 import sys
757 12 from pathlib import Path
758 13 from time import perf_counter
759 14 from typing import Any
760 15
761 16 import numpy as np
762 17
763 18
764 19 PROJECT_ROOT = Path(__file__).resolve().parents[2]
765 20 ALGORITHM_ROOT = PROJECT_ROOT / "ec_python_converted"
766 21 if str(ALGORITHM_ROOT) not in sys.path:
767 22     sys.path.insert(0, str(ALGORITHM_ROOT))
768 23
769 24 from omelet.analysis import run_analysis # noqa: E402
770 25
771 26
772 27 EVENT_PREFIX = "OMELET_EVENT "
773 28 KERNEL_KEYS = ("rbf_sigma_squared", "linear", "rbf_sigma", "polynomial_2")
774 29 _EVENT_CONTEXT: dict[str, Any] = {}
775 30
776 31
777 32 def emit(event_type: str, **payload: Any) -> None:
778 33     event = {"type": event_type, **_EVENT_CONTEXT, **payload}
779 34     print(
780 35         EVENT_PREFIX + json.dumps(event, ensure_ascii=False, allow_nan=False),
781 36         flush=True,
782 37     )
783 38
784 39
785 40 def _number(value: Any, digits: int = 10) -> float:
786 41     result = float(value)
787 42     if not np.isfinite(result):
```

```
backend/app/task_worker.py
```

```
43         raise ValueError("算法输出包含非有限数值")
44     return round(result, digits)
45
46
47 def _matrix_preview(matrix: np.ndarray, limit: int = 24) -> dict[str, Any]:
48     values = np.asarray(matrix, dtype=float)
49     if values.ndim != 2:
50         raise ValueError("矩阵产物必须是二维数组")
51     if not np.all(np.isfinite(values)):
52         raise ValueError("矩阵产物包含非有限数值")
53
54     row_indices = np.unique(np.linspace(0, values.shape[0] - 1, min(limit, values.shape[0]), dtype=int))
55     column_indices = np.unique(np.linspace(0, values.shape[1] - 1, min(limit, values.shape[1]), dtype=int))
56     sampled = values[np.ix_(row_indices, column_indices)]
57     diagonal = np.diag(values) if values.shape[0] == values.shape[1] else np.array([], dtype=float)
58     symmetry_error = (
59         float(np.max(np.abs(values - values.T)))
60         if values.shape[0] == values.shape[1] and values.size
61         else None
62     )
63
64     top_pairs: list[dict[str, Any]] = []
65     if values.shape[0] == values.shape[1] and values.shape[0] > 1:
66         upper_rows, upper_columns = np.triu_indices(values.shape[0], k=1)
67         upper_values = values[upper_rows, upper_columns]
68         count = min(20, upper_values.size)
69         if count:
70             positions = np.argpartition(upper_values, -count)[-count:]
71             positions = positions[np.argsort(upper_values[positions])[:-1]]
72             top_pairs = [
73                 {
74                     "row": int(upper_rows[position]) + 1,
75                     "column": int(upper_columns[position]) + 1,
76                     "value": _number(upper_values[position]),
77                 }
78                 for position in positions
79             ]
80
81     return {
82         "shape": [int(values.shape[0]), int(values.shape[1])],
83         "rowIndices": [int(index) + 1 for index in row_indices],
84         "columnIndices": [int(index) + 1 for index in column_indices],
85         "values": [_number(value, 8) for value in row] for row in sampled,
86         "stats": {
87             "min": _number(np.min(values)),
88             "max": _number(np.max(values)),
89             "mean": _number(np.mean(values)),
90             "std": _number(np.std(values)),
91             "diagonalMean": _number(np.mean(diagonal)) if diagonal.size else None,
92             "symmetryMaxError": _number(symmetry_error) if symmetry_error is not None else None,
```

```
backend/app/task_worker.py
```

```
93         "nonzeroRatio": _number(np.count_nonzero(values) / values.size),
94     },
95     "topPairs": top_pairs,
96 }
97
98
99 def _scatter_preview(
100     coordinates: np.ndarray,
101     predicted_labels: np.ndarray,
102     ground_truth: np.ndarray,
103     limit: int = 2000,
104 ) -> dict[str, Any]:
105     coordinates = np.asarray(coordinates, dtype=float)
106     predicted_labels = np.asarray(predicted_labels).reshape(-1)
107     ground_truth = np.asarray(ground_truth).reshape(-1)
108     sample_count = coordinates.shape[0]
109
110     if sample_count <= limit:
111         selected = np.arange(sample_count)
112     else:
113         selected_parts: list[np.ndarray] = []
114         unique_labels = np.unique(predicted_labels)
115         for label in unique_labels:
116             group = np.flatnonzero(predicted_labels == label)
117             allocation = max(1, round(limit * len(group) / sample_count))
118             positions = np.linspace(0, len(group) - 1, min(allocation, len(group)), dtype=int)
119             selected_parts.append(group[positions])
120         selected = np.unique(np.concatenate(selected_parts))[:limit]
121
122     points = [
123         {
124             "sampleIndex": int(index) + 1,
125             "x": _number(coordinates[index, 0], 8),
126             "y": _number(coordinates[index, 1], 8),
127             "predictedLabel": str(predicted_labels[index]),
128             "trueLabel": str(ground_truth[index]),
129         }
130         for index in selected
131     ]
132     return {"totalCount": int(sample_count), "sampled": len(points) < sample_count, "points": points}
133
134
135 def _metric_payload(run_results: list[dict[str, Any]]) -> dict[str, Any]:
136     aggregate: dict[str, Any] = {}
137     for key in ("acc", "nmi", "ari", "f1"):
138         values = np.array([result["metrics"][key] for result in run_results], dtype=float)
139         aggregate[key] = {
140             "mean": _number(np.mean(values)),
141             "std": _number(np.std(values)),
142             "min": _number(np.min(values)),
```


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```
143         "max": _number(np.max(values)),
144     }
145     return {
146         "aggregate": aggregate,
147         "runs": [
148             {
149                 "run": int(result["run"]),
150                 "seed": int(result["seed"]),
151                 "runtimeSeconds": _number(result["runtimeSeconds"], 6),
152                 **{key: _number(result["metrics"][key]) for key in ("acc", "nmi", "ari", "f1")},
153             }
154         for result in run_results
155     ],
156 }
157
158
159 def _kernel_payload(run_results: list[dict[str, Any]], representative_index: int) -> dict[str, Any]:
160     all_weights = np.stack([result["kernelWeights"] for result in run_results])
161     return {
162         "items": [
163             {
164                 "key": key,
165                 "mean": _number(np.mean(all_weights[:, index])),
166                 "std": _number(np.std(all_weights[:, index])),
167                 "representative": _number(all_weights[representative_index, index]),
168             }
169         for index, key in enumerate(KERNEL_KEYS)
170     ],
171     "runs": [
172         {
173             "run": int(result["run"]),
174             "values": [_number(value) for value in result["kernelWeights"]],
175         }
176     for result in run_results
177     ],
178 }
179
180
181 def _convergence_payload(run_results: list[dict[str, Any]], representative_index: int, max_iter: int) -> dict[str,
Any]:
182     runs: list[dict[str, Any]] = []
183     for result in run_results:
184         objective = np.asarray(result["objective"], dtype=float)
185         points = []
186         for index, value in enumerate(objective):
187             relative_change = None
188             if index:
189                 denominator = max(abs(float(objective[index - 1])), 1e-12)
190                 relative_change = _number(abs(float(value) - float(objective[index - 1])) / denominator)
191             points.append(
```

```
backend/app/task_worker.py
```

```
192         {
193             "iteration": index + 1,
194             "objective": _number(value),
195             "relativeChange": relative_change,
196         },
197     )
198     runs.append(
199         {
200             "run": int(result["run"]),
201             "converged": len(points) < max_iter,
202             "points": points,
203         },
204     )
205     return {"representativeRun": representative_index + 1, "runs": runs}
206
207
208 def _write_artifacts(
209     output_dir: Path,
210     run_results: list[dict[str, Any]],
211     representative_index: int,
212     ground_truth: np.ndarray,
213 ) -> dict[str, str]:
214     """同时保存机器可复用矩阵和用户可直接查看的 CSV，清单只记录相对文件名。"""
215     representative = run_results[representative_index]
216     labels = np.stack([result["labels"] for result in run_results])
217     np.savez_compressed(output_dir / "labels.npz", labels=labels, ground_truth=ground_truth)
218     np.savez_compressed(output_dir / "ca_matrix.npz", matrix=representative["ca"])
219     np.savez_compressed(output_dir / "s_matrix.npz", matrix=representative["s"])
220     np.savez_compressed(output_dir / "z_matrix.npz", matrix=representative["z"])
221
222     with (output_dir / "labels.csv").open("w", newline="", encoding="utf-8-sig") as stream:
223         writer = csv.writer(stream)
224         writer.writerow(["sample_index", "predicted_label", "true_label"])
225         for index, (predicted, actual) in enumerate(zip(representative["labels"], ground_truth), start=1):
226             writer.writerow([index, predicted, actual])
227
228     return {
229         "labels": "labels.npz",
230         "labelsCsv": "labels.csv",
231         "ca": "ca_matrix.npz",
232         "s": "s_matrix.npz",
233         "z": "z_matrix.npz",
234     }
235
236
237 def execute(job_path: Path) -> None:
238     """读取执行器生成的任务快照，完成计算后以 manifest 作为唯一完成信号。"""
239     job = json.loads(job_path.read_text(encoding="utf-8"))
240     _EVENT_CONTEXT.clear()
241     _EVENT_CONTEXT.update(
```

```
backend/app/task_worker.py
```

```
242         {
243             "taskId": job.get("taskId"),
244             "workerId": job.get("workerId"),
245         },
246     )
247     output_dir = Path(job["outputDir"]).resolve()
248     output_dir.mkdir(parents=True, exist_ok=True)
249     started = perf_counter()
250
251     def forward_progress(event: dict[str, Any]) -> None:
252         event_type = str(event.pop("type", "progress"))
253         emit(event_type, **event)
254
255     params = job["params"]
256     # 平台只向算法层传递已归一化参数; 算法实现不接触数据库、用户目录或认证上下文。
257     result = run_analysis(
258         job["datasetPath"],
259         mode=job["mode"],
260         n_base=int(params["nBase"]),
261         sigma=float(params["sigma"]),
262         lam=float(params["lambda"]),
263         gamma=float(params["gamma"]),
264         anchor=int(params.get("anchor") or 0),
265         runs=int(params["runs"]),
266         max_iter=int(params["maxIter"]),
267         random_seed=int(params.get("randomSeed") or 1),
268         progress_callback=forward_progress,
269     )
270     run_results = result["runs"]
271     representative_index = int(result["representativeIndex"])
272     representative = run_results[representative_index]
273     artifacts = _write_artifacts(output_dir, run_results, representative_index, result["groundTruth"])
274     metrics = _metric_payload(run_results)
275     kernels = _kernel_payload(run_results, representative_index)
276     convergence = _convergence_payload(run_results, representative_index, int(params["maxIter"]))
277     preview = {
278         "schemaVersion": 1,
279         "summary": {
280             "mode": result["mode"],
281             "sampleCount": result["sampleCount"],
282             "baseClusterCount": result["baseClusterCount"],
283             "classCount": result["classCount"],
284             "representativeRun": representative_index + 1,
285             "randomSeed": int(params.get("randomSeed") or 1),
286         },
287         "matrices": {
288             "ca": _matrix_preview(representative["ca"]),
289             "s": _matrix_preview(representative["s"]),
290             "z": _matrix_preview(representative["z"]),
291         },
```

```
backend/app/task_worker.py
```

```
292         "scatter": _scatter_preview(
293             representative["coordinates"],
294             representative["labels"],
295             result["groundTruth"],
296         ),
297     }
298     manifest = {
299         "schemaVersion": 1,
300         "runtimeSeconds": _number(perf_counter() - started, 6),
301         "parameters": params,
302         "metrics": metrics,
303         "kernelWeights": kernels,
304         "convergence": convergence,
305         "preview": preview,
306         "artifacts": artifacts,
307     }
308     manifest_path = output_dir / "manifest.json"
309     manifest_path.write_text(
310         json.dumps(manifest, ensure_ascii=False, allow_nan=False),
311         encoding="utf-8",
312     )
313     emit("completed", manifestPath=str(manifest_path), progress=100.0)
314
315
316 def main() -> None:
317     parser = argparse.ArgumentParser(description="OMELET 任务子进程")
318     parser.add_argument("--job", required=True, type=Path)
319     args = parser.parse_args()
320     try:
321         execute(args.job.resolve())
322     except Exception as exc:
323         emit("error", message=str(exc), errorType=type(exc).__name__)
324         raise
325
326
327 if __name__ == "__main__":
328     main()
329
330
331 # ===== 文件: backend/app/tasks.py =====
332 1 """分析任务业务接口。
333 2
334 3 本模块负责用户权限、任务状态转换、结果与导出访问；实际子进程调度由 task_executor 独立承担。
335 4 """
336 5
337 6 from __future__ import annotations
338 7
339 8 import json
340 9 import io
341 10 import math
```

backend/app/tasks.py

```
11 import zipfile
12 from datetime import datetime, timedelta
13 from pathlib import Path
14 from typing import Any
15
16 from fastapi import APIRouter, Depends, HTTPException, Query
17 from fastapi.responses import FileResponse
18 from sqlalchemy import and_, desc, func, or_, select, update
19 from sqlalchemy.orm import Session
20
21 from .auth import get_current_user
22 from .config import get_settings
23 from .database import get_session
24 from .models import (
25     AnalysisTask,
26     Dataset,
27     DatasetQuality,
28     OperationLog,
29     TaskExport,
30     TaskResult,
31     TaskTemplate,
32     User,
33 )
34 from .schemas import (
35     AnalysisTaskBulkRequest,
36     AnalysisTaskCreateRequest,
37     AnalysisTaskLogItem,
38     AnalysisTaskLogPageResponse,
39     AnalysisTaskMetricsSummary,
40     AnalysisTaskPageResponse,
41     AnalysisTaskParams,
42     AnalysisTaskResponse,
43     AnalysisTaskStatsResponse,
44     AnalysisTaskUpdateRequest,
45     MessageResponse,
46     TaskExportCreateRequest,
47     TaskExportListResponse,
48     TaskExportResponse,
49     TaskResultEnvelope,
50     TaskTemplateCreateRequest,
51     TaskTemplateListResponse,
52     TaskTemplateResponse,
53 )
54 from .storage_cleanup import enqueue_cleanups, process_pending_cleanup_jobs
55 from .task_executor import task_execution_manager
56 from .time_utils import format_utc_iso, utc_now
57
58 router = APIRouter(prefix="/api/tasks", tags=["tasks"])
59
60 TASK_STATUSES = ("draft", "queued", "running", "succeeded", "failed", "cancelled")
```

```
backend/app/tasks.py
```

```
61 DELETABLE_STATUSES = {"draft", "failed", "cancelled", "succeeded"}
62 STARTABLE_STATUSES = {"draft", "failed", "cancelled"}
63 CANCELABLE_STATUSES = {"queued", "running"}
64 MODES = {"OMELET", "OMELET-SV"}
65
66 # 五个固定分析阶段，前端流水线与进度映射共用。
67 PIPELINE_STAGES = [
68     ("select_base", "选择基础聚类结果"),
69     ("build_ca", "GBE 编码与 CA 构建"),
70     ("multi_kernel", "多核相似性学习"),
71     ("evaluate", "谱聚类与指标评估"),
72     ("persist", "结果落盘"),
73 ]
74
75
76 def _now() -> datetime:
77     return utc_now().replace(microsecond=0)
78
79
80 def _format_dt(value: datetime | None) -> str | None:
81     return format_utc_iso(value)
82
83
84 def _load_json(value: str | None, default: Any) -> Any:
85     if not value:
86         return default
87     try:
88         return json.loads(value)
89     except json.JSONDecodeError:
90         return default
91
92
93 def _dump_json(value: Any) -> str:
94     return json.dumps(value, ensure_ascii=False)
95
96
97 def _default_params(mode: str = "OMELET-SV") -> dict[str, Any]:
98     params = AnalysisTaskParams().model_dump(by_alias=True)
99     if mode != "OMELET-SV":
100         params["anchor"] = 0
101     return params
102
103
104 def _normalize_params(mode: str, params: AnalysisTaskParams | dict | None) -> dict[str, Any]:
105     """把前端参数归一化为执行器可直接消费的稳定结构，并在入口处收紧数值范围。"""
106     if params is None:
107         data = _default_params(mode)
108     elif isinstance(params, AnalysisTaskParams):
109         data = params.model_dump(by_alias=True)
110     else:
```

backend/app/tasks.py

```
111         data = {**_default_params(mode), **params}
112
113         if "lambdaValue" in data and "lambda" not in data:
114             data["lambda"] = data.pop("lambdaValue")
115         data.setdefault("nBase", 20)
116         data.setdefault("sigma", 1.0)
117         data.setdefault("lambda", 5.0)
118         data.setdefault("gamma", 5.0)
119         data.setdefault("anchor", 10 if mode == "OMELET-SV" else 0)
120         data.setdefault("runs", 10)
121         data.setdefault("maxIter", 10)
122         data.setdefault("randomSeed", 1)
123
124         data["nBase"] = max(1, int(data["nBase"]))
125         data["sigma"] = float(data["sigma"])
126         data["lambda"] = float(data["lambda"])
127         data["gamma"] = float(data["gamma"])
128         data["anchor"] = max(0, int(data["anchor"]))
129         data["runs"] = max(1, int(data["runs"]))
130         data["maxIter"] = max(1, int(data["maxIter"]))
131         data["randomSeed"] = max(0, int(data["randomSeed"]))
132         if mode != "OMELET-SV":
133             data["anchor"] = 0
134         return data
135
136
137 def _stage_from_progress(progress: float, status: str) -> str | None:
138     if status == "draft":
139         return None
140     if status == "queued":
141         return PIPELINE_STAGES[0][0]
142     if status == "succeeded":
143         return PIPELINE_STAGES[-1][0]
144     if status == "cancelled":
145         return None
146     ratio = max(0.0, min(float(progress), 100.0)) / 100.0
147     index = min(len(PIPELINE_STAGES) - 1, int(ratio * len(PIPELINE_STAGES)))
148     if ratio >= 1:
149         index = len(PIPELINE_STAGES) - 1
150     return PIPELINE_STAGES[index][0]
151
152
153 def _get_user_dataset(session: Session, dataset_id: int, user: User) -> Dataset:
154     dataset = session.scalar(select(Dataset).where(Dataset.id == dataset_id, Dataset.user_id == user.id))
155     if dataset is None:
156         raise HTTPException(404, "数据集不存在")
157     return dataset
158
159
160 def _ensure_taskable(session: Session, dataset: Dataset) -> None:
```

backend/app/tasks.py

```
161     """任务创建和启动共用的数据质量门槛，避免不可评估的数据进入执行队列。"""
162     quality = session.scalar(select(DatasetQuality).where(DatasetQuality.dataset_id == dataset.id))
163     if quality and quality.status == "error":
164         raise HTTPException(422, "数据集质量检查未通过，无法创建或启动任务")
165     if not dataset.has_ground_truth or not dataset.cluster_count or dataset.cluster_count < 2:
166         raise HTTPException(422, "OMELET 分析需要包含至少两个类别的 y 标签")
167
168
169 def _get_user_task(session: Session, task_id: int, user: User) -> AnalysisTask:
170     task = session.scalar(select(AnalysisTask).where(AnalysisTask.id == task_id, AnalysisTask.user_id == user.id))
171     if task is None:
172         raise HTTPException(404, "任务不存在")
173     return task
174
175
176 def _add_log(
177     session: Session,
178     *,
179     user_id: int | None,
180     task_id: int | None,
181     action: str,
182     message: str,
183     level: str = "info",
184     detail: dict | None = None,
185 ) -> None:
186     session.add(
187         OperationLog(
188             user_id=user_id,
189             task_id=task_id,
190             action=action,
191             level=level,
192             message=message[:500],
193             detail_json=_dump_json(detail) if detail is not None else None,
194         ),
195     )
196
197
198 def _metrics_summary(session: Session, task_id: int) -> AnalysisTaskMetricsSummary | None:
199     result = session.scalar(select(TaskResult).where(TaskResult.task_id == task_id))
200     if result is None or result.schema_version != 1:
201         return None
202     metrics = _load_json(result.metrics_json, {})
203     aggregate = metrics.get("aggregate") if isinstance(metrics, dict) else None
204     if not isinstance(aggregate, dict):
205         return None
206     return AnalysisTaskMetricsSummary(
207         acc=aggregate.get("acc", {}).get("mean"),
208         nmi=aggregate.get("nmi", {}).get("mean"),
209         ari=aggregate.get("ari", {}).get("mean"),
210         f1=aggregate.get("f1", {}).get("mean"),
```


backend/app/tasks.py

```
211     )
212
213
214 def _runtime_seconds(task: AnalysisTask, session: Session | None = None) -> float | None:
215     if session is not None:
216         result = session.scalar(select(TaskResult).where(TaskResult.task_id == task.id))
217         if result and result.runtime_seconds is not None:
218             return float(result.runtime_seconds)
219     if task.started_at and task.finished_at:
220         return max(0.0, (task.finished_at - task.started_at).total_seconds())
221     if task.started_at and task.status == "running":
222         return max(0.0, (_now() - task.started_at).total_seconds())
223     return None
224
225
226 def _task_response(session: Session, task: AnalysisTask, dataset_name: str) -> AnalysisTaskResponse:
227     queue_position = None
228     if task.status == "queued" and task.queued_at is not None:
229         queue_position = int(
230             session.scalar(
231                 select(func.count(AnalysisTask.id)).where(
232                     AnalysisTask.status == "queued",
233                     or_(
234                         AnalysisTask.queued_at < task.queued_at,
235                         and_(AnalysisTask.queued_at == task.queued_at, AnalysisTask.id <= task.id),
236                     ),
237                 ),
238             )
239         or 1,
240     )
241     params = _load_json(task.params_json, {})
242     return AnalysisTaskResponse(
243         id=task.id,
244         name=task.name,
245         mode=task.mode,
246         status=task.status,
247         progress=float(task.progress or 0),
248         currentRun=int(task.current_run or 0),
249         totalRuns=max(1, int(params.get("runs") or 1)),
250         queuePosition=queue_position,
251         currentIter=int(task.current_iter or 0),
252         maxIter=int(task.max_iter or 0),
253         currentStage=task.current_stage,
254         datasetId=task.dataset_id,
255         datasetName=dataset_name,
256         params=params,
257         errorMessage=task.error_message,
258         failureReason=task.failure_reason,
259         metricsSummary=_metrics_summary(session, task.id),
260         runtimeSeconds=_runtime_seconds(task, session),
```

```
backend/app/tasks.py
```

```
261         createdAt=_format_dt(task.created_at) or "",
262         startedAt=_format_dt(task.started_at),
263         finishedAt=_format_dt(task.finished_at),
264         updatedAt=_format_dt(task.updated_at) or "",
265     )
266
267
268 def _refresh_user_running_tasks(session: Session, user_id: int, task_ids: list[int] | None = None) -> None:
269     """执行器独立写入进度，保留旧调用点以兼容任务列表与统计接口。"""
270     return None
271
272
273 def _queue_task(
274     session: Session,
275     task: AnalysisTask,
276     user: User,
277     *,
278     increment_retry: bool = False,
279 ) -> None:
280     """仅负责持久化排队状态；后台执行器再通过数据库事务原子领取任务。"""
281     if task.status not in STARTABLE_STATUSES and task.status != "queued":
282         raise HTTPException(422, f"当前状态不可启动: {task.status}")
283
284     dataset = _get_user_dataset(session, task.dataset_id, user)
285     _ensure_taskable(session, dataset)
286
287     params = _load_json(task.params_json, {})
288     n_base = int(params.get("nBase") or 0)
289     if n_base > dataset.base_cluster_count:
290         raise HTTPException(422, f"n_base 不能超过数据集基础聚类数 {dataset.base_cluster_count}")
291
292     task.status = "queued"
293     if increment_retry:
294         task.retry_count = int(task.retry_count or 0) + 1
295     task.progress = 0
296     task.current_run = 0
297     task.current_iter = 0
298     task.error_message = None
299     task.failure_reason = None
300     task.queued_at = _now()
301     task.started_at = None
302     task.finished_at = None
303     task.worker_id = None
304     task.heartbeat_at = None
305     task.current_stage = PIPELINE_STAGES[0][0]
306     _add_log(
307         session,
308         user_id=user.id,
309         task_id=task.id,
310         action="task_queued",
```

backend/app/tasks.py

```
311         message="任务已进入执行队列",
312         detail={"mode": task.mode, "datasetId": task.dataset_id},
313     )
314
315
316 def _task_cleanup_paths(session: Session, task_id: int) -> set[tuple[str, Path]]:
317     """收集任务结果和导出文件，调用方必须在删除业务记录的事务中入队。"""
318     paths: set[tuple[str, Path]] = set()
319     results = session.scalars(select(TaskResult).where(TaskResult.task_id == task_id)).all()
320     for result in results:
321         for artifact_path in (
322             result.labels_path,
323             result.ca_matrix_path,
324             result.s_matrix_path,
325             result.z_matrix_path,
326         ):
327             if artifact_path:
328                 paths.add(("result", Path(artifact_path).parent))
329
330     exports = session.scalars(select(TaskExport).where(TaskExport.task_id == task_id)).all()
331     for export in exports:
332         paths.add(("export", Path(export.storage_path)))
333     return paths
334
335
336 def _detach_task_logs(session: Session, task_id: int) -> None:
337     """SQLite 测试库未必启用外键时，也保持 SET NULL 的业务语义。"""
338     session.execute(
339         update(OperationLog)
340         .where(OperationLog.task_id == task_id)
341         .values(task_id=None),
342     )
343
344
345 def _template_response(template: TaskTemplate) -> TaskTemplateResponse:
346     return TaskTemplateResponse(
347         id=template.id,
348         name=template.name,
349         mode=template.mode,
350         params=_load_json(template.params_json, {}),
351         createdAt=_format_dt(template.created_at) or "",
352     )
353
354
355 @router.get("/stats", response_model=AnalysisTaskStatsResponse)
356 def get_task_stats(
357     user: User = Depends(get_current_user),
358     session: Session = Depends(get_session),
359 ):
360     _refresh_user_running_tasks(session, user.id)
```

backend/app/tasks.py

```
361
362     rows = session.execute(
363         select(AnalysisTask.status, func.count(AnalysisTask.id))
364         .where(AnalysisTask.user_id == user.id)
365         .group_by(AnalysisTask.status),
366     ).all()
367     counts = {status: 0 for status in TASK_STATUSES}
368     total = 0
369     for status, count in rows:
370         counts[status] = int(count)
371         total += int(count)
372
373     today = _now().date()
374     today_completed = int(
375         session.scalar(
376             select(func.count(AnalysisTask.id)).where(
377                 AnalysisTask.user_id == user.id,
378                 AnalysisTask.status == "succeeded",
379                 AnalysisTask.finished_at.is_not(None),
380                 func.date(AnalysisTask.finished_at) == today,
381             ),
382         )
383         or 0,
384     )
385
386     runtime_values = session.scalars(
387         select(TaskResult.runtime_seconds)
388         .join(AnalysisTask, AnalysisTask.id == TaskResult.task_id)
389         .where(AnalysisTask.user_id == user.id, TaskResult.runtime_seconds.is_not(None)),
390     ).all()
391     average_runtime = round(sum(float(v) for v in runtime_values) / len(runtime_values), 2) if runtime_values else
None
392
393     finished = counts["succeeded"] + counts["failed"]
394     failure_rate = round(counts["failed"] / finished, 4) if finished else 0.0
395
396     return AnalysisTaskStatsResponse(
397         total=total,
398         draft=counts["draft"],
399         queued=counts["queued"],
400         running=counts["running"],
401         succeeded=counts["succeeded"],
402         failed=counts["failed"],
403         cancelled=counts["cancelled"],
404         todayCompleted=today_completed,
405         averageRuntimeSeconds=average_runtime,
406         failureRate=failure_rate,
407     )
408
409
```

backend/app/tasks.py

```
410 @router.get("/templates", response_model=TaskTemplateListResponse)
411 def list_templates(
412     user: User = Depends(get_current_user),
413     session: Session = Depends(get_session),
414 ):
415     rows = session.scalars(
416         select(TaskTemplate).where(TaskTemplate.user_id == user.id).order_by(desc(TaskTemplate.created_at)),
417     ).all()
418     return TaskTemplateListResponse(items=[_template_response(item) for item in rows])
419
420
421 @router.post("/templates", response_model=TaskTemplateResponse)
422 def create_template(
423     payload: TaskTemplateCreateRequest,
424     user: User = Depends(get_current_user),
425     session: Session = Depends(get_session),
426 ):
427     mode = payload.mode if payload.mode in MODES else "OMELET-SV"
428     template = TaskTemplate(
429         user_id=user.id,
430         name=payload.name.strip()[:128],
431         mode=mode,
432         params_json=_dump_json(_normalize_params(mode, payload.params)),
433     )
434     session.add(template)
435     session.commit()
436     session.refresh(template)
437     return _template_response(template)
438
439
440 @router.delete("/templates/{template_id}", response_model=MessageResponse)
441 def delete_template(
442     template_id: int,
443     user: User = Depends(get_current_user),
444     session: Session = Depends(get_session),
445 ):
446     template = session.scalar(
447         select(TaskTemplate).where(TaskTemplate.id == template_id, TaskTemplate.user_id == user.id),
448     )
449     if template is None:
450         raise HTTPException(404, "模板不存在")
451     session.delete(template)
452     session.commit()
453     return MessageResponse(message="模板已删除")
454
455
456 @router.get("", response_model=AnalysisTaskPageResponse)
457 def list_tasks(
458     page: int = Query(default=1, ge=1),
459     page_size: int = Query(default=20, ge=1, le=100, alias="pageSize"),
```

backend/app/tasks.py

```
460     status: str | None = Query(default=None),
461     mode: str | None = Query(default=None),
462     dataset_id: int | None = Query(default=None, alias="datasetId"),
463     keyword: str | None = Query(default=None),
464     created_from: str | None = Query(default=None, alias="createdFrom"),
465     created_to: str | None = Query(default=None, alias="createdTo"),
466     user: User = Depends(get_current_user),
467     session: Session = Depends(get_session),
468 ):
469     _refresh_user_running_tasks(session, user.id)
470
471     stmt = (
472         select(AnalysisTask, Dataset.name)
473         .join(Dataset, Dataset.id == AnalysisTask.dataset_id)
474         .where(AnalysisTask.user_id == user.id)
475     )
476
477     if status:
478         statuses = [item.strip() for item in status.split(",") if item.strip()]
479         if statuses:
480             stmt = stmt.where(AnalysisTask.status.in_(statuses))
481
482     if mode:
483         stmt = stmt.where(AnalysisTask.mode == mode)
484
485     if dataset_id is not None:
486         stmt = stmt.where(AnalysisTask.dataset_id == dataset_id)
487
488     if keyword:
489         like = f"%{keyword.strip()}%"
490         stmt = stmt.where(
491             or_(
492                 AnalysisTask.name.like(like),
493                 Dataset.name.like(like),
494                 AnalysisTask.mode.like(like),
495             ),
496         )
497
498     if created_from:
499         try:
500             start = datetime.strptime(created_from, "%Y-%m-%d")
501             stmt = stmt.where(AnalysisTask.created_at >= start)
502         except ValueError as exc:
503             raise HTTPException(422, "createdFrom 格式应为 YYYY-MM-DD") from exc
504
505     if created_to:
506         try:
507             end = datetime.strptime(created_to, "%Y-%m-%d") + timedelta(days=1)
508             stmt = stmt.where(AnalysisTask.created_at < end)
509         except ValueError as exc:
510             raise HTTPException(422, "createdTo 格式应为 YYYY-MM-DD") from exc
511
512     total = int(session.scalar(select(func.count()).select_from(stmt.order_by(None).subquery()))) or 0)
513     rows = session.execute(
514         stmt.order_by(desc(AnalysisTask.updated_at), desc(AnalysisTask.id))
515     )
```

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```
510         .offset((page - 1) * page_size)
511         .limit(page_size),
512     ).all()
513
514     total_pages = max(1, math.ceil(total / page_size)) if total else 0
515     return AnalysisTaskPageResponse(
516         items=[_task_response(session, task, dataset_name) for task, dataset_name in rows],
517         total=total,
518         page=page,
519         pageSize=page_size,
520         totalPages=total_pages,
521     )
522
523
524 @router.post("", response_model=AnalysisTaskResponse)
525 def create_task(
526     payload: AnalysisTaskCreateRequest,
527     user: User = Depends(get_current_user),
528     session: Session = Depends(get_session),
529 ):
530     dataset = _get_user_dataset(session, payload.datasetId, user)
531     _ensure_taskable(session, dataset)
532
533     mode = payload.mode if payload.mode in MODES else "OMELET-SV"
534     params = _normalize_params(mode, payload.params)
535
536     if payload.templateId is not None:
537         template = session.scalar(
538             select(TaskTemplate).where(TaskTemplate.id == payload.templateId, TaskTemplate.user_id == user.id),
539         )
540         if template is None:
541             raise HTTPException(404, "模板不存在")
542         mode = template.mode if template.mode in MODES else mode
543         params = _normalize_params(mode, _load_json(template.params_json, {}))
544
545     # 未显式给出过大 nBase 时，自动裁到数据集可用上限；显式超限仍返回 422。
546     explicit_n_base = None
547     if payload.params is not None:
548         explicit_n_base = payload.params.nBase
549     if explicit_n_base is not None and int(explicit_n_base) > dataset.base_cluster_count:
550         raise HTTPException(422, f"n_base 不能超过数据集基础聚类数 {dataset.base_cluster_count}")
551     params["nBase"] = min(int(params["nBase"]), int(dataset.base_cluster_count))
552
553     task_name = (payload.name or f"{dataset.name} {mode} 任务").strip()[:128]
554     task = AnalysisTask(
555         user_id=user.id,
556         dataset_id=dataset.id,
557         name=task_name,
558         mode=mode,
559         status="draft",
```

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```
560         progress=0,
561         current_iter=0,
562         max_iter=int(params["maxIter"]),
563         params_json=_dump_json(params),
564         current_stage=None,
565     )
566     session.add(task)
567     session.flush()
568     _add_log(
569         session,
570         user_id=user.id,
571         task_id=task.id,
572         action="task_created",
573         message=f"创建任务: {task.name}",
574         detail={"mode": mode, "datasetId": dataset.id, "params": params},
575     )
576
577     should_notify_executor = bool(payload.startImmediately)
578     if should_notify_executor:
579         _queue_task(session, task, user)
580
581     session.commit()
582     session.refresh(task)
583     if should_notify_executor:
584         task_execution_manager.notify()
585     return _task_response(session, task, dataset.name)
586
587
588 def _result_artifact_paths(result: TaskResult) -> dict[str, Path]:
589     paths = {
590         "labels": result.labels_path,
591         "ca": result.ca_matrix_path,
592         "s": result.s_matrix_path,
593         "z": result.z_matrix_path,
594     }
595     resolved = {key: Path(value).resolve() for key, value in paths.items() if value}
596     labels_path = resolved.get("labels")
597     if labels_path is not None:
598         csv_path = labels_path.parent / "labels.csv"
599         if csv_path.is_file():
600             resolved["labels-csv"] = csv_path
601     return resolved
602
603
604 def _safe_artifact_path(result: TaskResult, artifact_key: str) -> Path:
605     """只允许访问结果清单登记且位于任务目录内的文件，阻断路径穿越。"""
606     path = _result_artifact_paths(result).get(artifact_key)
607     if path is None or not path.is_file():
608         raise HTTPException(404, "请求的结果产物不存在")
609     root = Path(get_settings().result_storage_dir).resolve()
```


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```
610     try:
611         path.relative_to(root)
612     except ValueError as exc:
613         raise HTTPException(404, "请求的结果产物不可访问") from exc
614     return path
615
616
617 def _artifact_metadata(task_id: int, result: TaskResult) -> list[dict[str, Any]]:
618     labels = {
619         "labels": ("全部轮次标签", "NPZ"),
620         "labels-csv": ("代表轮次标签", "CSV"),
621         "ca": ("CA 协关联矩阵", "NPZ"),
622         "s": ("S 共识矩阵", "NPZ"),
623         "z": ("Z 拓扑矩阵", "NPZ"),
624     }
625     return [
626         {
627             "key": key,
628             "name": labels[key][0],
629             "format": labels[key][1],
630             "size": path.stat().st_size,
631             "downloadUrl": f"/api/tasks/{task_id}/artifacts/{key}",
632         }
633         for key, path in _result_artifact_paths(result).items()
634         if key in labels and path.is_file()
635     ]
636
637
638 def _result_envelope(session: Session, task: AnalysisTask, dataset_name: str) -> TaskResultEnvelope:
639     """将任务状态和已持久化结果合并为前端唯一的结果读取契约。"""
640     response_task = _task_response(session, task, dataset_name)
641     if task.status != "succeeded":
642         return TaskResultEnvelope(state=task.status, task=response_task)
643     result = session.scalar(select(TaskResult).where(TaskResult.task_id == task.id))
644     if result is None or result.schema_version != 1:
645         return TaskResultEnvelope(state="legacy", task=response_task)
646     metrics = _load_json(result.metrics_json, {})
647     kernels = _load_json(result.kernel_weights_json, {})
648     convergence = _load_json(result.convergence_json, {})
649     preview = _load_json(result.preview_json, {})
650     if not isinstance(preview, dict) or preview.get("schemaVersion") != 1:
651         return TaskResultEnvelope(state="legacy", task=response_task)
652     return TaskResultEnvelope(
653         state="ready",
654         task=response_task,
655         result={
656             "schemaVersion": 1,
657             "parameters": _load_json(task.params_json, {}),
658             "runtimeSeconds": result.runtime_seconds,
659             "metrics": metrics,
```

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```
660         "kernelWeights": kernels,
661         "convergence": convergence,
662         "preview": preview,
663         "artifacts": _artifact_metadata(task.id, result),
664     },
665 )
666
667
668 @router.get("/results/latest", response_model=TaskResultEnvelope)
669 def get_latest_result(
670     user: User = Depends(get_current_user),
671     session: Session = Depends(get_session),
672 ):
673     task = session.scalar(
674         select(AnalysisTask)
675         .join(TaskResult, TaskResult.task_id == AnalysisTask.id)
676         .where(
677             AnalysisTask.user_id == user.id,
678             AnalysisTask.status == "succeeded",
679             TaskResult.schema_version == 1,
680         )
681         .order_by(desc(AnalysisTask.finished_at), desc(AnalysisTask.id))
682         .limit(1),
683     )
684     if task is None:
685         return TaskResultEnvelope(state="empty")
686     dataset = _get_user_dataset(session, task.dataset_id, user)
687     return _result_envelope(session, task, dataset.name)
688
689
690 @router.get("/{task_id}/result", response_model=TaskResultEnvelope)
691 def get_task_result(
692     task_id: int,
693     user: User = Depends(get_current_user),
694     session: Session = Depends(get_session),
695 ):
696     task = _get_user_task(session, task_id, user)
697     dataset = _get_user_dataset(session, task.dataset_id, user)
698     return _result_envelope(session, task, dataset.name)
699
700
701 @router.get("/{task_id}/artifacts/{artifact_key}")
702 def download_task_artifact(
703     task_id: int,
704     artifact_key: str,
705     user: User = Depends(get_current_user),
706     session: Session = Depends(get_session),
707 ):
708     _get_user_task(session, task_id, user)
709     result = session.scalar(select(TaskResult).where(TaskResult.task_id == task_id, TaskResult.schema_version == 1))
```

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```
710     if result is None:
711         raise HTTPException(404, "任务结果不存在")
712     path = _safe_artifact_path(result, artifact_key)
713     media_type = "text/csv; charset=utf-8" if path.suffix == ".csv" else "application/octet-stream"
714     return FileResponse(path, media_type=media_type, filename=path.name)
715
716
717 def _export_response(item: TaskExport) -> TaskExportResponse:
718     selected = _load_json(item.items_json, [])
719     if not isinstance(selected, list):
720         selected = []
721     return TaskExportResponse(
722         id=item.id,
723         name=(item.name or item.filename).strip(),
724         items=[str(value) for value in selected],
725         itemCount=len(selected),
726         status=item.status or "ready",
727         filename=item.filename,
728         fileSize=int(item.file_size or 0),
729         createdAt=_format_dt(item.created_at) or "",
730         downloadUrl=f"/api/tasks/{item.task_id}/exports/{item.id}/download",
731     )
732
733
734 @router.get("/{task_id}/exports", response_model=TaskExportListResponse)
735 def list_task_exports(
736     task_id: int,
737     user: User = Depends(get_current_user),
738     session: Session = Depends(get_session),
739 ):
740     _get_user_task(session, task_id, user)
741     rows = session.scalars(
742         select(TaskExport)
743         .where(TaskExport.task_id == task_id)
744         .order_by(desc(TaskExport.created_at), desc(TaskExport.id)),
745     ).all()
746     return TaskExportListResponse(items=[_export_response(item) for item in rows])
747
748
749 @router.post("/{task_id}/exports", response_model=TaskExportResponse)
750 def create_task_export(
751     task_id: int,
752     payload: TaskExportCreateRequest,
753     user: User = Depends(get_current_user),
754     session: Session = Depends(get_session),
755 ):
756     """先生成受控临时档案再提交数据库，任何失败都不会留下未登记文件。"""
757     task = _get_user_task(session, task_id, user)
758     result = session.scalar(select(TaskResult).where(TaskResult.task_id == task_id, TaskResult.schema_version == 1))
759     if result is None:
```

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```
760         raise HTTPException(409, "仅已完成的任務可以导出")
761     allowed = {"metrics", "parameters", "result", "labels", "ca", "s", "z"}
762     selected = list(dict.fromkeys(item.strip().lower() for item in payload.items if item.strip()))
763     if not selected or any(item not in allowed for item in selected):
764         raise HTTPException(422, "导出项包含不支持的类型")
765
766     root = Path(get_settings().result_storage_dir).resolve()
767     source_dir = _safe_artifact_path(result, "labels").parent
768     export_dir = source_dir / "exports"
769     export_dir.mkdir(parents=True, exist_ok=True)
770     created_at = utc_now()
771     archive_name = (payload.name or "").strip() or f"任务 #{task_id} 交付档案"
772     # 精确到微秒, 避免同一任务在同一秒内导出时覆盖前一个档案。
773     filename = f"omelet-task-{task_id}-{created_at.strftime('%Y%m%d%H%M%S%f')}.zip"
774     export_path = export_dir / filename
775     metrics = _load_json(result.metrics_json, {})
776     preview = _load_json(result.preview_json, {})
777     try:
778         with zipfile.ZipFile(export_path, "w", compression=zipfile.ZIP_DEFLATED) as archive:
779             archive.writestr(
780                 "manifest.json",
781                 _dump_json({
782                     "schemaVersion": 1,
783                     "archiveName": archive_name,
784                     "taskId": task_id,
785                     "datasetId": task.dataset_id,
786                     "mode": task.mode,
787                     "items": selected,
788                     "createdAt": format_utc_iso(created_at),
789                 }),
790             )
791             if "metrics" in selected:
792                 output = io.StringIO()
793                 output.write("metric,mean,std,min,max\n")
794                 for key, values in metrics.get("aggregate", {}).items():
795                     output.write(
796                         f"{key}, {values.get('mean', '')}, {values.get('std', '')}, {values.get('min', '')}, {values.get('max', '')}\n",
797                     )
798                 archive.writestr("metrics.csv", output.getvalue().encode("utf-8-sig"))
799             if "parameters" in selected:
800                 archive.writestr("parameters.json", _dump_json(_load_json(task.params_json, {})))
801             if "result" in selected:
802                 archive.writestr(
803                     "result.json",
804                     _dump_json({
805                         "metrics": metrics,
806                         "kernelWeights": _load_json(result.kernel_weights_json, {}),
807                         "convergence": _load_json(result.convergence_json, {}),
808                         "preview": preview,
```

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```
809         }),
810     )
811     artifact_map = {"labels": "labels-csv", "ca": "ca", "s": "s", "z": "z"}
812     for item in selected:
813         key = artifact_map.get(item)
814         if key:
815             path = _safe_artifact_path(result, key)
816             archive.write(path, arcname=path.name)
817     try:
818         export_path.resolve().relative_to(root)
819     except ValueError as exc:
820         raise HTTPException(500, "导出文件路径异常") from exc
821 except Exception:
822     # 文件先写入、数据库后提交;生成或校验失败时不留下孤立档案。
823     export_path.unlink(missing_ok=True)
824     raise
825 item = TaskExport(
826     task_id=task_id,
827     export_type="zip",
828     name=archive_name,
829     items_json=_dump_json(selected),
830     status="ready",
831     filename=filename,
832     storage_path=str(export_path),
833     file_size=export_path.stat().st_size,
834 )
835 session.add(item)
836 try:
837     session.commit()
838 except Exception:
839     session.rollback()
840     # 数据库记录未提交时删除已生成文件,保持两侧状态一致。
841     export_path.unlink(missing_ok=True)
842     raise
843 session.refresh(item)
844 return _export_response(item)
845
846
847 @router.get("/{task_id}/exports/{export_id}/download")
848 def download_task_export(
849     task_id: int,
850     export_id: int,
851     user: User = Depends(get_current_user),
852     session: Session = Depends(get_session),
853 ):
854     _get_user_task(session, task_id, user)
855     item = session.scalar(select(TaskExport).where(TaskExport.id == export_id, TaskExport.task_id == task_id))
856     if item is None:
857         raise HTTPException(404, "导出文件不存在")
858     path = Path(item.storage_path).resolve()
```

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```
859     root = Path(get_settings().result_storage_dir).resolve()
860     try:
861         path.relative_to(root)
862     except ValueError as exc:
863         raise HTTPException(404, "导出文件不可访问") from exc
864     if not path.is_file():
865         raise HTTPException(404, "导出文件已被清理")
866     return FileResponse(path, media_type="application/zip", filename=item.filename)
867
868
869 @router.get("/{task_id}", response_model=AnalysisTaskResponse)
870 def get_task(
871     task_id: int,
872     user: User = Depends(get_current_user),
873     session: Session = Depends(get_session),
874 ):
875     _refresh_user_running_tasks(session, user.id, [task_id])
876     task = _get_user_task(session, task_id, user)
877     dataset = _get_user_dataset(session, task.dataset_id, user)
878     return _task_response(session, task, dataset.name)
879
880
881 @router.patch("/{task_id}", response_model=AnalysisTaskResponse)
882 def update_task(
883     task_id: int,
884     payload: AnalysisTaskUpdateRequest,
885     user: User = Depends(get_current_user),
886     session: Session = Depends(get_session),
887 ):
888     task = _get_user_task(session, task_id, user)
889     if task.status not in {"draft", "failed", "cancelled"}:
890         raise HTTPException(422, "仅草稿/失败/取消状态可修改参数")
891
892     dataset = _get_user_dataset(session, task.dataset_id, user)
893     if payload.name is not None:
894         task.name = payload.name.strip()[:128] or task.name
895     if payload.mode is not None:
896         if payload.mode not in MODES:
897             raise HTTPException(422, "不支持的算法模式")
898         task.mode = payload.mode
899     if payload.params is not None or payload.mode is not None:
900         params = _normalize_params(task.mode, payload.params or _load_json(task.params_json, {}))
901         if int(params["nBase"]) > dataset.base_cluster_count:
902             raise HTTPException(422, f"n_base 不能超过数据集基础聚类数 {dataset.base_cluster_count}")
903         task.params_json = _dump_json(params)
904         task.max_iter = int(params["maxIter"])
905
906     _add_log(
907         session,
908         user_id=user.id,
```

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```
909         task_id=task.id,
910         action="task_updated",
911         message="任务参数已更新",
912     )
913     session.commit()
914     session.refresh(task)
915     return _task_response(session, task, dataset.name)
916
917
918 @router.post("/{task_id}/start", response_model=AnalysisTaskResponse)
919 def start_task(
920     task_id: int,
921     user: User = Depends(get_current_user),
922     session: Session = Depends(get_session),
923 ):
924     task = _get_user_task(session, task_id, user)
925     if task.status not in STARTABLE_STATUSES:
926         raise HTTPException(422, f"当前状态不可启动: {task.status}")
927     dataset = _get_user_dataset(session, task.dataset_id, user)
928     _queue_task(session, task, user)
929     session.commit()
930     session.refresh(task)
931     task_execution_manager.notify()
932     return _task_response(session, task, dataset.name)
933
934
935 @router.post("/{task_id}/cancel", response_model=AnalysisTaskResponse)
936 def cancel_task(
937     task_id: int,
938     user: User = Depends(get_current_user),
939     session: Session = Depends(get_session),
940 ):
941     task = _get_user_task(session, task_id, user)
942     if task.status not in CANCELABLE_STATUSES:
943         raise HTTPException(422, f"当前状态不可取消: {task.status}")
944     dataset = _get_user_dataset(session, task.dataset_id, user)
945     task.status = "cancelled"
946     task.finished_at = _now()
947     task.current_stage = task.current_stage or PIPELINE_STAGES[0][0]
948     task.worker_id = None
949     task.heartbeat_at = None
950     _add_log(
951         session,
952         user_id=user.id,
953         task_id=task.id,
954         action="task_cancelled",
955         level="warning",
956         message="任务已取消",
957     )
958     session.commit()
```

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```
959     session.refresh(task)
960     task_execution_manager.cancel_active_process(task.id)
961     return _task_response(session, task, dataset.name)
962
963
964 @router.post("/{task_id}/retry", response_model=AnalysisTaskResponse)
965 def retry_task(
966     task_id: int,
967     user: User = Depends(get_current_user),
968     session: Session = Depends(get_session),
969 ):
970     task = _get_user_task(session, task_id, user)
971     if task.status not in {"failed", "cancelled"}:
972         raise HTTPException(422, "仅失败或取消的任务可重试")
973     dataset = _get_user_dataset(session, task.dataset_id, user)
974     cleanup_paths = _task_cleanup_paths(session, task.id)
975     enqueue_cleanups(session, cleanup_paths)
976     for export in session.scalars(select(TaskExport).where(TaskExport.task_id == task.id)).all():
977         session.delete(export)
978     for result in session.scalars(select(TaskResult).where(TaskResult.task_id == task.id)).all():
979         session.delete(result)
980     _queue_task(session, task, user, increment_retry=True)
981     _add_log(
982         session,
983         user_id=user.id,
984         task_id=task.id,
985         action="task retried",
986         message="任务已重新执行",
987     )
988     session.commit()
989     process_pending_cleanup_jobs(limit=max(20, len(cleanup_paths)), bind=session.get_bind())
990     session.refresh(task)
991     task_execution_manager.notify()
992     return _task_response(session, task, dataset.name)
993
994
995 @router.post("/{task_id}/clone", response_model=AnalysisTaskResponse)
996 def clone_task(
997     task_id: int,
998     user: User = Depends(get_current_user),
999     session: Session = Depends(get_session),
1000 ):
1001     source = _get_user_task(session, task_id, user)
1002     dataset = _get_user_dataset(session, source.dataset_id, user)
1003     _ensure_taskable(session, dataset)
1004
1005     cloned = AnalysisTask(
1006         user_id=user.id,
1007         dataset_id=source.dataset_id,
1008         name=f"{source.name} 副本"[:128],
```


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```
1009         mode=source.mode,
1010         status="draft",
1011         progress=0,
1012         current_iter=0,
1013         max_iter=source.max_iter,
1014         params_json=source.params_json,
1015         current_stage=None,
1016     )
1017     session.add(cloned)
1018     session.flush()
1019     _add_log(
1020         session,
1021         user_id=user.id,
1022         task_id=cloned.id,
1023         action="task_cloned",
1024         message=f"从任务 #{source.id} 克隆",
1025         detail={"sourceTaskId": source.id},
1026     )
1027     session.commit()
1028     session.refresh(cloned)
1029     return _task_response(session, cloned, dataset.name)
1030
1031
1032 @router.delete("/{task_id}", response_model=MessageResponse)
1033 def delete_task(
1034     task_id: int,
1035     user: User = Depends(get_current_user),
1036     session: Session = Depends(get_session),
1037 ):
1038     """业务记录在事务内删除，结果和导出文件由提交后的清理队列处理。"""
1039     task = _get_user_task(session, task_id, user)
1040     if task.status not in DELETABLE_STATUSES:
1041         raise HTTPException(422, "运行中或排队中的任务不可删除，请先取消")
1042     cleanup_paths = _task_cleanup_paths(session, task.id)
1043     enqueue_cleanups(session, cleanup_paths)
1044     for export in session.scalars(select(TaskExport).where(TaskExport.task_id == task.id)).all():
1045         session.delete(export)
1046     for result in session.scalars(select(TaskResult).where(TaskResult.task_id == task.id)).all():
1047         session.delete(result)
1048     _detach_task_logs(session, task.id)
1049     session.delete(task)
1050     _add_log(
1051         session,
1052         user_id=user.id,
1053         task_id=None,
1054         action="task_deleted",
1055         level="warning",
1056         message=f"删除任务 #{task_id}",
1057         detail={"taskId": task_id},
1058     )
```

backend/app/tasks.py

```
1059     session.commit()
1060     process_pending_cleanup_jobs(limit=max(20, len(cleanup_paths)), bind=session.get_bind())
1061     return MessageResponse(message="任务已删除，相关结果和导出文件已进入清理流程")
1062
1063
1064 @router.get("/{task_id}/logs", response_model=AnalysisTaskLogPageResponse)
1065 def list_task_logs(
1066     task_id: int,
1067     user: User = Depends(get_current_user),
1068     session: Session = Depends(get_session),
1069 ):
1070     _get_user_task(session, task_id, user)
1071     rows = session.scalars(
1072         select(OperationLog)
1073         .where(OperationLog.task_id == task_id, or_(OperationLog.user_id == user.id, OperationLog.user_id.is_(None)))
1074         .order_by(desc(OperationLog.created_at), desc(OperationLog.id))
1075         .limit(200),
1076     ).all()
1077     items = [
1078         AnalysisTaskLogItem(
1079             id=row.id,
1080             action=row.action,
1081             level=row.level,
1082             message=row.message,
1083             createdAt=format_dt(row.created_at) or "",
1084             detail=_load_json(row.detail_json, None),
1085         )
1086         for row in rows
1087     ]
1088     return AnalysisTaskLogPageResponse(items=items, total=len(items))
1089
1090
1091 @router.post("/bulk", response_model=MessageResponse)
1092 def bulk_tasks(
1093     payload: AnalysisTaskBulkRequest,
1094     user: User = Depends(get_current_user),
1095     session: Session = Depends(get_session),
1096 ):
1097     action = payload.action.strip().lower()
1098     if action not in {"retry", "cancel", "delete"}:
1099         raise HTTPException(422, "不支持的批量操作")
1100
1101     tasks = session.scalars(
1102         select(AnalysisTask).where(AnalysisTask.user_id == user.id, AnalysisTask.id.in_(payload.taskIds)),
1103     ).all()
1104     if not tasks:
1105         raise HTTPException(404, "未找到可操作的任务")
1106
1107     affected = 0
1108     notify_executor = False
```

backend/app/tasks.py

```
1109     active_cancellations: list[int] = []
1110     cleanup_paths: set[tuple[str, Path]] = set()
1111     for task in tasks:
1112         try:
1113             if action == "retry" and task.status in {"failed", "cancelled"}:
1114                 task_paths = _task_cleanup_paths(session, task.id)
1115                 cleanup_paths.update(task_paths)
1116                 enqueue_cleanups(session, task_paths)
1117                 for export in session.scalars(select(TaskExport).where(TaskExport.task_id == task.id)).all():
1118                     session.delete(export)
1119                 for result in session.scalars(select(TaskResult).where(TaskResult.task_id == task.id)).all():
1120                     session.delete(result)
1121                 _queue_task(session, task, user, increment_retry=True)
1122                 affected += 1
1123                 notify_executor = True
1124             elif action == "cancel" and task.status in CANCELABLE_STATUSES:
1125                 task.status = "cancelled"
1126                 task.finished_at = _now()
1127                 task.worker_id = None
1128                 task.heartbeat_at = None
1129                 affected += 1
1130                 active_cancellations.append(task.id)
1131             elif action == "delete" and task.status in DELETABLE_STATUSES:
1132                 task_paths = _task_cleanup_paths(session, task.id)
1133                 cleanup_paths.update(task_paths)
1134                 enqueue_cleanups(session, task_paths)
1135                 for export in session.scalars(select(TaskExport).where(TaskExport.task_id == task.id)).all():
1136                     session.delete(export)
1137                 for result in session.scalars(select(TaskResult).where(TaskResult.task_id == task.id)).all():
1138                     session.delete(result)
1139                 _detach_task_logs(session, task.id)
1140                 session.delete(task)
1141                 affected += 1
1142         except HTTPException:
1143             # 批量场景跳过单条失败，尽量处理其余任务。
1144             continue
1145
1146     session.commit()
1147     process_pending_cleanup_jobs(limit=max(20, len(cleanup_paths)), bind=session.get_bind())
1148     if notify_executor:
1149         task_execution_manager.notify()
1150     for task_id in active_cancellations:
1151         task_execution_manager.cancel_active_process(task_id)
1152     return MessageResponse(message=f"已处理 {affected} 个任务")
1153
1154
1155 # ===== 文件: backend/app/time_utils.py =====
1156 1 """应用统一使用的 UTC 时间工具。
1157 2
1158 3 MySQL 的 DATETIME 本身不携带时区信息，因此这里在 ORM 边界明确约定：
```

backend/app/time_utils.py

```
4 数据库中的所有时间值都代表 UTC。写入时去掉时区标记，读取时重新附加
5 UTC，避免业务层出现一部分 aware、一部分 naive 的混合状态。
6 """
7
8 from datetime import datetime, timezone
9
10 from sqlalchemy import DateTime
11 from sqlalchemy.types import TypeDecorator
12
13
14 UTC = timezone.utc
15
16
17 def utc_now() -> datetime:
18     """返回当前带 UTC 时区的时间。"""
19     return datetime.now(UTC)
20
21
22 def ensure_aware_utc(value: datetime) -> datetime:
23     """校验并转换为 UTC aware datetime，拒绝没有时区语义的值。"""
24     if value.tzinfo is None or value.utcoffset() is None:
25         raise ValueError("时间值必须包含明确的时区信息")
26     return value.astimezone(UTC)
27
28
29 def format_utc_iso(value: datetime | None) -> str | None:
30     """把时间序列化为 API 统一使用的 ISO 8601 UTC 格式。"""
31     if value is None:
32         return None
33     return ensure_aware_utc(value).isoformat(timespec="seconds").replace("+00:00", "Z")
34
35
36 class UTCDateTime(TypeDecorator[datetime]):
37     """将 Python aware UTC datetime 适配到无时区的数据库 DATETIME。"""
38
39     impl = DateTime
40     cache_ok = True
41
42     def process_bind_param(self, value: datetime | None, dialect) -> datetime | None:
43         if value is None:
44             return None
45         return ensure_aware_utc(value).replace(tzinfo=None)
46
47     def process_result_value(self, value: datetime | None, dialect) -> datetime | None:
48         if value is None:
49             return None
50         if value.tzinfo is None or value.utcoffset() is None:
51             return value.replace(tzinfo=UTC)
52         return value.astimezone(UTC)
53
```

backend/app/time_utils.py

```
# ===== 文件: backend/app/turnstile.py =====
1 import logging
2
3 import httpx
4 from fastapi import HTTPException, status
5
6 from .config import get_settings
7 from .security import decode_turnstile_pass_token
8
9
10 logger = logging.getLogger(__name__)
11
12
13 def verify_turnstile_token(token: str | None, expected_action: str) -> None:
14     settings = get_settings()
15     secret_key = settings.turnstile_secret_key.strip()
16
17     if not secret_key:
18         return
19
20     if not token:
21         raise HTTPException(
22             status_code=status.HTTP_400_BAD_REQUEST,
23             detail="请先完成人机验证",
24         )
25
26     # Turnstile token 只能使用一次, 认证前先完成服务端校验, 避免创建无效会话。
27     try:
28         with httpx.Client(timeout=settings.turnstile_timeout_seconds) as client:
29             response = client.post(
30                 settings.turnstile_verify_url,
31                 data={
32                     "secret": secret_key,
33                     "response": token,
34                 },
35             )
36             response.raise_for_status()
37             result = response.json()
38     except (httpx.HTTPError, ValueError) as exc:
39         logger.warning("Cloudflare Turnstile verification request failed: %s", exc)
40         raise HTTPException(
41             status_code=status.HTTP_503_SERVICE_UNAVAILABLE,
42             detail="人机验证服务暂不可用",
43         ) from exc
44
45     if not result.get("success"):
46         raise HTTPException(
47             status_code=status.HTTP_400_BAD_REQUEST,
48             detail="人机验证失败, 请重新验证",
```

backend/app/turnstile.py

```
49         )
50
51     if result.get("action") != expected_action:
52         raise HTTPException(
53             status_code=status.HTTP_400_BAD_REQUEST,
54             detail="人机验证场景不匹配, 请重新验证",
55         )
56
57
58 def verify_turnstile_access(
59     turnstile_token: str | None,
60     turnstile_pass_token: str | None,
61     expected_action: str,
62 ) -> None:
63     settings = get_settings()
64     secret_key = settings.turnstile_secret_key.strip()
65
66     if not secret_key:
67         return
68
69     if turnstile_pass_token:
70         if decode_turnstile_pass_token(turnstile_pass_token):
71             return
72         raise HTTPException(
73             status_code=status.HTTP_400_BAD_REQUEST,
74             detail="人机验证已过期, 请重新验证",
75         )
76
77     verify_turnstile_token(turnstile_token, expected_action)
78
```

===== 文件: backend/main.py =====

```
1 import logging
2 import sys
3
4 from fastapi import FastAPI
5 from fastapi.middleware.cors import CORSMiddleware
6
7 from app.auth import router as auth_router
8 from app.datasets import router as datasets_router
9 from app.config import Settings, get_settings, validate_jwt_secret_key
10 from app.database import engine
11 from app.migrations import ensure_database_is_current
12 from app.product import SOFTWARE_NAME, SOFTWARE_VERSION
13 from app.task_executor import task_execution_manager
14 from app.tasks import router as tasks_router
15 from app.storage_cleanup import process_pending_cleanup_jobs
16
17
18 settings = get_settings()
```

backend/main.py

```
19 logger = logging.getLogger(__name__)
20
21
22 def validate_startup_configuration(current_settings: Settings) -> None:
23     """启动前再次校验安全配置，防止绕过 Settings 校验后继续提供服务。"""
24     try:
25         validate_jwt_secret_key(current_settings.jwt_secret_key, current_settings.app_env)
26     except ValueError as exc:
27         raise RuntimeError(f"启动配置无效: {exc}") from exc
28
29
30 validate_startup_configuration(settings)
31
32 app = FastAPI(title=f"{SOFTWARE_NAME} API {SOFTWARE_VERSION}")
33
34 app.add_middleware(
35     CORSMiddleware,
36     allow_origins=settings.allowed_origins,
37     allow_credentials=True,
38     allow_methods=["*"],
39     allow_headers=["*"],
40 )
41
42 app.include_router(auth_router)
43 app.include_router(datasets_router)
44 app.include_router(tasks_router)
45 @app.on_event("startup")
46 def validate_database_schema():
47     """启动时只读核对迁移版本，数据库结构变更必须在部署步骤中完成。"""
48     ensure_database_is_current(engine)
49     try:
50         process_pending_cleanup_jobs(limit=100)
51     except Exception:
52         logger.exception("启动时处理文件清理队列失败，调度 Worker 将继续重试")
53     # pytest 使用独立内存库和依赖覆盖，不启动后台线程以免触碰本地开发数据库。
54     if "pytest" not in sys.modules:
55         task_execution_manager.start()
56
57
58 @app.on_event("shutdown")
59 def stop_task_executor():
60     if "pytest" not in sys.modules:
61         task_execution_manager.stop()
62
63
64 @app.get("/api/health")
65 def health():
66     return {"status": "ok", "service": "soft_web_backend"}
67
```

front/src/App.tsx

```
// ===== 文件: front/src/App.tsx =====
1 import { useCallback, useEffect, useRef, useState } from 'react';
2 import {
3   ChevronLeft,
4   ChevronRight,
5   Logout,
6   Menu,
7   Play,
8   Upload,
9 } from 'lucide-react';
10 import { Navigate, NavLink, Route, Routes, useLocation, useNavigate } from 'react-router-dom';
11 import './landing.css';
12 import { SOFTWARE_SHORT_NAME } from './appMeta';
13 import sidebarBrandLogo from './images/Logo.svg';
14 import { LandingPage } from './LandingPage';
15 import { AuthFlipCard } from './AuthPages';
16 import { AppBackground } from './AppBackground';
17 import { DatasetManagementPage } from './DatasetManagementPage';
18 import { DataQualityPage } from './workbench/data-quality/DataQualityPage';
19 import { DatasetVersionsPage } from './workbench/dataset-versions/DatasetVersionsPage';
20 import { TaskCenterPage } from './workbench/tasks/TaskCenterPage';
21 import { TaskResultViews } from './workbench/results/TaskResultViews';
22 import { useTaskResult } from './workbench/results/useTaskResult';
23 import {
24   clearAuthSession,
25   logout as logoutFromApi,
26   restoreAuthSession,
27   subscribeAuthSession,
28   type AuthUser,
29 } from './api/auth';
30 import { publicFetch } from './api/client';
31 import {
32   dashboardGroups,
33   dashboardNavItems,
34   defaultWorkbenchPath,
35   getActiveWorkbenchSection,
36   legacyWorkbenchRedirects,
37 } from './dashboard/navigation';
38
39 const resultWorkbenchSections = new Set([
40   'analysis',
41   'ca-matrix',
42   'kernel-config',
43   'mkl',
44   'evaluation',
45   'visualization',
46   'results',
47   'export',
48   'reports',
49   'logs',
```


front/src/App.tsx

```
50  });
51
52  type BackendStatus = 'checking' | 'online' | 'offline';
53
54  function Sidebar({
55    activeSection,
56    collapsed,
57    open,
58    onClose,
59  }): {
60    activeSection: string;
61    collapsed: boolean;
62    open: boolean;
63    onClose: () => void;
64  } {
65    const location = useLocation();
66    const taskId = new URLSearchParams(location.search).get('taskId');
67    const resultPaths = new Set([
68      '/workbench/analysis',
69      '/workbench/ca-matrix',
70      '/workbench/kernel-config',
71      '/workbench/mkl',
72      '/workbench/evaluation',
73      '/workbench/visualization',
74      '/workbench/results',
75      '/workbench/export',
76      '/workbench/reports',
77      '/workbench/logs',
78    ]);
79    return (
80      <>
81        <button
82          type="button"
83          className={`sidebar-backdrop ${open ? 'visible' : ''}`}
84          aria-label="关闭导航"
85          aria-hidden={!open}
86          tabIndex={open ? 0 : -1}
87          onClick={onClose}
88        />
89        <aside className={`sidebar ${open ? 'open' : ''} ${collapsed ? 'collapsed' : ''}`}>
90          <div className="sidebar-brand" title={SOFTWARE_SHORT_NAME}>
91            <span className="brand-logo" aria-hidden="true">
92              <img
93                className="sidebar-brand-logo-image"
94                src={sidebarBrandLogo}
95                width={30}
96                height={30}
97                alt=""
98                aria-hidden="true"
99              />
```

front/src/App.tsx

```
100     </span>
101     <span className="brand-copy">
102 <strong>{SOFTWARE_SHORT_NAME}</strong>
103     <small>新材料聚类分析</small>
104     </span>
105 </div>
106
107 <nav className="nav-list" aria-label="主导航">
108   {dashboardGroups.map((group) => (
109     <div className="nav-group" key={group.key}>
110       <span className="nav-group-label">{group.label}</span>
111       {dashboardNavItems
112         .filter((item) => item.group === group.key)
113         .map((item) => {
114           const Icon = item.icon;
115           const isActive = activeSection === item.label;
116
117           return (
118             <NavLink
119               key={item.label}
120               to={resultPaths.has(item.path) && taskId ? `${item.path}?taskId=${encodeURIComponent(taskId)}` :
121               item.path}
122               aria-current={isActive ? 'page' : undefined}
123               className={({ isActive: routeIsActive }) =>
124                 `nav-item ${routeIsActive ? 'active' : ''}`
125               }
126               onClick={onClose}
127               title={item.label}
128             >
129               <span className="nav-icon" aria-hidden="true">
130                 <Icon size={20} />
131               </span>
132               <span className="nav-label">{item.label}</span>
133             </NavLink>
134           );
135         })}
136       </div>
137     </nav>
138   </aside>
139 </>
140 );
141 }
142
143 function TopHeader({
144   activeSection,
145   currentUser,
146   onToggleMobileNav,
147   onLogout,
148 }): {
```

front/src/App.tsx

```
149   activeSection: string;
150   currentUser: AuthUser | null;
151   onToggleMobileNav: () => void;
152   onLogout: () => void;
153 }) {
154   const username = currentUser?.username || '研究用户';
155
156   return (
157     <header className="main-header">
158       <div className="navbar-left">
159         <button
160           type="button"
161           className="header-icon-button mobile-nav-button"
162           aria-label="打开导航"
163           onClick={onToggleMobileNav}>
164           <
165             <Menu size={18} />
166           </button>
167           <div className="header-context">
168             <span>{SOFTWARE_SHORT_NAME} 工作台</span>
169             <ChevronRight size={14} aria-hidden="true" />
170             <div className="header-section-title" aria-label="当前模块">
171               {activeSection}
172             </div>
173           </div>
174         </div>
175
176         <div className="navbar-right">
177           <div className="header-user" title={`当前用户: ${username}`} aria-label={`当前用户: ${username}`}>
178             <span className="header-user-avatar" aria-hidden="true">
179               {username.slice(0, 1).toUpperCase()}
180             </span>
181           </div>
182           <button
183             type="button"
184             className="header-icon-button"
185             aria-label="退出登录"
186             title="退出登录"
187             onClick={onLogout}>
188             <
189               <LogOut size={16} aria-hidden="true" />
190             </button>
191           </div>
192         </div>
193       </header>
194     );
195
196   function StatusHeader({
197     status,
198     onImportData,
```

front/src/App.tsx

```
199   onCreateTask,
200 } : {
201   status: BackendStatus;
202   onImportData: () => void;
203   onCreateTask: () => void;
204 }) {
205   const statusText =
206     status === 'online' ? 'FastAPI 已连接' : status === 'offline' ? 'FastAPI 连接异常' : '正在检查 FastAPI...';
207   const statusClass = status === 'online' ? 'success' : status === 'offline' ? 'error' : 'warning';
208
209   return (
210     <section className="status-header" aria-label="系统连接状态">
211       <div className="status-left">
212         <span className={`status-dot ${statusClass}`} aria-hidden="true" />
213         <div>
214           <span className="status-label">Python 服务</span>
215           <strong>{statusText}</strong>
216         </div>
217       </div>
218       <div className="status-actions">
219         <button type="button" className="btn btn-secondary" onClick={onImportData}>
220           <Upload size={16} aria-hidden="true" />
221           导入数据
222         </button>
223         <button type="button" className="btn btn-primary" onClick={onCreateTask}>
224           <Play size={16} aria-hidden="true" />
225           创建任务
226         </button>
227       </div>
228     </section>
229   );
230 }
231
232 function WorkbenchPageContent({
233   activeSection,
234   sectionKey,
235   resultResource,
236 }) : {
237   activeSection: string;
238   sectionKey: string;
239   resultResource: ReturnType<typeof useTaskResult>;
240 }) {
241   if (resultWorkbenchSections.has(sectionKey)) {
242     return <TaskResultViews section={sectionKey} resource={resultResource} />;
243   }
244
245   switch (activeSection) {
246     case '数据管理':
247       return <DatasetManagementPage />;
248     case '数据质量检查':
```

front/src/App.tsx

```
249     return <DataQualityPage />;
250   case '数据版本记录':
251     return <DatasetVersionsPage />;
252   case '任务中心':
253     return <TaskCenterPage />;
254   default:
255     return <Navigate to={defaultWorkbenchPath} replace />;
256 }
257 }
258
259 function App() {
260   const navigate = useNavigate();
261   const location = useLocation();
262   const workbenchSection = getActiveWorkbenchSection(location.pathname);
263   const workbenchKey = location.pathname.split('/')[2] || 'analysis';
264   const resultResource = useTaskResult(Boolean(workbenchSection) && resultWorkbenchSections.has(workbenchKey));
265   const activeSection = workbenchSection ?? '分析工作台';
266   const showStatusHeader = location.pathname === defaultWorkbenchPath;
267   const initialPathRef = useRef(location.pathname);
268   const sessionCheckedRef = useRef(false);
269   const [sidebarOpen, setSidebarOpen] = useState(false);
270   const [sidebarCollapsed, setSidebarCollapsed] = useState(false);
271   const [workbenchEntryPending, setWorkbenchEntryPending] = useState(false);
272   const [currentUser, setCurrentUser] = useState<AuthUser | null>(() => {
273     if (typeof localStorage === 'undefined') return null;
274     const storedUser = localStorage.getItem('soft_web_user');
275     if (!storedUser) return null;
276     try {
277       return JSON.parse(storedUser) as AuthUser;
278     } catch {
279       return null;
280     }
281   });
282   const [backendStatus, setBackendStatus] = useState<BackendStatus>('checking');
283
284   const openWorkbench = useCallback(() => {
285     navigate(defaultWorkbenchPath);
286     setSidebarOpen(false);
287   }, [navigate]);
288
289   useEffect(() => subscribeAuthSession((event) => {
290     setCurrentUser(null);
291     setSidebarOpen(false);
292
293     if (event.reason === 'logout') {
294       navigate('/', { replace: true });
295       return;
296     }
297
298     if (!['/login', '/register'].includes(location.pathname)) {
```

front/src/App.tsx

```
299     navigate('/login', { replace: true });
300   }
301 }, [location.pathname, navigate]);
302
303 const enterWorkbench = useCallback(async () => {
304   if (workbenchEntryPending) return;
305
306   setWorkbenchEntryPending(true);
307
308   try {
309     const user = await restoreAuthSession();
310     setCurrentUser(user);
311     openWorkbench();
312   } catch {
313     clearAuthSession();
314     setCurrentUser(null);
315     setSidebarOpen(false);
316     navigate('/login');
317   } finally {
318     setWorkbenchEntryPending(false);
319   }
320 }, [navigate, openWorkbench, workbenchEntryPending]);
321
322 const refreshBackendStatus = useCallback(async () => {
323   setBackendStatus('checking');
324   if (typeof fetch === 'undefined') {
325     setBackendStatus('offline');
326     return;
327   }
328   try {
329     const response = await publicFetch('/health');
330     if (!response.ok) {
331       throw new Error('health check failed');
332     }
333     setBackendStatus('online');
334   } catch {
335     setBackendStatus('offline');
336   }
337 }, []);
338
339 useEffect(() => {
340   if (sessionCheckedRef.current) {
341     return;
342   }
343
344   sessionCheckedRef.current = true;
345
346   const hasCachedUser = Boolean(localStorage.getItem('soft_web_user'));
347   const isWorkbenchPath = initialPathRef.current.startsWith('/workbench');
348   if (!hasCachedUser && !isWorkbenchPath) {
```

front/src/App.tsx

```
349     return;
350   }
351
352   let cancelled = false;
353
354   restoreAuthSession()
355     .then((user) => {
356       if (cancelled) return;
357       setCurrentUser(user);
358       if (['/', '/login', '/register'].includes(initialPathRef.current)) {
359         navigate(defaultWorkbenchPath, { replace: true });
360       }
361     })
362     .catch(() => {
363       if (cancelled) return;
364       clearAuthSession();
365       setCurrentUser(null);
366       if (isWorkbenchPath) {
367         navigate('/login', { replace: true });
368       }
369     });
370
371   return () => {
372     cancelled = true;
373   };
374 }, [navigate]);
375
376 useEffect(() => {
377   if (!workbenchSection) return;
378   void refreshBackendStatus();
379 }, [refreshBackendStatus, workbenchSection]);
380
381 const handleAuthSuccess = () => {
382   const storedUser = localStorage.getItem('soft_web_user');
383   if (storedUser) {
384     try {
385       setCurrentUser(JSON.parse(storedUser) as AuthUser);
386     } catch {
387       setCurrentUser(null);
388     }
389   }
390   openWorkbench();
391 };
392
393 const handleLogout = async () => {
394   await logoutFromApi();
395   setCurrentUser(null);
396   setSidebarOpen(false);
397   navigate('/', { replace: true });
398 };
```

front/src/App.tsx

```
399
400   const dashboardShell = workbenchSection ? (
401     <div className={`app-shell ${sidebarCollapsed ? 'sidebar-collapsed' : ''}`}>
402       <AppBackground />
403       <a className="skip-link" href="#main-content">
404         跳转到主要内容
405       </a>
406
407       <Sidebar
408         activeSection={activeSection}
409         collapsed={sidebarCollapsed}
410         open={sidebarOpen}
411         onClose={() => setSidebarOpen(false)}
412       />
413
414       <TopHeader
415         activeSection={activeSection}
416         currentUser={currentUser}
417         onToggleMobileNav={() => setSidebarOpen((value) => !value)}
418         onLogout={handleLogout}
419       />
420
421       <button
422         type="button"
423         className="sidebar-toggle-floating"
424         aria-label={sidebarCollapsed ? '展开侧栏' : '收起侧栏'}
425         title={sidebarCollapsed ? '展开侧栏' : '收起侧栏'}
426         onClick={() => setSidebarCollapsed((value) => !value)}
427       >
428         {sidebarCollapsed ? <ChevronRight size={15} /> : <ChevronLeft size={15} />}
429     </button>
430
431     <main className={`dashboard-content${workbenchKey === 'logs' ? ' is-log-layout' : ''}`} id="main-content">
432       {showStatusHeader ? (
433         <StatusHeader
434           status={backendStatus}
435           onImportData={() => navigate('/workbench/datasets')}
436           onCreateTask={() => navigate('/workbench/tasks')}
437         />
438       ) : null}
439       <WorkbenchPageContent
440         activeSection={activeSection}
441         sectionKey={workbenchKey}
442         resultResource={resultResource}
443       />
444     </main>
445   </div>
446 ) : (
447   <Navigate to={defaultWorkbenchPath} replace />
448 );
```


front/src/App.tsx

```
449
450   const isAuthRoute = location.pathname === '/login' || location.pathname === '/register';
451
452   return (
453     <>
454       <Routes>
455         <Route
456           path="/"
457           element={
458             <LandingPage
459               onLogin={() => navigate('/login')}
460               onRegister={() => navigate('/register')}
461               onEnterWorkbench={enterWorkbench}
462               isEnteringWorkbench={workbenchEntryPending}
463             />
464           }
465         />
466         <Route path="/login" element={null} />
467         <Route path="/register" element={null} />
468         <Route path="/workbench" element={<Navigate to={defaultWorkbenchPath} replace />} />
469         {Object.entries(legacyWorkbenchRedirects).map(([from, to]) => (
470           <Route key={from} path={from} element={<Navigate to={to} replace />} />
471         ))}
472         <Route path="/workbench/:section" element={dashboardShell} />
473         <Route path="*" element={<Navigate to="/" replace />} />
474       </Routes>
475       {isAuthRoute ? (
476         <AuthFlipCard
477           mode={location.pathname === '/register' ? 'register' : 'login'}
478           onSuccess={handleAuthSuccess}
479           onShowLogin={() => navigate('/login')}
480           onShowRegister={() => navigate('/register')}
481           onBack={() => navigate('/') }
482         />
483       ) : null}
484     </>
485   );
486 }
487
488 export default App;
489
```

// ===== 文件: front/src/AppBackground.tsx =====

```
1 export function AppBackground() {
2   return (
3     <div className="app-background" aria-hidden="true">
4       <svg className="app-background__shape app-background__shape-1" viewBox="0 0 200 200">
5         <defs>
6           <linearGradient id="app-bg-grad-1" x1="0%" y1="0%" x2="100%" y2="100%">
7             <stop offset="0%" stopColor="var(--app-bg-blob-1-start)" stopOpacity="1" />
```

```
front/src/AppBackground.tsx
```

```
8         <stop offset="100%" stopColor="var(--app-bg-blob-1-end)" stopOpacity="1" />
9       </linearGradient>
10     </defs>
11     <path
12       fill="url(#app-bg-grad-1)"
13       d="M44.7,-76.4C58.9,-69.2,71.8,-59.1,79.6,-46.9C87.4,-34.7,90.1,-20.4,85.8,-7.8C81.5,4.8,70.2,15.7,59.9,25.4
C49.6,35.1,40.3,43.6,30.3,51.8C20.3,60.9,6,67.9,-2.7,72.6C-15,77.3,-29,78.8,-41.8,73.5C-54.6,68.2,-66.2,56.1,-73.4
,42.1C-80.6,28.1,-83.4,12.2,-79.9,-2.1C-76.4,-16.4,-66.6,-29.1,-56.3,-39.8C-46,-50.5,-35.3,-59.2,-23.8,-68C-12.3,-
76.8,-0.1,-85.7,13.2,-87.3C26.5,-88.9,30.5,-63.6,44.7,-76.4Z"
14       transform="translate(100 100)"
15     />
16   </svg>
17   <svg className="app-background__shape app-background__shape-2" viewBox="0 0 200 200">
18     <defs>
19       <linearGradient id="app-bg-grad-2" x1="0%" y1="0%" x2="100%" y2="0%">
20         <stop offset="0%" stopColor="var(--app-bg-blob-2-start)" stopOpacity="1" />
21         <stop offset="100%" stopColor="var(--app-bg-blob-2-end)" stopOpacity="1" />
22       </linearGradient>
23     </defs>
24     <path
25       fill="url(#app-bg-grad-2)"
26       d="M41.2,-68.8C52.4,-63.9,60.6,-50.7,68.4,-37.8C76.2,-24.9,83.6,-12.3,81.3,-0.7C78.9,10.9,66.8,21.5,56.7,31.
7C46.6,41.9,38.5,51.7,28.4,59.5C18.3,67.3,6.2,73.1,-5.6,76.3C-17.4,79.5,-29,80.1,-39.8,74.7C-50.6,69.3,-60.6,57.9,
-67.9,45.4C-75.2,32.9,-79.8,19.3,-77.8,6.8C-75.8,-5.7,-67.2,-17.1,-57.8,-27.1C-48.4,-37.1,-38.2,-45.7,-27.6,-51C-1
7,-56.3,-6,-58.3,5.6,-61.5C17.2,-64.7,30,-63.7,41.2,-68.8Z"
27       transform="translate(100 100)"
28     />
29   </svg>
30 </div>
31 );
32 }
33
```

```
// ===== 文件: front/src/AuthPages.tsx =====
```

```
1 import { useCallback, useEffect, useRef, useState } from 'react';
2 import type { ReactNode } from 'react';
3 import { ArrowLeft, LoaderCircle, LogIn, MailCheck, UserPlus } from 'lucide-react';
4 import {
5   createTurnstilePass,
6   login,
7   register,
8   saveAuthSession,
9   sendRegisterEmailCode,
10 } from './api/auth';
11 import type { TurnstileAction } from './api/auth';
12 import { TURNSTILE_SITE_KEY } from './api/config';
13 import { SOFTWARE_SHORT_NAME } from './appMeta';
14 import { TurnstileWidget, type TurnstileWidgetHandle } from './TurnstileWidget';
15
16 const AUTH_SUBMIT_FEEDBACK_MS = 260;
```

front/src/AuthPages.tsx

```
17
18 interface AuthProps {
19   onSuccess: () => void;
20   onSwitch: () => void;
21   onBack: () => void;
22   turnstileEnabled?: boolean;
23   hasTurnstilePass?: boolean;
24   turnstilePassToken?: string;
25   onTurnstileVerify?: (token: string, action: TurnstileAction) => Promise<void>;
26 }
27
28 type TurnstilePassState = {
29   token: string;
30   expiresAtMs: number;
31 };
32
33 function waitForAuthSubmitFeedback(startedAt: number) {
34   const remaining = AUTH_SUBMIT_FEEDBACK_MS - (Date.now() - startedAt);
35
36   if (remaining <= 0) {
37     return Promise.resolve();
38   }
39
40   // 极快响应也保留短暂加载态，避免用户误以为点击没有生效。
41   return new Promise<void>((resolve) => {
42     setTimeout(resolve, remaining);
43   });
44 }
45
46 function getTurnstilePassExpiresAt(expiresInSeconds: number) {
47   const seconds = Number.isFinite(expiresInSeconds) && expiresInSeconds > 0 ? expiresInSeconds : 0;
48   return Date.now() + seconds * 1000;
49 }
50
51 function useAuthTurnstilePass() {
52   const [turnstilePass, setTurnstilePass] = useState<TurnstilePassState | null>(null);
53
54   useEffect(() => {
55     if (!turnstilePass) {
56       return undefined;
57     }
58
59     const remainingMs = turnstilePass.expiresAtMs - Date.now();
60     if (remainingMs <= 0) {
61       setTurnstilePass(null);
62       return undefined;
63     }
64
65     const timer = window.setTimeout(() => {
66       setTurnstilePass(null);
```

```
front/src/AuthPages.tsx
```

```
67     }, remainingMs);
68
69     return () => window.clearTimeout(timer);
70 }, [turnstilePass]);
71
72 const exchangeTurnstilePass = useCallback(async (token: string, action: TurnstileAction) => {
73     const response = await createTurnstilePass(token, action);
74     if (!response.turnstile_pass_token || response.expires_in_seconds <= 0) {
75         throw new Error('人机验证已过期，请重新验证');
76     }
77
78     setTurnstilePass({
79         token: response.turnstile_pass_token,
80         expiresAtMs: getTurnstilePassExpiresAt(response.expires_in_seconds),
81     });
82 }, []);
83
84 const hasTurnstilePass = Boolean(turnstilePass && turnstilePass.expiresAtMs > Date.now());
85
86 return {
87     hasTurnstilePass,
88     turnstilePassToken: hasTurnstilePass && turnstilePass ? turnstilePass.token : '',
89     onTurnstileVerify: exchangeTurnstilePass,
90 };
91 }
92
93 function AuthField({
94     id,
95     name,
96     label,
97     type,
98     value,
99     onChange,
100     autoComplete,
101     placeholder,
102 }): {
103     id: string;
104     name: string;
105     label: string;
106     type: string;
107     value: string;
108     onChange: (value: string) => void;
109     autoComplete: string;
110     placeholder?: string;
111 }) {
112     return (
113         <div className="auth-field">
114             <label htmlFor={id}>{label}</label>
115             <input
116                 id={id}
```

front/src/AuthPages.tsx

```
117     name={name}
118     className="auth-input"
119     type={type}
120     value={value}
121     onChange={(e) => onChange(e.target.value)}
122     autoComplete={autoComplete}
123     placeholder={placeholder}
124     spellCheck={type === 'text' ? false : undefined}
125     required
126   />
127 </div>
128 );
129 }
130
131 function LoginCard({
132   onSuccess,
133   onSwitch,
134   onBack,
135   turnstileEnabled = true,
136   hasTurnstilePass = false,
137   turnstilePassToken = '',
138   onTurnstileVerify,
139 }: AuthProps) {
140   const [username, setUsername] = useState('');
141   const [password, setPassword] = useState('');
142   const [turnstileToken, setTurnstileToken] = useState('');
143   const [error, setError] = useState('');
144   const [submitting, setSubmitting] = useState(false);
145   const turnstileRef = useRef<TurnstileWidgetHandle | null>(null);
146   const turnstileRequired = Boolean(TURNSTILE_SITE_KEY) && turnstileEnabled && !hasTurnstilePass;
147
148   const handleTurnstileVerify = useCallback((token: string) => {
149     setTurnstileToken(token);
150     if (!token) {
151       return;
152     }
153
154     setError('');
155     void onTurnstileVerify?.(token, 'login')
156       .then(() => {
157         setTurnstileToken('');
158       })
159       .catch((err) => {
160         setTurnstileToken('');
161         setError(err instanceof Error ? err.message : '人机验证失败, 请重新验证');
162         turnstileRef.current?.reset();
163       });
164   }, [onTurnstileVerify]);
165
166   const handleTurnstileError = useCallback((message: string) => {
```

front/src/AuthPages.tsx

```
167     setTurnstileToken('');
168     setError(message);
169   }, []);
170
171   const resetTurnstile = useCallback(() => {
172     if (!turnstileRequired) {
173       return;
174     }
175     turnstileRef.current?.reset();
176     setTurnstileToken('');
177   }, [turnstileRequired]);
178
179   const handleSubmit = async () => {
180     const startedAt = Date.now();
181     setError('');
182
183     if (turnstileRequired && !turnstileToken) {
184       setError('请先完成人机验证');
185       return;
186     }
187
188     setSubmitting(true);
189     try {
190       const auth = await login(
191         username.trim(),
192         password,
193         turnstileRequired ? turnstileToken || null : null,
194         hasTurnstilePass ? turnstilePassToken : null,
195       );
196       saveAuthSession(auth);
197       await waitForAuthSubmitFeedback(startedAt);
198       onSuccess();
199     } catch (err) {
200       setError(err instanceof Error ? err.message : '登录失败');
201       resetTurnstile();
202     } finally {
203       setSubmitting(false);
204     }
205   };
206
207   return (
208     <div className="auth-card">
209       <h1 className="auth-card-title">登录</h1>
210       <p className="auth-card-subtitle">登录到 {SOFTWARE_SHORT_NAME}</p>
211       <form
212         className="auth-form"
213         noValidate
214         onSubmit={async (e) => {
215           e.preventDefault();
216           await handleSubmit();
```

front/src/AuthPages.tsx

```
217     }}
218   >
219   <AuthField
220     id="login-username"
221     name="username"
222     label="用户名或邮箱"
223     type="text"
224     value={username}
225     onChange={setUsername}
226     autoComplete="username"
227     placeholder="请输入用户名或邮箱"
228   />
229   <AuthField
230     id="login-password"
231     name="password"
232     label="密码"
233     type="password"
234     value={password}
235     onChange={setPassword}
236     autoComplete="current-password"
237     placeholder="请输入密码"
238   />
239   <TurnstileWidget
240     ref={turnstileRef}
241     siteKey={turnstileRequired ? TURNSTILE_SITE_KEY : ''}
242     action="login"
243     onVerify={handleTurnstileVerify}
244     onError={handleTurnstileError}
245   />
246   {error ? <p className="auth-error" role="alert">{error}</p> : null}
247   <button
248     type="submit"
249     className={`btn btn-primary auth-submit ${submitting ? 'is-loading' : ''}`}
250     disabled={submitting}
251     aria-busy={submitting}
252   >
253     {submitting ? (
254       <LoaderCircle className="auth-submit-spinner" size={16} aria-hidden="true" />
255     ) : (
256       <LogIn size={16} aria-hidden="true" />
257     )}
258     {submitting ? '登录中...' : '登录'}
259   </button>
260 </form>
261 <p className="auth-switch">
262   还没有账号?
263   <button type="button" onClick={onSwitch}>去注册</button>
264 </p>
265 <button type="button" className="auth-back" onClick={onBack}>
266   <ArrowLeft size={12} aria-hidden="true" /> 返回首页
```

front/src/AuthPages.tsx

```
267     </button>
268   </div>
269 );
270 }
271
272 function RegisterCard({
273   onSuccess,
274   onSwitch,
275   onBack,
276   turnstileEnabled = true,
277   hasTurnstilePass = false,
278   turnstilePassToken = '',
279   onTurnstileVerify,
280 }: AuthProps) {
281   const [username, setUsername] = useState('');
282   const [email, setEmail] = useState('');
283   const [password, setPassword] = useState('');
284   const [confirm, setConfirm] = useState('');
285   const [emailCode, setEmailCode] = useState('');
286   const [turnstileToken, setTurnstileToken] = useState('');
287   const [error, setError] = useState('');
288   const [notice, setNotice] = useState('');
289   const [sendingEmailCode, setSendingEmailCode] = useState(false);
290   const [emailCodeCooldown, setEmailCodeCooldown] = useState(0);
291   const [submitting, setSubmitting] = useState(false);
292   const turnstileRef = useRef<TurnstileWidgetHandle | null>(null);
293   const turnstileRequired = Boolean(TURNSTILE_SITE_KEY) && turnstileEnabled && !hasTurnstilePass;
294
295   const handleTurnstileVerify = useCallback((token: string) => {
296     setTurnstileToken(token);
297     if (!token) {
298       return;
299     }
300
301     setError('');
302     void onTurnstileVerify?.(token, 'register')
303       .then(() => {
304         setTurnstileToken('');
305       })
306       .catch((err) => {
307         setTurnstileToken('');
308         setError(err instanceof Error ? err.message : '人机验证失败, 请重新验证');
309         turnstileRef.current?.reset();
310       });
311   }, [onTurnstileVerify]);
312
313   const handleTurnstileError = useCallback((message: string) => {
314     setTurnstileToken('');
315     setError(message);
316   }, []);
```


front/src/AuthPages.tsx

```
317
318   const resetTurnstile = useCallback(() => {
319     if (!turnstileRequired) {
320       return;
321     }
322     turnstileRef.current?.reset();
323     setTurnstileToken('');
324   }, [turnstileRequired]);
325
326   useEffect(() => {
327     if (emailCodeCooldown <= 0) {
328       return undefined;
329     }
330
331     const timer = window.setInterval(() => {
332       setEmailCodeCooldown((current) => Math.max(current - 1, 0));
333     }, 1000);
334
335     return () => window.clearInterval(timer);
336   }, [emailCodeCooldown]);
337
338   const handleSendEmailCode = async () => {
339     setError('');
340     setNotice('');
341
342     if (!email.trim()) {
343       setError('请先填写邮箱');
344       return;
345     }
346
347     if (turnstileRequired && !turnstileToken) {
348       setError('请先完成人机验证');
349       return;
350     }
351
352     setSendingEmailCode(true);
353     try {
354       await sendRegisterEmailCode(
355         email.trim(),
356         turnstileRequired ? turnstileToken || null : null,
357         hasTurnstilePass ? turnstilePassToken : null,
358       );
359       setNotice('验证码已发送，请查看邮箱');
360       setEmailCodeCooldown(60);
361       resetTurnstile();
362     } catch (err) {
363       setError(err instanceof Error ? err.message : '验证码发送失败');
364       resetTurnstile();
365     } finally {
366       setSendingEmailCode(false);
```

front/src/AuthPages.tsx

```
367     }
368   };
369
370   const handleSubmit = async () => {
371     const startedAt = Date.now();
372     setError('');
373     setNotice('');
374
375     if (password !== confirm) {
376       setError('两次密码不一致');
377       return;
378     }
379
380     if (!email.trim()) {
381       setError('请先填写邮箱');
382       return;
383     }
384
385     if (!emailCode.trim()) {
386       setError('请填写邮箱验证码');
387       return;
388     }
389
390     setSubmitting(true);
391     try {
392       const auth = await register(username.trim(), email.trim(), password, confirm, emailCode.trim());
393       saveAuthSession(auth);
394       await waitForAuthSubmitFeedback(startedAt);
395       onSuccess();
396     } catch (err) {
397       setError(err instanceof Error ? err.message : '注册失败');
398       resetTurnstile();
399     } finally {
400       setSubmitting(false);
401     }
402   };
403
404   return (
405     <div className="auth-card">
406       <h1 className="auth-card-title">注册</h1>
407       <p className="auth-card-subtitle">创建 {SOFTWARE_SHORT_NAME} 账号</p>
408       <form
409         className="auth-form"
410         noValidate
411         onSubmit={async (e) => {
412           e.preventDefault();
413           await handleSubmit();
414         }}
415       >
416         <AuthField
```

front/src/AuthPages.tsx

```
417         id="register-username"
418         name="username"
419         label="用户名"
420         type="text"
421         value={username}
422         onChange={setUsername}
423         autoComplete="username"
424         placeholder="请输入用户名"
425     />
426     <AuthField
427         id="register-email"
428         name="email"
429         label="邮箱"
430         type="email"
431         value={email}
432         onChange={setEmail}
433         autoComplete="email"
434         placeholder="请输入邮箱"
435     />
436     <AuthField
437         id="register-password"
438         name="password"
439         label="密码"
440         type="password"
441         value={password}
442         onChange={setPassword}
443         autoComplete="new-password"
444         placeholder="请输入密码"
445     />
446     <AuthField
447         id="register-confirm"
448         name="confirmPassword"
449         label="确认密码"
450         type="password"
451         value={confirm}
452         onChange={setConfirm}
453         autoComplete="new-password"
454         placeholder="请再次输入密码"
455     />
456     <TurnstileWidget
457         ref={turnstileRef}
458         siteKey={turnstileRequired ? TURNSTILE_SITE_KEY : ''}
459         action="register"
460         onVerify={handleTurnstileVerify}
461         onError={handleTurnstileError}
462     />
463     <div className="auth-code-row">
464         <AuthField
465             id="register-email-code"
466             name="emailCode"
```

front/src/AuthPages.tsx

```
467         label="邮箱验证码"
468         type="text"
469         value={emailCode}
470         onChange={setEmailCode}
471         autoComplete="one-time-code"
472         placeholder="请输入验证码"
473       />
474       <button
475         type="button"
476         className="auth-code-button"
477         disabled={sendingEmailCode || emailCodeCooldown > 0}
478         aria-busy={sendingEmailCode}
479         onClick={handleSendEmailCode}
480       >
481         {sendingEmailCode ? (
482           <LoaderCircle className="auth-submit-spinner" size={15} aria-hidden="true" />
483         ) : (
484           <MailCheck size={15} aria-hidden="true" />
485         )}
486         {emailCodeCooldown > 0 ? `${emailCodeCooldown}s` : '发送验证码'}
487       </button>
488     </div>
489     {notice ? <p className="auth-notice" role="status">{notice}</p> : null}
490     {error ? <p className="auth-error" role="alert">{error}</p> : null}
491     <button
492       type="submit"
493       className={`btn btn-primary auth-submit ${submitting ? 'is-loading' : ''}`}
494       disabled={submitting}
495       aria-busy={submitting}
496     >
497       {submitting ? (
498         <LoaderCircle className="auth-submit-spinner" size={16} aria-hidden="true" />
499       ) : (
500         <UserPlus size={16} aria-hidden="true" />
501       )}
502       {submitting ? '注册中...' : '注册'}
503     </button>
504   </form>
505   <p className="auth-switch">
506     已有账号?
507     <button type="button" onClick={onSwitch}>去登录</button>
508   </p>
509   <button type="button" className="auth-back" onClick={onBack}>
510     <ArrowLeft size={12} aria-hidden="true" /> 返回首页
511   </button>
512 </div>
513 );
514 }
515
516 export function LoginPage({ onSuccess, onSwitch, onBack }: AuthProps) {
```

front/src/AuthPages.tsx

```
517   const turnstilePass = useAuthTurnstilePass();
518
519   return (
520     <div className="auth-page">
521       <div className="blur-ball blur-ball-amber" aria-hidden="true" />
522       <LoginCard onSuccess={onSuccess} onSwitch={onSwitch} onBack={onBack} {...turnstilePass} />
523     </div>
524   );
525 }
526
527 export function RegisterPage({ onSuccess, onSwitch, onBack }: AuthProps) {
528   const turnstilePass = useAuthTurnstilePass();
529
530   return (
531     <div className="auth-page">
532       <div className="blur-ball blur-ball-amber" aria-hidden="true" />
533       <RegisterCard onSuccess={onSuccess} onSwitch={onSwitch} onBack={onBack} {...turnstilePass} />
534     </div>
535   );
536 }
537
538 interface AuthFlipCardProps {
539   mode: 'login' | 'register';
540   onSuccess: () => void;
541   onShowLogin: () => void;
542   onShowRegister: () => void;
543   onBack: () => void;
544 }
545
546 function AuthFlipFace({
547   active,
548   className,
549   children,
550 }): {
551   active: boolean;
552   className: string;
553   children: ReactNode;
554 }) {
555   const faceRef = useRef<HTMLDivElement | null>(null);
556
557   useEffect(() => {
558     const face = faceRef.current;
559
560     if (!face) {
561       return;
562     }
563
564     if (active) {
565       face.removeAttribute('inert');
566     } else {
```

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```
567     face.setAttribute('inert', '');
568   }
569 }, [active]);
570
571 return (
572   <div ref={faceRef} className={className} aria-hidden={!active}>
573     {children}
574   </div>
575 );
576 }
577
578 export function AuthFlipCard({
579   mode,
580   onSuccess,
581   onShowLogin,
582   onShowRegister,
583   onBack,
584 }: AuthFlipCardProps) {
585   const flipped = mode === 'register';
586   const turnstilePass = useAuthTurnstilePass();
587
588   return (
589     <div className="auth-page">
590       <div className="blur-ball blur-ball-amber" aria-hidden="true" />
591       <div className={`auth-flip${flipped ? ' is-flipped' : ''}`>
592         <div className="auth-flip-inner">
593           <AuthFlipFace active={!flipped} className="auth-flip-face auth-flip-front">
594             <LoginCard
595               onSuccess={onSuccess}
596               onSwitch={onShowRegister}
597               onBack={onBack}
598               turnstileEnabled={!flipped}
599               {...turnstilePass}
600             />
594             </AuthFlipFace>
601             <AuthFlipFace active={flipped} className="auth-flip-face auth-flip-back">
602               <RegisterCard
603                 onSuccess={onSuccess}
604                 onSwitch={onShowLogin}
605                 onBack={onBack}
606                 turnstileEnabled={flipped}
607                 {...turnstilePass}
608               />
609             </AuthFlipFace>
610           </div>
611         </div>
612       </div>
613     </div>
614   );
615 }
616
```

front/src/AuthPages.tsx

```
// ===== 文件: front/src/DatasetManagementPage.tsx =====
1 import { createPortal } from 'react-dom';
2 import { useEffect, useMemo, useRef, useState } from 'react';
3 import type { ChangeEvent, FormEvent } from 'react';
4 import { useSearchParams } from 'react-router-dom';
5 import {
6   AlertTriangle,
7   ArrowLeft,
8   Download,
9   Eye,
10  FilePlus2,
11  Grid3x3,
12  List,
13  PencilLine,
14  Search,
15  Shuffle,
16  Trash2,
17  Upload,
18  X,
19 } from 'lucide-react';
20 /**
21  * 数据管理页面维护数据集目录与当前选中项，上传、追加、重命名和删除都通过后端接口完成。
22  * 页面只负责交互状态和展示格式化，版本、质量、任务引用和文件生命周期由后端决定。
23  */
24 import { apiBlob, apiJson, DATASET_UPLOAD_TIMEOUT_MS } from './api/client';
25 import { SelectField } from './components/SelectField';
26 import { formatLocalDateTime } from './utils/time';
27
28 const DATASET_RENDER_LIMIT = 60;
29 const DATASET_TYPE_OPTIONS = [
30   { value: '全部', label: '全部' },
31   { value: '数值', label: '数值' },
32   { value: '混合', label: '混合' },
33 ];
34
35 type ClusterStat = {
36   name: string;
37   clusterCount: number;
38   range: string;
39 };
40
41 type LabelDistributionItem = {
42   label: string;
43   count: number;
44   percent: number;
45 };
46
47 type DatasetCatalogItem = {
48   id: string;
```

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```
49   name: string;
50   createdAt: string;
51   fileSizeBytes: number;
52   sampleCount: number;
53   baseCount: number;
54   classCount: number;
55   hasLabels: boolean;
56   dataType: '数值' | '混合';
57   taskCount: number;
58   lastAnalysisAt: string | null;
59   version: number;
60   qualityStatus: 'ready' | 'warning' | 'error';
61   qualityIssues: string[];
62   matrixShape: string;
63   labelShape: string;
64   labelDistribution: LabelDistributionItem[];
65   clusterStats: ClusterStat[];
66 };
67
68 type RawDatasetCatalogItem = Partial<DatasetCatalogItem> & {
69   id: string | number;
70 };
71
72 type DatasetCatalogPayload = RawDatasetCatalogItem[] | {
73   items?: RawDatasetCatalogItem[];
74 };
75
76 function extractDatasetItems(payload: DatasetCatalogPayload) {
77   if (Array.isArray(payload)) {
78     return payload;
79   }
80
81   return Array.isArray(payload.items) ? payload.items : [];
82 }
83
84 function buildInitialSelectedBases(total: number) {
85   return Array.from({ length: Math.min(20, Math.max(0, total)) }, (_, index) => index + 1);
86 }
87
88 type RenameDialogProps = {
89   dataset: DatasetCatalogItem;
90   onClose: () => void;
91   onSubmit: (nextName: string) => Promise<void>;
92 };
93
94 function DatasetRenameDialog({ dataset, onClose, onSubmit }: RenameDialogProps) {
95   const inputRef = useRef<HTMLInputElement | null>(null);
96   const [draftName, setDraftName] = useState(dataset.name);
97   const [localError, setLocalError] = useState('');
98   const [submitting, setSubmitting] = useState(false);
```


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```
99
100   useEffect(() => {
101     setDraftName(dataset.name);
102     setLocalError('');
103     setSubmitting(false);
104   }, [dataset.id, dataset.name]);
105
106   useEffect(() => {
107     const raf = window.requestAnimationFrame(() => {
108       inputRef.current?.focus();
109       inputRef.current?.select();
110     });
111
112     return () => window.cancelAnimationFrame(raf);
113   }, [dataset.id]);
114
115   useEffect(() => {
116     const onKeyDown = (event: KeyboardEvent) => {
117       if (event.key === 'Escape' && !submitting) {
118         event.preventDefault();
119         onClose();
120       }
121     };
122
123     window.addEventListener('keydown', onKeyDown);
124     return () => window.removeEventListener('keydown', onKeyDown);
125   }, [onClose, submitting]);
126
127   async function handleSubmit(event: FormEvent<HTMLFormElement>) {
128     event.preventDefault();
129
130     const nextName = draftName.trim();
131     if (!nextName) {
132       setLocalError('请输入新的数据集名称');
133       return;
134     }
135
136     if (nextName === dataset.name) {
137       setLocalError('名称没有变化');
138       return;
139     }
140
141     setSubmitting(true);
142     setLocalError('');
143
144     try {
145       await onSubmit(nextName);
146     } catch (err) {
147       setSubmitting(false);
148       setLocalError(err instanceof Error ? err.message : '重命名失败');
```

```
front/src/DatasetManagementPage.tsx
```

```
149     }
150   }
151
152   if (typeof document === 'undefined') {
153     return null;
154   }
155
156   const trimmedName = draftName.trim();
157   const canSubmit = trimmedName.length > 0 && trimmedName !== dataset.name && !submitting;
158
159   return createPortal(
160     <div className="dataset-rename-modal" role="presentation" onClick={submitting ? undefined : onClose}>
161       <div
162         className="dataset-rename-dialog"
163         role="dialog"
164         aria-modal="true"
165         aria-labelledby="dataset-rename-title"
166         aria-describedby="dataset-rename-description"
167         onClick={(event) => event.stopPropagation()}
168       >
169         <div className="dataset-rename-header">
170           <span className="dataset-rename-icon" aria-hidden="true">
171             <PencilLine size={18} />
172           </span>
173           <div className="dataset-rename-copy">
174             <h3 id="dataset-rename-title">重命名数据集</h3>
175             <p id="dataset-rename-description">修改后会同步更新列表、详情页和后续任务引用。</p>
176           </div>
177         </div>
178
179         <div className="dataset-rename-summary" aria-label="当前数据集信息">
180           <span>{formatLocalDateTime(dataset.createdAt)}</span>
181           <span>{formatFileSize(dataset.fileSizeBytes)}</span>
182           <span>{dataset.dataType}</span>
183           <span className={`dataset-label-tag ${dataset.hasLabels ? 'has-labels' : ''}`>
184             <span className={`dataset-status-dot ${dataset.hasLabels ? 'has-labels' : ''}`} aria-hidden="true" />
185             {dataset.hasLabels ? '有标签' : '无标签'}
186           </span>
187         </div>
188
189         <form className="dataset-rename-form" onSubmit={handleSubmit}>
190           <label className="dataset-rename-field">
191             <span>新名称</span>
192             <input
193               ref={inputRef}
194               className="dataset-rename-input"
195               type="text"
196               value={draftName}
197               maxLength={80}
198               autoComplete="off"
```

```
front/src/DatasetManagementPage.tsx
```

```
199         spellCheck={false}
200         aria-invalid={Boolean(localError)}
201         aria-describedby={localError ? 'dataset-rename-error' : undefined}
202         onChange={(event) => {
203             setDraftName(event.target.value);
204             if (localError) {
205                 setLocalError('');
206             }
207         }}
208     />
209 </label>
210
211     {localError ? (
212         <p className="dataset-rename-error" id="dataset-rename-error" role="alert">
213             {localError}
214         </p>
215     ) : null}
216
217     <div className="dataset-rename-actions">
218         <button type="button" className="btn btn-secondary" onClick={onClose} disabled={submitting}>
219             取消
220         </button>
221         <button type="submit" className="btn btn-primary" disabled={!canSubmit}>
222             {submitting ? '保存中...' : '保存'}
223         </button>
224     </div>
225 </form>
226 </div>
227 </div>,
228     document.body,
229 );
230 }
231
232 type DeleteDialogProps = {
233     dataset: DatasetCatalogItem;
234     onClose: () => void;
235     onSubmit: () => Promise<void>;
236 };
237
238 function DatasetDeleteDialog({ dataset, onClose, onSubmit }: DeleteDialogProps) {
239     const confirmButtonRef = useRef<HTMLButtonElement | null>(null);
240     const [localError, setLocalError] = useState('');
241     const [submitting, setSubmitting] = useState(false);
242
243     useEffect(() => {
244         setLocalError('');
245         setSubmitting(false);
246     }, [dataset.id]);
247
248     useEffect(() => {
```

```
front/src/DatasetManagementPage.tsx
```

```
249     const raf = window.requestAnimationFrame(() => confirmButtonRef.current?.focus());
250
251     return () => window.cancelAnimationFrame(raf);
252 }, [dataset.id]);
253
254 useEffect(() => {
255     const onKeyDown = (event: KeyboardEvent) => {
256         if (event.key === 'Escape' && !submitting) {
257             event.preventDefault();
258             onClose();
259         }
260     };
261
262     window.addEventListener('keydown', onKeyDown);
263     return () => window.removeEventListener('keydown', onKeyDown);
264 }, [onClose, submitting]);
265
266 async function handleSubmit(event: FormEvent<HTMLFormElement>) {
267     event.preventDefault();
268     setSubmitting(true);
269     setLocalError('');
270
271     try {
272         await onSubmit();
273     } catch (err) {
274         setSubmitting(false);
275         setLocalError(err instanceof Error ? err.message : '删除失败，请稍后重试');
276     }
277 }
278
279 if (typeof document === 'undefined') {
280     return null;
281 }
282
283 return createPortal(
284     <div className="dataset-delete-modal" role="presentation" onClick={submitting ? undefined : onClose}>
285         <div
286             className="dataset-delete-dialog"
287             role="dialog"
288             aria-modal="true"
289             aria-labelledby="dataset-delete-title"
290             aria-describedby="dataset-delete-description"
291             onClick={(event) => event.stopPropagation()}
292         >
293             <div className="dataset-delete-header">
294                 <span className="dataset-delete-icon" aria-hidden="true">
295                     <Trash2 size={18} />
296                 </span>
297                 <div className="dataset-delete-copy">
298                     <h3 id="dataset-delete-title">删除数据集</h3>
```

```
front/src/DatasetManagementPage.tsx
```

```
299         <p id="dataset-delete-description">删除后将无法恢复，请确认目标数据集无误。</p>
300     </div>
301 </div>
302
303     <div className="dataset-delete-target" aria-label="待删除数据集">
304         <span>待删除数据集</span>
305         <strong>{dataset.name}</strong>
306         <small>{dataset.sampleCount} 个样本 • {dataset.baseCount} 个基础聚类</small>
307     </div>
308
309     {localError ? (
310         <p className="dataset-delete-error" role="alert">{localError}</p>
311     ) : null}
312
313     <form className="dataset-delete-actions" onSubmit={handleSubmit}>
314         <button type="button" className="btn btn-secondary" onClick={onClose} disabled={submitting}>
315             取消
316         </button>
317         <button ref={confirmButtonRef} type="submit" className="btn dataset-delete-confirm" disabled={submitting}>
318             <Trash2 size={15} aria-hidden="true" />
319             {submitting ? '删除中...' : '确认删除'}
320         </button>
321     </form>
322 </div>
323 </div>,
324 document.body,
325 );
326 }
327
328 function clamp(value: number, min: number, max: number) {
329     if (!Number.isFinite(value)) return min;
330     return Math.min(max, Math.max(min, Math.round(value)));
331 }
332
333 function selectBases(total: number, count: number, seed: number) {
334     const ids = Array.from({ length: total }, (_, index) => index + 1);
335     let state = Math.abs(Math.trunc(seed)) || 1;
336     for (let index = ids.length - 1; index > 0; index -= 1) {
337         state = (state * 1664525 + 1013904223) % 4294967296;
338         const swapIndex = state % (index + 1);
339         [ids[index], ids[swapIndex]] = [ids[swapIndex], ids[index]];
340     }
341     return ids.slice(0, count).sort((left, right) => left - right);
342 }
343
344 function formatBaseName(id: number) {
345     return `base_${id}`;
346 }
347
348 function formatFileSize(bytes: number) {
```

```
front/src/DatasetManagementPage.tsx
```

```
349   if (!Number.isFinite(bytes) || bytes <= 0) return '未知大小';
350   if (bytes < 1024 * 1024) {
351     return `${Math.max(1, Math.round(bytes / 1024))} KB`;
352   }
353   return `${(bytes / 1024 / 1024).toFixed(2)} MB`;
354 }
355
356 function formatMetricWidth(value: number, maximum: number) {
357   if (maximum <= 0 || value <= 0) return '0%';
358   return `${Math.min(100, Math.max(4, Math.round((value / maximum) * 100)))}%`;
359 }
360
361 function getDatasetMatrixSummary(dataset: DatasetCatalogItem) {
362   return dataset.matrixShape || `E: ${dataset.sampleCount} x ${dataset.baseCount}`;
363 }
364
365 function getDatasetLabelSummary(dataset: DatasetCatalogItem) {
366   if (!dataset.hasLabels) return '无标签';
367   return dataset.labelShape || (dataset.classCount > 0 ? `${dataset.classCount} 类` : '有标签');
368 }
369
370 function getDatasetUsageSummary(dataset: DatasetCatalogItem) {
371   if (dataset.taskCount <= 0) return '未使用';
372   return `${dataset.taskCount} 任务` + `${formatLocalDateTime(dataset.lastAnalysisAt)}`;
373 }
374
375 function getDatasetQualityPresentation(dataset: DatasetCatalogItem) {
376   if (dataset.qualityStatus === 'error') {
377     return { label: '需处理', description: dataset.qualityIssues.join('; ') || '数据检查未通过' };
378   }
379
380   if (dataset.qualityStatus === 'warning') {
381     return { label: '待确认', description: dataset.qualityIssues.join('; ') || '存在需要确认的数据项' };
382   }
383
384   return { label: '检查通过', description: '基础聚类矩阵可用于创建任务' };
385 }
386
387 function normalizeDataset(raw: RawDatasetCatalogItem): DatasetCatalogItem {
388   const hasLabels = Boolean(raw.hasLabels);
389   const sampleCount = Number(raw.sampleCount) || 0;
390   const baseCount = Number(raw.baseCount) || 0;
391
392   return {
393     id: String(raw.id),
394     name: raw.name ?? '未命名数据集',
395     createdAt: raw.createdAt ?? '',
396     fileSizeBytes: Number(raw.fileSizeBytes) || 0,
397     sampleCount,
398     baseCount,
```

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```
399     classCount: hasLabels ? Number(raw.classCount) || 0 : 0,
400     hasLabels,
401     dataType: raw.dataType === '混合' ? '混合' : '数值',
402     taskCount: Number(raw.taskCount) || 0,
403     lastAnalysisAt: raw.lastAnalysisAt ?? null,
404     version: Math.max(1, Number(raw.version) || 1),
405     qualityStatus: raw.qualityStatus === 'warning' || raw.qualityStatus === 'error' ? raw.qualityStatus : 'ready',
406     qualityIssues: Array.isArray(raw.qualityIssues) ? raw.qualityIssues : [],
407     matrixShape: raw.matrixShape ?? `E: ${sampleCount} x ${baseCount}`,
408     labelShape: hasLabels ? raw.labelShape ?? `y: ${sampleCount}` : '',
409     labelDistribution: Array.isArray(raw.labelDistribution) ? raw.labelDistribution : [],
410     clusterStats: Array.isArray(raw.clusterStats) ? raw.clusterStats : [],
411   });
412 }
413
414 function normalizeTypeFilter(value: string | null): '全部' | '数值' | '混合' {
415   return value === '数值' || value === '混合' ? value : '全部';
416 }
417
418 function normalizeViewMode(value: string | null): 'list' | 'card' {
419   return value === 'card' ? 'card' : 'list';
420 }
421
422 /* ===== 卡片网格视图 ===== */
423
424 function DatasetCardGrid({
425   datasets,
426   onSelect,
427   onAppend,
428   onDelete,
429   emptyMessage,
430   actionDisabled,
431 }): {
432   datasets: DatasetCatalogItem[];
433   onSelect: (id: string) => void;
434   onAppend: (dataset: DatasetCatalogItem) => void;
435   onDelete: (dataset: DatasetCatalogItem) => void;
436   emptyMessage: string;
437   actionDisabled: boolean;
438 }) {
439   const metricMaximums = useMemo(
440     () =>
441       datasets.reduce(
442         (max, dataset) => ({
443           sampleCount: Math.max(max.sampleCount, dataset.sampleCount),
444           baseCount: Math.max(max.baseCount, dataset.baseCount),
445           classCount: Math.max(max.classCount, dataset.classCount),
446         }),
447         { sampleCount: 0, baseCount: 0, classCount: 0 },
448       ),
```

```
front/src/DatasetManagementPage.tsx
```

```
449     [datasets],
450   );
451
452   return (
453     <div className="dataset-card-grid">
454       {datasets.length === 0 ? (
455         <div className="dataset-table-empty">{emptyMessage}</div>
456       ) : (
457         datasets.map((d) => {
458           const metrics = [
459             { key: 'n', value: d.sampleCount, maximum: metricMaximums.sampleCount },
460             { key: 'm', value: d.baseCount, maximum: metricMaximums.baseCount },
461             { key: 'c', value: d.classCount, maximum: metricMaximums.classCount },
462           ];
463           const facts = [
464             { label: '矩阵', value: getDatasetMatrixSummary(d) },
465             { label: '标签', value: getDatasetLabelSummary(d) },
466             { label: '任务', value: getDatasetUsageSummary(d) },
467           ];
468
469           const quality = getDatasetQualityPresentation(d);
470
471           return (
472             <div
473               key={d.id}
474               className="dataset-card"
475             >
476               <div className="dataset-card-top">
477                 <div className="dataset-card-title-block">
478                   <strong>{d.name}</strong>
479                   <small>{formatLocalDateTime(d.createdAt)} • {formatFileSize(d.fileSizeBytes)}</small>
480                 </div>
481                 <span className="dataset-type-tag dataset-card-type">{d.dataType}</span>
482               </div>
483               <div className="dataset-card-body">
484                 <div className="dataset-card-metrics" aria-label={` ${d.name} 数据规模`}>
485                   {metrics.map((metric) => (
486                     <div className="dataset-card-metric" key={metric.key}>
487                       <div className="dataset-card-metric-head">
488                         <span>{metric.key}</span>
489                         <strong>{metric.value}</strong>
490                       </div>
491                       <div className="dataset-metric-bar" aria-hidden="true">
492                         <span
493                           className="dataset-metric-fill"
494                           style={{ width: formatMetricWidth(metric.value, metric.maximum) }}
495                         />
496                       </div>
497                     </div>
498                   ))}
499                 </div>
500               </div>
501             </div>
502           );
503         })
504       )
505     </div>
506   );
507 }
```


front/src/DatasetManagementPage.tsx

```
499         </div>
500         <div className="dataset-card-facts" aria-label={` ${d.name} 元数据`} >
501             {facts.map((fact) => (
502                 <div className="dataset-card-fact" key={fact.label}>
503                     <span>{fact.label}</span>
504                     <strong>{fact.value}</strong>
505                 </div>
506             ))}
507         </div>
508     </div>
509     <div className="dataset-card-footer">
510         <div className="dataset-card-statuses">
511             <span className={` dataset-label-tag ${d.hasLabels ? 'has-labels' : ''}`} >
512                 <span className={` dataset-status-dot ${d.hasLabels ? 'has-labels' : ''}`} aria-hidden="true" />
513                 {d.hasLabels ? '有标签' : '无标签'}
514             </span>
515             <span className={` dataset-quality-status ${d.qualityStatus}`} title={quality.description}>
516                 {quality.label}
517             </span>
518         </div>
519         <div className="dataset-card-actions" aria-label={` ${d.name} 操作`} >
520             <button
521                 type="button"
522                 className="dataset-list-action view"
523                 aria-label="查看"
524                 title="查看"
525                 onClick={() => onSelect(d.id)}
526             >
527                 <Eye size={15} aria-hidden="true" />
528             </button>
529             <button
530                 type="button"
531                 className="dataset-list-action append"
532                 aria-label="追加数据"
533                 title="向当前数据集追加数据"
534                 disabled={actionDisabled}
535                 onClick={() => onAppend(d)}
536             >
537                 <FilePlus2 size={15} aria-hidden="true" />
538             </button>
539             <button
540                 type="button"
541                 className="dataset-list-action delete"
542                 aria-label="删除"
543                 title="删除"
544                 disabled={actionDisabled}
545                 onClick={() => onDelete(d)}
546             >
547                 <Trash2 size={15} aria-hidden="true" />
548             </button>
```

front/src/DatasetManagementPage.tsx

```
549         </div>
550     </div>
551 </div>
552     );
553 })
554 })
555 </div>
556 );
557 }
558
559 /* ===== 目录视图 ===== */
560
561 function DatasetCatalogView({
562     datasets,
563     viewMode,
564     emptyMessage,
565     onSelect,
566     onAppend,
567     onDelete,
568     onViewModeChange,
569     onUpload,
570     uploading,
571 }): {
572     datasets: DatasetCatalogItem[];
573     viewMode: 'list' | 'card';
574     emptyMessage: string;
575     onSelect: (id: string) => void;
576     onAppend: (dataset: DatasetCatalogItem) => void;
577     onDelete: (dataset: DatasetCatalogItem) => void;
578     onViewModeChange: (mode: 'list' | 'card') => void;
579     onUpload: () => void;
580     uploading: boolean;
581 }) {
582     const [searchParams, setSearchParams] = useSearchParams();
583     const [search, setSearch] = useState(() => searchParams.get('q') ?? '');
584     const [typeFilter, setTypeFilter] = useState<'全部' | '数值' | '混合'>(() =>
585         normalizeTypeFilter(searchParams.get('type')),
586     );
587
588     useEffect(() => {
589         const nextSearch = searchParams.get('q') ?? '';
590         const nextType = normalizeTypeFilter(searchParams.get('type'));
591         setSearch((current) => (current === nextSearch ? current : nextSearch));
592         setTypeFilter((current) => (current === nextType ? current : nextType));
593     }, [searchParams]);
594
595     function updateCatalogParam(key: 'q' | 'type', value: string) {
596         const next = new URLSearchParams(searchParams);
597         if (!value || value === '全部') {
598             next.delete(key);
```

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```
599     } else {
600       next.set(key, value);
601     }
602     setSearchParams(next, { replace: true });
603   }
604
605   function handleSearchChange(value: string) {
606     setSearch(value);
607     updateCatalogParam('q', value.trim());
608   }
609
610   function handleTypeChange(value: '全部' | '数值' | '混合') {
611     setTypeFilter(value);
612     updateCatalogParam('type', value);
613   }
614
615   const filtered = useMemo(() => {
616     let list = datasets;
617     if (search) {
618       const q = search.toLowerCase();
619       list = list.filter((d) => d.name.toLowerCase().includes(q));
620     }
621     if (typeFilter !== '全部') {
622       list = list.filter((d) => d.dataType === typeFilter);
623     }
624     return list;
625   }, [datasets, search, typeFilter]);
626   const visibleDatasets = filtered.slice(0, DATASET_RENDER_LIMIT);
627   const isLimited = filtered.length > visibleDatasets.length;
628   const tableEmptyMessage = datasets.length === 0 ? emptyMessage : '暂无匹配数据集';
629
630   return (
631     <div className="dataset-catalog">
632       <div className="dataset-toolbar">
633         <div className="dataset-toolbar-left">
634           <span className="dataset-search-icon" aria-hidden="true">
635             <Search size={15} />
636           </span>
637           <input
638             className="dataset-search-input"
639             type="text"
640             name="datasetSearch"
641             placeholder="搜索数据集..."
642             autoComplete="off"
643             aria-label="搜索数据集"
644             value={search}
645             onChange={(e) => handleSearchChange(e.target.value)}
646           />
647         </div>
648
```

front/src/DatasetManagementPage.tsx

```
649     <div className="dataset-toolbar-center">
650       <SelectField
651         className="dataset-type-select"
652         size="sm"
653         value={typeFilter}
654         ariaLabel="数据类型筛选"
655         options={DATASET_TYPE_OPTIONS}
656         onChange={(value) => handleTypeChange(value as typeof typeFilter)}
657       />
658     </div>
659
660     <div className="dataset-toolbar-segment">
661       <button
662         type="button"
663         className={`dataset-view-btn ${viewMode === 'list' ? 'active' : ''}`}
664         onClick={() => { if (viewMode !== 'list') onViewModeChange('list'); }}
665         aria-label="列表视图"
666       >
667         <List size={15} />
668       </button>
669       <button
670         type="button"
671         className={`dataset-view-btn ${viewMode === 'card' ? 'active' : ''}`}
672         onClick={() => { if (viewMode !== 'card') onViewModeChange('card'); }}
673         aria-label="卡片视图"
674       >
675         <Grid3x3 size={15} />
676       </button>
677     </div>
678
679     <div className="dataset-toolbar-right">
680       <button
681         type="button"
682         className="btn btn-primary"
683         onClick={onUpload}
684         disabled={uploading}
685       >
686         <Upload size={15} aria-hidden="true" />
687         {uploading ? '加载中...' : '上传数据集'}
688       </button>
689     </div>
690   </div>
691
692   {viewMode === 'list' ? (
693     <div className="dataset-list-grid">
694       <div className="dataset-list-header" aria-hidden="true">
695         <span>数据集</span>
696         <span>规模 n / m / c</span>
697         <span>标签 / 类型</span>
698         <span>质量 / 使用</span>
```

```
front/src/DatasetManagementPage.tsx
```

```
699         <span>操作</span>
700     </div>
701     {filtered.length === 0 ? (
702         <div className="dataset-table-empty">{tableEmptyMessage}</div>
703     ) : (
704         visibleDatasets.map((d) => (
705             <article key={d.id} className="dataset-list-item">
706                 <div className="dataset-list-item-name">
707                     <strong title={d.name}>{d.name}</strong>
708                     <small>
709                         {formatLocalDateTime(d.createdAt)} • {formatFileSize(d.fileSizeBytes)} • v{d.version}
710                     </small>
711                 </div>
712                 <span className="dataset-list-metric" aria-label="规模 n m c">
713                     <code>{d.sampleCount} / {d.baseCount} / {d.classCount}</code>
714                 </span>
715                 <div className="dataset-list-labels">
716                     <span className={`dataset-label-tag ${d.hasLabels ? 'has-labels' : ''}`}>
717                         <span className={`dataset-status-dot ${d.hasLabels ? 'has-labels' : ''}`} aria-hidden="true" />
718                         {d.hasLabels ? '有标签' : '无标签'}
719                     </span>
720                     <span className="dataset-type-tag">{d.dataType}</span>
721                 </div>
722                 <div className="dataset-list-lifecycle">
723                     <span
724                         className={`dataset-quality-status ${d.qualityStatus}`}
725                         title={getDatasetQualityPresentation(d).description}
726                     >
727                         {getDatasetQualityPresentation(d).label}
728                     </span>
729                     <span
730                         className={`dataset-usage-status ${d.taskCount > 0 ? 'used' : 'unused'}`}
731                         title={d.taskCount > 0 ? getDatasetUsageSummary(d) : undefined}
732                     >
733                         {getDatasetUsageSummary(d)}
734                     </span>
735                 </div>
736                 <div className="dataset-list-actions" aria-label={` ${d.name} 操作`} >
737                     <button
738                         type="button"
739                         className="dataset-list-action view"
740                         aria-label="查看"
741                         title="查看"
742                         onClick={() => onSelect(d.id)}
743                     >
744                         <Eye size={15} aria-hidden="true" />
745                     </button>
746                     <button
747                         type="button"
748                         className="dataset-list-action append"
```

```
front/src/DatasetManagementPage.tsx
```

```
749             aria-label="追加数据"
750             title="向当前数据集追加数据"
751             disabled={uploading}
752             onClick={() => onAppend(d)}
753         >
754             <FilePlus2 size={15} aria-hidden="true" />
755         </button>
756         <button
757             type="button"
758             className="dataset-list-action delete"
759             aria-label="删除"
760             title="删除"
761             disabled={uploading}
762             onClick={() => onDelete(d)}
763         >
764             <Trash2 size={15} aria-hidden="true" />
765         </button>
766     </div>
767 </article>
768 ))
769 }}
770 {isLimited ? (
771     <div className="dataset-render-limit-note">
772         当前显示前 {DATASET_RENDER_LIMIT} 条数据集，请搜索或筛选缩小范围。
773     </div>
774 ) : null}
775 </div>
776 ) : (
777     <DatasetCardGrid
778         datasets={visibleDatasets}
779         onSelect={onSelect}
780         onAppend={onAppend}
781         onDelete={onDelete}
782         emptyMessage={tableEmptyMessage}
783         actionDisabled={uploading}
784     />
785 )}
786 {viewMode === 'card' && isLimited ? (
787     <div className="dataset-render-limit-note">
788         当前显示前 {DATASET_RENDER_LIMIT} 条数据集，请搜索或筛选缩小范围。
789     </div>
790 ) : null}
791 </div>
792 );
793 }
794
795 /* ===== 详情视图 ===== */
796
797 function DatasetDetailView({
798     dataset,
```

```
front/src/DatasetManagementPage.tsx
```

```
799   onBack,
800   onExport,
801   onRename,
802   onDelete,
803 }: {
804   dataset: DatasetCatalogItem;
805   onBack: () => void;
806   onExport: () => void;
807   onRename: () => void;
808   onDelete: () => void;
809 }) {
810   const [nBaseInput, setNBaseInput] = useState(() => String(Math.min(20, Math.max(0, dataset.baseCount))));
811   const [seedInput, setSeedInput] = useState('1');
812   const [selectedBases, setSelectedBases] = useState(() => buildInitialSelectedBases(dataset.baseCount));
813
814   const nBase = dataset.baseCount > 0 ? clamp(Number(nBaseInput), 0, dataset.baseCount) : 0;
815   const seed = clamp(Number(seedInput), 1, 999999);
816   const selectedRatio = dataset.baseCount > 0
817     ? Math.round((selectedBases.length / dataset.baseCount) * 100)
818     : 0;
819
820   const selectedPreview = useMemo(
821     () => selectedBases.map((id) => formatBaseName(id)),
822     [selectedBases],
823   );
824   const selectedBaseNameSet = useMemo(() => new Set(selectedPreview), [selectedPreview]);
825   const quality = getDatasetQualityPresentation(dataset);
826   const detailMetrics = [
827     { label: '样本 n', value: dataset.sampleCount, note: '数据对象总量' },
828     { label: '基础聚类 m', value: dataset.baseCount, note: '候选聚类成员' },
829     { label: '真实类别 c', value: dataset.classCount, note: dataset.hasLabels ? '标签类别数量' : '未提供标签' },
830     { label: '已选基础聚类', value: selectedBases.length, note: `当前选择 / ${dataset.baseCount}` },
831   ];
832
833   function handleRandomSelect() {
834     setSelectedBases(selectBases(dataset.baseCount, nBase, seed));
835   }
836
837   function handleRemoveBase(baseId: number) {
838     setSelectedBases((current) => {
839       const nextSelectedBases = current.filter((id) => id !== baseId);
840       setNBaseInput(String(nextSelectedBases.length));
841       return nextSelectedBases;
842     });
843   }
844
845   return (
846     <div className="dataset-detail-shell">
847       <header className="dataset-detail-hero">
848         <div className="dataset-detail-title-row">
```

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```
849     <button type="button" className="dataset-detail-back" onClick={onBack}>
850       <ArrowLeft size={14} aria-hidden="true" /> 返回
851     </button>
852     <div className="dataset-detail-title-block">
853       <div className="dataset-detail-identity">
854         <h2>{dataset.name}</h2>
855         <div className="dataset-detail-meta" aria-label={` ${dataset.name} 基本信息`} >
856           <span>{formatLocalDateTime(dataset.createdAt)}</span>
857           <span>{formatFileSize(dataset.fileSizeBytes)}</span>
858           <span>{dataset.dataType}</span>
859           <span className={` dataset-label-tag ${dataset.hasLabels ? 'has-labels' : ''}`} >
860             <span className={` dataset-status-dot ${dataset.hasLabels ? 'has-labels' : ''}` aria-hidden="true"
861               />
862               {dataset.hasLabels ? '有标签' : '无标签'}
863             </span>
864             <span className={` dataset-quality-status ${dataset.qualityStatus}`} title={quality.description}>
865               {quality.label}
866             </span>
867           </div>
868         </div>
869       </div>
870     </div>
871     <div className="dataset-detail-actions" aria-label={` ${dataset.name} 操作`} >
872       <button type="button" className="btn btn-secondary" onClick={onExport}>
873         <Download size={14} aria-hidden="true" /> 导出数据
874       </button>
875       <button type="button" className="btn btn-secondary" onClick={onRename}>
876         <PencilLine size={14} aria-hidden="true" /> 重命名
877       </button>
878       <button type="button" className="btn btn-danger" onClick={onDelete}>
879         <Trash2 size={14} aria-hidden="true" /> 删除
880       </button>
881     </div>
882   </header>
883
884   <div className="dataset-detail-layout">
885     <main className="dataset-detail-main">
886       {dataset.qualityStatus !== 'ready' && dataset.qualityIssues.length > 0 ? (
887         <div className={` dataset-quality-note ${dataset.qualityStatus}`} role="status">
888           <AlertTriangle size={16} aria-hidden="true" />
889           <span>{quality.description}</span>
890         </div>
891       ) : null}
892     <section className="dataset-overview-grid" aria-label="数据集概览">
893       {detailMetrics.map((metric) => (
894         <div className="dataset-overview-card" key={metric.label}>
895           <span>{metric.label}</span>
896           <strong>{metric.value}</strong>
897           <small>{metric.note}</small>
```



```
front/src/DatasetManagementPage.tsx
```

```
898         </div>
899     }}}
900 </section>
901
902 <div className="dataset-detail-workspace">
903     <section className="dataset-detail-section" aria-labelledby="dataset-label-distribution-heading">
904         <div className="dataset-section-heading">
905             <div>
906                 <h3 id="dataset-label-distribution-heading">标签分布</h3>
907                 <p>按真实类别统计样本数量与占比</p>
908             </div>
909         </div>
910         <div className="label-distribution-panel">
911             {dataset.labelDistribution.length === 0 ? (
912                 <div className="dataset-empty-state">暂无标签分布数据</div>
913             ) : (
914                 <div className="label-distribution-list">
915                     {dataset.labelDistribution.map((item) => (
916                         <div className="label-distribution-row" key={item.label}>
917                             <strong>类别 {item.label}</strong>
918                             <span className="label-distribution-count">{item.count} 个</span>
919                             <div className="label-distribution-track" aria-hidden="true">
920                                 <i style={{ width: `${item.percent}%` }} />
921                             </div>
922                             <span className="label-distribution-percent">{item.percent}%</span>
923                         </div>
924                     ))}
925                 </div>
926             )}
927         </div>
928     </section>
929
930 <div className="dataset-detail-lower-grid">
931     <section className="dataset-detail-section" aria-labelledby="dataset-cluster-table-heading">
932         <div className="dataset-section-heading">
933             <div>
934                 <h3 id="dataset-cluster-table-heading">基础聚类统计</h3>
935                 <p>逐个基础聚类查看簇数量、标签范围和当前选择状态</p>
936             </div>
937         </div>
938         <div className="dataset-table-wrap">
939             <table className="dataset-stat-table">
940                 <thead>
941                     <tr>
942                         <th>基础聚类</th>
943                         <th>簇数量</th>
944                         <th>标签范围</th>
945                         <th>选择状态</th>
946                     </tr>
947                 </thead>
```

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```
948         <tbody>
949             {dataset.clusterStats.length === 0 ? (
950                 <tr>
951                     <td colSpan={4} className="dataset-table-empty-cell">
952                         暂无聚类统计
953                     </td>
954                 </tr>
955             ) : (
956                 dataset.clusterStats.map((stat) => {
957                     const selected = selectedBaseNameSet.has(stat.name);
958                     return (
959                         <tr key={stat.name}>
960                             <td>{stat.name}</td>
961                             <td>{stat.clusterCount}</td>
962                             <td>{stat.range}</td>
963                             <td>
964                                 <span className={`dataset-selection-status ${selected ? 'selected' : ''}`}>
965                                     {selected ? '已选' : '未选'}
966                                 </span>
967                             </td>
968                         </tr>
969                     );
970                 })
971             )}
972         </tbody>
973     </table>
974 </div>
975 </section>
976
977     <section className="dataset-detail-section dataset-selection-panel"
978       aria-labelledby="dataset-selection-heading">
979         <div className="selection-control-heading">
980             <h3 id="dataset-selection-heading">选择基础聚类</h3>
981             <strong>{selectedBases.length} / {dataset.baseCount}</strong>
982         </div>
983         <div className="selection-controls">
984             <div className="selection-inputs">
985                 <label>
986                     <span>n_base</span>
987                     <input
988                       type="number"
989                       min={0}
990                       max={Math.max(1, dataset.baseCount)}
991                       aria-label="n_base"
992                       value={nBaseInput}
993                       onBlur={() => setNBaseInput(String(nBase))}
994                       onChange={(e) => setNBaseInput(e.target.value)}
995                     />
996                 </label>
997             </div>
998         </div>
```

```
front/src/DatasetManagementPage.tsx
```

```
997         <span>随机种子</span>
998         <input
999             type="number"
1000             min={1}
1001             aria-label="随机种子"
1002             value={seedInput}
1003             onBlur={() => setSeedInput(String(seed))}
1004             onChange={(e) => setSeedInput(e.target.value)}
1005         />
1006     </label>
1007 </div>
1008 <button
1009     type="button"
1010     className="btn btn-secondary"
1011     onClick={handleRandomSelect}
1012     disabled={dataset.baseCount === 0}
1013 >
1014     <Shuffle size={16} aria-hidden="true" />
1015     随机选择
1016 </button>
1017 </div>
1018 <div className="selection-meter" aria-label="当前选择进度">
1019     <div className="selection-meter-heading">
1020         <span>选择进度</span>
1021         <strong>{selectedRatio}%</strong>
1022     </div>
1023     <div className="progress-track" aria-hidden="true">
1024         <span style={{ width: `${selectedRatio}%` }} />
1025     </div>
1026 </div>
1027 <div className="selected-base-panel" aria-label="已选基础聚类">
1028     <div className="selected-base-panel-heading">
1029         <span>候选集合</span>
1030         <strong>{selectedBases.length} 项</strong>
1031     </div>
1032     <div className="selected-base-grid">
1033         {selectedBases.length === 0 ? (
1034             <span className="selected-base-empty">暂无已选基础聚类</span>
1035         ) : (
1036             selectedBases.map((baseId) => {
1037                 const name = formatBaseName(baseId);
1038
1039                 return (
1040                     <span className="selected-base-chip" key={baseId}>
1041                         {name}
1042                     <button
1043                         type="button"
1044                         className="selected-base-remove"
1045                         aria-label={`移除 ${name}`}
1046                         onClick={() => handleRemoveBase(baseId)}

```

```
front/src/DatasetManagementPage.tsx
```

```
1047         >
1048         <X size={13} aria-hidden="true" />
1049       </button>
1050     </span>
1051   );
1052 })
1053 })
1054 </div>
1055 </div>
1056 </section>
1057 </div>
1058 </div>
1059 </main>
1060 </div>
1061 </div>
1062 );
1063 }
1064
1065 /* ===== 主组件 ===== */
1066
1067 export function DatasetManagementPage() {
1068   const [searchParams, setSearchParams] = useSearchParams();
1069   const [selectedId, setSelectedId] = useState<string | null>(null);
1070   const [viewMode, setViewMode] = useState<'list' | 'card'>(() =>
1071     normalizeViewMode(searchParams.get('view')),
1072   );
1073   const [datasets, setDatasets] = useState<DatasetCatalogItem[]>([]);
1074   const [loadingDatasets, setLoadingDatasets] = useState(true);
1075   const [loadError, setLoadError] = useState<string | null>(null);
1076   const [uploading, setUploading] = useState(false);
1077   const [uploadError, setUploadError] = useState<string | null>(null);
1078   const [renameTarget, setRenameTarget] = useState<DatasetCatalogItem | null>(null);
1079   const [deleteTarget, setDeleteTarget] = useState<DatasetCatalogItem | null>(null);
1080   const fileInputRef = useRef<HTMLInputElement>(null);
1081   const appendTargetIdRef = useRef<string | null>(null);
1082
1083   const activeDataset = selectedId
1084     ? datasets.find((d) => d.id === selectedId) ?? null
1085     : null;
1086   const catalogEmptyMessage = loadingDatasets ? '正在加载数据集...' : '暂无数据集，请上传数据集';
1087
1088   useEffect(() => {
1089     const nextViewMode = normalizeViewMode(searchParams.get('view'));
1090     setViewMode((current) => (current === nextViewMode ? current : nextViewMode));
1091   }, [searchParams]);
1092
1093   function handleViewModeChange(nextMode: 'list' | 'card') {
1094     setViewMode(nextMode);
1095     const next = new URLSearchParams(searchParams);
1096     if (nextMode === 'card') {
```

```
front/src/DatasetManagementPage.tsx
```

```
1097     next.set('view', 'card');
1098   } else {
1099     next.delete('view');
1100   }
1101   setSearchParams(next, { replace: true });
1102 }
1103
1104 useEffect(() => {
1105   let cancelled = false;
1106
1107   async function loadDatasets() {
1108     try {
1109       setLoadError(null);
1110       const data = await apiJson<DatasetCatalogPayload>('/datasets', {
1111         auth: true,
1112         unauthenticatedMessage: '请先登录后操作数据集',
1113       });
1114       if (!cancelled) {
1115         setDatasets(extractDatasetItems(data).map(normalizeDataset));
1116       }
1117     } catch (err) {
1118       if (!cancelled) {
1119         setLoadError(err instanceof Error ? err.message : '加载数据集失败');
1120       }
1121     } finally {
1122       if (!cancelled) {
1123         setLoadingDatasets(false);
1124       }
1125     }
1126   }
1127
1128   loadDatasets();
1129
1130   return () => {
1131     cancelled = true;
1132   };
1133 }, []);
1134
1135 function handleUploadClick() {
1136   appendTargetIdRef.current = null;
1137   fileInputRef.current?.click();
1138 }
1139
1140 function handleAppendClick(dataset: DatasetCatalogItem) {
1141   appendTargetIdRef.current = dataset.id;
1142   fileInputRef.current?.click();
1143 }
1144
1145 async function handleFileChange(event: ChangeEvent<HTMLInputElement>) {
1146   const file = event.target.files?.[0];
```

front/src/DatasetManagementPage.tsx

```
1147     if (!file) {
1148         appendTargetIdRef.current = null;
1149         return;
1150     }
1151     const appendTargetId = appendTargetIdRef.current;
1152
1153     setUploading(true);
1154     setLoadError(null);
1155     setUploadError(null);
1156
1157     try {
1158         // 先由后端分块暂存并校验文件，前端不读取文件内容，避免重复占用浏览器内存。
1159         const formData = new FormData();
1160         formData.append('file', file);
1161
1162         const data = await apiJson<DatasetCatalogItem>(appendTargetId ? `/datasets/${appendTargetId}/append` :
1163         '/datasets', {
1164             method: 'POST',
1165             body: formData,
1166             auth: true,
1167             timeoutMs: DATASET_UPLOAD_TIMEOUT_MS,
1168             unauthenticatedMessage: '请先登录后操作数据集',
1169         });
1170         const savedDataset = normalizeDataset(data);
1171
1172         setDataSets((prev) => {
1173             if (!appendTargetId) return [savedDataset, ...prev];
1174             return prev.map((d) => (d.id === appendTargetId ? savedDataset : d));
1175         });
1176         if (!appendTargetId) {
1177             setSelectedId(null);
1178         }
1179     } catch (err) {
1180         setUploadError(err instanceof Error ? err.message : appendTargetId ? '追加失败' : '上传失败');
1181     } finally {
1182         setUploading(false);
1183         appendTargetIdRef.current = null;
1184         // 重置 input 以便再次选择同一文件
1185         if (fileInputRef.current) fileInputRef.current.value = '';
1186     }
1187
1188     async function handleExport(dataset: DatasetCatalogItem) {
1189         // 导出必须由后端读取真实数据文件并生成 ZIP，目录摘要不能替代原始数据交付物。
1190         const blob = await apiBlob('/datasets/export', {
1191             method: 'POST',
1192             headers: {
1193                 'Content-Type': 'application/json',
1194             },
1195             body: JSON.stringify({ datasetIds: [Number(dataset.id)] }),
```

front/src/DatasetManagementPage.tsx

```
1196     auth: true,
1197     unauthenticatedMessage: '请先登录后导出数据集',
1198   });
1199   const url = URL.createObjectURL(blob);
1200   const a = document.createElement('a');
1201   a.href = url;
1202   a.download = `${dataset.name}-${dataset.id}.zip`;
1203   a.click();
1204   URL.revokeObjectURL(url);
1205 }
1206
1207 function openRenameDialog(dataset: DatasetCatalogItem) {
1208   setRenameTarget(dataset);
1209 }
1210
1211 function openDeleteDialog(dataset: DatasetCatalogItem) {
1212   setDeleteTarget(dataset);
1213 }
1214
1215 async function submitRename(dataset: DatasetCatalogItem, nextName: string) {
1216   const renamedDataset = normalizeDataset(await apiJson<DatasetCatalogItem>(` /datasets/${dataset.id}`, {
1217     method: 'PATCH',
1218     headers: {
1219       'Content-Type': 'application/json',
1220     },
1221     body: JSON.stringify({ name: nextName }),
1222     auth: true,
1223     unauthenticatedMessage: '请先登录后操作数据集',
1224   }));
1225   setDatasets((prev) => prev.map((item) => (item.id === dataset.id ? renamedDataset : item)));
1226   setRenameTarget(null);
1227 }
1228
1229 async function submitDelete(dataset: DatasetCatalogItem) {
1230   await apiJson<{ message: string }>(` /datasets/${dataset.id}`, {
1231     method: 'DELETE',
1232     auth: true,
1233     unauthenticatedMessage: '请先登录后操作数据集',
1234   });
1235
1236   setDatasets((prev) => prev.filter((item) => item.id !== dataset.id));
1237   setSelectedId(null);
1238   setDeleteTarget(null);
1239 }
1240
1241 const catalog = (
1242   <DatasetCatalogView
1243     datasets={datasets}
1244     viewMode={viewMode}
1245     emptyMessage={catalogEmptyMessage}
```

```
front/src/DatasetManagementPage.tsx
```

```
1246     onSelect={setSelectedId}
1247     onAppend={handleAppendClick}
1248     onDelete={openDeleteDialog}
1249     onViewModeChange={handleViewModeChange}
1250     onUpload={handleUploadClick}
1251     uploading={uploading}
1252   />
1253 );
1254
1255 return (
1256   <>
1257     <input
1258       ref={fileInputRef}
1259       type="file"
1260       accept=".mat"
1261       style={{ display: 'none' }}
1262       onChange={handleFileChange}
1263       aria-hidden="true"
1264     />
1265     {loadError || uploadError ? (
1266       <div className="dataset-upload-error">{loadError ?? uploadError}</div>
1267     ) : null}
1268     {selectedId === null ? (
1269       catalog
1270     ) : activeDataset ? (
1271       <DatasetDetailView
1272         dataset={activeDataset}
1273         onBack={() => setSelectedId(null)}
1274         onExport={() => void handleExport(activeDataset)}
1275         onRename={() => openRenameDialog(activeDataset)}
1276         onDelete={() => openDeleteDialog(activeDataset)}
1277       />
1278     ) : (
1279       catalog
1280     )}
1281     {renameTarget ? (
1282       <DatasetRenameDialog
1283         dataset={renameTarget}
1284         onClose={() => setRenameTarget(null)}
1285         onSubmit={(nextName) => submitRename(renameTarget!, nextName)}
1286       />
1287     ) : null}
1288     {deleteTarget ? (
1289       <DatasetDeleteDialog
1290         dataset={deleteTarget}
1291         onClose={() => setDeleteTarget(null)}
1292         onSubmit={() => submitDelete(deleteTarget)}
1293       />
1294     ) : null}
1295   </>
```



```
front/src/DatasetManagementPage.tsx
```

```
1296   });
1297 }
1298
// ===== 文件: front/src/LandingPage.tsx =====
1 import { type CSSProperties, useEffect, useState } from 'react';
2 import { Boxes, ClipboardCheck, Database, Download, Network, type LucideIcon } from 'lucide-react';
3 import { SOFTWARE_SHORT_NAME, SOFTWARE_VERSION } from './appMeta';
4 import loginIcon from './images/登录.svg';
5 import registerIcon from './images/注册.svg';
6 import copyIcon from './images/复制.svg';
7 import brandLogo from './images/Logo.svg';
8 import workbenchIcon from './images/工作台.svg';
9 import capabilityOmeletIcon from './images/capability-omelet.svg';
10 import capabilityCaMatrixIcon from './images/capability-ca-matrix.svg';
11 import capabilityOmeletSvIcon from './images/capability-omelet-sv.svg';
12 import capabilityPythonIcon from './images/capability-python.svg';
13 import capabilityFastapiIcon from './images/capability-fastapi.svg';
14 import capabilityNumpyIcon from './images/capability-numpy.svg';
15 import capabilityScipyIcon from './images/capability-scipy.svg';
16 import capabilityEchartsIcon from './images/capability-echarts.svg';
17
18 const DEFAULT_FRONTEND_ORIGIN = 'http://127.0.0.1:5173';
19 const ACCESS_PATHS = [
20   '/login',
21   '/register',
22 ];
23
24 function getFrontendOrigin() {
25   return typeof window === 'undefined' ? DEFAULT_FRONTEND_ORIGIN : window.location.origin;
26 }
27
28 async function copyUrl(value: string) {
29   if (typeof navigator !== 'undefined' && navigator.clipboard?.writeText) {
30     await navigator.clipboard.writeText(value);
31     return;
32   }
33
34   if (typeof document === 'undefined') {
35     throw new Error('当前环境不支持复制地址');
36   }
37
38   const input = document.createElement('textarea');
39   input.value = value;
40   input.setAttribute('readonly', '');
41   input.style.position = 'fixed';
42   input.style.opacity = '0';
43   document.body.appendChild(input);
44   input.select();
45   const copied = document.execCommand('copy');
```

```
front/src/LandingPage.tsx
```

```
46     document.body.removeChild(input);
47
48     if (!copied) {
49         throw new Error('复制地址失败');
50     }
51 }
52
53 // 首页能力条使用本地 SVG，避免首屏依赖第三方图标服务。
54 const CAPABILITY_MARKS = [
55     { label: '数据导入', short: 'DATA', display: 'OMELET', icon: capabilityOmeletIcon, tone: 'blue' },
56     { label: 'CA 矩阵', short: 'CA', display: 'CA Matrix', icon: capabilityCaMatrixIcon, tone: 'indigo' },
57     { label: 'OMELET-SV', short: 'SV', display: 'OMELET-SV', icon: capabilityOmeletSvIcon, tone: 'cyan' },
58     { label: '多核学习', short: 'MKL', display: 'Python', icon: capabilityPythonIcon, tone: 'violet' },
59     { label: 'Python 算法实现', short: 'PY', display: 'FastAPI', icon: capabilityFastapiIcon, tone: 'green' },
60     { label: 'FastAPI 服务编排', short: 'API', display: 'NumPy', icon: capabilityNumpyIcon, tone: 'orange' },
61     { label: '可视化看板', short: 'CHART', display: 'SciPy', icon: capabilityScipyIcon, tone: 'blue' },
62     { label: '结果导出', short: 'EXPORT', display: 'ECharts', icon: capabilityEchartsIcon, tone: 'slate' },
63 ] as const;
64
65 const DETAIL_CARDS = [
66     {
67         title: '数据管理',
68         description: '导入基础聚类结果、查看基础聚类统计、随机选择样本并创建聚类任务。',
69     },
70     {
71         title: '协关联矩阵分析',
72         description: '构建 CA 矩阵，并通过热力图查看样本间协关联结构。',
73     },
74     {
75         title: '多核相似性学习',
76         description: '选择核函数、设置参数、构建基础核矩阵并执行集成聚类算法。',
77     },
78     {
79         title: '性能评估',
80         description: '计算 ACC、NMI、ARI、F1-score，并支持多次运行统计。',
81     },
82     {
83         title: '可视化展示',
84         description: '展示 CA 矩阵热力图、核权重分布、拓朴亲和矩阵、聚类散点图和收敛曲线。',
85     },
86     {
87         title: '结果导出',
88         description: '导出聚类标签、指标表格、图像结果、运行日志，并生成聚类分析报告。',
89     },
90 ];
91
92 type WorkflowStage = {
93     order: string;
94     title: string;
95     description: string;
```

```
front/src/LandingPage.tsx
```

```
96   Icon: LucideIcon;
97   positionX: string;
98   nodeY: string;
99   placement: 'upper' | 'lower';
100 };
101
102 const ANALYSIS_WORKFLOW_STAGES: readonly WorkflowStage[] = [
103   {
104     order: '01',
105     title: '选择基础聚类结果',
106     description: '导入并确认基础结果',
107     Icon: Database,
108     positionX: '9%',
109     nodeY: '82px',
110     placement: 'upper',
111   },
112   {
113     order: '02',
114     title: 'GBE 编码与 CA 构建',
115     description: '生成协关联矩阵',
116     Icon: Boxes,
117     positionX: '29.5%',
118     nodeY: '154px',
119     placement: 'lower',
120   },
121   {
122     order: '03',
123     title: '多核相似性学习',
124     description: '优化融合权重',
125     Icon: Network,
126     positionX: '50%',
127     nodeY: '82px',
128     placement: 'upper',
129   },
130   {
131     order: '04',
132     title: '谱聚类与指标评估',
133     description: '形成标签并计算核心指标',
134     Icon: ClipboardCheck,
135     positionX: '70.5%',
136     nodeY: '154px',
137     placement: 'lower',
138   },
139   {
140     order: '05',
141     title: '结果落盘',
142     description: '保存标签、图表与运行产物',
143     Icon: Download,
144     positionX: '91%',
145     nodeY: '82px',
```

front/src/LandingPage.tsx

```
146     placement: 'upper',
147   },
148 ];
149
150 type WorkflowStageStyle = CSSProperties & {
151   '--workflow-stage-x': string;
152   '--workflow-node-y': string;
153 };
154
155 function TopBar() {
156   return (
157     <header className="landing-topbar">
158       <div className="landing-topbar-inner">
159         <div className="landing-brand">
160           <span className="brand-logo" aria-hidden="true">
161             <img className="brand-logo-image" src={brandLogo} width={32} height={32} alt="" aria-hidden="true" />
162           </span>
163           <span className="landing-brand-copy">
164             <strong>{SOFTWARE_SHORT_NAME}</strong>
165           </span>
166         </div>
167         <span className="landing-version-pill" aria-label={`当前版本 ${SOFTWARE_VERSION}`}>
168           {SOFTWARE_VERSION}
169         </span>
170       </div>
171     </header>
172   );
173 }
174
175 function BaseUrlRow() {
176   const [index, setIndex] = useState(0);
177   const [copied, setCopied] = useState(false);
178   const frontendOrigin = getFrontendOrigin();
179   const activePath = ACCESS_PATHS[index];
180   const fullAccessUrl = `${frontendOrigin}${activePath}`;
181
182   useEffect(() => {
183     if (typeof window === 'undefined' || typeof window.matchMedia === 'undefined') return;
184     const mq = window.matchMedia('(prefers-reduced-motion: reduce)');
185     if (mq.matches) return;
186     const timer = setInterval(() => {
187       setIndex((prev) => (prev + 1) % ACCESS_PATHS.length);
188     }, 3000);
189     return () => clearInterval(timer);
190   }, []);
191
192   const handleCopy = async () => {
193     try {
194       await copyUrl(fullAccessUrl);
195       setCopied(true);
196     }
```

```
front/src/LandingPage.tsx
```

```
196     setTimeout(() => setCopied(false), 1500);
197   } catch {
198     setCopied(false);
199   }
200 };
201
202 return (
203   <div className="base-url-row" role="group" aria-label="登录或注册页面地址">
204     <span className="base-url-address" title={frontendOrigin}>{frontendOrigin}</span>
205     <span key={activePath} className="base-url-endpoint" title={activePath}>
206       {activePath}
207     </span>
208     <button
209       type="button"
210       className="base-url-copy"
211       aria-label={copied ? '已复制地址' : '复制地址'}
212       onClick={handleCopy}
213     >
214       <img className="base-url-copy-icon" src={copyIcon} width={14} height={14} alt="" aria-hidden="true" />
215     </button>
216     <span className="sr-only" aria-live="polite">
217       {copied ? '地址已复制' : ''}
218     </span>
219   </div>
220 );
221 }
222
223 function Hero({
224   onLogin,
225   onRegister,
226 }): {
227   onLogin: () => void;
228   onRegister: () => void;
229 }) {
230   return (
231     <section className="landing-hero" aria-label="新材料数据分析平台介绍">
232       <div className="blur-ball blur-ball-indigo" aria-hidden="true" />
233       <div className="blur-ball blur-ball-teal" aria-hidden="true" />
234       <div className="landing-hero-inner">
235         <div className="landing-hero-copy">
236           <h1>
237             新材料
238           <br />
239           <span className="shine-text">数据分析统一平台</span>
240         </h1>
241         <p className="hero-subtitle">
242           登录或注册页面地址:
243         </p>
244       </div>
245     </section>
246   );
247 }
```

```
front/src/LandingPage.tsx
```

```
246     <BaseUrlRow />
247
248     <div className="hero-buttons" aria-label="首页账号操作">
249       <button type="button" className="btn btn-primary hero-btn" onClick={onLogin}>
250         <img className="hero-btn-icon" src={loginIcon} width={16} height={16} alt="" aria-hidden="true" />
251         登录
252       </button>
253       <button type="button" className="btn btn-secondary hero-btn" onClick={onRegister}>
254         <img className="hero-btn-icon" src={registerIcon} width={16} height={16} alt="" aria-hidden="true" />
255         注册
256       </button>
257     </div>
258
259     <CapabilityStrip />
260   </div>
261 </section>
262 );
263 }
264
265 function CapabilityStrip() {
266   return (
267     <section className="landing-capability-strip" id="datasets" aria-labelledby="capability-title">
268       <h2 id="capability-title">覆盖完整的分析链路</h2>
269       <div className="capability-mark-grid" aria-label="分析链路能力">
270         {CAPABILITY_MARKS.map((item) => {
271           return (
272             <span
273               key={item.label}
274               className={`capability-mark capability-mark-${item.tone}`}
275               aria-label={item.label}
276               title={item.label}
277             >
278               <img className="capability-mark-icon" src={item.icon} alt="" aria-hidden="true" />
279               <strong>{item.short}</strong>
280               <span className="capability-mark-label">{item.display}</span>
281             </span>
282           );
283         })}
284       </div>
285     </section>
286   );
287 }
288
289 function DetailSection() {
290   return (
291     <section className="landing-section" aria-labelledby="detail-title">
292       <div className="landing-section-inner">
293         <div className="landing-section-header">
294           <span className="landing-section-kicker">平台能力</span>
295           <h2 id="detail-title">从数据管理到分析报告的完整流程</h2>
```

```
front/src/LandingPage.tsx
```

```
296         <p>
297             围绕基础聚类结果、CA 矩阵、多核相似性学习、性能评估、可视化展示与结果导出，形成面向新材料数据分析的闭环工作流。
298         </p>
299     </div>
300
301     <div className="capability-grid">
302         {DETAIL_CARDS.map((item) => (
303             <article key={item.title} className="capability-card">
304                 <span className="capability-icon" aria-hidden="true" />
305                 <h3>{item.title}</h3>
306                 <p>{item.description}</p>
307             </article>
308         ))}
309     </div>
310 </div>
311 </section>
312 );
313 }
314
315 function WorkflowTimelineSection() {
316     return (
317         <section className="landing-workflow-section" aria-labelledby="workflow-timeline-title">
318             <div className="landing-section-inner">
319                 <div className="landing-workflow-card">
320                     <header className="landing-workflow-heading">
321                         <div>
322                             <span className="landing-workflow-kicker">执行流程</span>
323                             <h2 id="workflow-timeline-title">分析执行脉络</h2>
324                         </div>
325                         <p>从基础聚类结果到可交付分析产物</p>
326                     </header>
327
328                     <ol className="landing-workflow-timeline" aria-label="五阶段分析执行流程">
329                         <svg
330                             className="workflow-route"
331                             viewBox="0 0 1000 240"
332                             preserveAspectRatio="none"
333                             aria-hidden="true"
334                             focusable="false"
335                         >
336                             <defs>
337                                 <linearGradient id="workflow-route-gradient" x1="0%" y1="50%" x2="100%" y2="50%">
338                                     <stop offset="0%" stopColor="#60a5fa" />
339                                     <stop offset="48%" stopColor="#2dd4bf" />
340                                     <stop offset="100%" stopColor="#60a5fa" />
341                                 </linearGradient>
342                             </defs>
343                             <path
344                                 className="workflow-route-path"
345                                 d="M 90 82 C 190 82, 190 154, 295 154 S 400 82, 500 82 S 605 154, 705 154 S 810 82, 910 82"
```

front/src/LandingPage.tsx

```
346     />
347   </svg>
348   {ANALYSIS_WORKFLOW_STAGES.map(({ order, title, description, Icon, positionX, nodeY, placement }) => {
349     const stageStyle: WorkflowStageStyle = {
350       '--workflow-stage-x': positionX,
351       '--workflow-node-y': nodeY,
352     };
353
354     return (
355       <li key={title} className={`workflow-stage is-${placement}`} style={stageStyle}>
356         <div className="workflow-stage-copy">
357           <span className="workflow-stage-order">{order}</span>
358           <h3>{title}</h3>
359           <p>{description}</p>
360         </div>
361         <span className="workflow-stage-node" aria-hidden="true">
362           <Icon size={17} strokeWidth={2.1} />
363         </span>
364       </li>
365     );
366   })}
367 </ol>
368 </div>
369 </div>
370 </section>
371 );
372 }
373
374 function AboutSection({
375   onEnterWorkbench,
376   isEnteringWorkbench,
377 }): {
378   onEnterWorkbench: () => void | Promise<void>;
379   isEnteringWorkbench: boolean;
380 }) {
381   return (
382     <section className="landing-section landing-section-compact" id="about" aria-labelledby="about-title">
383       <div className="landing-section-inner landing-about-card">
384         <div className="landing-about-copy">
385           <span className="landing-section-kicker">系统概览</span>
386           <h2 id="about-title">拓扑感知多核集成聚类分析系统</h2>
387           <p>
388             系统以新材料数据分析为场景，串联数据管理、协关联矩阵分析、多核相似性学习、性能评估、可视化展示与结果导出，支撑从基础聚类结果导入到聚类分析报告
389           </p>
390         </div>
391         <button
392           type="button"
393           className="landing-link-button"
394           disabled={isEnteringWorkbench}
395           aria-busy={isEnteringWorkbench}
```



```
front/src/LandingPage.tsx
```

```
396         onClick={() => void onEnterWorkbench()}
397       >
398         {isEnteringWorkbench ? '正在验证登录状态' : '前往分析工作台'}
399       <img
400         className="landing-link-button-icon"
401         src={workbenchIcon}
402         width={16}
403         height={16}
404         alt=""
405         aria-hidden="true"
406       />
407     </button>
408   </div>
409 </section>
410 );
411 }
412
413 export function LandingPage({
414   onLogin,
415   onRegister,
416   onEnterWorkbench,
417   isEnteringWorkbench = false,
418 }): {
419   onLogin: () => void;
420   onRegister: () => void;
421   onEnterWorkbench: () => void | Promise<void>;
422   isEnteringWorkbench?: boolean;
423 }) {
424   // 首页隐藏视口滚动条（仅视觉），卸载时恢复其他页面的默认滚动条。
425   useEffect(() => {
426     document.body.classList.add('landing-scroll-hidden');
427     return () => {
428       document.body.classList.remove('landing-scroll-hidden');
429     };
430   }, []);
431
432   return (
433     <div className="landing-page">
434       <TopBar />
435       <main className="landing-main">
436         <Hero onLogin={onLogin} onRegister={onRegister} />
437         <DetailSection />
438         <WorkflowTimelineSection />
439         <AboutSection onEnterWorkbench={onEnterWorkbench} isEnteringWorkbench={isEnteringWorkbench} />
440       </main>
441     </div>
442   );
443 }
444
445 export default LandingPage;
```

front/src/LandingPage.tsx

446

```
// ===== 文件: front/src/TurnstileWidget.tsx =====
1 import { forwardRef, useEffect, useImperativeHandle, useRef, useState } from 'react';
2
3 const TURNSTILE_SCRIPT_ID = 'soft-web-turnstile-script';
4 const TURNSTILE_SCRIPT_URL = 'https://challenges.cloudflare.com/turnstile/v0/api.js?render=explicit';
5
6 type TurnstileAction = 'login' | 'register';
7 type TurnstileStatus = 'idle' | 'loading' | 'ready' | 'failed';
8
9 type TurnstileRenderOptions = {
10   sitekey: string;
11   action: TurnstileAction;
12   theme: 'light';
13   callback: (token: string) => void;
14   'expired-callback': () => void;
15   'error-callback': (code?: string) => void;
16   'timeout-callback': () => void;
17 };
18
19 type TurnstileApi = {
20   render: (container: HTMLElement, options: TurnstileRenderOptions) => string;
21   reset: (widgetId: string) => void;
22   remove?: (widgetId: string) => void;
23 };
24
25 declare global {
26   interface Window {
27     turnstile?: TurnstileApi;
28   }
29 }
30
31 export type TurnstileWidgetHandle = {
32   reset: () => void;
33 };
34
35 type TurnstileWidgetProps = {
36   siteKey: string;
37   action: TurnstileAction;
38   onVerify: (token: string) => void;
39   onError: (message: string) => void;
40 };
41
42 let turnstileScriptPromise: Promise<void> | null = null;
43
44 function loadTurnstileScript() {
45   if (typeof window === 'undefined') {
46     return Promise.reject(new Error('Turnstile 只能在浏览器中加载'));
47   }
```

```
front/src/TurnstileWidget.tsx
```

```
48
49   if (window.turnstile) {
50     return Promise.resolve();
51   }
52
53   if (turnstileScriptPromise) {
54     return turnstileScriptPromise;
55   }
56
57   turnstileScriptPromise = new Promise<void>((resolve, reject) => {
58     const existingScript = document.getElementById(TURNSTILE_SCRIPT_ID) as HTMLScriptElement | null;
59
60     if (existingScript) {
61       if (existingScript.dataset.loaded === 'true') {
62         reject(new Error('Turnstile 脚本未就绪'));
63         return;
64       }
65       existingScript.addEventListener('load', () => resolve(), { once: true });
66       existingScript.addEventListener('error', () => reject(new Error('Turnstile 脚本加载失败')), {
67         once: true,
68       });
69       return;
70     }
71
72     const script = document.createElement('script');
73     script.id = TURNSTILE_SCRIPT_ID;
74     script.src = TURNSTILE_SCRIPT_URL;
75     script.async = true;
76     script.defer = true;
77     script.onload = () => {
78       script.dataset.loaded = 'true';
79       resolve();
80     };
81     script.onerror = () => {
82       turnstileScriptPromise = null;
83       reject(new Error('Turnstile 脚本加载失败'));
84     };
85
86     document.head.appendChild(script);
87   });
88
89   return turnstileScriptPromise;
90 }
91
92 export const TurnstileWidget = forwardRef<TurnstileWidgetHandle, TurnstileWidgetProps>(
93   function TurnstileWidget({ siteKey, action, onVerify, onError }, ref) {
94     const containerRef = useRef<HTMLDivElement | null>(null);
95     const widgetIdRef = useRef<string | null>(null);
96     const [status, setStatus] = useState<TurnstileStatus>('idle');
97
```

```
front/src/TurnstileWidget.tsx
```

```
98     useImperativeHandle(ref, () => ({
99       reset() {
100         onVerify('');
101         if (widgetIdRef.current && window.turnstile) {
102           window.turnstile.reset(widgetIdRef.current);
103         }
104       },
105     }), [onVerify]);
106
107     useEffect(() => {
108       if (!siteKey) {
109         onVerify('');
110         setStatus('idle');
111         return undefined;
112       }
113
114       let cancelled = false;
115       setStatus('loading');
116
117       loadTurnstileScript()
118         .then(() => {
119           if (cancelled || !containerRef.current) {
120             return;
121           }
122
123           if (!window.turnstile) {
124             throw new Error('Turnstile 脚本未就绪');
125           }
126
127           widgetIdRef.current = window.turnstile.render(containerRef.current, {
128             sitekey: siteKey,
129             action,
130             theme: 'light',
131             callback(token) {
132               setStatus('ready');
133               onVerify(token);
134             },
135             'expired-callback'() {
136               onVerify('');
137               onError('人机验证已过期，请重新验证');
138             },
139             'error-callback'(code) {
140               onVerify('');
141               setStatus('failed');
142               onError(code ? `人机验证失败，请重新尝试（错误码：${code}）` : '人机验证失败，请重新尝试');
143             },
144             'timeout-callback'() {
145               onVerify('');
146               onError('人机验证超时，请重新验证');
147             },
148           });
149         })
150         .catch(() => {
151           setStatus('failed');
152           onError('Turnstile 脚本加载失败，请重新尝试');
153         });
154       return undefined;
155     }, [siteKey, action]);
156
157     return (
158       <div
159         ref={containerRef}
160         style={styles.container}
161       />
162     );
163   }, [siteKey, action]);
164
165   return (
166     <div
167       ref={ref}
168       style={styles.container}
169     />
170   );
171 }
172
173 export default TurnstileWidget;
```

```
front/src/TurnstileWidget.tsx
```

```
148         });
149         setStatus('ready');
150     })
151     .catch(() => {
152         if (cancelled) {
153             return;
154         }
155         setStatus('failed');
156         onVerify('');
157         onError('人机验证加载失败，请刷新页面重试');
158     });
159
160     return () => {
161         cancelled = true;
162         if (widgetIdRef.current && window.turnstile?.remove) {
163             window.turnstile.remove(widgetIdRef.current);
164         }
165         widgetIdRef.current = null;
166         if (containerRef.current) {
167             containerRef.current.replaceChildren();
168         }
169     };
170     }, [action, onError, onVerify, siteKey]);
171
172     if (!siteKey) {
173         return null;
174     }
175
176     return (
177         <div className="auth-turnstile" aria-label="Cloudflare 人机验证">
178             <div className="auth-turnstile-shell" ref={containerRef} />
179             {status === 'loading' ? (
180                 <span className="auth-turnstile-loading">正在加载人机验证...</span>
181             ) : null}
182             {status === 'failed' ? (
183                 <span className="auth-turnstile-error">人机验证加载失败</span>
184             ) : null}
185         </div>
186     );
187 },
188 );
189
```

```
// ===== 文件: front/src/api/auth-session.ts =====
```

```
1 /**
2  * 认证会话的前端生命周期管理。
3  * access token 只保存在内存中；refresh token 由后端通过 HttpOnly Cookie 管理，页面不能读取或持久化它。
4  */
5 import { API_BASE_URL } from './config';
6 import {
```

```
front/src/api/auth-session.ts
```

```
7   ApiError,
8   createAuthenticationError,
9   createHttpError,
10  createNetworkError,
11  createTimeoutError,
12 } from './errors';
13
14 export const AUTH_STORAGE_KEYS = {
15   user: 'soft_web_user',
16 } as const;
17
18 const LEGACY_ACCESS_TOKEN_KEY = 'soft_web_access_token';
19 const LEGACY_REFRESH_TOKEN_KEY = 'soft_web_refresh_token';
20 const REFRESH_TIMEOUT_MS = 30_000;
21
22 export type AuthSessionEventReason = 'refresh_failed' | 'unauthorized' | 'logout';
23
24 export type AuthSessionEvent = {
25   reason: AuthSessionEventReason;
26   error?: ApiError;
27 };
28
29 type AuthSessionPayload = {
30   access_token: string;
31   user?: unknown;
32 };
33
34 type AuthSessionListener = (event: AuthSessionEvent) => void;
35
36 let accessToken: string | null = null;
37 let refreshPromise: Promise<string | null> | null = null;
38 let lastRefreshError: ApiError | null = null;
39 let logoutNotified = false;
40 const listeners = new Set<AuthSessionListener>();
41
42 function removeStoredAuth() {
43   if (typeof localStorage === 'undefined') return;
44   localStorage.removeItem(AUTH_STORAGE_KEYS.user);
45   localStorage.removeItem(LEGACY_ACCESS_TOKEN_KEY);
46   localStorage.removeItem(LEGACY_REFRESH_TOKEN_KEY);
47 }
48
49 function notifyLogout(reason: AuthSessionEventReason, error?: ApiError) {
50   if (logoutNotified) return;
51   logoutNotified = true;
52   const event = { reason, error } satisfies AuthSessionEvent;
53   listeners.forEach((listener) => listener(event));
54 }
55
56 export function subscribeAuthSession(listener: AuthSessionListener) {
```

front/src/api/auth-session.ts

```
57   listeners.add(listener);
58   return () => {
59     listeners.delete(listener);
60   };
61 }
62
63 export function getAccessToken() {
64   return accessToken;
65 }
66
67 export function setAccessToken(token: string | null) {
68   accessToken = token;
69   if (token) {
70     logoutNotified = false;
71     lastRefreshError = null;
72   }
73 }
74
75 export function saveAuthSession(auth: AuthSessionPayload) {
76   setAccessToken(auth.access_token);
77   if (typeof localStorage !== 'undefined' && auth.user !== undefined) {
78     localStorage.setItem(AUTH_STORAGE_KEYS.user, JSON.stringify(auth.user));
79   }
80   removeLegacyTokens();
81 }
82
83 function removeLegacyTokens() {
84   if (typeof localStorage === 'undefined') return;
85   localStorage.removeItem(LEGACY_ACCESS_TOKEN_KEY);
86   localStorage.removeItem(LEGACY_REFRESH_TOKEN_KEY);
87 }
88
89 export function getCachedAuthUser<T>() {
90   if (typeof localStorage === 'undefined') return null;
91   const storedUser = localStorage.getItem(AUTH_STORAGE_KEYS.user);
92   if (!storedUser) return null;
93   try {
94     return JSON.parse(storedUser) as T;
95   } catch {
96     return null;
97   }
98 }
99
100 export function cacheAuthUser(user: unknown) {
101   if (typeof localStorage === 'undefined') return;
102   localStorage.setItem(AUTH_STORAGE_KEYS.user, JSON.stringify(user));
103 }
104
105 export function clearAuthSession(options: {
106   notify?: boolean;
```

front/src/api/auth-session.ts

```
107   reason?: AuthSessionEventReason;
108   error?: ApiError;
109 } = {}) {
110   accessToken = null;
111   if (!options.error) {
112     lastRefreshError = null;
113   }
114   removeStoredAuth();
115   if (options.notify) {
116     notifyLogout(options.reason ?? 'logout', options.error);
117   }
118 }
119
120 function fetchRefreshToken() {
121   const controller = new AbortController();
122   let timedOut = false;
123   const timeoutId = setTimeout(() => {
124     timedOut = true;
125     controller.abort();
126   }, REFRESH_TIMEOUT_MS);
127
128   return (async () => {
129     try {
130       const response = await fetch(`${API_BASE_URL}/auth/refresh`, {
131         method: 'POST',
132         credentials: 'include',
133         signal: controller.signal,
134       });
135       const body = await response.json().catch(() => ({}));
136       if (!response.ok) {
137         throw createHttpError(response.status, body, '刷新登录状态失败');
138       }
139       if (!body || typeof body.access_token !== 'string' || !body.access_token) {
140         throw createAuthenticationError('刷新登录状态失败', body);
141       }
142       return body.access_token as string;
143     } catch (error: unknown) {
144       if (error instanceof ApiError) {
145         throw error;
146       }
147       if (timedOut) {
148         throw createTimeoutError(REFRESH_TIMEOUT_MS);
149       }
150       throw createNetworkError(error);
151     } finally {
152       clearTimeout(timeoutId);
153     }
154   })();
155 }
156
```


front/src/api/auth-session.ts

```
157 export async function refreshAccessToken(): Promise<string | null> {
158   // 刷新是跨请求的临界区，复用同一个 Promise 可避免多个 401 同时轮换 refresh token。
159   if (refreshPromise) {
160     return refreshPromise;
161   }
162
163   refreshPromise = (async () => {
164     try {
165       const token = await fetchRefreshToken();
166       setAccessToken(token);
167       return token;
168     } catch (error) {
169       lastRefreshError = error instanceof ApiError ? error : createNetworkError(error);
170       clearAuthSession({ notify: true, reason: 'refresh_failed', error: lastRefreshError });
171       return null;
172     } finally {
173       refreshPromise = null;
174     }
175   })();
176
177   return refreshPromise;
178 }
179
180 export async function refreshAccessTokenOrThrow() {
181   const token = await refreshAccessToken();
182   if (token) return token;
183   throw lastRefreshError ?? createAuthenticationError();
184 }
185
186 export function notifyUnauthorized(error?: ApiError) {
187   clearAuthSession({ notify: true, reason: 'unauthorized', error });
188 }
189
190 // ===== 文件: front/src/api/auth.ts =====
191 import { apiFetch, apiJson, publicJson, type ApiRequestOptions } from './client';
192 import {
193   AUTH_STORAGE_KEYS,
194   cacheAuthUser,
195   clearAuthSession,
196   getAccessToken,
197   getCachedAuthUser,
198   notifyUnauthorized,
199   refreshAccessToken,
200   saveAuthSession,
201   setAccessToken,
202 } from './auth-session';
203 import type { ApiError } from './errors';
204
205 export type AuthUser = {
```

front/src/api/auth.ts

```
16   id: number;
17   username: string;
18   email?: string | null;
19   role: string;
20   status: string;
21   last_login_at?: string | null;
22 };
23
24 export type AuthResponse = {
25   user: AuthUser;
26   access_token: string;
27   token_type: string;
28 };
29
30 export type TurnstileAction = 'login' | 'register';
31
32 export type TurnstilePassResponse = {
33   turnstile_pass_token: string;
34   expires_at: string;
35   expires_in_seconds: number;
36 };
37
38 export { AUTH_STORAGE_KEYS, clearAuthSession, getAccessToken, saveAuthSession, setAccessToken };
39 export {
40   subscribeAuthSession,
41   type AuthSessionEvent,
42   type AuthSessionEventReason,
43 } from './auth-session';
44
45 async function requestJson<T>(path: string, options: ApiRequestOptions = {}) {
46   return publicJson<T>(path, options);
47 }
48
49 export function authorizedFetch(
50   path: string,
51   options: RequestInit = {},
52   unauthenticatedMessage = '请先登录后再操作',
53 ) {
54   return apiFetch(path, {
55     ...options,
56     auth: true,
57     unauthenticatedMessage,
58   });
59 }
60
61 export function authorizedJson<T>(
62   path: string,
63   options: RequestInit = {},
64   unauthenticatedMessage = '请先登录后再操作',
65 ) {
```

```
front/src/api/auth.ts
```

```
66   return apiJson<T>(path, {
67     ...options,
68     auth: true,
69     unauthenticatedMessage,
70   });
71 }
72
73 export async function restoreAuthSession(): Promise<AuthUser | null> {
74   const user = await authorizedJson<AuthUser>('/auth/me', {}, '请先登录后再进入工作台');
75   cacheAuthUser(user);
76   return user;
77 }
78
79 function buildTurnstilePayload(
80   turnstileToken: string | null,
81   turnstilePassToken: string | null,
82 ) {
83   const payload: {
84     turnstile_token: string | null;
85     turnstile_pass_token?: string;
86   } = {
87     turnstile_token: turnstilePassToken ? null : turnstileToken,
88   };
89
90   if (turnstilePassToken) {
91     payload.turnstile_pass_token = turnstilePassToken;
92   }
93
94   return payload;
95 }
96
97 export function createTurnstilePass(turnstileToken: string, action: TurnstileAction) {
98   return requestJson<TurnstilePassResponse>('/auth/turnstile-pass', {
99     method: 'POST',
100     body: JSON.stringify({ turnstile_token: turnstileToken, action }),
101   });
102 }
103
104 export function login(
105   username: string,
106   password: string,
107   turnstileToken: string | null = null,
108   turnstilePassToken: string | null = null,
109 ) {
110   return requestJson<AuthResponse>('/auth/login', {
111     method: 'POST',
112     body: JSON.stringify({
113       username,
114       password,
115       ...buildTurnstilePayload(turnstileToken, turnstilePassToken),
```

```
front/src/api/auth.ts
```

```
116     }},
117   });
118 }
119
120 export function sendRegisterEmailCode(
121   email: string,
122   turnstileToken: string | null = null,
123   turnstilePassToken: string | null = null,
124 ) {
125   return requestJson<{ message: string }>('/auth/register/email-code', {
126     method: 'POST',
127     body: JSON.stringify({
128       email,
129       ...buildTurnstilePayload(turnstileToken, turnstilePassToken),
130     }),
131   });
132 }
133
134 export function register(
135   username: string,
136   email: string,
137   password: string,
138   confirmPassword: string,
139   emailCode: string,
140 ) {
141   return requestJson<AuthResponse>('/auth/register', {
142     method: 'POST',
143     body: JSON.stringify({
144       username,
145       email,
146       password,
147       confirm_password: confirmPassword,
148       email_code: emailCode,
149     }),
150   });
151 }
152
153 export function getCurrentUser() {
154   return authorizedJson<AuthUser>('/auth/me', {}, '请先登录后再获取用户信息');
155 }
156
157 export async function logout() {
158   try {
159     await requestJson<{ message: string }>('/auth/logout', {
160       method: 'POST',
161     });
162   } finally {
163     clearAuthSession({ notify: true, reason: 'logout' });
164   }
165 }
```

front/src/api/auth.ts

```
166
167 // 兼容旧页面和外部调用方：真正的刷新状态机位于 auth-session.ts。
168 export { refreshAccessToken };
169 export function notifyAuthUnauthorized(error?: ApiError) {
170   notifyUnauthorized(error);
171 }
172
173 export function getCachedUser() {
174   return getCachedAuthUser<AuthUser>();
175 }
176
// ===== 文件：front/src/api/client.ts =====
1 /**
2  * 前端统一请求边界：负责凭证注入、超时与取消、错误归类，以及 401 后的一次重试。
3  * 业务模块只描述接口路径和请求数据，不直接处理 Cookie、刷新并发或响应解析细节。
4  */
5 import { API_BASE_URL } from './config';
6 import {
7   ApiError,
8   createAbortedError,
9   createAuthenticationError,
10  createHttpError,
11  createNetworkError,
12  createTimeoutError,
13  toApiError,
14 } from './errors';
15 import {
16   getAccessToken,
17   notifyUnauthorized,
18   refreshAccessTokenOrThrow,
19 } from './auth-session';
20
21 export const DEFAULT_REQUEST_TIMEOUT_MS = 30_000;
22 export const DATASET_UPLOAD_TIMEOUT_MS = 5 * 60_000;
23 export const PROTECTED_DOWNLOAD_TIMEOUT_MS = 2 * 60_000;
24
25 export type ApiRequestOptions = RequestInit & {
26   timeoutMs?: number;
27   auth?: boolean;
28   retryOnUnauthorized?: boolean;
29   unauthenticatedMessage?: string;
30 };
31
32 function buildUrl(path: string) {
33   if (/^https?:\/\//i.test(path)) return path;
34   if (path === API_BASE_URL || path.startsWith(`${API_BASE_URL}/`)) return path;
35   return `${API_BASE_URL}${path.startsWith('/') ? path : `/${path}`}`;
36 }
37
```

```
front/src/api/client.ts
```

```
38 function buildHeaders(options: RequestInit, token: string | null) {
39   const headers = new Headers(options.headers);
40   if (
41     options.body
42     && typeof options.body === 'string'
43     && !headers.has('Content-Type')
44   ) {
45     headers.set('Content-Type', 'application/json');
46   }
47   if (token) {
48     headers.set('Authorization', `Bearer ${token}`);
49   }
50   return headers;
51 }
52
53 function createRequestSignal(timeoutMs: number, callerSignal?: AbortSignal) {
54   // 一个请求同时接受业务取消和超时取消, 合并后的 signal 交给 fetch, 避免悬挂请求占用页面资源。
55   const controller = new AbortController();
56   let timedOut = false;
57   const timeoutId = setTimeout(() => {
58     timedOut = true;
59     controller.abort();
60   }, timeoutMs);
61
62   const abortFromCaller = () => controller.abort();
63   if (callerSignal) {
64     if (callerSignal.aborted) {
65       controller.abort();
66     } else {
67       callerSignal.addEventListener('abort', abortFromCaller, { once: true });
68     }
69   }
70
71   return {
72     signal: controller.signal,
73     wasTimedOut: () => timedOut,
74     cleanup: () => {
75       clearTimeout(timeoutId);
76       callerSignal?.removeEventListener('abort', abortFromCaller);
77     },
78   };
79 }
80
81 async function fetchOnce(
82   path: string,
83   options: RequestInit,
84   token: string | null,
85   timeoutMs: number,
86 ) {
87   const requestSignal = createRequestSignal(timeoutMs, options.signal ?? undefined);
```

front/src/api/client.ts

```
88   try {
89     return await fetch(buildUrl(path), {
90       ...options,
91       credentials: 'include',
92       headers: buildHeaders(options, token),
93       signal: requestSignal.signal,
94     });
95   } catch (error) {
96     if (requestSignal.wasTimedOut()) {
97       throw createTimeoutError(timeoutMs);
98     }
99     if (options.signal?.aborted) {
100      throw createAbortedError();
101    }
102    throw createNetworkError(error);
103  } finally {
104    requestSignal.cleanup();
105  }
106 }
107
108 function isRefreshPath(path: string) {
109   return /\auth\/refresh(?:$|\?)/.test(path);
110 }
111
112 function throwIfAborted(signal?: AbortSignal) {
113   if (signal?.aborted) {
114     throw createAbortedError();
115   }
116 }
117
118 function withAuthenticationMessage(error: ApiError, message?: string) {
119   if (!message || error.category !== 'authentication') return error;
120   return new ApiError({
121     message,
122     status: error.status,
123     code: error.code,
124     category: error.category,
125     detail: error.detail,
126     isRetryable: error.isRetryable,
127   });
128 }
129
130 async function getTokenOrThrow(unauthenticatedMessage?: string) {
131   const existingToken = getAccessToken();
132   if (existingToken) return existingToken;
133
134   try {
135     return await refreshAccessTokenOrThrow();
136   } catch (error) {
137     throw withAuthenticationMessage(toApiError(error), unauthenticatedMessage);
138   }
139 }
```

```
front/src/api/client.ts
```

```
138   }
139 }
140
141 export async function apiFetch(path: string, options: ApiRequestOptions = {}) {
142   const {
143     timeoutMs = DEFAULT_REQUEST_TIMEOUT_MS,
144     auth = true,
145     retryOnUnauthorized = true,
146     unauthenticatedMessage,
147     ...requestInit
148   } = options;
149
150   throwIfAborted(requestInit.signal ?? undefined);
151
152   // refresh 端点永远走无认证分支, 避免没有 access token 时再次递归 refresh。
153   if (!auth || isRefreshPath(path)) {
154     return fetchOnce(path, requestInit, null, timeoutMs);
155   }
156
157   const token = await getTokenOrThrow(unauthenticatedMessage);
158   const response = await fetchOnce(path, requestInit, token, timeoutMs);
159   if (
160     response.status !== 401
161     || !retryOnUnauthorized
162     || isRefreshPath(path)
163   ) {
164     return response;
165   }
166
167   throwIfAborted(requestInit.signal ?? undefined);
168   // 所有并发请求共享 auth-session 中的刷新 Promise; 当前请求只在刷新成功后重试一次。
169   let refreshedToken: string;
170   try {
171     refreshedToken = await refreshAccessTokenOrThrow();
172   } catch (error) {
173     const apiError = toApiError(error);
174     if (apiError.category === 'authentication') {
175       throw new ApiError({
176         message: '登录已过期, 请重新登录',
177         status: 401,
178         code: 'AUTHENTICATION_EXPIRED',
179         category: 'authentication',
180         detail: apiError.detail,
181       });
182     }
183     throw apiError;
184   }
185
186   throwIfAborted(requestInit.signal ?? undefined);
187   const retriedResponse = await fetchOnce(path, requestInit, refreshedToken, timeoutMs);
```

front/src/api/client.ts

```
188   if (retriedResponse.status === 401) {
189     const error = createAuthenticationError('登录已过期，请重新登录');
190     notifyUnauthorized(error);
191   }
192   return retriedResponse;
193 }
194
195 async function readResponseBody(response: Response): Promise<unknown> {
196   if (response.status === 204) return undefined;
197
198   if (typeof response.json === 'function') {
199     try {
200       return await response.json();
201     } catch {
202       // 某些下载或空响应没有 JSON 正文，继续尝试文本读取。
203     }
204   }
205
206   if (typeof response.text === 'function') {
207     let text: string;
208     try {
209       text = await response.text();
210     } catch {
211       return undefined;
212     }
213     if (!text) return undefined;
214     try {
215       return JSON.parse(text) as unknown;
216     } catch {
217       return text;
218     }
219   }
220
221   return undefined;
222 }
223
224 export async function apiJson<T>(path: string, options: ApiRequestOptions = {}) {
225   const response = await apiFetch(path, options);
226   const body = await readResponseBody(response);
227   if (!response.ok) {
228     throw createHttpError(response.status, body);
229   }
230   return body as T;
231 }
232
233 export async function ensureApiResponse(response: Response, fallback = '请求失败') {
234   if (response.ok) return response;
235   const body = await readResponseBody(response);
236   throw createHttpError(response.status, body, fallback);
237 }
```

front/src/api/client.ts

```
238
239 export async function apiBlob(path: string, options: ApiRequestOptions = {}) {
240   const response = await apiFetch(path, options);
241   if (!response.ok) {
242     const body = await readResponseBody(response);
243     throw createHttpError(response.status, body);
244   }
245   return response.blob();
246 }
247
248 export function publicFetch(path: string, options: Omit<ApiRequestOptions, 'auth'> = {}) {
249   return apiFetch(path, { ...options, auth: false });
250 }
251
252 export function publicJson<T>(path: string, options: Omit<ApiRequestOptions, 'auth'> = {}) {
253   return apiJson<T>(path, { ...options, auth: false });
254 }
255
```

// ===== 文件: front/src/api/config.ts =====

```
1 export const API_BASE_URL = '/api';
2 export const BACKEND_ORIGIN = 'http://127.0.0.1:8000';
3 export const TURNSTILE_SITE_KEY =
4   import.meta.env.MODE === 'test' ? '' : import.meta.env.VITE_TURNSTILE_SITE_KEY ?? '';
5
```

// ===== 文件: front/src/api/datasets.ts =====

```
1 import { apiJson } from './client';
2
3 export type DatasetQualityStatus = 'ready' | 'warning' | 'error';
4
5 export type DatasetCatalogItem = {
6   id: number;
7   name: string;
8   createdAt: string;
9   fileSizeBytes: number;
10  sampleCount: number;
11  baseCount: number;
12  classCount: number;
13  hasLabels: boolean;
14  dataType: string;
15  taskCount: number;
16  lastAnalysisAt: string | null;
17  version: number;
18  qualityStatus: DatasetQualityStatus;
19  qualityIssues: string[];
20  matrixShape: string;
21  labelShape: string;
22 };
23
```

front/src/api/datasets.ts

```
24 export type DatasetCatalogPage = {
25   items: DatasetCatalogItem[];
26   total: number;
27   page: number;
28   pageSize: number;
29   totalPages: number;
30 };
31
32 export type DatasetRevision = {
33   id: number;
34   version: number;
35   action: string;
36   name: string;
37   originalFilename: string;
38   createdAt: string;
39   sampleCount: number;
40   baseCount: number;
41   classCount: number;
42   hasLabels: boolean;
43   qualityStatus: DatasetQualityStatus;
44   qualityIssues: string[];
45 };
46
47 async function requestJson<T>(path: string, options: RequestInit = {}): Promise<T> {
48   return apiJson<T>(path, {
49     ...options,
50     auth: true,
51     unauthenticatedMessage: '请先登录后再查看数据集信息',
52   });
53 }
54
55 export async function fetchDatasetCatalog(pageSize = 100): Promise<DatasetCatalogPage> {
56   const body = await requestJson<DatasetCatalogPage | DatasetCatalogItem[]>(
57     `/datasets?page=1&pageSize=${pageSize}`,
58   );
59
60   if (Array.isArray(body)) {
61     return {
62       items: body,
63       total: body.length,
64       page: 1,
65       pageSize,
66       totalPages: body.length > 0 ? 1 : 0,
67     };
68   }
69
70   return {
71     items: Array.isArray(body.items) ? body.items : [],
72     total: Number(body.total) || 0,
73     page: Number(body.page) || 1,
```

```
front/src/api/datasets.ts
```

```
74     pageSize: Number(body.pageSize) || pageSize,
75     totalPages: Number(body.totalPages) || 0,
76   };
77 }
78
79 export function recheckDatasetQuality(datasetId: number) {
80   return requestJson<DatasetCatalogItem>(`/datasets/${datasetId}/quality`, {
81     method: 'POST',
82   });
83 }
84
85 export function fetchDatasetVersions(datasetId: number) {
86   return requestJson<DatasetRevision[]>(`/datasets/${datasetId}/versions`);
87 }
88
```

```
// ===== 文件: front/src/api/errors.ts =====
```

```
1 export type ApiErrorCategory =
2   | 'network'
3   | 'timeout'
4   | 'aborted'
5   | 'authentication'
6   | 'authorization'
7   | 'business'
8   | 'validation'
9   | 'conflict'
10  | 'not_found'
11  | 'rate_limit'
12  | 'server'
13  | 'http';
14
15 export type ApiErrorInit = {
16   message: string;
17   status?: number | null;
18   code?: string;
19   category: ApiErrorCategory;
20   detail?: unknown;
21   isRetryable?: boolean;
22 };
23
24 export class ApiError extends Error {
25   readonly status: number | null;
26   readonly code: string;
27   readonly category: ApiErrorCategory;
28   readonly detail: unknown;
29   readonly isRetryable: boolean;
30
31   constructor({
32     message,
33     status = null,
```

front/src/api/errors.ts

```
34     code,
35     category,
36     detail,
37     isRetryable,
38   }: ApiErrorInit) {
39     super(message);
40     this.name = 'ApiError';
41     this.status = status;
42     this.code = code ?? (status === null ? 'API_ERROR' : `HTTP_${status}`);
43     this.category = category;
44     this.detail = detail;
45     this.isRetryable = isRetryable ?? isRetryableCategory(category, status);
46   }
47 }
48
49 function isRetryableCategory(category: ApiErrorCategory, status: number | null) {
50   return (
51     category === 'network'
52     || category === 'timeout'
53     || category === 'server'
54     || category === 'rate_limit'
55     || status === 408
56   );
57 }
58
59 function statusCategory(status: number): ApiErrorCategory {
60   if (status === 401) return 'authentication';
61   if (status === 403) return 'authorization';
62   if (status === 404) return 'not_found';
63   if (status === 409) return 'business';
64   if (status === 422 || status === 400) return 'validation';
65   if (status === 429) return 'rate_limit';
66   if (status >= 500) return 'server';
67   return 'http';
68 }
69
70 function readErrorDetail(body: unknown): { message?: string; code?: string; detail: unknown } {
71   if (body === null || body === undefined) {
72     return { detail: body };
73   }
74
75   if (typeof body === 'string') {
76     return { message: body, detail: body };
77   }
78
79   if (typeof body !== 'object') {
80     return { message: String(body), detail: body };
81   }
82
83   const record = body as Record<string, unknown>;
```

front/src/api/errors.ts

```
84   const detail = record.detail ?? body;
85   const code = typeof record.code === 'string' ? record.code : undefined;
86
87   if (typeof detail === 'string') {
88     return { message: detail, code, detail };
89   }
90
91   if (Array.isArray(detail)) {
92     const messages = detail
93       .map((item) => {
94         if (item && typeof item === 'object' && 'msg' in item) {
95           return String((item as { msg?: unknown }).msg ?? '');
96         }
97         return typeof item === 'string' ? item : '';
98       })
99       .filter(Boolean);
100    return {
101      message: messages.length > 0 ? messages.join('; ') : undefined,
102      code,
103      detail,
104    };
105  }
106
107  if (detail && typeof detail === 'object' && 'message' in detail) {
108    const detailRecord = detail as Record<string, unknown>;
109    return {
110      message: typeof detailRecord.message === 'string' ? detailRecord.message : undefined,
111      code: code ?? (typeof detailRecord.code === 'string' ? detailRecord.code : undefined),
112      detail,
113    };
114  }
115
116  return { code, detail };
117 }
118
119 export function createHttpError(status: number, body: unknown, fallback = '请求失败') {
120   const parsed = readErrorDetail(body);
121   return new ApiError({
122     status,
123     category: statusCategory(status),
124     code: parsed.code ?? `HTTP_${status}`,
125     message: parsed.message || fallback,
126     detail: parsed.detail,
127   });
128 }
129
130 export function createAuthenticationError(message = '登录已过期，请重新登录', detail?: unknown) {
131   return new ApiError({
132     status: 401,
133     category: 'authentication',
```

```
front/src/api/errors.ts
```

```
134     code: 'AUTHENTICATION_REQUIRED',
135     message,
136     detail,
137   });
138 }
139
140 export function createTimeoutError(timeoutMs: number) {
141   return new ApiError({
142     category: 'timeout',
143     code: 'REQUEST_TIMEOUT',
144     message: `请求超过 ${Math.ceil(timeoutMs / 1000)} 秒未完成`,
145     detail: { timeoutMs },
146   });
147 }
148
149 export function createAbortedError() {
150   return new ApiError({
151     category: 'aborted',
152     code: 'REQUEST_ABORTED',
153     message: '请求已取消',
154   });
155 }
156
157 export function createNetworkError(error: unknown) {
158   return new ApiError({
159     category: 'network',
160     code: 'NETWORK_ERROR',
161     message: error instanceof Error && error.message ? error.message : '网络连接失败，请稍后重试',
162     detail: error,
163   });
164 }
165
166 export function toApiError(error: unknown) {
167   if (error instanceof ApiError) {
168     return error;
169   }
170
171   return createNetworkError(error);
172 }
173
174
175 // ===== 文件: front/src/api/results.ts =====
176 1 import {
177   2   apiFetch,
178   3   apiJson,
179   4   ensureApiResponse,
180   5   PROTECTED_DOWNLOAD_TIMEOUT_MS,
181   6 } from './client';
182 7 import type { TaskExport, TaskResultEnvelope } from '../workbench/results/types';
183 8
```

front/src/api/results.ts

```
9 function requestJson<T>(path: string, options: RequestInit = {}) {
10   return apiJson<T>(path, {
11     ...options,
12     auth: true,
13     unauthenticatedMessage: '请先登录后再查看分析结果',
14   });
15 }
16
17 export function fetchLatestTaskResult(signal?: AbortSignal) {
18   return requestJson<TaskResultEnvelope>(`/tasks/results/latest`, { signal });
19 }
20
21 export function fetchTaskResult(taskId: number, signal?: AbortSignal) {
22   return requestJson<TaskResultEnvelope>(`/tasks/${taskId}/result`, { signal });
23 }
24
25 export function fetchTaskExports(taskId: number) {
26   return requestJson<{ items: TaskExport[] }>(`/tasks/${taskId}/exports`);
27 }
28
29 export function createTaskExport(taskId: number, items: string[], name?: string) {
30   return requestJson<TaskExport>(`/tasks/${taskId}/exports`, {
31     method: 'POST',
32     body: JSON.stringify({ items, name: name?.trim() || undefined }),
33   });
34 }
35
36 export async function downloadProtectedFile(path: string, fallbackName: string) {
37   const response = await apiFetch(path, {
38     auth: true,
39     timeoutMs: PROTECTED_DOWNLOAD_TIMEOUT_MS,
40     unauthenticatedMessage: '请先登录后再下载文件',
41   });
42   await ensureApiResponse(response, '下载文件失败');
43   const blob = await response.blob();
44   const header = response.headers.get('content-disposition') || '';
45   const matched = /filename="?([^"]+)"?/i.exec(header);
46   const anchor = document.createElement('a');
47   const url = URL.createObjectURL(blob);
48   anchor.href = url;
49   anchor.download = matched?.[1] || fallbackName;
50   anchor.click();
51   URL.revokeObjectURL(url);
52 }
53
54 // ===== 文件: front/src/api/tasks.ts =====
55 1 import { apiJson } from './client';
56 2 import type {
57 3   AnalysisTask,
```



```
front/src/api/tasks.ts
```

```
4   AnalysisTaskLog,
5   AnalysisTaskPage,
6   AnalysisTaskStats,
7   CreateTaskPayload,
8   TaskListQuery,
9   TaskTemplate,
10 } from '../workbench/tasks/types';
11
12 async function requestJson<T>(path: string, options: RequestInit = {}): Promise<T> {
13   return apiJson<T>(path, {
14     ...options,
15     auth: true,
16     unauthenticatedMessage: '请先登录后再操作任务中心',
17   });
18 }
19
20 function toQuery(params: Record<string, string | number | undefined | null>) {
21   const search = new URLSearchParams();
22   Object.entries(params).forEach(([key, value]) => {
23     if (value === undefined || value === null || value === '') return;
24     search.set(key, String(value));
25   });
26   const text = search.toString();
27   return text ? `?${text}` : '';
28 }
29
30 export function fetchTaskStats() {
31   return requestJson<AnalysisTaskStats>('/tasks/stats');
32 }
33
34 export function fetchTasks(query: TaskListQuery = {}) {
35   return requestJson<AnalysisTaskPage>(
36     `/tasks${toQuery({
37       page: query.page,
38       pageSize: query.pageSize,
39       status: query.status,
40       mode: query.mode,
41       datasetId: query.datasetId,
42       keyword: query.keyword,
43       createdFrom: query.createdFrom,
44       createdTo: query.createdTo,
45     })}`,
46   );
47 }
48
49 export function fetchTaskDetail(taskId: number) {
50   return requestJson<AnalysisTask>(`/tasks/${taskId}`);
51 }
52
53 export function createTask(payload: CreateTaskPayload) {
```

```
front/src/api/tasks.ts
```

```
54   return requestJson<AnalysisTask>(`/tasks`, {
55     method: 'POST',
56     body: JSON.stringify(payload),
57   });
58 }
59
60 export function updateTask(
61   taskId: number,
62   payload: { name?: string; mode?: string; params?: CreateTaskPayload['params'] },
63 ) {
64   return requestJson<AnalysisTask>(`/tasks/${taskId}`, {
65     method: 'PATCH',
66     body: JSON.stringify(payload),
67   });
68 }
69
70 export function startTask(taskId: number) {
71   return requestJson<AnalysisTask>(`/tasks/${taskId}/start`, { method: 'POST' });
72 }
73
74 export function cancelTask(taskId: number) {
75   return requestJson<AnalysisTask>(`/tasks/${taskId}/cancel`, { method: 'POST' });
76 }
77
78 export function retryTask(taskId: number) {
79   return requestJson<AnalysisTask>(`/tasks/${taskId}/retry`, { method: 'POST' });
80 }
81
82 export function cloneTask(taskId: number) {
83   return requestJson<AnalysisTask>(`/tasks/${taskId}/clone`, { method: 'POST' });
84 }
85
86 export function deleteTask(taskId: number) {
87   return requestJson<{ message: string }>(`/tasks/${taskId}`, { method: 'DELETE' });
88 }
89
90 export function fetchTaskLogs(taskId: number, signal?: AbortSignal) {
91   return requestJson<{ items: AnalysisTaskLog[]; total: number }>(`/tasks/${taskId}/logs`, { signal });
92 }
93
94 export function bulkTasks(taskIds: number[], action: 'retry' | 'cancel' | 'delete') {
95   return requestJson<{ message: string }>(`/tasks/bulk`, {
96     method: 'POST',
97     body: JSON.stringify({ taskIds, action }),
98   });
99 }
100
101 export function fetchTaskTemplates() {
102   return requestJson<{ items: TaskTemplate[] }>(`/tasks/templates`);
103 }
```

front/src/api/tasks.ts

```
104
105 export function createTaskTemplate(payload: {
106   name: string;
107   mode: string;
108   params?: CreateTaskPayload['params'];
109 }) {
110   return requestJson<TaskTemplate>('/tasks/templates', {
111     method: 'POST',
112     body: JSON.stringify(payload),
113   });
114 }
115
116 export function deleteTaskTemplate(templateId: number) {
117   return requestJson<{ message: string }>(`/tasks/templates/${templateId}`, { method: 'DELETE' });
118 }
119
120 export async function fetchDatasetOptions() {
121   const body = await apiJson<{
122     items?: Array<Record<string, unknown>>;
123   } | Array<Record<string, unknown>>>(<
124     '/datasets?page=1&pageSize=100',
125     {
126       auth: true,
127       unauthenticatedMessage: '请先登录后再操作任务中心',
128     },
129   );
130   const items = Array.isArray(body) ? body : Array.isArray(body.items) ? body.items : [];
131   return items.map((item: any) => ({
132     id: Number(item.id),
133     name: String(item.name ?? ''),
134     baseCount: Number(item.baseCount) || 0,
135     sampleCount: Number(item.sampleCount) || 0,
136     qualityStatus: String(item.qualityStatus ?? 'ready'),
137     qualityIssues: Array.isArray(item.qualityIssues) ? item.qualityIssues : [],
138     hasLabels: Boolean(item.hasLabels),
139     classCount: Number(item.classCount) || 0,
140   }));
141 }
142
// ===== 文件: front/src/appMeta.ts =====
1 /** 软件登记名称和版本的唯一来源，页面标题、页眉和截图材料统一引用。 */
2 export const SOFTWARE_NAME = '新材料数据分析——拓扑感知多核集成聚类分析系统';
3 export const SOFTWARE_SHORT_NAME = '新材料数据分析系统';
4 export const SOFTWARE_VERSION = 'V1.0';
5
// ===== 文件: front/src/components/MathFormula.tsx =====
1 import { useMemo } from 'react';
2 import katex from 'katex';
```

```
front/src/components/MathFormula.tsx
```

```
3
4 type Props = {
5   latex: string;
6   label: string;
7   displayMode?: boolean;
8   className?: string;
9 };
10
11 export function MathFormula({ latex, label, displayMode = false, className = '' }: Props) {
12   const markup = useMemo(() => {
13     try {
14       return katex.renderToString(latex, {
15         displayMode,
16         output: 'htmlAndMathml',
17         throwOnError: false,
18         strict: 'warn',
19         trust: false,
20       });
21     } catch {
22       return null;
23     }
24   }, [displayMode, latex]);
25
26   if (!markup) {
27     return <span className={`math-formula math-formula-fallback ${className}`} aria-label={label}>{label}</span>;
28   }
29
30   return (
31     <span
32       className={`math-formula ${displayMode ? 'math-formula-display' : ''} ${className}`}
33       aria-label={label}
34       dangerouslySetInnerHTML={{ __html: markup }}
35     />
36   );
37 }
38
```

```
// ===== 文件: front/src/components/SelectField.tsx =====
```

```
1 import { useEffect, useId, useLayoutEffect, useMemo, useRef, useState } from 'react';
2 import { createPortal } from 'react-dom';
3 import { ChevronDown } from 'lucide-react';
4
5 export type SelectFieldOption = {
6   value: string;
7   label: string;
8   disabled?: boolean;
9 };
10
11 type SelectFieldProps = {
12   value: string;
```

```
front/src/components/SelectField.tsx
```

```
13   options: SelectFieldOption[];
14   onChange: (value: string) => void;
15   label?: string;
16   ariaLabel?: string;
17   placeholder?: string;
18   disabled?: boolean;
19   className?: string;
20   size?: 'sm' | 'md';
21 };
22
23 type MenuPosition = {
24   top: number;
25   left: number;
26   width: number;
27   maxHeight: number;
28 };
29
30 const MENU_GAP = 6;
31 const MENU_MAX_HEIGHT = 260;
32 const VIEWPORT_PADDING = 12;
33
34 function getNextEnabledIndex(options: SelectFieldOption[], startIndex: number, direction: 1 | -1) {
35   if (options.length === 0) return -1;
36
37   for (let offset = 1; offset <= options.length; offset += 1) {
38     const index = (startIndex + offset * direction + options.length) % options.length;
39     if (!options[index].disabled) return index;
40   }
41
42   return -1;
43 }
44
45 export function SelectField({
46   value,
47   options,
48   onChange,
49   label,
50   ariaLabel,
51   placeholder = '请选择',
52   disabled = false,
53   className,
54   size = 'md',
55 }: SelectFieldProps) {
56   const reactId = useId();
57   const rootRef = useRef<HTMLDivElement | null>(null);
58   const triggerRef = useRef<HTMLButtonElement | null>(null);
59   const menuRef = useRef<HTMLDivElement | null>(null);
60   const [open, setOpen] = useState(false);
61   const [activeIndex, setActiveIndex] = useState(-1);
62   const [menuPosition, setMenuPosition] = useState<MenuPosition | null>(null);
```

```
front/src/components/SelectField.tsx
```

```
63
64   const selectedIndex = useMemo(
65     () => options.findIndex((option) => option.value === value),
66     [options, value],
67   );
68   const selectedOption = selectedIndex >= 0 ? options[selectedIndex] : null;
69   const listBoxId = `select-field-${reactId}-listbox`;
70   const labelId = label ? `select-field-${reactId}-label` : undefined;
71   const activeOption = activeIndex >= 0 ? options[activeIndex] : null;
72   const activeOptionId = activeOption ? `${listboxId}-option-${activeIndex}` : undefined;
73   const isDisabled = disabled || options.length === 0;
74
75   function getInitialActiveIndex() {
76     if (selectedIndex >= 0 && !options[selectedIndex].disabled) return selectedIndex;
77     return options.findIndex((option) => !option.disabled);
78   }
79
80   function updateMenuPosition() {
81     const trigger = triggerRef.current;
82     if (!trigger) return;
83
84     const rect = trigger.getBoundingClientRect();
85     const availableBelow = window.innerHeight - rect.bottom - MENU_GAP - VIEWPORT_PADDING;
86     const availableAbove = rect.top - MENU_GAP - VIEWPORT_PADDING;
87     const shouldOpenAbove = availableBelow < 160 && availableAbove > availableBelow;
88     const availableHeight = shouldOpenAbove ? availableAbove : availableBelow;
89     const maxHeight = Math.min(MENU_MAX_HEIGHT, Math.max(120, availableHeight));
90
91     setMenuPosition({
92       top: shouldOpenAbove
93         ? Math.max(VIEWPORT_PADDING, rect.top - MENU_GAP - maxHeight)
94         : rect.bottom + MENU_GAP,
95       left: Math.max(VIEWPORT_PADDING, rect.left),
96       width: rect.width,
97       maxHeight,
98     });
99   }
100
101   function openMenu() {
102     if (isDisabled) return;
103     setActiveIndex(getInitialActiveIndex());
104     setOpen(true);
105   }
106
107   function closeMenu() {
108     setOpen(false);
109     setMenuPosition(null);
110   }
111
112   function commitOption(option: SelectFieldOption) {
```

```
front/src/components/SelectField.tsx
```

```
113     if (option.disabled) return;
114     if (option.value !== value) {
115         onChange(option.value);
116     }
117     closeMenu();
118     triggerRef.current?.focus();
119 }
120
121 function moveActiveIndex(direction: 1 | -1) {
122     const baseIndex = activeIndex >= 0 ? activeIndex : getInitialActiveIndex();
123     const nextIndex = getNextEnabledIndex(options, baseIndex, direction);
124     if (nextIndex >= 0) setActiveIndex(nextIndex);
125 }
126
127 useEffect(() => {
128     if (!open) return;
129     updateMenuPosition();
130 }, [open, options.length]);
131
132 useEffect(() => {
133     if (!open) return undefined;
134
135     function handlePointerDown(event: PointerEvent) {
136         const target = event.target as Node;
137         if (rootRef.current?.contains(target) || menuRef.current?.contains(target)) return;
138         closeMenu();
139     }
140
141     function handleViewportChange() {
142         updateMenuPosition();
143     }
144
145     document.addEventListener('pointerdown', handlePointerDown);
146     window.addEventListener('resize', handleViewportChange);
147     window.addEventListener('scroll', handleViewportChange, true);
148
149     return () => {
150         document.removeEventListener('pointerdown', handlePointerDown);
151         window.removeEventListener('resize', handleViewportChange);
152         window.removeEventListener('scroll', handleViewportChange, true);
153     };
154 }, [open]);
155
156 useEffect(() => {
157     if (!open) return;
158     setActiveIndex(getInitialActiveIndex());
159 }, [open, value, options]);
160
161 const rootClassName = [
162     'select-field',
```

```
front/src/components/SelectField.tsx
```

```
163     `select-field-${size}`,
164     open ? 'is-open' : '',
165     isDisabled ? 'is-disabled' : '',
166     selectedOption ? '' : 'is-empty',
167     className ?? '',
168   ]
169   .filter(Boolean)
170   .join(' ');
171
172   const menu = open && menuPosition
173     ? createPortal(
174       <div
175         ref={menuRef}
176         id={listboxId}
177         className="select-field-menu"
178         role="listbox"
179         aria-label={ariaLabel ?? label}
180         style={{
181           top: menuPosition.top,
182           left: menuPosition.left,
183           width: menuPosition.width,
184           maxHeight: menuPosition.maxHeight,
185         }}
186       >
187         {options.map((option, index) => (
188           <div
189             key={option.value || `empty-${index}`}
190             id={` ${listboxId}-option-${index}`}
191             className={[
192               'select-field-option',
193               option.value === value ? 'is-selected' : '',
194               activeIndex === index ? 'is-active' : '',
195               option.disabled ? 'is-disabled' : '',
196             ]}
197             .filter(Boolean)
198             .join(' ')}
199             role="option"
200             aria-selected={option.value === value}
201             aria-disabled={option.disabled || undefined}
202             title={option.label}
203             onMouseEnter={() => {
204               if (!option.disabled) setActiveIndex(index);
205             }}
206             onClick={() => commitOption(option)}
207           >
208             {option.label}
209           </div>
210         ))}
211       </div>,
212       document.body,
```


front/src/components/SelectField.tsx

```
213     )
214     : null;
215
216   return (
217     <div ref={rootRef} className={rootClassName}>
218       {label ? (
219         <span id={labelId} className="select-field-label">
220           {label}
221         </span>
222       ) : null}
223       <button
224         ref={triggerRef}
225         type="button"
226         className="select-field-trigger"
227         role="combobox"
228         aria-label={ariaLabel ?? label}
229         aria-labelledby={ariaLabel ? undefined : labelId}
230         aria-haspopup="listbox"
231         aria-expanded={open}
232         aria-controls={open ? listBoxId : undefined}
233         aria-activedescendant={open ? activeOptionId : undefined}
234         disabled={isDisabled}
235         onClick={() => {
236           if (open) {
237             closeMenu();
238           } else {
239             openMenu();
240           }
241         }}
242         onKeyDown={(event) => {
243           if (isDisabled) return;
244
245           if (event.key === 'ArrowDown') {
246             event.preventDefault();
247             if (!open) {
248               openMenu();
249               return;
250             }
251             moveActiveIndex(1);
252           }
253
254           if (event.key === 'ArrowUp') {
255             event.preventDefault();
256             if (!open) {
257               openMenu();
258               return;
259             }
260             moveActiveIndex(-1);
261           }
262         }}
```

```
front/src/components/SelectField.tsx
```

```
263         if (event.key === 'Enter' || event.key === ' ') {
264             event.preventDefault();
265             if (!open) {
266                 openMenu();
267                 return;
268             }
269             const option = options[activeIndex];
270             if (option) commitOption(option);
271         }
272
273         if (event.key === 'Escape') {
274             event.preventDefault();
275             closeMenu();
276         }
277     }}
278     >
279     <span className="select-field-trigger-value" title={selectedOption?.label}>
280         {selectedOption?.label ?? placeholder}
281     </span>
282     <ChevronDown size={16} aria-hidden="true" />
283 </button>
284     {menu}
285 </div>
286 );
287 }
288
```

```
// ===== 文件: front/src/dashboard/navigation.ts =====
```

```
1 import type { LucideIcon } from 'lucide-react';
2 import {
3     Activity,
4     BarChart3,
5     Boxes,
6     ClipboardCheck,
7     ClipboardList,
8     Database,
9     Download,
10    FileText,
11    History,
12    Images,
13    LayoutDashboard,
14    Network,
15    ShieldCheck,
16    SlidersHorizontal,
17 } from 'lucide-react';
18
19 export type DashboardGroupKey = 'workspace' | 'data' | 'model' | 'output' | 'audit';
20
21 export type DashboardNavItem = {
22     label: string;
```

```
front/src/dashboard/navigation.ts
```

```
23   path: string;
24   icon: LucideIcon;
25   group: DashboardGroupKey;
26 };
27
28 export const dashboardGroups: Array<{ key: DashboardGroupKey; label: string }> = [
29   { key: 'workspace', label: '工作区' },
30   { key: 'data', label: '数据中心' },
31   { key: 'model', label: '算法流程' },
32   { key: 'output', label: '结果中心' },
33   { key: 'audit', label: '审计记录' },
34 ];
35
36 export const dashboardNavItems: DashboardNavItem[] = [
37   { label: '分析工作台', path: '/workbench/analysis', icon: LayoutDashboard, group: 'workspace' },
38   { label: '任务中心', path: '/workbench/tasks', icon: ClipboardList, group: 'workspace' },
39   { label: '数据管理', path: '/workbench/datasets', icon: Database, group: 'data' },
40   { label: '数据质量检查', path: '/workbench/data-quality', icon: ShieldCheck, group: 'data' },
41   { label: '数据版本记录', path: '/workbench/dataset-versions', icon: History, group: 'data' },
42   { label: 'CA 协关联矩阵', path: '/workbench/ca-matrix', icon: Boxes, group: 'model' },
43   { label: '核函数配置', path: '/workbench/kernel-config', icon: SlidersHorizontal, group: 'model' },
44   { label: '多核相似性学习', path: '/workbench/mkl', icon: Network, group: 'model' },
45   { label: '性能评估', path: '/workbench/evaluation', icon: ClipboardCheck, group: 'output' },
46   { label: '可视化展示', path: '/workbench/visualization', icon: Images, group: 'output' },
47   { label: '结果分析', path: '/workbench/results', icon: BarChart3, group: 'output' },
48   { label: '结果导出', path: '/workbench/export', icon: Download, group: 'output' },
49   { label: '分析报告', path: '/workbench/reports', icon: FileText, group: 'output' },
50   { label: '运行日志', path: '/workbench/logs', icon: Activity, group: 'audit' },
51 ];
52
53 export const defaultWorkbenchPath = '/workbench/analysis';
54
55 export const legacyWorkbenchRedirects: Record<string, string> = {};
56
57 export function getActiveWorkbenchSection(pathname: string) {
58   return dashboardNavItems.find((item) => item.path === pathname)?.label ?? null;
59 }
60
61 // ===== 文件: front/src/landing.css =====
62 1 /* Landing page styled after new-api classic composition while keeping OMELET-specific content. */
63 2
64 3 .landing-page {
65 4   min-height: 100vh;
66 5   font-family: Lato, 'Helvetica Neue', Arial, Helvetica, 'Microsoft YaHei', sans-serif;
67 6   /* 对齐 new-api: 不设额外纯白背景, 继承 body 的 var(--app-bg)=#f8fafc 统一底色 */
68 7   background: var(--app-bg);
69 8   color: var(--app-text-primary);
70 9 }
71 10
```

```
front/src/landing.css
```

```
11 .landing-page.theme-dark {
12   background:
13     radial-gradient(circle at 50% 20%, rgba(129, 140, 248, 0.2), transparent 24%),
14     radial-gradient(circle at 38% 36%, rgba(45, 212, 191, 0.14), transparent 20%),
15     #2c3e50;
16   color: #e2e8f0;
17 }
18
19 .landing-page.theme-dark .landing-topbar {
20   background: transparent;
21   backdrop-filter: blur(16px);
22   -webkit-backdrop-filter: blur(16px);
23   border-bottom: 0;
24 }
25
26 .landing-page.theme-dark .hero-subtitle,
27 .landing-page.theme-dark .landing-section-header p,
28 .landing-page.theme-dark .capability-card p,
29 .landing-page.theme-dark .landing-about-copy p,
30 .landing-page.theme-dark .landing-workflow-heading p,
31 .landing-page.theme-dark .workflow-stage p {
32   color: rgba(226, 232, 240, 0.72);
33 }
34
35 .landing-page.theme-dark .base-url-row,
36 .landing-page.theme-dark .capability-card,
37 .landing-page.theme-dark .landing-about-card,
38 .landing-page.theme-dark .stack-chip {
39   background: rgba(15, 23, 42, 0.62);
40   border-color: rgba(148, 163, 184, 0.14);
41   color: #e2e8f0;
42 }
43
44 .landing-page.theme-dark .landing-link-button {
45   background: var(--app-accent);
46   color: #fff;
47 }
48
49 .landing-page.theme-dark .base-url-copy {
50   color: #e2e8f0;
51   background: transparent;
52 }
53
54 .landing-page.theme-dark .base-url-copy:hover,
55 .landing-page.theme-dark .base-url-copy:focus-visible {
56   color: #f8fafc;
57   background: rgba(148, 163, 184, 0.16);
58 }
59
60 .landing-main {
```

front/src/landing.css

```
61   position: relative;
62   z-index: 1;
63 }
64
65 .landing-topbar {
66   position: sticky;
67   top: 0;
68   z-index: 50;
69   backdrop-filter: blur(16px);
70   -webkit-backdrop-filter: blur(16px);
71   background: transparent;
72   border-bottom: 0;
73 }
74
75 .landing-topbar-inner {
76   width: 100%;
77   min-height: 64px;
78   padding: 0 16px;
79   display: grid;
80   grid-template-columns: auto 1fr;
81   align-items: center;
82   gap: 24px;
83 }
84
85 .landing-brand {
86   display: flex;
87   align-items: center;
88   gap: 10px;
89   min-width: 0;
90 }
91
92 .landing-brand .brand-logo {
93   display: flex;
94   align-items: center;
95   justify-content: center;
96   width: 32px;
97   height: 32px;
98   border-radius: 10px;
99   overflow: hidden;
100  background: rgba(255, 255, 255, 0.72);
101  box-shadow: 0 10px 24px -16px rgba(0, 98, 214, 0.8);
102 }
103
104 .brand-logo-image {
105   width: 100%;
106   height: 100%;
107   display: block;
108   object-fit: cover;
109   object-position: center;
110   transform: scale(1.18);
```

front/src/landing.css

```
111 }
112
113 .landing-brand-copy strong {
114   display: block;
115   /* 对齐 new-api: text-lg (18px) font-semibold */
116   font-size: 18px;
117   font-weight: 600;
118 }
119
120 .landing-version-pill {
121   justify-self: end;
122   min-width: 72px;
123   height: 28px;
124   padding: 0 14px;
125   border: 1px solid rgba(64, 158, 255, 0.24);
126   border-radius: 999px;
127   display: inline-flex;
128   align-items: center;
129   justify-content: center;
130   background: rgba(64, 158, 255, 0.12);
131   box-shadow: 0 3px 8px -6px rgba(51, 126, 204, 0.45);
132   color: #245f9f;
133   font-size: 13px;
134   font-weight: 600;
135   line-height: 1;
136   white-space: nowrap;
137 }
138
139 .landing-page.theme-dark .landing-version-pill {
140   border-color: rgba(121, 187, 255, 0.36);
141   background: rgba(64, 158, 255, 0.2);
142   box-shadow: 0 3px 8px -6px rgba(0, 0, 0, 0.42);
143   color: #eaf4ff;
144 }
145
146 .landing-hero {
147   position: relative;
148   min-height: calc(100vh - 64px);
149   display: flex;
150   align-items: flex-start;
151   justify-content: center;
152   padding: 92px 24px 36px;
153 }
154
155 .blur-ball {
156   position: absolute;
157   width: 360px;
158   height: 360px;
159   border-radius: 50%;
160   filter: blur(120px);
```

front/src/landing.css

```
161   pointer-events: none;
162   z-index: -1;
163 }
164
165 .blur-ball-indigo {
166   top: 40px;
167   left: 50%;
168   transform: translateX(-50%);
169   background: #6366f1;
170   opacity: 0.25;
171 }
172
173 .blur-ball-teal {
174   top: 200px;
175   left: 30%;
176   background: #14b8a6;
177   opacity: 0.2;
178 }
179
180 .landing-hero-inner {
181   position: relative;
182   z-index: 1;
183   width: min(920px, 100%);
184   display: flex;
185   flex-direction: column;
186   align-items: center;
187   text-align: center;
188 }
189
190 .landing-hero-copy {
191   max-width: 760px;
192 }
193
194 .landing-hero h1 {
195   margin: 0;
196   /* 对齐 new-api: text-4xl md:text-5xl lg:text-6xl xl:text-7xl (36/48/60/72px) */
197   font-size: clamp(36px, 2vw + 30px, 72px);
198   font-weight: 700;
199   line-height: 1.25; /* leading-tight */
200   letter-spacing: 0.025em; /* 中文 tracking-wide */
201   color: var(--text-primary);
202 }
203
204 @media (min-width: 768px) {
205   .landing-hero h1 {
206     font-size: clamp(48px, 2vw + 36px, 72px);
207   }
208 }
209
210 @media (min-width: 1024px) {
```

```
front/src/landing.css
```

```
211   .landing-hero h1 {
212     font-size: clamp(60px, 2vw + 42px, 72px);
213   }
214 }
215
216 @media (min-width: 1280px) {
217   .landing-hero h1 {
218     font-size: 72px;
219   }
220 }
221
222 .landing-page.theme-dark .landing-hero h1,
223 .landing-page.theme-dark .landing-section-header h2,
224 .landing-page.theme-dark .capability-card h3,
225 .landing-page.theme-dark .landing-about-copy h2,
226 .landing-page.theme-dark .landing-workflow-heading h2,
227 .landing-page.theme-dark .workflow-stage h3 {
228   color: #f8fafc;
229 }
230
231 @keyframes sweep-shine {
232   0% {
233     background-position: 200% 0;
234   }
235
236   100% {
237     background-position: -200% 0;
238   }
239 }
240
241 .shine-text {
242   background: linear-gradient(
243     90deg,
244     currentColor 0%,
245     currentColor 38%,
246     rgba(255, 255, 255, 0.95) 50%,
247     currentColor 62%,
248     currentColor 100%
249   );
250   background-size: 200% 100%;
251   -webkit-background-clip: text;
252   background-clip: text;
253   -webkit-text-fill-color: transparent;
254   color: var(--text-primary);
255   animation: sweep-shine 4s linear infinite;
256 }
257
258 .hero-subtitle {
259   margin: 18px auto 0;
260   max-width: 640px;
```


front/src/landing.css

```
261   color: var(--app-text-regular);
262   /* 对齐 new-api: text-base md:text-lg lg:text-xl (16/18/20px) */
263   font-size: clamp(16px, 0.6vw + 14px, 20px);
264   line-height: 1.65;
265 }
266
267 @keyframes base-url-endpoint-enter {
268   from {
269     opacity: 0;
270     filter: blur(2px);
271   }
272
273   to {
274     opacity: 1;
275     filter: blur(0);
276   }
277 }
278
279 @keyframes auth-page-enter {
280   from {
281     opacity: 0;
282     filter: blur(6px);
283   }
284
285   to {
286     opacity: 1;
287     filter: blur(0);
288   }
289 }
290
291 @keyframes auth-card-enter {
292   from {
293     opacity: 0;
294     transform: translateY(10px) scale(0.985);
295   }
296
297   to {
298     opacity: 1;
299     transform: none;
300   }
301 }
302
303 @keyframes auth-submit-spin {
304   to {
305     transform: rotate(360deg);
306   }
307 }
308
309 .base-url-row {
310   margin-top: 24px;
```

```
front/src/landing.css
```

```
311 position: relative;
312 z-index: 1;
313 max-width: 448px;
314 width: 100%;
315 min-height: 40px;
316 padding: 6px 8px 6px 16px;
317 display: grid;
318 grid-template-columns: minmax(0, 1fr) auto 28px;
319 align-items: center;
320 gap: 8px;
321 border-radius: 999px;
322 /* 对齐 Semi-UI Input: 无可见边框, 靠底色差(比页面浅灰略深)定义边界 */
323 border: 1px solid transparent;
324 background: rgba(0, 0, 0, 0.04);
325 font-family: Inter, -apple-system, BlinkMacSystemFont, 'Segoe UI', 'PingFang SC', 'Hiragino Sans GB', 'Microsoft
    YaHei', 'Helvetica Neue', Helvetica, Arial, sans-serif;
326 line-height: 1;
327 text-align: left;
328 }
329
330 .base-url-address {
331 min-width: 0;
332 overflow: visible;
333 text-overflow: clip;
334 white-space: nowrap;
335 color: var(--app-text-primary);
336 font-size: 15px;
337 font-weight: 400;
338 }
339
340 .base-url-endpoint {
341 min-width: 0;
342 overflow: visible;
343 text-overflow: clip;
344 white-space: nowrap;
345 color: #0064fa;
346 font-size: 14px;
347 font-style: normal;
348 font-weight: 600;
349 line-height: 1.25;
350 animation: base-url-endpoint-enter 0.28s cubic-bezier(0.22, 1, 0.36, 1) both;
351 will-change: opacity, filter;
352 }
353
354 .base-url-copy {
355 width: 28px;
356 height: 28px;
357 flex: 0 0 28px;
358 border: 0;
359 border-radius: 999px;
```

front/src/landing.css

```
360     display: inline-flex;
361     align-items: center;
362     justify-content: center;
363     background: transparent;
364     color: var(--app-text-muted);
365     cursor: pointer;
366     transition:
367         background-color 0.16s ease,
368         color 0.16s ease;
369 }
370
371 .base-url-copy-icon {
372     width: 14px;
373     height: 14px;
374     display: block;
375     opacity: 0.52;
376     transition: opacity 0.16s ease;
377 }
378
379 .base-url-copy:hover,
380 .base-url-copy:focus-visible {
381     color: var(--app-text-primary);
382     background: rgba(15, 23, 42, 0.08);
383 }
384
385 .base-url-copy:hover .base-url-copy-icon,
386 .base-url-copy:focus-visible .base-url-copy-icon {
387     opacity: 0.78;
388 }
389
390 .base-url-copy:focus-visible {
391     outline: 2px solid rgba(64, 158, 255, 0.32);
392     outline-offset: 2px;
393     box-shadow: 0 0 0 2px rgba(64, 158, 255, 0.2);
394 }
395
396 .hero-buttons {
397     margin-top: 28px;
398     display: flex;
399     align-items: center;
400     justify-content: center;
401     gap: 12px;
402     flex-wrap: wrap;
403 }
404
405 .hero-btn,
406 .btn.hero-btn {
407     min-width: 112px;
408     height: 44px;
409     padding: 0 26px;
```

```
front/src/landing.css
```

```
410   border-radius: 999px;
411   /* 对齐 new-api: large 按钮 16px/600 (双类选择器优先级高于 .btn, 覆盖 13px) */
412   font-size: 16px;
413   font-weight: 600;
414 }
415
416 .hero-btn {
417   transition: transform 0.2s ease, box-shadow 0.2s ease;
418 }
419
420 .hero-btn:active {
421   transform: translateY(1px) scale(0.985);
422 }
423
424 /* hero 登录按钮蓝色改为 #0062D6 */
425 .hero-btn.btn-primary {
426   background: #0062D6;
427   border-color: #0062D6;
428 }
429 .hero-btn.btn-primary:hover {
430   background: #0050b0;
431   border-color: #0050b0;
432 }
433
434 .hero-btn-icon {
435   width: 16px;
436   height: 16px;
437   display: block;
438   flex: 0 0 16px;
439   object-fit: contain;
440 }
441
442 .landing-capability-strip {
443   margin-top: 84px;
444   width: min(860px, 100%);
445   display: flex;
446   flex-direction: column;
447   align-items: center;
448 }
449
450 .landing-capability-strip h2 {
451   margin: 0 0 28px;
452   color: var(--app-text-muted);
453   font-size: 24px;
454   font-weight: 400;
455   line-height: 1.3;
456   letter-spacing: 0;
457 }
458
459 .capability-mark-grid {
```

front/src/landing.css

```
460   width: min(960px, 100%);
461   display: grid;
462   grid-template-columns: repeat(8, minmax(72px, 1fr));
463   align-items: start;
464   justify-items: center;
465   gap: 18px;
466 }
467
468 .capability-mark {
469   min-width: 0;
470   min-height: 88px;
471   display: inline-flex;
472   flex-direction: column;
473   align-items: center;
474   justify-content: flex-start;
475   gap: 16px;
476   position: relative;
477   color: var(--app-text-primary);
478 }
479
480 .capability-mark-icon {
481   width: 52px;
482   height: 52px;
483   flex: 0 0 52px;
484   display: block;
485   object-fit: contain;
486 }
487
488 .capability-mark-strong {
489   position: absolute;
490   width: 1px;
491   height: 1px;
492   padding: 0;
493   margin: -1px;
494   overflow: hidden;
495   clip: rect(0, 0, 0, 0);
496   white-space: nowrap;
497   border: 0;
498 }
499
500 .capability-mark-label {
501   position: static;
502   width: auto;
503   height: auto;
504   padding: 0;
505   margin: 0;
506   overflow: visible;
507   clip: auto;
508   white-space: nowrap;
509   border: 0;
```

```
front/src/landing.css
```

```
510   color: var(--app-text-primary);
511   font-size: 14px;
512   font-weight: 600;
513   line-height: 1.2;
514   text-align: center;
515 }
516
517 .capability-mark-blue {
518   color: #409eff;
519 }
520
521 .capability-mark-indigo {
522   color: #4f46e5;
523 }
524
525 .capability-mark-cyan {
526   color: #0891b2;
527 }
528
529 .capability-mark-violet {
530   color: #7c3aed;
531 }
532
533 .capability-mark-green {
534   color: #047857;
535 }
536
537 .capability-mark-orange {
538   color: #ea580c;
539 }
540
541 .capability-mark-slate {
542   color: #2c3e50;
543 }
544
545 .landing-section {
546   padding: 0 24px 32px;
547 }
548
549 .landing-section-inner {
550   width: min(1200px, 100%);
551   margin: 0 auto;
552 }
553
554 .landing-section-header {
555   max-width: 720px;
556   margin: 0 auto 24px;
557   text-align: center;
558 }
559
```

front/src/landing.css

```
560 .landing-section-kicker {
561   display: inline-block;
562   margin-bottom: 12px;
563   padding: 6px 12px;
564   border-radius: 999px;
565   background: rgba(64, 158, 255, 0.1);
566   color: var(--app-accent-dark);
567   font-size: 12px;
568   font-weight: 700;
569 }
570
571 .landing-section-header h2,
572 .landing-about-copy h2 {
573   margin: 0;
574   font-size: clamp(1.8rem, 3vw, 2.5rem);
575   line-height: 1.15;
576   color: var(--text-primary);
577 }
578
579 .landing-section-header p,
580 .landing-about-copy p {
581   margin: 14px 0 0;
582   color: var(--app-text-regular);
583   font-size: 15px;
584   line-height: 1.8;
585 }
586
587 .capability-grid {
588   display: grid;
589   grid-template-columns: repeat(3, minmax(0, 1fr));
590   gap: 16px;
591 }
592
593 .capability-card,
594 .landing-about-card {
595   border: 1px solid rgba(15, 23, 42, 0.08);
596   border-radius: 24px;
597   background: rgba(255, 255, 255, 0.72);
598   backdrop-filter: blur(18px);
599   -webkit-backdrop-filter: blur(18px);
600   box-shadow: 0 24px 60px -42px rgba(15, 23, 42, 0.4);
601 }
602
603 .capability-card {
604   min-height: 220px;
605   padding: 22px;
606 }
607
608 .capability-icon {
609   width: 12px;
```

front/src/landing.css

```
610 height: 12px;
611 border-radius: 999px;
612 display: inline-block;
613 flex: none;
614 background: radial-gradient(
615   circle at 36% 32%,
616   #7ffe7 0 12%,
617   #35f4c1 34%,
618   #10d9a6 68%,
619   #08ad80 100%
620 );
621 box-shadow:
622   0 0 12px rgba(16, 217, 166, 0.58),
623   0 0 26px rgba(16, 217, 166, 0.2),
624   0 6px 14px -8px rgba(6, 95, 70, 0.72);
625 }
626
627 .capability-card h3 {
628   margin: 26px 0 10px;
629   font-size: 19px;
630   line-height: 1.25;
631   color: var(--text-primary);
632 }
633
634 .capability-card p {
635   margin: 0;
636   color: var(--app-text-regular);
637   font-size: 14px;
638   line-height: 1.8;
639 }
640
641 .landing-workflow-section {
642   padding: 0 24px 20px;
643 }
644
645 .landing-workflow-card {
646   position: relative;
647   overflow: visible;
648   padding: 4px 0 0;
649   border: 0;
650   border-radius: 0;
651   background: transparent;
652   box-shadow: none;
653 }
654
655 .landing-workflow-heading {
656   margin: 0 2px 8px;
657   display: flex;
658   align-items: baseline;
659   justify-content: space-between;
```


front/src/landing.css

```
660   gap: 20px;
661 }
662
663 .landing-workflow-heading > div {
664   min-width: 0;
665   display: flex;
666   align-items: baseline;
667   gap: 10px;
668 }
669
670 .landing-workflow-kicker {
671   flex: 0 0 auto;
672   padding: 3px 8px;
673   border: 1px solid rgba(64, 158, 255, 0.16);
674   border-radius: 999px;
675   background: rgba(64, 158, 255, 0.08);
676   color: var(--app-accent-dark);
677   font-size: 10px;
678   font-weight: 800;
679   letter-spacing: 0.08em;
680   line-height: 1.2;
681 }
682
683 .landing-workflow-heading h2 {
684   margin: 0;
685   color: var(--app-text-primary);
686   font-size: 17px;
687   line-height: 1.3;
688 }
689
690 .landing-workflow-heading p {
691   margin: 0;
692   color: var(--app-text-muted);
693   font-size: 12px;
694   line-height: 1.5;
695   text-align: right;
696 }
697
698 .landing-workflow-timeline {
699   position: relative;
700   isolation: isolate;
701   min-height: 240px;
702   margin: 8px 0 0;
703   padding: 0;
704   list-style: none;
705 }
706
707 .landing-workflow-timeline::before {
708   display: none;
709   content: '';
```

```
front/src/landing.css
```

```
710 }
711
712 .workflow-route {
713   position: absolute;
714   z-index: 0;
715   inset: 0;
716   width: 100%;
717   height: 100%;
718   overflow: visible;
719   pointer-events: none;
720 }
721
722 .workflow-route-path {
723   fill: none;
724   stroke: url('#workflow-route-gradient');
725   stroke-width: 4;
726   stroke-linecap: round;
727   stroke-linejoin: round;
728   filter: drop-shadow(0 0 7px rgba(45, 212, 191, 0.36));
729   vector-effect: non-scaling-stroke;
730 }
731
732 .workflow-stage {
733   position: absolute;
734   z-index: 1;
735   top: 0;
736   left: var(--workflow-stage-x);
737   min-width: 0;
738   width: 20%;
739   height: 240px;
740   transform: translateX(-50%);
741   text-align: center;
742 }
743
744 .workflow-stage:first-of-type,
745 .workflow-stage:last-of-type {
746   width: 18%;
747 }
748
749 .workflow-stage-copy {
750   position: absolute;
751   left: 50%;
752   width: min(200px, calc(100% - 8px));
753   padding: 8px 10px;
754   border: 1px solid rgba(148, 163, 184, 0.16);
755   border-radius: 12px;
756   background: rgba(255, 255, 255, 0.64);
757   box-shadow: 0 6px 8px -8px rgba(15, 23, 42, 0.8);
758   color: var(--app-text-primary);
759   transform: translateX(-50%);
```

front/src/landing.css

```
760   transition: transform 180ms cubic-bezier(0.22, 1, 0.36, 1), border-color 180ms ease, box-shadow 180ms ease;
761 }
762
763 .workflow-stage.is-upper .workflow-stage-copy {
764   top: 0;
765 }
766
767 .workflow-stage.is-lower .workflow-stage-copy {
768   bottom: 0;
769 }
770
771 .workflow-stage-copy::after {
772   position: absolute;
773   left: 50%;
774   width: 1px;
775   height: 18px;
776   background: linear-gradient(180deg, rgba(64, 158, 255, 0.2), rgba(45, 212, 191, 0.82));
777   content: '';
778   transform: translateX(-50%);
779 }
780
781 .workflow-stage.is-upper .workflow-stage-copy::after {
782   bottom: -19px;
783 }
784
785 .workflow-stage.is-lower .workflow-stage-copy::after {
786   top: -19px;
787   transform: translateX(-50%) rotate(180deg);
788 }
789
790 .workflow-stage-order {
791   display: block;
792   margin-bottom: 2px;
793   color: var(--app-accent-dark);
794   font-size: 9px;
795   font-weight: 800;
796   letter-spacing: 0.12em;
797   line-height: 1.2;
798 }
799
800 .workflow-stage h3 {
801   margin: 0;
802   color: var(--app-text-primary);
803   font-size: 12px;
804   line-height: 1.35;
805 }
806
807 .workflow-stage p {
808   margin: 4px 0 0;
809   color: var(--app-text-muted);
```

```
front/src/landing.css
```

```
810     font-size: 10px;
811     line-height: 1.45;
812 }
813
814 .workflow-stage-node {
815     position: absolute;
816     top: var(--workflow-node-y);
817     left: 50%;
818     width: 40px;
819     height: 40px;
820     display: grid;
821     place-items: center;
822     border: 1px solid rgba(64, 158, 255, 0.24);
823     border-radius: 50%;
824     background: linear-gradient(135deg, #fafdff, #d8f0ff);
825     box-shadow:
826         0 0 0 6px rgba(248, 250, 252, 0.86),
827         0 8px 18px -12px rgba(24, 119, 216, 0.82),
828         0 0 16px rgba(45, 212, 191, 0.2);
829     color: #2074c8;
830     transform: translate(-50%, -50%);
831     transition: transform 180ms cubic-bezier(0.22, 1, 0.36, 1), box-shadow 180ms ease, background-color 180ms ease;
832 }
833
834 .workflow-stage.is-lower .workflow-stage-node {
835     border-color: rgba(16, 185, 129, 0.26);
836     background: linear-gradient(135deg, #f7fffd, #cef7eb);
837     color: #0c9876;
838 }
839
840 .landing-page.theme-dark .workflow-stage-node {
841     border-color: rgba(96, 165, 250, 0.42);
842     background: linear-gradient(135deg, rgba(30, 41, 59, 0.98), rgba(30, 92, 126, 0.9));
843     box-shadow:
844         0 0 0 5px rgba(30, 41, 59, 0.9),
845         0 8px 18px -12px rgba(96, 165, 250, 0.9),
846         0 0 16px rgba(45, 212, 191, 0.24);
847     color: #7dd3fc;
848 }
849
850 .landing-page.theme-dark .workflow-stage-copy {
851     border-color: rgba(148, 163, 184, 0.16);
852     background: rgba(15, 23, 42, 0.46);
853     box-shadow: 0 6px 8px -8px rgba(2, 6, 23, 0.92);
854 }
855
856 .landing-page.theme-dark .workflow-stage.is-lower .workflow-stage-node {
857     border-color: rgba(45, 212, 191, 0.44);
858     background: linear-gradient(135deg, rgba(20, 57, 66, 0.98), rgba(15, 91, 77, 0.9));
859     color: #5eead4;
```

```
front/src/landing.css
```

```
860 }
861
862 .workflow-stage:is(:hover, :focus-within) .workflow-stage-copy {
863   border-color: rgba(45, 212, 191, 0.36);
864   box-shadow: 0 8px 8px -8px rgba(14, 116, 144, 0.65);
865   transform: translateX(-50%) translateY(-3px);
866 }
867
868 .workflow-stage:is(:hover, :focus-within) .workflow-stage-node {
869   transform: translate(-50%, -50%) translateY(-2px) scale(1.05);
870   box-shadow:
871     0 0 0 6px rgba(248, 250, 252, 0.86),
872     0 12px 22px -12px rgba(24, 119, 216, 0.9),
873     0 0 22px rgba(45, 212, 191, 0.3);
874 }
875
876 .landing-page.theme-dark .workflow-stage:is(:hover, :focus-within) .workflow-stage-node {
877   box-shadow:
878     0 0 0 6px rgba(30, 41, 59, 0.92),
879     0 12px 22px -12px rgba(96, 165, 250, 0.94),
880     0 0 22px rgba(45, 212, 191, 0.34);
881 }
882
883 .stack-cloud {
884   margin-top: 28px;
885   display: flex;
886   align-items: center;
887   justify-content: center;
888   gap: 10px;
889   flex-wrap: wrap;
890 }
891
892 .stack-chip {
893   padding: 6px 12px;
894   border-radius: 999px;
895   border: 0;
896   background: transparent;
897   font-size: 13px;
898   font-weight: 700;
899   color: var(--app-text-primary);
900 }
901
902 .landing-section-compact {
903   padding-bottom: 64px;
904 }
905
906 .landing-about-card {
907   padding: 28px;
908   display: flex;
909   align-items: center;
```

```
front/src/landing.css
```

```
910     justify-content: space-between;
911     gap: 24px;
912 }
913
914 .landing-about-copy {
915     max-width: 760px;
916 }
917
918 .landing-link-button {
919     flex: 0 0 auto;
920     height: 44px;
921     padding: 0 20px;
922     border: 0;
923     border-radius: 999px;
924     display: inline-flex;
925     align-items: center;
926     gap: 8px;
927     background: var(--app-accent);
928     color: #fff;
929     cursor: pointer;
930     font-size: 14px;
931     font-weight: 700;
932     box-shadow: 0 6px 14px -10px rgba(64, 158, 255, 0.72);
933     transition:
934         transform 180ms cubic-bezier(0.22, 1, 0.36, 1),
935         background-color 180ms cubic-bezier(0.22, 1, 0.36, 1),
936         box-shadow 220ms cubic-bezier(0.22, 1, 0.36, 1);
937 }
938
939 .landing-link-button:hover,
940 .landing-link-button:focus-visible {
941     background: var(--app-accent-dark);
942     box-shadow: 0 12px 20px -10px rgba(64, 158, 255, 0.7);
943     transform: translateY(-2px);
944 }
945
946 .landing-link-button:focus-visible {
947     outline: 2px solid rgba(64, 158, 255, 0.4);
948     outline-offset: 3px;
949 }
950
951 .landing-link-button:active {
952     box-shadow: 0 5px 10px -9px rgba(64, 158, 255, 0.72);
953     transform: translateY(0) scale(0.98);
954 }
955
956 .landing-link-button:disabled {
957     color: rgba(255, 255, 255, 0.9);
958     background: #6aa9ec;
959     box-shadow: none;
```

front/src/landing.css

```
960   cursor: wait;
961   transform: none;
962 }
963
964 .landing-link-button:disabled:hover,
965 .landing-link-button:disabled:focus-visible,
966 .landing-link-button:disabled:active {
967   background: #6aa9ec;
968   box-shadow: none;
969   transform: none;
970 }
971
972 .landing-link-button-icon {
973   width: 16px;
974   height: 16px;
975   flex: 0 0 16px;
976   display: block;
977   object-fit: contain;
978   filter: brightness(0) invert(1);
979   transition: transform 180ms cubic-bezier(0.22, 1, 0.36, 1);
980 }
981
982 .landing-link-button:hover .landing-link-button-icon,
983 .landing-link-button:focus-visible .landing-link-button-icon {
984   transform: translateX(3px);
985 }
986
987 .landing-link-button:disabled:hover .landing-link-button-icon,
988 .landing-link-button:disabled:focus-visible .landing-link-button-icon,
989 .landing-link-button:disabled:active .landing-link-button-icon {
990   transform: none;
991 }
992
993 .auth-page {
994   position: relative;
995   min-height: 100vh;
996   display: grid;
997   place-items: center;
998   padding: 32px;
999   background-color: #eef6fb;
1000  background-image:
1001    linear-gradient(180deg, rgba(248, 250, 252, 0.12), rgba(248, 250, 252, 0.1)),
1002    url('./images/登录注册背景图.png');
1003  background-size: cover;
1004  background-position: center center;
1005  background-repeat: no-repeat;
1006  color: var(--app-text-primary);
1007  overflow-x: hidden;
1008  animation: auth-page-enter 0.42s cubic-bezier(0.22, 1, 0.36, 1) both;
1009  will-change: opacity, filter;
```

front/src/landing.css

```
1010 }
1011
1012 .auth-page .blur-ball-indigo {
1013     width: 340px;
1014     height: 340px;
1015     top: -80px;
1016     right: -40px;
1017     left: auto;
1018 }
1019
1020 .auth-flip {
1021     position: relative;
1022     z-index: 1;
1023     width: 100%;
1024     max-width: 420px;
1025     perspective: 1600px;
1026 }
1027
1028 .auth-flip-inner {
1029     display: grid;
1030     width: 100%;
1031     transform-style: preserve-3d;
1032     transition: transform 620ms cubic-bezier(0.22, 1, 0.36, 1);
1033 }
1034
1035 .auth-flip.is-flipped .auth-flip-inner {
1036     transform: rotateY(180deg);
1037 }
1038
1039 .auth-flip-face {
1040     grid-area: 1 / 1;
1041     width: 100%;
1042     align-self: center;
1043     backface-visibility: hidden;
1044     -webkit-backface-visibility: hidden;
1045 }
1046
1047 .auth-flip-back {
1048     transform: rotateY(180deg);
1049 }
1050
1051 .auth-flip-face .auth-card {
1052     max-width: none;
1053     animation: none;
1054 }
1055
1056 .auth-flip-face[aria-hidden='true'] {
1057     pointer-events: none;
1058 }
1059
```



```
front/src/landing.css
```

```
1060 .auth-card {
1061   position: relative;
1062   z-index: 1;
1063   width: 100%;
1064   max-width: 420px;
1065   padding: 24px 32px;
1066   border: none;
1067   border-radius: 16px;
1068   background:
1069     linear-gradient(135deg, rgba(255, 255, 255, 0.4), rgba(255, 255, 255, 0.24)),
1070     rgba(255, 255, 255, 0.26);
1071   backdrop-filter: blur(40px) saturate(200%);
1072   -webkit-backdrop-filter: blur(40px) saturate(200%);
1073   box-shadow: 0 30px 76px -34px rgba(15, 23, 42, 0.46);
1074   animation: auth-card-enter 0.48s cubic-bezier(0.22, 1, 0.36, 1) both;
1075   will-change: opacity, transform;
1076 }
1077
1078 .auth-card-title {
1079   margin: 0 0 4px;
1080   font-size: 22px;
1081   font-weight: 800;
1082   text-align: center;
1083 }
1084
1085 .auth-card-subtitle {
1086   margin: 0 0 18px;
1087   text-align: center;
1088   color: rgba(71, 85, 105, 0.74);
1089   font-size: 13px;
1090 }
1091
1092 .auth-form {
1093   display: flex;
1094   flex-direction: column;
1095   gap: 10px;
1096 }
1097
1098 .auth-field {
1099   display: flex;
1100   flex-direction: column;
1101   gap: 6px;
1102 }
1103
1104 .auth-field label {
1105   font-size: 12px;
1106   font-weight: 700;
1107   color: var(--app-text-regular);
1108 }
1109
```

```
front/src/landing.css
```

```
1110 .auth-input {
1111     width: 100%;
1112     height: 40px;
1113     padding: 0 14px;
1114     border: 1px solid var(--app-border-strong);
1115     border-radius: 999px;
1116     background: rgba(248, 250, 252, 0.55);
1117     backdrop-filter: blur(14px) saturate(160%);
1118     -webkit-backdrop-filter: blur(14px) saturate(160%);
1119     color: var(--app-text-primary);
1120     font: inherit;
1121     font-size: 14px;
1122     transition: border-color 0.16s ease, box-shadow 0.16s ease;
1123 }
1124
1125 .auth-input:focus-visible {
1126     outline: 2px solid rgba(64, 158, 255, 0.32);
1127     outline-offset: 2px;
1128     border-color: rgba(64, 158, 255, 0.42);
1129     box-shadow: inset 0 0 0 1px rgba(64, 158, 255, 0.22);
1130 }
1131
1132 /* 输入框占位提示文字，颜色比正文更浅，避免与已填写内容混淆 */
1133 .auth-input::placeholder {
1134     color: var(--app-text-tertiary, rgba(100, 116, 139, 0.7));
1135     opacity: 1;
1136 }
1137
1138 .auth-turnstile {
1139     position: relative;
1140     min-height: 60px;
1141     display: flex;
1142     align-items: center;
1143     justify-content: center;
1144 }
1145
1146 .auth-turnstile-shell {
1147     min-height: 56px;
1148     display: flex;
1149     align-items: center;
1150     justify-content: center;
1151 }
1152
1153 .auth-turnstile-loading,
1154 .auth-turnstile-error {
1155     position: absolute;
1156     inset: 0;
1157     display: grid;
1158     place-items: center;
1159     color: rgba(71, 85, 105, 0.78);
```

front/src/landing.css

```
1160     font-size: 12px;
1161     font-weight: 700;
1162     line-height: 1.4;
1163     text-align: center;
1164     pointer-events: none;
1165 }
1166
1167 .auth-turnstile-error {
1168     color: #b91c1c;
1169 }
1170
1171 .auth-code-row {
1172     display: grid;
1173     grid-template-columns: minmax(0, 1fr) 122px;
1174     align-items: end;
1175     gap: 10px;
1176 }
1177
1178 .auth-code-button {
1179     height: 40px;
1180     display: inline-flex;
1181     align-items: center;
1182     justify-content: center;
1183     gap: 6px;
1184     padding: 0 12px;
1185     border: 1px solid rgba(64, 158, 255, 0.28);
1186     border-radius: 999px;
1187     background: rgba(64, 158, 255, 0.12);
1188     color: #245f9f;
1189     font: inherit;
1190     font-size: 13px;
1191     font-weight: 800;
1192     white-space: nowrap;
1193     cursor: pointer;
1194     transition: transform 0.16s ease, box-shadow 0.16s ease, background 0.16s ease;
1195 }
1196
1197 .auth-code-button:not(:disabled):hover,
1198 .auth-code-button:not(:disabled):focus-visible {
1199     background: rgba(64, 158, 255, 0.18);
1200     box-shadow: 0 10px 22px -16px rgba(37, 99, 235, 0.68);
1201     transform: translateY(-1px);
1202 }
1203
1204 .auth-code-button:disabled {
1205     cursor: not-allowed;
1206     opacity: 0.68;
1207 }
1208
1209 .auth-notice {
```

front/src/landing.css

```
1210     margin: 0;
1211     padding: 8px 12px;
1212     border-radius: 10px;
1213     background: rgba(34, 197, 94, 0.12);
1214     color: #15803d;
1215     font-size: 13px;
1216     font-weight: 700;
1217 }
1218
1219 .auth-error {
1220     margin: 0;
1221     padding: 8px 12px;
1222     border-radius: 10px;
1223     background: rgba(239, 68, 68, 0.1);
1224     color: #b91c1c;
1225     font-size: 13px;
1226     font-weight: 600;
1227 }
1228
1229 .auth-submit {
1230     min-height: 44px;
1231     height: 44px;
1232     width: 100%;
1233     margin-top: 4px;
1234     padding: 0 16px;
1235     border-radius: 8px;
1236     border-color: #2f8ff0;
1237     background: #2f8ff0;
1238     font-size: 14px;
1239     font-weight: 800;
1240     box-shadow: 0 4px 8px -5px rgba(51, 126, 204, 0.52);
1241     transition:
1242         transform 180ms cubic-bezier(0.22, 1, 0.36, 1),
1243         background-color 180ms cubic-bezier(0.22, 1, 0.36, 1),
1244         border-color 180ms cubic-bezier(0.22, 1, 0.36, 1),
1245         box-shadow 220ms cubic-bezier(0.22, 1, 0.36, 1);
1246 }
1247
1248 .auth-submit:not(:disabled):hover,
1249 .auth-submit:not(:disabled):focus-visible {
1250     border-color: #267fd9;
1251     background: #267fd9;
1252     box-shadow: 0 7px 12px -7px rgba(51, 126, 204, 0.58);
1253     transform: translateY(-1px);
1254 }
1255
1256 .auth-submit:not(:disabled):active {
1257     box-shadow: 0 2px 5px -4px rgba(51, 126, 204, 0.5);
1258     transform: translateY(1px) scale(0.985);
1259 }
```

front/src/landing.css

```
1260
1261 .auth-submit:disabled,
1262 .auth-submit:disabled:hover {
1263     cursor: wait;
1264     opacity: 0.82;
1265     border-color: #2f8ff0;
1266     background: #2f8ff0;
1267     box-shadow: none;
1268     transform: none;
1269 }
1270
1271 .auth-submit-spinner {
1272     flex: 0 0 16px;
1273     animation: auth-submit-spin 720ms linear infinite;
1274 }
1275
1276 .auth-switch {
1277     margin: 10px 0 0;
1278     text-align: center;
1279     font-size: 13px;
1280     color: rgba(71, 85, 105, 0.72);
1281 }
1282
1283 .auth-switch button {
1284     margin-left: 6px;
1285     padding: 0;
1286     border: 0;
1287     background: transparent;
1288     color: var(--app-accent-dark);
1289     font: inherit;
1290     font-size: 13px;
1291     font-weight: 700;
1292     cursor: pointer;
1293 }
1294
1295 .auth-switch button:hover {
1296     text-decoration: underline;
1297 }
1298
1299 .auth-back {
1300     display: block;
1301     margin-top: 8px;
1302     text-align: center;
1303     border: 0;
1304     background: transparent;
1305     color: rgba(71, 85, 105, 0.64);
1306     font: inherit;
1307     font-size: 12px;
1308     cursor: pointer;
1309     width: 100%;
```

```
front/src/landing.css
```

```
1310 }
1311
1312 .auth-back:hover {
1313     color: var(--app-text-primary);
1314 }
1315
1316 @media (max-width: 1120px) {
1317     .capability-grid {
1318         grid-template-columns: repeat(2, minmax(0, 1fr));
1319     }
1320 }
1321
1322 @media (max-width: 900px) {
1323     .landing-topbar-inner {
1324         grid-template-columns: auto 1fr;
1325     }
1326
1327     .landing-about-card {
1328         flex-direction: column;
1329         align-items: flex-start;
1330     }
1331
1332     .landing-workflow-section {
1333         padding: 0 16px 24px;
1334     }
1335
1336     .landing-workflow-card {
1337         padding: 0;
1338     }
1339
1340     .landing-workflow-heading {
1341         align-items: flex-start;
1342         flex-direction: column;
1343         gap: 4px;
1344         margin-bottom: 14px;
1345     }
1346
1347     .landing-workflow-heading > div {
1348         align-items: center;
1349     }
1350
1351     .landing-workflow-heading p {
1352         text-align: left;
1353     }
1354
1355     .landing-workflow-timeline {
1356         min-width: 0;
1357         min-height: 0;
1358         margin-top: 2px;
1359         padding: 0 0 0 6px;
```

front/src/landing.css

```
1360     display: flex;
1361     flex-direction: column;
1362     isolation: auto;
1363 }
1364
1365 .workflow-route {
1366     display: none;
1367 }
1368
1369 .landing-workflow-timeline::before {
1370     position: absolute;
1371     z-index: 0;
1372     top: 18px;
1373     right: auto;
1374     bottom: 18px;
1375     left: 25px;
1376     width: 2px;
1377     height: auto;
1378     display: block;
1379     border-radius: 999px;
1380     background: linear-gradient(180deg, rgba(64, 158, 255, 0.36), rgba(45, 212, 191, 0.82), rgba(64, 158, 255, 0.36));
1381     box-shadow: 0 0 14px rgba(45, 212, 191, 0.26), 0 0 0 5px rgba(45, 212, 191, 0.035);
1382 }
1383
1384 .workflow-stage {
1385     position: relative;
1386     top: auto;
1387     left: auto;
1388     width: auto;
1389     height: auto;
1390     min-height: 74px;
1391     display: grid;
1392     grid-template-columns: 40px minmax(0, 1fr);
1393     grid-template-rows: auto;
1394     column-gap: 10px;
1395     justify-items: stretch;
1396     transform: none;
1397     text-align: left;
1398 }
1399
1400 .workflow-stage:first-of-type,
1401 .workflow-stage:last-of-type {
1402     width: auto;
1403 }
1404
1405 .workflow-stage.is-upper .workflow-stage-copy,
1406 .workflow-stage.is-lower .workflow-stage-copy {
1407     position: relative;
1408     top: auto;
1409     bottom: auto;
```

front/src/landing.css

```
1410     left: auto;
1411     width: auto;
1412     grid-column: 2;
1413     grid-row: 1;
1414     align-self: center;
1415     margin: 0;
1416     padding: 0;
1417     border: 0;
1418     border-radius: 0;
1419     background: transparent;
1420     box-shadow: none;
1421     transform: none;
1422 }
1423
1424 .workflow-stage-copy::after {
1425     display: none;
1426 }
1427
1428 .workflow-stage-node,
1429 .workflow-stage.is-upper .workflow-stage-node,
1430 .workflow-stage.is-lower .workflow-stage-node {
1431     position: relative;
1432     top: auto;
1433     left: auto;
1434     grid-column: 1;
1435     grid-row: 1;
1436     place-self: start center;
1437     margin-top: 12px;
1438     transform: none;
1439 }
1440
1441 .workflow-stage.is(:hover, :focus-within) .workflow-stage-copy {
1442     transform: translateY(-2px);
1443 }
1444
1445 .workflow-stage.is(:hover, :focus-within) .workflow-stage-node,
1446 .landing-page.theme-dark .workflow-stage.is(:hover, :focus-within) .workflow-stage-node {
1447     transform: translateY(-2px) scale(1.05);
1448 }
1449
1450 .workflow-stage p {
1451     text-overflow: clip;
1452     white-space: normal;
1453 }
1454 }
1455
1456 @media (max-width: 760px) {
1457     .capability-mark-grid {
1458         grid-template-columns: repeat(4, minmax(68px, 1fr));
1459         gap: 22px 14px;
```



```
front/src/landing.css
```

```
1460     }
1461
1462     .capability-mark {
1463         min-height: 80px;
1464         gap: 14px;
1465     }
1466
1467     .capability-mark-icon {
1468         width: 46px;
1469         height: 46px;
1470         flex-basis: 46px;
1471     }
1472
1473     .capability-mark-label {
1474         font-size: 13px;
1475     }
1476 }
1477
1478 @media (max-width: 640px) {
1479     .landing-topbar-inner {
1480         width: min(100%, calc(100% - 24px));
1481         min-height: 68px;
1482         gap: 12px;
1483     }
1484
1485     .landing-version-pill {
1486         min-width: 62px;
1487         height: 26px;
1488         padding: 0 10px;
1489         font-size: 12px;
1490     }
1491
1492     .landing-hero {
1493         min-height: auto;
1494         padding-top: 64px;
1495     }
1496
1497     .landing-hero h1 {
1498         font-size: 36px;
1499         line-height: 1.15;
1500     }
1501
1502     .base-url-row {
1503         grid-template-columns: 1fr auto;
1504         padding: 12px;
1505         border-radius: 24px;
1506     }
1507
1508     .base-url-endpoint {
1509         grid-column: 1 / 2;
```

front/src/landing.css

```
1510     }
1511
1512     .base-url-copy {
1513         grid-column: 2 / 3;
1514         grid-row: 1 / 3;
1515         align-self: center;
1516         justify-self: end;
1517     }
1518
1519     .capability-grid {
1520         grid-template-columns: 1fr;
1521     }
1522 }
1523
1524 @media (prefers-reduced-motion: reduce) {
1525     .blur-ball {
1526         filter: none;
1527     }
1528
1529     .shine-text {
1530         animation: none;
1531     }
1532
1533     .base-url-endpoint {
1534         animation: none;
1535         transform: none;
1536         filter: none;
1537     }
1538
1539     .workflow-stage-copy,
1540     .workflow-stage-node {
1541         transition: none;
1542     }
1543
1544     .auth-page,
1545     .auth-card {
1546         animation: none;
1547         will-change: auto;
1548     }
1549
1550     .auth-flip-inner {
1551         transition: none;
1552         transform: none !important;
1553         will-change: auto;
1554     }
1555
1556     .auth-flip-face {
1557         transform: none;
1558     }
1559 }
```

front/src/landing.css

```
1560   .auth-flip-face[aria-hidden='true'] {
1561     display: none;
1562   }
1563
1564   .auth-submit,
1565   .auth-code-button,
1566   .auth-submit-spinner {
1567     transition: none;
1568     animation: none;
1569   }
1570
1571   .auth-code-button:not(:disabled):hover,
1572   .auth-code-button:not(:disabled):focus-visible,
1573   .auth-submit:not(:disabled):hover,
1574   .auth-submit:not(:disabled):focus-visible,
1575   .auth-submit:not(:disabled):active {
1576     transform: none;
1577   }
1578
1579   .hero-btn {
1580     transition: none;
1581   }
1582
1583   .hero-btn:active {
1584     transform: none;
1585   }
1586
1587   .landing-link-button,
1588   .landing-link-button-icon {
1589     transition: none;
1590   }
1591
1592   .landing-link-button:hover,
1593   .landing-link-button:focus-visible,
1594   .landing-link-button:active,
1595   .landing-link-button:hover .landing-link-button-icon,
1596   .landing-link-button:focus-visible .landing-link-button-icon {
1597     transform: none;
1598   }
1599 }
1600
1601 /* 首页隐藏视口滚动条（仅视觉），保留滚动能力。作用域由 body.landing-scroll-hidden 控制，
1602    LandingPage 卸载时会移除该 class，因此工作台/登录注册页的滚动条不受影响。 */
1603 body.landing-scroll-hidden {
1604   scrollbar-width: none; /* Firefox */
1605 }
1606
1607 html:has(body.landing-scroll-hidden) {
1608   -ms-overflow-style: none; /* 老 IE/Edge */
1609 }
```

front/src/landing.css

```
1610
1611 html:has(body, landing-scroll-hidden)::-webkit-scrollbar {
1612   width: 0;
1613   height: 0;
1614   display: none; /* Chromium / WebKit */
1615 }
1616

// ===== 文件: front/src/main.tsx =====
1 import React from 'react';
2 import ReactDOM from 'react-dom/client';
3 import { BrowserRouter } from 'react-router-dom';
4 import App from './App';
5 import 'katex/dist/katex.min.css';
6 import './styles.css';
7 import './workbench/workbench-polish.css';
8
9 ReactDOM.createRoot(document.getElementById('root')!).render(
10   <React.StrictMode>
11     <BrowserRouter future={{ v7_relativeSplatPath: true, v7_startTransition: true }}>
12       <App />
13     </BrowserRouter>
14   </React.StrictMode>,
15 );
16

// ===== 文件: front/src/styles.css =====
1 :root {
2   /* ===== CPA-Manager-Plus 体系 token ===== */
3   /* 行动色（主色） */
4   --primary-color: #409eff;
5   --primary-hover: #79bbff;
6   --primary-active: #337ecc;
7   --primary-contrast: #ffffff;
8
9   /* 背景层级 */
10  --bg-secondary: #f8fafc; /* 页面背景 */
11  --bg-primary: #ffffff; /* 卡片背景 */
12  --bg-tertiary: #f1f5f9; /* hover/次级 */
13  --bg-quinary: #f8fafc;
14  --bg-hover: var(--bg-tertiary);
15  --bg-error-light: rgba(124, 58, 237, 0.1);
16
17  /* 浮层（floating surface） */
18  --floating-surface: #ffffff;
19  --floating-border: #e5e7eb;
20  --floating-shadow: 0 12px 26px rgba(15, 23, 42, 0.12);
21
22  /* 文字层级 */
23  --text-primary: #2c3e50;
```

```
front/src/styles.css
```

```
24  --text-secondary: #5f6c7b;
25  --text-tertiary: #8b95a6;
26  --text-quaternary: #b5bdca;
27  --text-muted: var(--text-tertiary);
28
29  /* 边框层级 */
30  --border-color: #e5e7eb;
31  --border-secondary: var(--border-color);
32  --border-primary: #cbd5e1;
33  --border-hover: #94a3b8;
34
35  /* 状态色 */
36  --success-color: #67c23a;
37  --quota-medium-color: #e6a23c;
38  --amber-color: #e6a23c;
39  --amber-text: #b45309; /* amber-700 */
40  --warning-color: #e6a23c;
41  --error-color: #ef4444;
42  --danger-color: var(--error-color);
43  --info-color: var(--primary-color);
44
45  /* 警告/徽章半透明底 */
46  --warning-bg: rgba(230, 162, 60, 0.12);
47  --warning-border: rgba(230, 162, 60, 0.35);
48  --warning-text: var(--warning-color);
49  --success-badge-bg: rgba(103, 194, 58, 0.12);
50  --success-badge-text: #529b2e;
51  --success-badge-border: rgba(103, 194, 58, 0.35);
52  --failure-badge-bg: rgba(245, 108, 108, 0.14);
53  --failure-badge-text: #c45656;
54  --failure-badge-border: rgba(245, 108, 108, 0.35);
55  --count-badge-bg: rgba(64, 158, 255, 0.12);
56  --count-badge-text: var(--primary-color);
57
58  /* 阴影 / 圆角 */
59  --shadow: 0 1px 2px 0 rgb(15 23 42 / 0.08);
60  --shadow-lg: 0 10px 18px -3px rgb(15 23 42 / 0.1);
61  --radius-sm: 4px;
62  --radius-md: 8px;
63  --radius-lg: 12px;
64  --radius-full: 9999px;
65
66  /* 主色派生半透明 (color-mix) */
67  --primary-8: color-mix(in srgb, var(--primary-color) 8%, transparent);
68  --primary-10: color-mix(in srgb, var(--primary-color) 10%, transparent);
69  --primary-30: color-mix(in srgb, var(--primary-color) 30%, transparent);
70  --amber-10: color-mix(in srgb, var(--amber-color) 10%, transparent);
71  --amber-30: color-mix(in srgb, var(--amber-color) 30%, transparent);
72  --destructive-color: var(--error-color);
73  --destructive-10: color-mix(in srgb, var(--destructive-color) 10%, transparent);
```

```
front/src/styles.css
```

```
74   --destructive-30: color-mix(in srgb, var(--destructive-color) 30%, transparent);
75
76   /* 玻璃质感 */
77   --glass-blur: 12px;
78   --glass-backdrop-filter: blur(var(--glass-blur));
79   --glass-bg: color-mix(in srgb, var(--bg-primary) 82%, transparent);
80   --glass-bg-secondary: color-mix(in srgb, var(--bg-secondary) 82%, transparent);
81   --glass-border: color-mix(in srgb, var(--border-color) 60%, transparent);
82
83   /* ===== soft_web 既有 --app-* 别名（映射到上方 token，保持向后兼容）===== */
84   --app-bg: #eff2f7;
85   --app-bg-gradient: linear-gradient(120deg, #f0f7ff 0%, #e7f2ff 50%, #edf7ff 100%);
86   --app-bg-blob-1-start: #7aa2ff;
87   --app-bg-blob-1-end: #b6d4ff;
88   --app-bg-blob-2-start: #6bc5ff;
89   --app-bg-blob-2-end: #7debdc;
90   --app-bg-blob-1-opacity: 0.5;
91   --app-bg-blob-2-opacity: 0.5;
92   --app-bg-blob-blur: 60px;
93   --app-surface: rgba(255, 255, 255, 0.9);
94   --app-surface-strong: var(--floating-surface);
95   --app-surface-muted: rgba(255, 255, 255, 0.62);
96   --app-border: rgba(15, 23, 42, 0.08);
97   --app-border-strong: rgba(15, 23, 42, 0.14);
98   --app-text-primary: var(--text-primary);
99   --app-text-regular: var(--text-secondary);
100  --app-text-muted: var(--text-tertiary);
101  --app-accent: var(--primary-color);
102  --app-accent-dark: var(--primary-active);
103  --app-accent-soft: var(--primary-10);
104  --success: var(--success-color);
105  --warning: var(--amber-color);
106  --danger: var(--error-color);
107  --purple: var(--error-color);
108
109  --sidebar-width: 210px;
110  --sidebar-collapsed-width: 58px;
111  --header-height: 50px;
112  --app-gap: 20px;
113  --panel-radius: var(--radius-md);
114  --z-dropdown: 1100;
115  color: var(--app-text-primary);
116  background: var(--app-bg);
117  font-family:
118    Lato, 'Helvetica Neue', Arial, Helvetica, 'Microsoft YaHei', sans-serif;
119  font-synthesis: none;
120  text-rendering: optimizeLegibility;
121 }
122
123 * {
```

```
front/src/styles.css
```

```
124   box-sizing: border-box;
125 }
126
127 html,
128 body,
129 #root {
130   width: 100%;
131   min-height: 100%;
132 }
133
134 body {
135   margin: 0;
136   min-width: 320px;
137   min-height: 100vh;
138   background: var(--app-bg);
139   -webkit-tap-highlight-color: transparent;
140 }
141
142 button,
143 input {
144   font: inherit;
145 }
146
147 button {
148   touch-action: manipulation;
149 }
150
151 .sr-only {
152   position: absolute;
153   width: 1px;
154   height: 1px;
155   padding: 0;
156   margin: -1px;
157   overflow: hidden;
158   clip: rect(0, 0, 0, 0);
159   white-space: nowrap;
160   border: 0;
161 }
162
163 .select-field {
164   min-width: 0;
165   display: grid;
166   gap: 6px;
167 }
168
169 .select-field-label {
170   color: var(--app-text-muted);
171   font-size: 12px;
172   font-weight: 700;
173 }
```

front/src/styles.css

```
174
175 .select-field-trigger {
176   width: 100%;
177   height: 38px;
178   min-width: 0;
179   display: flex;
180   align-items: center;
181   justify-content: space-between;
182   gap: 8px;
183   padding: 0 10px 0 12px;
184   border: 1px solid var(--app-border);
185   border-radius: var(--panel-radius);
186   color: var(--app-text-primary);
187   background: var(--floating-surface);
188   cursor: pointer;
189   text-align: left;
190   transition:
191     border-color 0.16s ease,
192     box-shadow 0.16s ease,
193     background-color 0.16s ease;
194 }
195
196 .select-field-sm .select-field-trigger {
197   height: 34px;
198   padding-inline: 10px;
199 }
200
201 .select-field-trigger:hover {
202   border-color: var(--app-border-strong);
203   background: var(--floating-surface);
204 }
205
206 .select-field-trigger:focus-visible {
207   outline: 2px solid var(--primary-30);
208   outline-offset: 1px;
209   border-color: var(--app-accent);
210 }
211
212 .select-field-trigger:disabled {
213   color: var(--app-text-muted);
214   background: var(--bg-secondary);
215   cursor: not-allowed;
216 }
217
218 .select-field-trigger svg {
219   flex: 0 0 auto;
220   color: var(--app-text-muted);
221   transition: transform 0.16s ease;
222 }
223
```


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```
224 .select-field.is-open .select-field-trigger svg {
225   transform: rotate(180deg);
226 }
227
228 .select-field-trigger .select-field-trigger-value {
229   min-width: 0;
230   overflow: hidden;
231   color: var(--app-text-primary);
232   font-size: 13px;
233   font-weight: 600;
234   line-height: 1;
235   text-overflow: ellipsis;
236   white-space: nowrap;
237 }
238
239 .select-field.is-empty .select-field-trigger-value {
240   color: var(--app-text-muted);
241 }
242
243 .select-field-menu {
244   position: fixed;
245   z-index: var(--z-dropdown);
246   overflow-y: auto;
247   scrollbar-width: none;
248   padding: 4px;
249   border: 1px solid var(--app-border-strong);
250   border-radius: var(--panel-radius);
251   background: var(--floating-surface);
252   box-shadow: 0 8px 14px rgba(15, 23, 42, 0.12);
253 }
254
255 .select-field-menu::-webkit-scrollbar {
256   width: 0;
257   height: 0;
258 }
259
260 .select-field-option {
261   height: 34px;
262   overflow: hidden;
263   padding: 0 10px;
264   border-radius: calc(var(--panel-radius) - 2px);
265   color: var(--app-text-primary);
266   font-size: 13px;
267   font-weight: 600;
268   line-height: 34px;
269   text-overflow: ellipsis;
270   white-space: nowrap;
271   cursor: pointer;
272 }
273
```

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```
274 .select-field-option.is-active {
275   background: var(--bg-tertiary);
276 }
277
278 .select-field-option.is-selected {
279   color: var(--app-accent-dark);
280   background: var(--app-accent-soft);
281 }
282
283 .select-field-option.is-disabled {
284   color: var(--app-text-muted);
285   cursor: not-allowed;
286 }
287
288 @media (prefers-reduced-motion: reduce) {
289   .select-field-trigger,
290   .select-field-trigger svg {
291     transition: none;
292   }
293 }
294
295 h1,
296 h2,
297 p {
298   margin-top: 0;
299 }
300
301 .skip-link {
302   position: fixed;
303   top: 8px;
304   left: 8px;
305   z-index: 1000;
306   padding: 8px 12px;
307   border-radius: var(--panel-radius);
308   color: #ffffff;
309   background: var(--app-accent-dark);
310   text-decoration: none;
311   transform: translateY(-160%);
312   transition: transform 0.16s ease;
313 }
314
315 .skip-link:focus-visible {
316   transform: translateY(0);
317 }
318
319 .app-background {
320   position: fixed;
321   inset: 0;
322   z-index: 0;
323   width: 100%;
```

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```
324     height: 100%;
325     overflow: hidden;
326     pointer-events: none;
327     background-color: var(--app-bg);
328     background-image: var(--app-bg-gradient);
329 }
330
331 .app-background__shape {
332     position: absolute;
333     filter: blur(var(--app-bg-blob-blur));
334     opacity: var(--app-bg-blob-1-opacity);
335     animation: app-float 20s infinite ease-in-out alternate;
336 }
337
338 .app-background__shape-1 {
339     top: -10%;
340     right: -10%;
341     width: 600px;
342     height: 600px;
343     opacity: var(--app-bg-blob-1-opacity);
344 }
345
346 .app-background__shape-2 {
347     bottom: -10%;
348     left: -10%;
349     width: 500px;
350     height: 500px;
351     opacity: var(--app-bg-blob-2-opacity);
352     animation-delay: -5s;
353 }
354
355 @keyframes app-float {
356     0% {
357         transform: translate(0, 0) rotate(0deg);
358     }
359
360     50% {
361         transform: translate(30px, 20px) rotate(10deg);
362     }
363
364     100% {
365         transform: translate(0, 0) rotate(0deg);
366     }
367 }
368
369 @media (prefers-reduced-motion: reduce) {
370     .app-background__shape {
371         animation: none;
372     }
373 }
```

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```
374
375 @media (max-width: 768px) {
376   .app-background__shape-1 {
377     width: 420px;
378     height: 420px;
379   }
380
381   .app-background__shape-2 {
382     width: 360px;
383     height: 360px;
384   }
385 }
386
387 .app-shell {
388   position: relative;
389   min-height: 100vh;
390   background: transparent;
391   color: var(--app-text-primary);
392   font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI', 'PingFang SC', 'Hiragino Sans GB',
393     'Microsoft YaHei', 'Helvetica Neue', Arial, sans-serif;
394   font-weight: 600;
395 }
396
397 .sidebar {
398   position: fixed;
399   inset: 0 auto 0 0;
400   z-index: 30;
401   width: var(--sidebar-width);
402   overflow: hidden;
403   display: flex;
404   flex-direction: column;
405   border-right: 1px solid var(--app-border);
406   background: rgba(248, 250, 252, 0.58);
407   box-shadow: 2px 0 12px rgba(15, 23, 42, 0.1);
408   backdrop-filter: blur(20px);
409   transition:
410     width 0.2s ease,
411     transform 0.2s ease;
412 }
413
414 .sidebar.collapsed {
415   width: var(--sidebar-collapsed-width);
416 }
417
418 .sidebar-brand {
419   display: flex;
420   align-items: center;
421   gap: 10px;
422   height: var(--header-height);
423   min-height: var(--header-height);
```

```
front/src/styles.css
```

```
424   padding: 0 14px;
425   overflow: hidden;
426   background: rgba(248, 250, 252, 0.34);
427 }
428
429 .brand-logo {
430   width: 30px;
431   height: 30px;
432   flex: 0 0 30px;
433   display: grid;
434   place-items: center;
435   border-radius: 10px;
436   overflow: hidden;
437   color: #ffffff;
438   background: var(--app-accent);
439 }
440
441 .sidebar-brand-logo-image {
442   width: 100%;
443   height: 100%;
444   display: block;
445   object-fit: cover;
446   object-position: center;
447 }
448
449 .brand-copy {
450   min-width: 0;
451   display: grid;
452   gap: 1px;
453 }
454
455 .brand-copy strong,
456 .brand-copy small {
457   overflow: hidden;
458   text-overflow: ellipsis;
459   white-space: nowrap;
460 }
461
462 .brand-copy strong {
463   font-size: 15px;
464   line-height: 1.15;
465 }
466
467 .brand-copy small {
468   color: var(--app-text-muted);
469   font-size: 11px;
470 }
471
472 .sidebar.collapsed .brand-copy,
473 .sidebar.collapsed .nav-label,
```

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```
474 .sidebar.collapsed .nav-group-label {
475   display: none;
476 }
477
478 .nav-list {
479   flex: 1;
480   overflow-y: auto;
481   padding: 10px 0;
482   scrollbar-width: none;
483 }
484
485 .nav-list::-webkit-scrollbar {
486   display: none;
487 }
488
489 .nav-group {
490   display: grid;
491   gap: 2px;
492   padding: 6px 0;
493 }
494
495 .nav-group + .nav-group {
496   border-top: 1px solid rgba(15, 23, 42, 0.06);
497 }
498
499 .nav-group-label {
500   padding: 8px 18px 6px;
501   color: var(--app-text-muted);
502   font-size: 11px;
503   font-weight: 700;
504 }
505
506 .nav-item {
507   position: relative;
508   height: 42px;
509   width: calc(100% - 20px);
510   margin: 2px 10px;
511   padding: 0 10px;
512   display: flex;
513   align-items: center;
514   gap: 10px;
515   border: 1px solid transparent;
516   border-radius: var(--panel-radius);
517   color: var(--app-text-regular);
518   background: transparent;
519   cursor: pointer;
520   text-decoration: none;
521   text-align: left;
522   transition:
523     background-color 0.16s ease,
```

```
front/src/styles.css
```

```
524     border-color 0.16s ease,
525     color 0.16s ease;
526 }
527
528 .nav-item:hover,
529 .nav-item.active {
530     color: var(--app-text-primary);
531     background: var(--app-accent-soft);
532     border-color: rgba(64, 158, 255, 0.14);
533 }
534
535 .nav-item:focus-visible,
536 .header-icon-button:focus-visible,
537 .sidebar-toggle-floating:focus-visible,
538 .btn:focus-visible,
539 .panel-action:focus-visible,
540 .chart-segment:focus-visible,
541 .soft-segment:focus-visible,
542 .export-actions button:focus-visible {
543     outline: 2px solid rgba(64, 158, 255, 0.42);
544     outline-offset: 2px;
545     border-color: rgba(64, 158, 255, 0.42);
546     box-shadow: inset 0 0 0 1px rgba(64, 158, 255, 0.32);
547 }
548
549 .nav-icon {
550     width: 24px;
551     min-width: 24px;
552     display: inline-flex;
553     justify-content: center;
554 }
555
556 .nav-label {
557     min-width: 0;
558     overflow: hidden;
559     font-size: 15px;
560     font-weight: 600;
561     text-overflow: ellipsis;
562     white-space: nowrap;
563 }
564
565 .sidebar.collapsed .nav-item {
566     justify-content: center;
567     width: 42px;
568     padding: 0;
569     margin-inline: 8px;
570 }
571
572 .sidebar-backdrop {
573     display: none;
```

```
front/src/styles.css
```

```
574     position: fixed;
575     inset: 0;
576     z-index: 20;
577     border: 0;
578     padding: 0;
579     background: rgba(15, 23, 42, 0.34);
580     opacity: 0;
581     pointer-events: none;
582     transition: opacity 0.16s ease;
583 }
584
585 .main-header {
586     position: fixed;
587     top: 0;
588     right: 0;
589     left: var(--sidebar-width);
590     z-index: 25;
591     height: var(--header-height);
592     display: flex;
593     align-items: center;
594     justify-content: space-between;
595     border-bottom: 1px solid var(--app-border);
596     background: rgba(248, 250, 252, 0.9);
597     backdrop-filter: blur(20px);
598     transition: left 0.2s ease;
599 }
600
601 .sidebar-collapsed .main-header {
602     left: var(--sidebar-collapsed-width);
603 }
604
605 .navbar-left,
606 .navbar-right {
607     display: flex;
608     align-items: center;
609     min-width: 0;
610     height: 100%;
611 }
612
613 .navbar-left {
614     flex: 1 1 auto;
615 }
616
617 .navbar-right {
618     flex: 0 0 auto;
619     gap: 4px;
620     padding-right: 18px;
621 }
622
623 .header-icon-button {
```



```
front/src/styles.css
```

```
624   width: 34px;
625   min-width: 34px;
626   height: 34px;
627   padding: 0;
628   display: inline-flex;
629   align-items: center;
630   justify-content: center;
631   border: 1px solid transparent;
632   border-radius: var(--panel-radius);
633   color: var(--app-text-regular);
634   background: transparent;
635   cursor: pointer;
636   transition:
637     background-color 0.16s ease,
638     border-color 0.16s ease,
639     color 0.16s ease;
640 }
641
642 .header-icon-button:hover {
643   color: var(--app-accent-dark);
644   background: var(--app-accent-soft);
645 }
646
647 .sidebar-toggle-floating {
648   position: fixed;
649   top: 50%;
650   left: calc(var(--sidebar-width) - 12px);
651   z-index: 35;
652   width: 24px;
653   height: 52px;
654   padding: 0;
655   display: inline-flex;
656   align-items: center;
657   justify-content: center;
658   border: 1px solid var(--app-border-strong);
659   border-radius: 14px;
660   color: var(--app-text-regular);
661   background: rgba(248, 250, 252, 0.88);
662   box-shadow:
663     4px 0 14px rgba(15, 23, 42, 0.14),
664     inset 0 1px 0 rgba(248, 250, 252, 0.7);
665   backdrop-filter: blur(16px);
666   cursor: pointer;
667   transform: translateY(-50%);
668   transition:
669     left 0.2s ease,
670     color 0.16s ease,
671     background-color 0.16s ease,
672     border-color 0.16s ease,
673     transform 0.16s ease;
```

```
front/src/styles.css
```

```
674 }
675
676 .sidebar-toggle-floating:hover {
677   color: var(--app-accent-dark);
678   border-color: rgba(64, 158, 255, 0.32);
679   background: #ffffff;
680   transform: translateY(-50%) translateX(2px);
681 }
682
683 .sidebar-toggle-floating:active {
684   transform: translateY(-50%);
685 }
686
687 .sidebar-collapsed .sidebar-toggle-floating {
688   left: calc(var(--sidebar-collapsed-width) - 12px);
689 }
690
691 .mobile-nav-button {
692   display: none;
693 }
694
695 .header-section-title {
696   min-width: 0;
697   height: 100%;
698   display: inline-flex;
699   align-items: center;
700   margin-left: 10px;
701   overflow: hidden;
702   color: var(--app-text-primary);
703   font-size: 14px;
704   font-weight: 800;
705   text-overflow: ellipsis;
706   white-space: nowrap;
707 }
708
709 .dashboard-content {
710   position: relative;
711   z-index: 1;
712   min-height: 100vh;
713   margin-left: var(--sidebar-width);
714   padding: calc(var(--header-height) + var(--app-gap)) var(--app-gap) var(--app-gap);
715   display: flex;
716   flex-direction: column;
717   gap: var(--app-gap);
718   overflow-x: hidden;
719 }
720
721 .sidebar-collapsed .dashboard-content {
722   margin-left: var(--sidebar-collapsed-width);
723 }
```

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```
724
725 .status-header,
726 .panel,
727 .summary-card,
728 .metric-card {
729   border: 1px solid var(--app-border);
730   border-radius: var(--panel-radius);
731   background: var(--app-surface);
732   backdrop-filter: blur(20px);
733 }
734
735 .status-header {
736   min-height: 58px;
737   display: grid;
738   grid-template-columns: minmax(220px, 1fr) minmax(260px, 1fr) auto;
739   align-items: center;
740   gap: 14px;
741   padding: 10px 16px;
742 }
743
744 .status-left,
745 .status-meta {
746   min-width: 0;
747   display: flex;
748   align-items: center;
749   gap: 10px;
750 }
751
752 .status-left > div,
753 .status-meta {
754   display: grid;
755   gap: 2px;
756 }
757
758 .status-label,
759 .status-meta span {
760   color: var(--app-text-muted);
761   font-size: 11px;
762 }
763
764 .status-left strong,
765 .status-meta strong {
766   min-width: 0;
767   overflow: hidden;
768   color: var(--app-text-primary);
769   font-size: 13px;
770   text-overflow: ellipsis;
771   white-space: nowrap;
772 }
773
```

```
front/src/styles.css
```

```
774 .status-dot {  
775     width: 10px;  
776     height: 10px;  
777     border-radius: 999px;  
778 }  
779  
780 .status-dot.warning {  
781     background: var(--warning);  
782     box-shadow: 0 0 0 4px var(--warning-bg);  
783 }  
784  
785 .status-dot.success {  
786     background: var(--success);  
787     box-shadow: 0 0 0 4px var(--success-badge-bg);  
788 }  
789  
790 .status-dot.error {  
791     background: var(--error-color);  
792     box-shadow: 0 0 0 4px var(--failure-badge-bg);  
793 }  
794  
795 .status-actions {  
796     display: flex;  
797     justify-content: flex-end;  
798     gap: 8px;  
799 }  
800  
801 .btn {  
802     min-height: 34px;  
803     display: inline-flex;  
804     align-items: center;  
805     justify-content: center;  
806     gap: 7px;  
807     border: 1px solid transparent;  
808     border-radius: var(--panel-radius);  
809     padding: 0 12px;  
810     font-size: 13px;  
811     font-weight: 700;  
812     cursor: pointer;  
813     transition:  
814         background-color 0.16s ease,  
815         border-color 0.16s ease,  
816         color 0.16s ease,  
817         transform 0.16s ease;  
818 }  
819  
820 .btn:hover {  
821     transform: translateY(-1px);  
822 }  
823
```

```
front/src/styles.css
```

```
824 .btn-primary {
825   color: #ffffff;
826   border-color: var(--app-accent);
827   background: var(--app-accent);
828 }
829
830 .btn-primary:hover {
831   background: var(--primary-hover);
832 }
833
834 .btn-secondary {
835   color: var(--app-text-primary);
836   border-color: var(--app-border);
837   background: var(--app-surface-muted);
838 }
839
840 .btn-secondary:hover {
841   color: var(--app-accent-dark);
842   border-color: rgba(64, 158, 255, 0.28);
843   background: var(--app-accent-soft);
844 }
845
846 .panel-header span {
847   color: var(--app-text-muted);
848   font-size: 13px;
849 }
850
851 .summary-grid {
852   display: grid;
853   grid-template-columns: repeat(4, minmax(0, 1fr));
854   gap: 12px;
855 }
856
857 .summary-card {
858   min-width: 0;
859   padding: 14px;
860   display: flex;
861   gap: 12px;
862 }
863
864 .summary-icon {
865   width: 38px;
866   height: 38px;
867   flex: 0 0 38px;
868   display: grid;
869   place-items: center;
870   border-radius: var(--panel-radius);
871 }
872
873 .tone-neutral .summary-icon {
```

```
front/src/styles.css
```

```
874   color: var(--app-accent);
875   background: var(--app-accent-soft);
876 }
877
878 .tone-green .summary-icon {
879   color: var(--success);
880   background: rgba(103, 194, 58, 0.12);
881 }
882
883 .tone-amber .summary-icon {
884   color: var(--warning);
885   background: rgba(245, 158, 11, 0.14);
886 }
887
888 .tone-purple .summary-icon {
889   color: var(--purple);
890   background: rgba(239, 68, 68, 0.1);
891 }
892
893 .summary-card > div:last-child,
894 .metric-card {
895   min-width: 0;
896   display: grid;
897   gap: 5px;
898 }
899
900 .card-label,
901 .summary-card small,
902 .metric-note {
903   color: var(--app-text-muted);
904   font-size: 12px;
905 }
906
907 .summary-card strong {
908   min-width: 0;
909   overflow: hidden;
910   font-size: 22px;
911   font-variant-numeric: tabular-nums;
912   text-overflow: ellipsis;
913   white-space: nowrap;
914 }
915
916 .summary-card small,
917 .metric-note {
918   overflow: hidden;
919   text-overflow: ellipsis;
920   white-space: nowrap;
921 }
922
923 .metrics-grid {
```

```
front/src/styles.css
```

```
924   display: grid;
925   grid-template-columns: repeat(6, minmax(0, 1fr));
926   gap: 12px;
927 }
928
929 .metric-card {
930   padding: 14px;
931 }
932
933 .metric-head {
934   display: flex;
935   align-items: center;
936   justify-content: space-between;
937   gap: 8px;
938 }
939
940 .metric-head span {
941   min-width: 0;
942   overflow: hidden;
943   color: var(--app-text-muted);
944   font-size: 12px;
945   font-weight: 700;
946   text-overflow: ellipsis;
947   white-space: nowrap;
948 }
949
950 .metric-head small {
951   flex: 0 0 auto;
952   color: var(--success);
953   font-size: 11px;
954   font-weight: 800;
955 }
956
957 .metric-card strong {
958   min-width: 0;
959   overflow: hidden;
960   font-size: 25px;
961   font-weight: 800;
962   font-variant-numeric: tabular-nums;
963   line-height: 1.15;
964   text-overflow: ellipsis;
965   white-space: nowrap;
966 }
967
968 .top-insight-grid {
969   display: grid;
970   grid-template-columns: minmax(0, 4fr) minmax(0, 4fr) minmax(220px, 2fr);
971   gap: 12px;
972   align-items: stretch;
973 }
```

```
front/src/styles.css
```

```
974
975 .top-insight-grid > .panel {
976   height: 100%;
977 }
978
979 .top-insight-grid .timeline-matrix {
980   min-width: 0;
981 }
982
983 .top-insight-grid .timeline-cells,
984 .top-insight-grid .timeline-hour-axis {
985   gap: 3px;
986 }
987
988 .top-insight-grid .timeline-row {
989   gap: 7px;
990 }
991
992 .top-insight-grid .timeline-cell {
993   min-height: 10px;
994 }
995
996 .top-insight-grid .export-actions {
997   grid-template-columns: 1fr;
998 }
999
1000 .secondary-card-grid {
1001   display: grid;
1002   grid-template-columns: repeat(4, minmax(0, 1fr));
1003   gap: 12px;
1004   align-items: stretch;
1005 }
1006
1007 .secondary-card-grid > .panel {
1008   height: 100%;
1009 }
1010
1011 .secondary-card-grid .chart-panel {
1012   display: flex;
1013   flex-direction: column;
1014 }
1015
1016 .secondary-card-grid .chart-surface {
1017   flex: 1;
1018   min-height: 220px;
1019   height: auto;
1020 }
1021
1022
1023
```


front/src/styles.css

```
1024
1025
1026 .dataset-dropzone {
1027   position: relative;
1028   min-height: 118px;
1029   display: grid;
1030   place-items: center;
1031   align-content: center;
1032   gap: 8px;
1033   padding: 16px;
1034   border: 1px dashed var(--primary-30);
1035   border-radius: var(--panel-radius);
1036   background: var(--primary-8);
1037   color: var(--app-text-primary);
1038   cursor: pointer;
1039   text-align: center;
1040   transition:
1041     border-color 0.16s ease,
1042     background-color 0.16s ease,
1043     transform 0.16s ease;
1044 }
1045
1046 .dataset-dropzone:hover {
1047   border-color: color-mix(in srgb, var(--primary-color) 44%, transparent);
1048   background: var(--primary-10);
1049   transform: translateY(-1px);
1050 }
1051
1052 .dataset-dropzone input {
1053   position: absolute;
1054   width: 1px;
1055   height: 1px;
1056   opacity: 0;
1057   pointer-events: none;
1058 }
1059
1060 .dataset-dropzone-icon {
1061   width: 42px;
1062   height: 42px;
1063   display: grid;
1064   place-items: center;
1065   border-radius: var(--panel-radius);
1066   color: var(--app-accent);
1067   background: #ffffff;
1068   box-shadow: inset 0 0 0 1px var(--primary-10);
1069 }
1070
1071 .dataset-dropzone strong {
1072   font-size: 15px;
1073 }
```

front/src/styles.css

```
1074
1075 .dataset-dropzone small {
1076   color: var(--app-text-muted);
1077   font-size: 12px;
1078   font-weight: 700;
1079 }
1080
1081 .format-chip-row {
1082   display: flex;
1083   flex-wrap: wrap;
1084   gap: 8px;
1085   margin-top: 14px;
1086 }
1087
1088 .format-chip-row span {
1089   min-width: 0;
1090   overflow: hidden;
1091   padding: 4px 8px;
1092   border-radius: 999px;
1093   color: var(--app-accent-dark);
1094   background: var(--app-accent-soft);
1095   font-size: 11px;
1096   font-weight: 800;
1097   line-height: 1.35;
1098   text-overflow: ellipsis;
1099   white-space: nowrap;
1100 }
1101
1102 .dataset-file-summary {
1103   margin-top: 14px;
1104   display: grid;
1105   grid-template-columns: repeat(2, minmax(0, 1fr));
1106   gap: 10px;
1107 }
1108
1109 .dataset-file-summary > div,
1110 .dataset-stat-grid > div {
1111   min-width: 0;
1112   padding: 12px;
1113   border: 1px solid var(--app-border);
1114   border-radius: var(--panel-radius);
1115   background: var(--app-surface-muted);
1116 }
1117
1118 .dataset-file-summary span,
1119 .dataset-stat-grid span,
1120 .selection-meter span,
1121 .task-form-grid label span,
1122 .task-draft-card span,
1123 .task-draft-card dt {
```

```
front/src/styles.css
```

```
1124   color: var(--app-text-muted);
1125   font-size: 12px;
1126   font-weight: 700;
1127 }
1128
1129 .dataset-file-summary strong,
1130 .dataset-stat-grid strong {
1131   display: block;
1132   margin-top: 4px;
1133   font-size: 15px;
1134 }
1135
1136 .label-distribution-panel {
1137   min-width: 0;
1138 }
1139
1140 .label-distribution-list {
1141   display: grid;
1142   gap: 8px;
1143 }
1144
1145 .label-distribution-row {
1146   display: grid;
1147   grid-template-columns: minmax(96px, 0.8fr) 78px minmax(220px, 1.8fr) 54px;
1148   gap: 10px;
1149   align-items: center;
1150   min-height: 36px;
1151   padding: 7px 10px;
1152   border: 1px solid var(--app-border);
1153   border-radius: var(--panel-radius);
1154   background: var(--app-surface-muted);
1155 }
1156
1157 .label-distribution-row strong,
1158 .label-distribution-count,
1159 .label-distribution-percent {
1160   font-variant-numeric: tabular-nums;
1161   white-space: nowrap;
1162 }
1163
1164 .label-distribution-row strong {
1165   color: var(--app-text-primary);
1166   font-size: 13px;
1167   font-weight: 800;
1168 }
1169
1170 .label-distribution-count,
1171 .label-distribution-percent {
1172   color: var(--app-text-muted);
1173   font-size: 12px;
```

```
front/src/styles.css
```

```
1174   font-weight: 800;
1175 }
1176
1177 .label-distribution-percent {
1178   text-align: right;
1179 }
1180
1181 .label-distribution-track {
1182   height: 7px;
1183   overflow: hidden;
1184   border-radius: 999px;
1185   background: #e5e7eb;
1186 }
1187
1188 .label-distribution-track i {
1189   display: block;
1190   height: 100%;
1191   border-radius: inherit;
1192   background: var(--app-accent);
1193 }
1194
1195 .dataset-table-wrap {
1196   min-height: 0;
1197   max-height: 360px;
1198   overflow: auto;
1199   border: 1px solid var(--app-border);
1200   border-radius: var(--panel-radius);
1201   background: var(--app-surface-strong);
1202 }
1203
1204 .dataset-stat-table {
1205   width: 100%;
1206   border-collapse: separate;
1207   border-spacing: 0;
1208   font-size: 12px;
1209 }
1210
1211 .dataset-stat-table th,
1212 .dataset-stat-table td {
1213   padding: 8px 12px;
1214   border-bottom: 1px solid var(--app-border);
1215   text-align: left;
1216   white-space: nowrap;
1217   font-variant-numeric: tabular-nums;
1218 }
1219
1220 .dataset-stat-table th {
1221   position: sticky;
1222   top: 0;
1223   z-index: 1;
```

```
front/src/styles.css
```

```
1224   color: var(--app-text-muted);
1225   background: #f8fafc;
1226   font-weight: 800;
1227   box-shadow: inset 0 -1px 0 var(--app-border);
1228 }
1229
1230 .dataset-stat-table td {
1231   color: var(--app-text-regular);
1232   font-weight: 700;
1233 }
1234
1235 .dataset-stat-table tbody tr:hover td {
1236   background: var(--bg-quinary);
1237 }
1238
1239 .dataset-table-empty-cell {
1240   padding: 28px 12px !important;
1241   color: var(--app-text-muted) !important;
1242   text-align: center !important;
1243 }
1244
1245 .dataset-selection-status {
1246   display: inline-flex;
1247   align-items: center;
1248   min-height: 22px;
1249   padding: 0 8px;
1250   border: 1px solid var(--app-border);
1251   border-radius: var(--radius-full);
1252   background: var(--bg-tertiary);
1253   color: var(--app-text-muted);
1254   font-size: 11px;
1255   font-weight: 800;
1256 }
1257
1258 .dataset-selection-status.selected {
1259   border-color: var(--success-badge-border);
1260   background: var(--success-badge-bg);
1261   color: var(--success-badge-text);
1262 }
1263
1264 .selection-control-heading {
1265   display: flex;
1266   align-items: center;
1267   justify-content: space-between;
1268   gap: 12px;
1269 }
1270
1271 .selection-control-heading h3 {
1272   margin: 0;
1273   color: var(--app-text-primary);
```

```
front/src/styles.css
```

```
1274   font-size: 14px;
1275   font-weight: 800;
1276 }
1277
1278 .selection-control-heading strong {
1279   flex: 0 0 auto;
1280   color: var(--app-text-primary);
1281   font-size: 18px;
1282   font-weight: 800;
1283   font-variant-numeric: tabular-nums;
1284 }
1285
1286 .selection-controls {
1287   display: grid;
1288   grid-template-columns: minmax(0, 1fr) minmax(132px, auto);
1289   gap: 12px;
1290   align-items: end;
1291 }
1292
1293 .selection-inputs {
1294   display: grid;
1295   grid-template-columns: repeat(2, minmax(0, 1fr));
1296   gap: 12px;
1297 }
1298
1299 .selection-inputs label,
1300 .task-form-grid label {
1301   min-width: 0;
1302   display: grid;
1303   gap: 6px;
1304 }
1305
1306 .selection-inputs input,
1307 .task-form-grid input {
1308   min-width: 0;
1309   height: 38px;
1310   padding: 0 10px;
1311   border: 1px solid var(--app-border);
1312   border-radius: var(--panel-radius);
1313   background: var(--app-surface-muted);
1314   color: var(--app-text-primary);
1315   font: inherit;
1316   font-size: 13px;
1317   font-weight: 700;
1318 }
1319
1320 .selection-controls > .btn {
1321   min-height: 38px;
1322   min-width: 132px;
1323   justify-content: center;
```

```
front/src/styles.css
```

```
1324 }
1325
1326 .selection-meter {
1327     display: grid;
1328     gap: 8px;
1329 }
1330
1331 .selection-meter-heading {
1332     display: flex;
1333     justify-content: space-between;
1334     gap: 12px;
1335 }
1336
1337 .selection-meter strong {
1338     color: var(--app-text-primary);
1339     font-variant-numeric: tabular-nums;
1340 }
1341
1342 .selected-base-panel {
1343     min-height: 0;
1344     display: grid;
1345     grid-template-rows: auto minmax(0, 1fr);
1346     gap: 8px;
1347     overflow: hidden;
1348 }
1349
1350 .selected-base-panel-heading {
1351     display: flex;
1352     align-items: center;
1353     justify-content: space-between;
1354     gap: 12px;
1355     color: var(--app-text-muted);
1356     font-size: 12px;
1357     font-weight: 700;
1358 }
1359
1360 .selected-base-panel-heading strong {
1361     color: var(--app-text-primary);
1362     font-variant-numeric: tabular-nums;
1363 }
1364
1365 .selected-base-grid {
1366     min-height: 0;
1367     max-height: 240px;
1368     overflow-x: hidden;
1369     overflow-y: auto;
1370     display: flex;
1371     flex-wrap: wrap;
1372     align-content: flex-start;
1373     gap: 6px;
```

```
front/src/styles.css
```

```
1374 padding: 10px;
1375 border: 1px solid var(--app-border);
1376 border-radius: var(--panel-radius);
1377 background: var(--bg-quinary);
1378 }
1379
1380 .selected-base-chip {
1381   display: inline-flex;
1382   align-items: center;
1383   min-width: 0;
1384   max-width: 100%;
1385   overflow: hidden;
1386   padding: 3px 4px 3px 8px;
1387   border-radius: var(--radius-full);
1388   color: var(--app-accent-dark);
1389   background: var(--app-accent-soft);
1390   font-size: 11px;
1391   font-weight: 800;
1392   line-height: 1.35;
1393   white-space: nowrap;
1394 }
1395
1396 .selected-base-remove {
1397   width: 20px;
1398   height: 20px;
1399   flex: 0 0 auto;
1400   display: inline-grid;
1401   place-items: center;
1402   margin-left: 3px;
1403   padding: 0;
1404   border: 0;
1405   border-radius: var(--radius-full);
1406   background: transparent;
1407   color: currentColor;
1408   cursor: pointer;
1409   transition: background-color 0.12s ease, color 0.12s ease;
1410 }
1411
1412 .selected-base-remove:hover {
1413   background: rgba(64, 158, 255, 0.16);
1414   color: var(--app-text-primary);
1415 }
1416
1417 .selected-base-remove:focus-visible {
1418   outline: 2px solid var(--primary-color);
1419   outline-offset: 1px;
1420 }
1421
1422 .selected-base-empty {
1423   align-self: flex-start;
```



```
front/src/styles.css
```

```
1424 padding: 6px 8px;
1425 border-radius: var(--radius-full);
1426 color: var(--app-text-muted) !important;
1427 background: var(--bg-tertiary) !important;
1428 font-size: 12px;
1429 font-weight: 700;
1430 }
1431
1432 /* ===== 数据管理 - 目录视图 ===== */
1433
1434 .dataset-catalog {
1435   display: flex;
1436   flex-direction: column;
1437   gap: 0;
1438 }
1439
1440 .dataset-toolbar {
1441   display: flex;
1442   align-items: center;
1443   gap: 12px;
1444   padding: 10px 14px;
1445   background: var(--app-surface);
1446   border: 1px solid var(--app-border);
1447   border-bottom: none;
1448   border-radius: var(--panel-radius) var(--panel-radius) 0 0;
1449 }
1450
1451 .dataset-toolbar-left {
1452   flex: 1;
1453   display: flex;
1454   align-items: center;
1455   gap: 8px;
1456   min-width: 0;
1457 }
1458
1459 .dataset-search-icon {
1460   flex: 0 0 auto;
1461   display: flex;
1462   color: var(--app-text-muted);
1463 }
1464
1465 .dataset-search-input {
1466   flex: 1;
1467   min-width: 0;
1468   height: 34px;
1469   padding: 0;
1470   border: none;
1471   outline: 2px solid transparent;
1472   outline-offset: 2px;
1473   background: transparent;
```

```
front/src/styles.css
```

```
1474   color: var(--app-text-primary);
1475   font-size: 13px;
1476 }
1477
1478 .dataset-search-input::placeholder {
1479   color: var(--app-text-muted);
1480 }
1481
1482 .dataset-search-input:focus-visible {
1483   outline-color: rgba(64, 158, 255, 0.42);
1484 }
1485
1486 .dataset-toolbar-center {
1487   flex: 0 0 auto;
1488 }
1489
1490 .dataset-type-select {
1491   width: 104px;
1492 }
1493
1494 .dataset-toolbar-right {
1495   flex: 0 0 auto;
1496   display: flex;
1497   align-items: center;
1498   gap: 8px;
1499 }
1500
1501 .dataset-table {
1502   border: 1px solid var(--app-border);
1503   border-radius: 0 0 var(--panel-radius) var(--panel-radius);
1504   background: var(--app-surface);
1505   overflow: hidden;
1506 }
1507
1508 .dataset-table-header {
1509   display: grid;
1510   grid-template-columns: minmax(180px, 2fr) minmax(140px, auto) minmax(140px, auto) minmax(130px, 1fr) minmax(230px,
     auto);
1511   gap: 8px;
1512   padding: 10px 14px;
1513   border-bottom: 1px solid var(--app-border);
1514   color: var(--app-text-muted);
1515   font-size: 11px;
1516   font-weight: 700;
1517   text-transform: uppercase;
1518   letter-spacing: 0.04em;
1519 }
1520
1521 .dataset-row {
1522   display: grid;
```

```
front/src/styles.css
```

```
1523     grid-template-columns: minmax(180px, 2fr) minmax(140px, auto) minmax(140px, auto) minmax(130px, 1fr) minmax(230px,
1524         auto);
1525     gap: 8px;
1526     padding: 12px 14px;
1527     align-items: center;
1527     border-bottom: 1px solid var(--app-border);
1528     cursor: default;
1529     transition: background-color 0.12s ease;
1530     outline: 2px solid transparent;
1531     outline-offset: -2px;
1532 }
1533
1534 .dataset-row:last-child {
1535     border-bottom: none;
1536 }
1537
1538 .dataset-row:hover,
1539 .dataset-row:focus-visible {
1540     background: var(--primary-8);
1541 }
1542
1543 .dataset-row:focus-visible {
1544     outline-color: var(--primary-color);
1545 }
1546
1547 .dataset-col-name {
1548     display: flex;
1549     flex-direction: column;
1550     gap: 2px;
1551     min-width: 0;
1552     overflow: hidden;
1553 }
1554
1555 .dataset-col-name strong {
1556     font-size: 15px;
1557     font-weight: 800;
1558     color: var(--app-text-primary);
1559     overflow: hidden;
1560     text-overflow: ellipsis;
1561     white-space: nowrap;
1562 }
1563
1564 .dataset-col-name small {
1565     font-size: 11px;
1566     color: var(--app-text-muted);
1567 }
1568
1569 .dataset-col-size {
1570     font-size: 13px;
1571     color: var(--app-text-primary);
```

```
front/src/styles.css
```

```
1572 }
1573
1574 .dataset-col-size code {
1575   font-family: 'SF Mono', 'Cascadia Code', 'Consolas', monospace;
1576   font-size: 12px;
1577   font-variant-numeric: tabular-nums;
1578   color: var(--app-text-regular);
1579 }
1580
1581 .dataset-col-status {
1582   display: flex;
1583   align-items: center;
1584   gap: 6px;
1585   font-size: 13px;
1586   color: var(--app-text-regular);
1587 }
1588
1589 .dataset-status-dot {
1590   width: 6px;
1591   height: 6px;
1592   border-radius: 50%;
1593   background: var(--text-muted);
1594   flex: 0 0 6px;
1595 }
1596
1597 .dataset-status-dot.has-labels {
1598   background: var(--success-color);
1599 }
1600
1601 .dataset-type-tag {
1602   display: inline-flex;
1603   align-items: center;
1604   height: 22px;
1605   padding: 0 9px;
1606   border-radius: 999px;
1607   border: 1px solid rgba(64, 158, 255, 0.24);
1608   background: rgba(64, 158, 255, 0.13);
1609   color: #1b65b9;
1610   font-size: 11px;
1611   font-weight: 700;
1612   margin-left: 2px;
1613 }
1614
1615 .dataset-col-usage {
1616   font-size: 13px;
1617   color: var(--app-text-regular);
1618 }
1619
1620 .dataset-col-actions {
1621   display: flex;
```

```
front/src/styles.css
```

```
1622     align-items: center;
1623     justify-content: flex-end;
1624     gap: 8px;
1625     min-width: 0;
1626 }
1627
1628 .dataset-usage-empty {
1629     color: var(--app-text-muted);
1630     font-size: 12px;
1631 }
1632
1633 .dataset-col-enter {
1634     text-align: right;
1635     font-size: 16px;
1636     color: var(--text-quaternary);
1637     line-height: 1;
1638 }
1639
1640 .dataset-table-empty {
1641     grid-column: 1 / -1;
1642     padding: 40px 14px;
1643     text-align: center;
1644     color: var(--app-text-muted);
1645     font-size: 13px;
1646 }
1647
1648 .dataset-list-grid {
1649     --dataset-list-columns: minmax(320px, 1.15fr) minmax(150px, 0.45fr) minmax(180px, 0.6fr) minmax(220px, 0.8fr) auto;
1650     display: flex;
1651     flex-direction: column;
1652     gap: 0;
1653     padding: 0;
1654     border: 1px solid var(--app-border);
1655     border-radius: 0 0 var(--panel-radius) var(--panel-radius);
1656     background: var(--app-surface);
1657     overflow: hidden;
1658 }
1659
1660 .dataset-list-header {
1661     display: grid;
1662     grid-template-columns: var(--dataset-list-columns);
1663     align-items: center;
1664     gap: 16px;
1665     padding: 10px 14px;
1666     border-bottom: 1px solid var(--app-border);
1667     background: var(--bg-quinary);
1668     color: var(--app-text-muted);
1669     font-size: 11px;
1670     font-weight: 700;
1671 }
```

front/src/styles.css

```
1672
1673 .dataset-list-header > span {
1674     justify-self: start;
1675     text-align: left;
1676 }
1677
1678 .dataset-list-item {
1679     min-width: 0;
1680     display: grid;
1681     grid-template-columns: var(--dataset-list-columns);
1682     align-items: center;
1683     gap: 16px;
1684     min-height: 68px;
1685     padding: 10px 14px;
1686     border: 0;
1687     border-bottom: 1px solid var(--app-border);
1688     background: var(--app-surface);
1689     cursor: default;
1690     transition: background-color 0.12s ease;
1691 }
1692
1693 .dataset-list-item:hover,
1694 .dataset-list-item:focus-within {
1695     background: var(--primary-8);
1696 }
1697
1698 .dataset-list-item-name {
1699     min-width: 0;
1700     display: grid;
1701     gap: 3px;
1702 }
1703
1704 .dataset-list-item-name strong {
1705     overflow: hidden;
1706     color: var(--app-text-primary);
1707     font-size: 15px;
1708     font-weight: 800;
1709     text-overflow: ellipsis;
1710     white-space: nowrap;
1711 }
1712
1713 .dataset-list-item-name small {
1714     color: var(--app-text-muted);
1715     font-size: 11px;
1716 }
1717
1718 .dataset-list-actions {
1719     flex: 0 0 auto;
1720     display: flex;
1721     align-items: center;
```

```
front/src/styles.css
```

```
1722     justify-content: flex-end;
1723     gap: 6px;
1724 }
1725
1726 .dataset-list-action {
1727     width: 32px;
1728     height: 32px;
1729     display: inline-grid;
1730     place-items: center;
1731     padding: 0;
1732     border: 1px solid var(--app-border);
1733     border-radius: var(--panel-radius);
1734     background: var(--app-surface-muted);
1735     color: var(--app-text-regular);
1736     cursor: pointer;
1737     transition: border-color 0.12s ease, background-color 0.12s ease, color 0.12s ease;
1738 }
1739
1740 .dataset-list-action:hover:not(:disabled) {
1741     border-color: rgba(64, 158, 255, 0.3);
1742     background: var(--app-accent-soft);
1743     color: var(--app-accent-dark);
1744 }
1745
1746 .dataset-list-action:focus-visible {
1747     outline: 2px solid var(--primary-color);
1748     outline-offset: 2px;
1749 }
1750
1751 .dataset-list-action.append {
1752     border-color: rgba(64, 158, 255, 0.24);
1753     background: rgba(64, 158, 255, 0.12);
1754     color: var(--app-accent-dark);
1755 }
1756
1757 .dataset-list-action.append:hover:not(:disabled) {
1758     border-color: rgba(64, 158, 255, 0.42);
1759     background: rgba(64, 158, 255, 0.2);
1760 }
1761
1762 .dataset-list-action.delete {
1763     border-color: color-mix(in srgb, var(--error-color) 28%, var(--app-border));
1764     background: color-mix(in srgb, var(--error-color) 8%, #ffffff);
1765     color: var(--error-color);
1766 }
1767
1768 .dataset-list-action.delete:hover:not(:disabled) {
1769     border-color: color-mix(in srgb, var(--error-color) 54%, var(--app-border));
1770     background: color-mix(in srgb, var(--error-color) 15%, #ffffff);
1771     color: #dc2626;
```

```
front/src/styles.css
```

```
1772 }
1773
1774 .dataset-list-action:disabled {
1775   cursor: not-allowed;
1776   opacity: 0.55;
1777 }
1778
1779 .dataset-list-metric {
1780   min-width: 0;
1781 }
1782
1783 .dataset-list-metric code {
1784   color: var(--app-text-regular);
1785   font-family: 'SF Mono', 'Cascadia Code', 'Consolas', monospace;
1786   font-size: 12px;
1787   font-variant-numeric: tabular-nums;
1788 }
1789
1790 .dataset-list-labels,
1791 .dataset-list-lifecycle,
1792 .dataset-card-statuses {
1793   min-width: 0;
1794   display: flex;
1795   align-items: center;
1796   gap: 6px;
1797   white-space: nowrap;
1798 }
1799
1800 .dataset-list-lifecycle {
1801   overflow: hidden;
1802 }
1803
1804 .dataset-quality-status,
1805 .dataset-usage-status {
1806   min-width: 0;
1807   display: inline-flex;
1808   align-items: center;
1809   min-height: 24px;
1810   padding: 0 8px;
1811   border: 1px solid var(--app-border);
1812   border-radius: var(--radius-full);
1813   color: var(--app-text-regular);
1814   background: var(--bg-tertiary);
1815   font-size: 11px;
1816   font-weight: 800;
1817   line-height: 1;
1818   white-space: nowrap;
1819 }
1820
1821 .dataset-list-lifecycle .dataset-quality-status {
```



```
front/src/styles.css
```

```
1822     flex: 0 0 auto;
1823 }
1824
1825 .dataset-list-lifecycle .dataset-usage-status {
1826     flex: 0 1 auto;
1827     overflow: hidden;
1828     text-overflow: ellipsis;
1829 }
1830
1831 .dataset-quality-status.ready {
1832     border-color: var(--success-badge-border);
1833     color: var(--success-badge-text);
1834     background: var(--success-badge-bg);
1835 }
1836
1837 .dataset-quality-status.warning {
1838     border-color: var(--warning-border);
1839     color: var(--warning-text);
1840     background: var(--warning-bg);
1841 }
1842
1843 .dataset-quality-status.error {
1844     border-color: color-mix(in srgb, var(--error-color) 34%, var(--app-border));
1845     color: var(--error-color);
1846     background: color-mix(in srgb, var(--error-color) 8%, #ffffff);
1847 }
1848
1849 .dataset-usage-status.used {
1850     border-color: rgba(64, 158, 255, 0.24);
1851     color: var(--app-accent-dark);
1852     background: var(--app-accent-soft);
1853 }
1854
1855 .dataset-usage-status.unused {
1856     color: var(--app-text-muted);
1857 }
1858
1859 .dataset-list-usage {
1860     min-width: 0;
1861     overflow: hidden;
1862     color: var(--app-text-regular);
1863     font-size: 12px;
1864     text-overflow: ellipsis;
1865     white-space: nowrap;
1866 }
1867
1868 .dataset-render-limit-note {
1869     grid-column: 1 / -1;
1870     padding: 10px 14px;
1871     border-top: 1px solid var(--app-border);
```

```
front/src/styles.css
```

```
1872   color: var(--app-text-muted);
1873   background: var(--bg-tertiary);
1874   font-size: 12px;
1875   text-align: center;
1876 }
1877
1878 .dataset-toolbar-segment {
1879   display: flex;
1880   gap: 0;
1881   border: 1px solid var(--app-border);
1882   border-radius: var(--panel-radius);
1883   overflow: hidden;
1884 }
1885
1886 .dataset-view-btn {
1887   display: flex;
1888   align-items: center;
1889   justify-content: center;
1890   width: 32px;
1891   height: 32px;
1892   padding: 0;
1893   border: none;
1894   background: transparent;
1895   color: var(--app-text-muted);
1896   cursor: pointer;
1897   transition: background-color 0.12s ease, color 0.12s ease;
1898 }
1899
1900 .dataset-view-btn + .dataset-view-btn {
1901   border-left: 1px solid var(--app-border);
1902 }
1903
1904 .dataset-view-btn:hover {
1905   background: var(--primary-8);
1906   color: var(--app-text-primary);
1907 }
1908
1909 .dataset-view-btn.active {
1910   background: var(--primary-color);
1911   color: #ffffff;
1912 }
1913
1914 .dataset-card-grid {
1915   display: grid;
1916   grid-template-columns: repeat(4, minmax(0, 1fr));
1917   gap: 14px;
1918   padding: 14px;
1919   border: 1px solid var(--app-border);
1920   border-radius: 0 0 var(--panel-radius) var(--panel-radius);
1921   background: color-mix(in srgb, var(--app-surface) 84%, var(--bg-tertiary));
```

```
front/src/styles.css
```

```
1922 }
1923
1924 .dataset-card {
1925     display: flex;
1926     flex-direction: column;
1927     gap: 14px;
1928     min-height: 248px;
1929     padding: 16px 16px 14px;
1930     border: 1px solid rgba(148, 163, 184, 0.26);
1931     border-radius: var(--panel-radius);
1932     background: linear-gradient(180deg, #ffffff 0%, #fbfdff 100%);
1933     cursor: default;
1934     outline: 2px solid transparent;
1935     outline-offset: 2px;
1936     transition:
1937         border-color 0.16s ease,
1938         box-shadow 0.16s ease,
1939         transform 0.16s ease;
1940 }
1941
1942 .dataset-card:hover,
1943 .dataset-card:focus-visible {
1944     border-color: rgba(64, 158, 255, 0.28);
1945     box-shadow: 0 4px 8px rgba(15, 23, 42, 0.07);
1946     transform: translateY(-1px);
1947 }
1948
1949 .dataset-card:focus-visible {
1950     outline-color: var(--primary-color);
1951 }
1952
1953 .dataset-card-top {
1954     display: flex;
1955     align-items: flex-start;
1956     justify-content: space-between;
1957     gap: 12px;
1958     min-width: 0;
1959     padding-bottom: 11px;
1960     border-bottom: 1px solid var(--app-border);
1961 }
1962
1963 .dataset-card-title-block {
1964     display: flex;
1965     flex-direction: column;
1966     gap: 4px;
1967     min-width: 0;
1968 }
1969
1970 .dataset-card-title-block strong {
1971     display: -webkit-box;
```

```
front/src/styles.css
```

```
1972     overflow: hidden;
1973     color: var(--app-text-primary);
1974     font-size: 17px;
1975     font-weight: 800;
1976     line-height: 1.32;
1977     text-overflow: ellipsis;
1978     -webkit-line-clamp: 2;
1979     -webkit-box-orient: vertical;
1980 }
1981
1982 .dataset-card-title-block small {
1983     overflow: hidden;
1984     color: var(--app-text-muted);
1985     font-size: 11px;
1986     font-weight: 600;
1987     text-overflow: ellipsis;
1988     white-space: nowrap;
1989 }
1990
1991 .dataset-card-type {
1992     flex: 0 0 auto;
1993     margin-left: 0;
1994 }
1995
1996 .dataset-card-body {
1997     flex: 1;
1998     display: flex;
1999     flex-direction: column;
2000     align-items: stretch;
2001     gap: 10px;
2002 }
2003
2004 .dataset-card-metrics {
2005     width: 100%;
2006     display: grid;
2007     grid-template-columns: repeat(3, minmax(0, 1fr));
2008     gap: 10px;
2009 }
2010
2011 .dataset-card-facts {
2012     display: grid;
2013     grid-template-columns: repeat(3, minmax(0, 1fr));
2014     gap: 8px;
2015 }
2016
2017 .dataset-card-fact {
2018     min-width: 0;
2019     display: flex;
2020     flex-direction: column;
2021     gap: 4px;
```

```
front/src/styles.css
```

```
2022   padding: 9px 10px;
2023   border-radius: var(--panel-radius);
2024   background: var(--bg-tertiary);
2025 }
2026
2027 .dataset-card-fact span {
2028   color: var(--app-text-muted);
2029   font-size: 10px;
2030   font-weight: 700;
2031 }
2032
2033 .dataset-card-fact strong {
2034   overflow: hidden;
2035   color: var(--app-text-primary);
2036   font-size: 12px;
2037   font-weight: 800;
2038   text-overflow: ellipsis;
2039   white-space: nowrap;
2040 }
2041
2042 .dataset-card-metric {
2043   display: flex;
2044   flex-direction: column;
2045   gap: 8px;
2046   min-width: 0;
2047   padding: 10px;
2048   border: 1px solid rgba(34, 197, 94, 0.16);
2049   border-radius: var(--panel-radius);
2050   background: color-mix(in srgb, var(--success-color) 7%, #ffffff);
2051 }
2052
2053 .dataset-card-metric-head {
2054   display: flex;
2055   align-items: center;
2056   justify-content: space-between;
2057   gap: 8px;
2058 }
2059
2060 .dataset-card-metric-head span {
2061   color: #3f7a36;
2062   font-size: 12px;
2063   font-weight: 800;
2064   letter-spacing: 0;
2065 }
2066
2067 .dataset-card-metric-head strong {
2068   min-width: 0;
2069   overflow: hidden;
2070   color: var(--app-text-primary);
2071   font-size: 16px;
```

```
front/src/styles.css
```

```
2072   font-weight: 800;
2073   font-variant-numeric: tabular-nums;
2074   line-height: 1;
2075   text-overflow: ellipsis;
2076 }
2077
2078 .dataset-metric-bar {
2079   width: 100%;
2080   height: 5px;
2081   overflow: hidden;
2082   border-radius: var(--radius-full);
2083   background: color-mix(in srgb, var(--success-color) 13%, #ffffff);
2084 }
2085
2086 .dataset-metric-fill {
2087   display: block;
2088   height: 100%;
2089   border-radius: inherit;
2090   background: #63c354;
2091   box-shadow: inset 0 0 0 1px color-mix(in srgb, #ffffff 28%, transparent);
2092 }
2093
2094 .dataset-label-tag {
2095   display: inline-flex;
2096   align-items: center;
2097   gap: 6px;
2098   height: 24px;
2099   padding: 0 9px;
2100   border: 1px solid var(--app-border);
2101   border-radius: var(--radius-full);
2102   background: var(--bg-tertiary);
2103   color: var(--app-text-regular);
2104   font-size: 12px;
2105   font-weight: 700;
2106 }
2107
2108 .dataset-label-tag.has-labels {
2109   border-color: var(--success-badge-border);
2110   background: var(--success-badge-bg);
2111   color: var(--success-badge-text);
2112 }
2113
2114 .dataset-card-footer {
2115   display: flex;
2116   align-items: center;
2117   justify-content: space-between;
2118   gap: 12px;
2119   padding-top: 12px;
2120   border-top: 1px solid var(--app-border);
2121 }
```

```
front/src/styles.css
```

```
2122
2123 .dataset-card-statuses {
2124     flex: 1 1 auto;
2125 }
2126
2127 .dataset-card-actions {
2128     display: flex;
2129     align-items: center;
2130     justify-content: flex-end;
2131     gap: 6px;
2132     flex: 0 0 auto;
2133 }
2134
2135 .dataset-card-task-count {
2136     margin-left: auto;
2137     color: var(--app-text-muted);
2138 }
2139
2140 .dataset-upload-error {
2141     margin-bottom: 12px;
2142     padding: 10px 14px;
2143     border: 1px solid var(--error-color);
2144     border-radius: var(--panel-radius);
2145     background: color-mix(in srgb, var(--error-color) 8%, transparent);
2146     color: var(--error-color);
2147     font-size: 13px;
2148 }
2149
2150 /* ===== 数据管理 - 详情视图 ===== */
2151
2152 .dataset-detail-shell {
2153     display: flex;
2154     flex-direction: column;
2155     gap: 14px;
2156 }
2157
2158 .dataset-detail-hero {
2159     display: grid;
2160     grid-template-columns: minmax(0, 1fr) auto;
2161     align-items: center;
2162     gap: 16px;
2163     padding: 14px 16px;
2164     border: 1px solid var(--app-border);
2165     border-radius: var(--panel-radius);
2166     background: var(--app-surface);
2167 }
2168
2169 .dataset-detail-title-row {
2170     min-width: 0;
2171     display: flex;
```

```
front/src/styles.css
```

```
2172     align-items: center;
2173     gap: 12px;
2174 }
2175
2176 .dataset-detail-back {
2177     flex: 0 0 auto;
2178     display: inline-flex;
2179     align-items: center;
2180     gap: 4px;
2181     min-height: 32px;
2182     padding: 0 10px;
2183     border: 1px solid var(--app-border);
2184     border-radius: var(--panel-radius);
2185     background: var(--app-surface-muted);
2186     color: var(--app-accent);
2187     font-size: 12px;
2188     font-weight: 800;
2189     cursor: pointer;
2190     transition:
2191         background-color 0.12s ease,
2192         border-color 0.12s ease,
2193         color 0.12s ease;
2194 }
2195
2196 .dataset-detail-back:hover {
2197     border-color: rgba(64, 158, 255, 0.28);
2198     background: var(--primary-8);
2199     color: var(--app-accent-dark);
2200 }
2201
2202 .dataset-detail-title-block {
2203     min-width: 0;
2204 }
2205
2206 .dataset-detail-identity {
2207     min-width: 0;
2208     display: flex;
2209     flex-wrap: wrap;
2210     align-items: center;
2211     row-gap: 8px;
2212 }
2213
2214 .dataset-detail-title-block h2 {
2215     min-width: 0;
2216     margin: 0;
2217     padding-right: 12px;
2218     color: var(--app-text-primary);
2219     font-size: 20px;
2220     font-weight: 800;
2221     line-height: 1.25;
```



```
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```

```
2222   overflow-wrap: anywhere;
2223 }
2224
2225 .dataset-detail-meta {
2226   min-width: 0;
2227   display: flex;
2228   flex-wrap: wrap;
2229   align-items: center;
2230   gap: 0;
2231   min-height: 22px;
2232   padding-left: 12px;
2233   border-left: 1px solid rgba(64, 158, 255, 0.72);
2234   color: var(--app-text-muted);
2235   font-size: 12px;
2236   font-weight: 700;
2237 }
2238
2239 .dataset-detail-meta > span {
2240   display: inline-flex;
2241   align-items: center;
2242   min-height: 22px;
2243   white-space: nowrap;
2244 }
2245
2246 .dataset-detail-meta > span + span::before {
2247   content: ' ';
2248   position: absolute;
2249   top: 50%;
2250   left: -10px;
2251   width: 1px;
2252   height: 14px;
2253   flex: 0 0 1px;
2254   background: rgba(64, 158, 255, 0.62);
2255   transform: translateY(-50%);
2256 }
2257
2258 .dataset-detail-meta > span + span {
2259   position: relative;
2260   margin-left: 20px;
2261 }
2262
2263 .dataset-detail-meta .dataset-label-tag,
2264 .dataset-detail-meta .dataset-quality-status {
2265   min-height: 22px;
2266   height: 22px;
2267   padding: 0 8px;
2268   font-size: 11px;
2269 }
2270
2271 .dataset-detail-actions {
```

```
front/src/styles.css
```

```
2272   display: flex;
2273   align-items: center;
2274   justify-content: flex-end;
2275   gap: 8px;
2276 }
2277
2278 .dataset-detail-actions .btn {
2279   min-height: 34px;
2280   justify-content: center;
2281   white-space: nowrap;
2282 }
2283
2284 .dataset-detail-layout {
2285   display: flex;
2286   flex-direction: column;
2287   gap: 14px;
2288 }
2289
2290 .dataset-detail-main {
2291   display: flex;
2292   flex-direction: column;
2293   gap: 14px;
2294   min-width: 0;
2295 }
2296
2297 .dataset-quality-note {
2298   display: flex;
2299   align-items: flex-start;
2300   gap: 8px;
2301   padding: 10px 12px;
2302   border: 1px solid var(--warning-border);
2303   border-radius: var(--panel-radius);
2304   color: var(--warning-text);
2305   background: var(--warning-bg);
2306   font-size: 12px;
2307   font-weight: 700;
2308   line-height: 1.55;
2309 }
2310
2311 .dataset-quality-note.error {
2312   border-color: color-mix(in srgb, var(--error-color) 34%, var(--app-border));
2313   color: var(--error-color);
2314   background: color-mix(in srgb, var(--error-color) 8%, #ffffff);
2315 }
2316
2317 .dataset-overview-grid {
2318   display: grid;
2319   grid-template-columns: repeat(4, minmax(0, 1fr));
2320   gap: 10px;
2321 }
```

```
front/src/styles.css
```

```
2322
2323 .dataset-overview-card {
2324   min-width: 0;
2325   display: grid;
2326   gap: 5px;
2327   padding: 12px 14px;
2328   border: 1px solid var(--app-border);
2329   border-radius: var(--panel-radius);
2330   background: var(--app-surface);
2331 }
2332
2333 .dataset-overview-card span,
2334 .dataset-overview-card small {
2335   color: var(--app-text-muted);
2336   font-size: 12px;
2337   font-weight: 700;
2338 }
2339
2340 .dataset-overview-card strong {
2341   color: var(--app-text-primary);
2342   font-size: 22px;
2343   font-weight: 800;
2344   line-height: 1;
2345   font-variant-numeric: tabular-nums;
2346 }
2347
2348 .dataset-detail-workspace {
2349   display: grid;
2350   gap: 14px;
2351 }
2352
2353 .dataset-detail-lower-grid {
2354   display: grid;
2355   grid-template-columns: minmax(0, 1.42fr) minmax(360px, 0.58fr);
2356   gap: 14px;
2357   align-items: stretch;
2358 }
2359
2360 .dataset-detail-section {
2361   display: flex;
2362   flex-direction: column;
2363   gap: 12px;
2364   min-width: 0;
2365   padding: 14px;
2366   border: 1px solid var(--app-border);
2367   border-radius: var(--panel-radius);
2368   background: var(--app-surface);
2369 }
2370
2371 .dataset-section-heading {
```

```
front/src/styles.css
```

```
2372   display: flex;
2373   justify-content: space-between;
2374   gap: 16px;
2375   min-width: 0;
2376 }
2377
2378 .dataset-section-heading h3 {
2379   margin: 0;
2380   font-size: 14px;
2381   font-weight: 800;
2382   color: var(--app-text-primary);
2383 }
2384
2385 .dataset-section-heading p {
2386   margin: 4px 0 0;
2387   color: var(--app-text-muted);
2388   font-size: 12px;
2389   font-weight: 700;
2390 }
2391
2392 .dataset-empty-state {
2393   display: grid;
2394   place-items: center;
2395   min-height: 86px;
2396   padding: 14px;
2397   border: 1px dashed var(--app-border);
2398   border-radius: var(--panel-radius);
2399   background: var(--bg-quinary);
2400   color: var(--app-text-muted);
2401   font-size: 13px;
2402   font-weight: 700;
2403 }
2404
2405 .dataset-selection-panel {
2406   min-height: 0;
2407   height: 100%;
2408   align-self: stretch;
2409   display: grid;
2410   grid-template-rows: auto auto auto minmax(0, 1fr);
2411   overflow: hidden;
2412 }
2413
2414 @keyframes dataset-rename-enter {
2415   from {
2416     opacity: 0;
2417     transform: translateY(8px) scale(0.98);
2418   }
2419
2420   to {
2421     opacity: 1;
```

```
front/src/styles.css
```

```
2422     transform: translateY(0) scale(1);
2423   }
2424 }
2425
2426 .dataset-rename-modal {
2427   position: fixed;
2428   inset: 0;
2429   z-index: 60;
2430   display: grid;
2431   place-items: center;
2432   padding: 24px;
2433   overflow: auto;
2434   background: rgba(15, 23, 42, 0.22);
2435   backdrop-filter: blur(6px);
2436   -webkit-backdrop-filter: blur(6px);
2437 }
2438
2439 .dataset-delete-modal {
2440   position: fixed;
2441   inset: 0;
2442   z-index: 60;
2443   display: grid;
2444   place-items: center;
2445   padding: 24px;
2446   overflow: auto;
2447   background: rgba(15, 23, 42, 0.28);
2448   backdrop-filter: blur(6px);
2449   -webkit-backdrop-filter: blur(6px);
2450 }
2451
2452 .dataset-rename-dialog {
2453   width: min(560px, 100%);
2454   display: grid;
2455   gap: 16px;
2456   padding: 20px;
2457   border: 1px solid var(--app-border-strong);
2458   border-radius: var(--panel-radius);
2459   background: linear-gradient(180deg, rgba(255, 255, 255, 0.98), rgba(248, 250, 252, 0.96));
2460   box-shadow: 0 6px 14px rgba(15, 23, 42, 0.12);
2461   animation: dataset-rename-enter 0.16s ease-out both;
2462 }
2463
2464 .dataset-rename-header {
2465   display: flex;
2466   align-items: flex-start;
2467   gap: 12px;
2468 }
2469
2470 .dataset-rename-icon {
2471   width: 42px;
```

```
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```

```
2472   height: 42px;
2473   flex: 0 0 42px;
2474   display: grid;
2475   place-items: center;
2476   border-radius: var(--panel-radius);
2477   color: var(--app-accent-dark);
2478   background: var(--app-accent-soft);
2479 }
2480
2481 .dataset-rename-copy {
2482   min-width: 0;
2483   display: grid;
2484   gap: 4px;
2485 }
2486
2487 .dataset-rename-copy h3 {
2488   margin: 0;
2489   color: var(--app-text-primary);
2490   font-size: 18px;
2491   font-weight: 800;
2492 }
2493
2494 .dataset-rename-copy p {
2495   margin: 0;
2496   color: var(--app-text-muted);
2497   font-size: 13px;
2498   font-weight: 700;
2499   line-height: 1.6;
2500 }
2501
2502 .dataset-rename-summary {
2503   display: flex;
2504   flex-wrap: wrap;
2505   gap: 8px;
2506 }
2507
2508 .dataset-rename-summary > span:not(.dataset-label-tag) {
2509   display: inline-flex;
2510   align-items: center;
2511   min-height: 24px;
2512   padding: 0 9px;
2513   border: 1px solid var(--app-border);
2514   border-radius: var(--radius-full);
2515   background: var(--bg-quinary);
2516   color: var(--app-text-muted);
2517   font-size: 12px;
2518   font-weight: 700;
2519 }
2520
2521 .dataset-rename-form {
```

```
front/src/styles.css
```

```
2522   display: grid;
2523   gap: 12px;
2524 }
2525
2526 .dataset-rename-field {
2527   display: grid;
2528   gap: 6px;
2529 }
2530
2531 .dataset-rename-field > span {
2532   color: var(--app-text-regular);
2533   font-size: 12px;
2534   font-weight: 700;
2535 }
2536
2537 .dataset-rename-input {
2538   width: 100%;
2539   min-width: 0;
2540   height: 42px;
2541   padding: 0 12px;
2542   border: 1px solid var(--app-border-strong);
2543   border-radius: var(--panel-radius);
2544   background: var(--app-surface-muted);
2545   color: var(--app-text-primary);
2546   font: inherit;
2547   font-size: 14px;
2548   font-weight: 700;
2549   transition:
2550     border-color 0.16s ease,
2551     background-color 0.16s ease,
2552     box-shadow 0.16s ease;
2553 }
2554
2555 .dataset-rename-input:hover {
2556   border-color: rgba(64, 158, 255, 0.28);
2557   background: var(--app-surface);
2558 }
2559
2560 .dataset-rename-input:focus-visible {
2561   outline: none;
2562   border-color: rgba(64, 158, 255, 0.46);
2563   background: var(--app-surface);
2564   box-shadow: 0 0 0 3px rgba(64, 158, 255, 0.16);
2565 }
2566
2567 .dataset-rename-input[aria-invalid='true'] {
2568   border-color: var(--error-color);
2569   box-shadow: 0 0 0 3px color-mix(in srgb, var(--error-color) 12%, transparent);
2570 }
2571
```

```
front/src/styles.css
```

```
2572 .dataset-rename-error {
2573   margin: 0;
2574   color: var(--error-color);
2575   font-size: 12px;
2576   font-weight: 700;
2577 }
2578
2579 .dataset-rename-actions {
2580   display: flex;
2581   justify-content: flex-end;
2582   gap: 8px;
2583 }
2584
2585 .dataset-rename-actions .btn {
2586   min-width: 88px;
2587 }
2588
2589 .dataset-delete-dialog {
2590   width: min(480px, 100%);
2591   display: grid;
2592   gap: 16px;
2593   padding: 20px;
2594   border: 1px solid var(--app-border-strong);
2595   border-radius: var(--panel-radius);
2596   background: var(--app-surface);
2597   box-shadow: 0 6px 14px rgba(15, 23, 42, 0.12);
2598   animation: dataset-rename-enter 0.16s ease-out both;
2599 }
2600
2601 .dataset-delete-header {
2602   display: flex;
2603   align-items: flex-start;
2604   gap: 12px;
2605 }
2606
2607 .dataset-delete-icon {
2608   width: 42px;
2609   height: 42px;
2610   flex: 0 0 42px;
2611   display: grid;
2612   place-items: center;
2613   border-radius: var(--panel-radius);
2614   color: var(--error-color);
2615   background: color-mix(in srgb, var(--error-color) 10%, #ffffff);
2616 }
2617
2618 .dataset-delete-copy {
2619   min-width: 0;
2620   display: grid;
2621   gap: 4px;
```


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```
2622 }
2623
2624 .dataset-delete-copy h3 {
2625   margin: 0;
2626   color: var(--app-text-primary);
2627   font-size: 18px;
2628   font-weight: 800;
2629 }
2630
2631 .dataset-delete-copy p {
2632   margin: 0;
2633   color: var(--app-text-muted);
2634   font-size: 13px;
2635   font-weight: 700;
2636   line-height: 1.6;
2637 }
2638
2639 .dataset-delete-target {
2640   display: grid;
2641   gap: 4px;
2642   padding: 12px;
2643   border: 1px solid color-mix(in srgb, var(--error-color) 24%, var(--app-border));
2644   border-radius: var(--panel-radius);
2645   background: color-mix(in srgb, var(--error-color) 5%, #ffffff);
2646 }
2647
2648 .dataset-delete-target span,
2649 .dataset-delete-target small {
2650   color: var(--app-text-muted);
2651   font-size: 12px;
2652   font-weight: 700;
2653 }
2654
2655 .dataset-delete-target strong {
2656   overflow-wrap: anywhere;
2657   color: var(--app-text-primary);
2658   font-size: 14px;
2659   font-weight: 800;
2660 }
2661
2662 .dataset-delete-error {
2663   margin: 0;
2664   color: var(--error-color);
2665   font-size: 12px;
2666   font-weight: 700;
2667 }
2668
2669 .dataset-delete-actions {
2670   display: flex;
2671   justify-content: flex-end;
```

```
front/src/styles.css
```

```
2672   gap: 8px;
2673 }
2674
2675 .dataset-delete-actions .btn {
2676   min-width: 96px;
2677 }
2678
2679 .dataset-delete-confirm {
2680   border-color: var(--error-color);
2681   background: var(--error-color);
2682   color: #ffffff;
2683 }
2684
2685 .dataset-delete-confirm:hover:not(:disabled) {
2686   border-color: #dc2626;
2687   background: #dc2626;
2688 }
2689
2690 @media (prefers-reduced-motion: reduce) {
2691   .dataset-rename-dialog,
2692   .dataset-delete-dialog {
2693     animation: none;
2694   }
2695 }
2696
2697 /* ===== 按钮变体 ===== */
2698
2699 .btn.btn-danger {
2700   border-color: var(--error-color);
2701   color: var(--error-color);
2702   background: transparent;
2703 }
2704
2705 .btn.btn-danger:hover {
2706   background: color-mix(in srgb, var(--error-color) 10%, transparent);
2707 }
2708
2709 .task-form-grid {
2710   display: grid;
2711   grid-template-columns: repeat(3, minmax(0, 1fr));
2712   gap: 10px;
2713 }
2714
2715 .task-form-wide,
2716 .task-mode-control {
2717   grid-column: 1 / -1;
2718 }
2719
2720 .task-mode-control {
2721   display: grid;
```

```
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```

```
2722   grid-template-columns: repeat(2, minmax(0, 1fr));
2723   gap: 8px;
2724 }
2725
2726 .task-mode-control button {
2727   height: 38px;
2728   border: 1px solid var(--app-border);
2729   border-radius: var(--panel-radius);
2730   color: var(--app-text-regular);
2731   background: var(--app-surface-muted);
2732   font: inherit;
2733   font-size: 13px;
2734   font-weight: 800;
2735   cursor: pointer;
2736 }
2737
2738 .task-mode-control button.active {
2739   color: #ffffff;
2740   border-color: var(--app-accent);
2741   background: var(--app-accent);
2742 }
2743
2744 .dataset-create-task {
2745   width: 100%;
2746   margin-top: 14px;
2747 }
2748
2749 .task-draft-card {
2750   margin-top: 14px;
2751   padding: 12px;
2752   border: 1px solid var(--primary-30);
2753   border-radius: var(--panel-radius);
2754   background: linear-gradient(135deg, var(--primary-10), rgba(255, 255, 255, 0.76));
2755 }
2756
2757 .task-draft-card > div:first-child {
2758   display: grid;
2759   gap: 4px;
2760   margin-bottom: 10px;
2761 }
2762
2763 .task-draft-card strong {
2764   font-size: 15px;
2765 }
2766
2767 .task-draft-card dl {
2768   display: grid;
2769   grid-template-columns: repeat(2, minmax(0, 1fr));
2770   gap: 8px;
2771   margin: 0;
```

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```
2772 }
2773
2774 .task-draft-card dl div {
2775     min-width: 0;
2776     display: grid;
2777     gap: 3px;
2778 }
2779
2780 .task-draft-card dd {
2781     min-width: 0;
2782     margin: 0;
2783     overflow: hidden;
2784     color: var(--app-text-primary);
2785     font-size: 12px;
2786     font-weight: 800;
2787     text-overflow: ellipsis;
2788     white-space: nowrap;
2789 }
2790
2791 .dataset-task-note {
2792     margin-top: 12px;
2793     display: flex;
2794     align-items: flex-start;
2795     gap: 8px;
2796     color: var(--app-text-muted);
2797     font-size: 12px;
2798     line-height: 1.5;
2799 }
2800
2801 .dataset-task-note svg {
2802     flex: 0 0 auto;
2803     color: var(--warning);
2804 }
2805
2806 .panel {
2807     min-width: 0;
2808     padding: 16px;
2809 }
2810
2811 .panel-header {
2812     display: flex;
2813     align-items: flex-start;
2814     justify-content: space-between;
2815     gap: 12px;
2816     margin-bottom: 14px;
2817 }
2818
2819 .panel-header h2 {
2820     margin: 0 0 5px;
2821     font-size: 16px;
```

```
front/src/styles.css
```

```
2822   font-weight: 700;
2823   line-height: 1.25;
2824 }
2825
2826 .panel-action {
2827   width: 32px;
2828   height: 32px;
2829   flex: 0 0 32px;
2830   display: inline-flex;
2831   align-items: center;
2832   justify-content: center;
2833   border: 1px solid var(--app-border);
2834   border-radius: var(--panel-radius);
2835   color: var(--app-accent-dark);
2836   background: var(--app-surface-muted);
2837   cursor: pointer;
2838   transition:
2839     background-color 0.16s ease,
2840     border-color 0.16s ease,
2841     color 0.16s ease;
2842 }
2843
2844 .panel-action:hover {
2845   border-color: rgba(64, 158, 255, 0.28);
2846   background: var(--app-accent-soft);
2847 }
2848
2849 .selectable-chart-header {
2850   align-items: flex-start;
2851 }
2852
2853 .chart-segmented {
2854   flex: 0 0 auto;
2855   display: inline-flex;
2856   gap: 2px;
2857   padding: 3px;
2858   border: 1px solid var(--app-border);
2859   border-radius: var(--panel-radius);
2860   background: var(--app-surface-muted);
2861 }
2862
2863 .chart-segment {
2864   display: inline-flex;
2865   align-items: center;
2866   gap: 6px;
2867   height: 26px;
2868   padding: 0 12px;
2869   border: 1px solid transparent;
2870   border-radius: 5px;
2871   color: var(--app-text-muted);
```

```
front/src/styles.css
```

```
2872   background: transparent;
2873   font-size: 12px;
2874   font-weight: 700;
2875   cursor: pointer;
2876   transition:
2877     background-color 0.16s ease,
2878     border-color 0.16s ease,
2879     color 0.16s ease;
2880 }
2881
2882 .chart-segment:hover:not(.active) {
2883   color: var(--app-text-primary);
2884   background: rgba(248, 250, 252, 0.55);
2885 }
2886
2887 .chart-segment.active {
2888   color: var(--app-accent-dark);
2889   background: #ffffff;
2890   border-color: rgba(64, 158, 255, 0.28);
2891   box-shadow: 0 1px 3px rgba(15, 23, 42, 0.08);
2892 }
2893
2894 .selectable-chart-panel .chart-surface {
2895   height: 252px;
2896 }
2897
2898 .chart-surface {
2899   width: 100%;
2900   height: 210px;
2901 }
2902
2903 .ca-matrix-panel {
2904   min-height: 360px;
2905   display: flex;
2906   flex-direction: column;
2907 }
2908
2909 .ca-panel-header {
2910   display: flex;
2911   align-items: flex-start;
2912   justify-content: space-between;
2913   gap: 16px;
2914   margin-bottom: 16px;
2915 }
2916
2917 .ca-panel-header h2 {
2918   margin: 0 0 8px;
2919   font-size: 18px;
2920   line-height: 1.25;
2921 }
```

```
front/src/styles.css
```

```
2922
2923 .ca-panel-header span {
2924     color: var(--app-text-muted);
2925     font-size: 13px;
2926 }
2927
2928 .ca-formula {
2929     flex: 0 0 auto;
2930     padding: 8px 10px;
2931     border: 1px solid var(--primary-30);
2932     border-radius: var(--panel-radius);
2933     color: var(--app-accent-dark);
2934     background: var(--app-accent-soft);
2935     font-family: ui-monospace, SFMono-Regular, Consolas, 'Liberation Mono', monospace;
2936     font-size: 12px;
2937     font-weight: 700;
2938     white-space: nowrap;
2939 }
2940
2941 .ca-panel-body {
2942     flex: 1;
2943     min-height: 0;
2944     display: grid;
2945     grid-template-columns: minmax(250px, 1fr) minmax(190px, 0.78fr);
2946     gap: 16px;
2947 }
2948
2949 .ca-visual-column {
2950     min-width: 0;
2951     display: flex;
2952     flex-direction: column;
2953 }
2954
2955 .ca-matrix-scroll {
2956     flex: 1;
2957     overflow: visible;
2958     min-height: 270px;
2959     padding: 14px 0 8px;
2960     display: flex;
2961     align-items: center;
2962     justify-content: center;
2963 }
2964
2965 .ca-matrix-preview {
2966     width: max-content;
2967     display: grid;
2968     gap: 5px;
2969     perspective: 720px;
2970 }
2971
```

```
front/src/styles.css
```

```
2972 .ca-axis-x {
2973     display: grid;
2974     grid-template-columns: 28px repeat(8, 30px);
2975     gap: 5px;
2976 }
2977
2978 .ca-axis-x span,
2979 .ca-axis-y {
2980     color: var(--app-text-muted);
2981     font-size: 11px;
2982     line-height: 1;
2983     text-align: center;
2984 }
2985
2986 .ca-row {
2987     display: grid;
2988     grid-template-columns: 28px max-content;
2989     gap: 5px;
2990     align-items: center;
2991 }
2992
2993 .ca-cells {
2994     display: grid;
2995     grid-template-columns: repeat(8, 30px);
2996     gap: 5px;
2997     perspective: 720px;
2998 }
2999
3000 .ca-cell {
3001     width: 30px;
3002     height: 30px;
3003     display: grid;
3004     place-items: center;
3005     border-radius: 6px;
3006     color: rgba(15, 23, 42, 0.72);
3007     font-size: 10px;
3008     font-weight: 700;
3009     font-variant-numeric: tabular-nums;
3010 }
3011
3012 .ca-matrix-cell {
3013     --tilt-x: 0deg;
3014     --tilt-y: 0deg;
3015     position: relative;
3016     isolation: isolate;
3017     cursor: default;
3018     transform: translate3d(0, 0, 0) rotateX(0deg) rotateY(0deg) scale(1);
3019     transform-style: preserve-3d;
3020     transition:
3021         transform 0.24s cubic-bezier(0.16, 1, 0.3, 1),
```



```
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```

```
3022     box-shadow 0.24s ease,
3023     filter 0.24s ease;
3024 }
3025
3026 .ca-matrix-cell:hover {
3027     z-index: 12;
3028     filter: saturate(1.12);
3029     transform:
3030         translateY(-3px)
3031         translateZ(14px)
3032         rotateX(var(--tilt-x))
3033         rotateY(var(--tilt-y))
3034         scale(1.1);
3035     box-shadow:
3036         0 14px 22px rgba(15, 23, 42, 0.16),
3037         0 6px 12px rgba(64, 158, 255, 0.16),
3038         inset 0 1px 0 rgba(255, 255, 255, 0.38);
3039 }
3040
3041 .ca-tooltip {
3042     position: absolute;
3043     left: 50%;
3044     bottom: calc(100% + 12px);
3045     z-index: 24;
3046     min-width: 104px;
3047     padding: 8px 10px;
3048     border: 1px solid rgba(148, 163, 184, 0.24);
3049     border-radius: 8px;
3050     display: grid;
3051     gap: 2px;
3052     color: var(--app-text-primary);
3053     background: #ffffff;
3054     box-shadow: 0 14px 28px rgba(15, 23, 42, 0.16);
3055     opacity: 0;
3056     pointer-events: none;
3057     transform: translate(-50%, 7px) scale(0.96);
3058     transform-origin: 50% 100%;
3059     transform-style: flat;
3060     transition:
3061         opacity 0.18s ease,
3062         transform 0.22s cubic-bezier(0.16, 1, 0.3, 1);
3063     white-space: nowrap;
3064 }
3065
3066 .ca-tooltip::after {
3067     position: absolute;
3068     left: 50%;
3069     bottom: -5px;
3070     width: 10px;
3071     height: 10px;
```

```
front/src/styles.css
```

```
3072   border-right: 1px solid rgba(148, 163, 184, 0.24);
3073   border-bottom: 1px solid rgba(148, 163, 184, 0.24);
3074   background: #ffffff;
3075   content: '';
3076   transform: translateX(-50%) rotate(45deg);
3077 }
3078
3079 .ca-tooltip strong,
3080 .ca-tooltip small,
3081 .ca-tooltip em {
3082   position: relative;
3083   z-index: 1;
3084 }
3085
3086 .ca-tooltip strong {
3087   font-size: 12px;
3088   line-height: 1.2;
3089 }
3090
3091 .ca-tooltip small {
3092   color: var(--app-accent-dark);
3093   font-size: 12px;
3094   font-weight: 800;
3095   line-height: 1.2;
3096 }
3097
3098 .ca-tooltip em {
3099   color: var(--app-text-muted);
3100   font-size: 11px;
3101   font-style: normal;
3102   line-height: 1.2;
3103 }
3104
3105 .ca-matrix-cell:hover .ca-tooltip {
3106   opacity: 1;
3107   transform: translate(-50%, 0) scale(1);
3108 }
3109
3110 .ca-cell.low {
3111   background: #ecf5ff;
3112 }
3113
3114 .ca-cell.mid {
3115   background: #a0cfff;
3116 }
3117
3118 .ca-cell.high {
3119   color: #ffffff;
3120   background: var(--primary-color);
3121 }
```

```
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```

```
3122
3123 .ca-cell.diagonal {
3124     color: #ffffff;
3125     background: #0f172a;
3126 }
3127
3128 .ca-legend {
3129     display: flex;
3130     justify-content: center;
3131     flex-wrap: wrap;
3132     gap: 10px;
3133     margin-top: 16px;
3134     color: var(--app-text-muted);
3135     font-size: 12px;
3136 }
3137
3138 .ca-legend span {
3139     display: inline-flex;
3140     align-items: center;
3141     gap: 5px;
3142 }
3143
3144 .ca-legend .ca-cell {
3145     width: 12px;
3146     height: 12px;
3147     min-height: 12px;
3148     flex: 0 0 12px;
3149     border-radius: 3px;
3150     color: transparent;
3151 }
3152
3153 .ca-info-panel {
3154     min-width: 0;
3155     padding-left: 16px;
3156     border-left: 1px solid rgba(148, 163, 184, 0.24);
3157     display: flex;
3158     flex-direction: column;
3159     justify-content: space-between;
3160     gap: 14px;
3161 }
3162
3163 .ca-stat-list {
3164     display: grid;
3165     gap: 0;
3166 }
3167
3168 .ca-stat-item {
3169     min-width: 0;
3170     display: flex;
3171     align-items: flex-start;
```

```
front/src/styles.css
```

```
3172   justify-content: space-between;
3173   gap: 12px;
3174   padding: 9px 0;
3175   border-bottom: 1px solid rgba(148, 163, 184, 0.18);
3176 }
3177
3178 .ca-stat-item:first-child {
3179   padding-top: 0;
3180 }
3181
3182 .ca-stat-item > div {
3183   min-width: 0;
3184   display: grid;
3185   gap: 3px;
3186 }
3187
3188 .ca-stat-item span,
3189 .ca-pair-heading span {
3190   color: var(--app-text-muted);
3191   font-size: 12px;
3192   font-weight: 700;
3193 }
3194
3195 .ca-stat-item small {
3196   overflow: hidden;
3197   color: var(--app-text-soft);
3198   font-size: 11px;
3199   line-height: 1.35;
3200   text-overflow: ellipsis;
3201   white-space: nowrap;
3202 }
3203
3204 .ca-stat-item strong {
3205   flex: 0 0 auto;
3206   color: var(--app-text);
3207   font-size: 14px;
3208   font-variant-numeric: tabular-nums;
3209   line-height: 1.35;
3210   text-align: right;
3211 }
3212
3213 .ca-pair-summary {
3214   display: grid;
3215   gap: 8px;
3216 }
3217
3218 .ca-pair-heading,
3219 .ca-pair-row {
3220   display: flex;
3221   align-items: center;
```

```
front/src/styles.css
```

```
3222     justify-content: space-between;
3223     gap: 10px;
3224 }
3225
3226 .ca-pair-heading strong {
3227     color: var(--app-accent);
3228     font-size: 13px;
3229 }
3230
3231 .ca-pair-list {
3232     display: grid;
3233     gap: 6px;
3234 }
3235
3236 .ca-pair-row {
3237     padding: 6px 0;
3238     color: var(--app-text);
3239     font-size: 12px;
3240 }
3241
3242 .ca-pair-row span {
3243     display: inline-flex;
3244     align-items: center;
3245     gap: 7px;
3246     min-width: 0;
3247 }
3248
3249 .ca-pair-row span::before {
3250     width: 6px;
3251     height: 6px;
3252     flex: 0 0 6px;
3253     border-radius: 999px;
3254     background: var(--app-accent);
3255     content: '';
3256 }
3257
3258 .ca-pair-row strong {
3259     color: var(--app-accent-dark);
3260     font-size: 12px;
3261     font-variant-numeric: tabular-nums;
3262 }
3263
3264 .health-timeline-panel {
3265     min-height: 360px;
3266     display: flex;
3267     flex-direction: column;
3268 }
3269
3270 .timeline-header {
3271     display: flex;
```

```
front/src/styles.css
```

```
3272   align-items: flex-start;
3273   justify-content: space-between;
3274   gap: 16px;
3275   margin-bottom: 18px;
3276 }
3277
3278 .timeline-header h2 {
3279   margin: 0 0 8px;
3280   font-size: 18px;
3281   line-height: 1.25;
3282 }
3283
3284 .timeline-header span {
3285   color: var(--app-text-muted);
3286   font-size: 13px;
3287 }
3288
3289 .timeline-summary {
3290   flex: 0 0 auto;
3291   display: grid;
3292   gap: 7px;
3293   justify-items: end;
3294 }
3295
3296 .timeline-summary strong {
3297   padding: 7px 12px;
3298   border-radius: var(--panel-radius);
3299   color: #10b981;
3300   background: rgba(103, 194, 58, 0.12);
3301   font-size: 15px;
3302   font-weight: 800;
3303   line-height: 1;
3304 }
3305
3306 .timeline-summary div {
3307   display: inline-flex;
3308   align-items: center;
3309   gap: 7px;
3310 }
3311
3312 .timeline-summary small {
3313   color: var(--app-text-muted);
3314   font-size: 12px;
3315   font-weight: 700;
3316   font-variant-numeric: tabular-nums;
3317 }
3318
3319 .summary-dot {
3320   width: 7px;
3321   height: 7px;
```

```
front/src/styles.css
```

```
3322     border-radius: 999px;
3323 }
3324
3325 .summary-dot.success {
3326     background: #10b981;
3327 }
3328
3329 .summary-dot.error {
3330     background: #ef4444;
3331 }
3332
3333 .timeline-scroll {
3334     flex: 1;
3335     overflow-x: auto;
3336     overflow-y: hidden;
3337     padding: 8px 0 2px;
3338 }
3339
3340 .timeline-matrix {
3341     min-width: 640px;
3342     display: grid;
3343     gap: 4px;
3344 }
3345
3346 .timeline-hour-axis {
3347     display: grid;
3348     grid-template-columns: 28px repeat(24, minmax(0, 1fr));
3349     gap: 4px;
3350     align-items: center;
3351     min-height: 16px;
3352     color: var(--app-text-muted);
3353     font-size: 11px;
3354     line-height: 1;
3355     text-align: center;
3356 }
3357
3358 .timeline-row {
3359     display: grid;
3360     grid-template-columns: 28px minmax(0, 1fr);
3361     gap: 8px;
3362     align-items: center;
3363 }
3364
3365 .timeline-row-label {
3366     color: var(--app-text-muted);
3367     font-size: 11px;
3368     text-align: left;
3369 }
3370
3371 .timeline-cells {
```

```
front/src/styles.css
```

```
3372   display: grid;
3373   grid-template-columns: repeat(24, minmax(0, 1fr));
3374   gap: 4px;
3375 }
3376
3377 .timeline-cell {
3378   display: block;
3379   width: 100%;
3380   aspect-ratio: 1;
3381   min-height: 12px;
3382   border-radius: 3px;
3383   background: #e5e7eb;
3384 }
3385
3386 .timeline-cell.empty {
3387   background: #e5e7eb;
3388 }
3389
3390 .timeline-cell.success {
3391   background: #10b981;
3392 }
3393
3394 .timeline-cell.warning {
3395   background: #f59e0b;
3396 }
3397
3398 .timeline-cell.error {
3399   background: #ef4444;
3400 }
3401
3402 .timeline-cell.future {
3403   border: 1px solid #cbd5e1;
3404   background: #ffffff;
3405 }
3406
3407 .timeline-legend {
3408   display: flex;
3409   justify-content: center;
3410   flex-wrap: wrap;
3411   gap: 10px;
3412   margin-top: 18px;
3413   color: var(--app-text-muted);
3414   font-size: 12px;
3415 }
3416
3417 .timeline-legend span {
3418   display: inline-flex;
3419   align-items: center;
3420   gap: 4px;
3421 }
```



```
front/src/styles.css
```

```
3422
3423 .timeline-legend .timeline-cell {
3424   width: 10px;
3425   min-height: 10px;
3426   flex: 0 0 10px;
3427   aspect-ratio: 1;
3428 }
3429
3430 .pipeline-list {
3431   display: grid;
3432   gap: 9px;
3433 }
3434
3435 .pipeline-step {
3436   min-width: 0;
3437   display: grid;
3438   grid-template-columns: 30px minmax(0, 1fr) auto;
3439   align-items: center;
3440   gap: 10px;
3441   padding: 10px;
3442   border: 1px solid var(--app-border);
3443   border-radius: var(--panel-radius);
3444   background: var(--app-surface-muted);
3445 }
3446
3447 .step-index {
3448   width: 26px;
3449   height: 26px;
3450   display: grid;
3451   place-items: center;
3452   border-radius: 999px;
3453   color: #ffffff;
3454   background: var(--app-accent);
3455   font-size: 12px;
3456   font-weight: 800;
3457 }
3458
3459 .pipeline-step.pending .step-index {
3460   color: var(--app-text-muted);
3461   background: #e5e7eb;
3462 }
3463
3464 .pipeline-step strong,
3465 .pipeline-step small {
3466   display: block;
3467   min-width: 0;
3468   overflow: hidden;
3469   text-overflow: ellipsis;
3470   white-space: nowrap;
3471 }
```

```
front/src/styles.css
```

```
3472
3473 .pipeline-step strong {
3474   font-size: 14px;
3475 }
3476
3477 .pipeline-step small,
3478 .step-status {
3479   color: var(--app-text-muted);
3480   font-size: 12px;
3481 }
3482
3483 .pipeline-step.done .step-status {
3484   color: var(--success);
3485 }
3486
3487 .pipeline-step.running .step-status {
3488   color: var(--warning);
3489 }
3490
3491 .task-grid {
3492   display: grid;
3493   grid-template-columns: repeat(3, minmax(0, 1fr));
3494   gap: 8px;
3495   margin: 0;
3496 }
3497
3498 .task-grid div {
3499   min-width: 0;
3500   padding: 10px;
3501   border: 1px solid var(--app-border);
3502   border-radius: var(--panel-radius);
3503   background: var(--app-surface-muted);
3504 }
3505
3506 .task-grid dt {
3507   color: var(--app-text-muted);
3508   font-size: 11px;
3509 }
3510
3511 .task-grid dd {
3512   margin: 4px 0 0;
3513   overflow: hidden;
3514   font-size: 17px;
3515   font-weight: 800;
3516   font-variant-numeric: tabular-nums;
3517   text-overflow: ellipsis;
3518   white-space: nowrap;
3519 }
3520
3521 .progress-block {
```

```
front/src/styles.css
```

```
3522     margin-top: 14px;
3523     display: grid;
3524     gap: 8px;
3525 }
3526
3527 .progress-block > div:first-child {
3528     display: flex;
3529     align-items: center;
3530     justify-content: space-between;
3531     gap: 10px;
3532     color: var(--app-text-muted);
3533     font-size: 12px;
3534 }
3535
3536 .progress-block strong {
3537     color: var(--app-text-primary);
3538     font-variant-numeric: tabular-nums;
3539 }
3540
3541 .progress-track {
3542     height: 8px;
3543     overflow: hidden;
3544     border-radius: 999px;
3545     background: #e5e7eb;
3546 }
3547
3548 .progress-track span {
3549     display: block;
3550     height: 100%;
3551     border-radius: inherit;
3552     background: var(--app-accent);
3553 }
3554
3555 .export-actions {
3556     display: grid;
3557     grid-template-columns: repeat(2, minmax(0, 1fr));
3558     gap: 8px;
3559 }
3560
3561 .export-actions button {
3562     min-width: 0;
3563     min-height: 42px;
3564     padding: 8px 10px;
3565     display: grid;
3566     grid-template-columns: auto minmax(0, 1fr) auto;
3567     align-items: center;
3568     gap: 8px;
3569     border: 1px solid var(--app-border);
3570     border-radius: var(--panel-radius);
3571     color: var(--app-text-primary);
```

```
front/src/styles.css
```

```
3572   background: var(--app-surface-muted);
3573   cursor: pointer;
3574   text-align: left;
3575   transition:
3576     background-color 0.16s ease,
3577     border-color 0.16s ease,
3578     color 0.16s ease;
3579 }
3580
3581 .export-actions button:hover {
3582   color: var(--app-accent-dark);
3583   border-color: rgba(64, 158, 255, 0.28);
3584   background: var(--app-accent-soft);
3585 }
3586
3587 .export-actions span,
3588 .export-actions small {
3589   min-width: 0;
3590   overflow: hidden;
3591   text-overflow: ellipsis;
3592   white-space: nowrap;
3593 }
3594
3595 .export-actions span {
3596   font-size: 13px;
3597   font-weight: 700;
3598 }
3599
3600 .export-actions small {
3601   color: var(--app-text-muted);
3602   font-size: 11px;
3603 }
3604
3605 .focused-analysis-grid,
3606 .task-board,
3607 .result-analysis-layout {
3608   display: grid;
3609   gap: 12px;
3610 }
3611
3612 .focused-analysis-grid {
3613   grid-template-columns: minmax(0, 1.6fr) minmax(320px, 1fr);
3614   align-items: stretch;
3615 }
3616
3617 .focused-analysis-grid > .panel {
3618   height: 100%;
3619 }
3620
3621 .task-board {
```

```
front/src/styles.css
```

```
3622   grid-template-columns: minmax(0, 1.1fr) minmax(320px, 0.9fr) minmax(360px, 1fr);
3623   align-items: stretch;
3624 }
3625
3626 .result-analysis-layout {
3627   grid-template-columns: 360px minmax(0, 1fr);
3628   align-items: start;
3629 }
3630
3631 .result-analysis-layout .secondary-card-grid {
3632   grid-template-columns: repeat(3, minmax(0, 1fr));
3633 }
3634
3635 .feature-list-panel {
3636   min-height: 100%;
3637 }
3638
3639 .feature-list {
3640   display: grid;
3641   gap: 9px;
3642 }
3643
3644 .feature-row {
3645   min-width: 0;
3646   display: grid;
3647   grid-template-columns: minmax(0, 1fr) auto;
3648   align-items: center;
3649   gap: 10px;
3650   padding: 10px;
3651   border: 1px solid var(--app-border);
3652   border-radius: var(--panel-radius);
3653   background: var(--app-surface-muted);
3654 }
3655
3656 .feature-row div {
3657   min-width: 0;
3658   display: grid;
3659   gap: 4px;
3660 }
3661
3662 .feature-row strong,
3663 .feature-row span {
3664   min-width: 0;
3665   overflow: hidden;
3666   text-overflow: ellipsis;
3667   white-space: nowrap;
3668 }
3669
3670 .feature-row strong {
3671   font-size: 14px;
```

front/src/styles.css

```
3672 }
3673
3674 .feature-row span,
3675 .feature-row small {
3676   color: var(--app-text-muted);
3677   font-size: 12px;
3678 }
3679
3680 .feature-row small {
3681   padding: 4px 7px;
3682   border-radius: 999px;
3683   color: var(--app-accent-dark);
3684   background: var(--app-accent-soft);
3685   font-weight: 700;
3686   white-space: nowrap;
3687 }
3688
3689 .config-panel {
3690   display: grid;
3691   grid-template-columns: repeat(2, minmax(0, 1fr));
3692   gap: 12px;
3693 }
3694
3695 .config-item {
3696   min-width: 0;
3697   display: flex;
3698   gap: 10px;
3699   padding: 12px;
3700   border: 1px solid var(--app-border);
3701   border-radius: var(--panel-radius);
3702   background: var(--app-surface-muted);
3703 }
3704
3705 .config-item svg {
3706   flex: 0 0 auto;
3707   color: var(--success);
3708 }
3709
3710 .config-item div {
3711   min-width: 0;
3712   display: grid;
3713   gap: 4px;
3714 }
3715
3716 .config-item strong,
3717 .config-item span {
3718   min-width: 0;
3719   overflow: hidden;
3720   text-overflow: ellipsis;
3721 }
```

front/src/styles.css

```
3722
3723 .config-item strong {
3724   font-size: 14px;
3725 }
3726
3727 .config-item span {
3728   color: var(--app-text-muted);
3729   font-size: 13px;
3730   line-height: 1.5;
3731 }
3732
3733 .soft-page {
3734   display: grid;
3735   gap: 12px;
3736 }
3737
3738 .soft-page-heading {
3739   min-width: 0;
3740   display: flex;
3741   align-items: center;
3742   gap: 12px;
3743   padding: 2px 2px 4px;
3744 }
3745
3746 .soft-page-icon {
3747   width: 38px;
3748   height: 38px;
3749   flex: 0 0 38px;
3750   display: grid;
3751   place-items: center;
3752   border-radius: var(--panel-radius);
3753   color: var(--app-accent-dark);
3754   background: var(--app-accent-soft);
3755 }
3756
3757 .soft-page-heading h1 {
3758   margin: 0 0 4px;
3759   color: var(--app-text-primary);
3760   font-size: 20px;
3761   font-weight: 800;
3762   line-height: 1.25;
3763 }
3764
3765 .soft-page-heading p {
3766   margin: 0;
3767   color: var(--app-text-muted);
3768   font-size: 13px;
3769   line-height: 1.45;
3770 }
3771
```

```
front/src/styles.css
```

```
3772 .soft-metric-strip {
3773   display: grid;
3774   grid-template-columns: repeat(4, minmax(0, 1fr));
3775   gap: 12px;
3776 }
3777
3778 .soft-metric-card {
3779   min-width: 0;
3780   padding: 14px;
3781   border: 1px solid var(--app-border);
3782   border-radius: var(--panel-radius);
3783   background: var(--app-surface);
3784   display: grid;
3785   gap: 5px;
3786 }
3787
3788 .soft-metric-card span,
3789 .soft-metric-card small {
3790   min-width: 0;
3791   overflow: hidden;
3792   color: var(--app-text-muted);
3793   font-size: 12px;
3794   text-overflow: ellipsis;
3795   white-space: nowrap;
3796 }
3797
3798 .soft-metric-card strong {
3799   min-width: 0;
3800   overflow: hidden;
3801   color: var(--app-text-primary);
3802   font-size: 24px;
3803   font-weight: 800;
3804   font-variant-numeric: tabular-nums;
3805   line-height: 1.15;
3806   text-overflow: ellipsis;
3807   white-space: nowrap;
3808 }
3809
3810 .soft-metric-card.green strong {
3811   color: var(--success);
3812 }
3813
3814 .soft-metric-card.amber strong {
3815   color: var(--warning);
3816 }
3817
3818 .soft-metric-card.red strong {
3819   color: var(--error-color);
3820 }
3821
```



```
front/src/styles.css
```

```
3822 .soft-metric-card.blue strong {
3823   color: var(--app-accent-dark);
3824 }
3825
3826 .soft-two-column {
3827   display: grid;
3828   grid-template-columns: minmax(0, 1.4fr) minmax(320px, 0.8fr);
3829   gap: 12px;
3830   align-items: stretch;
3831 }
3832
3833 .soft-two-column.wide-left {
3834   grid-template-columns: minmax(0, 1.7fr) minmax(320px, 0.8fr);
3835 }
3836
3837 .soft-two-column > .panel,
3838 .soft-chart-grid > .panel {
3839   height: 100%;
3840 }
3841
3842 .soft-status-badge {
3843   width: max-content;
3844   max-width: 100%;
3845   display: inline-flex;
3846   align-items: center;
3847   justify-content: center;
3848   min-height: 24px;
3849   padding: 3px 8px;
3850   border-radius: 999px;
3851   font-size: 12px;
3852   font-weight: 800;
3853   line-height: 1;
3854   white-space: nowrap;
3855 }
3856
3857 .soft-status-badge.ready,
3858 .soft-status-badge.info {
3859   color: #3f8f1f;
3860   border: 1px solid var(--success-badge-border);
3861   background: var(--success-badge-bg);
3862 }
3863
3864 .soft-status-badge.warning {
3865   color: var(--amber-text);
3866   border: 1px solid var(--warning-border);
3867   background: var(--warning-bg);
3868 }
3869
3870 .soft-status-badge.error {
3871   color: var(--error-color);
```

```
front/src/styles.css
```

```
3872   border: 1px solid var(--failure-badge-border);
3873   background: var(--failure-badge-bg);
3874 }
3875
3876 .soft-table-panel {
3877   overflow: hidden;
3878 }
3879
3880 .soft-data-table {
3881   display: grid;
3882   border: 1px solid var(--app-border);
3883   border-radius: var(--panel-radius);
3884   overflow: hidden;
3885 }
3886
3887 .soft-data-row {
3888   min-width: 0;
3889   display: grid;
3890   grid-template-columns: minmax(180px, 1.2fr) minmax(90px, 0.55fr) minmax(220px, 1.35fr) minmax(138px, 0.85fr)
3891     minmax(130px, 0.75fr);
3892   align-items: center;
3893   gap: 12px;
3894   padding: 11px 12px;
3895   border-top: 1px solid var(--app-border);
3896   background: rgba(255, 255, 255, 0.56);
3897 }
3898 .soft-data-row:first-child {
3899   border-top: 0;
3900 }
3901
3902 .soft-data-row.header {
3903   min-height: 36px;
3904   color: var(--app-text-muted);
3905   background: var(--app-surface-muted);
3906   font-size: 12px;
3907   font-weight: 800;
3908 }
3909
3910 .soft-data-row strong,
3911 .soft-data-row span {
3912   min-width: 0;
3913   overflow: hidden;
3914   font-size: 13px;
3915   text-overflow: ellipsis;
3916   white-space: nowrap;
3917 }
3918
3919 .soft-data-row strong {
3920   display: grid;
```

```
front/src/styles.css
```

```
3921   gap: 3px;
3922   color: var(--app-text-primary);
3923 }
3924
3925 .soft-data-row strong small {
3926   overflow: hidden;
3927   color: var(--app-text-muted);
3928   font-size: 11px;
3929   text-overflow: ellipsis;
3930   white-space: nowrap;
3931 }
3932
3933 .soft-data-table.evaluation .soft-data-row {
3934   grid-template-columns: repeat(6, minmax(0, 1fr));
3935 }
3936
3937 .soft-rule-list {
3938   display: grid;
3939   gap: 9px;
3940 }
3941
3942 .soft-rule-list.compact {
3943   margin-top: 12px;
3944   grid-template-columns: repeat(2, minmax(0, 1fr));
3945 }
3946
3947 .soft-rule-item {
3948   min-width: 0;
3949   display: flex;
3950   gap: 10px;
3951   padding: 10px;
3952   border: 1px solid var(--app-border);
3953   border-radius: var(--panel-radius);
3954   background: var(--app-surface-muted);
3955 }
3956
3957 .soft-rule-item svg {
3958   flex: 0 0 auto;
3959   color: var(--app-accent-dark);
3960 }
3961
3962 .soft-rule-item div {
3963   min-width: 0;
3964   display: grid;
3965   gap: 4px;
3966 }
3967
3968 .soft-rule-item strong,
3969 .soft-rule-item span {
3970   min-width: 0;
```

```
front/src/styles.css
```

```
3971     overflow: hidden;
3972     text-overflow: ellipsis;
3973 }
3974
3975 .soft-rule-item strong {
3976     color: var(--app-text-primary);
3977     font-size: 13px;
3978 }
3979
3980 .soft-rule-item span {
3981     color: var(--app-text-muted);
3982     font-size: 12px;
3983     line-height: 1.45;
3984 }
3985
3986 /* ===== 数据质量检查：待处理队列与规则检查器 ===== */
3987 .quality-workspace,
3988 .versions-workspace {
3989     --versions-accent: var(--app-accent);
3990     --versions-accent-dark: var(--app-accent-dark);
3991     --versions-accent-ink: #1d4ed8;
3992     --versions-accent-soft: #eff6ff;
3993     --versions-accent-border: #bfdbfe;
3994     --versions-accent-badge: #dbeafe;
3995     display: grid;
3996     gap: 14px;
3997 }
3998
3999 .quality-page-header,
4000 .versions-page-header {
4001     min-width: 0;
4002     display: flex;
4003     align-items: center;
4004     justify-content: space-between;
4005     gap: 18px;
4006     padding: 2px 2px 4px;
4007 }
4008
4009 .quality-page-header > div,
4010 .versions-page-header > div {
4011     min-width: 0;
4012     display: flex;
4013     align-items: center;
4014     gap: 12px;
4015 }
4016
4017 .quality-page-icon,
4018 .versions-page-icon {
4019     width: 40px;
4020     height: 40px;
```

front/src/styles.css

```
4021     flex: 0 0 40px;
4022     display: grid;
4023     place-items: center;
4024     border-radius: var(--panel-radius);
4025 }
4026
4027 .quality-page-icon {
4028     color: #166534;
4029     background: #dcfce7;
4030 }
4031
4032 .versions-page-icon {
4033     color: var(--versions-accent-dark);
4034     background: var(--versions-accent-soft);
4035 }
4036
4037 .quality-page-header h1,
4038 .versions-page-header h1 {
4039     margin: 0 0 4px;
4040     color: var(--app-text-primary);
4041     font-size: 20px;
4042     font-weight: 800;
4043     line-height: 1.25;
4044 }
4045
4046 .quality-page-header p,
4047 .versions-page-header p {
4048     max-width: 72ch;
4049     margin: 0;
4050     color: var(--app-text-regular);
4051     font-size: 13px;
4052     line-height: 1.45;
4053 }
4054
4055 .quality-page-header .btn,
4056 .versions-page-header .btn {
4057     flex: 0 0 auto;
4058 }
4059
4060 .quality-feedback,
4061 .versions-feedback {
4062     min-height: 40px;
4063     display: flex;
4064     align-items: center;
4065     justify-content: space-between;
4066     gap: 12px;
4067     padding: 9px 12px;
4068     border: 1px solid;
4069     border-radius: var(--panel-radius);
4070     font-size: 13px;
```

```
front/src/styles.css
```

```
4071   font-weight: 700;
4072 }
4073
4074 .quality-feedback.error,
4075 .versions-feedback.error {
4076   color: #991b1b;
4077   border-color: #fecaca;
4078   background: #fef2f2;
4079 }
4080
4081 .quality-feedback.success {
4082   color: #166534;
4083   border-color: #bbf7d0;
4084   background: #f0fdf4;
4085 }
4086
4087 .versions-feedback button {
4088   flex: 0 0 auto;
4089   display: inline-flex;
4090   align-items: center;
4091   gap: 5px;
4092   border: 0;
4093   padding: 3px 5px;
4094   color: inherit;
4095   background: transparent;
4096   font: inherit;
4097   cursor: pointer;
4098 }
4099
4100 .quality-layout {
4101   min-height: 650px;
4102   display: grid;
4103   grid-template-columns: minmax(310px, 340px) minmax(0, 1fr);
4104   gap: 12px;
4105   align-items: stretch;
4106 }
4107
4108 .quality-queue,
4109 .quality-inspector {
4110   min-height: 0;
4111   overflow: hidden;
4112   padding: 0;
4113 }
4114
4115 .quality-queue {
4116   display: grid;
4117   grid-template-rows: auto auto minmax(0, 1fr);
4118 }
4119
4120 .quality-queue-header {
```

```
front/src/styles.css
```

```
4121     display: grid;
4122     gap: 12px;
4123     padding: 16px;
4124     border-bottom: 1px solid var(--app-border);
4125     background: rgba(248, 250, 252, 0.72);
4126 }
4127
4128 .quality-queue-header > div:first-child {
4129     display: flex;
4130     align-items: baseline;
4131     justify-content: space-between;
4132     gap: 12px;
4133 }
4134
4135 .quality-queue-header strong {
4136     color: var(--app-text-primary);
4137     font-size: 15px;
4138 }
4139
4140 .quality-queue-header > div:first-child > span {
4141     color: var(--app-text-muted);
4142     font-size: 12px;
4143 }
4144
4145 .quality-search {
4146     height: 36px;
4147     display: flex;
4148     align-items: center;
4149     gap: 8px;
4150     padding: 0 10px;
4151     border: 1px solid var(--app-border-strong);
4152     border-radius: var(--panel-radius);
4153     background: var(--app-surface-strong);
4154     color: var(--app-text-muted);
4155 }
4156
4157 .quality-search:focus-within {
4158     border-color: var(--app-accent);
4159     box-shadow: 0 0 0 2px var(--primary-10);
4160 }
4161
4162 .quality-search input {
4163     min-width: 0;
4164     width: 100%;
4165     border: 0;
4166     outline: 0;
4167     padding: 0;
4168     color: var(--app-text-primary);
4169     background: transparent;
4170     font-size: 13px;
```

```
front/src/styles.css
```

```
4171 }
4172
4173 .quality-search input::placeholder {
4174   color: #64748b;
4175 }
4176
4177 .quality-filters {
4178   display: grid;
4179   grid-template-columns: repeat(2, minmax(0, 1fr));
4180   gap: 6px;
4181   padding: 10px 16px;
4182   border-bottom: 1px solid var(--app-border);
4183   background: rgba(255, 255, 255, 0.5);
4184 }
4185
4186 .quality-filters button {
4187   min-width: 0;
4188   min-height: 34px;
4189   display: flex;
4190   align-items: center;
4191   justify-content: space-between;
4192   gap: 8px;
4193   border: 1px solid transparent;
4194   border-radius: 7px;
4195   padding: 0 9px;
4196   color: var(--app-text-regular);
4197   background: transparent;
4198   font: inherit;
4199   font-size: 12px;
4200   font-weight: 700;
4201   cursor: pointer;
4202 }
4203
4204 .quality-filters button:hover {
4205   background: var(--bg-tertiary);
4206 }
4207
4208 .quality-filters button.active {
4209   color: var(--app-accent-dark);
4210   border-color: #bfdbfe;
4211   background: #eff6ff;
4212 }
4213
4214 .quality-filters strong {
4215   min-width: 23px;
4216   height: 22px;
4217   display: inline-grid;
4218   place-items: center;
4219   border-radius: 999px;
4220   padding: 0 6px;
```



```
front/src/styles.css
```

```
4221   color: var(--app-text-regular);
4222   background: #e2e8f0;
4223   font-size: 11px;
4224   font-variant-numeric: tabular-nums;
4225 }
4226
4227 .quality-filters button.active strong {
4228   color: #1d4ed8;
4229   background: #dbeafe;
4230 }
4231
4232 .quality-queue-list {
4233   min-height: 0;
4234   overflow-y: auto;
4235   padding: 8px;
4236   background: rgba(248, 250, 252, 0.42);
4237 }
4238
4239 .quality-queue-item {
4240   width: 100%;
4241   min-height: 72px;
4242   display: grid;
4243   grid-template-columns: minmax(0, 1fr) auto;
4244   align-items: center;
4245   gap: 12px;
4246   border: 1px solid transparent;
4247   border-radius: var(--panel-radius);
4248   padding: 10px;
4249   color: inherit;
4250   background: transparent;
4251   text-align: left;
4252   font: inherit;
4253   cursor: pointer;
4254 }
4255
4256 .quality-queue-item + .quality-queue-item {
4257   margin-top: 3px;
4258 }
4259
4260 .quality-queue-item:hover {
4261   background: rgba(255, 255, 255, 0.86);
4262 }
4263
4264 .quality-queue-item.active {
4265   border-color: #bfdbfe;
4266   background: #eff6ff;
4267   box-shadow: inset 0 0 0 1px rgba(64, 158, 255, 0.08);
4268 }
4269
4270 .quality-queue-item > div {
```

```
front/src/styles.css
```

```
4271   min-width: 0;
4272   display: grid;
4273   gap: 5px;
4274 }
4275
4276 .quality-queue-item > div:last-child {
4277   justify-items: end;
4278 }
4279
4280 .quality-queue-item strong,
4281 .quality-queue-item span,
4282 .quality-queue-item small {
4283   min-width: 0;
4284 }
4285
4286 .quality-queue-item > div:first-child > strong {
4287   overflow: hidden;
4288   color: var(--app-text-primary);
4289   font-size: 13px;
4290   text-overflow: ellipsis;
4291   white-space: nowrap;
4292 }
4293
4294 .quality-queue-item > div:first-child > span,
4295 .quality-queue-item small {
4296   color: var(--app-text-muted);
4297   font-size: 11px;
4298 }
4299
4300 .quality-status,
4301 .versions-quality-badge {
4302   width: max-content;
4303   display: inline-flex;
4304   align-items: center;
4305   justify-content: center;
4306   min-height: 24px;
4307   padding: 0 8px;
4308   border: 1px solid;
4309   border-radius: 999px;
4310   font-size: 11px;
4311   font-weight: 800;
4312   line-height: 1;
4313   white-space: nowrap;
4314 }
4315
4316 .quality-status.ready,
4317 .versions-quality-badge.ready {
4318   color: #166534;
4319   border-color: #bbf7d0;
4320   background: #f0fdf4;
```

```
front/src/styles.css
```

```
4321 }
4322
4323 .quality-status.warning,
4324 .versions-quality-badge.warning {
4325     color: #92400e;
4326     border-color: #fde68a;
4327     background: #fffbeb;
4328 }
4329
4330 .quality-status.error,
4331 .versions-quality-badge.error {
4332     color: #b91c1c;
4333     border-color: #fecaca;
4334     background: #fef2f2;
4335 }
4336
4337 .quality-loading,
4338 .versions-loading,
4339 .versions-rail-loading {
4340     display: grid;
4341     gap: 9px;
4342     padding: 14px;
4343 }
4344
4345 .quality-loading span,
4346 .versions-loading span,
4347 .versions-rail-loading span {
4348     height: 48px;
4349     border-radius: var(--panel-radius);
4350     background: #e2e8f0;
4351     animation: data-workspace-pulse 1.25s ease-in-out infinite alternate;
4352 }
4353
4354 .versions-loading span {
4355     height: 72px;
4356 }
4357
4358 .quality-empty,
4359 .versions-rail-empty,
4360 .versions-inspector-empty {
4361     min-height: 220px;
4362     display: grid;
4363     place-items: center;
4364     align-content: center;
4365     gap: 7px;
4366     padding: 24px;
4367     color: var(--app-text-muted);
4368     text-align: center;
4369 }
4370
```

```
front/src/styles.css
```

```
4371 .quality-empty strong,
4372 .versions-rail-empty strong,
4373 .versions-inspector-empty strong {
4374   color: var(--app-text-primary);
4375   font-size: 14px;
4376 }
4377
4378 .quality-empty span,
4379 .versions-rail-empty span,
4380 .versions-inspector-empty span {
4381   max-width: 48ch;
4382   font-size: 12px;
4383   line-height: 1.5;
4384 }
4385
4386 .quality-empty.inspector {
4387   min-height: 100%;
4388 }
4389
4390 .quality-inspector {
4391   display: flex;
4392   flex-direction: column;
4393   background: rgba(255, 255, 255, 0.86);
4394 }
4395
4396 .quality-inspector-header {
4397   display: flex;
4398   align-items: flex-start;
4399   justify-content: space-between;
4400   gap: 18px;
4401   padding: 18px 20px;
4402   border-bottom: 1px solid var(--app-border);
4403 }
4404
4405 .quality-inspector-header > div:first-child {
4406   min-width: 0;
4407 }
4408
4409 .quality-inspector-header > div:first-child > span,
4410 .quality-rule-heading span {
4411   color: var(--app-text-muted);
4412   font-size: 12px;
4413 }
4414
4415 .quality-inspector-header h2 {
4416   margin: 4px 0 5px;
4417   color: var(--app-text-primary);
4418   font-size: 20px;
4419   line-height: 1.3;
4420 }
```

```
front/src/styles.css
```

```
4421
4422 .quality-inspector-header p {
4423   margin: 0;
4424   color: var(--app-text-regular);
4425   font-size: 13px;
4426   line-height: 1.5;
4427 }
4428
4429 .quality-inspector-actions {
4430   flex: 0 0 auto;
4431   display: flex;
4432   align-items: center;
4433   gap: 8px;
4434 }
4435
4436 .quality-dataset-meta {
4437   display: grid;
4438   grid-template-columns: repeat(4, minmax(0, 1fr));
4439   margin: 0;
4440   padding: 14px 20px;
4441   border-bottom: 1px solid var(--app-border);
4442   background: rgba(248, 250, 252, 0.66);
4443 }
4444
4445 .quality-dataset-meta > div {
4446   min-width: 0;
4447   display: grid;
4448   gap: 4px;
4449   padding: 0 14px;
4450   border-right: 1px solid var(--app-border);
4451 }
4452
4453 .quality-dataset-meta > div:first-child {
4454   padding-left: 0;
4455 }
4456
4457 .quality-dataset-meta > div:last-child {
4458   padding-right: 0;
4459   border-right: 0;
4460 }
4461
4462 .quality-dataset-meta dt {
4463   color: var(--app-text-muted);
4464   font-size: 11px;
4465 }
4466
4467 .quality-dataset-meta dd {
4468   margin: 0;
4469   overflow: hidden;
4470   color: var(--app-text-primary);
```

```
front/src/styles.css
```

```
4471   font-size: 14px;
4472   font-weight: 800;
4473   text-overflow: ellipsis;
4474   white-space: nowrap;
4475 }
4476
4477 .quality-issue-summary {
4478   display: grid;
4479   grid-template-columns: auto minmax(0, 1fr);
4480   gap: 10px;
4481   margin: 16px 20px 4px;
4482   padding: 12px 14px;
4483   border: 1px solid;
4484   border-radius: var(--panel-radius);
4485 }
4486
4487 .quality-issue-summary.ready {
4488   color: #166534;
4489   border-color: #bbf7d0;
4490   background: #f0fdf4;
4491 }
4492
4493 .quality-issue-summary.warning {
4494   color: #92400e;
4495   border-color: #fde68a;
4496   background: #fffbeb;
4497 }
4498
4499 .quality-issue-summary.error {
4500   color: #991b1b;
4501   border-color: #fecaca;
4502   background: #fef2f2;
4503 }
4504
4505 .quality-issue-summary svg {
4506   margin-top: 1px;
4507 }
4508
4509 .quality-issue-summary div {
4510   min-width: 0;
4511   display: grid;
4512   gap: 4px;
4513 }
4514
4515 .quality-issue-summary strong {
4516   font-size: 13px;
4517 }
4518
4519 .quality-issue-summary span {
4520   font-size: 12px;
```

```
front/src/styles.css
```

```
4521     line-height: 1.55;
4522 }
4523
4524 .quality-rule-heading {
4525     display: flex;
4526     align-items: center;
4527     justify-content: space-between;
4528     gap: 12px;
4529     padding: 16px 20px 10px;
4530 }
4531
4532 .quality-rule-heading > div {
4533     min-width: 0;
4534 }
4535
4536 .quality-rule-heading h3 {
4537     margin: 0 0 3px;
4538     color: var(--app-text-primary);
4539     font-size: 14px;
4540 }
4541
4542 .quality-rule-heading > svg {
4543     color: #15803d;
4544 }
4545
4546 .quality-rule-results {
4547     display: grid;
4548     padding: 0 20px;
4549 }
4550
4551 .quality-rule-result {
4552     min-width: 0;
4553     display: grid;
4554     grid-template-columns: 32px minmax(0, 1fr) minmax(120px, auto);
4555     align-items: center;
4556     gap: 11px;
4557     min-height: 68px;
4558     border-top: 1px solid var(--app-border);
4559     padding: 10px 0;
4560 }
4561
4562 .quality-rule-icon {
4563     width: 30px;
4564     height: 30px;
4565     display: grid;
4566     place-items: center;
4567     border-radius: 50%;
4568 }
4569
4570 .quality-rule-result.pass .quality-rule-icon {
```

```
front/src/styles.css
```

```
4571     color: #15803d;
4572     background: #dcfce7;
4573 }
4574
4575 .quality-rule-result.warning .quality-rule-icon {
4576     color: #b45309;
4577     background: #fef3c7;
4578 }
4579
4580 .quality-rule-result.error .quality-rule-icon {
4581     color: #dc2626;
4582     background: #fee2e2;
4583 }
4584
4585 .quality-rule-result > div {
4586     min-width: 0;
4587     display: grid;
4588     gap: 4px;
4589 }
4590
4591 .quality-rule-result > div > strong {
4592     color: var(--app-text-primary);
4593     font-size: 13px;
4594 }
4595
4596 .quality-rule-result > div > span {
4597     color: var(--app-text-regular);
4598     font-size: 12px;
4599     line-height: 1.5;
4600 }
4601
4602 .quality-rule-result > b {
4603     max-width: 240px;
4604     color: var(--app-text-primary);
4605     font-size: 12px;
4606     text-align: right;
4607     overflow-wrap: anywhere;
4608 }
4609
4610 .quality-readiness {
4611     margin-top: auto;
4612     display: flex;
4613     align-items: center;
4614     justify-content: flex-start;
4615     gap: 18px;
4616     padding: 15px 20px;
4617     border-top: 1px solid var(--app-border);
4618     background: #f8fafc;
4619 }
4620
```



```
front/src/styles.css
```

```
4621 .quality-readiness > div {
4622   min-width: 0;
4623   display: grid;
4624   gap: 4px;
4625 }
4626
4627 .quality-readiness strong {
4628   color: var(--app-text-primary);
4629   font-size: 13px;
4630 }
4631
4632 .quality-readiness span {
4633   color: var(--app-text-regular);
4634   font-size: 12px;
4635 }
4636
4637 .quality-readiness.error strong {
4638   color: #b91c1c;
4639 }
4640
4641 /* ===== 数据版本记录：版本轨道与字段差异检查器 ===== */
4642 .versions-dataset-bar {
4643   min-height: 70px;
4644   display: flex;
4645   align-items: center;
4646   justify-content: space-between;
4647   gap: 20px;
4648   padding: 12px 16px;
4649 }
4650
4651 .versions-dataset-picker,
4652 .versions-compare-picker {
4653   min-width: 0;
4654   display: grid;
4655   gap: 5px;
4656 }
4657
4658 .versions-dataset-picker {
4659   width: min(390px, 36%);
4660 }
4661
4662 .versions-dataset-picker > span,
4663 .versions-compare-picker > span {
4664   color: var(--app-text-muted);
4665   font-size: 11px;
4666   font-weight: 700;
4667 }
4668
4669 .versions-dataset-picker .select-field-trigger,
4670 .versions-compare-picker .select-field-trigger {
```

```
front/src/styles.css
```

```
4671     width: 100%;
4672     height: 36px;
4673     border: 1px solid var(--app-border-strong);
4674     border-radius: var(--panel-radius);
4675     padding: 0 34px 0 10px;
4676     color: var(--app-text-primary);
4677     background: var(--app-surface-strong);
4678     font-size: 13px;
4679     font-weight: 700;
4680 }
4681
4682 .versions-dataset-picker .select-field-trigger-value,
4683 .versions-compare-picker .select-field-trigger-value {
4684     font-weight: 700;
4685 }
4686
4687 .versions-dataset-picker .select-field-trigger:focus-visible,
4688 .versions-compare-picker .select-field-trigger:focus-visible {
4689     outline: 2px solid var(--primary-30);
4690     outline-offset: 1px;
4691     border-color: var(--app-accent);
4692 }
4693
4694 .versions-dataset-context {
4695     min-width: 0;
4696     display: flex;
4697     align-items: stretch;
4698 }
4699
4700 .versions-dataset-context > div {
4701     min-width: 112px;
4702     display: grid;
4703     align-content: center;
4704     gap: 4px;
4705     padding: 0 18px;
4706     border-right: 1px solid var(--app-border);
4707 }
4708
4709 .versions-dataset-context > div:last-child {
4710     padding-right: 0;
4711     border-right: 0;
4712 }
4713
4714 .versions-dataset-context span {
4715     color: var(--app-text-muted);
4716     font-size: 11px;
4717 }
4718
4719 .versions-dataset-context strong {
4720     color: var(--app-text-primary);
```

```
front/src/styles.css
```

```
4721     font-size: 14px;
4722     font-weight: 800;
4723     font-variant-numeric: tabular-nums;
4724 }
4725
4726 .versions-loading {
4727     min-height: 430px;
4728     align-content: center;
4729 }
4730
4731 .versions-empty {
4732     min-height: 430px;
4733     display: grid;
4734     place-items: center;
4735     align-content: center;
4736     gap: 9px;
4737     padding: 48px;
4738     text-align: center;
4739 }
4740
4741 .versions-empty > svg {
4742     color: var(--versions-accent);
4743 }
4744
4745 .versions-empty h2 {
4746     margin: 2px 0 0;
4747     color: var(--app-text-primary);
4748     font-size: 18px;
4749 }
4750
4751 .versions-empty p {
4752     max-width: 58ch;
4753     margin: 0 0 5px;
4754     color: var(--app-text-regular);
4755     font-size: 13px;
4756     line-height: 1.6;
4757 }
4758
4759 .versions-explorer {
4760     min-height: 650px;
4761     display: grid;
4762     grid-template-columns: minmax(260px, 292px) minmax(0, 1fr);
4763     overflow: hidden;
4764     padding: 0;
4765     background: rgba(255, 255, 255, 0.88);
4766 }
4767
4768 .versions-rail {
4769     min-width: 0;
4770     min-height: 0;
```

```
front/src/styles.css
```

```
4771   display: grid;
4772   grid-template-rows: auto minmax(0, 1fr);
4773   border-right: 1px solid var(--app-border);
4774   background: #f8fafc;
4775 }
4776
4777 .versions-rail-header {
4778   min-width: 0;
4779   display: flex;
4780   align-items: center;
4781   justify-content: space-between;
4782   gap: 12px;
4783   min-height: 66px;
4784   padding: 13px 16px;
4785   border-bottom: 1px solid var(--app-border);
4786 }
4787
4788 .versions-rail-header > div {
4789   min-width: 0;
4790   display: grid;
4791   gap: 4px;
4792 }
4793
4794 .versions-rail-header strong {
4795   color: var(--app-text-primary);
4796   font-size: 14px;
4797 }
4798
4799 .versions-rail-header span {
4800   overflow: hidden;
4801   color: var(--app-text-muted);
4802   font-size: 11px;
4803   text-overflow: ellipsis;
4804   white-space: nowrap;
4805 }
4806
4807 .versions-rail-header b {
4808   min-width: 27px;
4809   height: 25px;
4810   display: inline-grid;
4811   place-items: center;
4812   border-radius: 999px;
4813   padding: 0 7px;
4814   color: var(--versions-accent-ink);
4815   background: var(--versions-accent-badge);
4816   font-size: 11px;
4817   font-variant-numeric: tabular-nums;
4818 }
4819
4820 .versions-rail-list {
```

front/src/styles.css

```
4821     position: relative;
4822     min-height: 0;
4823     overflow-y: auto;
4824     padding: 10px 8px;
4825 }
4826
4827 .versions-rail-list::before {
4828     position: absolute;
4829     top: 24px;
4830     bottom: 24px;
4831     left: 30px;
4832     width: 1px;
4833     background: #cbd5e1;
4834     content: '';
4835 }
4836
4837 .versions-rail-item {
4838     position: relative;
4839     z-index: 1;
4840     width: 100%;
4841     min-height: 70px;
4842     display: grid;
4843     grid-template-columns: 34px minmax(0, 1fr) auto;
4844     align-items: center;
4845     gap: 9px;
4846     border: 1px solid transparent;
4847     border-radius: var(--panel-radius);
4848     padding: 8px 8px 8px 5px;
4849     color: inherit;
4850     background: transparent;
4851     text-align: left;
4852     font: inherit;
4853     cursor: pointer;
4854 }
4855
4856 .versions-rail-item + .versions-rail-item {
4857     margin-top: 3px;
4858 }
4859
4860 .versions-rail-item:hover {
4861     background: rgba(255, 255, 255, 0.9);
4862 }
4863
4864 .versions-rail-item.active {
4865     border-color: var(--versions-accent-border);
4866     background: var(--versions-accent-soft);
4867 }
4868
4869 .versions-rail-node {
4870     width: 30px;
```

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```
4871   height: 30px;
4872   display: grid;
4873   place-items: center;
4874   border: 1px solid #cbd5e1;
4875   border-radius: 50%;
4876   color: #475569;
4877   background: #ffffff;
4878 }
4879
4880 .versions-rail-item.active .versions-rail-node {
4881   color: #ffffff;
4882   border-color: var(--versions-accent);
4883   background: var(--versions-accent);
4884 }
4885
4886 .versions-rail-copy {
4887   min-width: 0;
4888   display: grid;
4889   gap: 4px;
4890 }
4891
4892 .versions-rail-copy strong,
4893 .versions-rail-copy time {
4894   min-width: 0;
4895   overflow: hidden;
4896   text-overflow: ellipsis;
4897   white-space: nowrap;
4898 }
4899
4900 .versions-rail-copy strong {
4901   color: var(--app-text-primary);
4902   font-size: 12px;
4903 }
4904
4905 .versions-rail-copy time {
4906   color: var(--app-text-muted);
4907   font-size: 11px;
4908 }
4909
4910 .versions-rail-state {
4911   min-width: 36px;
4912   padding: 4px 6px;
4913   border-radius: 999px;
4914   color: #475569;
4915   background: #e2e8f0;
4916   font-size: 10px;
4917   font-weight: 800;
4918   text-align: center;
4919   white-space: nowrap;
4920 }
```

```
front/src/styles.css
```

```
4921
4922 .versions-rail-state.ready {
4923     color: #166534;
4924     background: #dcfce7;
4925 }
4926
4927 .versions-rail-state.warning {
4928     color: #92400e;
4929     background: #fff3cd;
4930 }
4931
4932 .versions-rail-state.error {
4933     color: #b91c1c;
4934     background: #f8d7da;
4935 }
4936
4937 .versions-rail-item:first-of-type .versions-rail-state {
4938     color: var(--versions-accent-ink);
4939     background: var(--versions-accent-badge);
4940 }
4941
4942 .versions-rail-loading {
4943     position: relative;
4944     z-index: 1;
4945 }
4946
4947 .versions-rail-loading span {
4948     height: 58px;
4949     background: #e2e8f0;
4950 }
4951
4952 .versions-rail-empty {
4953     position: relative;
4954     z-index: 1;
4955     min-height: 100%;
4956     background: #f8f9fa;
4957 }
4958
4959 .versions-inspector {
4960     min-width: 0;
4961     display: flex;
4962     flex-direction: column;
4963 }
4964
4965 .versions-inspector-header {
4966     display: flex;
4967     align-items: flex-start;
4968     justify-content: space-between;
4969     gap: 24px;
4970     padding: 18px 20px 14px;
```

```
front/src/styles.css
```

```
4971 }
4972
4973 .versions-inspector-header > div {
4974   min-width: 0;
4975 }
4976
4977 .versions-inspector-header > div > span {
4978   color: var(--versions-accent-dark);
4979   font-size: 11px;
4980   font-weight: 800;
4981 }
4982
4983 .versions-inspector-header h2 {
4984   margin: 5px 0 5px;
4985   color: var(--app-text-primary);
4986   font-size: 20px;
4987   line-height: 1.3;
4988 }
4989
4990 .versions-inspector-header p {
4991   margin: 0;
4992   color: var(--app-text-regular);
4993   font-size: 13px;
4994   line-height: 1.5;
4995 }
4996
4997 .versions-compare-picker {
4998   flex: 0 0 250px;
4999 }
5000
5001 .versions-revision-meta {
5002   min-height: 42px;
5003   display: flex;
5004   align-items: center;
5005   gap: 10px;
5006   padding: 0 20px 12px;
5007   border-bottom: 1px solid var(--app-border);
5008 }
5009
5010 .versions-revision-meta > span:not(.versions-quality-badge) {
5011   min-width: 0;
5012   display: inline-flex;
5013   align-items: center;
5014   gap: 5px;
5015   overflow: hidden;
5016   color: var(--app-text-muted);
5017   font-size: 11px;
5018   text-overflow: ellipsis;
5019   white-space: nowrap;
5020 }
```


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```
5021
5022 .versions-revision-meta > span:last-child {
5023     flex: 1;
5024 }
5025
5026 .versions-change-summary {
5027     display: grid;
5028     grid-template-columns: auto minmax(0, 1fr) auto;
5029     align-items: center;
5030     gap: 11px;
5031     margin: 16px 20px 12px;
5032     padding: 12px 14px;
5033     border: 1px solid var(--versions-accent-border);
5034     border-radius: var(--panel-radius);
5035     color: var(--versions-accent-ink);
5036     background: var(--versions-accent-soft);
5037 }
5038
5039 .versions-change-summary.snapshot {
5040     color: #475569;
5041     border-color: #cbd5e1;
5042     background: #f8fafc;
5043 }
5044
5045 .versions-change-summary > div {
5046     min-width: 0;
5047     display: grid;
5048     gap: 4px;
5049 }
5050
5051 .versions-change-summary strong {
5052     font-size: 13px;
5053 }
5054
5055 .versions-change-summary span {
5056     color: color-mix(in srgb, currentColor 84%, #ffffff);
5057     font-size: 12px;
5058     line-height: 1.5;
5059 }
5060
5061 .versions-change-summary > b {
5062     min-width: 54px;
5063     color: inherit;
5064     font-size: 12px;
5065     text-align: right;
5066     font-variant-numeric: tabular-nums;
5067 }
5068
5069 .versions-diff-table {
5070     margin: 0 20px 20px;
```

```
front/src/styles.css
```

```
5071     overflow: hidden;
5072     border: 1px solid var(--app-border);
5073     border-radius: var(--panel-radius);
5074 }
5075
5076 .versions-diff-row {
5077     min-width: 0;
5078     display: grid;
5079     grid-template-columns: minmax(110px, 0.58fr) minmax(180px, 1fr) 28px minmax(180px, 1fr);
5080     align-items: center;
5081     gap: 10px;
5082     min-height: 52px;
5083     padding: 9px 12px;
5084     border-top: 1px solid var(--app-border);
5085     background: rgba(255, 255, 255, 0.72);
5086 }
5087
5088 .versions-diff-row:first-child {
5089     border-top: 0;
5090 }
5091
5092 .versions-diff-row.header {
5093     min-height: 38px;
5094     color: var(--app-text-muted);
5095     background: #f8fafc;
5096     font-size: 11px;
5097     font-weight: 800;
5098 }
5099
5100 .versions-diff-row.changed {
5101     background: #fffbeb;
5102 }
5103
5104 .versions-diff-row > strong {
5105     min-width: 0;
5106     color: var(--app-text-primary);
5107     font-size: 12px;
5108 }
5109
5110 .versions-diff-row > span {
5111     min-width: 0;
5112     color: var(--app-text-regular);
5113     font-size: 12px;
5114     line-height: 1.5;
5115     overflow-wrap: anywhere;
5116 }
5117
5118 .versions-diff-row > svg {
5119     color: #94a3b8;
5120 }
```

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```
5121
5122 .versions-diff-row.changed > svg {
5123   color: #b45309;
5124 }
5125
5126 .versions-diff-row > span:last-child {
5127   display: flex;
5128   align-items: flex-start;
5129   justify-content: space-between;
5130   gap: 8px;
5131   color: var(--app-text-primary);
5132   font-weight: 700;
5133 }
5134
5135 .versions-diff-row > span:last-child > b {
5136   flex: 0 0 auto;
5137   border-radius: 999px;
5138   padding: 2px 6px;
5139   color: #92400e;
5140   background: #fef3c7;
5141   font-size: 10px;
5142   white-space: nowrap;
5143 }
5144
5145 .versions-inspector-empty {
5146   flex: 1;
5147   min-height: 430px;
5148 }
5149
5150 .quality-queue-item:focus-visible,
5151 .quality-filters button:focus-visible,
5152 .versions-rail-item:focus-visible,
5153 .versions-feedback button:focus-visible {
5154   outline: 2px solid var(--app-accent);
5155   outline-offset: 1px;
5156 }
5157
5158 .quality-page-header .btn:disabled,
5159 .quality-inspector .btn:disabled {
5160   color: #94a3b8;
5161   border-color: #e2e8f0;
5162   background: #f1f5f9;
5163   cursor: not-allowed;
5164   transform: none;
5165 }
5166
5167 @keyframes data-workspace-pulse {
5168   from {
5169     opacity: 0.48;
5170   }
```

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```
5171
5172   to {
5173     opacity: 0.9;
5174   }
5175 }
5176
5177 @media (max-width: 1280px) {
5178   .quality-layout {
5179     grid-template-columns: minmax(290px, 310px) minmax(0, 1fr);
5180   }
5181
5182   .versions-dataset-context > div {
5183     min-width: 96px;
5184     padding-inline: 12px;
5185   }
5186
5187   .versions-explorer {
5188     grid-template-columns: 260px minmax(0, 1fr);
5189   }
5190
5191   .versions-diff-row {
5192     grid-template-columns: 100px minmax(150px, 1fr) 24px minmax(150px, 1fr);
5193   }
5194 }
5195
5196 .soft-segmented {
5197   display: inline-flex;
5198   width: max-content;
5199   max-width: 100%;
5200   gap: 2px;
5201   padding: 3px;
5202   border: 1px solid var(--app-border);
5203   border-radius: var(--panel-radius);
5204   background: var(--app-surface-muted);
5205 }
5206
5207 .soft-segment {
5208   min-height: 30px;
5209   padding: 0 12px;
5210   border: 1px solid transparent;
5211   border-radius: 5px;
5212   color: var(--app-text-muted);
5213   background: transparent;
5214   font-size: 12px;
5215   font-weight: 800;
5216   cursor: pointer;
5217 }
5218
5219 .soft-segment.active {
5220   color: var(--app-accent-dark);
```

```
front/src/styles.css
```

```
5221   border-color: rgba(64, 158, 255, 0.28);
5222   background: #ffffff;
5223 }
5224
5225 .soft-formula-box {
5226   margin-top: 14px;
5227   padding: 12px;
5228   border: 1px solid var(--primary-30);
5229   border-radius: var(--panel-radius);
5230   background: var(--app-accent-soft);
5231   display: grid;
5232   gap: 6px;
5233 }
5234
5235 .soft-formula-box span {
5236   color: var(--app-text-muted);
5237   font-size: 12px;
5238 }
5239
5240 .soft-formula-box strong {
5241   overflow: hidden;
5242   color: var(--app-accent-dark);
5243   font-family: ui-monospace, SFMono-Regular, Consolas, 'Liberation Mono', monospace;
5244   font-size: 13px;
5245   text-overflow: ellipsis;
5246   white-space: nowrap;
5247 }
5248
5249 .soft-param-grid {
5250   margin-top: 12px;
5251   display: grid;
5252   grid-template-columns: repeat(2, minmax(0, 1fr));
5253   gap: 10px;
5254 }
5255
5256 .soft-param-grid label {
5257   display: grid;
5258   gap: 6px;
5259   color: var(--app-text-muted);
5260   font-size: 12px;
5261   font-weight: 800;
5262 }
5263
5264 .soft-param-grid input {
5265   min-width: 0;
5266   height: 34px;
5267   border: 1px solid var(--app-border);
5268   border-radius: var(--panel-radius);
5269   padding: 0 10px;
5270   color: var(--app-text-primary);
```

```
front/src/styles.css
```

```
5271     background: #ffffff;
5272     font-size: 13px;
5273     font-weight: 800;
5274 }
5275
5276 .soft-full-button {
5277     width: 100%;
5278     margin-top: 12px;
5279 }
5280
5281 .soft-kernel-grid {
5282     display: grid;
5283     grid-template-columns: repeat(6, minmax(0, 1fr));
5284     gap: 5px;
5285     padding: 14px;
5286     border: 1px solid var(--app-border);
5287     border-radius: var(--panel-radius);
5288     background: var(--app-surface-muted);
5289 }
5290
5291 .soft-kernel-grid span {
5292     display: block;
5293     aspect-ratio: 1;
5294     border-radius: 5px;
5295     background: var(--app-accent);
5296 }
5297
5298 .soft-chart-grid {
5299     display: grid;
5300     grid-template-columns: repeat(2, minmax(0, 1fr));
5301     gap: 12px;
5302 }
5303
5304 .soft-chart-grid .soft-chart-panel:last-child {
5305     grid-column: 1 / -1;
5306 }
5307
5308 .soft-chart-panel .chart-surface {
5309     height: 252px;
5310 }
5311
5312 .soft-check-list {
5313     display: grid;
5314     gap: 9px;
5315 }
5316
5317 .soft-check-list label {
5318     min-width: 0;
5319     display: grid;
5320     grid-template-columns: auto minmax(0, 1fr);
```

```
front/src/styles.css
```

```
5321     align-items: flex-start;
5322     gap: 10px;
5323     padding: 11px;
5324     border: 1px solid var(--app-border);
5325     border-radius: var(--panel-radius);
5326     background: var(--app-surface-muted);
5327 }
5328
5329 .soft-check-list input {
5330     width: 16px;
5331     height: 16px;
5332     margin: 2px 0 0;
5333     accent-color: var(--app-accent);
5334 }
5335
5336 .soft-check-list span {
5337     min-width: 0;
5338     display: grid;
5339     gap: 4px;
5340 }
5341
5342 .soft-check-list strong,
5343 .soft-check-list small {
5344     min-width: 0;
5345     overflow: hidden;
5346     text-overflow: ellipsis;
5347 }
5348
5349 .soft-check-list strong {
5350     color: var(--app-text-primary);
5351     font-size: 13px;
5352 }
5353
5354 .soft-check-list small {
5355     color: var(--app-text-muted);
5356     font-size: 12px;
5357     line-height: 1.45;
5358 }
5359
5360 .soft-document-preview {
5361     min-height: 286px;
5362     padding: 16px;
5363     border: 1px solid var(--app-border);
5364     border-radius: var(--panel-radius);
5365     background: #ffffff;
5366     display: grid;
5367     align-content: start;
5368     gap: 12px;
5369 }
5370
```

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```
5371 .soft-document-preview strong {
5372   color: var(--app-text-primary);
5373   font-size: 17px;
5374 }
5375
5376 .soft-document-preview span {
5377   padding: 9px 10px;
5378   border-radius: var(--panel-radius);
5379   color: var(--app-text-regular);
5380   background: var(--app-surface-muted);
5381   font-size: 12px;
5382 }
5383
5384 .soft-log-list {
5385   display: grid;
5386   gap: 9px;
5387 }
5388
5389 .soft-log-item {
5390   min-width: 0;
5391   display: grid;
5392   grid-template-columns: 72px minmax(0, 1fr) 148px;
5393   align-items: center;
5394   gap: 12px;
5395   padding: 11px 12px;
5396   border: 1px solid var(--app-border);
5397   border-radius: var(--panel-radius);
5398   background: var(--app-surface-muted);
5399 }
5400
5401 .soft-log-item div {
5402   min-width: 0;
5403   display: grid;
5404   gap: 4px;
5405 }
5406
5407 .soft-log-item strong,
5408 .soft-log-item span,
5409 .soft-log-item time {
5410   min-width: 0;
5411   overflow: hidden;
5412   text-overflow: ellipsis;
5413   white-space: nowrap;
5414 }
5415
5416 .soft-log-item strong {
5417   color: var(--app-text-primary);
5418   font-size: 13px;
5419 }
5420
```



```
front/src/styles.css
```

```
5421 .soft-log-item span,
5422 .soft-log-item time {
5423   color: var(--app-text-muted);
5424   font-size: 12px;
5425 }
5426
5427 @media (max-width: 1280px) {
5428   .metrics-grid {
5429     grid-template-columns: repeat(3, minmax(0, 1fr));
5430   }
5431
5432   .summary-grid {
5433     grid-template-columns: repeat(2, minmax(0, 1fr));
5434   }
5435
5436   .top-insight-grid {
5437     grid-template-columns: 1fr;
5438   }
5439
5440   .secondary-card-grid {
5441     grid-template-columns: repeat(2, minmax(0, 1fr));
5442   }
5443
5444   .top-insight-grid .timeline-matrix {
5445     min-width: 640px;
5446   }
5447
5448   .soft-metric-strip {
5449     grid-template-columns: repeat(2, minmax(0, 1fr));
5450   }
5451
5452   .soft-two-column,
5453   .soft-two-column.wide-left,
5454   .soft-chart-grid {
5455     grid-template-columns: 1fr;
5456   }
5457
5458   .soft-chart-grid .soft-chart-panel:last-child {
5459     grid-column: auto;
5460   }
5461 }
5462
5463 @media (max-width: 920px) {
5464   .sidebar,
5465   .sidebar.collapsed {
5466     top: var(--header-height);
5467     width: min(280px, calc(100vw - 24px));
5468     transform: translateX(calc(-100% - 24px));
5469   }
5470 }
```

front/src/styles.css

```
5471 .sidebar.open {
5472     transform: translateX(0);
5473 }
5474
5475 .sidebar.collapsed .brand-copy,
5476 .sidebar.collapsed .nav-label,
5477 .sidebar.collapsed .nav-group-label {
5478     display: initial;
5479 }
5480
5481 .sidebar.collapsed .nav-item {
5482     justify-content: flex-start;
5483     width: calc(100% - 20px);
5484     padding: 0 10px;
5485     margin: 2px 10px;
5486 }
5487
5488 .sidebar-backdrop {
5489     display: block;
5490 }
5491
5492 .sidebar-backdrop.visible {
5493     opacity: 1;
5494     pointer-events: auto;
5495 }
5496
5497 .main-header,
5498 .sidebar.collapsed .main-header {
5499     left: 0;
5500 }
5501
5502 .dashboard-content,
5503 .sidebar.collapsed .dashboard-content {
5504     margin-left: 0;
5505 }
5506
5507 .mobile-nav-button {
5508     display: inline-flex;
5509     margin-left: 8px;
5510 }
5511
5512 .sidebar-toggle-floating {
5513     display: none;
5514 }
5515
5516 .status-header {
5517     grid-template-columns: 1fr;
5518 }
5519
5520 .status-actions {
```

```
front/src/styles.css
```

```
5521     justify-content: flex-start;
5522     flex-wrap: wrap;
5523   }
5524 }
5525
5526 @media (max-width: 640px) {
5527   :root {
5528     --app-gap: 12px;
5529   }
5530
5531   .navbar-right {
5532     padding-right: 8px;
5533   }
5534
5535   .header-section-title {
5536     margin-left: 6px;
5537   }
5538
5539   .dashboard-content {
5540     padding: calc(var(--header-height) + var(--app-gap)) 12px var(--app-gap);
5541   }
5542
5543   .summary-grid,
5544   .metrics-grid,
5545   .secondary-card-grid,
5546   .config-panel {
5547     grid-template-columns: 1fr;
5548   }
5549
5550   .chart-surface {
5551     height: 230px;
5552   }
5553
5554   .secondary-card-grid .chart-surface {
5555     flex: none;
5556     min-height: 0;
5557     height: 230px;
5558   }
5559
5560   .selectable-chart-header {
5561     align-items: stretch;
5562     flex-direction: column;
5563   }
5564
5565   .chart-segmented {
5566     width: 100%;
5567   }
5568
5569   .chart-segment {
5570     flex: 1;
```

```
front/src/styles.css
```

```
5571     justify-content: center;
5572   }
5573
5574   .selectable-chart-panel .chart-surface {
5575     height: 230px;
5576   }
5577
5578   .health-timeline-panel {
5579     min-height: 336px;
5580   }
5581
5582   .ca-matrix-panel {
5583     min-height: 336px;
5584   }
5585
5586   .ca-panel-header {
5587     flex-direction: column;
5588     margin-bottom: 12px;
5589   }
5590
5591   .ca-formula {
5592     white-space: normal;
5593   }
5594
5595   .ca-panel-body {
5596     grid-template-columns: 1fr;
5597     gap: 12px;
5598   }
5599
5600   .ca-matrix-scroll {
5601     min-height: 200px;
5602     justify-content: flex-start;
5603   }
5604
5605   .ca-matrix-preview {
5606     min-width: 0;
5607   }
5608
5609   .ca-info-panel {
5610     padding-top: 12px;
5611     padding-left: 0;
5612     border-top: 1px solid rgba(148, 163, 184, 0.24);
5613     border-left: 0;
5614   }
5615
5616   .timeline-header {
5617     flex-direction: column;
5618     margin-bottom: 12px;
5619   }
5620
```

```
front/src/styles.css
```

```
5621   .timeline-summary {
5622     justify-items: start;
5623   }
5624
5625   .timeline-matrix {
5626     min-width: 560px;
5627   }
5628
5629   .task-grid,
5630   .export-actions {
5631     grid-template-columns: 1fr;
5632   }
5633
5634   .pipeline-step {
5635     grid-template-columns: 30px minmax(0, 1fr);
5636   }
5637
5638   .step-status {
5639     grid-column: 2;
5640   }
5641 }
5642
5643
5644
5645 /* ===== Task Center ===== */
5646 .task-center-page {
5647   display: grid;
5648   gap: 12px;
5649 }
5650
5651 .task-center-stats {
5652   display: grid;
5653   grid-template-columns: repeat(7, minmax(0, 1fr));
5654   gap: 10px;
5655 }
5656
5657 .task-stat-card {
5658   min-width: 0;
5659   display: grid;
5660   gap: 8px;
5661   padding: 12px 14px;
5662   border: 1px solid var(--app-border);
5663   border-radius: var(--panel-radius);
5664   background: var(--app-surface);
5665   text-align: left;
5666   color: var(--app-text-primary);
5667   font: inherit;
5668   cursor: default;
5669   transition: border-color 0.16s ease, box-shadow 0.16s ease, transform 0.16s ease;
5670 }
```

```
front/src/styles.css
```

```
5671
5672 .task-stat-card.is-actionable {
5673   cursor: pointer;
5674 }
5675
5676 .task-stat-card:hover,
5677 .task-stat-card.active {
5678   border-color: var(--app-accent);
5679   box-shadow: 0 4px 8px rgba(37, 99, 235, 0.1);
5680   transform: translateY(-1px);
5681 }
5682
5683 .task-stat-card span {
5684   color: var(--app-text-muted);
5685   font-size: 12px;
5686   font-weight: 700;
5687 }
5688
5689 .task-stat-card strong {
5690   font-size: 22px;
5691   font-weight: 800;
5692   font-variant-numeric: tabular-nums;
5693 }
5694
5695 .task-stat-card.tone-info strong { color: #2563eb; }
5696 .task-stat-card.tone-running strong { color: #0284c7; }
5697 .task-stat-card.tone-success strong { color: #15803d; }
5698 .task-stat-card.tone-danger strong { color: #dc2626; }
5699 .task-stat-card.meta strong { font-size: 14px; line-height: 1.4; }
5700
5701 .task-center-toolbar {
5702   display: grid;
5703   gap: 12px;
5704 }
5705
5706 .task-filter-fields {
5707   display: grid;
5708   grid-template-columns: minmax(300px, 1.8fr) repeat(3, minmax(150px, 0.8fr)) minmax(132px, auto);
5709   gap: 12px;
5710 }
5711
5712 .task-filter-fields label,
5713 .task-filter-fields > .select-field,
5714 .task-form-field {
5715   display: grid;
5716   gap: 6px;
5717   min-width: 0;
5718 }
5719
5720 .task-filter-fields span,
```

```
front/src/styles.css
```

```
5721 .task-form-field span,
5722 .task-check-field {
5723   color: var(--app-text-muted);
5724   font-size: 12px;
5725   font-weight: 700;
5726 }
5727
5728 .task-filter-fields input,
5729 .task-filter-fields .select-field-trigger,
5730 .task-form-field input,
5731 .task-form-field .select-field-trigger,
5732 .task-create-grid input {
5733   width: 100%;
5734   height: 38px;
5735   border: 1px solid var(--app-border);
5736   border-radius: var(--panel-radius);
5737   background: var(--app-surface-muted);
5738   color: var(--app-text-primary);
5739   padding: 0 10px;
5740   font: inherit;
5741 }
5742
5743 .task-toolbar-actions {
5744   display: flex;
5745   justify-content: flex-end;
5746   gap: 8px;
5747   flex-wrap: wrap;
5748 }
5749
5750 .task-center-alert {
5751   padding: 10px 12px;
5752   border-radius: var(--panel-radius);
5753   font-size: 13px;
5754   font-weight: 600;
5755 }
5756
5757 .task-center-alert.error {
5758   color: #991b1b;
5759   background: #fef2f2;
5760   border: 1px solid #fecaca;
5761 }
5762
5763 .task-center-alert.success {
5764   color: #166534;
5765   background: #f0fdf4;
5766   border: 1px solid #bbf7d0;
5767 }
5768
5769 .task-bulk-bar {
5770   display: flex;
```

```
front/src/styles.css
```

```
5771     align-items: center;
5772     justify-content: space-between;
5773     gap: 12px;
5774     padding: 10px 12px;
5775     border: 1px solid var(--primary-30);
5776     border-radius: var(--panel-radius);
5777     background: linear-gradient(135deg, var(--primary-10), rgba(255, 255, 255, 0.9));
5778 }
5779
5780 .task-bulk-bar > div {
5781     display: flex;
5782     gap: 8px;
5783     flex-wrap: wrap;
5784 }
5785
5786 .task-table-wrap {
5787     overflow: auto;
5788 }
5789
5790 .task-table {
5791     width: 100%;
5792     border-collapse: collapse;
5793     min-width: 1240px;
5794 }
5795
5796 .task-table th,
5797 .task-table td {
5798     padding: 12px 10px;
5799     border-bottom: 1px solid var(--app-border);
5800     text-align: left;
5801     vertical-align: top;
5802     font-size: 13px;
5803 }
5804
5805 .task-table th {
5806     color: var(--app-text-muted);
5807     font-size: 12px;
5808     font-weight: 700;
5809     white-space: nowrap;
5810 }
5811
5812 .task-table tr.is-failed {
5813     background: rgba(254, 242, 242, 0.55);
5814 }
5815
5816 .task-table-panel.density-compact .task-table th,
5817 .task-table-panel.density-compact .task-table td {
5818     padding: 8px 8px;
5819 }
5820
```



```
front/src/styles.css
```

```
5821 .task-table-empty {
5822   text-align: center !important;
5823   color: var(--app-text-muted);
5824   padding: 28px 12px !important;
5825 }
5826
5827 .task-name-btn {
5828   display: grid;
5829   gap: 2px;
5830   border: 0;
5831   background: transparent;
5832   padding: 0;
5833   text-align: left;
5834   color: inherit;
5835   cursor: pointer;
5836   font: inherit;
5837 }
5838
5839 .task-name-btn strong {
5840   color: var(--app-text-primary);
5841 }
5842
5843 .task-name-btn small,
5844 .task-fail-hint {
5845   color: var(--app-text-muted);
5846   font-size: 12px;
5847 }
5848
5849 .task-fail-hint {
5850   display: block;
5851   margin-top: 4px;
5852   color: #dc2626;
5853 }
5854
5855 .task-mode-tag,
5856 .task-status-pill {
5857   display: inline-flex;
5858   align-items: center;
5859   min-height: 24px;
5860   padding: 0 8px;
5861   border-radius: 999px;
5862   font-size: 12px;
5863   font-weight: 700;
5864   white-space: nowrap;
5865 }
5866
5867 .task-mode-tag {
5868   color: var(--app-accent-dark);
5869   background: var(--primary-10);
5870   border: 1px solid var(--primary-30);
```

```
front/src/styles.css
```

```
5871 }
5872
5873 .task-status-pill.tone-neutral,
5874 .task-status-pill.tone-muted {
5875     color: #475569;
5876     background: #f8fafc;
5877     border: 1px solid #e2e8f0;
5878 }
5879
5880 .task-status-pill.tone-info {
5881     color: #1d4ed8;
5882     background: #eff6ff;
5883     border: 1px solid #bfdbfe;
5884 }
5885
5886 .task-status-pill.tone-running {
5887     color: #0369a1;
5888     background: #f0f9ff;
5889     border: 1px solid #bae6fd;
5890 }
5891
5892 .task-status-pill.tone-success {
5893     color: #166534;
5894     background: #f0fdf4;
5895     border: 1px solid #bbf7d0;
5896 }
5897
5898 .task-status-pill.tone-danger {
5899     color: #b91c1c;
5900     background: #fef2f2;
5901     border: 1px solid #fecaca;
5902 }
5903
5904 .task-progress-cell {
5905     display: grid;
5906     gap: 6px;
5907     min-width: 120px;
5908 }
5909
5910 .task-stage-cell {
5911     min-width: 92px;
5912     color: var(--app-text-regular);
5913     font-weight: 700;
5914     white-space: nowrap;
5915 }
5916
5917 .task-progress-cell .progress-track {
5918     height: 8px;
5919     border-radius: 999px;
5920     overflow: hidden;
```

```
front/src/styles.css
```

```
5921   background: var(--app-surface-muted);
5922   border: 1px solid var(--app-border);
5923 }
5924
5925 .task-progress-cell .progress-track > span {
5926   display: block;
5927   height: 100%;
5928   background: linear-gradient(90deg, var(--app-accent), #60a5fa);
5929 }
5930
5931 .task-progress-cell .progress-track > span.is-failed {
5932   background: linear-gradient(90deg, #ef4444, #f97316);
5933 }
5934
5935 .task-row-actions {
5936   display: flex;
5937   flex-wrap: wrap;
5938   justify-content: flex-start;
5939   gap: 6px 5px;
5940   min-width: 232px;
5941   max-width: 260px;
5942 }
5943
5944 .task-row-actions button {
5945   height: 28px;
5946   min-width: 44px;
5947   display: inline-flex;
5948   align-items: center;
5949   justify-content: center;
5950   gap: 4px;
5951   padding: 0 7px;
5952   border: 1px solid var(--app-border);
5953   border-radius: 7px;
5954   background: var(--app-surface-muted);
5955   color: var(--app-text-regular);
5956   font: inherit;
5957   font-size: 11px;
5958   font-weight: 700;
5959   line-height: 1;
5960   white-space: nowrap;
5961   cursor: pointer;
5962 }
5963
5964 .task-row-actions button.danger {
5965   color: #b91c1c;
5966   border-color: #fecaca;
5967   background: #fef2f2;
5968 }
5969
5970 .task-pagination {
```

```
front/src/styles.css
```

```
5971   display: flex;
5972   align-items: center;
5973   justify-content: space-between;
5974   gap: 12px;
5975   color: var(--app-text-muted);
5976   font-size: 13px;
5977 }
5978
5979 .task-pagination > div {
5980   display: flex;
5981   gap: 8px;
5982 }
5983
5984 .btn-danger {
5985   color: #fff !important;
5986   background: #dc2626 !important;
5987   border-color: #dc2626 !important;
5988 }
5989
5990 .task-drawer-root {
5991   position: fixed;
5992   inset: 0;
5993   z-index: 1000;
5994   display: grid;
5995   grid-template-columns: 1fr minmax(420px, 520px);
5996 }
5997
5998 .task-drawer-mask {
5999   border: 0;
6000   background: rgba(15, 23, 42, 0.28);
6001   cursor: pointer;
6002 }
6003
6004 .task-drawer-panel {
6005   display: grid;
6006   grid-template-rows: auto auto minmax(0, 1fr) auto;
6007   height: 100dvh;
6008   min-height: 0;
6009   background: var(--app-surface);
6010   border-left: 1px solid var(--app-border);
6011   box-shadow: -18px 0 40px rgba(15, 23, 42, 0.12);
6012 }
6013
6014 .task-drawer-panel.create {
6015   grid-template-rows: auto auto minmax(0, 1fr) auto;
6016   background: #ffffff;
6017 }
6018
6019 .task-drawer-panel.create .task-drawer-header,
6020 .task-drawer-panel.create .task-drawer-footer {
```

```
front/src/styles.css
```

```
6021     background: #ffffff;
6022 }
6023
6024 .task-drawer-panel.create .task-drawer-body {
6025     background: #ffffff;
6026 }
6027
6028 .task-drawer-panel.create .task-create-stepper {
6029     background: #f8fafc;
6030 }
6031
6032 .task-drawer-panel.create .task-template-picker,
6033 .task-drawer-panel.create .task-dataset-snapshot,
6034 .task-drawer-panel.create .task-create-summary {
6035     background: #ffffff;
6036 }
6037
6038 .task-drawer-panel.create .task-dataset-snapshot.quality-error {
6039     background: #fef2f2;
6040 }
6041
6042 .task-drawer-panel.create .task-dataset-snapshot.quality-warning {
6043     background: #fffbeb;
6044 }
6045
6046 .task-drawer-panel.create .task-parameter-note {
6047     background: #eff6ff;
6048 }
6049
6050 .task-drawer-header,
6051 .task-drawer-footer {
6052     display: flex;
6053     align-items: flex-start;
6054     justify-content: space-between;
6055     gap: 12px;
6056     padding: 16px;
6057     border-bottom: 1px solid var(--app-border);
6058 }
6059
6060 .task-drawer-footer {
6061     border-bottom: 0;
6062     border-top: 1px solid var(--app-border);
6063     justify-content: flex-end;
6064     flex-wrap: wrap;
6065 }
6066
6067 .task-drawer-header h2 {
6068     margin: 0 0 4px;
6069     font-size: 18px;
6070 }
```

```
front/src/styles.css
```

```
6071
6072 .task-drawer-header span,
6073 .task-detail-block h3 {
6074   color: var(--app-text-muted);
6075 }
6076
6077 .task-drawer-body {
6078   min-height: 0;
6079   overflow-x: hidden;
6080   overflow-y: auto;
6081   overscroll-behavior: contain;
6082   scroll-padding-block-end: 24px;
6083   padding: 16px 16px 24px;
6084   display: grid;
6085   gap: 16px;
6086 }
6087
6088 .task-detail-block {
6089   display: grid;
6090   gap: 10px;
6091 }
6092
6093 .task-detail-block h3 {
6094   margin: 0;
6095   font-size: 13px;
6096   font-weight: 800;
6097 }
6098
6099 .task-detail-grid,
6100 .task-param-grid,
6101 .task-metric-grid,
6102 .task-create-grid {
6103   display: grid;
6104   grid-template-columns: repeat(2, minmax(0, 1fr));
6105   gap: 8px;
6106 }
6107
6108 .task-detail-grid > div,
6109 .task-param-grid > div,
6110 .task-metric-grid > div {
6111   min-width: 0;
6112   padding: 10px;
6113   border: 1px solid var(--app-border);
6114   border-radius: var(--panel-radius);
6115   background: var(--app-surface-muted);
6116   display: grid;
6117   gap: 4px;
6118 }
6119
6120 .task-detail-grid span,
```

```
front/src/styles.css
```

```
6121 .task-param-grid dt,
6122 .task-metric-grid span {
6123     color: var(--app-text-muted);
6124     font-size: 12px;
6125 }
6126
6127 .task-detail-grid strong,
6128 .task-param-grid dd,
6129 .task-metric-grid strong {
6130     margin: 0;
6131     font-size: 14px;
6132     font-weight: 800;
6133 }
6134
6135 .task-error-box {
6136     padding: 10px 12px;
6137     border-radius: var(--panel-radius);
6138     background: #fef2f2;
6139     border: 1px solid #fecaca;
6140     color: #991b1b;
6141 }
6142
6143 .task-error-box p {
6144     margin: 6px 0 0;
6145 }
6146
6147 .task-pipeline-list,
6148 .task-log-list {
6149     display: grid;
6150     gap: 8px;
6151 }
6152
6153 .task-pipeline-item,
6154 .task-log-item {
6155     display: grid;
6156     grid-template-columns: 28px minmax(0, 1fr);
6157     gap: 10px;
6158     padding: 10px;
6159     border: 1px solid var(--app-border);
6160     border-radius: var(--panel-radius);
6161     background: var(--app-surface-muted);
6162 }
6163
6164 .task-pipeline-item span {
6165     width: 28px;
6166     height: 28px;
6167     display: inline-flex;
6168     align-items: center;
6169     justify-content: center;
6170     border-radius: 999px;
```

```
front/src/styles.css
```

```
6171   background: #e2e8f0;
6172   font-size: 12px;
6173   font-weight: 800;
6174 }
6175
6176 .task-pipeline-item.state-done span { background: #bbf7d0; color: #166534; }
6177 .task-pipeline-item.state-running span { background: #bae6fd; color: #0369a1; }
6178 .task-pipeline-item.state-failed span { background: #fecaca; color: #b91c1c; }
6179
6180 .task-pipeline-item strong,
6181 .task-log-item strong {
6182   display: block;
6183 }
6184
6185 .task-pipeline-item small,
6186 .task-log-item span,
6187 .task-log-item p {
6188   color: var(--app-text-muted);
6189 }
6190
6191 .task-log-item {
6192   grid-template-columns: 1fr;
6193 }
6194
6195 .task-log-item header {
6196   display: flex;
6197   justify-content: space-between;
6198   gap: 10px;
6199 }
6200
6201 .task-log-item.level-error,
6202 .task-log-item.level-warning {
6203   border-color: #fecaca;
6204   background: #fff7f7;
6205 }
6206
6207 .task-check-field {
6208   display: flex;
6209   align-items: center;
6210   gap: 8px;
6211 }
6212
6213 .task-focus-grid {
6214   display: grid;
6215   grid-template-columns: repeat(2, minmax(0, 1fr));
6216   gap: 12px;
6217 }
6218
6219 .task-focus-card {
6220   display: grid;
```



```
front/src/styles.css
```

```
6221     gap: 10px;
6222     min-width: 0;
6223     padding: 15px 16px;
6224     border: 1px solid var(--app-border-strong);
6225     border-radius: var(--panel-radius);
6226     background: var(--app-surface-strong);
6227 }
6228
6229 .task-focus-card.primary {
6230     border-color: #bfdbfe;
6231     background: #f5f9ff;
6232 }
6233
6234 .task-focus-card.attention {
6235     border-color: #fde68a;
6236     background: #fffbeb;
6237 }
6238
6239 .task-focus-card header,
6240 .task-compare-header,
6241 .task-toolbar-header,
6242 .task-dataset-snapshot header,
6243 .task-create-summary header,
6244 .task-template-picker-heading,
6245 .task-local-meta-heading {
6246     display: flex;
6247     align-items: center;
6248     justify-content: space-between;
6249     gap: 10px;
6250 }
6251
6252 .task-focus-card header > span,
6253 .task-compare-kicker,
6254 .task-template-picker-heading,
6255 .task-local-meta-heading > div > span {
6256     display: inline-flex;
6257     align-items: center;
6258     gap: 6px;
6259     color: var(--app-text-regular);
6260     font-size: 12px;
6261     font-weight: 800;
6262 }
6263
6264 .task-focus-card header button {
6265     display: inline-flex;
6266     align-items: center;
6267     gap: 4px;
6268     border: 0;
6269     padding: 0;
6270     background: transparent;
```

```
front/src/styles.css
```

```
6271   color: var(--app-accent-dark);
6272   font: inherit;
6273   font-size: 12px;
6274   font-weight: 800;
6275   cursor: pointer;
6276 }
6277
6278 .task-focus-card > strong {
6279   overflow: hidden;
6280   color: var(--app-text-primary);
6281   font-size: 16px;
6282   line-height: 1.35;
6283   text-overflow: ellipsis;
6284   white-space: nowrap;
6285 }
6286
6287 .task-focus-card p,
6288 .task-empty-copy p,
6289 .task-empty-steps small,
6290 .task-compare-note,
6291 .task-parameter-note,
6292 .task-local-meta-heading p {
6293   margin: 0;
6294   color: var(--app-text-regular);
6295   font-size: 13px;
6296   line-height: 1.55;
6297 }
6298
6299 .task-focus-progress {
6300   height: 7px;
6301   overflow: hidden;
6302   border-radius: 999px;
6303   background: #dbeafe;
6304 }
6305
6306 .task-focus-progress span {
6307   display: block;
6308   width: 100%;
6309   height: 100%;
6310   border-radius: inherit;
6311   background: var(--app-accent);
6312   transform-origin: left center;
6313   transition: transform 200ms ease-out;
6314 }
6315
6316 .task-focus-start {
6317   justify-self: start;
6318 }
6319
6320 .task-toolbar-header {
```

```
front/src/styles.css
```

```
6321     display: flex;
6322     align-items: center;
6323     justify-content: space-between;
6324     gap: 20px;
6325 }
6326
6327 .task-toolbar-title {
6328     min-width: 0;
6329     display: flex;
6330     align-items: center;
6331     gap: 10px;
6332 }
6333
6334 .task-toolbar-title-icon {
6335     width: 34px;
6336     height: 34px;
6337     flex: 0 0 34px;
6338     display: inline-flex;
6339     align-items: center;
6340     justify-content: center;
6341     border-radius: 7px;
6342     background: var(--app-accent-soft);
6343     color: var(--app-accent-dark);
6344 }
6345
6346 .task-toolbar-title h2 {
6347     margin: 0 0 3px;
6348     color: var(--app-text-primary);
6349     font-size: 15px;
6350     line-height: 1.35;
6351 }
6352
6353 .task-toolbar-title > div > span {
6354     display: block;
6355     color: var(--app-text-muted);
6356     font-size: 12px;
6357 }
6358
6359 .task-icon-button,
6360 .task-row-favorite,
6361 .task-template-list button,
6362 .task-add-tag button,
6363 .task-compare-item header button {
6364     display: inline-flex;
6365     align-items: center;
6366     justify-content: center;
6367     width: 30px;
6368     height: 30px;
6369     border: 1px solid var(--app-border);
6370     border-radius: 7px;
```

```
front/src/styles.css
```

```
6371   background: var(--app-surface-strong);
6372   color: var(--app-text-regular);
6373   cursor: pointer;
6374 }
6375
6376 .task-icon-button:hover,
6377 .task-row-favorite:hover,
6378 .task-template-list button:hover,
6379 .task-add-tag button:hover,
6380 .task-compare-item header button:hover {
6381   border-color: var(--app-accent);
6382   color: var(--app-accent-dark);
6383 }
6384
6385 .task-icon-button.is-active,
6386 .task-row-favorite.is-active {
6387   border-color: #fbbf24;
6388   background: #fffbeb;
6389   color: #b45309;
6390 }
6391
6392 .task-filter-fields {
6393   display: grid;
6394   grid-template-columns: minmax(300px, 1.8fr) repeat(3, minmax(150px, 0.8fr)) minmax(132px, auto);
6395   align-items: end;
6396   gap: 12px;
6397 }
6398
6399 .task-filter-fields label,
6400 .task-filter-fields > .select-field {
6401   display: grid;
6402   gap: 6px;
6403   min-width: 0;
6404 }
6405
6406 .task-filter-fields label > span:first-child,
6407 .task-filter-fields > .select-field > .select-field-label,
6408 .task-filter-date-range label > span {
6409   color: var(--app-text-muted);
6410   font-size: 12px;
6411   font-weight: 700;
6412 }
6413
6414 .task-filter-keyword {
6415   grid-column: auto;
6416 }
6417
6418 .task-filter-search {
6419   position: relative;
6420   min-width: 0;
```

front/src/styles.css

```
6421     display: flex;
6422     align-items: center;
6423     height: 38px;
6424     border: 1px solid var(--app-border);
6425     border-radius: var(--panel-radius);
6426     background: var(--app-surface-muted);
6427     transition:
6428         border-color 160ms ease,
6429         background-color 160ms ease,
6430         box-shadow 160ms ease;
6431 }
6432
6433 .task-filter-search:focus-within {
6434     border-color: var(--app-accent);
6435     background: var(--app-surface-strong);
6436     box-shadow: 0 0 0 3px var(--app-accent-soft);
6437 }
6438
6439 .task-filter-search > svg {
6440     flex: 0 0 auto;
6441     margin-left: 11px;
6442     color: var(--app-text-muted);
6443     transition: color 160ms ease;
6444 }
6445
6446 .task-filter-search:focus-within > svg,
6447 .task-filter-search.has-value > svg {
6448     color: var(--app-accent-dark);
6449 }
6450
6451 .task-filter-search input {
6452     min-width: 0;
6453     height: 36px;
6454     flex: 1;
6455     padding: 0 8px;
6456     border: 0;
6457     outline: 0;
6458     background: transparent;
6459     color: var(--app-text-primary);
6460     font: inherit;
6461 }
6462
6463 .task-filter-search button {
6464     width: 28px;
6465     height: 28px;
6466     flex: 0 0 28px;
6467     display: inline-flex;
6468     align-items: center;
6469     justify-content: center;
6470     margin-right: 4px;
```

```
front/src/styles.css
```

```
6471   border: 0;
6472   border-radius: 6px;
6473   background: transparent;
6474   color: var(--app-text-muted);
6475   cursor: pointer;
6476   transition: background-color 150ms ease, color 150ms ease, transform 120ms ease;
6477 }
6478
6479 .task-filter-search button:hover {
6480   background: var(--app-accent-soft);
6481   color: var(--app-accent-dark);
6482 }
6483
6484 .task-filter-search button:active {
6485   transform: scale(0.94);
6486 }
6487
6488 .task-filter-advanced-toggle {
6489   min-width: 132px;
6490   height: 38px;
6491   display: inline-flex;
6492   align-items: center;
6493   justify-content: center;
6494   gap: 7px;
6495   padding: 0 10px;
6496   border: 1px solid var(--app-border);
6497   border-radius: var(--panel-radius);
6498   background: var(--app-surface-strong);
6499   color: var(--app-text-regular);
6500   font: inherit;
6501   font-size: 13px;
6502   font-weight: 700;
6503   cursor: pointer;
6504   transition:
6505     border-color 160ms ease,
6506     background-color 160ms ease,
6507     color 160ms ease,
6508     box-shadow 160ms ease;
6509 }
6510
6511 .task-filter-advanced-toggle:hover,
6512 .task-filter-advanced-toggle.is-open {
6513   border-color: var(--app-accent);
6514   background: var(--app-accent-soft);
6515   color: var(--app-accent-dark);
6516 }
6517
6518 .task-filter-advanced-toggle:focus-visible,
6519 .task-filter-search button:focus-visible,
6520 .task-filter-clear-dates:focus-visible,
```

front/src/styles.css

```
6521 .task-filter-reset:focus-visible,
6522 .task-filter-chips button:focus-visible,
6523 .task-density-control button:focus-visible {
6524     outline: 2px solid var(--app-accent);
6525     outline-offset: 2px;
6526 }
6527
6528 .task-filter-advanced-toggle strong {
6529     min-width: 18px;
6530     height: 18px;
6531     padding-inline: 5px;
6532     border-radius: 999px;
6533     background: var(--app-accent);
6534     color: #ffffff;
6535     font-size: 10px;
6536     line-height: 18px;
6537     text-align: center;
6538 }
6539
6540 .task-filter-advanced-chevron {
6541     margin-left: auto;
6542     transition: transform 200ms cubic-bezier(0.22, 1, 0.36, 1);
6543 }
6544
6545 .task-filter-advanced-toggle.is-open .task-filter-advanced-chevron {
6546     transform: rotate(180deg);
6547 }
6548
6549 .task-filter-advanced {
6550     display: grid;
6551     grid-template-rows: 0fr;
6552     border-block: 0 solid transparent;
6553     background: #f8fafc;
6554     opacity: 0;
6555     transition:
6556         grid-template-rows 220ms cubic-bezier(0.22, 1, 0.36, 1),
6557         border-width 220ms cubic-bezier(0.22, 1, 0.36, 1),
6558         opacity 160ms ease-out;
6559 }
6560
6561 .task-filter-advanced.is-open {
6562     grid-template-rows: 1fr;
6563     border-block-width: 1px;
6564     border-color: var(--app-border);
6565     opacity: 1;
6566 }
6567
6568 .task-filter-advanced-inner {
6569     min-height: 0;
6570     display: grid;
```

```
front/src/styles.css
```

```
6571   grid-template-columns: minmax(190px, 0.6fr) auto minmax(80px, 1fr);
6572   align-items: end;
6573   gap: 22px;
6574   overflow: hidden;
6575   padding: 0 2px;
6576   transform: translateY(-5px);
6577   transition:
6578     padding 220ms cubic-bezier(0.22, 1, 0.36, 1),
6579     transform 220ms cubic-bezier(0.22, 1, 0.36, 1);
6580 }
6581
6582 .task-filter-advanced.is-open .task-filter-advanced-inner {
6583   padding-block: 14px;
6584   transform: translateY(0);
6585 }
6586
6587 .task-filter-advanced-heading {
6588   align-self: center;
6589   display: flex;
6590   align-items: center;
6591   gap: 9px;
6592   color: var(--app-accent-dark);
6593 }
6594
6595 .task-filter-advanced-heading > svg {
6596   flex: 0 0 auto;
6597 }
6598
6599 .task-filter-advanced-heading div {
6600   display: grid;
6601   gap: 2px;
6602 }
6603
6604 .task-filter-advanced-heading strong {
6605   color: var(--app-text-primary);
6606   font-size: 13px;
6607 }
6608
6609 .task-filter-advanced-heading span {
6610   color: var(--app-text-muted);
6611   font-size: 11px;
6612 }
6613
6614 .task-filter-date-range {
6615   position: relative;
6616   display: grid;
6617   grid-template-columns: repeat(2, minmax(0, 210px));
6618   align-items: end;
6619   gap: 28px;
6620   width: min(100%, 448px);
```



```
front/src/styles.css
```

```
6621 }
6622
6623 .task-filter-date-range label {
6624   position: relative;
6625   display: grid;
6626   gap: 6px;
6627   min-width: 0;
6628 }
6629
6630 .task-filter-date-range input {
6631   width: 100%;
6632   height: 36px;
6633   padding: 0 10px;
6634   border: 1px solid var(--app-border);
6635   border-radius: var(--panel-radius);
6636   background: var(--app-surface-strong);
6637   color: var(--app-text-primary);
6638   font: inherit;
6639   transition: border-color 160ms ease, box-shadow 160ms ease;
6640 }
6641
6642 .task-filter-date-range input:focus {
6643   outline: 0;
6644   border-color: var(--app-accent);
6645   box-shadow: 0 0 0 3px var(--app-accent-soft);
6646 }
6647
6648 .task-filter-date-separator {
6649   position: absolute;
6650   left: 50%;
6651   bottom: 10px;
6652   color: var(--app-text-muted);
6653   font-size: 11px;
6654   transform: translateX(-50%);
6655 }
6656
6657 .task-filter-clear-dates,
6658 .task-filter-reset {
6659   border: 0;
6660   background: transparent;
6661   color: var(--app-accent-dark);
6662   font: inherit;
6663   font-size: 12px;
6664   font-weight: 700;
6665   cursor: pointer;
6666 }
6667
6668 .task-filter-clear-dates {
6669   min-height: 36px;
6670   justify-self: start;
```

```
front/src/styles.css
```

```
6671   padding: 0 8px;
6672 }
6673
6674 .task-filter-clear-dates:disabled {
6675   color: var(--app-text-muted);
6676   cursor: default;
6677   opacity: 0.55;
6678 }
6679
6680 .task-filter-summary {
6681   display: grid;
6682   grid-template-rows: 0fr;
6683   opacity: 0;
6684   transition:
6685     grid-template-rows 180ms cubic-bezier(0.22, 1, 0.36, 1),
6686     opacity 140ms ease-out;
6687 }
6688
6689 .task-filter-summary.has-filters {
6690   grid-template-rows: 1fr;
6691   opacity: 1;
6692 }
6693
6694 .task-filter-summary-inner {
6695   min-height: 0;
6696   display: flex;
6697   align-items: center;
6698   gap: 9px;
6699   overflow: hidden;
6700 }
6701
6702 .task-filter-summary.has-filters .task-filter-summary-inner {
6703   padding-top: 1px;
6704 }
6705
6706 .task-filter-summary-label {
6707   flex: 0 0 auto;
6708   color: var(--app-text-muted);
6709   font-size: 11px;
6710   font-weight: 700;
6711 }
6712
6713 .task-filter-chips {
6714   min-width: 0;
6715   display: flex;
6716   align-items: center;
6717   gap: 6px;
6718   flex-wrap: wrap;
6719 }
6720
```

```
front/src/styles.css
```

```
6721 .task-filter-chips button {
6722     max-width: 230px;
6723     height: 26px;
6724     display: inline-flex;
6725     align-items: center;
6726     gap: 5px;
6727     padding: 0 8px;
6728     border: 1px solid #bfdbfe;
6729     border-radius: 999px;
6730     background: #eff6ff;
6731     color: #1d4f91;
6732     font: inherit;
6733     font-size: 11px;
6734     font-weight: 650;
6735     cursor: pointer;
6736     transition: border-color 150ms ease, background-color 150ms ease, transform 120ms ease;
6737 }
6738
6739 .task-filter-chips button span {
6740     overflow: hidden;
6741     text-overflow: ellipsis;
6742     white-space: nowrap;
6743 }
6744
6745 .task-filter-chips button:hover {
6746     border-color: var(--app-accent);
6747     background: #dbeafe;
6748 }
6749
6750 .task-filter-chips button:active {
6751     transform: scale(0.97);
6752 }
6753
6754 .task-filter-reset {
6755     flex: 0 0 auto;
6756     margin-left: auto;
6757     padding: 4px 2px;
6758 }
6759
6760 .task-toolbar-actions {
6761     display: flex;
6762     align-items: center;
6763     justify-content: flex-end;
6764     gap: 8px;
6765     flex-wrap: wrap;
6766     padding-top: 0;
6767 }
6768
6769 .task-toolbar-actions .btn,
6770 .task-bulk-bar .btn,
```

```
front/src/styles.css
```

```
6771 .task-drawer-footer .btn {
6772   display: inline-flex;
6773   align-items: center;
6774   gap: 6px;
6775 }
6776
6777 .task-toolbar-actions .btn {
6778   min-height: 36px;
6779   transition:
6780     border-color 160ms ease,
6781     background-color 160ms ease,
6782     color 160ms ease,
6783     transform 120ms ease,
6784     box-shadow 160ms ease;
6785 }
6786
6787 .task-toolbar-actions .btn:active:not(:disabled) {
6788   transform: scale(0.97);
6789 }
6790
6791 .task-toolbar-actions .btn:disabled {
6792   cursor: wait;
6793   opacity: 0.72;
6794 }
6795
6796 .task-toolbar-actions svg.is-spinning {
6797   animation: task-toolbar-spin 800ms linear infinite;
6798 }
6799
6800 @keyframes task-toolbar-spin {
6801   to { transform: rotate(360deg); }
6802 }
6803
6804 .task-table-header {
6805   align-items: center;
6806 }
6807
6808 .task-density-control {
6809   display: inline-flex;
6810   align-items: center;
6811   gap: 2px;
6812   padding: 3px;
6813   border: 1px solid var(--app-border);
6814   border-radius: 7px;
6815   background: var(--app-surface-muted);
6816 }
6817
6818 .task-density-control button {
6819   height: 28px;
6820   padding: 0 11px;
```

```
front/src/styles.css
```

```
6821   border: 0;
6822   border-radius: 5px;
6823   background: transparent;
6824   color: var(--app-text-muted);
6825   font: inherit;
6826   font-size: 12px;
6827   font-weight: 700;
6828   cursor: pointer;
6829   transition: background-color 160ms ease, color 160ms ease, box-shadow 160ms ease;
6830 }
6831
6832 .task-density-control button:hover:not(.is-active) {
6833   color: var(--app-text-primary);
6834 }
6835
6836 .task-density-control button.is-active {
6837   background: var(--app-surface-strong);
6838   color: var(--app-accent-dark);
6839   box-shadow: 0 1px 3px rgba(15, 23, 42, 0.12);
6840 }
6841
6842 .task-bulk-bar {
6843   border-color: #bfdbfe;
6844   background: #eff6ff;
6845 }
6846
6847 .task-name-cell {
6848   display: grid;
6849   gap: 6px;
6850   min-width: 180px;
6851 }
6852
6853 .task-row-meta,
6854 .task-tag-list {
6855   display: flex;
6856   align-items: center;
6857   gap: 5px;
6858   flex-wrap: wrap;
6859 }
6860
6861 .task-row-meta > span,
6862 .task-local-tag {
6863   display: inline-flex;
6864   align-items: center;
6865   min-height: 22px;
6866   max-width: 120px;
6867   overflow: hidden;
6868   padding: 0 7px;
6869   border: 1px solid #c7d2fe;
6870   border-radius: 999px;
```

```
front/src/styles.css
```

```
6871   background: #eef2ff;
6872   color: #4338ca;
6873   font-size: 11px;
6874   font-weight: 700;
6875   text-overflow: ellipsis;
6876   white-space: nowrap;
6877 }
6878
6879 .task-row-favorite {
6880   width: 22px;
6881   height: 22px;
6882   border-radius: 6px;
6883 }
6884
6885 .task-row-actions button.is-active {
6886   border-color: #a5b4fc;
6887   background: #eef2ff;
6888   color: #4338ca;
6889 }
6890
6891 .task-empty-state {
6892   display: grid;
6893   grid-template-columns: minmax(280px, 0.88fr) minmax(420px, 1.12fr);
6894   gap: 28px;
6895   align-items: center;
6896   padding: 28px;
6897 }
6898
6899 .task-empty-copy {
6900   display: grid;
6901   gap: 12px;
6902   align-content: center;
6903 }
6904
6905 .task-empty-kicker {
6906   color: var(--app-accent-dark);
6907   font-size: 12px;
6908   font-weight: 800;
6909 }
6910
6911 .task-empty-copy h2,
6912 .task-compare-header h2 {
6913   margin: 0;
6914   color: var(--app-text-primary);
6915   font-size: 21px;
6916   line-height: 1.28;
6917   text-wrap: balance;
6918 }
6919
6920 .task-empty-actions {
```

```
front/src/styles.css
```

```
6921   display: flex;
6922   flex-wrap: wrap;
6923   gap: 8px;
6924   padding-top: 4px;
6925 }
6926
6927 .task-empty-actions .btn {
6928   display: inline-flex;
6929   align-items: center;
6930   gap: 7px;
6931 }
6932
6933 .task-empty-steps {
6934   display: grid;
6935   gap: 0;
6936   padding: 0;
6937   margin: 0;
6938   list-style: none;
6939 }
6940
6941 .task-empty-steps li {
6942   display: grid;
6943   grid-template-columns: 30px minmax(0, 1fr) 30px;
6944   gap: 12px;
6945   align-items: center;
6946   padding: 13px 0;
6947   border-bottom: 1px solid var(--app-border);
6948 }
6949
6950 .task-empty-steps li:last-child {
6951   border-bottom: 0;
6952 }
6953
6954 .task-empty-steps > li > span {
6955   display: inline-grid;
6956   width: 30px;
6957   height: 30px;
6958   place-items: center;
6959   border-radius: 50%;
6960   background: var(--primary-10);
6961   color: var(--app-accent-dark);
6962   font-size: 12px;
6963   font-weight: 800;
6964 }
6965
6966 .task-empty-steps strong,
6967 .task-empty-steps small {
6968   display: block;
6969 }
6970
```

```
front/src/styles.css
```

```
6971 .task-empty-steps strong {
6972   margin-bottom: 3px;
6973   color: var(--app-text-primary);
6974   font-size: 13px;
6975 }
6976
6977 .task-empty-steps small {
6978   font-size: 12px;
6979 }
6980
6981 .task-empty-steps button {
6982   display: inline-flex;
6983   align-items: center;
6984   justify-content: center;
6985   width: 30px;
6986   height: 30px;
6987   border: 0;
6988   border-radius: 7px;
6989   background: transparent;
6990   color: var(--app-accent-dark);
6991   cursor: pointer;
6992 }
6993
6994 .task-compare-tray {
6995   display: grid;
6996   gap: 14px;
6997   padding: 16px;
6998 }
6999
7000 .task-compare-header {
7001   align-items: flex-end;
7002 }
7003
7004 .task-compare-kicker {
7005   margin-bottom: 4px;
7006   color: var(--app-accent-dark);
7007 }
7008
7009 .task-compare-grid {
7010   display: grid;
7011   grid-template-columns: repeat(3, minmax(0, 1fr));
7012   gap: 10px;
7013 }
7014
7015 .task-compare-item {
7016   display: grid;
7017   gap: 12px;
7018   min-width: 0;
7019   padding: 12px;
7020   border: 1px solid var(--app-border);
```


front/src/styles.css

```
7021     border-radius: var(--panel-radius);
7022     background: var(--app-surface-muted);
7023 }
7024
7025 .task-compare-item header {
7026     display: flex;
7027     align-items: flex-start;
7028     justify-content: space-between;
7029     gap: 8px;
7030 }
7031
7032 .task-compare-item header > div {
7033     display: grid;
7034     gap: 3px;
7035     min-width: 0;
7036 }
7037
7038 .task-compare-item strong {
7039     overflow: hidden;
7040     color: var(--app-text-primary);
7041     font-size: 13px;
7042     text-overflow: ellipsis;
7043     white-space: nowrap;
7044 }
7045
7046 .task-compare-item header span {
7047     overflow: hidden;
7048     color: var(--app-text-muted);
7049     font-size: 11px;
7050     text-overflow: ellipsis;
7051     white-space: nowrap;
7052 }
7053
7054 .task-compare-item header button {
7055     flex: 0 0 auto;
7056     width: 24px;
7057     height: 24px;
7058 }
7059
7060 .task-compare-item dl,
7061 .task-dataset-snapshot dl,
7062 .task-create-summary dl {
7063     display: grid;
7064     grid-template-columns: repeat(2, minmax(0, 1fr));
7065     gap: 8px;
7066     margin: 0;
7067 }
7068
7069 .task-compare-item dl div,
7070 .task-dataset-snapshot dl div,
```

front/src/styles.css

```
7071 .task-create-summary dl div {
7072   display: grid;
7073   gap: 3px;
7074 }
7075
7076 .task-compare-item dt,
7077 .task-dataset-snapshot dt,
7078 .task-create-summary dt {
7079   color: var(--app-text-muted);
7080   font-size: 11px;
7081 }
7082
7083 .task-compare-item dd,
7084 .task-dataset-snapshot dd,
7085 .task-create-summary dd {
7086   margin: 0;
7087   color: var(--app-text-primary);
7088   font-size: 12px;
7089   font-weight: 800;
7090 }
7091
7092 .task-compare-note {
7093   display: flex;
7094   align-items: flex-start;
7095   gap: 6px;
7096   color: var(--app-text-muted);
7097   font-size: 12px;
7098 }
7099
7100 .task-drawer-title-row {
7101   display: flex;
7102   align-items: center;
7103   gap: 8px;
7104 }
7105
7106 .task-drawer-title-row .task-icon-button {
7107   width: 28px;
7108   height: 28px;
7109 }
7110
7111 .task-detail-tabs {
7112   display: flex;
7113   gap: 2px;
7114   overflow-x: auto;
7115   padding: 0 16px;
7116   border-bottom: 1px solid var(--app-border);
7117   background: var(--app-surface-strong);
7118 }
7119
7120 .task-detail-tabs button {
```

```
front/src/styles.css
```

```
7121     flex: 0 0 auto;
7122     min-height: 42px;
7123     border: 0;
7124     border-bottom: 2px solid transparent;
7125     background: transparent;
7126     color: var(--app-text-muted);
7127     font: inherit;
7128     font-size: 13px;
7129     font-weight: 700;
7130     cursor: pointer;
7131 }
7132
7133 .task-detail-tabs button + button {
7134     margin-left: 16px;
7135 }
7136
7137 .task-detail-tabs button.active {
7138     border-bottom-color: var(--app-accent);
7139     color: var(--app-accent-dark);
7140 }
7141
7142 .task-local-meta-block {
7143     padding: 14px;
7144     border: 1px solid #c7d2fe;
7145     border-radius: var(--panel-radius);
7146     background: #f8faff;
7147 }
7148
7149 .task-local-meta-heading {
7150     align-items: flex-start;
7151 }
7152
7153 .task-local-meta-heading > div {
7154     display: grid;
7155     gap: 5px;
7156 }
7157
7158 .task-local-meta-heading > div > span {
7159     color: #4338ca;
7160 }
7161
7162 .task-local-meta-heading > svg {
7163     color: #6366f1;
7164 }
7165
7166 .task-tag-editor {
7167     display: grid;
7168     gap: 8px;
7169 }
7170
```

front/src/styles.css

```
7171 .task-tag-editor > span {
7172   display: inline-flex;
7173   align-items: center;
7174   gap: 6px;
7175   color: var(--app-text-muted);
7176   font-size: 12px;
7177   font-weight: 700;
7178 }
7179
7180 .task-local-tag {
7181   gap: 4px;
7182   max-width: none;
7183 }
7184
7185 .task-local-tag button {
7186   display: inline-flex;
7187   align-items: center;
7188   justify-content: center;
7189   width: 14px;
7190   height: 14px;
7191   border: 0;
7192   padding: 0;
7193   background: transparent;
7194   color: inherit;
7195   cursor: pointer;
7196 }
7197
7198 .task-add-tag {
7199   display: flex;
7200   gap: 7px;
7201 }
7202
7203 .task-add-tag input {
7204   min-width: 0;
7205   flex: 1;
7206   height: 36px;
7207   border: 1px solid var(--app-border);
7208   border-radius: var(--panel-radius);
7209   padding: 0 10px;
7210   background: var(--app-surface-strong);
7211   color: var(--app-text-primary);
7212   font: inherit;
7213 }
7214
7215 .task-add-tag button {
7216   flex: 0 0 auto;
7217   height: 36px;
7218 }
7219
7220 .task-form-field textarea {
```

```
front/src/styles.css
```

```
7221   width: 100%;
7222   min-height: 132px;
7223   resize: vertical;
7224   border: 1px solid var(--app-border);
7225   border-radius: var(--panel-radius);
7226   padding: 10px;
7227   background: var(--app-surface-strong);
7228   color: var(--app-text-primary);
7229   font: inherit;
7230   line-height: 1.5;
7231 }
7232
7233 .task-create-stepper {
7234   display: grid;
7235   grid-template-columns: repeat(3, minmax(0, 1fr));
7236   gap: 8px;
7237   padding: 14px 16px;
7238   border-bottom: 1px solid var(--app-border);
7239   background: var(--app-surface-muted);
7240 }
7241
7242 .task-create-stepper > div {
7243   display: flex;
7244   align-items: center;
7245   gap: 8px;
7246   min-width: 0;
7247   color: var(--app-text-muted);
7248 }
7249
7250 .task-create-stepper > div > span {
7251   display: inline-flex;
7252   align-items: center;
7253   justify-content: center;
7254   width: 28px;
7255   height: 28px;
7256   flex: 0 0 auto;
7257   border: 1px solid var(--app-border);
7258   border-radius: 50%;
7259   background: var(--app-surface-strong);
7260 }
7261
7262 .task-create-stepper strong {
7263   overflow: hidden;
7264   font-size: 12px;
7265   text-overflow: ellipsis;
7266   white-space: nowrap;
7267 }
7268
7269 .task-create-stepper > div.active,
7270 .task-create-stepper > div.done {
```

```
front/src/styles.css
```

```
7271   color: var(--app-accent-dark);
7272 }
7273
7274 .task-create-stepper > div.active > span {
7275   border-color: var(--app-accent);
7276   background: var(--app-accent);
7277   color: #fff;
7278 }
7279
7280 .task-create-stepper > div.done > span {
7281   border-color: #86efac;
7282   background: #f0fdf4;
7283   color: #15803d;
7284 }
7285
7286 .task-dataset-snapshot,
7287 .task-template-picker,
7288 .task-create-summary {
7289   display: grid;
7290   gap: 12px;
7291   padding: 13px;
7292   border: 1px solid var(--app-border);
7293   border-radius: var(--panel-radius);
7294   background: var(--app-surface-muted);
7295 }
7296
7297 .task-dataset-snapshot header,
7298 .task-create-summary header {
7299   justify-content: flex-start;
7300   color: var(--app-text-primary);
7301   font-size: 13px;
7302 }
7303
7304 .task-dataset-snapshot.quality-error {
7305   border-color: #fecaca;
7306   background: #fef2f2;
7307 }
7308
7309 .task-dataset-snapshot.quality-warning {
7310   border-color: #fde68a;
7311   background: #fffbeb;
7312 }
7313
7314 .task-dataset-snapshot p {
7315   margin: 0;
7316   color: var(--app-text-regular);
7317   font-size: 12px;
7318   line-height: 1.5;
7319 }
7320
```

front/src/styles.css

```
7321 .task-template-list {
7322     display: grid;
7323     gap: 6px;
7324 }
7325
7326 .task-template-list > div {
7327     display: flex;
7328     align-items: center;
7329     justify-content: space-between;
7330     gap: 8px;
7331     padding: 7px 0;
7332     border-top: 1px solid var(--app-border);
7333     color: var(--app-text-regular);
7334     font-size: 12px;
7335 }
7336
7337 .task-template-list button {
7338     width: 26px;
7339     height: 26px;
7340     color: #b91c1c;
7341 }
7342
7343 .task-template-picker > small {
7344     color: var(--app-text-muted);
7345     font-size: 12px;
7346 }
7347
7348 .task-parameter-note {
7349     padding: 9px 10px;
7350     border-radius: 7px;
7351     background: #eff6ff;
7352     color: #1e40af;
7353     font-size: 12px;
7354 }
7355
7356 .task-create-summary dl {
7357     grid-template-columns: repeat(3, minmax(0, 1fr));
7358 }
7359
7360 .task-create-summary dd {
7361     overflow: hidden;
7362     text-overflow: ellipsis;
7363     white-space: nowrap;
7364 }
7365
7366 .task-row-actions button:focus-visible,
7367 .task-icon-button:focus-visible,
7368 .task-row-favorite:focus-visible,
7369 .task-detail-tabs button:focus-visible,
7370 .task-empty-steps button:focus-visible,
```

```
front/src/styles.css
```

```
7371 .task-template-list button:focus-visible,
7372 .task-add-tag button:focus-visible {
7373     outline: 2px solid var(--app-accent);
7374     outline-offset: 2px;
7375 }
7376
7377 @media (max-width: 1280px) {
7378     .task-center-stats {
7379         grid-template-columns: repeat(4, minmax(0, 1fr));
7380     }
7381
7382     .task-filter-fields {
7383         grid-template-columns: repeat(3, minmax(0, 1fr));
7384     }
7385 }
7386
7387 @media (prefers-reduced-motion: reduce) {
7388     *,
7389     *:before,
7390     *:after {
7391         scroll-behavior: auto !important;
7392         transition-duration: 0.01ms !important;
7393         animation-duration: 0.01ms !important;
7394         animation-iteration-count: 1 !important;
7395     }
7396
7397     .ca-matrix-cell:hover {
7398         transform: none;
7399     }
7400
7401     .ca-matrix-cell:hover .ca-tooltip {
7402         transform: translate(-50%, 0) scale(1);
7403     }
7404 }
7405
```

```
// ===== 文件: front/src/utils/time.ts =====
```

```
1 const ISO_WITH_TIMEZONE = /(?:Z|[\+-]\d{2}:?\d{2})$/i;
2
3 type LocalDateTimeOptions = {
4     locale?: string | string[];
5     timeZone?: string;
6 };
7
8 function parseUtcDate(value: string | null | undefined) {
9     const normalized = typeof value === 'string' ? value.trim() : '';
10    if (!normalized || !ISO_WITH_TIMEZONE.test(normalized)) {
11        return null;
12    }
13
```



```
front/src/utils/time.ts
```

```
14   const date = new Date(normalized);
15   return Number.isNaN(date.getTime()) ? null : date;
16 }
17
18 export function utcTimestamp(value: string | null | undefined) {
19   return parseUtcDate(value)?.getTime() ?? null;
20 }
21
22 export function formatLocalDateTime(
23   value: string | null | undefined,
24   options: LocalDateTimeOptions = {},
25 ) {
26   const date = parseUtcDate(value);
27   if (!date) {
28     return '—';
29   }
30
31   return new Intl.DateTimeFormat(options.locale, {
32     timeZone: options.timeZone,
33     year: 'numeric',
34     month: '2-digit',
35     day: '2-digit',
36     hour: '2-digit',
37     minute: '2-digit',
38     second: '2-digit',
39     hourCycle: 'h23',
40   }).format(date);
41 }
42
43 export function formatLocalDate(value: string | null | undefined, options: LocalDateTimeOptions = {}) {
44   const date = parseUtcDate(value);
45   if (!date) {
46     return '—';
47   }
48
49   return new Intl.DateTimeFormat(options.locale, {
50     timeZone: options.timeZone,
51     year: 'numeric',
52     month: '2-digit',
53     day: '2-digit',
54   }).format(date);
55 }
56
57 export function formatLocalClock(value: string | null | undefined, options: LocalDateTimeOptions = {}) {
58   const date = parseUtcDate(value);
59   if (!date) {
60     return '—';
61   }
62
63   return new Intl.DateTimeFormat(options.locale, {
```

```
front/src/utils/time.ts
```

```
64     timeZone: options.timeZone,
65     hour: '2-digit',
66     minute: '2-digit',
67     second: '2-digit',
68     hourCycle: 'h23',
69   }).format(date);
70 }
71
72 export function formatLocalDateTimeOr(
73   value: string | null | undefined,
74   fallback: string,
75   options: LocalDateTimeOptions = {},
76 ) {
77   const formatted = formatLocalDateTime(value, options);
78   return formatted === '—' ? fallback : formatted;
79 }
80
81 export function utcDateTimeAttribute(value: string | null | undefined) {
82   return parseUtcDate(value) ? value ?? undefined : undefined;
83 }
84
85
86 // ===== 文件: front/src/workbench/WorkbenchUi.tsx =====
87 1 import type { ReactNode } from 'react';
88 2 import type { LucideIcon } from 'lucide-react';
89 3
90 4 export type WorkbenchMetricTone = 'blue' | 'teal' | 'green' | 'amber' | 'red' | 'neutral';
91 5
92 6 export type WorkbenchMetric = {
93 7   label: string;
94 8   value: string;
95 9   note: string;
96 10  tone?: WorkbenchMetricTone;
97 11  icon?: LucideIcon;
98 12 };
99 13
100 14 type WorkbenchPageHeaderProps = {
101 15   icon: LucideIcon;
102 16   title: string;
103 17   context: string;
104 18   status?: ReactNode;
105 19   actions?: ReactNode;
106 20   backAction?: ReactNode;
107 21 };
108 22
109 23 type WorkbenchSectionHeaderProps = {
110 24   title: string;
111 25   meta?: string;
112 26   actions?: ReactNode;
113 27 };
```

```
front/src/workbench/WorkbenchUi.tsx
```

```
28
29 type WorkbenchStatusProps = {
30   tone: 'success' | 'warning' | 'error' | 'info' | 'neutral';
31   children: ReactNode;
32   pulse?: boolean;
33 };
34
35 export function WorkbenchPageHeader({
36   icon: Icon,
37   title,
38   context,
39   status,
40   actions,
41   backAction,
42 }: WorkbenchPageHeaderProps) {
43   return (
44     <header className="workbench-page-header">
45       <div className="workbench-page-title-block">
46         {backAction ? <div className="workbench-page-back">{backAction}</div> : null}
47         <span className="workbench-page-icon" aria-hidden="true">
48           <Icon size={20} />
49         </span>
50         <div>
51           <div className="workbench-page-title-line">
52             <h1>{title}</h1>
53             {status}
54           </div>
55           <p>{context}</p>
56         </div>
57       </div>
58       {actions ? <div className="workbench-page-actions">{actions}</div> : null}
59     </header>
60   );
61 }
62
63 export function WorkbenchMetricStrip({
64   label,
65   metrics,
66 }: {
67   label: string;
68   metrics: WorkbenchMetric[];
69 }) {
70   return (
71     <section className="workbench-metric-strip" aria-label={label}>
72       {metrics.map((metric) => {
73         const Icon = metric.icon;
74         return (
75           <article className={`workbench-metric tone-${metric.tone ?? 'neutral'}`} key={metric.label}>
76             <div className="workbench-metric-label">
77               {Icon ? <Icon size={15} aria-hidden="true" /> : null}
```

```
front/src/workbench/WorkbenchUi.tsx
```

```
78         <span>{metric.label}</span>
79       </div>
80       <strong>{metric.value}</strong>
81       <small>{metric.note}</small>
82     </article>
83   );
84   }}}
85 </section>
86 );
87 }
88
89 export function WorkbenchSectionHeader({ title, meta, actions }: WorkbenchSectionHeaderProps) {
90   return (
91     <header className="workbench-section-header">
92       <div>
93         <h2>{title}</h2>
94         {meta ? <span>{meta}</span> : null}
95       </div>
96       {actions ? <div className="workbench-section-actions">{actions}</div> : null}
97     </header>
98   );
99 }
100
101 export function WorkbenchStatus({ tone, children, pulse = false }: WorkbenchStatusProps) {
102   return (
103     <span className={`workbench-status ${tone}${pulse ? ' is-pulsing' : ''}`}>
104       <i aria-hidden="true" />
105       {children}
106     </span>
107   );
108 }
109
110 export function WorkbenchProgress({
111   value,
112   label,
113   tone = 'blue',
114   animated = false,
115 }): {
116   value: number;
117   label: string;
118   tone?: 'blue' | 'teal' | 'green' | 'amber';
119   animated?: boolean;
120 }) {
121   const safeValue = Math.max(0, Math.min(100, value));
122
123   return (
124     <div className={`workbench-progress tone-${tone}${animated ? ' is-animated' : ''}`}>
125       <div>
126         <span>{label}</span>
127         <strong>{safeValue}%</strong>
```

```
front/src/workbench/WorkbenchUi.tsx
```

```
128     </div>
129     <div
130       className="workbench-progress-track"
131       role="progressbar"
132       aria-label={label}
133       aria-valuemin={0}
134       aria-valuemax={100}
135       aria-valuenow={safeValue}
136     >
137       <span style={{ transform: `scaleX(${safeValue / 100})` }} />
138     </div>
139   </div>
140 );
141 }
142
143 export function WorkbenchNotice({
144   tone,
145   icon: Icon,
146   title,
147   detail,
148   action,
149 }): {
150   tone: 'info' | 'success' | 'warning' | 'error';
151   icon: LucideIcon;
152   title: string;
153   detail: string;
154   action?: ReactNode;
155 }) {
156   return (
157     <div className={`workbench-notice ${tone}`} role={tone === 'error' ? 'alert' : 'status'}>
158       <Icon size={17} aria-hidden="true" />
159       <div>
160         <strong>{title}</strong>
161         <span>{detail}</span>
162       </div>
163       {action ? <div className="workbench-notice-action">{action}</div> : null}
164     </div>
165   );
166 }
167
168 // ===== 文件: front/src/workbench/data-quality/DataQualityPage.tsx =====
169 1 import { useCallback, useEffect, useMemo, useState } from 'react';
170 2 import {
171 3   AlertTriangle,
172 4   CheckCircle2,
173 5   CircleAlert,
174 6   Database,
175 7   RefreshCw,
176 8   Search,
```

```
front/src/workbench/data-quality/DataQualityPage.tsx
```

```
9   ShieldCheck,
10 } from 'lucide-react';
11 import { useSearchParams } from 'react-router-dom';
12 import {
13   fetchDatasetCatalog,
14   recheckDatasetQuality,
15   type DatasetCatalogItem,
16   type DatasetQualityStatus,
17 } from '../../api/datasets';
18
19 type QualityFilter = 'all' | DatasetQualityStatus;
20 type RuleTone = 'pass' | 'warning' | 'error';
21
22 const FILTERS: Array<{ key: QualityFilter; label: string }> = [
23   { key: 'all', label: '全部' },
24   { key: 'error', label: '需处理' },
25   { key: 'warning', label: '待确认' },
26   { key: 'ready', label: '检查通过' },
27 ];
28
29 const STATUS_META: Record<
30   DatasetQualityStatus,
31   { label: string; description: string; tone: 'ready' | 'warning' | 'error' }
32 > = {
33   ready: {
34     label: '检查通过',
35     description: '数据结构完整, 当前未发现阻塞质量问题。',
36     tone: 'ready',
37   },
38   warning: {
39     label: '待确认',
40     description: '存在待确认项, 建议复检或核对标签信息。',
41     tone: 'warning',
42   },
43   error: {
44     label: '需处理',
45     description: '存在质量问题, 请修复后重新复检。',
46     tone: 'error',
47   },
48 };
49
50 function normalizeFilter(value: string | null): QualityFilter {
51   return value === 'ready' || value === 'warning' || value === 'error' ? value : 'all';
52 }
53
54 function pickDefaultDataset(items: DatasetCatalogItem[]) {
55   return (
56     items.find((item) => item.qualityStatus === 'error') ??
57     items.find((item) => item.qualityStatus === 'warning') ??
58     items[0] ??
```

front/src/workbench/data-quality/DataQualityPage.tsx

```
59     null
60   );
61 }
62
63 function includesIssue(dataset: DatasetCatalogItem, phrase: string) {
64   return dataset.qualityIssues.some((issue) => issue.includes(phrase));
65 }
66
67 function buildQualityRules(dataset: DatasetCatalogItem) {
68   const matrixInvalid = includesIssue(dataset, '未检测到有效的基础聚类矩阵');
69   const labelsMissing = !dataset.hasLabels || includesIssue(dataset, '未提供真实标签');
70   const labelCountMismatch = includesIssue(dataset, '标签数量与样本数量不一致');
71
72   return [
73     {
74       key: 'matrix',
75       title: '基础聚类矩阵',
76       tone: matrixInvalid ? ('error' as RuleTone) : ('pass' as RuleTone),
77       result: matrixInvalid ? '未检测到有效矩阵' : dataset.matrixShape || `${dataset.sampleCount} x ${dataset.baseCount}`,
78       detail: matrixInvalid
79         ? '请重新上传包含有效基础聚类矩阵的数据文件。'
80         : '样本数和基础聚类数量均已识别。',
81     },
82     {
83       key: 'labels',
84       title: '真实标签可用性',
85       tone: labelsMissing ? ('warning' as RuleTone) : ('pass' as RuleTone),
86       result: labelsMissing ? '未提供真实标签' : dataset.labelShape || `${dataset.sampleCount} 个标签`,
87       detail: labelsMissing
88         ? '聚类计算仍可执行, 但 ACC、NMI、ARI 和 F1 等监督指标不可用。'
89         : '真实标签可以用于性能评估。',
90     },
91     {
92       key: 'label-count',
93       title: '标签数量一致性',
94       tone: labelCountMismatch ? ('warning' as RuleTone) : ('pass' as RuleTone),
95       result: labelCountMismatch ? '标签数量与样本数不一致' : '数量一致',
96       detail: labelCountMismatch
97         ? '建议检查标签变量和主矩阵是否来自同一批样本。'
98         : '标签数量与基础聚类矩阵样本数一致。',
99     },
100    {
101      key: 'readiness',
102      title: '质量检查结论',
103      tone:
104        dataset.qualityStatus === 'error'
105          ? ('error' as RuleTone)
106          : dataset.qualityStatus === 'warning'
107            ? ('warning' as RuleTone)
108            : ('pass' as RuleTone),
```

```
front/src/workbench/data-quality/DataQualityPage.tsx
```

```
109     result:
110       dataset.qualityStatus === 'error'
111       ? '需要处理'
112       : dataset.qualityStatus === 'warning'
113       ? '建议确认'
114       : '检查通过',
115     detail: STATUS_META[dataset.qualityStatus].description,
116   },
117 ];
118 }
119
120 function QualityStatus({ status }: { status: DatasetQualityStatus }) {
121   const meta = STATUS_META[status];
122   return <span className={`quality-status ${meta.tone}`}>{meta.label}</span>;
123 }
124
125 function RuleIcon({ tone }: { tone: RuleTone }) {
126   if (tone === 'pass') {
127     return <CheckCircle2 size={18} aria-hidden="true" />;
128   }
129   if (tone === 'warning') {
130     return <CircleAlert size={18} aria-hidden="true" />;
131   }
132   return <AlertTriangle size={18} aria-hidden="true" />;
133 }
134
135 export function DataQualityPage() {
136   const [searchParams, setSearchParams] = useSearchParams();
137   const [datasets, setDatasets] = useState<DatasetCatalogItem[]>([]);
138   const [filter, setFilter] = useState<QualityFilter>(() => normalizeFilter(searchParams.get('status')));
139   const [keyword, setKeyword] = useState(searchParams.get('q') || '');
140   const [selectedId, setSelectedId] = useState<number | null>(() => {
141     const value = Number(searchParams.get('datasetId'));
142     return Number.isFinite(value) && value > 0 ? value : null;
143   });
144   const [loading, setLoading] = useState(true);
145   const [error, setError] = useState('');
146   const [message, setMessage] = useState('');
147   const [recheckingIds, setRecheckingIds] = useState<number[]>([]);
148
149   // 每次交互只改动对应的查询字段，避免搜索、筛选和当前数据集互相覆盖。
150   const syncQuery = useCallback(
151     (next: { status?: QualityFilter; q?: string; datasetId?: number | null }) => {
152       setSearchParams(
153         (current) => {
154           const params = new URLSearchParams(current);
155
156           if (next.status !== undefined) {
157             if (next.status === 'all') params.delete('status');
158             else params.set('status', next.status);
```



```
front/src/workbench/data-quality/DataQualityPage.tsx
```

```
159         }
160         if (next.q !== undefined) {
161             if (next.q.trim()) params.set('q', next.q.trim());
162             else params.delete('q');
163         }
164         if (next.datasetId !== undefined) {
165             if (next.datasetId) params.set('datasetId', String(next.datasetId));
166             else params.delete('datasetId');
167         }
168         return params;
169     },
170     { replace: true },
171 );
172 },
173 [setSearchParams],
174 );
175
176 const loadDatasets = useCallback(async () => {
177     setLoading(true);
178     setError('');
179     try {
180         const page = await fetchDatasetCatalog();
181         setDatasets(page.items);
182         setSelectedId((current) => {
183             if (current && page.items.some((item) => item.id === current)) return current;
184             return pickDefaultDataset(page.items)?.id ?? null;
185         });
186     } catch (err) {
187         setError(err instanceof Error ? err.message : '加载数据质量信息失败');
188     } finally {
189         setLoading(false);
190     }
191 }, []);
192
193 useEffect(() => {
194     void loadDatasets();
195 }, [loadDatasets]);
196
197 const counts = useMemo(
198     () => ({
199         all: datasets.length,
200         error: datasets.filter((item) => item.qualityStatus === 'error').length,
201         warning: datasets.filter((item) => item.qualityStatus === 'warning').length,
202         ready: datasets.filter((item) => item.qualityStatus === 'ready').length,
203     }),
204     [datasets],
205 );
206
207 const filteredDatasets = useMemo(() => {
208     const normalizedKeyword = keyword.trim().toLowerCase();
```

```
front/src/workbench/data-quality/DataQualityPage.tsx
```

```
209     return datasets.filter((dataset) => {
210       if (filter !== 'all' && dataset.qualityStatus !== filter) return false;
211       if (!normalizedKeyword) return true;
212       return dataset.name.toLowerCase().includes(normalizedKeyword);
213     });
214   }, [datasets, filter, keyword]);
215
216   useEffect(() => {
217     if (filteredDatasets.length === 0) return;
218     if (selectedId && filteredDatasets.some((item) => item.id === selectedId)) return;
219     const nextId = filteredDatasets[0].id;
220     setSelectedId(nextId);
221     syncQuery({ datasetId: nextId });
222   }, [filteredDatasets, selectedId, syncQuery]);
223
224   const selectedDataset = datasets.find((item) => item.id === selectedId) ?? null;
225   const rules = selectedDataset ? buildQualityRules(selectedDataset) : [];
226   const abnormalIds = datasets
227     .filter((item) => item.qualityStatus !== 'ready')
228     .map((item) => item.id);
229   const isRechecking = recheckingIds.length > 0;
230
231   async function handleRecheck(ids: number[]) {
232     if (ids.length === 0 || isRechecking) return;
233
234     setRecheckingIds(ids);
235     setError('');
236     setMessage('');
237
238     // 批量复检允许部分成功，成功结果立即回写列表，失败项统一给出数量提示。
239     const results = await Promise.allSettled(ids.map((id) => recheckDatasetQuality(id)));
240     const updated = results.flatMap((result) => (result.status === 'fulfilled' ? [result.value] : []));
241     const failureCount = results.length - updated.length;
242
243     if (updated.length > 0) {
244       const byId = new Map(updated.map((item) => [item.id, item]));
245       setDatasets((current) => current.map((item) => byId.get(item.id) ?? item));
246     }
247
248     if (failureCount > 0) {
249       setError(`完成 ${updated.length} 项复检，${failureCount} 项失败，请稍后重试。`);
250     } else {
251       setMessage(`已完成 ${updated.length} 个数据集的质量复检。`);
252     }
253
254     setRecheckingIds([]);
255   }
256
257   function handleFilterChange(nextFilter: QualityFilter) {
258     setFilter(nextFilter);
```

```
front/src/workbench/data-quality/DataQualityPage.tsx
```

```
259     syncQuery({ status: nextFilter });
260   }
261
262   function handleSelect(datasetId: number) {
263     setSelectedId(datasetId);
264     syncQuery({ datasetId });
265   }
266
267   return (
268     <section className="quality-workspace" aria-label="数据质量检查">
269       <header className="quality-page-header">
270         <div>
271           <span className="quality-page-icon" aria-hidden="true">
272             <ShieldCheck size={20} />
273           </span>
274           <div>
275             <h1>数据质量检查</h1>
276             <p>优先处理基础数据结构问题，再复核标签相关提醒。</p>
277           </div>
278         </div>
279         <button
280           type="button"
281           className="btn btn-secondary"
282           disabled={abnormalIds.length === 0 || isRechecking}
283           onClick={() => void handleRecheck(abnormalIds)}
284         >
285           <RefreshCw size={15} aria-hidden="true" />
286           {isRechecking ? '正在复检' : `复检异常项 (${abnormalIds.length})`}
287         </button>
288       </header>
289
290       {error ? <div className="quality-feedback error" role="alert">{error}</div> : null}
291       {message ? <div className="quality-feedback success" role="status">{message}</div> : null}
292
293       <div className="quality-layout">
294         <aside className="quality-queue panel" aria-label="数据集质量队列">
295           <div className="quality-queue-header">
296             <div>
297               <strong>检查队列</strong>
298               <span>{datasets.length} 个数据集</span>
299             </div>
300             <div className="quality-search">
301               <Search size={15} aria-hidden="true" />
302               <input
303                 type="search"
304                 value={keyword}
305                 placeholder="搜索数据集"
306                 aria-label="搜索质量检查数据集"
307                 onChange={(event) => {
308                   setKeyword(event.target.value);
```

```
front/src/workbench/data-quality/DataQualityPage.tsx
```

```
309         syncQuery({ q: event.target.value });
310     }}
311     />
312 </div>
313 </div>
314
315 <div className="quality-filters" aria-label="质量状态筛选">
316     {FILTERS.map((item) => (
317         <button
318             type="button"
319             key={item.key}
320             className={filter === item.key ? 'active' : ''}
321             aria-pressed={filter === item.key}
322             onClick={() => handleFilterChange(item.key)}
323         >
324             <span>{item.label}</span>
325             <strong>{counts[item.key]}</strong>
326         </button>
327     ))}
328 </div>
329
330 <div className="quality-queue-list">
331     {loading ? (
332         <div className="quality-loading" aria-label="正在加载数据质量信息">
333             <span />
334             <span />
335             <span />
336         </div>
337     ) : filteredDatasets.length === 0 ? (
338         <div className="quality-empty">
339             <Database size={22} aria-hidden="true" />
340             <strong>没有匹配的数据集</strong>
341             <span>调整状态筛选或搜索关键词.</span>
342         </div>
343     ) : (
344         filteredDatasets.map((dataset) => (
345             <button
346                 type="button"
347                 className={`quality-queue-item${dataset.id === selectedId ? ' active' : ''}`}
348                 key={dataset.id}
349                 onClick={() => handleSelect(dataset.id)}
350             >
351                 <div>
352                     <strong>{dataset.name}</strong>
353                     <span>
354                         n {dataset.sampleCount} • m {dataset.baseCount} • v {dataset.version}
355                     </span>
356                 </div>
357                 <div>
358                     <QualityStatus status={dataset.qualityStatus} />
```

```
front/src/workbench/data-quality/DataQualityPage.tsx
```

```
359         <small>{dataset.qualityIssues.length} 个问题</small>
360     </div>
361 </button>
362 ))
363 }}
364 </div>
365 </aside>
366
367 <section className="quality-inspector panel" aria-label="质量检查详情">
368     {selectedDataset ? (
369         <div className="quality-inspector-header">
370             <div>
371                 <span>当前数据集</span>
372                 <h2>{selectedDataset.name}</h2>
373                 <p>{STATUS_META[selectedDataset.qualityStatus].description}</p>
374             </div>
375             <div className="quality-inspector-actions">
376                 <QualityStatus status={selectedDataset.qualityStatus} />
377                 <button
378                     type="button"
379                     className="btn btn-secondary"
380                     disabled={isRechecking}
381                     onClick={() => void handleRecheck([selectedDataset.id])}
382                 >
383                     <RefreshCw size={15} aria-hidden="true" />
384                     复检当前
385                 </button>
386             </div>
387         </div>
388     )}
389
390     <dl className="quality-dataset-meta" aria-label="数据集质量元数据">
391         <div>
392             <dt>矩阵规模</dt>
393             <dd>{selectedDataset.matrixShape} || `${selectedDataset.sampleCount} x
394             ${selectedDataset.baseCount}`</dd>
395         </div>
396         <div>
397             <dt>真实类别</dt>
398             <dd>{selectedDataset.hasLabels ? `${selectedDataset.classCount} 类` : '未提供'}</dd>
399         </div>
400         <div>
401             <dt>当前版本</dt>
402             <dd>v{selectedDataset.version}</dd>
403         </div>
404         <div>
405             <dt>关联任务</dt>
406             <dd>{selectedDataset.taskCount} 个</dd>
407         </div>
408     </dl>
```

front/src/workbench/data-quality/DataQualityPage.tsx

```
408
409     <section className={`quality-issue-summary ${selectedDataset.qualityStatus}`}>
410       {selectedDataset.qualityStatus === 'ready' ? (
411         <CheckCircle2 size={19} aria-hidden="true" />
412       ) : (
413         <AlertTriangle size={19} aria-hidden="true" />
414       )}
415     <div>
416       <strong>
417         {selectedDataset.qualityIssues.length > 0
418           ? `发现 ${selectedDataset.qualityIssues.length} 个质量提醒`
419           : '未发现质量问题'}
420       </strong>
421       <span>
422         {selectedDataset.qualityIssues.length > 0
423           ? selectedDataset.qualityIssues.join('; ')
424           : '基础聚类矩阵和标签数据均通过当前质量规则。'}
425       </span>
426     </div>
427   </section>
428
429   <div className="quality-rule-heading">
430     <div>
431       <h3>检查结果</h3>
432       <span>依据后端实际质量规则与数据结构检查结果生成。</span>
433     </div>
434     <ShieldCheck size={18} aria-hidden="true" />
435   </div>
436
437   <div className="quality-rule-results">
438     {rules.map((rule) => (
439       <article className={`quality-rule-result ${rule.tone}`} key={rule.key}>
440         <span className="quality-rule-icon">
441           <RuleIcon tone={rule.tone} />
442         </span>
443         <div>
444           <strong>{rule.title}</strong>
445           <span>{rule.detail}</span>
446         </div>
447         <b>{rule.result}</b>
448       </article>
449     ))}
450   </div>
451
452   <footer className={`quality-readiness ${selectedDataset.qualityStatus}`}>
453     <div>
454       <strong>
455         {selectedDataset.qualityStatus === 'error'
456           ? '存在质量问题'
457           : selectedDataset.qualityStatus === 'warning'
```

```
front/src/workbench/data-quality/DataQualityPage.tsx

458             ? '存在待确认项'
459             : '质量检查通过'})
460         </strong>
461         <span>{STATUS_META[selectedDataset.qualityStatus].description}</span>
462     </div>
463 </footer>
464 </>
465 ) : (
466     <div className="quality-empty inspector">
467         <ShieldCheck size={26} aria-hidden="true" />
468         <strong>{loading ? '正在加载检查详情' : '暂无可检查数据集'}</strong>
469         <span>{loading ? '请稍候。' : '请先在数据管理中上传基础聚类结果。'}</span>
470     </div>
471   )}
472 </section>
473 </div>
474 </section>
475   );
476 }
477
478 export default DataQualityPage;
479
// ===== 文件: front/src/workbench/dataset-versions/DatasetVersionsPage.tsx =====
1 import { useCallback, useEffect, useMemo, useRef, useState } from 'react';
2 import {
3   ArrowRight,
4   Database,
5   FileClock,
6   FilePlus2,
7   GitCompareArrows,
8   History,
9   PencilLine,
10  RefreshCw,
11  Replace,
12  UploadCloud,
13 } from 'lucide-react';
14 import type { LucideIcon } from 'lucide-react';
15 import { useNavigate, useSearchParams } from 'react-router-dom';
16 import {
17   fetchDatasetCatalog,
18   fetchDatasetVersions,
19   type DatasetCatalogItem,
20   type DatasetQualityStatus,
21   type DatasetRevision,
22 } from '../api/datasets';
23 import { SelectField } from '../components/SelectField';
24 import { formatLocalDateTime } from '../utils/time';
25
26 type QueryPatch = {
```

```
front/src/workbench/dataset-versions/DatasetVersionsPage.tsx
```

```
27   datasetId?: number | null;
28   version?: number | null;
29   compare?: number | null;
30 };
31
32 type RevisionActionMeta = {
33   label: string;
34   description: string;
35   icon: LucideIcon;
36 };
37
38 type RevisionDiffField = {
39   key: string;
40   label: string;
41   baselineValue: string;
42   currentValue: string;
43   changed: boolean;
44 };
45
46 const QUALITY_LABELS: Record<DatasetQualityStatus, string> = {
47   ready: '检查通过',
48   warning: '待确认',
49   error: '需处理',
50 };
51
52 const ACTION_META: Record<string, RevisionActionMeta> = {
53   uploaded: {
54     label: '初始上传',
55     description: '创建数据集并保存首个可追踪版本。',
56     icon: UploadCloud,
57   },
58   replaced: {
59     label: '替换文件',
60     description: '更新源文件，并记录替换后的结构与质量状态。',
61     icon: Replace,
62   },
63   appended: {
64     label: '追加数据',
65     description: '向当前数据集追加样本，并记录合并后的结构与质量状态。',
66     icon: FilePlus2,
67   },
68   renamed: {
69     label: '重命名',
70     description: '调整数据集名称，文件内容保持不变。',
71     icon: PencilLine,
72   },
73 };
74
75 function parsePositiveNumber(value: string | null) {
76   const parsed = Number(value);
```



```
front/src/workbench/dataset-versions/DatasetVersionsPage.tsx
```

```
77   return Number.isInteger(parsed) && parsed > 0 ? parsed : null;
78 }
79
80 function getActionMeta(action: string): RevisionActionMeta {
81   return (
82     ACTION_META[action] ?? {
83       label: action || '数据变更',
84       description: '记录本次数据集元数据变更。',
85       icon: FileClock,
86     }
87   );
88 }
89
90 function getDefaultBaseline(revisions: DatasetRevision[], selectedVersion: number) {
91   const selectedIndex = revisions.findIndex((revision) => revision.version === selectedVersion);
92   return selectedIndex >= 0 ? revisions[selectedIndex + 1] ?? null : null;
93 }
94
95 function formatLabels(revision: DatasetRevision) {
96   return revision.hasLabels ? `已提供 (${revision.classCount} 类)` : '未提供';
97 }
98
99 function formatIssues(revision: DatasetRevision) {
100   return revision.qualityIssues.length > 0 ? revision.qualityIssues.join('; ') : '无质量问题';
101 }
102
103 function buildDiffFields(
104   current: DatasetRevision,
105   baseline: DatasetRevision | null,
106 ): RevisionDiffField[] {
107   // 版本接口保存的是可追踪元数据，差异表固定字段顺序，便于不同版本之间逐项核对。
108   const values = [
109     ['name', '数据集名称', current.name, baseline?.name ?? '首次记录'],
110     ['filename', '原始文件', current.originalFilename, baseline?.originalFilename ?? '首次记录'],
111     ['samples', '样本数 n', String(current.sampleCount), baseline ? String(baseline.sampleCount) : '首次记录'],
112     ['bases', '基础聚类数 m', String(current.baseCount), baseline ? String(baseline.baseCount) : '首次记录'],
113     ['classes', '类别数 c', current.hasLabels ? String(current.classCount) : '未提供', baseline ? (baseline.hasLabels ?
114       String(baseline.classCount) : '未提供') : '首次记录'],
115     ['labels', '真实标签', formatLabels(current), baseline ? formatLabels(baseline) : '首次记录'],
116     ['quality', '质量状态', QUALITY_LABELS[current.qualityStatus], baseline ? QUALITY_LABELS[baseline.qualityStatus] :
117       '首次记录'],
118     ['issues', '质量问题', formatIssues(current), baseline ? formatIssues(baseline) : '首次记录'],
119   ] as const;
120
121   return values.map(([key, label, currentValue, baselineValue]) => ({
122     key,
123     label,
124     currentValue,
125     baselineValue,
126     changed: Boolean(baseline && currentValue !== baselineValue),
```

front/src/workbench/dataset-versions/DatasetVersionsPage.tsx

```
125   }));
126 }
127
128 function RevisionQualityBadge({ status }: { status: DatasetQualityStatus }) {
129   return <span className={`versions-quality-badge ${status}`}>{QUALITY_LABELS[status]}</span>;
130 }
131
132 export function DatasetVersionsPage() {
133   const navigate = useNavigate();
134   const [searchParams, setSearchParams] = useSearchParams();
135   // 目录和版本记录分两次异步加载, ref 用于保留地址栏指定的恢复目标。
136   const requestedVersionRef = useRef(parsePositiveNumber(searchParams.get('version')));
137   const requestedCompareRef = useRef(parsePositiveNumber(searchParams.get('compare')));
138
139   const [datasets, setDatasets] = useState<DatasetCatalogItem[]>([]);
140   const [selectedDatasetId, setSelectedDatasetId] = useState<number | null>(() =>
141     parsePositiveNumber(searchParams.get('datasetId')),
142   );
143   const [revisions, setRevisions] = useState<DatasetRevision[]>([]);
144   const [selectedVersion, setSelectedVersion] = useState<number | null>(
145     requestedVersionRef.current,
146   );
147   const [compareVersion, setCompareVersion] = useState<number | null>(
148     requestedCompareRef.current,
149   );
150   const [catalogLoading, setCatalogLoading] = useState(true);
151   const [revisionsLoading, setRevisionsLoading] = useState(false);
152   const [revisionReloadKey, setRevisionReloadKey] = useState(0);
153   const [error, setError] = useState('');
154
155   const updateQuery = useCallback(
156     (patch: QueryPatch) => {
157       setSearchParams(
158         (current) => {
159           const params = new URLSearchParams(current);
160           const assign = (key: keyof QueryPatch, value: number | null | undefined) => {
161             if (value) params.set(key, String(value));
162             else params.delete(key);
163           };
164
165           if (Object.prototype.hasOwnProperty.call(patch, 'datasetId')) {
166             assign('datasetId', patch.datasetId);
167           }
168           if (Object.prototype.hasOwnProperty.call(patch, 'version')) {
169             assign('version', patch.version);
170           }
171           if (Object.prototype.hasOwnProperty.call(patch, 'compare')) {
172             assign('compare', patch.compare);
173           }
174           return params;
175         }
176       );
177     },
178   );
```

front/src/workbench/dataset-versions/DatasetVersionsPage.tsx

```
175         },
176         { replace: true },
177       );
178     },
179     [setSearchParams],
180   );
181
182   const loadCatalog = useCallback(async () => {
183     setCatalogLoading(true);
184     setError('');
185     try {
186       const page = await fetchDatasetCatalog();
187       setDatasets(page.items);
188       setSelectedDatasetId((current) => {
189         if (current && page.items.some((dataset) => dataset.id === current)) return current;
190         return page.items[0]?.id ?? null;
191       });
192     } catch (err) {
193       setError(err instanceof Error ? err.message : '加载数据集目录失败');
194     } finally {
195       setCatalogLoading(false);
196     }
197   }, []);
198
199   useEffect(() => {
200     void loadCatalog();
201   }, [loadCatalog]);
202
203   // 仅在数据集切换或主动重试时请求版本，比较基准变化不会重复访问后端。
204   useEffect(() => {
205     if (!selectedDatasetId) {
206       setRevisions([]);
207       setSelectedVersion(null);
208       setCompareVersion(null);
209       return;
210     }
211
212     let cancelled = false;
213     setRevisionsLoading(true);
214     setError('');
215
216     void fetchDatasetVersions(selectedDatasetId)
217       .then((items) => {
218         if (cancelled) return;
219
220         setRevisions(items);
221         const selected =
222           items.find((revision) => revision.version === requestedVersionRef.current) ??
223           items[0] ??
224           null;
```

front/src/workbench/dataset-versions/DatasetVersionsPage.tsx

```
225     const requestedCompare = items.find(  
226       (revision) =>  
227         revision.version === requestedCompareRef.current &&  
228         revision.version !== selected?.version,  
229     );  
230     const baseline = selected  
231       ? requestedCompare ?? getDefaultBaseline(items, selected.version)  
232       : null;  
233  
234     setSelectedVersion(selected?.version ?? null);  
235     setCompareVersion(baseline?.version ?? null);  
236     requestedVersionRef.current = selected?.version ?? null;  
237     requestedCompareRef.current = baseline?.version ?? null;  
238   })  
239   .catch((err) => {  
240     if (!cancelled) {  
241       setError(err instanceof Error ? err.message : '加载版本记录失败');  
242       setRevisions([]);  
243     }  
244   })  
245   .finally(() => {  
246     if (!cancelled) setRevisionsLoading(false);  
247   });  
248  
249   return () => {  
250     cancelled = true;  
251   };  
252 }, [revisionReloadKey, selectedDatasetId]);  
253  
254 // 地址栏同步与数据请求分离, 防止查询参数更新覆盖用户刚选择的版本。  
255 useEffect(() => {  
256   if (!selectedDatasetId) return;  
257   updateQuery({  
258     datasetId: selectedDatasetId,  
259     version: selectedVersion,  
260     compare: compareVersion,  
261   });  
262 }, [compareVersion, selectedDatasetId, selectedVersion, updateQuery]);  
263  
264 const selectedDataset =  
265   datasets.find((dataset) => dataset.id === selectedDatasetId) ?? null;  
266 const selectedRevision =  
267   revisions.find((revision) => revision.version === selectedVersion) ?? null;  
268 const compareRevision =  
269   revisions.find((revision) => revision.version === compareVersion) ?? null;  
270 const diffFields = useMemo(  
271   () => (selectedRevision ? buildDiffFields(selectedRevision, compareRevision) : []),  
272   [compareRevision, selectedRevision],  
273 );  
274 const changedCount = diffFields.filter((field) => field.changed).length;
```

front/src/workbench/dataset-versions/DatasetVersionsPage.tsx

```
275
276 function handleDatasetChange(datasetId: number) {
277     setSelectedDatasetId(datasetId);
278     setRevisions([]);
279     setSelectedVersion(null);
280     setCompareVersion(null);
281     requestedVersionRef.current = null;
282     requestedCompareRef.current = null;
283 }
284
285 function handleVersionChange(version: number) {
286     const baseline = getDefaultBaseline(revisions, version);
287     setSelectedVersion(version);
288     setCompareVersion(baseline?.version ?? null);
289     requestedVersionRef.current = version;
290     requestedCompareRef.current = baseline?.version ?? null;
291 }
292
293 function handleCompareChange(version: number | null) {
294     setCompareVersion(version);
295     requestedCompareRef.current = version;
296 }
297
298 const selectedAction = selectedRevision ? getActionMeta(selectedRevision.action) : null;
299 const datasetSelectOptions = useMemo(
300     () => (datasets.length === 0
301         ? [{ value: '', label: '暂无数据集' }]
302         : datasets.map((dataset) => ({
303             value: String(dataset.id),
304             label: dataset.name,
305         }))),
306     [datasets],
307 );
308 const compareSelectOptions = useMemo(
309     () => [
310         { value: '', label: '不比较 (查看快照)' },
311         ...revisions
312             .filter((revision) => revision.version !== selectedRevision?.version)
313             .map((revision) => ({
314                 value: String(revision.version),
315                 label: `v${revision.version} • ${getActionMeta(revision.action).label}`,
316             })),
317     ],
318     [revisions, selectedRevision?.version],
319 );
320
321 return (
322     <section className="versions-workspace" aria-label="数据版本记录">
323         <header className="versions-page-header">
324             <div>
```

```
front/src/workbench/dataset-versions/DatasetVersionsPage.tsx
```

```
325         <span className="versions-page-icon" aria-hidden="true">
326             <History size={20} />
327         </span>
328         <div>
329             <h1>数据版本记录</h1>
330             <p>按数据集查看上传、追加、替换和重命名历史，并核对每次变更影响的元数据。</p>
331         </div>
332     </div>
333     <button
334         type="button"
335         className="btn btn-secondary"
336         onClick={() => navigate('/workbench/datasets')}
337     >
338         <Database size={15} aria-hidden="true" />
339         进入数据管理
340     </button>
341 </header>
342
343 {error ? (
344     <div className="versions-feedback error" role="alert">
345         <span>{error}</span>
346         <button
347             type="button"
348             onClick={() =>
349                 selectedDatasetId
350                 ? setRevisionReloadKey((current) => current + 1)
351                 : void loadCatalog()
352             }
353         >
354             <RefreshCw size={14} aria-hidden="true" />
355             重试
356         </button>
357     </div>
358 ) : null}
359
360 <section className="versions-dataset-bar panel" aria-label="版本数据集选择">
361     <SelectField
362         className="versions-dataset-picker"
363         label="选择数据集"
364         ariaLabel="选择版本数据集"
365         disabled={catalogLoading || datasets.length === 0}
366         value={selectedDatasetId ? String(selectedDatasetId) : ''}
367         options={datasetSelectOptions}
368         onChange={(value) => handleDatasetChange(Number(value))}
369     />
370
371     <div className="versions-dataset-context" aria-label="所选数据集版本摘要">
372         <div>
373             <span>当前版本</span>
374             <strong>{selectedDataset ? `v${selectedDataset.version}` : '---'}</strong>
```

```
front/src/workbench/dataset-versions/DatasetVersionsPage.tsx
```

```
375     </div>
376   <div>
377     <span>历史记录</span>
378     <strong>{revisionsLoading ? '--' : `${revisions.length} 条`}</strong>
379   </div>
380 <div>
381   <span>数据规模</span>
382   <strong>
383     {selectedDataset
384       ? `${selectedDataset.sampleCount} x ${selectedDataset.baseCount}`
385       : '--'}
386   </strong>
387 </div>
388 <div>
389   <span>当前质量</span>
390   {selectedDataset ? (
391     <RevisionQualityBadge status={selectedDataset.qualityStatus} />
392   ) : (
393     <strong>--</strong>
394   )}
395 </div>
396 </div>
397 </section>
398
399 {catalogLoading ? (
400   <div className="versions-loading panel" aria-label="正在加载版本数据">
401     <span />
402     <span />
403     <span />
404   </div>
405 ) : datasets.length === 0 ? (
406   <section className="versions-empty panel">
407     <Database size={28} aria-hidden="true" />
408     <h2>暂无可追踪的数据集</h2>
409     <p>上传数据集后，系统会自动保存初始版本，并持续记录追加、替换和重命名操作。</p>
410     <button
411       type="button"
412       className="btn btn-primary"
413       onClick={() => navigate('/workbench/datasets')}
414     >
415       <Database size={15} aria-hidden="true" />
416       前往数据管理
417     </button>
418   </section>
419 ) : (
420   <section className="versions-explorer panel" aria-label="数据版本浏览器">
421     <aside className="versions-rail" aria-label="版本轨道">
422       <div className="versions-rail-header">
423         <div>
424           <strong>版本轨道</strong>
```

```
front/src/workbench/dataset-versions/DatasetVersionsPage.tsx
```

```
425         <span>{selectedDataset?. name}</span>
426     </div>
427     <b>{revisions.length}</b>
428 </div>
429
430 <div className="versions-rail-list">
431     {revisionsLoading ? (
432         <div className="versions-rail-loading" aria-label="正在加载版本轨道">
433             <span />
434             <span />
435             <span />
436         </div>
437     ) : revisions.length === 0 ? (
438         <div className="versions-rail-empty">
439             <FileClock size={21} aria-hidden="true" />
440             <strong>暂无版本记录</strong>
441             <span>该数据集尚未生成可追踪版本。</span>
442         </div>
443     ) : (
444         revisions.map((revision, index) => {
445             const action = getActionMeta(revision.action);
446             const ActionIcon = action.icon;
447             const active = revision.version === selectedVersion;
448
449             return (
450                 <button
451                     type="button"
452                     className={`versions-rail-item${active ? ' active' : ''}`}
453                     aria-current={active ? 'true' : undefined}
454                     key={revision.id}
455                     onClick={() => handleVersionChange(revision.version)}
456                 >
457                     <span className="versions-rail-node" aria-hidden="true">
458                         <ActionIcon size={14} />
459                     </span>
460                     <span className="versions-rail-copy">
461                         <strong>
462                             v {revision.version} • {action.label}
463                         </strong>
464                         <time>{formatLocalDateTime(revision.createdAt)}</time>
465                     </span>
466                     <span className={`versions-rail-state ${revision.qualityStatus}`}>
467                         {index === 0 ? '当前' : QUALITY_LABELS[revision.qualityStatus]}
468                     </span>
469                 </button>
470             );
471         })
472     )}
473 </div>
474 </aside>
```


front/src/workbench/dataset-versions/DatasetVersionsPage.tsx

```
475
476     <section className="versions-inspector" aria-label="版本差异详情">
477       {selectedRevision && selectedAction ? (
478         <>
479           <header className="versions-inspector-header">
480             <div>
481               <span>查看版本</span>
482               <h2>
483                 v{selectedRevision.version} • {selectedAction.label}
484               </h2>
485               <p>{selectedAction.description}</p>
486             </div>
487             <SelectField
488               className="versions-compare-picker"
489               label="比较基准"
490               ariaLabel="选择比较基准"
491               value={compareVersion ? String(compareVersion) : ''}
492               options={compareSelectOptions}
493               onChange={(value) => handleCompareChange(value ? Number(value) : null)}
494             />
495           </header>
496
497           <div className="versions-revision-meta">
498             <span>
499               <FileClock size={14} aria-hidden="true" />
500               {formatLocalDateTime(selectedRevision.createdAt)}
501             </span>
502             <RevisionQualityBadge status={selectedRevision.qualityStatus} />
503             <span>{selectedRevision.originalFilename}</span>
504           </div>
505
506           <section className={`versions-change-summary${compareRevision ? '' : ' snapshot'}`}>
507             {compareRevision ? (
508               <GitCompareArrows size={19} aria-hidden="true" />
509             ) : (
510               <FileClock size={19} aria-hidden="true" />
511             )}
512             <div>
513               <strong>
514                 {compareRevision
515                   ? `与 v${compareRevision.version} 相比, ${changedCount} 项元数据发生变化`
516                   : '当前以版本快照方式查看'}
517               </strong>
518               <span>
519                 {compareRevision
520                   ? changedCount > 0
521                     ? '变更字段已在下方突出显示, 未变化字段保持普通状态。'
522                     : '两个版本的可追踪元数据一致。'
523                   : '选择其他版本作为比较基准, 可查看字段级差异。'}
524               </span>
```

```
front/src/workbench/dataset-versions/DatasetVersionsPage.tsx
```

```
525         </div>
526         <b>{compareRevision ? `${changedCount} / ${diffFields.length}` : '快照'}</b>
527     </section>
528
529     <div className="versions-diff-table" role="table" aria-label="版本元数据差异">
530         <div className="versions-diff-row header" role="row">
531             <span role="columnheader">字段</span>
532             <span role="columnheader">
533                 {compareRevision ? `基准 v${compareRevision.version}` : '基准'}
534             </span>
535             <span aria-hidden="true" />
536             <span role="columnheader">查看 v {selectedRevision.version}</span>
537         </div>
538         {diffFields.map((field) => (
539             <div
540                 className={`versions-diff-row${field.changed ? ' changed' : ''}`}
541                 role="row"
542                 key={field.key}
543             >
544                 <strong role="rowheader">{field.label}</strong>
545                 <span role="cell">{field.baselineValue}</span>
546                 <ArrowRight size={15} aria-hidden="true" />
547                 <span role="cell">
548                     {field.currentValue}
549                     {field.changed ? <b>已变更</b> : null}
550                 </span>
551             </div>
552         )))
553     </div>
554 </>
555 ) : (
556     <div className="versions-inspector-empty">
557         <FileClock size={27} aria-hidden="true" />
558         <strong>{revisionsLoading ? '正在加载版本详情' : '暂无版本详情'}</strong>
559         <span>
560             {revisionsLoading
561                 ? '正在读取所选数据集的历史记录。'
562                 : '该数据集还没有可比较的版本记录。'}
563         </span>
564     </div>
565 )}
566 </section>
567 </section>
568 )}
569 </section>
570 );
571 }
572
573 export default DatasetVersionsPage;
574
```

front/src/workbench/dataset-versions/DatasetVersionsPage.tsx

```
// ===== 文件: front/src/workbench/results/AnalysisOverviewPage.tsx =====
1 import { useMemo } from 'react';
2 import { useNavigate } from 'react-router-dom';
3 import {
4   Activity,
5   ArrowRight,
6   BarChart3,
7   Boxes,
8   CheckCircle2,
9   ClipboardCheck,
10  Download,
11  FileText,
12  Images,
13  LayoutDashboard,
14  LayoutGrid,
15  Network,
16  RefreshCw,
17  SlidersHorizontal,
18  Target,
19  type LucideIcon,
20 } from 'lucide-react';
21 import {
22   WorkbenchMetricStrip,
23   WorkbenchNotice,
24   WorkbenchPageHeader,
25   WorkbenchSectionHeader,
26 } from '../WorkbenchUi';
27 import {
28   ConvergencePanel,
29   MatrixPanel,
30   ScatterPanel,
31   WeightPanel,
32 } from './ResultChartPanels';
33 import {
34   formatNumber,
35   formatPercent,
36   metricLabels,
37   navigateToTaskResult,
38   type MetricKey,
39 } from './resultPresentation';
40 import type { TaskResultResource } from './types';
41
42 type OverviewStage = {
43   label: string;
44   detail: string;
45   status: string;
46   complete: boolean;
47   section: string;
48   icon: LucideIcon;
```

front/src/workbench/results/AnalysisOverviewPage.tsx

```
49  };
50
51  export function AnalysisOverviewPage({ resource }: { resource: TaskResultResource }) {
52    const navigate = useNavigate();
53    const task = resource.envelope!.task!;
54    const result = resource.envelope!.result!;
55    const summary = result.preview.summary;
56    const representativeRun = result.convergence.runs.find(
57      (item) => item.run === result.convergence.representativeRun,
58    );
59
60    // 总览只组合后端已经持久化的结果字段，不在浏览器端推断算法质量结论。
61    const stages = useMemo<OverviewStage[]>(() => [
62      {
63        label: 'CA 矩阵',
64        detail: `${result.preview.matrices.ca.shape[0]} × ${result.preview.matrices.ca.shape[1]}`,
65        status: '已构建',
66        complete: result.preview.matrices.ca.values.length > 0,
67        section: 'ca-matrix',
68        icon: LayoutGrid,
69      },
70      {
71        label: '核函数',
72        detail: `${result.kernelWeights.items.length} 个核函数`,
73        status: result.kernelWeights.items.length ? '已计算' : '无权重',
74        complete: result.kernelWeights.items.length > 0,
75        section: 'kernel-config',
76        icon: SlidersHorizontal,
77      },
78      {
79        label: '多核学习',
80        detail: representativeRun
81          ? `代表轮次 ${representativeRun.points.length} 次迭代`
82          : '暂无迭代轨迹',
83        status: !representativeRun
84          ? '无轨迹'
85          : representativeRun.converged
86            ? '已收敛'
87            : '迭代结束',
88        complete: Boolean(representativeRun),
89        section: 'mkl',
90        icon: Network,
91      },
92      {
93        label: '性能评估',
94        detail: `${result.metrics.runs.length} 轮重复实验`,
95        status: result.metrics.runs.length ? '已汇总' : '无明细',
96        complete: result.metrics.runs.length > 0,
97        section: 'evaluation',
98        icon: ClipboardCheck,
```

front/src/workbench/results/AnalysisOverviewPage.tsx

```
99     },
100     {
101       label: '结果持久化',
102       detail: `${result.artifacts.length} 项完整产物`,
103       status: result.artifacts.length ? '可交付' : '无产物',
104       complete: result.artifacts.length > 0,
105       section: 'export',
106       icon: CheckCircle2,
107     },
108   ], [representativeRun, result]);
109
110   const recentRuns = useMemo(
111     () => [...result.metrics.runs].sort((left, right) => right.run - left.run).slice(0, 5),
112     [result.metrics.runs],
113   );
114
115   const metricIcons = [Target, Network, Boxes, ClipboardCheck];
116   const metricTones = ['green', 'blue', 'teal', 'neutral'] as const;
117   const followUpActions = [
118     {
119       label: '完整可视化',
120       detail: `${result.preview.scatter.points.length} 个预览点`,
121       section: 'visualization',
122       icon: Images,
123     },
124     {
125       label: '分析报告',
126       detail: `任务 #${task.id}`,
127       section: 'reports',
128       icon: FileText,
129     },
130     {
131       label: '结果导出',
132       detail: `${result.artifacts.length} 项产物`,
133       section: 'export',
134       icon: Download,
135     },
136     {
137       label: '运行日志',
138       detail: `${resource.logs.length} 条记录`,
139       section: 'logs',
140       icon: Activity,
141     },
142   ];
143
144   const goToResult = (section: string) => navigateToTaskResult(navigate, section, task.id);
145
146   return (
147     <section className="soft-page result-overview-page analysis-overview-page">
148       <WorkbenchPageHeader
```

```
front/src/workbench/results/AnalysisOverviewPage.tsx
```

```
149     icon={LayoutDashboard}
150     title="分析工作台"
151     context={`任务 #${task.id} • ${task.datasetName} • ${task.mode} • 第 ${summary.representativeRun} 轮代表结果`}
152     actions={
153       <button type="button" className="btn btn-secondary" onClick={resource.refresh}>
154         <RefreshCw size={15} aria-hidden="true" />
155         刷新
156       </button>
157     }
158   />
159
160   <WorkbenchMetricStrip
161     label="重复实验指标"
162     metrics={(Object.keys(metricLabels) as MetricKey[]).map((key, index) => ({
163       label: metricLabels[key],
164       value: formatPercent(result.metrics.aggregate[key].mean),
165       note: `标准差 ${formatPercent(result.metrics.aggregate[key].std)}`,
166       icon: metricIcons[index],
167       tone: metricTones[index],
168     })))}
169   />
170
171   <div className="analysis-overview-featured">
172     <MatrixPanel
173       matrix={result.preview.matrices.ca}
174       title="CA 协关联矩阵"
175       description="由本次代表运行的基础聚类结果计算"
176       actions={
177         <button
178           type="button"
179           className="text-action"
180           onClick={() => goToResult('ca-matrix')}
181         >
182           完整矩阵
183         <ArrowRight size={14} aria-hidden="true" />
184       </button>
185     }
186   />
187
188   <section className="panel analysis-overview-summary" aria-label="任务摘要">
189     <WorkbenchSectionHeader
190       title="任务摘要"
191       meta="本次持久化结果的运行上下文"
192       actions={
193         <button
194           type="button"
195           className="text-action"
196           onClick={() => navigate('/workbench/tasks')}
197         >
198           任务中心
```

```
front/src/workbench/results/AnalysisOverviewPage.tsx
```

```
199         <ArrowRight size={14} aria-hidden="true" />
200       </button>
201     }
202   />
203   <dl className="analysis-overview-summary-list">
204     <div className="is-wide"><dt>任务名称</dt><dd>{task.name}</dd></div>
205     <div><dt>数据集</dt><dd>{task.datasetName}</dd></div>
206     <div><dt>算法模式</dt><dd>{task.mode}</dd></div>
207     <div><dt>样本数量</dt><dd>{summary.sampleCount}</dd></div>
208     <div><dt>真实类别</dt><dd>{summary.classCount}</dd></div>
209     <div><dt>基础聚类</dt><dd>{summary.baseClusterCount}</dd></div>
210     <div><dt>运行耗时</dt><dd>{formatNumber(result.runtimeSeconds, 2)} s</dd></div>
211     <div><dt>随机种子</dt><dd>{summary.randomSeed}</dd></div>
212     <div><dt>代表轮次</dt><dd>第 {summary.representativeRun} 轮</dd></div>
213   </dl>
214   <WorkbenchNotice
215     tone="success"
216     icon={CheckCircle2}
217     title="结果已持久化"
218     detail={
219       result.artifacts.length
220       ? `已有 ${result.artifacts.length} 项完整产物，可在结果导出页面获取。`
221       : '任务结果已经保存，当前没有可下载完整产物。'
222     }
223   />
224   </section>
225 </div>
226
227 <section className="panel analysis-overview-flow" aria-label="分析链路">
228   <WorkbenchSectionHeader
229     title="分析链路"
230     meta="从协关联构建到结果交付，点击阶段查看完整细节"
231   />
232   <div className="analysis-overview-stage-list">
233     {stages.map((stage) => {
234       const Icon = stage.icon;
235       return (
236         <button
237           type="button"
238           className={`analysis-overview-stage${stage.complete ? '' : ' is-incomplete'}`}
239           key={stage.section}
240           onClick={() => goToResult(stage.section)}
241         >
242           <span className="analysis-overview-stage-icon" aria-hidden="true"><Icon size={17} /></span>
243           <span className="analysis-overview-stage-copy">
244             <strong>{stage.label}</strong>
245             <small>{stage.detail}</small>
246           </span>
247           <span className="analysis-overview-stage-status">
248             {stage.complete
```

```
front/src/workbench/results/AnalysisOverviewPage.tsx
```

```
249         ? <CheckCircle2 size={13} aria-hidden="true" />
250         : <Activity size={13} aria-hidden="true" />
251         {stage.status}
252       </span>
253       <ArrowRight className="analysis-overview-stage-arrow" size={14} aria-hidden="true" />
254     </button>
255   );
256   }}}
257 </div>
258 </section>
259
260 <div className="analysis-overview-behavior">
261   <ScatterPanel
262     result={result}
263     compact
264     actions={
265       <button
266         type="button"
267         className="text-action"
268         onClick={() => goToResult('visualization')}
269       >
270         完整视图
271         <ArrowRight size={14} aria-hidden="true" />
272       </button>
273     }
274   />
275   <ConvergencePanel result={result} compact />
276 </div>
277
278 <div className="analysis-overview-details">
279   <WeightPanel
280     result={result}
281     compact
282     actions={
283       <button type="button" className="text-action" onClick={() => goToResult('mk1')}>
284         学习详情
285         <ArrowRight size={14} aria-hidden="true" />
286       </button>
287     }
288   />
289
290   <section className="panel analysis-overview-runs" aria-label="重复实验明细">
291     <WorkbenchSectionHeader
292       title="最近五轮实验"
293       meta={`共 ${result.metrics.runs.length} 轮运行，按轮次倒序展示`}
294       actions={
295         <button
296           type="button"
297           className="text-action"
298           onClick={() => goToResult('evaluation')}

```



```
front/src/workbench/results/AnalysisOverviewPage.tsx
```

```
299         >
300         查看全部
301         <ArrowRight size={14} aria-hidden="true" />
302     </button>
303     }
304 />
305 {recentRuns.length ? (
306     <div className="analysis-overview-run-table" role="table" aria-label="最近五轮实验指标">
307         <div className="analysis-overview-run-row header" role="row">
308             <span role="columnheader">轮次</span>
309             <span role="columnheader">ACC</span>
310             <span role="columnheader">NMI</span>
311             <span role="columnheader">ARI</span>
312             <span role="columnheader">F1-score</span>
313             <span role="columnheader">耗时</span>
314         </div>
315         {recentRuns.map((run) => (
316             <div className="analysis-overview-run-row" role="row" key={run.run}>
317                 <strong role="cell">第 {run.run} 轮</strong>
318                 <span role="cell">{formatPercent(run.acc)}</span>
319                 <span role="cell">{formatPercent(run.nmi)}</span>
320                 <span role="cell">{formatPercent(run.ari)}</span>
321                 <span role="cell">{formatPercent(run.f1)}</span>
322                 <span role="cell">{formatNumber(run.runtimeSeconds, 2)} s</span>
323             </div>
324         )))
325     </div>
326 ) : (
327     <div className="analysis-overview-inline-empty" role="status">
328         当前结果没有重复实验明细。
329     </div>
330 )}
331 </section>
332 </div>
333
334 <section className="panel analysis-overview-follow-up" aria-label="后续处理">
335     <div className="analysis-overview-follow-up-heading">
336         <BarChart3 size={18} aria-hidden="true" />
337     <div>
338         <strong>继续处理当前结果</strong>
339         <span>{result.artifacts.length} 项产物 • {resource.logs.length} 条运行记录</span>
340     </div>
341 </div>
342 <div className="analysis-overview-follow-up-actions">
343     {followUpActions.map((action) => {
344         const Icon = action.icon;
345         return (
346             <button type="button" key={action.section} onClick={() => goToResult(action.section)}>
347                 <Icon size={17} aria-hidden="true" />
348                 <span><strong>{action.label}</strong><small>{action.detail}</small></span>
```

```
front/src/workbench/results/AnalysisOverviewPage.tsx
```

```
349         <ArrowRight size={14} aria-hidden="true" />
350       </button>
351     );
352   }}}
353 </div>
354 </section>
355 </section>
356 );
357 }
358
// ===== 文件: front/src/workbench/results/ResultChartPanels.tsx =====
1 import { useEffect, useMemo, useRef, useState, type ReactNode } from 'react';
2 import * as echarts from 'echarts';
3 import type { EChartsOption } from 'echarts';
4 import { Activity } from 'lucide-react';
5 import { SelectField } from '../../components/SelectField';
6 import { WorkbenchSectionHeader, WorkbenchStatus } from '../../WorkbenchUi';
7 import type { AnalysisResult, MatrixPreview } from './types';
8 import { formatNumber } from './resultPresentation';
9
10 type PanelProps = {
11   compact?: boolean;
12   actions?: ReactNode;
13 };
14
15 function ResultPanelEmpty({ title, detail }: { title: string; detail: string }) {
16   return (
17     <div className="result-panel-empty" role="status">
18       <Activity size={19} aria-hidden="true" />
19       <strong>{title}</strong>
20       <span>{detail}</span>
21     </div>
22   );
23 }
24
25 type HeatmapValueChangeHandler = (value: number | null) => void;
26
27 export function getMatrixScalePosition(value: number, minimum: number, maximum: number) {
28   if (![value, minimum, maximum].every(Number.isFinite)) return 0;
29   if (maximum <= minimum) return 50;
30   return Math.min(100, Math.max(0, ((value - minimum) / (maximum - minimum)) * 100));
31 }
32
33 export function getHeatmapValue(params: unknown) {
34   if (!params || typeof params !== 'object') return null;
35   const event = params as { seriesType?: string; value?: unknown };
36   if (event.seriesType !== 'heatmap' || !Array.isArray(event.value)) return null;
37   const value = Number(event.value[2]);
38   return Number.isFinite(value) ? value : null;
```

```
front/src/workbench/results/ResultChartPanels.tsx
```

```
39 }
40
41 // 所有结果图表统一使用 SVG 渲染。组件卸载时同步销毁实例和交互监听器，避免切换结果页面后残留状态。
42 export function useResultChart(
43   option: EChartsOption,
44   onHeatmapValueChange?: HeatmapValueChangeHandler,
45 ) {
46   const ref = useRef<HTMLDivElement | null>(null);
47   const heatmapValueHandlerRef = useRef(onHeatmapValueChange);
48
49   useEffect(() => {
50     heatmapValueHandlerRef.current = onHeatmapValueChange;
51   }, [onHeatmapValueChange]);
52
53   useEffect(() => {
54     if (!ref.current || (typeof navigator !== 'undefined' && navigator.userAgent.includes('jsdom'))) {
55       return undefined;
56     }
57
58     const chart = echarts.init(ref.current, undefined, { renderer: 'svg' });
59     chart.setOption(option, { notMerge: true });
60     const resize = () => chart.resize();
61     const handleMouseOver = (params: unknown) => {
62       const value = getHeatmapValue(params);
63       if (value !== null) heatmapValueHandlerRef.current?.(value);
64     };
65     const handleMouseOut = (params: unknown) => {
66       if (getHeatmapValue(params) !== null) heatmapValueHandlerRef.current?.(null);
67     };
68     const handleGlobalOut = () => heatmapValueHandlerRef.current?.(null);
69
70     heatmapValueHandlerRef.current?.(null);
71     chart.on('mouseover', handleMouseOver);
72     chart.on('mouseout', handleMouseOut);
73     chart.getZr().on('globalout', handleGlobalOut);
74     window.addEventListener('resize', resize);
75
76     return () => {
77       window.removeEventListener('resize', resize);
78       chart.off('mouseover', handleMouseOver);
79       chart.off('mouseout', handleMouseOut);
80       chart.getZr().off('globalout', handleGlobalOut);
81       chart.dispose();
82     };
83   }, [option]);
84
85   return ref;
86 }
87
88 export function buildMatrixOption(matrix: MatrixPreview, title: string): EChartsOption {
```

```
front/src/workbench/results/ResultChartPanels.tsx
```

```
89   const data = matrix.values.flatMap((row, rowIndex) =>
90     row.map((value, columnIndex) => [columnIndex, rowIndex, value]),
91   );
92
93   return {
94     animationDuration: 240,
95     animationDurationUpdate: 220,
96     tooltip: {
97       formatter: (params: any) =>
98         `${title}<br/>样本 ${matrix.rowIndices[params.value[1]]} /
99         ${matrix.columnIndices[params.value[0]]}<br/>${formatNumber(params.value[2], 4)}`,
100     },
101     grid: { top: 12, right: 18, bottom: 32, left: 46 },
102     xAxis: {
103       type: 'category',
104       data: matrix.columnIndices.map(String),
105       axisTick: { show: false },
106       axisLabel: { fontSize: 10, color: '#59677d' },
107     },
108     yAxis: {
109       type: 'category',
110       data: matrix.rowIndices.map(String),
111       axisTick: { show: false },
112       axisLabel: { fontSize: 10, color: '#59677d' },
113     },
114     visualMap: {
115       show: false,
116       min: matrix.stats.min,
117       max: matrix.stats.max,
118       calculable: false,
119       orient: 'horizontal',
120       left: 'center',
121       bottom: 0,
122       inRange: { color: ['#edf4fa', '#78a9d4', '#1c5f91'] },
123     },
124     series: [
125       {
126         type: 'heatmap',
127         data,
128         emphasis: { itemStyle: { borderColor: '#173d5e', borderWidth: 1 } },
129       },
130     ],
131   };
132
133 export function MatrixScaleLegend({
134   matrix,
135   compact = false,
136   activeValue = null,
137 }): {
```

front/src/workbench/results/ResultChartPanels.tsx

```
138   matrix: MatrixPreview;
139   compact?: boolean;
140   activeValue?: number | null;
141 }) {
142   const minimum = formatNumber(matrix.stats.min, 4);
143   const maximum = formatNumber(matrix.stats.max, 4);
144   const hasActiveValue = activeValue !== null && Number.isFinite(activeValue);
145   const position = hasActiveValue
146     ? getMatrixScalePosition(activeValue, matrix.stats.min, matrix.stats.max)
147     : 50;
148   const alignment = position <= 12 ? ' is-start' : position >= 88 ? ' is-end' : '';
149   const activeLabel = hasActiveValue ? formatNumber(activeValue, 4) : '';
150
151   return (
152     <div
153       className={`result-matrix-scale${compact ? ' is-compact' : ''}`}
154       role="img"
155       aria-label={`矩阵色阶，最小值 ${minimum}，最大值 ${maximum}${hasActiveValue ? `，当前值 ${activeLabel}` : ''}`}
156     >
157       <span className="result-matrix-scale-bound is-minimum">{minimum}</span>
158       <span className="result-matrix-scale-track" aria-hidden="true">
159         <span
160           className={`result-matrix-scale-indicator${hasActiveValue ? ' is-active' : ''}${alignment}`}
161           style={{ left: `${position}%` }}
162         >
163           <output>{activeLabel}</output>
164         <i />
165       </span>
166     </span>
167     <span className="result-matrix-scale-bound">{maximum}</span>
168   </div>
169 );
170 }
171
172 export function buildWeightsOption(result: AnalysisResult): EChartsOption {
173   return {
174     animationDuration: 240,
175     animationDurationUpdate: 220,
176     tooltip: {
177       trigger: 'axis',
178       valueFormatter: (value: unknown) => formatNumber(Number(value), 4),
179     },
180     grid: { top: 18, right: 18, bottom: 44, left: 48 },
181     xAxis: {
182       type: 'category',
183       data: result.kernelWeights.items.map((item) => item.key.replace(/_/g, ' ')),
184       axisLabel: { interval: 0, rotate: 18, color: '#59677d', fontSize: 10 },
185       axisTick: { show: false },
186     },
187     yAxis: {
```

```
front/src/workbench/results/ResultChartPanels.tsx
```

```
188     type: 'value',
189     axisLabel: { color: '#59677d' },
190     splitLine: { lineStyle: { color: '#e4eaf0' } },
191   },
192   series: [
193     {
194       type: 'bar',
195       data: result.kernelWeights.items.map((item) => item.mean),
196       barWidth: 28,
197       itemStyle: { color: '#2d769f', borderRadius: [4, 4, 0, 0] },
198     },
199   ],
200 };
201 }
202
203 export type ConvergenceChartMode = 'objective' | 'relative' | 'log';
204
205 export function getConvergenceRelativeValue(value: number, initial: number) {
206   if (!Number.isFinite(value) || !Number.isFinite(initial) || initial === 0) return null;
207   return Number(((value / initial) * 100).toFixed(6));
208 }
209
210 export function buildConvergenceOption(
211   result: AnalysisResult,
212   run: number,
213   mode: ConvergenceChartMode = 'objective',
214 ): EChartsOption {
215   const current = result.convergence.runs.find((item) => item.run === run)
216     ?? result.convergence.runs[0];
217   const points = current?.points ?? [];
218   const initialObjective = points[0]?.objective;
219   const relativeValues = points.map((point) => (
220     initialObjective === undefined
221       ? null
222       : getConvergenceRelativeValue(point.objective, initialObjective)
223   ));
224   const seriesData = mode === 'relative'
225     ? relativeValues
226     : points.map((point) => (mode === 'log' && point.objective <= 0 ? null : point.objective));
227   const axisName = mode === 'relative'
228     ? '相对初始值'
229     : mode === 'log'
230       ? '目标函数（对数）'
231       : '目标函数';
232
233   return {
234     animationDuration: 240,
235     animationDurationUpdate: 220,
236     tooltip: {
237       trigger: 'axis',
```

front/src/workbench/results/ResultChartPanels.tsx

```
238     confine: true,
239     borderWidth: 0,
240     padding: [9, 11],
241     formatter: (params) => {
242       const item = Array.isArray(params) ? params[0] : params;
243       const dataIndex = Number(item?.dataIndex ?? 0);
244       const point = points[dataIndex];
245       if (!point) return '暂无迭代数据';
246       const relativeValue = relativeValues[dataIndex];
247       const displayedValue = mode === 'relative'
248         ? `${formatNumber(relativeValue, 4)}%`
249         : formatNumber(point.objective, 5);
250       const modeLabel = mode === 'relative' ? '相对初始值' : mode === 'log' ? '对数轴目标' : '目标函数';
251
252       return [
253         `<strong>第 ${point.iteration} 次迭代</strong>`,
254         `${modeLabel}: ${displayedValue}`,
255         mode === 'relative' ? `原始目标: ${formatNumber(point.objective, 5)}%` : `相对初始: ${formatNumber(relativeValue,
256 4)}%`,
257         `本轮变化: ${point.relativeChange == null ? '—' : `${formatNumber(point.relativeChange * 100, 4)}%`}`,
258       ].join('<br/>');
259     },
260     grid: { top: 40, right: 18, bottom: 32, left: 72 },
261     xAxis: {
262       type: 'category',
263       name: '迭代次数',
264       nameLocation: 'middle',
265       nameGap: 24,
266       data: points.map((point) => String(point.iteration)),
267       axisTick: { show: false },
268       axisLabel: { color: '#59677d' },
269     },
270     yAxis: {
271       type: mode === 'log' ? 'log' : 'value',
272       logBase: 10,
273       scale: mode === 'log',
274       name: axisName,
275       nameLocation: 'end',
276       nameGap: 10,
277       nameTextStyle: {
278         color: '#59677d',
279         align: 'left',
280         padding: [0, 0, 4, 0],
281       },
282       axisLabel: {
283         color: '#59677d',
284         formatter: mode === 'relative'
285           ? (value: number) => `${formatNumber(value, 1)}%`
286           : (value: number) => (value === 0 ? '0' : value.toExponential(1)),
```

```
front/src/workbench/results/ResultChartPanels.tsx
```

```
287     },
288     splitLine: { lineStyle: { color: '#e4eaf0' } },
289   },
290   series: [
291     {
292       type: 'line',
293       smooth: true,
294       symbolSize: 6,
295       data: seriesData,
296       lineStyle: { color: '#227e71', width: 3 },
297       itemStyle: { color: '#227e71' },
298       areaStyle: { color: 'rgba(34, 126, 113, 0.12)' },
299       emphasis: { focus: 'series', scale: 1.35 },
300       ... (mode === 'relative' ? {
301         markLine: {
302           silent: true,
303           symbol: 'none',
304           lineStyle: { color: '#91a0b1', type: 'dashed', width: 1 },
305           label: { show: true, formatter: '初始基线', color: '#68798c', fontSize: 9 },
306           data: [{ yAxis: 100 }],
307         },
308       } : {}),
309     },
310   ],
311 };
312 }
313
314 export function buildScatterOption(result: AnalysisResult): EChartsOption {
315   const groups = new Map<string, Array<[number, number, number]>>();
316   result.preview.scatter.points.forEach((point) => {
317     const items = groups.get(point.predictedLabel) ?? [];
318     items.push([point.x, point.y, point.sampleIndex]);
319     groups.set(point.predictedLabel, items);
320   });
321
322   const colors = ['#2476a4', '#27796e', '#c47a29', '#9a4e77', '#6559a7'];
323   return {
324     animationDuration: 240,
325     animationDurationUpdate: 220,
326     tooltip: {
327       formatter: (params: any) => `样本 ${params.value[2]}<br/>预测簇 ${params.seriesName}`,
328     },
329     grid: { top: 18, right: 20, bottom: 38, left: 48 },
330     xAxis: {
331       type: 'value',
332       axisLabel: { color: '#59677d' },
333       splitLine: { lineStyle: { color: '#e4eaf0' } },
334     },
335     yAxis: {
336       type: 'value',
```



```
front/src/workbench/results/ResultChartPanels.tsx
```

```
337     axisLabel: { color: '#59677d' },
338     splitLine: { lineStyle: { color: '#e4eaf0' } },
339   },
340   series: [...groups.entries()].map(([label, data], index) => ({
341     type: 'scatter',
342     name: `簇 ${label}`,
343     data,
344     symbolSize: 7,
345     itemStyle: { color: colors[index % colors.length], opacity: 0.76 },
346   })),
347 ];
348 }
349
350 export function buildMetricRunsOption(result: AnalysisResult): EChartsOption {
351   const metricKeys = ['acc', 'nmi', 'ari', 'f1'] as const;
352   const colors = ['#23865f', '#2f7de1', '#14796f', '#59677d'];
353   const metricColors = Object.fromEntries(metricKeys.map((key, index) => [key === 'f1' ? 'F1-score' :
     key.toUpperCase(), colors[index]]));
354
355   return {
356     animationDuration: 240,
357     animationDurationUpdate: 220,
358     color: colors,
359     tooltip: {
360       trigger: 'axis',
361       confine: true,
362       borderWidth: 0,
363       padding: [8, 10],
364       extraCssText: 'border-radius: 6px; box-shadow: 0 4px 8px rgba(35, 55, 78, 0.18);',
365       formatter: (params: any) => {
366         const items = Array.isArray(params) ? params : [params];
367         const run = items[0]?.axisValue ?? '—';
368         const cells = items.map((item: any) => {
369           const color = metricColors[item.seriesName] ?? '#59677d';
370           return `<span style="display:flex;align-items:center;justify-content:space-between;gap:12px;min-width:88px;"
><span style="display:flex;align-items:center;gap:6px;color:#59677d;"><i
style="width:7px;height:7px;border-radius:50%;background:${color};"></i>${item.seriesName}</span><strong
style="color:#2f4054;font-variant-numeric:tabular-nums;">${Number(item.value).toFixed(2)}%</strong></span>`;
371         }).join('');
372         return `<div style="min-width:210px;"><div style="margin-bottom:7px;color:#34495d;font-size:11px;font-weight:7
00;">第 ${run} 轮</div><div style="display:grid;grid-template-columns:repeat(2,minmax(88px,1fr));gap:6px
14px;font-size:10px;line-height:1.35;">${cells}</div></div>`;
373       },
374       position: (point, _params, _dom, _rect, size) => {
375         const [contentWidth, contentHeight] = size.contentSize;
376         const [viewWidth, viewHeight] = size.viewSize;
377         const preferredX = point[0] + contentWidth + 12 <= viewWidth ? point[0] + 12 : point[0] - contentWidth - 12;
378         const maxX = Math.max(8, viewWidth - contentWidth - 8);
379         const maxY = Math.max(8, viewHeight - contentHeight - 8);
380         const minY = Math.min(36, maxY);
```

```
front/src/workbench/results/ResultChartPanels.tsx
```

```
381         return [
382             Math.max(8, Math.min(preferredX, maxX)),
383             Math.max(minY, Math.min(point[1] - contentHeight / 2, maxY)),
384         ];
385     },
386 },
387 legend: { top: 0, right: 12, textStyle: { color: '#59677d', fontSize: 11 } },
388 grid: { top: 40, right: 22, bottom: 44, left: 52 },
389 xAxis: {
390     type: 'category',
391     name: '运行轮次',
392     nameLocation: 'middle',
393     nameGap: 28,
394     data: result.metrics.runs.map((item) => String(item.run)),
395     axisTick: { show: false },
396     axisLabel: { color: '#59677d' },
397 },
398 yAxis: {
399     type: 'value',
400     axisLabel: { color: '#59677d', formatter: '{value}%' },
401     splitLine: { lineStyle: { color: '#e4eaf0' } },
402 },
403 series: metricKeys.map((key) => ({
404     type: 'line',
405     name: key === 'f1' ? 'F1-score' : key.toUpperCase(),
406     data: result.metrics.runs.map((item) => Number((item[key] * 100).toFixed(4))),
407     symbolSize: 6,
408     lineStyle: { width: 2 },
409     emphasis: { focus: 'series' },
410 }))),
411 });
412 }
413
414 export type EvaluationMetricKey = 'acc' | 'nmi' | 'ari' | 'f1';
415 export type EvaluationTrendMode = 'absolute' | 'relative';
416
417 const evaluationMetricLabels: Record<EvaluationMetricKey, string> = {
418     acc: 'ACC',
419     nmi: 'NMI',
420     ari: 'ARI',
421     f1: 'F1-score',
422 };
423
424 const evaluationMetricColors: Record<EvaluationMetricKey, string> = {
425     acc: '#23865f',
426     nmi: '#2f7de1',
427     ari: '#14796f',
428     f1: '#59677d',
429 };
430
```

front/src/workbench/results/ResultChartPanels.tsx

```
431 export function buildEvaluationTrendOption(  
432   result: AnalysisResult,  
433   metric: EvaluationMetricKey = 'acc',  
434   mode: EvaluationTrendMode = 'absolute',  
435 ): EChartsOption {  
436   const reduceMotion = typeof window !== 'undefined'  
437     && typeof window.matchMedia === 'function'  
438     && window.matchMedia('(prefers-reduced-motion: reduce)').matches;  
439   const label = evaluationMetricLabels[metric];  
440   const color = evaluationMetricColors[metric];  
441   const rows = result.metrics.runs;  
442   const values = rows  
443     .map((row) => Number(row[metric]) * 100)  
444     .filter((value) => Number.isFinite(value));  
445   const aggregate = result.metrics.aggregate[metric];  
446   const aggregateMean = Number(aggregate.mean) * 100;  
447   const displayedValues = mode === 'relative'  
448     ? values.map((value) => value - aggregateMean)  
449     : values;  
450   const minimum = displayedValues.length ? Math.min(...displayedValues) : mode === 'relative' ? -1 : 0;  
451   const maximum = displayedValues.length ? Math.max(...displayedValues) : mode === 'relative' ? 1 : 100;  
452   const spread = Math.max(maximum - minimum, 1);  
453   const padding = Math.max(spread * 0.3, 0.8);  
454   const axisMin = mode === 'relative'  
455     ? Math.floor((minimum - padding) * 10) / 10  
456     : Math.max(0, Math.floor((minimum - padding) * 10) / 10);  
457   const axisMax = mode === 'relative'  
458     ? Math.ceil((maximum + padding) * 10) / 10  
459     : Math.min(100, Math.ceil((maximum + padding) * 10) / 10);  
460   const markLineValue = mode === 'relative' ? 0 : aggregateMean;  
461  
462   return {  
463     animation: !reduceMotion,  
464     animationDuration: reduceMotion ? 0 : 220,  
465     animationDurationUpdate: reduceMotion ? 0 : 180,  
466     tooltip: {  
467       trigger: 'axis',  
468       confine: true,  
469       borderWidth: 0,  
470       padding: [9, 11],  
471       extraCssText: 'border-radius: 7px; box-shadow: 0 4px 10px rgba(35, 55, 78, 0.16);',  
472       formatter: (params: any) => {  
473         const item = Array.isArray(params) ? params[0] : params;  
474         const index = Number(item?.dataIndex ?? 0);  
475         const row = rows[index];  
476         if (!row) return '暂无实验数据';  
477         const value = Number(row[metric]) * 100;  
478         const delta = Number.isFinite(aggregateMean) && Number.isFinite(value)  
479           ? value - aggregateMean  
480           : null;
```

front/src/workbench/results/ResultChartPanels.tsx

```
481         return [
482             `
```

```
front/src/workbench/results/ResultChartPanels.tsx
```

```
531         silent: true,
532         symbol: 'none',
533         lineStyle: { color: '#9aa8b7', type: 'dashed', width: 1 },
534         label: {
535             show: true,
536             formatter: mode === 'relative' ? '均值基线 0' : `均值 ${aggregateMean.toFixed(2)}%`,
537             color: '#68798c',
538             fontSize: 9,
539         },
540         data: [{ yAxis: Number(markLineValue.toFixed(4)) }],
541     }
542     : undefined,
543 },
544 ],
545 };
546 }
547
548 export type KernelRunsMode = 'relative' | 'absolute';
549
550 const kernelRunLabels: Record<string, string> = {
551     rbf_sigma_squared: 'K1 • RBF /  $\sigma^2$ ',
552     linear: 'K2 • Linear',
553     rbf_sigma: 'K3 • RBF /  $\sigma$ ',
554     polynomial_2: 'K4 • Polynomial / 2',
555 };
556
557 export function getKernelRelativeChange(
558     value: number | null | undefined,
559     mean: number,
560     seriesValues: Array<number | null>,
561 ) {
562     if (value === null || value === undefined || !Number.isFinite(value) || !Number.isFinite(mean)) return null;
563     if (mean !== 0) return Number((((value - mean) / Math.abs(mean)) * 100).toFixed(6));
564     return seriesValues.every((current) => current !== null && Math.abs(current) <= Number.EPSILON) ? 0 : null;
565 }
566
567 export function buildKernelRunsOption(result: AnalysisResult, mode: KernelRunsMode = 'absolute'): EChartsOption {
568     const keys = result.kernelWeights.items.map((item) => item.key);
569     const colors = ['#2f7del', '#23865f', '#c97a17', '#7967a8'];
570     const absoluteSeries = keys.map((_, index) => result.kernelWeights.runs.map((run) => {
571         const value = run.values[index];
572         return value === undefined || !Number.isFinite(value) ? null : value;
573     }));
574     const relativeSeries = keys.map((_, index) => {
575         const mean = result.kernelWeights.items[index]?.mean;
576         return absoluteSeries[index].map((value) => getKernelRelativeChange(value, mean, absoluteSeries[index]));
577     });
578
579     return {
580         animationDuration: 240,
```

front/src/workbench/results/ResultChartPanels.tsx

```
581     animationDurationUpdate: 220,
582     color: colors,
583     tooltip: {
584       trigger: 'axis',
585       formatter: (params) => {
586         const items = Array.isArray(params) ? params : [params];
587         const first = items[0];
588         const dataIndex = Number(first?.dataIndex ?? 0);
589         const run = result.kernelWeights.runs[dataIndex];
590         const rows = items.map((item) => {
591           const seriesIndex = Number(item.seriesIndex ?? 0);
592           const absolute = absoluteSeries[seriesIndex]?.[dataIndex];
593           const relative = relativeSeries[seriesIndex]?.[dataIndex];
594           const relativeText = relative === null || relative === undefined ? '—' : `${formatNumber(relative, 3)}%`;
595           const primary = mode === 'relative' ? relativeText : formatNumber(absolute, 6);
596           const secondary = mode === 'relative' ? `alpha ${formatNumber(absolute, 6)}%` : `相对均值 ${relativeText}`;
597           return `${item.marker ?? ''}${item.seriesName}: <strong>${primary}</strong> <span
style="color:#758397">${secondary}</span>`;
598         });
599         return [ `第 ${run?.run ?? '—'} 轮`, ...rows ].join('<br/>');
600       },
601     },
602     legend: { top: 0, right: 12, textStyle: { color: '#59677d', fontSize: 10 } },
603     grid: { top: 44, right: 22, bottom: 44, left: 58 },
604     xAxis: {
605       type: 'category',
606       name: '运行轮次',
607       nameLocation: 'middle',
608       nameGap: 28,
609       data: result.kernelWeights.runs.map((item) => String(item.run)),
610       axisTick: { show: false },
611       axisLabel: { color: '#59677d' },
612     },
613     yAxis: {
614       type: 'value',
615       name: mode === 'relative' ? '相对均值变化' : '原始 alpha',
616       nameTextStyle: { color: '#738195', fontSize: 10, padding: [0, 0, 4, 0] },
617       scale: mode === 'relative',
618       axisLabel: {
619         color: '#59677d',
620         formatter: mode === 'relative' ? '{value}%' : undefined,
621       },
622       splitLine: { lineStyle: { color: '#e4eaf0' } },
623     },
624     series: keys.map((key, index) => ({
625       type: 'line',
626       name: kernelRunLabels[key] ?? key.replace(/_/g, ' '),
627       data: mode === 'relative' ? relativeSeries[index] : absoluteSeries[index],
628       connectNulls: false,
629       showSymbol: true,
```

```
front/src/workbench/results/ResultChartPanels.tsx
```

```
630     smooth: false,
631     symbol: 'circle',
632     symbolSize: 7,
633     lineStyle: { width: 2.2 },
634     emphasis: { focus: 'series' },
635     ...(mode === 'relative' && index === 0 ? {
636       markLine: {
637         silent: true,
638         symbol: 'none',
639         lineStyle: { color: '#95a3b3', type: 'dashed', width: 1 },
640         label: { show: true, formatter: '均值基线', color: '#758397', fontSize: 9 },
641         data: [{ yAxis: 0 }],
642       },
643     } : {}),
644   )),
645 };
646 }
647
648 export function OptionChartPanel({
649   title,
650   meta,
651   option,
652   hasData,
653   emptyTitle,
654   emptyDetail,
655   actions,
656   compact = false,
657 }): {
658   title: string;
659   meta: string;
660   option: EChartsOption;
661   hasData: boolean;
662   emptyTitle: string;
663   emptyDetail: string;
664 } & PanelProps) {
665   const ref = useResultChart(option);
666
667   return (
668     <section className={`panel result-chart-panel${compact ? ' is-compact' : ''}`}>
669       <WorkbenchSectionHeader title={title} meta={meta} actions={actions} />
670       {hasData ? (
671         <div className="result-chart" ref={ref} />
672       ) : (
673         <ResultPanelEmpty title={emptyTitle} detail={emptyDetail} />
674       )}
675     </section>
676   );
677 }
678
679 export function MatrixPanel({
```

```
front/src/workbench/results/ResultChartPanels.tsx
```

```
680   matrix,
681   title,
682   description,
683   compact = false,
684   actions,
685 }: {
686   matrix: MatrixPreview;
687   title: string;
688   description: string;
689 } & PanelProps) {
690   const option = useMemo(() => buildMatrixOption(matrix, title), [matrix, title]);
691   const [activeValue, setActiveValue] = useState<number | null>(null);
692   const ref = useResultChart(option, setActiveValue);
693   const hasData = matrix.values.length > 0 && matrix.values.some((row) => row.length > 0);
694
695   return (
696     <section className={`panel result-chart-panel${compact ? ' is-compact' : ''}`}>
697       <WorkbenchSectionHeader
698         title={title}
699         meta={description}
700         actions={
701           <div className="result-panel-actions">
702             <WorkbenchStatus tone="info">{matrix.shape[0]} × {matrix.shape[1]}</WorkbenchStatus>
703             {actions}
704           </div>
705         }
706       </>
707       {hasData ? (
708         <>
709           <div className="result-chart" ref={ref} />
710           <MatrixScaleLegend matrix={matrix} compact={compact} activeValue={activeValue} />
711         </>
712       ) : (
713         <ResultPanelEmpty title="矩阵预览不可用" detail="结果中没有可展示的矩阵采样数据。" />
714       )}
715       <div className="result-stat-line">
716         <span>均值 {formatNumber(matrix.stats.mean, 4)}</span>
717         <span>标准差 {formatNumber(matrix.stats.std, 4)}</span>
718         <span>对称误差 {formatNumber(matrix.stats.symmetryMaxError, 6)}</span>
719       </div>
720     </section>
721   );
722 }
723
724 export function WeightPanel({ result, compact = false, actions }: { result: AnalysisResult } & PanelProps) {
725   const option = useMemo(() => buildWeightsOption(result), [result]);
726   const ref = useResultChart(option);
727   const sum = result.kernelWeights.items.reduce((total, item) => total + item.mean, 0);
728   const hasData = result.kernelWeights.items.length > 0;
729 }
```



```
front/src/workbench/results/ResultChartPanels.tsx
```

```
730   return (
731     <section className={`panel result-chart-panel${compact ? ' is-compact' : ''}`}>
732       <WorkbenchSectionHeader
733         title="核权重贡献"
734         meta="原始 alpha 的重复实验均值，不进行前端归一化"
735         actions={
736           <div className="result-panel-actions">
737             {hasData ? (
738               <WorkbenchStatus tone={Math.abs(sum - 1) < 0.01 ? 'success' : 'warning'}>
739                 总和 {formatNumber(sum, 4)}
740               </WorkbenchStatus>
741             ) : null}
742             {actions}
743           </div>
744         }
745       </>
746       {hasData ? (
747         <>
748           <div className="result-chart" ref={ref} />
749           <div className="result-kernel-list">
750             {result.kernelWeights.items.map((item) => (
751               <div key={item.key}>
752                 <span>{item.key}</span>
753                 <strong>{formatNumber(item.mean, 5)}</strong>
754                 <small>std {formatNumber(item.std, 5)}</small>
755               </div>
756             ))}
757           </div>
758         </>
759       ) : (
760         <ResultPanelEmpty title="暂无核权重" detail="当前结果没有记录可展示的核权重。" />
761       )}
762     </section>
763   );
764 }
765
766 export function ConvergencePanel({
767   result,
768   compact = false,
769   actions,
770 }): { result: AnalysisResult } & PanelProps) {
771   const [run, setRun] = useState(result.convergence.representativeRun);
772   useEffect(() => setRun(result.convergence.representativeRun), [result.convergence.representativeRun]);
773   const current = result.convergence.runs.find((item) => item.run === run)
774     ?? result.convergence.runs[0];
775   const option = useMemo(() => buildConvergenceOption(result, run), [result, run]);
776   const ref = useResultChart(option);
777   const finalPoint = current?.points[current.points.length - 1];
778   const hasData = Boolean(current?.points.length);
779 }
```

front/src/workbench/results/ResultChartPanels.tsx

```
780   return (
781     <section className={`panel result-chart-panel${compact ? ' is-compact' : ''}`}>
782       <WorkbenchSectionHeader
783         title="优化收敛"
784         meta={
785           !current
786             ? '当前结果没有收敛轨迹'
787             : current.converged
788               ? '达到求解器收敛阈值'
789               : '达到任务设置的最大迭代次数'
790         }
791         actions={
792           <div className="result-panel-actions">
793             {result.convergence.runs.length ? (
794               <SelectField
795                 value={String(current?.run ?? '')}
796                 options={result.convergence.runs.map((item) => ({ value: String(item.run), label: `第 ${item.run} 轮`
797                   onChange={(value) => setRun(Number(value))}
798                   ariaLabel="选择运行轮次"
799                   size="sm"
800                   className="result-select-field result-chart-run-select-field"
801                 />
802               ) : null}
803             {actions}
804           </div>
805         }
806       />
807       {hasData ? (
808         <div className="result-chart" ref={ref} />
809       ) : (
810         <ResultPanelEmpty title="暂无收敛轨迹" detail="该任务未返回可绘制的迭代目标函数。" />
811       )}
812       {current ? (
813         <div className="result-stat-line">
814           <span>最终目标函数 {formatNumber(finalPoint?.objective, 5)}</span>
815           <span>{current.converged ? '提前收敛' : '达到迭代上限'}</span>
816         </div>
817       ) : null}
818     </section>
819   );
820 }
821
822 export function ScatterPanel({
823   result,
824   compact = false,
825   actions,
826 }: { result: AnalysisResult } & PanelProps) {
827   const option = useMemo(() => buildScatterOption(result), [result]);
828   const ref = useResultChart(option);
```

```
front/src/workbench/results/ResultChartPanels.tsx
```

```
829   const scatter = result.preview.scatter;
830   const hasData = scatter.points.length > 0;
831   const meta = scatter.sampled
832     ? `为保证可读性，展示 ${scatter.points.length}/${scatter.totalCount} 个样本`
833     : `展示全部 ${scatter.totalCount} 个样本`;
834
835   return (
836     <section className={`panel result-chart-panel${compact ? ' is-compact' : ''}`}>
837       <WorkbenchSectionHeader title="代表轮次聚类分布" meta={meta} actions={actions} />
838       {hasData ? (
839         <div className="result-chart" ref={ref} />
840       ) : (
841         <ResultPanelEmpty title="暂无聚类分布" detail="当前结果没有可展示的二维投影点。" />
842       )}
843     </section>
844   );
845 }
846
```

```
// ===== 文件: front/src/workbench/results/ResultDetailPages.tsx =====
```

```
1 import { useEffect, useMemo, useState, type ReactNode } from 'react';
2 import { useNavigate } from 'react-router-dom';
3 import {
4   Activity,
5   Archive,
6   ArrowRight,
7   BarChart3,
8   CheckCircle2,
9   Clock3,
10  ChevronsDown,
11  ChevronsUp,
12  Download,
13  FileArchive,
14  FileJson,
15  FileSpreadsheet,
16  FileText,
17  Gauge,
18  HardDrive,
19  History,
20  Images,
21  LayoutGrid,
22  ListChecks,
23  PackageOpen,
24  Network,
25  PackageCheck,
26  SlidersHorizontal,
27  Target,
28  TrendingDown,
29 } from 'lucide-react';
30 import { MathFormula } from '../../components/MathFormula';
```

front/src/workbench/results/ResultDetailPages.tsx

```
31 import { SelectField } from '../components/SelectField';
32 import { createTaskExport, downloadProtectedFile, fetchTaskExports } from '../api/results';
33 import { SOFTWARE_SHORT_NAME } from '../appMeta';
34 import { formatLocalDateTime } from '../utils/time';
35 import {
36   WorkbenchMetricStrip,
37   WorkbenchNotice,
38   WorkbenchSectionHeader,
39   WorkbenchStatus,
40 } from './WorkbenchUi';
41 import {
42   buildConvergenceOption,
43   buildKernelRunsOption,
44   buildMatrixOption,
45   buildMetricRunsOption,
46   buildScatterOption,
47   buildWeightsOption,
48   ConvergencePanel,
49   MatrixScaleLegend,
50   MatrixPanel,
51   OptionChartPanel,
52   ScatterPanel,
53   useResultChart,
54   type ConvergenceChartMode,
55 } from './ResultChartPanels';
56 import { MetricStrip, ReadyHeader } from './ResultPageShared';
57 import {
58   downloadTextFile,
59   formatBytes,
60   formatNumber,
61   formatPercent,
62   metricLabels,
63   navigateToTaskResult,
64   type MetricKey,
65 } from './resultPresentation';
66 import type { AnalysisResult, TaskExport, TaskResultResource } from './types';
67
68 const kernelDefinitions = [
69   {
70     key: 'rbf_sigma_squared',
71     name: 'K1 • RBF 核',
72     short: 'RBF /  $\sigma^2$ ',
73     latex: String.raw`\mathbf{K}_1(\mathbf{c}_i, \mathbf{c}_j) = \exp\left(-\frac{1}{2} \sqrt{\frac{\|\mathbf{c}_i - \mathbf{c}_j\|^2}{\sigma^2 + \varepsilon}}\right)`,
74     label: 'K1 RBF 核, 带宽分母为  $\sigma^2$  平方加  $\varepsilon$ ',
75   },
76   {
77     key: 'linear',
78     name: 'K2 • 线性核',
79     short: 'Linear',
```

```
front/src/workbench/results/ResultDetailPages.tsx
```

```
80     latex: String.raw`\mathbf{K}_2(\mathbf{c}_i,\mathbf{c}_j)=\mathbf{c}_i^{\mathsf{T}}\mathbf{c}_j`,
81     label: 'K2 线性核',
82   },
83   {
84     key: 'rbf_sigma',
85     name: 'K3 • RBF 核',
86     short: 'RBF / sigma',
87     latex: String.raw`\mathbf{K}_3(\mathbf{c}_i,\mathbf{c}_j)=\exp\left(-\frac{\lVert\mathbf{c}_i-\mathbf{c}_j\rVert^2}{2(\sigma+\varepsilon)}\right)`,
88     label: 'K3 RBF 核, 带宽分母为 sigma 加 epsilon',
89   },
90   {
91     key: 'polynomial_2',
92     name: 'K4 • 二次多项式核',
93     short: 'Polynomial / 2',
94     latex: String.raw`\mathbf{K}_4(\mathbf{c}_i,\mathbf{c}_j)=\left(\mathbf{c}_i^{\mathsf{T}}\mathbf{c}_j+1\right)^2`,
95     label: 'K4 二次多项式核',
96   },
97 ] as const;
98
99 type KernelKey = (typeof kernelDefinitions)[number]['key'];
100 type KernelTrendMode = 'relative' | 'absolute';
101
102 const topPairCountPresets = [5, 10, 20] as const;
103
104 function getTopPairCountOptions(availableCount: number) {
105   if (availableCount <= 0) return [];
106   const options = topPairCountPresets.filter((count) => count <= availableCount);
107   if (!options.includes(availableCount as (typeof topPairCountPresets)[number])) {
108     return [...options, availableCount];
109   }
110   return options;
111 }
112
113 const exportOptions = [
114   { key: 'metrics', label: '性能指标', detail: '聚合统计与逐轮结果', icon: BarChart3 },
115   { key: 'parameters', label: '任务参数', detail: '算法配置与随机种子', icon: SlidersHorizontal },
116   { key: 'result', label: '结果清单', detail: '任务和产物索引', icon: ListChecks },
117   { key: 'labels', label: '聚类标签', detail: '代表轮次及全部轮次标签', icon: FileSpreadsheet },
118   { key: 'ca', label: 'CA 矩阵', detail: '完整协关联矩阵', icon: LayoutGrid },
119   { key: 's', label: 'S 矩阵', detail: '多核相似性结果', icon: Network },
120   { key: 'z', label: 'Z 矩阵', detail: '拓朴关系结果', icon: Target },
121 ] as const;
122
123 const exportPresets = [
124   { key: 'review', label: '评审交付', detail: '指标、参数、结果清单与标签', items: ['metrics', 'parameters', 'result', 'labels'] },
125   { key: 'reproduce', label: '复现实验', detail: '任务参数、结果清单与全部矩阵', items: ['parameters', 'result', 'labels', 'ca', 's', 'z'] },
126   { key: 'complete', label: '完整归档', detail: '保存当前任务的全部可交付内容', items: exportOptions.map((item) => item.key) },
127 ] as const;
```

front/src/workbench/results/ResultDetailPages.tsx

```
128
129 function exportItemLabels(items: string[]) {
130   return items.map((key) => exportOptions.find((item) => item.key === key)?.label ?? key);
131 }
132
133 type ReportSection = 'summary' | 'metrics' | 'parameters' | 'runs' | 'artifacts';
134
135 const reportSectionOptions: Array<{ key: ReportSection; label: string; detail: string }> = [
136   { key: 'summary', label: '任务摘要', detail: '数据集、算法、样本和耗时' },
137   { key: 'metrics', label: '性能指标', detail: '均值、标准差与取值范围' },
138   { key: 'parameters', label: '算法参数', detail: '本次执行参数' },
139   { key: 'runs', label: '重复实验', detail: '逐轮指标和随机种子' },
140   { key: 'artifacts', label: '结果产物', detail: '文件格式与大小清单' },
141 ];
142
143 type ReportPreviewSectionProps = {
144   section: ReportSection;
145   title: string;
146   meta: string;
147   expanded: boolean;
148   onToggle: (section: ReportSection, expanded: boolean) => void;
149   children: ReactNode;
150 };
151
152 function ReportPreviewSection({ section, title, meta, expanded, onToggle, children }: ReportPreviewSectionProps) {
153   const bodyId = `report-preview-${section}`;
154
155   return (
156     <details
157       className="report-preview-section"
158       open={expanded}
159       onToggle={(event) => onToggle(section, event.currentTarget.open)}
160     >
161       <summary aria-controls={bodyId} aria-expanded={expanded}>
162         <span>
163           <h3>{title}</h3>
164           <small>{meta}</small>
165         </span>
166         <ChevronsDown size={16} aria-hidden="true" />
167       </summary>
168       <div className="report-preview-section-body" id={bodyId}>
169         {children}
170       </div>
171     </details>
172   );
173 }
174
175 function DetailLink({ label, onClick }: { label: string; onClick: () => void }) {
176   return (
177     <button type="button" className="text-action" onClick={onClick}>
```

front/src/workbench/results/ResultDetailPages.tsx

```
178     {label}
179     <ArrowRight size={14} aria-hidden="true" />
180   </button>
181 );
182 }
183
184 function matrixStatus(result: AnalysisResult) {
185   const error = result.preview.matrices.ca.stats.symmetryMaxError;
186   if (error === null || !Number.isFinite(error)) return { tone: 'neutral' as const, label: '未计算对称误差' };
187   return error <= 1e-8
188     ? { tone: 'success' as const, label: '矩阵对称' }
189     : { tone: 'warning' as const, label: '存在对称偏差' };
190 }
191
192 export function CaResultPage({ resource }: { resource: TaskResultResource }) {
193   const navigate = useNavigate();
194   const task = resource.envelope!.task!;
195   const result = resource.envelope!.result!;
196   const matrix = result.preview.matrices.ca;
197   const status = matrixStatus(result);
198   const availableTopPairCount = matrix.topPairs.length;
199   const topPairCountOptions = useMemo(
200     () => getTopPairCountOptions(availableTopPairCount),
201     [availableTopPairCount],
202   );
203   const [topPairCount, setTopPairCount] = useState(() => Math.min(10, availableTopPairCount));
204   const visibleTopPairs = matrix.topPairs.slice(0, topPairCount);
205
206   useEffect(() => {
207     setTopPairCount(Math.min(10, availableTopPairCount));
208   }, [availableTopPairCount]);
209
210   return (
211     <section className="soft-page result-detail-page result-ca-page">
212       <ReadyHeader title="CA 协关联矩阵" icon={LayoutGrid} resource={resource} />
213       <MetricStrip result={result} />
214
215       <div className="result-detail-primary result-ca-primary">
216         <MatrixPanel
217           matrix={matrix}
218           title="CA 矩阵预览"
219           description="等距抽取完整矩阵，坐标对应原始样本编号"
220           actions={
221             <DetailLink
222               label="导出完整矩阵"
223               onClick={() => navigateToTaskResult(navigate, 'export', task.id)}
224             />
225           }
226         />
227       </div>
228     </section>
229   );
230 }
```

```
front/src/workbench/results/ResultDetailPages.tsx
```

```
228     <section className="panel result-diagnostic-panel" aria-label="矩阵诊断">
229       <WorkbenchSectionHeader
230         title="矩阵诊断"
231         meta="基于持久化矩阵预览的统计"
232         actions={<WorkbenchStatus tone={status.tone}>{status.label}</WorkbenchStatus>}
233       />
234       <dl className="result-diagnostic-list">
235         <div><dt>完整尺寸</dt><dd>{matrix.shape[0]} × {matrix.shape[1]}</dd></div>
236         <div><dt>预览采样</dt><dd>{matrix.rowIndices.length} × {matrix.columnIndices.length}</dd></div>
237         <div><dt>对角均值</dt><dd>{formatNumber(matrix.stats.diagonalMean, 5)}</dd></div>
238         <div><dt>非零比例</dt><dd>{formatPercent(matrix.stats.nonzeroRatio)}</dd></div>
239         <div><dt>最小 / 最大</dt><dd>{formatNumber(matrix.stats.min, 4)} / {formatNumber(matrix.stats.max,
40         4)}</dd></div>
240         <div><dt>对称误差</dt><dd>{formatNumber(matrix.stats.symmetryMaxError, 8)}</dd></div>
241       </dl>
242       <div className="result-formula-inline">
243         <span>构造公式</span>
244         <MathFormula
245           className="result-ca-formula"
246           displayMode
247           latex={String.raw`\mathbf{CA}=\frac{\mathbf{M}\mathbf{M}^{\mathsf{T}}}{n_{\mathrm{base}}}`}
248           label="CA 等于 M 乘 M 转置除以基础聚类数量"
249         />
250       </div>
251       <WorkbenchNotice
252         tone="info"
253         icon={LayoutGrid}
254         title="预览经过等距抽样"
255         detail="统计值来自完整矩阵，热力图仅降低浏览器渲染密度。"
256       />
257     </section>
258   </div>
259
260   <section className="panel result-ranked-table-panel" aria-label="高协关联样本对">
261     <WorkbenchSectionHeader
262       title="高协关联样本对"
263       meta={`忽略对角线后按 CA 数值降序展示，当前显示 ${visibleTopPairs.length} / ${availableTopPairCount} 组`}
264       actions={availableTopPairCount ? (
265         <div className="result-panel-actions">
266           <div className="result-top-pair-control">
267             <span>显示数量</span>
268             <SelectField
269               value={String(topPairCount)}
270               options={topPairCountOptions.map((count) => ({ value: String(count), label: `Top ${count}` })))}
271               onChange={(value) => setTopPairCount(Number(value))}
272               ariaLabel="选择高协关联样本对显示数量"
273               size="sm"
274               className="result-select-field result-top-pair-select-field"
275             />
276           </div>
277         </div>
278       )}
```



```
front/src/workbench/results/ResultDetailPages.tsx
```

```
277         </div>
278       ) : null}
279     />
280     {matrix.topPairs.length ? (
281       <div className="result-ranked-table" role="table" aria-label="高协关联样本对排名">
282         <div className="result-ranked-row header" role="row">
283           <span role="columnheader">排名</span><span role="columnheader">样本 A</span><span role="columnheader">样本
284           B</span><span role="columnheader">CA 数值</span><span role="columnheader">矩阵位置</span>
285         </div>
286         {visibleTopPairs.map((pair, index) => (
287           <div className="result-ranked-row" role="row" key={` ${pair.row}-${pair.column}`} >
288             <strong role="cell">{String(index + 1).padStart(2, '0')}</strong>
289             <span role="cell">S{pair.row}</span>
290             <span role="cell">S{pair.column}</span>
291             <strong role="cell">{formatNumber(pair.value, 5)}</strong>
292             <span role="cell">{pair.row}, {pair.column}</span>
293           </div>
294         ))}
295       </div>
296     ) : (
297       <WorkbenchNotice tone="info" icon={LayoutGrid} title="暂无样本对" detail="当前矩阵预览没有可排序的非对角元素。" />
298     )}
299   </section>
300 </section>
301 );
302 }
303 export function KernelResultPage({ resource }: { resource: TaskResultResource }) {
304   const result = resource.envelope!.result!;
305   const [active, setActive] = useState<KernelKey>(kernelDefinitions[0].key);
306   const [trendMode, setTrendMode] = useState<KernelTrendMode>('relative');
307   const definition = kernelDefinitions.find((item) => item.key === active) ?? kernelDefinitions[0];
308   const weight = result.kernelWeights.items.find((item) => item.key === active);
309   const runsOption = useMemo(() => buildKernelRunsOption(result, trendMode), [result, trendMode]);
310   const runsRef = useResultChart(runsOption);
311   const hasRunWeights = result.kernelWeights.runs.length > 0 && result.kernelWeights.items.length > 0;
312   const parameters: Array<[string, unknown]> = [
313     ['sigma', result.parameters.sigma],
314     ['lambda', result.parameters.lambda],
315     ['gamma', result.parameters.gamma],
316     ['anchor', result.parameters.anchor],
317     ['nBase', result.parameters.nBase],
318     ['runs', result.parameters.runs],
319   ];
320
321   return (
322     <section className="soft-page result-detail-page result-kernel-page">
323       <ReadyHeader title="核函数配置" icon={SlidersHorizontal} resource={resource} />
324       <section className="panel result-kernel-workbench">
325         <WorkbenchSectionHeader title="核函数工作区" meta="选择核函数，核对公式、权重与本次任务参数" />
```

```
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```

```
326     <div className="result-kernel-workspace-body">
327       <nav className="result-kernel-catalog" role="tablist" aria-label="选择核函数">
328         <span>可用核函数</span>
329         {kernelDefinitions.map((item) => (
330           <button
331             key={item.key}
332             type="button"
333             role="tab"
334             className={active === item.key ? 'active' : ''}
335             aria-selected={active === item.key}
336             aria-controls="kernel-definition-panel"
337             onClick={() => setActive(item.key)}
338           >
339             <span><strong>{item.name}</strong><small>{item.short}</small></span>
340             <em>{formatNumber(result.kernelWeights.items.find((current) => current.key ===
item.key)?.representative, 5)}</em>
341           </button>
342         ))}
343       </nav>
344       <article className="result-kernel-definition" id="kernel-definition-panel" role="tabpanel">
345         <header>
346           <div><span>当前定义</span><strong>{definition.name}</strong></div>
347           <WorkbenchStatus tone="info">{definition.short}</WorkbenchStatus>
348         </header>
349         <MathFormula className="result-kernel-formula" displayMode latex={definition.latex}
label={definition.label} />
350         <p>{definition.label}。公式与后端 Python 核矩阵实现保持一致。</p>
351       </article>
352       <aside className="result-kernel-selected-weight" aria-label="当前核权重">
353         <header><span>当前核权重</span><strong>{formatNumber(weight?.representative, 5)}</strong></header>
354         <dl>
355           <div><dt>重复实验均值</dt><dd>{formatNumber(weight?.mean, 5)}</dd></div>
356           <div><dt>标准差</dt><dd>{formatNumber(weight?.std, 5)}</dd></div>
357           <div><dt>代表轮次</dt><dd>{formatNumber(weight?.representative, 5)}</dd></div>
358         </dl>
359         <small>该数值为代表轮次的原始 alpha，不在前端重新归一化。</small>
360       </aside>
361     </div>
362     <div className="result-parameter-strip" aria-label="算法参数">
363       {parameters.map([[label, value]] => (
364         <div key={String(label)}><span>{label}</span><strong>{String(value ?? '—')}</strong></div>
365       ))}
366     </div>
367   </section>
368
369   <section className="panel result-kernel-stability-panel">
370     <WorkbenchSectionHeader
371       title="跨轮核权重稳定性"
372       meta={trendMode === 'relative'
373         ? ` ${result.kernelWeights.runs.length} 轮数据，以各核自身均值为基线放大波动`
```

```
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```

```
374       : `${result.kernelWeights.runs.length} 轮 alpha, 保留原始权重尺度`
375       actions={
376         <div className="result-kernel-stability-actions">
377           <div className="result-kernel-trend-toggle" role="group" aria-label="切换核权重趋势视图">
378             <button type="button" className={trendMode === 'relative' ? 'active' : ''} aria-pressed={trendMode ===
'relative'} disabled={!hasRunWeights} onClick={() => setTrendMode('relative')}>相对变化</button>
379             <button type="button" className={trendMode === 'absolute' ? 'active' : ''} aria-pressed={trendMode ===
'absolute'} disabled={!hasRunWeights} onClick={() => setTrendMode('absolute')}>绝对权重</button>
380           </div>
381           <WorkbenchStatus tone={hasRunWeights ? 'success' : 'warning'}><{hasRunWeights ? '数据完整' :
'暂无趋势'}</WorkbenchStatus>
382         </div>
383       )}
384     />
385     {hasRunWeights ? <div className="result-chart result-kernel-stability-chart" ref={runsRef} /> :
<WorkbenchNotice tone="info" icon={Activity} title="暂无跨轮权重" detail="当前任务没有可绘制的逐轮核权重。" />}
386     <div className="result-kernel-comparison" role="table" aria-label="核贡献对照表">
387       <div className="result-kernel-comparison-row header" role="row">
388         <span>核函数</span><span>定义</span><span>均值</span><span>标准差</span><span>代表轮次</span>
389       </div>
390       {kernelDefinitions.map((item) => {
391         const current = result.kernelWeights.items.find((weightItem) => weightItem.key === item.key);
392         return (
393           <div className="result-kernel-comparison-row" role="row" key={item.key}>
394             <strong>{item.name}</strong><span>{item.short}</span><span>{formatNumber(current?.mean,
5)}</span><span>{formatNumber(current?.std, 5)}</span><span>{formatNumber(current?.representative, 5)}</span>
395           </div>
396         );
397       })}
398     </div>
399   </section>
400 </section>
401 );
402 }
403
404 function objectiveReduction(initial?: number, final?: number) {
405   if (initial == null || final == null || !Number.isFinite(initial) || !Number.isFinite(final) || initial === 0)
     return null;
406   return (initial - final) / Math.abs(initial);
407 }
408
409 function formatObjective(value?: number) {
410   if (value == null || !Number.isFinite(value)) return '—';
411   const absolute = Math.abs(value);
412   if (absolute >= 1_000_000 || (absolute > 0 && absolute < 0.001)) return value.toExponential(3);
413   return formatNumber(value, 5);
414 }
415
416 function formatDuration(seconds?: number | null) {
417   if (seconds == null || !Number.isFinite(seconds)) return '—';
```

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```
418   if (seconds < 60) return `${formatNumber(seconds, 2)} s`;
419   const minutes = Math.floor(seconds / 60);
420   return `${minutes}m ${formatNumber(seconds - minutes * 60, 1)}s`;
421 }
422
423 function relativeObjectiveText(value: number | undefined, initial: number | undefined) {
424   if (value == null || initial == null || !Number.isFinite(value) || !Number.isFinite(initial) || initial === 0)
425     return '—';
426   return `${formatNumber((value / initial) * 100, 2)}%`;
427 }
428
429 function variationText(mean: number, std: number) {
430   if (!Number.isFinite(mean) || !Number.isFinite(std) || mean === 0) return '—';
431   return formatPercent(std / Math.abs(mean));
432 }
433
434 export function MklResultPage({ resource }: { resource: TaskResultResource }) {
435   const result = resource.envelope!.result!;
436   const [run, setRun] = useState(result.convergence.representativeRun);
437   const [chartMode, setChartMode] = useState<ConvergenceChartMode>('objective');
438   useEffect(() => setRun(result.convergence.representativeRun), [result.convergence.representativeRun]);
439   const current = result.convergence.runs.find((item) => item.run === run) ?? result.convergence.runs[0];
440   const points = current?.points ?? [];
441   const firstPoint = points[0];
442   const finalPoint = points[points.length - 1];
443   const reduction = objectiveReduction(firstPoint?.objective, finalPoint?.objective);
444   const canUseLogMode = points.some((point) => point.objective > 0);
445   useEffect(() => {
446     if (chartMode === 'log' && !canUseLogMode) setChartMode('objective');
447   }, [canUseLogMode, chartMode]);
448   const option = useMemo(
449     () => buildConvergenceOption(result, current?.run ?? run, chartMode),
450     [chartMode, current?.run, result, run],
451   );
452   const chartRef = useResultChart(option);
453   const weights = result.kernelWeights.items;
454   const sortedWeights = [...weights].sort((left, right) => Math.abs(right.representative) -
455     Math.abs(left.representative));
456   const largestWeight = Math.max(0, ...weights.map((item) => Math.abs(item.representative)));
457   const dominantWeight = sortedWeights[0];
458   const dominantDefinition = kernelDefinitions.find((item) => item.key === dominantWeight?.key);
459   const representativeWeightSum = weights.reduce((sum, item) => sum + item.representative, 0);
460   const statusText = !current ? '暂无可求解轨迹' : current.converged ? '已达到收敛阈值' : '已完成预设迭代';
461   const chartModes: Array<{ key: ConvergenceChartMode; label: string; disabled?: boolean }> = [
462     { key: 'objective', label: '原始目标' },
463     { key: 'relative', label: '相对目标' },
464     { key: 'log', label: '对数目标', disabled: !canUseLogMode },
465   ];
466   return (
```

```
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```

```
466 <section className="soft-page result-detail-page result-mkl-page">
467   <ReadyHeader title="多核相似性学习" icon={Network} resource={resource} />
468   <section className="result-mkl-summary" aria-label="求解结论">
469     <div className="result-mkl-summary-lead">
470       <span className="result-mkl-summary-icon" aria-hidden="true"><Gauge size={21} /></span>
471       <div>
472         <span>求解结论</span>
473         <strong>{statusText}</strong>
474         <small>{current ? `第 ${current.run} 轮 • ${points.length} 个迭代点` : '当前结果没有可展示的轮次'}</small>
475       </div>
476       <WorkbenchStatus tone={!current ? 'warning' : current.converged ? 'success' : 'info'}>
477         {!current ? '无数据' : current.converged ? '提前收敛' : '达到迭代上限'}
478       </WorkbenchStatus>
479     </div>
480     <dl className="result-mkl-summary-metrics">
481       <div><dt><History size={13} aria-hidden="true" />当前轮次</dt><dd>第 {current?.run ?? '—'} 轮</dd></div>
482       <div><dt><ListChecks size={13} aria-hidden="true" />迭代点数</dt><dd>{points.length || '—'}</dd></div>
483       <div><dt><TrendingDown size={13} aria-hidden="true" />目标下降</dt><dd>{formatPercent(reduction)}</dd></div>
484       <div><dt><Target size={13} aria-hidden="true" />最终目标</dt><dd>{formatObjective(finalPoint?.objective)}</dd></div>
485     </div>
486     <div><dt><Clock3 size={13} aria-hidden="true" />运行耗时</dt><dd>{formatDuration(result.runtimeSeconds)}</dd></div>
487     <div><dt><Network size={13} aria-hidden="true" />主导核</dt><dd>{dominantDefinition?.name ?? '—'}</dd><small>{dominantWeight ? `alpha ${formatNumber(dominantWeight.representative, 5)}` : '暂无权重'}</small></div>
488   </dl>
489 </section>
490 <section className="panel result-mkl-solver-panel">
491   <WorkbenchSectionHeader
492     title="求解轨迹"
493     meta="选择轮次与显示尺度，查看目标函数下降和终止条件"
494     actions={
495       <div className="result-mkl-toolbar">
496         <div className="result-mkl-view-toggle" role="group" aria-label="收敛图表视图">
497           {chartModes.map((item) => (
498             <button
499               type="button"
500               key={item.key}
501               className={chartMode === item.key ? 'active' : undefined}
502               aria-pressed={chartMode === item.key}
503               disabled={item.disabled}
504               onClick={() => setChartMode(item.key)}
505             >
506               {item.label}
507             </button>
508           ))}
509         </div>
510         {result.convergence.runs.length ? (
511           <div className="result-mkl-run-control">
```

```
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```

```
512         <span>求解轮次</span>
513         <SelectField
514             value={String(current?.run ?? '')}
515             options={result.convergence.runs.map((item) => ({ value: String(item.run), label: `第 ${item.run}
轮` })))}
516             onChange={(value) => setRun(Number(value))}
517             ariaLabel="选择求解轮次"
518             size="sm"
519             className="result-select-field result-mkl-run-select-field"
520         />
521     </div>
522     ) : null}
523 </div>
524 }
525 />
526 <div className="result-mkl-solver-workspace">
527     <div className="result-mkl-chart-area">
528         {points.length ? <div className="result-chart result-mkl-convergence-chart" ref={chartRef} /> :
<WorkbenchNotice tone="info" icon={Activity} title="暂无收敛轨迹" detail="当前任务没有返回可绘制的目标函数序列。" />}
529     </div>
530     <aside className="result-mkl-diagnostics" aria-label="本轮求解诊断">
531         <header>
532             <div><span>当前诊断</span><strong>{current ? `第 ${current.run} 轮` : '暂无轮次'}</strong></div>
533             <WorkbenchStatus tone={!current ? 'warning' : current.converged ? 'success' : 'info'}>{!current ? '无数据'
: current.converged ? '提前收敛' : '达到迭代上限'}</WorkbenchStatus>
534         </header>
535         <dl>
536             <div><dt>迭代点数</dt><dd>{points.length}</dd></div>
537             <div><dt>初始目标函数</dt><dd>{formatObjective(firstPoint?. objective)}</dd></div>
538             <div><dt>最终目标函数</dt><dd>{formatObjective(finalPoint?. objective)}</dd></div>
539             <div><dt>目标下降幅度</dt><dd>{formatPercent(reduction)}</dd></div>
540             <div><dt>最终相对变化</dt><dd>{formatPercent(finalPoint?. relativeChange)}</dd></div>
541             <div><dt>代表轮次</dt><dd>第 {result.convergence.representativeRun} 轮</dd></div>
542         </dl>
543         <div className="result-mkl-diagnostic-note">
544             <CheckCircle2 size={14} aria-hidden="true" />
545             <span>{current?.converged ? '终止依据：达到求解器收敛阈值。' : '终止依据：达到任务设置的最大迭代次数。'}</span>
546         </div>
547         <p>诊断值直接取自求解器轨迹，不使用性能指标替代优化过程。</p>
548     </aside>
549 </div>
550 </section>
551
552 <div className="result-mkl-detail-grid">
553     <section className="panel result-mkl-iteration-panel">
554         <WorkbenchSectionHeader title="迭代明细" meta={current ? `第 ${current.run} 轮的完整目标函数序列` : '当前没有可展示的迭代记录'}
actions={(<WorkbenchStatus tone={points.length ? 'info' : 'warning'}>{points.length ? `${points.length} 个节点` :
'暂无数据'}</WorkbenchStatus>)} />
555         {points.length ? (
556             <div className="result-mkl-iteration-table" role="table" aria-label="迭代目标函数明细">
```

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```
557         <div className="result-mkl-iteration-row header" role="row"><span role="columnheader">迭代</span><span
558         role="columnheader">目标函数</span><span role="columnheader">相对初始</span><span role="columnheader">本轮变化</span><span
559         role="columnheader">状态</span></div>
560         {points.map((point, index) => (
561         <div className={`result-mkl-iteration-row${index === points.length - 1 ? ' is-terminal' : ''}`}
562         role="row" key={point.iteration}>
563         <strong role="cell">第 {point.iteration} 次</strong>
564         <span role="cell">{formatObjective(point.objective)}</span>
565         <span role="cell">{relativeObjectiveText(point.objective, firstPoint?.objective)}</span>
566         <span role="cell">{formatPercent(point.relativeChange)}</span>
567         <span role="cell">{index === points.length - 1 ? (current?.converged ? '收敛点' : '终止点') :
568         '迭代中'}</span>
569         </div>
570         )))
571         </div>
572         ) : <WorkbenchNotice tone="info" icon={Activity} title="暂无迭代明细" detail="选择包含收敛轨迹的任务后可查看逐次变化。" />
573     </section>
574
575     <section className="panel result-mkl-mixture-panel">
576       <WorkbenchSectionHeader title="代表轮次核组合" meta="按代表轮次原始 alpha 排序, 保留重复实验稳定性" actions={<WorkbenchStatus
577       tone={weights.length ? 'success' : 'warning'}>{weights.length} 个核</WorkbenchStatus>} />
578       {weights.length ? (
579         <div className="result-mkl-mixture-list">
580           {sortedWeights.map((item, index) => {
581             const definition = kernelDefinitions.find((currentItem) => currentItem.key === item.key);
582             const width = largestWeight > 0 ? Math.max(1, Math.abs(item.representative) / largestWeight * 100) :
583             0;
584             return (
585               <div className={`result-mkl-mixture-row tone-${index + 1}${index === 0 ? ' is-dominant' : ''}`}
586               key={item.key}>
587                 <span className="result-mkl-mixture-rank">{String(index + 1).padStart(2, '0')}</span>
588                 <div className="result-mkl-mixture-body">
589                   <header>
590                     <div><strong>{definition?.name ?? item.key}</strong><small>{definition?.short ??
591                     item.key}</small></div>
592                     <em>{formatNumber(item.representative, 5)}</em>
593                   </header>
594                   <i aria-hidden="true"><b style={{ width: `${width}%` }} /></i>
595                   <small className="result-mkl-mixture-meta"><span>均值 {formatNumber(item.mean, 5)}</span><span>std
596                   {formatNumber(item.std, 5)}</span><span>波动率 {variationText(item.mean, item.std)}</span></small>
597                 </div>
598               </div>
599             );
600           })}
601         </div>
602         ) : <WorkbenchNotice tone="info" icon={Network} title="暂无核组合" detail="当前结果没有记录代表轮次的核权重。" />
603         <p className="result-mkl-mixture-note">条形长度按代表 alpha 的绝对值缩放; 总和 {formatNumber(representativeWeightSum,
604         5)}, 不在前端重新归一化。</p>
605       </section>
606     </div>
```

```
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```

```
597
598     <section className="panel result-mkl-run-panel" aria-label="多轮运行对比">
599         <WorkbenchSectionHeader title="多轮运行对比" meta="比较每一轮的终止状态与目标函数结果，代表轮次已高亮" actions={<WorkbenchStatus
tone="info">{result.convergence.runs.length} 轮记录</WorkbenchStatus>} />
600         {result.convergence.runs.length ? (
601             <div className="result-mkl-run-table" role="table" aria-label="多轮运行对比表">
602                 <div className="result-mkl-run-row header" role="row"><span role="columnheader">轮次</span><span
role="columnheader">状态</span><span role="columnheader">迭代点</span><span role="columnheader">初始目标</span><span
role="columnheader">最终目标</span><span role="columnheader">目标下降</span><span role="columnheader">标记</span></div>
603                 {result.convergence.runs.map((runItem) => {
604                     const runPoints = runItem.points ?? [];
605                     const runInitial = runPoints[0]?.objective;
606                     const runFinal = runPoints[runPoints.length - 1]?.objective;
607                     const isRepresentative = runItem.run === result.convergence.representativeRun;
608                     return (
609                         <div className={`result-mkl-run-row${isRepresentative ? ' is-representative' : ''}`} role="row"
key={runItem.run}>
610                             <strong role="cell">第 {runItem.run} 轮</strong>
611                             <span role="cell"><WorkbenchStatus tone={runItem.converged ? 'success' : 'info'}>{runItem.converged
? '提前收敛' : '达到上限'}</WorkbenchStatus></span>
612                             <span role="cell">{runPoints.length}</span>
613                             <span role="cell">{formatObjective(runInitial)}</span>
614                             <span role="cell">{formatObjective(runFinal)}</span>
615                             <span role="cell">{formatPercent(objectiveReduction(runInitial, runFinal))}</span>
616                             <span role="cell">{isRepresentative ? '当前代表' : '—'}</span>
617                         </div>
618                     );
619                 })}
620             </div>
621         ) : <WorkbenchNotice tone="info" icon={History} title="暂无多轮记录" detail="任务结果没有返回可用于比较的运行轮次。" />
622     </section>
623 </section>
624 );
625 }
626
627 function selectedViewSummary(result: AnalysisResult, selected: string) {
628     const matrix = result.preview.matrices.ca;
629     const representative = result.convergence.runs.find((item) => item.run === result.convergence.representativeRun);
630     const finalPoint = representative?.points[representative.points.length - 1];
631     const weightSum = result.kernelWeights.items.reduce((sum, item) => sum + item.mean, 0);
632     const clusterCount = new Set(result.preview.scatter.points.map((item) => item.predictedLabel)).size;
633
634     if (selected === 'weights') return [['核数量', result.kernelWeights.items.length], ['权重总和', formatNumber(weightSum,
5)], ['重复轮次', result.kernelWeights.runs.length], ['代表轮次', result.convergence.representativeRun]];
635     if (selected === 'scatter') return [['预览点', result.preview.scatter.points.length], ['完整样本',
result.preview.scatter.totalCount], ['预测簇', clusterCount], ['是否抽样', result.preview.scatter.sampled ? '是' : '否']];
636     if (selected === 'convergence') return [['代表轮次', result.convergence.representativeRun], ['迭代点',
representative?.points.length ?? 0], ['最终目标', formatNumber(finalPoint?.objective, 5)], ['求解状态',
representative?.converged ? '提前收敛' : '达到上限']];
637     return [['完整尺寸', `${matrix.shape[0]} × ${matrix.shape[1]}`, ['预览尺寸', `${matrix.rowIndices.length} ×
```


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```
    ${matrix.columnIndices.length}`], ['矩阵均值', formatNumber(matrix.stats.mean, 5)], ['对称误差',
    formatNumber(matrix.stats.symmetryMaxError, 8)]];
638 }
639
640 export function VisualizationResultPage({ resource }: { resource: TaskResultResource }) {
641   const result = resource.envelope!.result!;
642   const options = useMemo(() => [
643     { key: 'ca', title: 'CA 热力图', description: '观察样本协关联结构', option: buildMatrixOption(result.preview.matrices.ca,
        'CA'), icon: LayoutGrid, section: 'ca-matrix' },
644     { key: 'weights', title: '核权重', description: '比较多核贡献差异', option: buildWeightsOption(result), icon: BarChart3,
        section: 'kernel-config' },
645     { key: 'scatter', title: '聚类散点', description: '检查代表轮次标签分布', option: buildScatterOption(result), icon: Target,
        section: 'results' },
646     { key: 'convergence', title: '收敛曲线', description: '跟踪目标函数变化', option: buildConvergenceOption(result,
        result.convergence.representativeRun), icon: TrendingDown, section: 'mkl' },
647   ], [result]);
648   const [selected, setSelected] = useState(options[0].key);
649   const current = options.find((item) => item.key === selected) ?? options[0];
650   const [activeMatrixValue, setActiveMatrixValue] = useState<number | null>(null);
651   const ref = useResultChart(current.option, setActiveMatrixValue);
652   const summary = selectedViewSummary(result, selected);
653   const task = resource.envelope!.task!;
654   const navigate = useNavigate();
655
656   const exportSvg = () => {
657     const svg = ref.current?.querySelector('svg')?.outerHTML;
658     if (svg) downloadTextFile(`${current.key}-task-${task.id}.svg`, svg, 'image/svg+xml;charset=utf-8');
659   };
660
661   return (
662     <section className="soft-page result-detail-page result-visualization-page">
663       <ReadyHeader title="可视化展示" icon={Images} resource={resource} />
664       <MetricStrip result={result} />
665       <div className="result-visual-workspace">
666         <section className="panel result-chart-panel result-visual-canvas">
667           <WorkbenchSectionHeader
668             title={current.title}
669             meta={` ${current.description} • 浏览器端 SVG 渲染` }
670             actions={<button type="button" className="btn btn-secondary" onClick={exportSvg}><Download size={15} />导出
        SVG</button>}
671           />
672           <div className="result-chart large" ref={ref} />
673           {current.key === 'ca' ? (
674             <MatrixScaleLegend
675               matrix={result.preview.matrices.ca}
676               activeValue={activeMatrixValue}
677             />
678           ) : null}
679         </section>
680         <aside className="panel result-visual-toolbox" aria-label="结果视图工具箱">
```

```
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```

```
681         <WorkbenchSectionHeader title="结果视图" meta="切换当前任务的结果图" />
682         <div className="result-visualization-list">
683             {options.map((item) => {
684                 const Icon = item.icon;
685                 return <button key={item.key} type="button" className={selected === item.key ? 'active' : ''}
onClick={() => setSelected(item.key)}><Icon size={16} /><span><strong>{item.title}</strong><small>{item.description}</small></span></button>;
686             })}
687         </div>
688         <dl className="result-view-summary">
689             {summary.map(([label, value]) => <div key={String(label)}><dt>{label}</dt><dd>{String(value)}</dd></div>)}
690         </dl>
691         <button type="button" className="btn btn-primary result-view-detail-button" onClick={() =>
navigateToTaskResult(navigate, current.section, task.id)}>
692             打开专业分析页 <ArrowRight size={15} />
693         </button>
694     </aside>
695 </div>
696 <section className="panel result-view-catalog" aria-label="可视化目录">
697     <WorkbenchSectionHeader title="可视化目录" meta="四类结果视图对应不同分析环节" />
698     <div>
699         {options.map((item) => {
700             const Icon = item.icon;
701             return <button type="button" key={item.key} onClick={() => setSelected(item.key)}><Icon size={17}
/><span><strong>{item.title}</strong><small>{item.description}</small></span><ArrowRight size={14} /></button>;
702         })}
703     </div>
704 </section>
705 </section>
706 );
707 }
708
709 function distributionRows(result: AnalysisResult, field: 'predictedLabel' | 'trueLabel') {
710     const counts = new Map<string, number>();
711     result.preview.scatter.points.forEach((point) => counts.set(point[field], (counts.get(point[field]) ?? 0) + 1));
712     const total = Math.max(1, result.preview.scatter.points.length);
713     return [...counts.entries()].sort((left, right) => right[1] - left[1]).map(([label, count]) => ({ label, count,
percent: count / total * 100 }));
714 }
715
716 export function ResultsAnalysisPage({ resource }: { resource: TaskResultResource }) {
717     const result = resource.envelope!.result!;
718     const metricTrend = useMemo(() => buildMetricRunsOption(result), [result]);
719     const runs = result.metrics.runs;
720     const bestRun = runs.length ? [...runs].sort((left, right) => right.acc - left.acc)[0] : null;
721     const stableMetric = (Object.keys(metricLabels) as MetricKey[]).sort((left, right) =>
result.metrics.aggregate[left].std - result.metrics.aggregate[right].std)[0];
722     const convergedCount = result.convergence.runs.filter((item) => item.converged).length;
723     const averageRuntime = runs.length ? runs.reduce((sum, item) => sum + item.runtimeSeconds, 0) / runs.length : null;
724     const predicted = distributionRows(result, 'predictedLabel');
```

```
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```

```
725   const truth = distributionRows(result, 'trueLabel');
726
727   return (
728     <section className="soft-page result-detail-page result-analysis-page">
729       <ReadyHeader title="结果分析" icon={BarChart3} resource={resource} />
730       <MetricStrip result={result} />
731       <section className="panel result-analysis-summary-panel" aria-label="实验诊断摘要">
732         <div className="result-analysis-summary-heading">
733           <WorkbenchSectionHeader title="实验诊断摘要" meta="先查看结论，再结合下方图表核对证据" />
734           <WorkbenchStatus tone={convergedCount > 0 ? 'success' : 'warning'}>
735             {convergedCount > 0 ? ` ${convergedCount} 轮提前收敛` : '全部达到迭代上限'}
736           </WorkbenchStatus>
737         </div>
738         <div className="result-analysis-summary-body">
739           <dl className="result-analysis-facts">
740             <div className="is-primary"><dt>最高 ACC 轮次</dt><dd>{bestRun ? `第 ${bestRun.run} 轮` :
741               `${formatPercent(bestRun.acc)}` : '—'}</dd></div>
742             <div className="is-primary"><dt>最稳定指标</dt><dd>{metricLabels[stableMetric]} • std
743               {formatPercent(result.metrics.aggregate[stableMetric].std)}</dd></div>
744             <div><dt>提前收敛轮次</dt><dd>{convergedCount} / {result.convergence.runs.length}</dd></div>
745             <div><dt>平均单轮耗时</dt><dd>{formatNumber(averageRuntime, 2)} s</dd></div>
746             <div><dt>代表轮次</dt><dd>第 {result.preview.summary.representativeRun} 轮</dd></div>
747             <div><dt>随机种子起点</dt><dd>{result.preview.summary.randomSeed}</dd></div>
748           </dl>
749           <WorkbenchNotice tone="info" icon={Activity} title="代表轮次不是最优轮次" detail="后端选择最接近各指标均值的轮次，用于降低单次极值偏差。" />
750         </div>
751       </section>
752       <div className="result-analysis-evidence" role="region" aria-label="实验图表证据">
753         <div className="result-analysis-scatter">
754           <ScatterPanel result={result} />
755         </div>
756         <div className="result-analysis-convergence">
757           <ConvergencePanel result={result} />
758         </div>
759         <div className="result-analysis-trend">
760           <OptionChartPanel
761             title="重复实验指标趋势"
762             meta={` ${runs.length} 轮运行，统一按百分比展示` }
763             option={metricTrend}
764             hasData={runs.length > 0}
765             emptyTitle="暂无指标趋势"
766             emptyDetail="当前任务没有逐轮指标数据。"
767           </div>
768         </div>
769       </div>
770       <section className="panel result-distribution-panel" aria-label="标签分布对照">
771         <WorkbenchSectionHeader title="标签分布对照" meta={result.preview.scatter.sampled ? '基于当前抽样预览点统计' : '基于全部预览点统计'} />
772         <div className="result-distribution-columns">
773           {[['预测簇', predicted], ['真实标签', truth]].map(([title, rows]) => (
774             <div className="result-distribution-column" key={title as string}>
```

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```
773         <strong>{title as string}</strong>
774         {(rows as ReturnType<typeof distributionRows>).map((row) => (
775             <div className="result-distribution-row" key={row.label}>
776                 <span>类别 {row.label}</span><i><b style={{ width: `${row.percent}%` }}
777                 /></i><strong>{row.count}</strong><small>{row.percent.toFixed(1)}%</small>
778             </div>
779         ))}
780     </div>
781 </section>
782 </section>
783 </section>
784 );
785 }
786
787 export function ExportResultPage({ resource }: { resource: TaskResultResource }) {
788     const task = resource.envelope!.task!;
789     const result = resource.envelope!.result!;
790     const [selected, setSelected] = useState<string[]>(['metrics', 'parameters', 'labels']);
791     const [exports, setExports] = useState<TaskExport[]>([]);
792     const [archiveName, setArchiveName] = useState(`${task.datasetName} • ${task.mode} 交付档案`);
793     const [submitting, setSubmitting] = useState(false);
794     const [message, setMessage] = useState<string | null>(null);
795     const totalSize = result.artifacts.reduce((sum, item) => sum + item.size, 0);
796     const reload = () => void fetchTaskExports(task.id).then((response) => setExports(response.items)).catch(() =>
797         undefined);
798     useEffect(reload, [task.id]);
799
800     const create = async () => {
801         setSubmitting(true);
802         setMessage(null);
803         try {
804             const item = await createTaskExport(task.id, selected, archiveName);
805             setMessage(`已生成 ${item.filename}`);
806             reload();
807         } catch (error) {
808             setMessage(error instanceof Error ? error.message : '导出失败');
809         } finally {
810             setSubmitting(false);
811         }
812     };
813     const toggle = (value: string) => setSelected((current) => current.includes(value) ? current.filter((item) => item
814         !== value) : [...current, value]);
815
816     return (
817         <section className="soft-page result-detail-page result-export-page">
818             <ReadyHeader title="结果导出" icon={Download} resource={resource} />
819             <WorkbenchMetricStrip
820                 label="交付摘要"
```

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```
820     metrics=[  
821       { label: '完整产物', value: `${result.artifacts.length} 项`, note: '后端持久化文件', icon: Archive, tone: 'blue' },  
822       { label: '产物体积', value: formatBytes(totalSize), note: '不含新生成 ZIP', icon: HardDrive, tone: 'neutral' },  
823       { label: '当前选择', value: `${selected.length} 项`, note: '组合导出内容', icon: ListChecks, tone: 'teal' },  
824       { label: '导出记录', value: `${exports.length} 条`, note: '当前任务历史', icon: History, tone: 'green' },  
825     ]  
826   />  
827   <section className="panel result-export-builder result-export-composer" aria-label="创建交付档案">  
828     <WorkbenchSectionHeader title="创建交付档案" meta="为本次交付命名、选择预设，并确认需要归档的数据" actions={<WorkbenchStatus  
tone="info">自动附带 manifest.json</WorkbenchStatus>} />  
829     <div className="result-export-compose-grid">  
830       <div className="result-export-setup">  
831         <label className="result-export-name-field">  
832           <span>档案名称</span>  
833           <input value={archiveName} maxLength={128} onChange={(event) => setArchiveName(event.target.value)}  
placeholder="例如：论文复现实验交付包" />  
834         </label>  
835         <div className="result-export-presets" aria-label="交付预设">  
836           {exportPresets.map((preset) => (  
837             <button key={preset.key} type="button" onClick={() => setSelected([...preset.items])}>  
838               <span><strong>{preset.label}</strong><small>{preset.detail}</small></span>  
839               <span>{preset.items.length} 项</span>  
840             </button>  
841           ))}  
842         </div>  
843         <dl className="result-export-manifest">  
844           <div><dt>任务</dt><dd>#{task.id} • {task.name}</dd></div>  
845           <div><dt>数据集</dt><dd>{task.datasetName}</dd></div>  
846           <div><dt>算法</dt><dd>{task.mode}</dd></div>  
847           <div><dt>代表轮次</dt><dd>第 {result.preview.summary.representativeRun} 轮</dd></div>  
848         </dl>  
849       </div>  
850       <div className="result-export-selection">  
851         <header><strong>归档内容</strong><span>已选择 {selected.length} / {exportOptions.length}</span></header>  
852         <div className="result-export-option-list">  
853           {exportOptions.map((item) => (  
854             <const Icon = item.icon;  
855             <const checked = selected.includes(item.key);  
856             <return (  
857               <label className={checked ? 'selected' : ''} key={item.key}>  
858                 <input type="checkbox" checked={checked} onChange={() => toggle(item.key)} />  
859                 <Icon size={16} aria-hidden="true" />  
860                 <span><strong>{item.label}</strong><small>{item.detail}</small></span>  
861               </label>  
862             );  
863           )}}  
864         </div>  
865         <button type="button" className="btn btn-primary result-export-submit" disabled={!selected.length ||  
!archiveName.trim() || submitting} onClick={() => void create()}><PackageCheck size={16} />{submitting ? '正在生成档案...'  
: '创建交付档案 (${selected.length} 项)'}</button>
```

```
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```

```
866     </div>
867   </div>
868   {message ? <WorkbenchNotice tone="info" icon={CheckCircle2} title="导出状态" detail={message} /> : null}
869 </section>
870 <div className="result-export-library">
871   <section className="panel result-export-history-panel">
872     <WorkbenchSectionHeader title="交付档案" meta={`任务 #${task.id} 的可追溯 ZIP 档案`} actions={<button type="button"
873       className="text-action" onClick={reload}>刷新记录</button>} />
874     {exports.length ? (
875       <div className="result-export-archive-list" aria-label="交付档案记录">
876         {exports.map((item) => (
877           <article className="result-export-archive" key={item.id}>
878             <span className="result-export-archive-icon"><PackageOpen size={18} /></span>
879             <div className="result-export-archive-main">
880               <strong>{item.name}</strong>
881               <small>{item.filename}</small>
882               <div>{exportItemLabels(item.items).map((label) => <span key={label}>{label}</span>)}</div>
883             </div>
884             <div className="result-export-archive-meta">
885               <span>{item.itemCount} || '旧版' 项</span><strong>{formatBytes(item.fileSize)}</strong><time>{formatL
886                 ocalDateTime(item.createdAt)}</time>
887             </div>
888             <WorkbenchStatus tone="success">{item.status === 'ready' ? '可下载' : item.status}</WorkbenchStatus>
889             <button type="button" className="btn btn-secondary" onClick={() => void
890               downloadProtectedFile(item.downloadUrl, item.filename)}><Download size={14} />下载</button>
891           </article>
892         )))
893       </div>
894     ) : (
895       <div className="result-export-empty">
896         <PackageOpen size={24} aria-hidden="true" />
897         <strong>还没有交付档案</strong>
898         <span>使用上方预设创建第一份带清单的 ZIP，之后可在这里重复下载。</span>
899       </div>
900     )
901   </section>
902   <section className="panel result-artifact-panel">
903     <WorkbenchSectionHeader title="原始产物" meta="由任务直接生成，可单独下载" actions={<WorkbenchStatus
904       tone="success">{result.artifacts.length} 项可用</WorkbenchStatus>} />
905     <div className="result-artifact-list">
906       {result.artifacts.map((artifact) => (
907         <article className="result-artifact-item" key={artifact.key}>
908           <span className="result-artifact-icon"><FileArchive size={17} /></span>
909           <span><strong>{artifact.name}</strong><small>{artifact.key} • {artifact.format}</small></span>
910           <span><small>文件大小</small><strong>{formatBytes(artifact.size)}</strong></span>
911           <button type="button" className="btn btn-secondary" title={`下载${artifact.name}`}
912             aria-label={`下载${artifact.name}`} onClick={() => void downloadProtectedFile(artifact.downloadUrl,
913               `${artifact.key}.${artifact.format.toLowerCase()}`)}><Download size={14} /></button>
914         </article>
915       )))
916     </div>
917   </section>
918 </div>
```

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```
910     </div>
911   </section>
912 </div>
913 </section>
914 );
915 }
916
917 function buildReportMarkdown(data: Record<string, unknown>, selected: ReportSection[]) {
918   const lines = [ `# ${String(data.title)}`, '', `任务: #${String(data.taskId)}`, `数据集: ${String(data.dataset)}`,
    `算法: ${String(data.mode)}`, '' ];
919   const record = data as any;
920   if (selected.includes('summary')) lines.push('## 任务摘要', '', ` - 样本: ${record.summary.sampleCount}`, ` -
    类别: ${record.summary.classCount}`, ` - 基础聚类: ${record.summary.baseClusterCount}`, ` -
    运行耗时: ${formatNumber(record.runtimeSeconds, 2)} s`, '');
921   if (selected.includes('metrics')) lines.push('## 性能指标', '', ... (Object.keys(metricLabels) as MetricKey[]).map((key)
    => ` - ${metricLabels[key]}: ${formatPercent(record.metrics[key].mean)} (标准差
    ${formatPercent(record.metrics[key].std)})`, '' );
922   if (selected.includes('parameters')) lines.push('## 算法参数', '', ...Object.entries(record.parameters).map(([key,
    value]) => ` - ${key}: ${String(value)}`, '' );
923   if (selected.includes('runs')) lines.push('## 重复实验', '', ...record.runs.map((run: any) => ` - 第 ${run.run} 轮: ACC
    ${formatPercent(run.acc)}, NMI ${formatPercent(run.nmi)}, ARI ${formatPercent(run.ari)}, F1
    ${formatPercent(run.f1)}`, '' );
924   if (selected.includes('artifacts')) lines.push('## 结果产物', '', ...record.artifacts.map((item: any) => ` -
    ${item.name} (${item.format}, ${formatBytes(item.size)})`, '' );
925   return lines.join('\n');
926 }
927
928 export function ReportResultPage({ resource }: { resource: TaskResultResource }) {
929   const task = resource.envelope!.task!;
930   const result = resource.envelope!.result!;
931   const [title, setTitle] = useState(` ${task.datasetName} 分析报告`);
932   const [format, setFormat] = useState<'markdown' | 'json'>('markdown');
933   const [sections, setSections] = useState<ReportSection[]>(['summary', 'metrics', 'parameters', 'runs',
    'artifacts']);
934   const [expandedSections, setExpandedSections] = useState<ReportSection[]>(['summary', 'metrics']);
935   const parameters = Object.entries(result.parameters);
936   const previewRuns = result.metrics.runs.slice(0, 5);
937   const expandedSectionCount = sections.filter((section) => expandedSections.includes(section)).length;
938   const reportSectionMeta: Record<ReportSection, string> = {
939     summary: '6 项信息',
940     metrics: ` ${Object.keys(metricLabels).length} 项指标`,
941     parameters: ` ${parameters.length} 项参数`,
942     runs: ` ${result.metrics.runs.length} 轮数据`,
943     artifacts: ` ${result.artifacts.length} 项产物`,
944   };
945   const report = useMemo(() => ({
946     title,
947     taskId: task.id,
948     dataset: task.datasetName,
949     mode: task.mode,
```

```
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```

```
950     finishedAt: formatLocalDateTime(task.finishedAt),
951     runtimeSeconds: result.runtimeSeconds,
952     summary: result.preview.summary,
953     metrics: result.metrics.aggregate,
954     parameters: result.parameters,
955     runs: result.metrics.runs,
956     artifacts: result.artifacts,
957   }}, [result, task, title]);
958
959   const toggleSection = (section: ReportSection) => setSections((current) => current.includes(section) ?
    current.filter((item) => item !== section) : [...current, section]);
960   const togglePreviewSection = (section: ReportSection, expanded: boolean) => {
961     setExpandedSections((current) => expanded
962       ? (current.includes(section) ? current : [...current, section])
963       : current.filter((item) => item !== section));
964   };
965   const expandAllPreviewSections = () => setExpandedSections(sections);
966   const collapseAllPreviewSections = () => setExpandedSections([]);
967   const exportReport = () => {
968     const selectedData = Object.fromEntries(Object.entries(report).filter(([key]) => ![ 'summary', 'metrics',
969       'parameters', 'runs', 'artifacts'].includes(key) || sections.includes(key as ReportSection)));
970     const content = format === 'json' ? JSON.stringify(selectedData, null, 2) : buildReportMarkdown(report, sections);
971     downloadTextFile(`${title} || 'omelet-report'}.${format === 'json' ? 'json' : 'md'}`, content, format === 'json' ?
972       'application/json' : 'text/markdown;charset=utf-8');
973   };
974
975   return (
976     <section className="soft-page result-detail-page result-report-page">
977       <ReadyHeader title="分析报告" icon={FileText} resource={resource} />
978       <WorkbenchMetricStrip
979         label="报告摘要"
980         metrics={[
981           { label: '已选章节', value: `${sections.length} / ${reportSectionOptions.length}`, note: '控制预览和导出内容', icon:
982             ListChecks, tone: 'blue' },
983           { label: '重复实验', value: `${result.metrics.runs.length} 轮`, note: '运行明细', icon: Activity, tone: 'teal' },
984           { label: '算法参数', value: `${parameters.length} 项`, note: '来自任务持久化结果', icon: SlidersHorizontal, tone:
985             'neutral' },
986           { label: '结果产物', value: `${result.artifacts.length} 项`, note: '纳入交付清单', icon: Archive, tone: 'green' },
987         ]}
988       />
989       <div className="result-report-workspace">
990         <section className="panel result-report-controls">
991           <WorkbenchSectionHeader title="报告设置" meta="调整名称、格式和交付章节" />
992           <label className="task-form-field"><span>报告名称</span><input value={title} onChange={(event) =>
993             setTitle(event.target.value)} /></label>
994           <fieldset className="result-report-format"><legend>文件格式</legend><div className="soft-segmented"><button
995             type="button" className={`soft-segment ${format === 'markdown' ? 'active' : ''}`} onClick={() =>
996               setFormat('markdown')}><FileText size={14} />Markdown</button><button type="button" className={`soft-segment
997               ${format === 'json' ? 'active' : ''}`} onClick={() => setFormat('json')}><FileJson size={14}
998               />JSON</button></div></fieldset>
```



```
front/src/workbench/results/ResultDetailPages.tsx
```

```
990      <fieldset className="result-report-sections"><legend>报告章节</legend>{reportSectionOptions.map((item) => <label
  className={sections.includes(item.key) ? 'selected' : ''} key={item.key}><input type="checkbox"
  checked={sections.includes(item.key)} onChange={() => toggleSection(item.key)}
  /><span><strong>{item.label}</strong><small>{item.detail}</small></span></label></div></fieldset>
991      <button type="button" className="btn btn-primary result-report-submit" disabled={!sections.length}
  onClick={exportReport}><Download size={15} />生成 {format === 'json' ? 'JSON' : 'Markdown'} 报告</button>
992    </section>
993    <section className="panel result-report-preview">
994      <WorkbenchSectionHeader
995        title="报告预览"
996        meta="所有内容均来自当前任务持久化结果"
997        actions={
998          <div className="result-report-preview-actions">
999            <WorkbenchStatus tone="info">{expandedSectionCount} / {sections.length} 展开</WorkbenchStatus>
1000            <button
1001              type="button"
1002              className="result-report-action"
1003              title="全部展开"
1004              aria-label="全部展开报告章节"
1005              disabled={!sections.length}
1006              onClick={expandAllPreviewSections}
1007            >
1008              <ChevronsDown size={15} aria-hidden="true" />
1009            </button>
1010            <button
1011              type="button"
1012              className="result-report-action"
1013              title="全部收起"
1014              aria-label="全部收起报告章节"
1015              disabled={!sections.length}
1016              onClick={collapseAllPreviewSections}
1017            >
1018              <ChevronsUp size={15} aria-hidden="true" />
1019            </button>
1020          </div>
1021        )}
1022      />
1023      <article className="report-document result-report-document">
1024        <header><span>{SOFTWARE.SHORT_NAME} • 分析报告</span><strong>{title || '未命名报告'}</strong><small>任务 #{task.id} •
  {task.finishedAt ? formatLocalDateTime(task.finishedAt) : '未完成'}</small></header>
1025        {sections.includes('summary') ? (
1026          <ReportPreviewSection section="summary" title="任务摘要" meta={reportSectionMeta.summary}
  expanded={expandedSections.includes('summary')} onToggle={togglePreviewSection}>
1027            <div className="report-document-summary"><div><span>数据集</span><strong>{task.datasetName}</strong></div>
  <div><span>算法</span><strong>{task.mode}</strong></div><div><span>样本</span><strong>{result.preview.summary.sampleC
  ount}</strong></div><div><span>类别</span><strong>{result.preview.summary.classCount}</strong></div><div><span>基础聚类<
  /span><strong>{result.preview.summary.baseClusterCount}</strong></div><div><span>耗时</span><strong>{formatNumber(re
  sult.runtimeSeconds, 2)} s</strong></div></div>
1028          </ReportPreviewSection>
1029        ) : null}
```

```
front/src/workbench/results/ResultDetailPages.tsx
```

```
1030         {sections.includes('metrics') ? (  
1031             <ReportPreviewSection section="metrics" title="性能指标" meta={reportSectionMeta.metrics}  
expanded={expandedSections.includes('metrics')} onToggle={togglePreviewSection}>  
1032                 <div className="report-document-metrics">{(Object.keys(metricLabels) as MetricKey[]).map((key) => <div  
key={key}><span>{metricLabels[key]}</span><strong>{formatPercent(result.metrics.aggregate[key].mean)}</strong><small>  
std {formatPercent(result.metrics.aggregate[key].std)} •  
{formatPercent(result.metrics.aggregate[key].min)} - {formatPercent(result.metrics.aggregate[key].max)}</small></div  
>)}</div>  
1033             </ReportPreviewSection>  
1034         ) : null}  
1035         {sections.includes('parameters') ? (  
1036             <ReportPreviewSection section="parameters" title="算法参数" meta={reportSectionMeta.parameters}  
expanded={expandedSections.includes('parameters')} onToggle={togglePreviewSection}>  
1037                 <div className="report-parameter-list">{parameters.map([(key, value)] => <div  
key={key}><span>{key}</span><strong>{String(value)}</strong></div>)}</div>  
1038             </ReportPreviewSection>  
1039         ) : null}  
1040         {sections.includes('runs') ? (  
1041             <ReportPreviewSection section="runs" title="重复实验" meta={reportSectionMeta.runs}  
expanded={expandedSections.includes('runs')} onToggle={togglePreviewSection}>  
1042                 <div className="report-run-table"><div className="header"><span>轮次</span><span>ACC</span><span>NMI</sp  
an><span>ARI</span><span>F1</span></div>{previewRuns.map((run) => <div key={run.run}><strong>第 {run.run}  
轮</strong><span>{formatPercent(run.acc)}</span><span>{formatPercent(run.nmi)}</span><span>{formatPercent(run.ari)}  
</span><span>{formatPercent(run.f1)}</span></div>)}</div>{result.metrics.runs.length > previewRuns.length ? <small  
className="report-preview-note">预览前 {previewRuns.length} 轮, 导出文件包含全部 {result.metrics.runs.length} 轮.</small> :  
null}  
1043             </ReportPreviewSection>  
1044         ) : null}  
1045         {sections.includes('artifacts') ? (  
1046             <ReportPreviewSection section="artifacts" title="结果产物" meta={reportSectionMeta.artifacts}  
expanded={expandedSections.includes('artifacts')} onToggle={togglePreviewSection}>  
1047                 {result.artifacts.length ? <div className="report-artifact-list">{result.artifacts.map((item) => <div  
key={item.key}><FileArchive size={14} /><span>{item.name}</span><strong>{item.format} •  
{formatBytes(item.size)}</strong></div>)}</div> : <p className="report-preview-empty">当前任务没有可交付的结果产物.</p>  
1048             </ReportPreviewSection>  
1049         ) : null}  
1050     </article>  
1051 </section>  
1052 </div>  
1053 </section>  
1054 );  
1055 }  
1056  
  
// ===== 文件: front/src/workbench/results/ResultPageShared.tsx =====  
1 import type { ReactNode } from 'react';  
2 import { useLocation, useNavigate } from 'react-router-dom';  
3 import { ArrowLeft, Boxes, ClipboardCheck, Network, RefreshCw, Target, type LucideIcon } from 'lucide-react';  
4 import {  
5     WorkbenchMetricStrip,
```

front/src/workbench/results/ResultPageShared.tsx

```
6   WorkbenchPageHeader,
7 } from '../WorkbenchUi';
8 import { formatPercent, hasResultBackState, metricLabels, type MetricKey } from './resultPresentation';
9 import type { AnalysisResult, TaskResultResource } from './types';
10
11 export function ReadyHeader({
12   title,
13   icon,
14   resource,
15   actions,
16 }): {
17   title: string;
18   icon: LucideIcon;
19   resource: TaskResultResource;
20   actions?: ReactNode;
21 }) {
22   const showBackButton = useShouldShowResultBack();
23   const task = resource.envelope!.task!;
24   const result = resource.envelope!.result!;
25
26   return (
27     <WorkbenchPageHeader
28       icon={icon}
29       title={title}
30       context={`任务 #${task.id} • ${task.datasetName} • ${task.mode} • 第 ${result.preview.summary.representativeRun}
轮代表结果`}
31       backAction={showBackButton ? <ResultBackButton taskId={task.id} /> : undefined}
32       actions={
33         actions ?? (
34           <button type="button" className="btn btn-secondary" onClick={resource.refresh}>
35             <RefreshCw size={15} aria-hidden="true" />
36             刷新
37           </button>
38         )
39       }
40     />
41   );
42 }
43
44 export function useShouldShowResultBack() {
45   const location = useLocation();
46   return hasResultBackState(location.state);
47 }
48
49 export function ResultBackButton({ taskId }: { taskId?: number | null }) {
50   const navigate = useNavigate();
51
52   function goBack() {
53     if (typeof window !== 'undefined' && window.history.length > 1) {
54       navigate(-1);
55     }
56   }
57 }
```

front/src/workbench/results/ResultPageShared.tsx

```
55     return;
56   }
57   navigate(taskId ? `/workbench/analysis?taskId=${encodeURIComponent(taskId)}` : '/workbench/analysis');
58 }
59
60 return (
61   <button type="button" className="workbench-page-back-button" onClick={goBack}>
62     <ArrowLeft size={14} aria-hidden="true" />
63     返回
64   </button>
65 );
66 }
67
68 export function MetricStrip({ result }: { result: AnalysisResult }) {
69   const icons = [Target, Network, Boxes, ClipboardCheck];
70   const tones = ['green', 'blue', 'teal', 'neutral'] as const;
71
72   return (
73     <WorkbenchMetricStrip
74       label="重复实验指标"
75       metrics={(Object.keys(metricLabels) as MetricKey[]).map((key, index) => ({
76         label: metricLabels[key],
77         value: formatPercent(result.metrics.aggregate[key].mean),
78         note: `标准差 ${formatPercent(result.metrics.aggregate[key].std)}`,
79         icon: icons[index],
80         tone: tones[index],
81       })))}
82     />
83   );
84 }
85
```

// ===== 文件: front/src/workbench/results/TaskLogsPage.tsx =====

```
1 import { useEffect, useMemo, useRef, useState } from 'react';
2 import {
3   Activity,
4   AlertTriangle,
5   ArrowDown,
6   ArrowUp,
7   Clock3,
8   Download,
9   Info,
10  RefreshCw,
11  Search,
12  X,
13  XCircle,
14 } from 'lucide-react';
15 import {
16   WorkbenchMetricStrip,
17   WorkbenchNotice,
```

```
front/src/workbench/results/TaskLogsPage.tsx
```

```
18   WorkbenchPageHeader,
19   WorkbenchSectionHeader,
20   WorkbenchStatus,
21 } from '../WorkbenchUi';
22 import { formatRuntime, getStatusMeta, getTaskRunProgress } from '../tasks/taskStatus';
23 import type { AnalysisTaskLog } from '../tasks/types';
24 import {
25   formatLocalClock,
26   formatLocalDate,
27   formatLocalDateTime,
28   utcDateTimeAttribute,
29   utcTimestamp,
30 } from '../utils/time';
31 import { downloadTextFile } from './resultPresentation';
32 import { ResultBackButton, useShouldShowResultBack } from './ResultPageShared';
33 import type { TaskResultResource } from './types';
34
35 type LogLevelFilter = 'all' | 'info' | 'warning' | 'error';
36 type LogOrder = 'asc' | 'desc';
37 type StatusTone = 'success' | 'warning' | 'error' | 'info' | 'neutral';
38
39 const ACTION_LABELS: Record<string, string> = {
40   task_cloned: '克隆任务',
41   task_created: '创建任务',
42   task_queued: '进入执行队列',
43   task_started: '启动计算进程',
44   task_run_started: '开始实验轮次',
45   task_run_completed: '完成实验轮次',
46   task_succeeded: '分析任务完成',
47   task_failed: '分析任务失败',
48   task_cancelled: '分析任务取消',
49   result_persisted: '保存分析结果',
50   export_created: '创建导出文件',
51 };
52
53 function normalizeLevel(level: string): Exclude<LogLevelFilter, 'all'> | 'neutral' {
54   const normalized = level.trim().toLowerCase();
55   if (normalized === 'error') return 'error';
56   if (normalized === 'warning' || normalized === 'warn') return 'warning';
57   if (normalized === 'info') return 'info';
58   return 'neutral';
59 }
60
61 function getLevelPresentation(level: string) {
62   const normalized = normalizeLevel(level);
63   if (normalized === 'error') return { key: normalized, label: '异常', tone: 'error' as const };
64   if (normalized === 'warning') return { key: normalized, label: '提醒', tone: 'warning' as const };
65   if (normalized === 'info') return { key: normalized, label: '信息', tone: 'info' as const };
66   return { key: normalized, label: level || '记录', tone: 'neutral' as const };
67 }
```

front/src/workbench/results/TaskLogsPage.tsx

```
68
69 function getTaskTone(status: string): StatusTone {
70   if (status === 'succeeded') return 'success';
71   if (status === 'failed') return 'error';
72   if (status === 'queued' || status === 'running') return 'warning';
73   return 'neutral';
74 }
75
76 function getTimestamp(value: string) {
77   return utcTimestamp(value);
78 }
79
80 function compareLogs(left: AnalysisTaskLog, right: AnalysisTaskLog) {
81   const leftTimestamp = getTimestamp(left.createdAt);
82   const rightTimestamp = getTimestamp(right.createdAt);
83   if (leftTimestamp !== null && rightTimestamp !== null && leftTimestamp !== rightTimestamp) {
84     return leftTimestamp - rightTimestamp;
85   }
86   return left.id - right.id;
87 }
88
89 function formatCoverage(logs: AnalysisTaskLog[]) {
90   if (logs.length < 2) return logs.length ? '单个时间点' : '等待事件写入';
91   const sorted = [...logs].sort(compareLogs);
92   const first = getTimestamp(sorted[0].createdAt);
93   const last = getTimestamp(sorted[sorted.length - 1].createdAt);
94   if (first === null || last === null) return '完整执行链路';
95   return `覆盖 ${formatRuntime(Math.max(0, (last - first) / 1000))}`;
96 }
97
98 function extractRunNumber(log: AnalysisTaskLog) {
99   const detailRun = Number(log.detail?.run ?? log.detail?.currentRun ?? log.detail?.runNumber);
100   if (Number.isInteger(detailRun) && detailRun > 0) return detailRun;
101   const match = log.message.match(/第\s*(\d+)\s*轮/);
102   return match ? Number(match[1]) : null;
103 }
104
105 function escapeCsv(value: unknown) {
106   return `${String(value ?? '').replace(/"/g, '""')}`;
107 }
108
109 export function buildTaskLogsCsv(logs: AnalysisTaskLog[]) {
110   const rows = [
111     ['级别', '事件', '内容', '时间'],
112     ...logs.map((log) => [log.level, log.action, log.message, formatLocalDateTime(log.createdAt)]),
113   ];
114   return `\uFEFF${rows.map((row) => row.map(escapeCsv).join(',')).join('\n')}`;
115 }
116
117 function LogTimelineSkeleton() {
```

front/src/workbench/results/TaskLogsPage.tsx

```
118   return (  
119     <div className="task-log-skeleton" aria-label="日志加载中" aria-busy="true">  
120       {Array.from({ length: 5 }, (_, index) => (  
121         <div key={index}>  
122           <i />  
123           <span />  
124           <span />  
125         </div>  
126       )  
127     </div>  
128   );  
129 }  
130  
131 export function TaskLogsPage({ resource }: { resource: TaskResultResource }) {  
132   const showBackButton = useShouldShowResultBack();  
133   const task = resource.envelope!.task!;  
134   const runProgress = getTaskRunProgress(task);  
135   const [level, setLevel] = useState<LogLevelFilter>('all');  
136   const [query, setQuery] = useState('');  
137   const [order, setOrder] = useState<LogOrder>('asc');  
138   const logListRef = useRef<HTMLDivElement>(null);  
139   const normalizedQuery = query.trim().toLowerCase();  
140  
141   // 筛选和排序会改变列表的首条事件，回到顶部可以避免用户停留在旧列表的中间位置。  
142   useEffect(() => {  
143     if (logListRef.current) logListRef.current.scrollTop = 0;  
144   }, [level, normalizedQuery, order]);  
145  
146   const summary = useMemo(() => resource.logs.reduce(  
147     (counts, log) => {  
148       const normalized = normalizeLevel(log.level);  
149       counts.total += 1;  
150       if (normalized === 'info') counts.info += 1;  
151       if (normalized === 'warning') counts.warning += 1;  
152       if (normalized === 'error') counts.error += 1;  
153       return counts;  
154     },  
155     { total: 0, info: 0, warning: 0, error: 0 },  
156   ), [resource.logs]);  
157  
158   const logs = useMemo(() => {  
159     const filtered = resource.logs.filter((log) => {  
160       const presentation = getLevelPresentation(log.level);  
161       const matchesLevel = level === 'all' || presentation.key === level;  
162       const eventLabel = ACTION_LABELS[log.action] ?? log.action;  
163       const matchesQuery = !normalizedQuery  
164         || `${eventLabel} ${log.action} ${log.message}`.toLowerCase().includes(normalizedQuery);  
165       return matchesLevel && matchesQuery;  
166     });  
167     filtered.sort((left, right) => (order === 'asc' ? compareLogs(left, right) : compareLogs(right, left)));
```

front/src/workbench/results/TaskLogsPage.tsx

```
168     return filtered;
169   }, [level, normalizedQuery, order, resource.logs]);
170
171   const sortedLogs = useMemo(() => [...resource.logs].sort(compareLogs), [resource.logs]);
172   const firstLog = sortedLogs[0];
173   const latestLog = sortedLogs[sortedLogs.length - 1];
174   const statusMeta = getStatusMeta(task.status);
175
176   const resetFilters = () => {
177     setLevel('all');
178     setQuery('');
179   };
180
181   const exportLogs = () => {
182     if (!logs.length) return;
183     downloadTextFile(
184       `task-${task.id}-logs.csv`,
185       buildTaskLogsCsv(logs),
186       'text/csv;charset=utf-8',
187     );
188   };
189
190   const levelOptions: Array<{ value: LogLevelFilter; label: string; count: number }> = [
191     { value: 'all', label: '全部', count: summary.total },
192     { value: 'info', label: '信息', count: summary.info },
193     { value: 'warning', label: '提醒', count: summary.warning },
194     { value: 'error', label: '异常', count: summary.error },
195   ];
196
197   return (
198     <section className="soft-page task-logs-page" aria-label="任务运行日志">
199       <WorkbenchPageHeader
200         icon={Activity}
201         title="运行日志"
202         context={`任务 #${task.id} • ${task.datasetName} • ${task.mode} • ${runProgress.total} 轮实验`}
203         status={{
204           <WorkbenchStatus tone={getTaskTone(task.status)} pulse={task.status === 'running'}>
205             {statusMeta.label}
206           </WorkbenchStatus>
207         }}
208         backAction={showBackButton ? <ResultBackButton taskId={task.id} /> : undefined}
209         actions={{
210           <>
211             <button
212               type="button"
213               className="btn btn-secondary task-log-refresh"
214               disabled={resource.logsLoading}
215               onClick={resource.refreshLogs}
216             >
217               <RefreshCw className={resource.logsLoading ? 'is-spinning' : ''} size={15} aria-hidden="true" />
```



```
front/src/workbench/results/TaskLogsPage.tsx
```

```
218         {resource.logsLoading ? '同步中' : '刷新日志'}
219       </button>
220       <button type="button" className="btn btn-primary" disabled={!logs.length} onClick={exportLogs}>
221         <Download size={15} aria-hidden="true" />
222         导出日志
223       </button>
224     </>
225   ))
226 />
227
228 <WorkbenchMetricStrip
229   label="日志摘要"
230   metrics=[
231     { label: '全部事件', value: String(summary.total), note: formatCoverage(resource.logs), icon: Activity, tone:
232       'blue' },
233     { label: '信息', value: String(summary.info), note: '正常执行记录', icon: Info, tone: 'teal' },
234     { label: '提醒', value: String(summary.warning), note: summary.warning ? '建议复核' : '当前无提醒', icon:
235       AlertTriangle, tone: 'amber' },
236     { label: '异常', value: String(summary.error), note: summary.error ? '需要处理' : '当前无异常', icon: XCircle, tone:
237       'red' },
238   ]
239 />
240
241 <div className="task-log-workspace">
242   <section className="panel task-log-stream-panel" aria-label="执行事件">
243     <WorkbenchSectionHeader
244       title="执行事件"
245       meta={` ${logs.length} / ${summary.total} 条 • ${order === 'asc' ? '按执行顺序' : '最新事件优先'} `}
246     />
247
248     <div className="task-log-toolbar">
249       <div className="task-log-search-row">
250         <label className="task-log-search">
251           <Search size={15} aria-hidden="true" />
252           <input
253             value={query}
254             aria-label="搜索任务日志"
255             placeholder="搜索事件名称、标识或日志内容"
256             onChange={(event) => setQuery(event.target.value)}
257           />
258           {query ? (
259             <button type="button" aria-label="清除日志搜索" onClick={() => setQuery('')}>
260               <X size={14} aria-hidden="true" />
261             </button>
262           ) : null}
263         </label>
264         <span className="task-log-result-count" aria-live="polite">当前显示 {logs.length} 条</span>
265       </div>
266
267       <div className="task-log-filter-row">
```

```
front/src/workbench/results/TaskLogsPage.tsx
```

```
265         <div className="task-log-segmented" role="group" aria-label="筛选日志级别">
266             {levelOptions.map((option) => (
267                 <button
268                     type="button"
269                     className={level === option.value ? 'is-active' : ''}
270                     aria-pressed={level === option.value}
271                     key={option.value}
272                     onClick={() => setLevel(option.value)}
273                 >
274                     <span>{option.label}</span>
275                     <small>{option.count}</small>
276                 </button>
277             ))}
278         </div>
279
280         <div className="task-log-order" role="group" aria-label="日志排序方式">
281             <button
282                 type="button"
283                 className={order === 'asc' ? 'is-active' : ''}
284                 aria-pressed={order === 'asc'}
285                 onClick={() => setOrder('asc')}
286             >
287                 <ArrowDown size={13} aria-hidden="true" />
288                 执行顺序
289             </button>
290             <button
291                 type="button"
292                 className={order === 'desc' ? 'is-active' : ''}
293                 aria-pressed={order === 'desc'}
294                 onClick={() => setOrder('desc')}
295             >
296                 <ArrowUp size={13} aria-hidden="true" />
297                 最新优先
298             </button>
299         </div>
300     </div>
301 </div>
302
303 <div
304     ref={logListRef}
305     className="task-log-list-region"
306     role="region"
307     aria-label="可滚动的任务执行日志"
308     tabIndex={0}
309 >
310     {resource.logsError ? (
311         <WorkbenchNotice
312             tone="error"
313             icon={XCircle}
314             title="日志同步失败"
```

```
front/src/workbench/results/TaskLogsPage.tsx
```

```
315         detail={resource.logsError}
316         action={<button type="button" className="btn btn-secondary"
onClick={resource.refreshLogs}>重新加载</button>}
317     />
318   ) : null}
319
320   {resource.logsLoading && !resource.logs.length ? <LogTimelineSkeleton /> : null}
321
322   {!resource.logsLoading && !resource.logsError && !resource.logs.length ? (
323     <div className="task-log-empty" role="status">
324       <Clock3 size={24} aria-hidden="true" />
325       <strong>暂时还没有执行事件</strong>
326       <span>任务进入队列后，启动、实验轮次和结果保存记录会显示在这里。</span>
327     </div>
328   ) : null}
329
330   {resource.logs.length && !logs.length ? (
331     <div className="task-log-empty" role="status">
332       <Search size={24} aria-hidden="true" />
333       <strong>没有匹配的日志</strong>
334       <span>调整搜索词或日志级别，查看其他执行事件。</span>
335       <button type="button" className="btn btn-secondary" onClick={resetFilters}>清除筛选</button>
336     </div>
337   ) : null}
338
339   {logs.length ? (
340     <ol className="task-log-timeline" aria-label="任务执行事件列表">
341       {logs.map((log) => {
342         const levelPresentation = getLevelPresentation(log.level);
343         const eventLabel = ACTION_LABELS[log.action] ?? log.action;
344         const runNumber = extractRunNumber(log);
345         return (
346           <li className={`task-log-event level-${levelPresentation.key}`} key={log.id}>
347             <span className="task-log-event-marker" aria-hidden="true"><i /></span>
348             <div className="task-log-event-content">
349               <div className="task-log-event-heading">
350                 <strong>{eventLabel}</strong>
351                 {eventLabel !== log.action ? <code>{log.action}</code> : null}
352               </div>
353               <p>{log.message}</p>
354               <div className="task-log-event-meta">
355                 <WorkbenchStatus tone={levelPresentation.tone}>{levelPresentation.label}</WorkbenchStatus>
356                 {runNumber ? (
357                   <span className="task-log-run">
358                     第 {runNumber} {runNumber <= runProgress.total ? ` / ${runProgress.total}` : ''} 轮
359                   </span>
360                 ) : null}
361               </div>
362             </div>
363             <time dateTime={utcDateTimeAttribute(log.createdAt)}>
```

```
front/src/workbench/results/TaskLogsPage.tsx
```

```
364         <span>{formatLocalDate(log.createdAt)}</span>
365         <strong>{formatLocalClock(log.createdAt)}</strong>
366       </time>
367     </li>
368   );
369   }}}
370 </ol>
371   ) : null}
372 </div>
373 </section>
374
375 <aside className="panel task-log-context-panel" aria-label="任务上下文">
376   <WorkbenchSectionHeader title="任务上下文" meta="当前执行记录对应的任务信息" />
377   <dl className="task-log-context-list">
378     <div><dt>任务名称</dt><dd>{task.name || `任务 #${task.id}`}</dd></div>
379     <div><dt>数据集</dt><dd>{task.datasetName}</dd></div>
380     <div><dt>算法模式</dt><dd><code>{task.mode}</code></dd></div>
381     <div><dt>任务状态</dt><dd><WorkbenchStatus tone={getTaskTone(task.status)}>{statusMeta.label}</WorkbenchStatus
></dd></div>
382     <div><dt>实验轮数</dt><dd>{runProgress.current} / {runProgress.total} 轮</dd></div>
383     <div><dt>运行耗时</dt><dd>{formatRuntime(task.runtimeSeconds)}</dd></div>
384   </dl>
385
386   <div className="task-log-time-range">
387     <h3>日志覆盖范围</h3>
388     <div><span>首个事件</span><time>{formatLocalDateTime(firstLog?.createdAt)}</time></div>
389     <div><span>最新事件</span><time>{formatLocalDateTime(latestLog?.createdAt)}</time></div>
390     <div><span>任务开始</span><time>{formatLocalDateTime(task.startedAt)}</time></div>
391     <div><span>任务结束</span><time>{formatLocalDateTime(task.finishedAt)}</time></div>
392   </div>
393 </aside>
394 </div>
395 </section>
396   );
397 }
398
```

```
// ===== 文件: front/src/workbench/results/TaskResultViews.tsx =====
```

```
1 /**
2  * 结果页面只消费任务结果接口返回的 envelope，不在前端生成演示指标、矩阵或运行日志。
3  * 状态页处理未完成任务，完成后的 section 再交给对应的真实结果视图。
4  */
5 import { useMemo, useState } from 'react';
6 import { useNavigate } from 'react-router-dom';
7 import {
8   Activity,
9   Award,
10  BarChart3,
11  CheckCircle2,
12  ClipboardCheck,
```

```
front/src/workbench/results/TaskResultViews.tsx
```

```
13   Clock3,
14   Download,
15   Gauge,
16   Info,
17   RefreshCw,
18   Target,
19   XCircle,
20 } from 'lucide-react';
21 import { retryTask } from '../../api/tasks';
22 import { SelectField } from '../../components/SelectField';
23 import {
24   WorkbenchNotice,
25   WorkbenchPageHeader,
26   WorkbenchSectionHeader,
27   WorkbenchStatus,
28 } from '../WorkbenchUi';
29 import { getTaskRunProgress } from '../tasks/taskStatus';
30 import { AnalysisOverviewPage } from './AnalysisOverviewPage';
31 import {
32   CaResultPage,
33   ExportResultPage,
34   KernelResultPage,
35   MklResultPage,
36   ReportResultPage,
37   ResultsAnalysisPage,
38   VisualizationResultPage,
39 } from './ResultDetailPages';
40 import {
41   buildEvaluationTrendOption,
42   type EvaluationMetricKey,
43   type EvaluationTrendMode,
44   useResultChart,
45 } from './ResultChartPanels';
46 import { MetricStrip, ReadyHeader, ResultBackButton, useShouldShowResultBack } from './ResultPageShared';
47 import {
48   buildEvaluationCsv,
49   deriveEvaluationSummary,
50   evaluationMetricKeys,
51 } from './evaluationPresentation';
52 import {
53   downloadTextFile,
54   formatNumber,
55   formatPercent,
56   metricLabels,
57 } from './resultPresentation';
58 import { TaskLogsPage } from './TaskLogsPage';
59 import type { TaskResultResource } from './types';
60
61 type Props = {
62   section: string;
```

```
front/src/workbench/results/TaskResultViews.tsx
```

```
63   resource: TaskResultResource;
64 };
65
66 function ResultState({ resource }: { resource: TaskResultResource }) {
67   const navigate = useNavigate();
68   const showBackButton = useShouldShowResultBack();
69   const envelope = resource.envelope;
70   const task = envelope?.task;
71   const state = envelope?.state;
72
73   const retry = async () => {
74     if (!task) return;
75     await retryTask(task.id);
76     resource.refresh();
77   };
78
79   let title = '正在加载分析结果';
80   let detail = '正在读取任务、结果和可视化预览。';
81   let tone: 'info' | 'warning' | 'error' | 'success' = 'info';
82
83   if (resource.error) {
84     title = '结果加载失败';
85     detail = resource.error;
86     tone = 'error';
87   }
88   if (state === 'empty') {
89     title = '还没有可用的分析结果';
90     detail = '请先创建并完成一个带 y 标签的数据集分析任务。';
91   }
92   if (state === 'draft') {
93     title = '任务尚未启动';
94     detail = '启动任务后，这里会显示计算过程和结果。';
95   }
96   if (state === 'queued') {
97     title = '任务正在排队';
98     detail = `队列位置 ${task?.queuePosition ?? '—'}，轮到后会自动开始。`;
99     tone = 'warning';
100  }
101  if (state === 'running' && task) {
102    const run = getTaskRunProgress(task);
103    title = 'OMELET 正在运行';
104    detail = `第 ${run.current}/${run.total} 轮，迭代 ${task.currentIter ?? 0}/${task.maxIter ?? 0}，${Math.round(task.progress ?? 0)}%。`;
105    tone = 'warning';
106  }
107  if (state === 'failed') {
108    title = '任务执行失败';
109    detail = task?.errorMessage || '算法子进程未能完成，请查看日志后重试。';
110    tone = 'error';
111  }
```

```
front/src/workbench/results/TaskResultViews.tsx
```

```
112   if (state === 'cancelled') {
113     title = '任务已取消';
114     detail = '该任务没有完整结果，可以调整参数后重新执行。';
115     tone = 'warning';
116   }
117   if (state === 'legacy') {
118     title = '结果结构需要更新';
119     detail = '当前任务结果缺少完整算法产物，重新执行任务后可生成新版分析结果。';
120     tone = 'warning';
121   }
122
123   return (
124     <section className="soft-page result-state-page" aria-live="polite">
125       <WorkbenchPageHeader
126         icon={Activity}
127         title="分析结果"
128         context={task ? `任务 #${task.id} • ${task.datasetName} • ${task.mode}` : '结果上下文'}
129         status={<WorkbenchStatus tone={tone} pulse={state === 'running'}>{title}</WorkbenchStatus>}
130         backAction={showBackButton ? <ResultBackButton taskId={task?.id} /> : undefined}
131       />
132       <section className="panel result-state-panel">
133         <WorkbenchNotice tone={tone} icon={tone === 'error' ? XCircle : Activity} title={title} detail={detail} />
134         <div className="result-state-actions">
135           {(state === 'failed' || state === 'cancelled' || state === 'legacy') && task ? (
136             <button type="button" className="btn btn-primary" onClick={() => void retry()}>重新执行</button>
137           ) : null}
138           <button type="button" className="btn btn-secondary" onClick={() => navigate('/workbench/tasks')}>
139             {state === 'empty' ? '创建任务' : '打开任务中心'}
140           </button>
141           {resource.error ? (
142             <button type="button" className="btn btn-secondary" onClick={resource.refresh}>
143               <RefreshCw size={15} />重试加载
144             </button>
145           ) : null}
146         </div>
147       </section>
148     </section>
149   );
150 }
151
152 function EvaluationPage({ resource }: { resource: TaskResultResource }) {
153   const task = resource.envelope!.task!;
154   const result = resource.envelope!.result!;
155   const rows = result.metrics.runs;
156   const [focusMetric, setFocusMetric] = useState<EvaluationMetricKey>('acc');
157   const [trendMode, setTrendMode] = useState<EvaluationTrendMode>('absolute');
158   const summary = useMemo(() => deriveEvaluationSummary(result), [result]);
159   const trendOption = useMemo(
160     () => buildEvaluationTrendOption(result, focusMetric, trendMode),
161     [focusMetric, result, trendMode],
```

```
front/src/workbench/results/TaskResultViews.tsx
```

```
162   );
163   const trendRef = useResultChart(trendOption);
164   const focusInsight = summary.metrics.find((metric) => metric.key === focusMetric)
165     ?? summary.metrics[0];
166   const accInsight = summary.metrics.find((metric) => metric.key === 'acc');
167   const representativeRun = result.preview.summary.representativeRun;
168   const overallTone = summary.runCount < 2
169     ? 'neutral'
170     : summary.metrics.some((metric) => metric.stabilityTone === 'warning')
171       ? 'warning'
172       : 'success';
173   const overallLabel = summary.runCount < 2
174     ? '数据不足'
175     : overallTone === 'success'
176       ? '整体波动较低'
177       : '存在波动项';
178
179   function exportMetrics() {
180     downloadTextFile(
181       `task-${task.id}-performance.csv`,
182       buildEvaluationCsv(result),
183       'text/csv;charset=utf-8',
184     );
185   }
186
187   return (
188     <section className="soft-page result-evaluation-page">
189       <ReadyHeader
190         title="性能评估"
191         icon={ClipboardCheck}
192         resource={resource}
193         actions={
194           <div className="result-evaluation-header-actions">
195             <button type="button" className="btn btn-secondary" onClick={resource.refresh}>
196               <RefreshCw size={15} aria-hidden="true" />
197               刷新
198             </button>
199             <button
200               type="button"
201               className="btn btn-primary"
202               disabled={!rows.length}
203               onClick={exportMetrics}
204             >
205               <Download size={15} aria-hidden="true" />
206               导出 CSV
207             </button>
208           </div>
209         }
210       />
211     <MetricStrip result={result} />
```



```

212 {rows.length ? (
213   <
214     <section className="panel result-evaluation-conclusion" aria-label="评估结论">
215       <WorkbenchSectionHeader
216         title="评估结论"
217         meta="基于当前重复实验的汇总结果，未引入外部方法基线"
218         actions=<WorkbenchStatus tone={overallTone}>{overallLabel}</WorkbenchStatus>
219       />
220     <div className="result-evaluation-conclusion-body">
221       <div className="result-evaluation-conclusion-copy">
222         <span className="result-evaluation-conclusion-state">
223           <CheckCircle2 size={16} aria-hidden="true" />
224           {summary.runCount} 轮重复实验已完成
225         </span>
226         <strong>四项聚类指标均已完成跨轮次统计</strong>
227         <p>
228           ACC 均值为 {formatPercent(accInsight?.mean)}, 最高值出现在第 {accInsight?.bestRun?.run ?? '—'} 轮。
229           {summary.mostStableMetric
230             ? ` ${summary.mostStableMetric.label} 的相对波动最低，可作为本次结果稳定性的主要参考。`
231             : `当前缺少足够轮次，暂时无法判断跨轮次稳定性。`}
232         </p>
233       </div>
234     </div>
235     <dl className="result-evaluation-conclusion-facts">
236       <div>
237         <dt><Target size={14} aria-hidden="true" />代表轮次</dt>
238         <dd>第 {representativeRun} 轮</dd>
239         <small>用于结果预览与后续分析</small>
240       </div>
241       <div>
242         <dt><Clock3 size={14} aria-hidden="true" />平均耗时</dt>
243         <dd>{formatNumber(summary.meanRuntime, 2)} s</dd>
244         <small>累计 {formatNumber(summary.totalRuntime, 2)} s</small>
245       </div>
246       <div>
247         <dt><Gauge size={14} aria-hidden="true" />波动最低</dt>
248         <dd>{summary.mostStableMetric?.label ?? '—'}</dd>
249         <small>变异系数 {summary.mostStableMetric?.variationRatio == null ? '—' :
formatPercent(summary.mostStableMetric.variationRatio)}</small>
250       </div>
251       <div>
252         <dt><BarChart3 size={14} aria-hidden="true" />波动最高</dt>
253         <dd>{summary.mostVariableMetric?.label ?? '—'}</dd>
254         <small>变异系数 {summary.mostVariableMetric?.variationRatio == null ? '—' :
formatPercent(summary.mostVariableMetric.variationRatio)}</small>
255       </div>
256     </dl>
257   </div>
258 </section>
259

```

```
front/src/workbench/results/TaskResultViews.tsx
```

```
260     <div className="result-evaluation-analysis-grid">
261       <section className="panel result-evaluation-trend-panel" aria-label="指标趋势">
262         <WorkbenchSectionHeader
263           title="指标趋势"
264           meta={trendMode === 'relative'
265             ? ` ${summary.runCount} 个观测点 • 以指标均值为 0 基线显示百分点变化`
266             : ` ${summary.runCount} 个观测点 • 纵轴按 ${metricLabels[focusMetric]} 区间自动缩放` }
267           actions={
268             <div className="result-evaluation-trend-controls">
269               <div className="result-evaluation-trend-mode" role="group" aria-label="趋势显示方式">
270                 <button
271                   type="button"
272                   aria-pressed={trendMode === 'absolute'}
273                   onClick={() => setTrendMode('absolute')}
274                 >
275                   原始值
276                 </button>
277                 <button
278                   type="button"
279                   aria-pressed={trendMode === 'relative'}
280                   onClick={() => setTrendMode('relative')}
281                 >
282                   相对均值
283                 </button>
284               </div>
285               <SelectField
286                 value={focusMetric}
287                 options={evaluationMetricKeys.map((key) => ({ value: key, label: metricLabels[key] })))}
288                 ariaLabel="突出评估指标"
289                 className="result-evaluation-metric-select"
290                 size="sm"
291                 onChange={(value) => setFocusMetric(value as EvaluationMetricKey)}
292               />
293             </div>
294           }
295         />
296       <div className="result-evaluation-trend-summary" aria-label={` ${metricLabels[focusMetric]} 统计摘要`}>
297         <span>均值 <strong>{formatPercent(focusInsight?.mean)}</strong></span>
298         <span>标准差 <strong>{formatPercent(focusInsight?.std)}</strong></span>
299         <span>最佳轮次 <strong>第 {focusInsight?.bestRun?.run ?? '—'} 轮</strong></span>
300       </div>
301       <div
302         className="result-evaluation-trend-chart"
303         ref={trendRef}
304         role="img"
305         aria-label={` ${metricLabels[focusMetric]} ${trendMode === 'relative' ? '相对均值变化趋势' : '跨轮次变化趋势'} `}
306       />
307     </section>
308
309     <section className="panel result-evaluation-stability" aria-label="稳定性诊断">
```

```
front/src/workbench/results/TaskResultViews.tsx

310      <WorkbenchSectionHeader
311        title="稳定性诊断"
312        meta="变异系数 = 标准差 / 指标均值"
313        actions={<WorkbenchStatus tone={overallTone}>{overallLabel}</WorkbenchStatus>}
314      />
315      <div className="result-evaluation-stability-list">
316        {summary.metrics.map((metric) => (
317          <div className={`result-evaluation-stability-row is-${metric.key}`} key={metric.key}>
318            <div>
319              <strong>{metric.label}</strong>
320              <small>标准差 {formatPercent(metric.std)}</small>
321            </div>
322            <span className={`result-evaluation-stability-status is-${metric.stabilityTone}`}>
323              {metric.stabilityLabel}
324            </span>
325            <b>{metric.variationRatio == null ? '—' : formatPercent(metric.variationRatio)}</b>
326          </div>
327        ))}
328      </div>
329      <div className="result-evaluation-runtime-band">
330        <Clock3 size={16} aria-hidden="true" />
331        <div>
332          <span>单轮耗时区间</span>
333          <strong>{formatNumber(summary.minimumRuntime, 2)} - {formatNumber(summary.maximumRuntime, 2)}
334        s</strong>
335        </div>
336        <small>平均 {formatNumber(summary.meanRuntime, 2)} s</small>
337      </div>
338      <div className="result-evaluation-method-note">
339        <Info size={15} aria-hidden="true" />
340        <span>稳定性标签仅用于页面内快速筛查，正式结论以标准差、区间和完整轮次明细为准。</span>
341      </div>
342    </section>
343  </div>
344  <section className="panel result-evaluation-distribution" aria-label="指标分布">
345    <WorkbenchSectionHeader
346      title="指标分布"
347      meta="逐项核对均值、标准差、跨轮次区间与最高值轮次"
348    />
349    <div className="result-evaluation-distribution-list">
350      {summary.metrics.map((metric) => (
351        <div className={`result-evaluation-distribution-row is-${metric.key}`} key={metric.key}>
352          <div className="result-evaluation-distribution-name">
353            <strong>{metric.label}</strong>
354            <span className={`result-evaluation-stability-status is-${metric.stabilityTone}`}>
355              {metric.stabilityLabel}
356            </span>
357          </div>
358          <div className="result-evaluation-distribution-values">
```

```
front/src/workbench/results/TaskResultViews.tsx
```

```
359         <span>均值<strong>{formatPercent(metric.mean)}</strong></span>
360         <span>标准差<strong>{formatPercent(metric.std)}</strong></span>
361         <span>区间宽度<strong>{formatPercent(metric.spread)}</strong></span>
362     </div>
363     <div
364         className="result-evaluation-range"
365         role="img"
366         aria-label={` ${metric.label} 最小值${formatPercent(metric.min)}, 均值${formatPercent(metric.mean)}, 最大值${f
formatPercent(metric.max)} `}
367     >
368         <span>{formatPercent(metric.min)}</span>
369         <div className="result-evaluation-range-track" aria-hidden="true">
370             <i style={{ left: ` ${metric.meanPosition}%` }} />
371         </div>
372         <span>{formatPercent(metric.max)}</span>
373     </div>
374     <div className="result-evaluation-best-run">
375         <Award size={15} aria-hidden="true" />
376         <span>最高值<strong>第 {metric.bestRun?.run ?? '—'} 轮</strong></span>
377     </div>
378 </div>
379 )))
380 </div>
381 </section>
382
383 <section className="panel result-evaluation-table-panel" aria-label="重复实验明细">
384     <WorkbenchSectionHeader
385         title="重复实验明细"
386         meta={` ${rows.length} 次运行 • 当前突出 ${metricLabels[focusMetric]} • 随机种子按任务参数派生` }
387         actions=<WorkbenchStatus tone="info">完整记录</WorkbenchStatus>
388     />
389     <div className="result-evaluation-table" role="table" aria-label="重复实验明细表">
390         <div className="result-evaluation-table-row header" role="row">
391             <span role="columnheader">轮次</span>
392             <span role="columnheader">随机种子</span>
393             {evaluationMetricKeys.map((key) => (
394                 <span className={key === focusMetric ? 'is-focused' : ''} role="columnheader" key={key}>
395                     {metricLabels[key]}
396                 </span>
397             ))}
398             <span role="columnheader">耗时</span>
399             <span role="columnheader">标记</span>
400         </div>
401         {rows.map((row) => {
402             const isRepresentative = row.run === representativeRun;
403             const isBest = row.run === focusInsight?.bestRun?.run;
404             return (
405                 <div
406                     className={`result-evaluation-table-row${isBest ? ' is-best' : ''}${isRepresentative ? '
is-representative' : ''}`}
407                 >
```

```
front/src/workbench/results/TaskResultViews.tsx
```

```
407         role="row"
408         key={row.run}
409       >
410         <strong role="cell">第 {row.run} 轮</strong>
411         <span role="cell">{row.seed}</span>
412         {evaluationMetricKeys.map((key) => (
413           <span className={key === focusMetric ? 'is-focused' : ''} role="cell" key={key}>
414             {formatPercent(row[key])}
415           </span>
416         ))}
417         <span role="cell">{formatNumber(row.runtimeSeconds, 2)} s</span>
418         <span className="result-evaluation-row-tags" role="cell">
419           {isBest ? <b className="is-best">最高 {metricLabels[focusMetric]}</b> : null}
420           {isRepresentative ? <b className="is-representative">代表轮次</b> : null}
421           {!isBest && !isRepresentative ? <small>常规轮次</small> : null}
422         </span>
423       </div>
424     );
425   }}}
426 </div>
427 </section>
428 </>
429 ) : (
430   <section className="panel result-evaluation-empty" role="status">
431     <Activity size={20} aria-hidden="true" />
432     <div>
433       <strong>当前结果没有重复实验明细</strong>
434       <span>汇总指标仍会保留；完成至少两轮运行后，趋势与稳定性诊断会自动出现。</span>
435     </div>
436   </section>
437 )}
438
439 <section className="panel result-evaluation-context" aria-label="评估口径">
440   <WorkbenchSectionHeader
441     title="评估口径"
442     meta="当前任务的结果复核上下文"
443   />
444   <dl className="result-evaluation-context-grid">
445     <div><dt>数据集</dt><dd>{task.datasetName}</dd></div>
446     <div><dt>算法模式</dt><dd>{task.mode}</dd></div>
447     <div><dt>样本数量</dt><dd>{result.preview.summary.sampleCount}</dd></div>
448     <div><dt>真实类别</dt><dd>{result.preview.summary.classCount}</dd></div>
449     <div><dt>基础聚类</dt><dd>{result.preview.summary.baseClusterCount}</dd></div>
450     <div><dt>重复轮次</dt><dd>{summary.runCount}</dd></div>
451     <div><dt>随机种子起点</dt><dd>{result.preview.summary.randomSeed}</dd></div>
452     <div><dt>代表轮次</dt><dd>第 {representativeRun} 轮</dd></div>
453     <div><dt>任务总耗时</dt><dd>{formatNumber(result.runtimeSeconds, 2)} s</dd></div>
454     <div><dt>实验累计耗时</dt><dd>{formatNumber(summary.totalRuntime, 2)} s</dd></div>
455   </dl>
456 </section>
```

```
front/src/workbench/results/TaskResultViews.tsx
```

```
457     </section>
458   );
459 }
460
461 export function TaskResultViews({ section, resource }: Props) {
462   const logTask = resource.envelope?.task;
463   const isSelectedLogTask = logTask && (!resource.taskId || logTask.id === resource.taskId);
464   if (!resource.error && section === 'logs' && isSelectedLogTask) {
465     return <TaskLogsPage resource={resource} />;
466   }
467
468   if (
469     resource.loading
470     || resource.error
471     || resource.envelope?.state !== 'ready'
472     || !resource.envelope.result
473     || !resource.envelope.task
474   ) {
475     return <ResultState resource={resource} />;
476   }
477
478   // 路由 section 是展示意图，数据来源始终是同一个后端结果资源，避免各页面各自拼装假数据。
479   if (section === 'analysis') return <AnalysisOverviewPage resource={resource} />;
480   if (section === 'ca-matrix') return <CaResultPage resource={resource} />;
481   if (section === 'kernel-config') return <KernelResultPage resource={resource} />;
482   if (section === 'mkl') return <MklResultPage resource={resource} />;
483   if (section === 'evaluation') return <EvaluationPage resource={resource} />;
484   if (section === 'visualization') return <VisualizationResultPage resource={resource} />;
485   if (section === 'results') return <ResultsAnalysisPage resource={resource} />;
486   if (section === 'export') return <ExportResultPage resource={resource} />;
487   if (section === 'reports') return <ReportResultPage resource={resource} />;
488   return <AnalysisOverviewPage resource={resource} />;
489 }
490
491
492 // ===== 文件: front/src/workbench/results/evaluationPresentation.ts =====
493 1 import { metricLabels, type MetricKey } from './resultPresentation';
494 2 import type { AnalysisResult, RunMetric } from './types';
495 3
496 4 export const evaluationMetricKeys: MetricKey[] = ['acc', 'nmi', 'ari', 'f1'];
497 5
498 6 export type EvaluationStabilityTone = 'success' | 'warning' | 'neutral';
499 7
500 8 export type EvaluationMetricInsight = {
501 9   key: MetricKey;
502 10  label: string;
503 11  mean: number;
504 12  std: number;
505 13  min: number;
506 14  max: number;
```

```
front/src/workbench/results/evaluationPresentation.ts
```

```
15   spread: number;
16   variationRatio: number | null;
17   meanPosition: number;
18   bestRun: RunMetric | null;
19   stabilityLabel: string;
20   stabilityTone: EvaluationStabilityTone;
21 };
22
23 export type EvaluationSummary = {
24   metrics: EvaluationMetricInsight[];
25   runCount: number;
26   meanRuntime: number | null;
27   totalRuntime: number | null;
28   minimumRuntime: number | null;
29   maximumRuntime: number | null;
30   mostStableMetric: EvaluationMetricInsight | null;
31   mostVariableMetric: EvaluationMetricInsight | null;
32 };
33
34 function getStability(
35   runCount: number,
36   variationRatio: number | null,
37 ): Pick<EvaluationMetricInsight, 'stabilityLabel' | 'stabilityTone'> {
38   if (runCount < 2 || variationRatio === null) {
39     return { stabilityLabel: '数据不足', stabilityTone: 'neutral' };
40   }
41   if (variationRatio <= 0.03) {
42     return { stabilityLabel: '波动较低', stabilityTone: 'success' };
43   }
44   if (variationRatio <= 0.08) {
45     return { stabilityLabel: '建议关注', stabilityTone: 'warning' };
46   }
47   return { stabilityLabel: '波动明显', stabilityTone: 'warning' };
48 }
49
50 function getBestRun(rows: RunMetric[], key: MetricKey) {
51   return rows.reduce<RunMetric | null>((best, row) => {
52     const value = Number(row[key]);
53     if (!Number.isFinite(value)) return best;
54     if (!best || value > Number(best[key])) return row;
55     return best;
56   }, null);
57 }
58
59 function getMeanPosition(mean: number, minimum: number, maximum: number) {
60   if (![mean, minimum, maximum].every(Number.isFinite) || maximum <= minimum) return 50;
61   return Math.min(100, Math.max(0, ((mean - minimum) / (maximum - minimum)) * 100));
62 }
63
64 export function deriveEvaluationSummary(result: AnalysisResult): EvaluationSummary {
```

```
front/src/workbench/results/evaluationPresentation.ts
```

```
65     const rows = result.metrics.runs;
66     const metrics = evaluationMetricKeys.map<EvaluationMetricInsight>((key) => {
67         const aggregate = result.metrics.aggregate[key];
68         const mean = Number(aggregate.mean);
69         const std = Number(aggregate.std);
70         const min = Number(aggregate.min);
71         const max = Number(aggregate.max);
72         const variationRatio = Number.isFinite(mean) && mean !== 0 && Number.isFinite(std)
73             ? Math.abs(std / mean)
74             : null;
75
76         return {
77             key,
78             label: metricLabels[key],
79             mean,
80             std,
81             min,
82             max,
83             spread: Number.isFinite(min) && Number.isFinite(max) ? Math.max(0, max - min) : 0,
84             variationRatio,
85             meanPosition: getMeanPosition(mean, min, max),
86             bestRun: getBestRun(rows, key),
87             ...getStability(rows.length, variationRatio),
88         };
89     });
90
91     const comparableMetrics = metrics.filter((metric) => metric.variationRatio !== null);
92     const byVariation = [...comparableMetrics].sort(
93         (left, right) => (left.variationRatio ?? 0) - (right.variationRatio ?? 0),
94     );
95     const runtimes = rows
96         .map((row) => Number(row.runtimeSeconds))
97         .filter((value) => Number.isFinite(value) && value >= 0);
98     const totalRuntime = runtimes.length
99         ? runtimes.reduce((total, value) => total + value, 0)
100         : null;
101
102     return {
103         metrics,
104         runCount: rows.length,
105         meanRuntime: totalRuntime === null ? null : totalRuntime / runtimes.length,
106         totalRuntime,
107         minimumRuntime: runtimes.length ? Math.min(...runtimes) : null,
108         maximumRuntime: runtimes.length ? Math.max(...runtimes) : null,
109         mostStableMetric: byVariation[0] ?? null,
110         mostVariableMetric: byVariation[byVariation.length - 1] ?? null,
111     };
112 }
113
114 export function buildEvaluationCsv(result: AnalysisResult) {
```



```
front/src/workbench/results/evaluationPresentation.ts
```

```
115   const header = 'run,seed,ACC,NMI,ARI,F1-score,runtimeSeconds';
116   const rows = result.metrics.runs.map((row) => [
117     row.run,
118     row.seed,
119     row.acc,
120     row.nmi,
121     row.ari,
122     row.f1,
123     row.runtimeSeconds,
124   ].join(', '));
125   return [header, ...rows].join('\n');
126 }
127
// ===== 文件: front/src/workbench/results/resultPresentation.ts =====
1 import type { NavigateFunction } from 'react-router-dom';
2
3 export type MetricKey = 'acc' | 'nmi' | 'ari' | 'f1';
4
5 export type ResultBackLocationState = {
6   resultBack?: boolean;
7 };
8
9 export const resultBackState = { resultBack: true } satisfies ResultBackLocationState;
10
11 export const metricLabels: Record<MetricKey, string> = {
12   acc: 'ACC',
13   nmi: 'NMI',
14   ari: 'ARI',
15   f1: 'F1-score',
16 };
17
18 export function formatPercent(value: number | null | undefined, digits = 2) {
19   if (value === null || value === undefined || !Number.isFinite(value)) return '—';
20   return `${(value * 100).toFixed(digits)}%`;
21 }
22
23 export function formatNumber(value: number | null | undefined, digits = 3) {
24   if (value === null || value === undefined || !Number.isFinite(value)) return '—';
25   return Number(value).toFixed(digits);
26 }
27
28 export function formatBytes(value: number) {
29   if (value < 1024) return `${value} B`;
30   if (value < 1024 * 1024) return `${(value / 1024).toFixed(1)} KB`;
31   return `${(value / (1024 * 1024)).toFixed(1)} MB`;
32 }
33
34 export function downloadTextFile(filename: string, content: string, type: string) {
35   const url = URL.createObjectURL(new Blob([content], { type }));
```

```
front/src/workbench/results/resultPresentation.ts
```

```
36   const anchor = document.createElement('a');
37   anchor.href = url;
38   anchor.download = filename;
39   anchor.click();
40   URL.revokeObjectURL(url);
41 }
42
43 export function taskResultPath(section: string, taskId: number) {
44   return `/workbench/${section}?taskId=${encodeURIComponent(taskId)}`;
45 }
46
47 export function hasResultBackState(state: unknown): state is ResultBackLocationState {
48   return Boolean(
49     state &&
50     typeof state === 'object' &&
51     (state as ResultBackLocationState).resultBack === true,
52   );
53 }
54
55 export function navigateToTaskResult(navigate: NavigateFunction, section: string, taskId: number) {
56   navigate(taskResultPath(section, taskId), { state: resultBackState });
57 }
58
```

```
// ===== 文件: front/src/workbench/results/types.ts =====
```

```
1 import type { AnalysisTask, AnalysisTaskLog } from '../tasks/types';
2
3 export type MetricSummary = {
4   mean: number;
5   std: number;
6   min: number;
7   max: number;
8 };
9
10 export type RunMetric = {
11   run: number;
12   seed: number;
13   runtimeSeconds: number;
14   acc: number;
15   nmi: number;
16   ari: number;
17   fl: number;
18 };
19
20 export type MatrixPreview = {
21   shape: [number, number];
22   rowIndices: number[];
23   columnIndices: number[];
24   values: number[][];
25   stats: {
```

```
front/src/workbench/results/types.ts
```

```
26     min: number;
27     max: number;
28     mean: number;
29     std: number;
30     diagonalMean: number | null;
31     symmetryMaxError: number | null;
32     nonzeroRatio: number;
33   };
34   topPairs: Array<{ row: number; column: number; value: number }>;
35 };
36
37 export type AnalysisResult = {
38   schemaVersion: 1;
39   parameters: Record<string, unknown>;
40   runtimeSeconds: number | null;
41   metrics: {
42     aggregate: Record<'acc' | 'nmi' | 'ari' | 'f1', MetricSummary>;
43     runs: RunMetric[];
44   };
45   kernelWeights: {
46     items: Array<{ key: string; mean: number; std: number; representative: number }>;
47     runs: Array<{ run: number; values: number[] }>;
48   };
49   convergence: {
50     representativeRun: number;
51     runs: Array<{
52       run: number;
53       converged: boolean;
54       points: Array<{ iteration: number; objective: number; relativeChange: number | null }>;
55     }>;
56   };
57   preview: {
58     schemaVersion: 1;
59     summary: {
60       mode: string;
61       sampleCount: number;
62       baseClusterCount: number;
63       classCount: number;
64       representativeRun: number;
65       randomSeed: number;
66     };
67     matrices: { ca: MatrixPreview; s: MatrixPreview; z: MatrixPreview };
68     scatter: {
69       totalCount: number;
70       sampled: boolean;
71       points: Array<{
72         sampleIndex: number;
73         x: number;
74         y: number;
75         predictedLabel: string;
```

front/src/workbench/results/types.ts

```
76     trueLabel: string;
77   }>;
78   };
79 };
80 artifacts: Array<{
81   key: string;
82   name: string;
83   format: string;
84   size: number;
85   downloadUrl: string;
86 }>;
87 };
88
89 export type TaskResultEnvelope = {
90   state: 'empty' | 'draft' | 'queued' | 'running' | 'failed' | 'cancelled' | 'ready' | 'legacy';
91   task?: AnalysisTask | null;
92   result?: AnalysisResult | null;
93 };
94
95 export type TaskExport = {
96   id: number;
97   name: string;
98   items: string[];
99   itemCount: number;
100  status: string;
101  filename: string;
102  fileSize: number;
103  createdAt: string;
104  downloadUrl: string;
105 };
106
107 export type TaskResultResource = {
108   loading: boolean;
109   error: string | null;
110   envelope: TaskResultEnvelope | null;
111   taskId: number | null;
112   refresh: () => void;
113   selectTask: (taskId: number) => void;
114   logs: AnalysisTaskLog[];
115   logsLoading: boolean;
116   logsError: string | null;
117   refreshLogs: () => void;
118 };
119
```

// ===== 文件: front/src/workbench/results/useTaskResult.ts =====

```
1 import { useCallback, useEffect, useMemo, useState } from 'react';
2 import { useLocation, useNavigate } from 'react-router-dom';
3 import { fetchTaskLogs } from '../../api/tasks';
4 import { fetchLatestTaskResult, fetchTaskResult } from '../../api/results';
```

```
front/src/workbench/results/useTaskResult.ts
```

```
5 import type { TaskResultEnvelope, TaskResultResource } from './types';
6
7 function parseTaskId(search: string) {
8   const value = Number(new URLSearchParams(search).get('taskId'));
9   return Number.isInteger(value) && value > 0 ? value : null;
10 }
11
12 export function useTaskResult(enabled = true): TaskResultResource {
13   const location = useLocation();
14   const navigate = useNavigate();
15   const taskId = useMemo(() => parseTaskId(location.search), [location.search]);
16   const [loading, setLoading] = useState(true);
17   const [error, setError] = useState<string | null>(null);
18   const [envelope, setEnvelope] = useState<TaskResultEnvelope | null>(null);
19   const [logs, setLogs] = useState<TaskResultResource['logs']>([]);
20   const [logsLoading, setLogsLoading] = useState(false);
21   const [logsError, setLogsError] = useState<string | null>(null);
22   const [refreshVersion, setRefreshVersion] = useState(0);
23   const [logsRefreshVersion, setLogsRefreshVersion] = useState(0);
24
25   const refresh = useCallback(() => {
26     setRefreshVersion((value) => value + 1);
27     setLogsRefreshVersion((value) => value + 1);
28   }, []);
29   const refreshLogs = useCallback(() => setLogsRefreshVersion((value) => value + 1), []);
30   const selectTask = useCallback((nextTaskId: number) => {
31     const query = new URLSearchParams(location.search);
32     query.set('taskId', String(nextTaskId));
33     navigate({ pathname: location.pathname, search: `?${query.toString()}` });
34   }, [location.pathname, location.search, navigate]);
35
36   useEffect(() => {
37     if (!enabled) {
38       setLoading(false);
39       setEnvelope(null);
40       setError(null);
41       setLogs([]);
42       setLogsLoading(false);
43       setLogsError(null);
44       return undefined;
45     }
46     const controller = new AbortController();
47     setLoading(true);
48     setError(null);
49     const request = taskId ? fetchTaskResult(taskId, controller.signal) : fetchLatestTaskResult(controller.signal);
50     request
51       .then((nextEnvelope) => {
52         if (controller.signal.aborted) return;
53         setEnvelope(nextEnvelope);
54         setLoading(false);
```

```
front/src/workbench/results/useTaskResult.ts
```

```
55     })
56     .catch((reason: unknown) => {
57       if (controller.signal.aborted) return;
58       setError(reason instanceof Error ? reason.message : '加载任务结果失败');
59       setLoading(false);
60     });
61     return () => controller.abort();
62   }, [enabled, refreshVersion, taskId]);
63
64   useEffect(() => {
65     const selectedTaskId = envelope?.task?.id;
66     if (!enabled || !selectedTaskId) {
67       setLogs([]);
68       setLogsLoading(false);
69       setLogsError(null);
70       return undefined;
71     }
72
73     const controller = new AbortController();
74     setLogsLoading(true);
75     setLogsError(null);
76     void fetchTaskLogs(selectedTaskId, controller.signal)
77       .then((response) => {
78         if (!controller.signal.aborted) setLogs(response.items);
79       })
80       .catch((reason: unknown) => {
81         if (controller.signal.aborted) return;
82         setLogsError(reason instanceof Error ? reason.message : '加载任务日志失败');
83       })
84       .finally(() => {
85         if (!controller.signal.aborted) setLogsLoading(false);
86       });
87
88     return () => controller.abort();
89   }, [enabled, envelope?.task?.id, logsRefreshVersion]);
90
91   useEffect(() => {
92     if (!enabled || (envelope?.state !== 'queued' && envelope?.state !== 'running')) return undefined;
93     const timer = window.setInterval(refresh, 2000);
94     return () => window.clearInterval(timer);
95   }, [enabled, envelope?.state, refresh]);
96
97   useEffect(() => {
98     if (!enabled) return undefined;
99     const onFocus = () => refresh();
100    window.addEventListener('focus', onFocus);
101    return () => window.removeEventListener('focus', onFocus);
102  }, [enabled, refresh]);
103
104  return {
```

```
front/src/workbench/results/useTaskResult.ts
```

```
105     loading,
106     error,
107     envelope,
108     taskId,
109     refresh,
110     selectTask,
111     logs,
112     logsLoading,
113     logsError,
114     refreshLogs,
115   };
116 }
117
// ===== 文件: front/src/workbench/tasks/TaskCenterPage.tsx =====
1 import { useCallback, useEffect, useMemo, useState } from 'react';
2 import { useNavigate, useSearchParams } from 'react-router-dom';
3 import {
4   bulkTasks,
5   cancelTask,
6   cloneTask,
7   createTask,
8   createTaskTemplate,
9   deleteTask,
10  deleteTaskTemplate,
11  fetchDatasetOptions,
12  fetchTaskDetail,
13  fetchTaskLogs,
14  fetchTaskStats,
15  fetchTaskTemplates,
16  fetchTasks,
17  retryTask,
18  startTask,
19 } from '../../api/tasks';
20 import { TaskBulkBar } from './components/TaskBulkBar';
21 import { formatLocalDateTime } from '../../utils/time';
22 import { TaskCompareTray } from './components/TaskCompareTray';
23 import { TaskCreateDrawer } from './components/TaskCreateDrawer';
24 import { TaskDetailDrawer } from './components/TaskDetailDrawer';
25 import { TaskEmptyState } from './components/TaskEmptyState';
26 import { TaskFiltersBar } from './components/TaskFiltersBar';
27 import { TaskFocusPanel } from './components/TaskFocusPanel';
28 import { TaskStatsBar } from './components/TaskStatsBar';
29 import { TaskTable } from './components/TaskTable';
30 import { navigateToTaskResult } from '../../results/resultPresentation';
31 import { loadTaskLocalMeta, saveTaskLocalMeta, type TaskLocalMeta, type TaskLocalMetaMap } from './taskLocalMeta';
32 import type {
33   AnalysisTask,
34   AnalysisTaskLog,
35   AnalysisTaskStats,
```

front/src/workbench/tasks/TaskCenterPage.tsx

```
36   CreateTaskPayload,
37   DatasetOption,
38   TaskTemplate,
39 } from './types';
40
41 const PAGE_SIZE = 20;
42
43 export function TaskCenterPage() {
44   const navigate = useNavigate();
45   const [searchParams, setSearchParams] = useSearchParams();
46
47   const [stats, setStats] = useState<AnalysisTaskStats | null>(null);
48   const [tasks, setTasks] = useState<AnalysisTask[]>([]);
49   const [total, setTotal] = useState(0);
50   const [page, setPage] = useState(1);
51   const [loading, setLoading] = useState(true);
52   const [error, setError] = useState('');
53   const [message, setMessage] = useState('');
54
55   const [keyword, setKeyword] = useState(searchParams.get('q') || '');
56   const [status, setStatus] = useState(searchParams.get('status') || '');
57   const [mode, setMode] = useState(searchParams.get('mode') || '');
58   const [datasetId, setDatasetId] = useState(searchParams.get('datasetId') || '');
59   const [createdFrom, setCreatedFrom] = useState(searchParams.get('createdFrom') || '');
60   const [createdTo, setCreatedTo] = useState(searchParams.get('createdTo') || '');
61   const [density, setDensity] = useState<'comfortable' | 'compact'>('comfortable');
62
63   const [datasets, setDatasets] = useState<DatasetOption[]>([]);
64   const [templates, setTemplates] = useState<TaskTemplate[]>([]);
65   const [selectedIds, setSelectedIds] = useState<number[]>([]);
66   const [compareIds, setCompareIds] = useState<number[]>([]);
67   const [taskLocalMeta, setTaskLocalMeta] = useState<TaskLocalMetaMap>({} => loadTaskLocalMeta());
68
69   const [createOpen, setCreateOpen] = useState(searchParams.get('create') === '1');
70   const [submitting, setSubmitting] = useState(false);
71   const [detailTask, setDetailTask] = useState<AnalysisTask | null>(null);
72   const [detailLogs, setDetailLogs] = useState<AnalysisTaskLog[]>([]);
73   const [loadingLogs, setLoadingLogs] = useState(false);
74
75   const initialDatasetId = useMemo(() => {
76     const raw = searchParams.get('datasetId');
77     return raw ? Number(raw) : null;
78   }, [searchParams]);
79
80   const syncQuery = useCallback(
81     (next: { q?: string; status?: string; mode?: string; datasetId?: string; createdFrom?: string; createdTo?: string;
82       create?: string }) => {
83       const params = new URLSearchParams(searchParams);
84       const assign = (key: string, value?: string) => {
85         if (!value) params.delete(key);
```


front/src/workbench/tasks/TaskCenterPage.tsx

```
85         else params.set(key, value);
86     };
87     assign('q', next.q);
88     assign('status', next.status);
89     assign('mode', next.mode);
90     assign('datasetId', next.datasetId);
91     assign('createdFrom', next.createdFrom);
92     assign('createdTo', next.createdTo);
93     assign('create', next.create);
94     setSearchParams(params, { replace: true });
95 },
96 [searchParams, setSearchParams],
97 );
98
99 const loadAll = useCallback(async () => {
100     setLoading(true);
101     setError('');
102     try {
103         const [statsData, pageData, datasetData, templateData] = await Promise.all([
104             fetchTaskStats(),
105             fetchTasks({
106                 page,
107                 pageSize: PAGE_SIZE,
108                 keyword: keyword.trim() || undefined,
109                 status: status || undefined,
110                 mode: mode || undefined,
111                 datasetId: datasetId ? Number(datasetId) : undefined,
112                 createdFrom: createdFrom || undefined,
113                 createdTo: createdTo || undefined,
114             }),
115             fetchDatasetOptions(),
116             fetchTaskTemplates(),
117         ]);
118         setStats(statsData);
119         setTasks(pageData.items);
120         setTotal(pageData.total);
121         setDatasets(datasetData);
122         setTemplates(templateData.items || []);
123     } catch (err) {
124         setError(err instanceof Error ? err.message : '加载任务中心失败');
125     } finally {
126         setLoading(false);
127     }
128 }, [page, keyword, status, mode, datasetId, createdFrom, createdTo]);
129
130 useEffect(() => {
131     void loadAll();
132 }, [loadAll]);
133
134 useEffect(() => {
```

```
front/src/workbench/tasks/TaskCenterPage.tsx
```

```
135     const hasRunning = tasks.some((task) => task.status === 'running' || task.status === 'queued');
136     if (!hasRunning) return undefined;
137     const timer = window.setInterval(() => {
138         void loadAll();
139         if (detailTask && (detailTask.status === 'running' || detailTask.status === 'queued')) {
140             void openDetail(detailTask.id, false);
141         }
142     }, 3000);
143     return () => window.clearInterval(timer);
144 }, [tasks, detailTask, loadAll]);
145
146 async function openDetail(taskId: number, withLoading = true) {
147     if (withLoading) setLoadingLogs(true);
148     try {
149         const [task, logs] = await Promise.all([fetchTaskDetail(taskId), fetchTaskLogs(taskId)]);
150         setDetailTask(task);
151         setDetailLogs(logs.items || []);
152     } catch (err) {
153         setError(err instanceof Error ? err.message : '加载任务详情失败');
154     } finally {
155         if (withLoading) setLoadingLogs(false);
156     }
157 }
158
159 async function runAction(action: () => Promise<unknown>, successText: string) {
160     setError('');
161     setMessage('');
162     try {
163         await action();
164         setMessage(successText);
165         await loadAll();
166         if (detailTask) {
167             await openDetail(detailTask.id, false);
168         }
169     } catch (err) {
170         setError(err instanceof Error ? err.message : '操作失败');
171     }
172 }
173
174 function exportCsv() {
175     const header = ['id', 'name', 'mode', 'status', 'dataset', 'progress', 'createdAt', 'runtimeSeconds'];
176     const lines = tasks.map((task) =>
177         [
178             task.id,
179             JSON.stringify(task.name),
180             task.mode,
181             task.status,
182             JSON.stringify(task.datasetName),
183             task.progress,
184             formatLocalDateTime(task.createdAt),
```

front/src/workbench/tasks/TaskCenterPage.tsx

```
185     task.runtimeSeconds ?? '',
186     ].join(','),
187   );
188   const blob = new Blob([[header.join(','), ...lines].join('\n')], { type: 'text/csv;charset=utf-8' });
189   const url = URL.createObjectURL(blob);
190   const anchor = document.createElement('a');
191   anchor.href = url;
192   anchor.download = `tasks-${Date.now()}.csv`;
193   anchor.click();
194   URL.revokeObjectURL(url);
195 }
196
197 async function handleCreate(payload: CreateTaskPayload, saveAsTemplateName?: string) {
198   setSubmitting(true);
199   setError('');
200   try {
201     if (saveAsTemplateName) {
202       await createTaskTemplate({
203         name: saveAsTemplateName,
204         mode: String(payload.mode || 'OMELET-SV'),
205         params: payload.params,
206       });
207     }
208     await createTask(payload);
209     setCreateOpen(false);
210     setMessage('任务创建成功');
211     syncQuery({
212       q: keyword,
213       status,
214       mode,
215       datasetId,
216       create: undefined,
217     });
218     await loadAll();
219   } finally {
220     setSubmitting(false);
221   }
222 }
223
224 const totalPages = Math.max(1, Math.ceil(total / PAGE_SIZE));
225 const hasActiveFilters = Boolean(keyword || status || mode || datasetId || createdFrom || createdTo);
226 const comparedTasks = tasks.filter((task) => compareIds.includes(task.id));
227
228 function toggleCompare(task: AnalysisTask) {
229   setError('');
230   setCompareIds((current) => {
231     if (current.includes(task.id)) return current.filter((id) => id !== task.id);
232     if (current.length >= 3) {
233       setError('一次最多比较 3 个任务，请先移除已选任务。');
234       return current;
```

```
front/src/workbench/tasks/TaskCenterPage.tsx
```

```
235     }
236     return [...current, task.id];
237   });
238 }
239
240 async function handleDeleteTemplate(template: TaskTemplate) {
241   if (!window.confirm(`确认删除模板 "${template.name}" ?`)) return;
242   setError('');
243   try {
244     await deleteTaskTemplate(template.id);
245     setTemplates((current) => current.filter((item) => item.id !== template.id));
246     setMessage(`模板 "${template.name}" 已删除`);
247   } catch (err) {
248     setError(err instanceof Error ? err.message : '删除模板失败');
249   }
250 }
251
252 function updateTaskLocalMeta(taskId: number, patch: Partial<TaskLocalMeta>) {
253   setTaskLocalMeta((current) => {
254     const next = {
255       ...current,
256       [taskId]: {
257         ...current[taskId],
258         ...patch,
259       },
260     };
261     saveTaskLocalMeta(next);
262     return next;
263   });
264 }
265
266 return (
267   <div className="task-center-page" aria-label="任务中心">
268     <TaskStatsBar
269       stats={stats}
270       loading={loading}
271       activeStatus={status}
272       onQuickFilter={(nextStatus) => {
273         setPage(1);
274         setStatus(nextStatus);
275         syncQuery({ q: keyword, status: nextStatus, mode, datasetId, createdFrom, createdTo });
276       }}
277     />
278
279     <TaskFocusPanel
280       tasks={tasks}
281       onOpenTask={(task) => void openDetail(task.id)}
282       onStartTask={(task) => void runAction(() => startTask(task.id), `任务 #${task.id} 已启动`)}
283     />
284
```

front/src/workbench/tasks/TaskCenterPage.tsx

```
285     <TaskFiltersBar
286       keyword={keyword}
287       status={status}
288       mode={mode}
289       datasetId={datasetId}
290       createdFrom={createdFrom}
291       createdTo={createdTo}
292       datasets={datasets}
293       total={total}
294       loading={loading}
295       onKeywordChange={(value) => {
296         setKeyword(value);
297         setPage(1);
298         syncQuery({ q: value, status, mode, datasetId, createdFrom, createdTo, create: createOpen ? '1' : undefined
    });
299       }}
300       onStatusChange={(value) => {
301         setStatus(value);
302         setPage(1);
303         syncQuery({ q: keyword, status: value, mode, datasetId, createdFrom, createdTo });
304       }}
305       onModeChange={(value) => {
306         setMode(value);
307         setPage(1);
308         syncQuery({ q: keyword, status, mode: value, datasetId, createdFrom, createdTo });
309       }}
310       onDatasetChange={(value) => {
311         setDatasetId(value);
312         setPage(1);
313         syncQuery({ q: keyword, status, mode, datasetId: value, createdFrom, createdTo });
314       }}
315       onCreatedFromChange={(value) => {
316         setCreatedFrom(value);
317         setPage(1);
318         syncQuery({ q: keyword, status, mode, datasetId, createdFrom: value, createdTo });
319       }}
320       onCreatedToChange={(value) => {
321         setCreatedTo(value);
322         setPage(1);
323         syncQuery({ q: keyword, status, mode, datasetId, createdFrom, createdTo: value });
324       }}
325       onClearDates={() => {
326         setCreatedFrom('');
327         setCreatedTo('');
328         setPage(1);
329         syncQuery({ q: keyword, status, mode, datasetId, createdFrom: '', createdTo: '' });
330       }}
331       onRefresh={() => void loadAll()}
332       onReset={() => {
333         setKeyword('');
```

```
front/src/workbench/tasks/TaskCenterPage.tsx
```

```
334         setStatus('');
335         setMode('');
336         setDatasetId('');
337         setCreatedFrom('');
338         setCreatedTo('');
339         setPage(1);
340         syncQuery({ q: '', status: '', mode: '', datasetId: '', createdFrom: '', createdTo: '' });
341     }}
342     onCreate={() => {
343         setCreateOpen(true);
344         syncQuery({ q: keyword, status, mode, datasetId, createdFrom, createdTo, create: '1' });
345     }}
346     onExport={exportCsv}
347 />
348
349 {error ? <div className="task-center-alert error">{error}</div> : null}
350 {message ? <div className="task-center-alert success">{message}</div> : null}
351
352 <TaskBulkBar
353     count={selectedIds.length}
354     onRetry={() =>
355         void runAction(() => bulkTasks(selectedIds, 'retry', '批量重试已提交')).then(() => setSelectedIds([]))
356     }
357     onCancel={() =>
358         void runAction(() => bulkTasks(selectedIds, 'cancel', '批量取消已提交')).then(() => setSelectedIds([]))
359     }
360     onDelete={() => {
361         if (!window.confirm(`确认删除选中的 ${selectedIds.length} 个任务?`)) return;
362         void runAction(() => bulkTasks(selectedIds, 'delete', '批量删除完成')).then(() => setSelectedIds([]));
363     }}
364     onFavorite={() => {
365         selectedIds.forEach((taskId) => updateTaskLocalMeta(taskId, { favorite: true }));
366         setMessage(`已关注 ${selectedIds.length} 个任务`);
367     }}
368     onCompare={() => {
369         const newIds = tasks
370             .filter((task) => selectedIds.includes(task.id))
371             .map((task) => task.id)
372             .filter((taskId) => !compareIds.includes(taskId));
373         if (compareIds.length + newIds.length > 3) {
374             setError('比较栏最多保留 3 个任务, 请减少选择后重试。');
375             return;
376         }
377         setCompareIds((current) => [...current, ...newIds]);
378     }}
379     onClear={() => setSelectedIds([])}
380 />
381
382 <TaskCompareTray
383     tasks={comparedTasks}
```

```
front/src/workbench/tasks/TaskCenterPage.tsx
```

```
384         onRemove={() => setCompareIds((current) => current.filter((id) => id !== taskId))}
385         onClear={() => setCompareIds([])}
386     />
387
388     {!loading && total === 0 && !hasActiveFilters ? (
389         <TaskEmptyState
390             onCreate={() => setCreateOpen(true)}
391             onUseTemplate={() => setCreateOpen(true)}
392             onOpenDatasets={() => navigate('/workbench/datasets')}
393         />
394     ) : null}
395
396     <TaskTable
397         tasks={tasks}
398         loading={loading}
399         density={density}
400         onDensityChange={setDensity}
401         selectedIds={selectedIds}
402         compareIds={compareIds}
403         taskLocalMeta={taskLocalMeta}
404         onToggleSelect={(id) =>
405             setSelectedIds((prev) => (prev.includes(id) ? prev.filter((item) => item !== id) : [...prev, id]))
406         }
407         onToggleSelectAll={() => {
408             if (tasks.every((task) => selectedIds.includes(task.id))) {
409                 setSelectedIds((prev) => prev.filter((id) => !tasks.some((task) => task.id === id)));
410             } else {
411                 setSelectedIds((prev) => Array.from(new Set([...prev, ...tasks.map((task) => task.id)]));
412             }
413         }}
414         onOpenDetail={(task) => void openDetail(task.id)}
415         onStart={(task) => void runAction(() => startTask(task.id), `任务 #${task.id} 已启动`)}
416         onCancel={(task) => void runAction(() => cancelTask(task.id), `任务 #${task.id} 已取消`)}
417         onRetry={(task) => void runAction(() => retryTask(task.id), `任务 #${task.id} 已重试`)}
418         onClone={(task) => void runAction(() => cloneTask(task.id), `任务 #${task.id} 已克隆`)}
419         onDelete={(task) => {
420             if (!window.confirm(`确认删除任务 #${task.id}?`)) return;
421             void runAction(async () => {
422                 await deleteTask(task.id);
423                 if (detailTask?.id === task.id) setDetailTask(null);
424             }, `任务 #${task.id} 已删除`);
425         }}
426         onOpenResults={(task) => navigateToTaskResult(navigate, 'results', task.id)}
427         onToggleCompare={toggleCompare}
428         onToggleFavorite={(task) => updateTaskLocalMeta(task.id, { favorite: !taskLocalMeta[task.id]?.favorite })}
429     />
430
431     <div className="task-pagination">
432         <span>
433             共 {total} 条, 第 {page}/{totalPages} 页
```

```
front/src/workbench/tasks/TaskCenterPage.tsx
```

```
434         </span>
435     </div>
436     <button type="button" className="btn btn-secondary" disabled={page <= 1} onClick={() => setPage((p) => p -
1)}>
437         上一页
438     </button>
439     <button
440         type="button"
441         className="btn btn-secondary"
442         disabled={page >= totalPages}
443         onClick={() => setPage((p) => p + 1)}
444     >
445         下一页
446     </button>
447 </div>
448 </div>
449
450 <TaskDetailDrawer
451     open={!!detailTask}
452     task={detailTask}
453     logs={detailLogs}
454     localMeta={detailTask ? taskLocalMeta[detailTask.id] || {} : {}}
455     loadingLogs={loadingLogs}
456     onClose={() => setDetailTask(null)}
457     onStart={(task) => void runAction(() => startTask(task.id, `任务 #${task.id} 已启动`)}
458     onCancel={(task) => void runAction(() => cancelTask(task.id, `任务 #${task.id} 已取消`)}
459     onRetry={(task) => void runAction(() => retryTask(task.id, `任务 #${task.id} 已重试`)}
460     onClone={(task) => void runAction(() => cloneTask(task.id, `任务 #${task.id} 已克隆`)}
461     onDelete={(task) => {
462         if (!window.confirm(`确认删除任务 #${task.id} ? `)) return;
463         void runAction(async () => {
464             await deleteTask(task.id);
465             setDetailTask(null);
466         }, `任务 #${task.id} 已删除`);
467     }}
468     onOpenResults={(task) => navigateToTaskResult(navigate, 'results', task.id)}
469     onOpenExport={(task) => navigateToTaskResult(navigate, 'export', task.id)}
470     onOpenReport={(task) => navigateToTaskResult(navigate, 'reports', task.id)}
471     onUpdateLocalMeta=(patch) => {
472         if (detailTask) updateTaskLocalMeta(detailTask.id, patch);
473     }
474 </>
475
476 <TaskCreateDrawer
477     open={createOpen}
478     datasets={datasets}
479     templates={templates}
480     initialDatasetId={initialDatasetId}
481     submitting={submitting}
482     onClose={() => {
```



```
front/src/workbench/tasks/TaskCenterPage.tsx
```

```
483         setCreateOpen(false);
484         syncQuery({ q: keyword, status, mode, datasetId });
485     }}
486     onSubmit={handleCreate}
487     onDeleteTemplate={(template) => void handleDeleteTemplate(template)}
488     />
489 </div>
490 );
491 }
492
493 export default TaskCenterPage;
494
```

```
// ===== 文件: front/src/workbench/tasks/components/TaskBulkBar.tsx =====
```

```
1 import { GitCompareArrows, Star } from 'lucide-react';
2
3 type Props = {
4   count: number;
5   onRetry: () => void;
6   onCancel: () => void;
7   onDelete: () => void;
8   onFavorite: () => void;
9   onCompare: () => void;
10  onClear: () => void;
11 };
12
13 export function TaskBulkBar({ count, onRetry, onCancel, onDelete, onFavorite, onCompare, onClear }: Props) {
14   if (count <= 0) return null;
15   return (
16     <section className="task-bulk-bar" aria-label="批量操作">
17       <span>已选 {count} 个任务</span>
18       <div>
19         <button type="button" className="btn btn-secondary" onClick={onFavorite}>
20           <Star size={15} aria-hidden="true" /> 关注
21         </button>
22         <button type="button" className="btn btn-secondary" onClick={onCompare}>
23           <GitCompareArrows size={15} aria-hidden="true" /> 加入对比
24         </button>
25         <button type="button" className="btn btn-secondary" onClick={onRetry}>批量重试</button>
26         <button type="button" className="btn btn-secondary" onClick={onCancel}>批量取消</button>
27         <button type="button" className="btn btn-danger" onClick={onDelete}>批量删除</button>
28         <button type="button" className="btn btn-secondary" onClick={onClear}>清空选择</button>
29       </div>
30     </section>
31   );
32 }
33
```

```
// ===== 文件: front/src/workbench/tasks/components/TaskCompareTray.tsx =====
```

```
1 import { BarChart3, GitCompareArrows, X } from 'lucide-react';
```

```
front/src/workbench/tasks/components/TaskCompareTray.tsx
```

```
2 import type { AnalysisTask } from '../types';
3 import { formatPercent, formatRuntime } from '../taskStatus';
4
5 type Props = {
6   tasks: AnalysisTask[];
7   onRemove: (taskId: number) => void;
8   onClear: () => void;
9 };
10
11 export function TaskCompareTray({ tasks, onRemove, onClear }: Props) {
12   if (tasks.length === 0) return null;
13
14   return (
15     <section className="task-compare-tray panel" aria-label="任务对比栏">
16       <header className="task-compare-header">
17         <div>
18           <span className="task-compare-kicker">
19             <GitCompareArrows size={15} aria-hidden="true" />
20             任务对比
21           </span>
22           <h2>已选 {tasks.length} / 3 个任务</h2>
23         </div>
24         <button type="button" className="btn btn-secondary" onClick={onClear}>
25           清空对比
26         </button>
27       </header>
28
29       <div className="task-compare-grid">
30         {tasks.map((task) => (
31           <article key={task.id} className="task-compare-item">
32             <header>
33               <div>
34                 <strong>{task.name}</strong>
35                 <span>#{task.id} • {task.datasetName}</span>
36               </div>
37               <button
38                 type="button"
39                 onClick={() => onRemove(task.id)}
40                 title="移出对比"
41                 aria-label={`移出任务 ${task.id} 对比`}
42               >
43                 <X size={15} aria-hidden="true" />
44               </button>
45             </header>
46             <dl>
47               <div><dt>模式</dt><dd>{task.mode}</dd></div>
48               <div><dt>状态</dt><dd>{task.status}</dd></div>
49               <div><dt>耗时</dt><dd>{formatRuntime(task.runtimeSeconds)}</dd></div>
50               <div><dt>ACC</dt><dd>{formatPercent(task.metricsSummary?.acc)}</dd></div>
51               <div><dt>NMI</dt><dd>{formatPercent(task.metricsSummary?.nmi)}</dd></div>
```

```
front/src/workbench/tasks/components/TaskCompareTray.tsx
```

```
52         <div><dt>F1</dt><dd>{formatPercent(task.metricsSummary?.f1)}</dd></div>
53     </dl>
54 </article>
55 )))
56 </div>
57
58 <p className="task-compare-note">
59     <BarChart3 size={15} aria-hidden="true" />
60     仅比较当前已加载任务的任务元数据和已有指标；图表与结果文件将在算法服务接入后显示。
61 </p>
62 </section>
63 );
64 }
65
```

```
// ===== 文件: front/src/workbench/tasks/components/TaskCreateDrawer.tsx =====
```

```
1 import { useEffect, useMemo, useState } from 'react';
2 import { createPortal } from 'react-dom';
3 import { CheckCircle2, ChevronLeft, ChevronRight, Database, FileStack, Settings2, Trash2 } from 'lucide-react';
4 import { SelectField } from '../../../components/SelectField';
5 import type { CreateTaskPayload, DatasetOption, TaskMode, TaskParams, TaskTemplate } from '../../../types';
6
7 type CreateStep = 1 | 2 | 3;
8
9 type Props = {
10   open: boolean;
11   datasets: DatasetOption[];
12   templates: TaskTemplate[];
13   initialDatasetId?: number | null;
14   submitting?: boolean;
15   onClose: () => void;
16   onSubmit: (payload: CreateTaskPayload, saveAsTemplateName?: string) => Promise<void> | void;
17   onDeleteTemplate: (template: TaskTemplate) => void;
18 };
19
20 const DEFAULT_PARAMS: TaskParams = {
21   nBase: 20,
22   sigma: 1,
23   lambda: 5,
24   gamma: 5,
25   anchor: 10,
26   runs: 10,
27   maxIter: 10,
28   randomSeed: 1,
29 };
30
31 const STEPS: Array<{ key: CreateStep; label: string; icon: typeof Database }> = [
32   { key: 1, label: '选择数据', icon: Database },
33   { key: 2, label: '配置算法', icon: Settings2 },
34   { key: 3, label: '确认执行', icon: CheckCircle2 },
```

front/src/workbench/tasks/components/TaskCreateDrawer.tsx

```
35 ];
36
37 export function TaskCreateDrawer({
38   open,
39   datasets,
40   templates,
41   initialDatasetId,
42   submitting,
43   onClose,
44   onSubmit,
45   onDeleteTemplate,
46 }: Props) {
47   const [step, setStep] = useState<CreateStep>(1);
48   const [name, setName] = useState('');
49   const [mode, setMode] = useState<TaskMode>('OMELET-SV');
50   const [datasetId, setDatasetId] = useState('');
51   const [params, setParams] = useState<TaskParams>(DEFAULT_PARAMS);
52   const [startImmediately, setStartImmediately] = useState(true);
53   const [templateId, setTemplateId] = useState('');
54   const [templateName, setTemplateName] = useState('');
55   const [error, setError] = useState('');
56
57   const selectedDataset = useMemo(
58     () => datasets.find((item) => String(item.id) === datasetId) || null,
59     [datasets, datasetId],
60   );
61
62   useEffect(() => {
63     if (!open) return;
64     setStep(1);
65     setError('');
66     setTemplateName('');
67     if (initialDatasetId) {
68       setDatasetId(String(initialDatasetId));
69     } else if (datasets[0]) {
70       setDatasetId(String(datasets[0].id));
71     }
72   }, [open, initialDatasetId, datasets]);
73
74   useEffect(() => {
75     if (!selectedDataset) return;
76     setParams((prev) => ({ ...prev, nBase: Math.min(prev.nBase || 20, selectedDataset.baseCount || 20) }));
77     if (!name.trim()) setName(`${selectedDataset.name} ${mode} 任务`);
78   }, [selectedDataset?.id, mode]);
79
80   function applyTemplate(id: string) {
81     setTemplateId(id);
82     const template = templates.find((item) => String(item.id) === id);
83     if (!template) return;
84     setMode((template.mode as TaskMode) || 'OMELET-SV');
```

front/src/workbench/tasks/components/TaskCreateDrawer.tsx

```
85     setParams({
86       ...DEFAULT_PARAMS,
87       ...template.params,
88       lambda: Number(template.params.lambda ?? DEFAULT_PARAMS.lambda),
89       nBase: Number(template.params.nBase ?? DEFAULT_PARAMS.nBase),
90       sigma: Number(template.params.sigma ?? DEFAULT_PARAMS.sigma),
91       gamma: Number(template.params.gamma ?? DEFAULT_PARAMS.gamma),
92       anchor: Number(template.params.anchor ?? DEFAULT_PARAMS.anchor),
93       runs: Number(template.params.runs ?? DEFAULT_PARAMS.runs),
94       maxIter: Number(template.params.maxIter ?? DEFAULT_PARAMS.maxIter),
95       randomSeed: Number(template.params.randomSeed ?? DEFAULT_PARAMS.randomSeed),
96     });
97   }
98
99   function validateData() {
100     if (!datasetId) {
101       setError('请选择数据集');
102       return false;
103     }
104     if (selectedDataset?.qualityStatus === 'error') {
105       setError('该数据集质量检查未通过，无法创建任务');
106       return false;
107     }
108     if (selectedDataset?.hasLabels === false || (selectedDataset?.classCount ?? 0) < 2) {
109       setError('OMELET 分析需要包含至少两个类别的 y 标签');
110       return false;
111     }
112     return true;
113   }
114
115   function validateParams() {
116     if (!selectedDataset) return false;
117     if (!Number.isFinite(params.nBase) || params.nBase < 1 || params.nBase > selectedDataset.baseCount) {
118       setError(`n_base 应为 1 到 ${selectedDataset.baseCount} 之间的整数`);
119       return false;
120     }
121     if ([params.sigma, params.lambda, params.gamma, params.runs, params.maxIter].some((value) =>
122       !Number.isFinite(value) || value <= 0)) {
123       setError('sigma、lambda、gamma、runs 与 max_iter 必须大于 0');
124       return false;
125     }
126     if (!Number.isInteger(params.randomSeed) || params.randomSeed < 0) {
127       setError('random_seed 必须是大于或等于 0 的整数');
128       return false;
129     }
130     if (mode === 'OMELET-SV' && (!Number.isFinite(params.anchor) || params.anchor < 1)) {
131       setError('OMELET-SV 的 anchor 必须大于 0');
132       return false;
133     }
134     return true;
135   }
```

```
front/src/workbench/tasks/components/TaskCreateDrawer.tsx
```

```
134   }
135
136   function nextStep() {
137     setError('');
138     if (step === 1 && !validateData()) return;
139     if (step === 2 && !validateParams()) return;
140     setStep((current) => Math.min(3, current + 1) as CreateStep);
141   }
142
143   async function handleSubmit() {
144     setError('');
145     if (!validateData() || !validateParams()) return;
146
147     const payload: CreateTaskPayload = {
148       datasetId: Number(datasetId),
149       name: name.trim() || undefined,
150       mode,
151       startImmediately,
152       templateId: templateId ? Number(templateId) : undefined,
153       params: {
154         nBase: Number(params.nBase),
155         sigma: Number(params.sigma),
156         lambda: Number(params.lambda),
157         gamma: Number(params.gamma),
158         anchor: mode === 'OMELET-SV' ? Number(params.anchor) : 0,
159         runs: Number(params.runs),
160         maxIter: Number(params.maxIter),
161         randomSeed: Number(params.randomSeed),
162       },
163     };
164
165     try {
166       await onSubmit(payload, templateName.trim() || undefined);
167     } catch (err) {
168       setError(err instanceof Error ? err.message : '创建失败');
169     }
170   }
171
172   if (!open) return null;
173
174   return createPortal(
175     <div className="task-drawer-root" role="dialog" aria-modal="true" aria-label="新建任务">
176       <button type="button" className="task-drawer-mask" aria-label="关闭新建任务" onClick={onClose} />
177       <aside className="task-drawer-panel create">
178         <header className="task-drawer-header">
179           <div>
180             <h2>新建分析任务</h2>
181             <span>将数据集、算法参数和执行计划整理为可追踪的实验任务。</span>
182           </div>
183           <button type="button" className="btn btn-secondary" onClick={onClose}>关闭</button>
```

```
front/src/workbench/tasks/components/TaskCreateDrawer.tsx
```

```
184         </header>
185
186         <div className="task-create-stepper" aria-label="创建任务步骤">
187             {STEPS.map((item) => {
188                 const Icon = item.icon;
189                 return (
190                     <div key={item.key} className={step === item.key ? 'active' : step > item.key ? 'done' : ''}>
191                         <span><Icon size={15} aria-hidden="true" /></span>
192                         <strong>{item.key} {item.label}</strong>
193                     </div>
194                 );
195             })}
196         </div>
197
198         <div className="task-drawer-body">
199             {step === 1 ? (
200                 <>
201                     <label className="task-form-field">
202                         <span>任务名称</span>
203                         <input value={name} onChange={(event) => setName(event.target.value)} placeholder="请输入便于识别的任务名称" />
204                     </label>
205
206                     <SelectField
207                         className="task-form-field"
208                         label="选择数据集"
209                         value={datasetId}
210                         options={[
211                             { value: '', label: '请选择数据集' },
212                             ...datasets.map((item) => ({
213                                 value: String(item.id),
214                                 label: `${item.name} (m=${item.baseCount}, n=${item.sampleCount})${item.hasLabels ?
215                                     `, ${item.classCount} 类标签` : ', 无标签'})${item.qualityStatus === 'error' ? ', 质量异常' : ''})`),
216                             ]}
217                         onChange={setDatasetId}
218                     />
219
220                     {selectedDataset ? (
221                         <section className={`task-dataset-snapshot quality-${selectedDataset.qualityStatus}`}>
222                             <header><Database size={16} aria-hidden="true" /><strong>{selectedDataset.name}</strong></header>
223                             <dl>
224                                 <div><dt>样本数量</dt><dd>{selectedDataset.sampleCount}</dd></div>
225                                 <div><dt>基础聚类</dt><dd>{selectedDataset.baseCount}</dd></div>
226                                 <div><dt>质量状态</dt><dd>{selectedDataset.qualityStatus === 'error' ? '需处理' :
selectedDataset.qualityStatus === 'warning' ? '待确认' : '检查通过'}</dd></div>
227                             </dl>
228                             {selectedDataset.qualityIssues?.length ? <p>{selectedDataset.qualityIssues.join('; ')}</p> : null}
229                         </section>
230                     ) : null}
231
```

```
front/src/workbench/tasks/components/TaskCreateDrawer.tsx
```

```
232         <section className="task-template-picker">
233             <div className="task-template-picker-heading"><FileStack size={16} aria-hidden="true"
        /><strong>参数模板</strong></div>
234             <SelectField
235                 className="task-form-field"
236                 label="从模板填充"
237                 value={templateId}
238                 options=[
239                     { value: '', label: '不使用模板' },
240                     ...templates.map((item) => ({
241                         value: String(item.id),
242                         label: `${item.name} (${item.mode})`,
243                     })),
244                 ]
245                 onChange={applyTemplate}
246             />
247             {templates.length > 0 ? (
248                 <div className="task-template-list">
249                     {templates.map((item) => (
250                         <div key={item.id}>
251                             <span>{item.name} • {item.mode}</span>
252                             <button type="button" title="删除模板" aria-label={`删除模板 ${item.name}`} onClick={() =>
onDeleteTemplate(item)}>
253                                 <Trash2 size={14} aria-hidden="true" />
254                             </button>
255                         </div>
256                     ))}
257                 </div>
258             ) : <small>尚无已保存模板，可在最后一步保存当前参数。</small>}
259         </section>
260     </>
261 ) : null}
262
263 {step === 2 ? (
264     <>
265         <section className="task-mode-control" aria-label="算法模式">
266             {[ 'OMELET', 'OMELET-SV' ] as TaskMode[]}.map((item) => (
267                 <button key={item} type="button" className={mode === item ? 'active' : ''} onClick={() =>
setMode(item)}>{item}</button>
268             ))}
269         </section>
270         <p className="task-parameter-note">OMELET-SV 适用于大规模数据并启用 anchor；OMELET 使用完整相似性计算。</p>
271         <div className="task-create-grid">
272             {(
273                 [
274                     ['nBase', 'n_base', 1, 1],
275                     ['sigma', 'sigma', 0.1, 0.1],
276                     ['lambda', 'lambda', 0.1, 0.1],
277                     ['gamma', 'gamma', 0.1, 0.1],
278                     ['runs', 'runs', 1, 1],
```



```
front/src/workbench/tasks/components/TaskCreateDrawer.tsx
```

```
279         ['maxIter', 'max_iter', 1, 1],
280         ['randomSeed', 'random_seed', 0, 1],
281     ] as const
282   ).map(([key, label, min, stepValue]) => (
283     <label key={key} className="task-form-field">
284       <span>{label}</span>
285       <input type="number" min={min} step={stepValue} value={params[key]} onChange={(event) =>
setParams((prev) => ({ ...prev, [key]: Number(event.target.value) }))) />
286     </label>
287   ))}
288   {mode === 'OMELET-SV' ? (
289     <label className="task-form-field">
290       <span>anchor</span>
291       <input type="number" min={1} step={1} value={params.anchor} onChange={(event) => setParams((prev)
=> ({ ...prev, anchor: Number(event.target.value) }))) />
292     </label>
293   ) : null}
294 </div>
295 </>
296 ) : null}
297
298 {step === 3 ? (
299   <>
300     <section className="task-create-summary">
301       <header><CheckCircle2 size={17} aria-hidden="true" /><strong>执行摘要</strong></header>
302       <dl>
303         <div><dt>任务</dt><dd>{name || '未命名任务'}</dd></div>
304         <div><dt>数据集</dt><dd>{selectedDataset?.name || '-'}</dd></div>
305         <div><dt>算法</dt><dd>{mode}</dd></div>
306         <div><dt>运行次数</dt><dd>{params.runs}</dd></div>
307         <div><dt>迭代上限</dt><dd>{params.maxIter}</dd></div>
308         <div><dt>基础聚类</dt><dd>{params.nBase}</dd></div>
309       </dl>
310     </section>
311     <label className="task-check-field">
312       <input type="checkbox" checked={startImmediately} onChange={(event) =>
setStartImmediately(event.target.checked)} />
313       <span>创建后立即启动</span>
314     </label>
315     <label className="task-form-field">
316       <span>同时保存为模板（可选）</span>
317       <input value={templateName} onChange={(event) => setTemplateName(event.target.value)} placeholder="例如
默认 OMELET-SV 参数" />
318     </label>
319   </>
320 ) : null}
321
322 {error ? <div className="task-error-box">{error}</div> : null}
323 </div>
324
```

```
front/src/workbench/tasks/components/TaskCreateDrawer.tsx
```

```
325     <footer className="task-drawer-footer">
326       {step > 1 ? (
327         <button type="button" className="btn btn-secondary" onClick={() => setStep((current) => Math.max(1,
current - 1) as CreateStep)}>
328         <ChevronLeft size={16} aria-hidden="true" /> 上一步
329       </button>
330     ) : null}
331     <button type="button" className="btn btn-secondary" onClick={onClose}>取消</button>
332     {step < 3 ? (
333       <button type="button" className="btn btn-primary" onClick={nextStep}>
334       下一步 <ChevronRight size={16} aria-hidden="true" />
335     </button>
336     ) : (
337       <button type="button" className="btn btn-primary" disabled={!submitting} onClick={handleSubmit}>
338       {submitting ? '提交中...' : startImmediately ? '创建并启动任务' : '创建任务草稿'}
339     </button>
340   )}
341 </footer>
342 </aside>
343 </div>,
344 document.body,
345 );
346 }
347
```

```
// ===== 文件: front/src/workbench/tasks/components/TaskDetailDrawer.tsx =====
```

```
1 import { useEffect, useState } from 'react';
2 import { createPortal } from 'react-dom';
3 import { Bookmark, NotebookPen, Plus, Star, Tag, X } from 'lucide-react';
4 import { SelectField } from '../../../components/SelectField';
5 import { formatLocalDateTime } from '../../../utils/time';
6 import type { TaskLocalMeta } from '../../../taskLocalMeta';
7 import type { AnalysisTask, AnalysisTaskLog } from '../../../types';
8 import {
9   buildPipelineState,
10  canCancel,
11  canDelete,
12  canRetry,
13  canStart,
14  failureReasonLabel,
15  formatPercent,
16  formatRuntime,
17  getTaskRunProgress,
18  getStatusMeta,
19  stageLabel,
20 } from '../../../taskStatus';
21
22 type DetailTab = 'overview' | 'params' | 'logs' | 'notes';
23
24 type Props = {
```

```
front/src/workbench/tasks/components/TaskDetailDrawer.tsx
```

```
25   open: boolean;
26   task: AnalysisTask | null;
27   logs: AnalysisTaskLog[];
28   localMeta: TaskLocalMeta;
29   loadingLogs?: boolean;
30   onClose: () => void;
31   onStart: (task: AnalysisTask) => void;
32   onCancel: (task: AnalysisTask) => void;
33   onRetry: (task: AnalysisTask) => void;
34   onClone: (task: AnalysisTask) => void;
35   onDelete: (task: AnalysisTask) => void;
36   onOpenResults: (task: AnalysisTask) => void;
37   onOpenExport: (task: AnalysisTask) => void;
38   onOpenReport: (task: AnalysisTask) => void;
39   onUpdateLocalMeta: (patch: Partial<TaskLocalMeta>) => void;
40 };
41
42 const TABS: Array<{ key: DetailTab; label: string }> = [
43   { key: 'overview', label: '任务概览' },
44   { key: 'params', label: '参数与数据' },
45   { key: 'logs', label: '执行记录' },
46   { key: 'notes', label: '研究备注' },
47 ];
48
49 export function TaskDetailDrawer({
50   open,
51   task,
52   logs,
53   localMeta,
54   loadingLogs,
55   onClose,
56   onStart,
57   onCancel,
58   onRetry,
59   onClone,
60   onDelete,
61   onOpenResults,
62   onOpenExport,
63   onOpenReport,
64   onUpdateLocalMeta,
65 }: Props) {
66   const [activeTab, setActiveTab] = useState<DetailTab>('overview');
67   const [newTag, setNewTag] = useState('');
68
69   useEffect(() => {
70     setActiveTab('overview');
71     setNewTag('');
72   }, [task?.id]);
73
74   if (!open || !task) return null;
```

```
front/src/workbench/tasks/components/TaskDetailDrawer.tsx
```

```
75
76   const meta = getStatusMeta(task.status);
77   const runProgress = getTaskRunProgress(task);
78   const pipeline = buildPipelineState(task);
79   const params = task.params || {};
80   const tags = localMeta.tags || [];
81
82   function addTag() {
83     const tag = newTag.trim().slice(0, 20);
84     if (!tag || tags.includes(tag)) return;
85     onUpdateLocalMeta({ tags: [...tags, tag] });
86     setNewTag('');
87   }
88
89   return createPortal(
90     <div className="task-drawer-root" role="dialog" aria-modal="true" aria-label="任务详情">
91       <button type="button" className="task-drawer-mask" aria-label="关闭详情" onClick={onClose} />
92       <aside className="task-drawer-panel">
93         <header className="task-drawer-header">
94           <div>
95             <div className="task-drawer-title-row">
96               <h2>{task.name}</h2>
97               <button
98                 type="button"
99                 className={`task-icon-button ${localMeta.favorite ? 'is-active' : ''}`}
100                 onClick={() => onUpdateLocalMeta({ favorite: !localMeta.favorite })}
101                 title={localMeta.favorite ? '取消关注' : '关注任务'}
102                 aria-label={localMeta.favorite ? '取消关注任务' : '关注任务'}
103               >
104                 <Star size={16} fill={localMeta.favorite ? 'currentColor' : 'none'} aria-hidden="true" />
105               </button>
106             </div>
107             <span>#{task.id} • {task.mode} • {task.datasetName}</span>
108           </div>
109           <button type="button" className="btn btn-secondary" onClick={onClose}>
110             关闭
111           </button>
112         </header>
113
114         <div className="task-detail-tabs" role="tablist" aria-label="任务详情分区">
115           {TABS.map((tab) => (
116             <button
117               key={tab.key}
118               type="button"
119               role="tab"
120               aria-selected={activeTab === tab.key}
121               className={activeTab === tab.key ? 'active' : ''}
122               onClick={() => setActiveTab(tab.key)}
123             >
124               {tab.label}
```

```
front/src/workbench/tasks/components/TaskDetailDrawer.tsx
```

```
125         </button>
126     }}}
127 </div>
128
129     <div className="task-drawer-body">
130         {activeTab === 'overview' ? (
131             <>
132                 <section className="task-detail-block">
133                     <h3>状态总览</h3>
134                     <div className="task-detail-grid">
135                         <div><span>状态</span><strong className={`task-status-pill
tone-${meta.tone}`}>{meta.label}</strong></div>
136                         <div><span>总体进度</span><strong>{Math.round(task.progress || 0)}%</strong></div>
137                         <div><span>当前轮次</span><strong>{runProgress.current > 0 ? `第
${runProgress.current}/${runProgress.total} 轮` : `等待开始, 共 ${runProgress.total} 轮`}</strong></div>
138                         <div><span>当前迭代</span><strong>{task.currentIter}/${task.maxIter}</strong></div>
139                         <div><span>当前阶段</span><strong>{stageLabel(task.currentStage)}</strong></div>
140                         <div><span>耗时</span><strong>{formatRuntime(task.runtimeSeconds)}</strong></div>
141                         <div><span>创建时间</span><strong>{formatLocalDateTime(task.createdAt)}</strong></div>
142                         <div><span>开始 / 结束</span><strong>{formatLocalDateTime(task.startedAt)} /
{formatLocalDateTime(task.finishedAt)}</strong></div>
143                     </div>
144                     {task.errorMessage ? (
145                         <div className="task-error-box">
146                             <strong>{failureReasonLabel(task.failureReason)}</strong>
147                             <p>{task.errorMessage}</p>
148                         </div>
149                     ) : null}
150                 </section>
151
152                 <section className="task-detail-block">
153                     <h3>执行流水线</h3>
154                     <div className="task-pipeline-list">
155                         {pipeline.map((step, index) => (
156                             <article key={step.key} className={`task-pipeline-item state-${step.state}`}>
157                                 <span>{index + 1}</span>
158                                 <div>
159                                     <strong>{step.label}</strong>
160                                     <small>{step.state === 'done' ? '完成' : step.state === 'running' ? '进行中' : step.state ===
'failed' ? '失败' : '等待'}</small>
161                                 </div>
162                             </article>
163                         ))}
164                     </div>
165                 </section>
166
167                 {task.metricsSummary ? (
168                     <section className="task-detail-block">
169                         <h3>指标摘要</h3>
170                         <div className="task-metric-grid">
```

```
front/src/workbench/tasks/components/TaskDetailDrawer.tsx
```

```
171         <div><span>ACC</span><strong>{formatPercent(task.metricsSummary.acc)}</strong></div>
172         <div><span>NMI</span><strong>{formatPercent(task.metricsSummary.nmi)}</strong></div>
173         <div><span>ARI</span><strong>{formatPercent(task.metricsSummary.ari)}</strong></div>
174         <div><span>F1</span><strong>{formatPercent(task.metricsSummary.f1)}</strong></div>
175     </div>
176 </section>
177 ) : null}
178 </>
179 ) : null}
180
181 {activeTab === 'params' ? (
182     <>
183         <section className="task-detail-block">
184             <h3>任务参数</h3>
185             <dl className="task-param-grid">
186                 {Object.entries(params).map(([key, value]) => (
187                     <div key={key}><dt>{key}</dt><dd>{String(value)}</dd></div>
188                 ))}
189             </dl>
190         </section>
191         <section className="task-detail-block">
192             <h3>数据集快照</h3>
193             <div className="task-detail-grid">
194                 <div><span>数据集</span><strong>{task.datasetName}</strong></div>
195                 <div><span>算法模式</span><strong>{task.mode}</strong></div>
196                 <div><span>基础聚类</span><strong>n_base = {String(params.nBase ?? '-')}</strong></div>
197                 <div><span>运行次数</span><strong>{String(params.runs ?? '-')}</strong></div>
198             </div>
199         </section>
200     </>
201 ) : null}
202
203 {activeTab === 'logs' ? (
204     <section className="task-detail-block">
205         <h3>运行日志</h3>
206         <div className="task-log-list">
207             {loadingLogs ? <p>日志加载中...</p> : null}
208             {!loadingLogs && logs.length === 0 ? <p>暂无日志</p> : null}
209             {logs.map((log) => (
210                 <article key={log.id} className={`task-log-item level-${log.level}`}>
211                     <header><strong>{log.action}</strong><span>{formatLocalDateTime(log.createdAt)}</span></header>
212                     <p>{log.message}</p>
213                 </article>
214             ))}
215         </div>
216     </section>
217 ) : null}
218
219 {activeTab === 'notes' ? (
220     <section className="task-detail-block task-local-meta-block">
```

```
front/src/workbench/tasks/components/TaskDetailDrawer.tsx
```

```
221         <div className="task-local-meta-heading">
222             <div>
223                 <span><NotebookPen size={16} aria-hidden="true" /> 本地管理信息</span>
224                 <p>仅保存在当前浏览器会话，不会影响服务端任务调度。</p>
225             </div>
226             <Bookmark size={18} aria-hidden="true" />
227         </div>
228
229         <SelectField
230             className="task-form-field"
231             label="本地优先级"
232             value={localMeta.priority || 'normal'}
233             options={[
234                 { value: 'high', label: '高优先级' },
235                 { value: 'normal', label: '常规优先级' },
236                 { value: 'low', label: '低优先级' },
237             ]}
238             onChange={(value) => onUpdateLocalMeta({ priority: value as TaskLocalMeta['priority'] })}
239         />
240
241         <div className="task-tag-editor">
242             <span><Tag size={15} aria-hidden="true" /> 任务标签</span>
243             <div className="task-tag-list">
244                 {tags.length === 0 ? <small>尚未添加标签</small> : null}
245                 {tags.map((tag) => (
246                     <span key={tag} className="task-local-tag">
247                         {tag}
248                         <button type="button" onClick={() => onUpdateLocalMeta({ tags: tags.filter((item) => item !==
tag) })} aria-label={`删除标签 ${tag}`}>
249                             <X size={12} aria-hidden="true" />
250                         </button>
251                     </span>
252                 ))}
253             </div>
254             <div className="task-add-tag">
255                 <input
256                     aria-label="新增任务标签"
257                     value={newTag}
258                     maxLength={20}
259                     placeholder="例如：待复核"
260                     onChange={(event) => setNewTag(event.target.value)}
261                     onKeyDown={(event) => {
262                         if (event.key === 'Enter') {
263                             event.preventDefault();
264                             addTag();
265                         }
266                     }}
267                 />
268                 <button type="button" onClick={addTag} aria-label="添加任务标签" title="添加任务标签">
269                     <Plus size={15} aria-hidden="true" />
```

```
front/src/workbench/tasks/components/TaskDetailDrawer.tsx
```

```
270         </button>
271     </div>
272 </div>
273
274     <label className="task-form-field">
275         <span>研究备注</span>
276         <textarea
277             aria-label="研究备注内容"
278             value={localMeta.note || ''}
279             placeholder="记录本次实验的假设、观察或后续复核事项。"
280             onChange={(event) => onUpdateLocalMeta({ note: event.target.value.slice(0, 1000) })}
281         />
282     </label>
283 </section>
284 ) : null}
285 </div>
286
287     <footer className="task-drawer-footer">
288         {canStart(task.status) ? <button type="button" className="btn btn-primary" onClick={() =>
onStart(task)}>启动</button> : null}
289         {canCancel(task.status) ? <button type="button" className="btn btn-secondary" onClick={() =>
onCancel(task)}>取消</button> : null}
290         {canRetry(task.status) ? <button type="button" className="btn btn-primary" onClick={() =>
onRetry(task)}>重试</button> : null}
291         <button type="button" className="btn btn-secondary" onClick={() => onClone(task)}>克隆</button>
292         {task.status === 'succeeded' ? (
293             <>
294                 <button type="button" className="btn btn-secondary" onClick={() => onOpenResults(task)}>结果分析</button>
295                 <button type="button" className="btn btn-secondary" onClick={() => onOpenExport(task)}>结果导出</button>
296                 <button type="button" className="btn btn-secondary" onClick={() => onOpenReport(task)}>分析报告</button>
297             </>
298         ) : null}
299         {canDelete(task.status) ? <button type="button" className="btn btn-danger" onClick={() =>
onDelete(task)}>删除</button> : null}
300     </footer>
301 </aside>
302 </div>,
303     document.body,
304 );
305 }
306
```

```
// ===== 文件: front/src/workbench/tasks/components/TaskEmptyState.tsx =====
```

```
1 import { ArrowRight, CirclePlus, Database, FileStack } from 'lucide-react';
2
3 type Props = {
4     onCreate: () => void;
5     onUseTemplate: () => void;
6     onOpenDatasets: () => void;
7 };
```



```
front/src/workbench/tasks/components/TaskEmptyState.tsx
```

```
8
9 export function TaskEmptyState({ onCreate, onUseTemplate, onOpenDatasets }: Props) {
10   return (
11     <section className="task-empty-state panel" aria-label="任务创建引导">
12       <div className="task-empty-copy">
13         <span className="task-empty-kicker">实验工作流从这里开始</span>
14         <h2>从第一个分析任务开始</h2>
15         <p>任务会固定本次的数据集、算法模式和核心参数，后续可在同一处跟踪进度、诊断异常和查看结果。</p>
16         <div className="task-empty-actions">
17           <button type="button" className="btn btn-primary" onClick={onCreate}>
18             <CirclePlus size={16} aria-hidden="true" />
19             新建分析任务
20           </button>
21           <button type="button" className="btn btn-secondary" onClick={onUseTemplate}>
22             <FileStack size={16} aria-hidden="true" />
23             从模板创建
24           </button>
25         </div>
26       </div>
27
28       <ol className="task-empty-steps">
29         <li>
30           <span>1</span>
31           <div>
32             <strong>选择数据集并确认质量状态</strong>
33             <small>导入基础聚类结果后，先核查样本、标签与基础聚类数量。</small>
34           </div>
35           <button type="button" onClick={onOpenDatasets} title="前往数据管理" aria-label="前往数据管理">
36             <Database size={16} aria-hidden="true" />
37           </button>
38         </li>
39         <li>
40           <span>2</span>
41           <div>
42             <strong>配置 OMELET 分析参数</strong>
43             <small>选择算法模式，设置 n_base、核函数相关参数、运行次数与迭代上限。</small>
44           </div>
45           <ArrowRight size={16} aria-hidden="true" />
46         </li>
47         <li>
48           <span>3</span>
49           <div>
50             <strong>追踪运行并交付结果</strong>
51             <small>在任务详情中查看进度、日志、指标摘要和后续结果入口。</small>
52           </div>
53           <ArrowRight size={16} aria-hidden="true" />
54         </li>
55       </ol>
56     </section>
57   );
```

front/src/workbench/tasks/components/TaskEmptyState.tsx

```
58 }
59
// ===== 文件: front/src/workbench/tasks/components/TaskFiltersBar.tsx =====
1 import { useEffect, useId, useMemo, useState } from 'react';
2 import {
3   CalendarRange,
4   ChevronDown,
5   Download,
6   Plus,
7   RefreshCw,
8   Search,
9   SlidersHorizontal,
10  X,
11 } from 'lucide-react';
12 import { SelectField } from '../../../components/SelectField';
13 import type { DatasetOption, TaskMode } from '../types';
14
15 type Props = {
16   keyword: string;
17   status: string;
18   mode: string;
19   datasetId: string;
20   createdFrom: string;
21   createdTo: string;
22   datasets: DatasetOption[];
23   total: number;
24   loading: boolean;
25   onKeywordChange: (value: string) => void;
26   onStatusChange: (value: string) => void;
27   onModeChange: (value: string) => void;
28   onDatasetChange: (value: string) => void;
29   onCreatedFromChange: (value: string) => void;
30   onCreatedToChange: (value: string) => void;
31   onClearDates: () => void;
32   onRefresh: () => void;
33   onReset: () => void;
34   onCreate: () => void;
35   onExport: () => void;
36 };
37
38 const STATUS_OPTIONS = [
39   { value: '', label: '全部状态' },
40   { value: 'draft', label: '草稿' },
41   { value: 'queued', label: '排队中' },
42   { value: 'running', label: '运行中' },
43   { value: 'succeeded', label: '成功' },
44   { value: 'failed', label: '失败' },
45   { value: 'cancelled', label: '已取消' },
46   { value: 'running,failed', label: '运行中 + 失败' },
```

front/src/workbench/tasks/components/TaskFiltersBar.tsx

```
47 ];
48
49 const MODE_OPTIONS: Array<{ value: '' | TaskMode; label: string }> = [
50   { value: '', label: '全部模式' },
51   { value: 'OMELET', label: 'OMELET' },
52   { value: 'OMELET-SV', label: 'OMELET-SV' },
53 ];
54
55 export function TaskFiltersBar({
56   keyword,
57   status,
58   mode,
59   datasetId,
60   createdFrom,
61   createdTo,
62   datasets,
63   total,
64   loading,
65   onKeywordChange,
66   onStatusChange,
67   onModeChange,
68   onDatasetChange,
69   onCreatedFromChange,
70   onCreatedToChange,
71   onClearDates,
72   onRefresh,
73   onReset,
74   onCreate,
75   onExport,
76 }: Props) {
77   const advancedId = useId();
78   const [advancedOpen, setAdvancedOpen] = useState(Boolean(createdFrom || createdTo));
79   const advancedFilterCount = Number(Boolean(createdFrom)) + Number(Boolean(createdTo));
80
81   useEffect(() => {
82     if (createdFrom || createdTo) setAdvancedOpen(true);
83   }, [createdFrom, createdTo]);
84
85   // 摘要项与筛选值共用同一组回调，保证移除单项条件时仍沿用任务中心的分页重置和 URL 同步逻辑。
86   const activeFilters = useMemo(() => {
87     const filters: Array<{ key: string; label: string; onRemove: () => void }> = [];
88     const selectedStatus = STATUS_OPTIONS.find((option) => option.value === status);
89     const selectedMode = MODE_OPTIONS.find((option) => option.value === mode);
90     const selectedDataset = datasets.find((dataset) => String(dataset.id) === datasetId);
91
92     if (keyword) filters.push({ key: 'keyword', label: `关键词: ${keyword}`, onRemove: () => onKeywordChange('') });
93     if (selectedStatus?.value) filters.push({ key: 'status', label: selectedStatus.label, onRemove: () =>
94       onStatusChange('') });
95     if (selectedMode?.value) filters.push({ key: 'mode', label: selectedMode.label, onRemove: () => onModeChange('')
96     });
97   }, [keyword, status, mode, datasetId, createdFrom, createdTo, datasets, STATUS_OPTIONS, MODE_OPTIONS]);
```

```
front/src/workbench/tasks/components/TaskFiltersBar.tsx
```

```
95     if (selectedDataset) filters.push({ key: 'dataset', label: selectedDataset.name, onRemove: () =>
96       onDatasetChange('') });
97     if (createdFrom || createdTo) {
98       filters.push({
99         key: 'date',
100         label: `${createdFrom || '不限'} 至 ${createdTo || '不限'}`,
101         onRemove: onClearDates,
102       });
103     }
104     return filters;
105   }, [
106     createdFrom,
107     createdTo,
108     datasetId,
109     datasets,
110     keyword,
111     mode,
112     onCreatedFromChange,
113     onCreatedToChange,
114     onClearDates,
115     onDatasetChange,
116     onKeywordChange,
117     onModeChange,
118     onStatusChange,
119     status,
120   ]);
121
122   const resetFilters = () => {
123     setAdvancedOpen(false);
124     onReset();
125   };
126
127   return (
128     <section className="task-center-toolbar panel" aria-label="任务筛选与操作">
129       <div className="task-toolbar-header">
130         <div className="task-toolbar-title">
131           <span className="task-toolbar-title-icon" aria-hidden="true"><Search size={17} /></span>
132           <div>
133             <h2>任务检索</h2>
134             <span>{loading ? '正在更新任务记录...' : `已匹配 ${total} 条任务记录`}</span>
135           </div>
136         </div>
137         <div className="task-toolbar-actions">
138           <button type="button" className="btn btn-secondary" onClick={onRefresh} disabled={loading}
139             aria-busy={loading}>
140             <RefreshCw className={loading ? 'is-spinning' : ''} size={15} aria-hidden="true" />
141             {loading ? '更新中' : '刷新'}
142           </button>
143           <button type="button" className="btn btn-secondary" onClick={onExport}>
144             <Download size={15} aria-hidden="true" /> 导出清单
145         </div>
146       </div>
147     </section>
148   );
149 }
```

```
front/src/workbench/tasks/components/TaskFiltersBar.tsx
```

```
143         </button>
144         <button type="button" className="btn btn-primary" onClick={onCreate}>
145             <Plus size={16} aria-hidden="true" /> 新建任务
146         </button>
147     </div>
148 </div>
149
150 <div className="task-filter-fields">
151     <label className="task-filter-keyword">
152         <span>关键词</span>
153         <span className={`task-filter-search${keyword ? ' has-value' : ''}`}>
154             <Search size={15} aria-hidden="true" />
155             <input value={keyword} placeholder="搜索任务名称、数据集或模式" onChange={(event) =>
onKeywordChange(event.target.value)} />
156             {keyword ? (
157                 <button type="button" onClick={() => onKeywordChange('')} aria-label="清空关键词" title="清空关键词">
158                     <X size={14} aria-hidden="true" />
159                 </button>
160             ) : null}
161         </span>
162     </label>
163     <SelectField
164         label="状态"
165         value={status}
166         options={STATUS_OPTIONS}
167         onChange={onStatusChange}
168     />
169     <SelectField
170         label="算法模式"
171         value={mode}
172         options={MODE_OPTIONS}
173         onChange={onModeChange}
174     />
175     <SelectField
176         label="数据集"
177         value={datasetId}
178         options={[
179             { value: '', label: '全部数据集' },
180             ...datasets.map((item) => ({ value: String(item.id), label: item.name })),
181         ]}
182         onChange={onDatasetChange}
183     />
184     <button
185         type="button"
186         className={`task-filter-advanced-toggle${advancedOpen ? ' is-open' : ''}${advancedFilterCount ? ' has-value'
: ''}`}
187         aria-expanded={advancedOpen}
188         aria-controls={advancedId}
189         onClick={() => setAdvancedOpen((open) => !open)}
190     >
```

```
front/src/workbench/tasks/components/TaskFiltersBar.tsx
```

```
191         <SlidersHorizontal size={15} aria-hidden="true" />
192         <span>高级筛选</span>
193         {advancedFilterCount ? <strong>{advancedFilterCount}</strong> : null}
194         <ChevronDown className="task-filter-advanced-chevron" size={15} aria-hidden="true" />
195     </button>
196 </div>
197
198 <div
199     id={advancedId}
200     className={`task-filter-advanced${advancedOpen ? ' is-open' : ''}`}
201     aria-hidden={!advancedOpen}
202 >
203     <div className="task-filter-advanced-inner">
204         <div className="task-filter-advanced-heading">
205             <CalendarRange size={16} aria-hidden="true" />
206             <div><strong>创建日期</strong><span>限定任务记录的创建时间范围</span></div>
207         </div>
208         <div className="task-filter-date-range" role="group" aria-label="创建日期范围">
209             <label>
210                 <span>开始日期</span>
211                 <input aria-label="创建开始日期" type="date" value={createdFrom} disabled={!advancedOpen} onChange={(event) =>
onCreatedFromChange(event, target.value)} />
212             </label>
213             <span className="task-filter-date-separator" aria-hidden="true">至</span>
214             <label>
215                 <span>结束日期</span>
216                 <input aria-label="创建结束日期" type="date" value={createdTo} disabled={!advancedOpen} onChange={(event) =>
onCreatedToChange(event, target.value)} />
217             </label>
218         </div>
219         <button type="button" className="task-filter-clear-dates" onClick={onClearDates}
disabled={!advancedFilterCount || !advancedOpen}>
220             清除日期
221         </button>
222     </div>
223 </div>
224
225 <div className={`task-filter-summary${activeFilters.length ? ' has-filters' : ''}`} aria-live="polite">
226     <div className="task-filter-summary-inner">
227         <span className="task-filter-summary-label">当前条件</span>
228         <div className="task-filter-chips">
229             {activeFilters.map((filter) => (
230                 <button type="button" key={filter.key} onClick={filter.onRemove} title={`移除${filter.label}`}>
231                     <span>{filter.label}</span><X size={12} aria-hidden="true" />
232                 </button>
233             ))}
234         </div>
235         <button type="button" className="task-filter-reset" onClick={resetFilters}>清除全部</button>
236     </div>
237 </div>
```

```
front/src/workbench/tasks/components/TaskFiltersBar.tsx
```

```
238     </section>
239   );
240 }
241
// ===== 文件: front/src/workbench/tasks/components/TaskFocusPanel.tsx =====
1 import { ArrowUpRight, CircleAlert, Play, RadioTower } from 'lucide-react';
2 import type { AnalysisTask } from '../types';
3 import { formatRuntime, getStatusMeta, getTaskRunProgress, stageLabel } from '../taskStatus';
4
5 type Props = {
6   tasks: AnalysisTask[];
7   onOpenTask: (task: AnalysisTask) => void;
8   onStartTask: (task: AnalysisTask) => void;
9 };
10
11 export function TaskFocusPanel({ tasks, onOpenTask, onStartTask }: Props) {
12   const runningTask = tasks.find((task) => task.status === 'running' || task.status === 'queued');
13   const attentionTask = tasks.find((task) => task.status === 'failed') || tasks.find((task) => task.status ===
    'draft');
14   const runProgress = runningTask ? getTaskRunProgress(runningTask) : null;
15
16   if (!runningTask && !attentionTask) return null;
17
18   return (
19     <section className="task-focus-grid" aria-label="任务运行概览">
20       {runningTask ? (
21         <article className="task-focus-card primary">
22           <header>
23             <span><RadioTower size={16} aria-hidden="true" /> 当前执行</span>
24             <button type="button" onClick={() => onOpenTask(runningTask)}>
25               查看详情 <ArrowUpRight size={15} aria-hidden="true" />
26             </button>
27           </header>
28           <strong>{runningTask.name}</strong>
29           <p>
30             {stageLabel(runningTask.currentStage)}
31             { ' • ' }
32             {runningTask.status === 'queued'
33               ? `等待执行, 共 ${runProgress?.total ?? 1} 轮`
34               : `第 ${runProgress?.current ?? 1}/${runProgress?.total ?? 1} 轮 • 迭代
35                 ${runningTask.currentIter}/${runningTask.maxIter}`}
36             { ' • ' } 已运行 {formatRuntime(runningTask.runtimeSeconds)}
37           </p>
38           <div className="task-focus-progress" aria-label={`任务进度 ${Math.round(runningTask.progress)}%`}>
39             <span style={{ transform: `scaleX(${Math.max(0, Math.min(100, runningTask.progress || 0)) / 100})` }} />
40           </div>
41         </article>
42       ) : null}
```

```
front/src/workbench/tasks/components/TaskFocusPanel.tsx
```

```
43     {attentionTask ? (  
44       <article className="task-focus-card attention">  
45         <header>  
46           <span><CircleAlert size={16} aria-hidden="true" /> 需要处理</span>  
47           <button type="button" onClick={() => onOpenTask(attentionTask)}>  
48             查看详情 <ArrowUpRight size={15} aria-hidden="true" />  
49           </button>  
50         </header>  
51         <strong>{attentionTask.name}</strong>  
52         <p>{getStatusMeta(attentionTask.status).description}</p>  
53         {attentionTask.status === 'draft' ? (  
54           <button type="button" className="btn btn-primary task-focus-start" onClick={() =>  
onStartTask(attentionTask)}>  
55             <Play size={15} aria-hidden="true" /> 启动草稿任务  
56           </button>  
57         ) : null}  
58       </article>  
59     ) : null}  
60 </section>  
61 );  
62 }  
63
```

```
// ===== 文件: front/src/workbench/tasks/components/TaskStatsBar.tsx =====
```

```
1 import type { AnalysisTaskStats } from '../types';  
2 import { formatRuntime } from '../taskStatus';  
3  
4 type Props = {  
5   stats: AnalysisTaskStats | null;  
6   loading?: boolean;  
7   onQuickFilter?: (status: string) => void;  
8   activeStatus?: string;  
9 };  
10  
11 const CARDS: Array<{ key: keyof AnalysisTaskStats; label: string; status?: string; tone: string }> = [  
12   { key: 'total', label: '全部任务', tone: 'neutral' },  
13   { key: 'queued', label: '排队中', status: 'queued', tone: 'info' },  
14   { key: 'running', label: '运行中', status: 'running', tone: 'running' },  
15   { key: 'succeeded', label: '成功', status: 'succeeded', tone: 'success' },  
16   { key: 'failed', label: '失败', status: 'failed', tone: 'danger' },  
17   { key: 'todayCompleted', label: '今日完成', tone: 'success' },  
18 ];  
19  
20 export function TaskStatsBar({ stats, loading, onQuickFilter, activeStatus }: Props) {  
21   return (  
22     <section className="task-center-stats" aria-label="任务统计">  
23       {CARDS.map((card) => {  
24         const value = stats ? Number(stats[card.key] ?? 0) : loading ? '...' : 0;  
25         const active = card.status && activeStatus === card.status;  
26         const actionable = Boolean(card.status && onQuickFilter);
```



```
front/src/workbench/tasks/components/TaskStatsBar.tsx
```

```
27     return (
28       <button
29         key={card.key}
30         type="button"
31         className={`task-stat-card tone-${card.tone}${actionable ? ' is-actionable' : ''}${active ? ' active' :
    ''}`}
32         onClick={() => {
33           if (card.status && onQuickFilter) onQuickFilter(card.status);
34         }}
35         disabled={!actionable}
36       >
37         <span>{card.label}</span>
38         <strong>{value}</strong>
39       </button>
40     );
41   }}}
42   <article className="task-stat-card tone-muted meta">
43     <span>平均耗时 / 失败率</span>
44     <strong>
45       {formatRuntime(stats?.averageRuntimeSeconds)} / {stats ? `${(stats.failureRate * 100).toFixed(1)}%` : '-'}
46     </strong>
47   </article>
48 </section>
49 );
50 }
51
```

```
// ===== 文件: front/src/workbench/tasks/components/TaskTable.tsx =====
```

```
1 import { GitCompareArrows, Star } from 'lucide-react';
2 import { formatLocalDateTime } from '../../utils/time';
3 import { getTaskPriorityLabel, type TaskLocalMetaMap } from '../../taskLocalMeta';
4 import type { AnalysisTask } from '../../types';
5 import {
6   canCancel,
7   canDelete,
8   canRetry,
9   canStart,
10  failureReasonLabel,
11  formatRuntime,
12  getStatusMeta,
13  stageLabel,
14 } from '../../taskStatus';
15
16 type Props = {
17   tasks: AnalysisTask[];
18   loading?: boolean;
19   density: 'comfortable' | 'compact';
20   onDensityChange: (value: 'comfortable' | 'compact') => void;
21   selectedIds: number[];
22   compareIds: number[];
```

```
front/src/workbench/tasks/components/TaskTable.tsx
```

```
23   taskLocalMeta: TaskLocalMetaMap;
24   onToggleSelect: (id: number) => void;
25   onToggleSelectAll: () => void;
26   onOpenDetail: (task: AnalysisTask) => void;
27   onStart: (task: AnalysisTask) => void;
28   onCancel: (task: AnalysisTask) => void;
29   onRetry: (task: AnalysisTask) => void;
30   onClone: (task: AnalysisTask) => void;
31   onDelete: (task: AnalysisTask) => void;
32   onOpenResults: (task: AnalysisTask) => void;
33   onToggleCompare: (task: AnalysisTask) => void;
34   onToggleFavorite: (task: AnalysisTask) => void;
35 };
36
37 export function TaskTable({
38   tasks,
39   loading,
40   density,
41   onDensityChange,
42   selectedIds,
43   compareIds,
44   taskLocalMeta,
45   onToggleSelect,
46   onToggleSelectAll,
47   onOpenDetail,
48   onStart,
49   onCancel,
50   onRetry,
51   onClone,
52   onDelete,
53   onOpenResults,
54   onToggleCompare,
55   onToggleFavorite,
56 }: Props) {
57   const allSelected = tasks.length > 0 && tasks.every((task) => selectedIds.includes(task.id));
58
59   return (
60     <section className={`panel task-table-panel density-${density}`} aria-label="任务列表">
61       <div className="panel-header task-table-header">
62         <div>
63           <h2>任务列表</h2>
64           <span>支持筛选、批量操作、详情钻取与结果跳转。</span>
65         </div>
66         <div className="task-density-control" role="group" aria-label="列表密度">
67           <button
68             type="button"
69             className={density === 'comfortable' ? 'is-active' : ''}
70             aria-pressed={density === 'comfortable'}
71             onClick={() => onDensityChange('comfortable')}
72           >
```

front/src/workbench/tasks/components/TaskTable.tsx

```
73      舒适
74    </button>
75    <button
76      type="button"
77      className={density === 'compact' ? 'is-active' : ''}
78      aria-pressed={density === 'compact'}
79      onClick={() => onDensityChange('compact')}
80    >
81      紧凑
82    </button>
83  </div>
84</div>
85
86<div className="task-table-wrap">
87  <table className="task-table">
88    <thead>
89      <tr>
90        <th>
91          <input type="checkbox" checked={allSelected} onChange={onToggleSelectAll} aria-label="全选任务" />
92        </th>
93        <th>任务</th>
94        <th>模式</th>
95        <th>数据集</th>
96        <th>状态</th>
97        <th>进度</th>
98        <th>阶段</th>
99        <th>创建时间</th>
100       <th>耗时</th>
101       <th>操作</th>
102     </tr>
103   </thead>
104   <tbody>
105     {loading ? (
106       <tr>
107         <td colspan={10} className="task-table-empty">
108           正在加载任务...
109         </td>
110       </tr>
111     ) : tasks.length === 0 ? (
112       <tr>
113         <td colspan={10} className="task-table-empty">
114           暂无任务。可以从数据集创建，或点击右上角“新建任务”。
115         </td>
116       </tr>
117     ) : (
118       tasks.map((task) => {
119         const meta = getStatusMeta(task.status);
120         const localMeta = taskLocalMeta[task.id] || {};
121         return (
122           <tr key={task.id} className={task.status === 'failed' ? 'is-failed' : ''}>
```

```
front/src/workbench/tasks/components/TaskTable.tsx
```

```
123         <td>
124             <input
125                 type="checkbox"
126                 checked={selectedIds.includes(task.id)}
127                 onChange={() => onToggleSelect(task.id)}
128                 aria-label={`选择任务 ${task.id}`}
129             />
130         </td>
131         <td>
132             <div className="task-name-cell">
133                 <button type="button" className="task-name-btn" onClick={() => onOpenDetail(task)}>
134                     <strong>{task.name}</strong>
135                     <small>#{task.id} • {getTaskPriorityLabel(localMeta.priority)}</small>
136                 </button>
137                 <div className="task-row-meta">
138                     <button
139                         type="button"
140                         className={`task-row-favorite ${localMeta.favorite ? 'is-active' : ''}`}
141                         onClick={() => onToggleFavorite(task)}
142                         title={localMeta.favorite ? '取消关注' : '关注任务'}
143                         aria-label={localMeta.favorite ? `取消关注任务 ${task.id}` : `关注任务 ${task.id}`}
144                     >
145                         <Star size={14} fill={localMeta.favorite ? 'currentColor' : 'none'} aria-hidden="true" />
146                     </button>
147                     {(localMeta.tags || []).slice(0, 2).map((tag) => <span key={tag}>{tag}</span>)}
148                 </div>
149             </div>
150         </td>
151         <td>
152             <span className="task-mode-tag">{task.mode}</span>
153         </td>
154         <td title={task.datasetName}>{task.datasetName}</td>
155         <td>
156             <span className={`task-status-pill tone-${meta.tone}`}>{meta.label}</span>
157             {task.status === 'failed' ? (
158                 <small className="task-fail-hint">{failureReasonLabel(task.failureReason)}</small>
159             ) : null}
160         </td>
161         <td>
162             <div className="task-progress-cell">
163                 <div className="progress-track">
164                     <span
165                         className={task.status === 'failed' ? 'is-failed' : ''}
166                         style={{ width: `${Math.max(0, Math.min(100, task.progress || 0))}%` }}
167                     />
168                 </div>
169                 <strong>
170                     {Math.round(task.progress || 0)}%
171                     {task.maxIter ? ` • ${task.currentIter}/${task.maxIter}` : ''}
172                 </strong>

```

```
front/src/workbench/tasks/components/TaskTable.tsx
```

```
173         </div>
174     </td>
175     <td className="task-stage-cell">{stageLabel(task.currentStage)}</td>
176     <td>{formatLocalDateTime(task.createdAt)}</td>
177     <td>{formatRuntime(task.runtimeSeconds)}</td>
178     <td>
179         <div className="task-row-actions">
180             <button type="button" onClick={() => onOpenDetail(task)}>
181                 详情
182             </button>
183             {canStart(task.status) ? (
184                 <button type="button" onClick={() => onStart(task)}>
185                     启动
186                 </button>
187             ) : null}
188             {canCancel(task.status) ? (
189                 <button type="button" onClick={() => onCancel(task)}>
190                     取消
191                 </button>
192             ) : null}
193             {canRetry(task.status) ? (
194                 <button type="button" onClick={() => onRetry(task)}>
195                     重试
196                 </button>
197             ) : null}
198             {task.status === 'succeeded' ? (
199                 <button type="button" onClick={() => onOpenResults(task)}>
200                     结果
201                 </button>
202             ) : null}
203             <button
204                 type="button"
205                 className={compareIds.includes(task.id) ? 'is-active' : ''}
206                 onClick={() => onToggleCompare(task)}
207                 title={compareIds.includes(task.id) ? '移出任务对比' : '加入任务对比'}
208                 aria-label={compareIds.includes(task.id) ? `将任务 ${task.id} 移出比较` : `将任务 ${task.id} 加入比较`}
209             >
210                 <GitCompareArrows size={14} aria-hidden="true" />
211                 对比
212             </button>
213             <button type="button" onClick={() => onClone(task)}>
214                 克隆
215             </button>
216             {canDelete(task.status) ? (
217                 <button type="button" className="danger" onClick={() => onDelete(task)}>
218                     删除
219                 </button>
220             ) : null}
221         </div>
222     </td>
```

```
front/src/workbench/tasks/components/TaskTable.tsx
```

```
223         </tr>
224     );
225 })
226 })
227 </tbody>
228 </table>
229 </div>
230 </section>
231 );
232 }
233
```

```
// ===== 文件: front/src/workbench/tasks/taskLocalMeta.ts =====
```

```
1 export type TaskPriority = 'high' | 'normal' | 'low';
2
3 export type TaskLocalMeta = {
4   favorite?: boolean;
5   priority?: TaskPriority;
6   tags?: string[];
7   note?: string;
8 };
9
10 export type TaskLocalMetaMap = Record<number, TaskLocalMeta>;
11
12 const STORAGE_KEY = 'soft_web_task_center_local_meta';
13
14 export function loadTaskLocalMeta(): TaskLocalMetaMap {
15   if (typeof window === 'undefined') return {};
16
17   try {
18     const saved = window.sessionStorage.getItem(STORAGE_KEY);
19     const parsed = saved ? JSON.parse(saved) : {};
20     return parsed && typeof parsed === 'object' ? parsed : {};
21   } catch {
22     return {};
23   }
24 }
25
26 export function saveTaskLocalMeta(meta: TaskLocalMetaMap) {
27   if (typeof window === 'undefined') return;
28
29   try {
30     window.sessionStorage.setItem(STORAGE_KEY, JSON.stringify(meta));
31   } catch {
32     // 本地辅助信息写入失败时不影响任务操作。
33   }
34 }
35
36 export function getTaskPriorityLabel(priority?: TaskPriority) {
37   if (priority === 'high') return '高优先级';
38 }
```

```
front/src/workbench/tasks/taskLocalMeta.ts
```

```
38   if (priority === 'low') return '低优先级';
39   return '常规优先级';
40 }
41
// ===== 文件: front/src/workbench/tasks/taskStatus.ts =====
1 import type { AnalysisTask, TaskStatus } from './types';
2
3 export const TASK_STATUS_META: Record<
4   TaskStatus,
5   { label: string; tone: string; description: string }
6 > = {
7   draft: { label: '草稿', tone: 'neutral', description: '已创建, 待启动' },
8   queued: { label: '排队中', tone: 'info', description: '等待执行资源' },
9   running: { label: '运行中', tone: 'running', description: '正在执行分析' },
10  succeeded: { label: '成功', tone: 'success', description: '已完成并可查看结果' },
11  failed: { label: '失败', tone: 'danger', description: '执行失败, 可重试' },
12  cancelled: { label: '已取消', tone: 'muted', description: '用户取消' },
13 };
14
15 export const PIPELINE_STAGES = [
16   { key: 'select_base', label: '选择基础聚类结果' },
17   { key: 'build_ca', label: 'GBE 编码与 CA 构建' },
18   { key: 'multi_kernel', label: '多核相似性学习' },
19   { key: 'evaluate', label: '谱聚类与指标评估' },
20   { key: 'persist', label: '结果落盘' },
21 ] as const;
22
23 export function getStatusMeta(status: string) {
24   return TASK_STATUS_META[(status as TaskStatus) in TASK_STATUS_META ? (status as TaskStatus) : 'draft'];
25 }
26
27 export function canStart(status: string) {
28   return status === 'draft' || status === 'failed' || status === 'cancelled';
29 }
30
31 export function canCancel(status: string) {
32   return status === 'queued' || status === 'running';
33 }
34
35 export function canRetry(status: string) {
36   return status === 'failed' || status === 'cancelled';
37 }
38
39 export function canDelete(status: string) {
40   return status === 'draft' || status === 'failed' || status === 'cancelled' || status === 'succeeded';
41 }
42
43 export function canEdit(status: string) {
44   return status === 'draft' || status === 'failed' || status === 'cancelled';
```

```
front/src/workbench/tasks/taskStatus.ts
```

```
45 }
46
47 export function formatRuntime(seconds?: number | null) {
48   if (seconds == null || Number.isNaN(seconds)) return '-';
49   if (seconds < 60) return `${seconds.toFixed(1)}s`;
50   const mins = Math.floor(seconds / 60);
51   const rest = seconds - mins * 60;
52   return `${mins}m ${rest.toFixed(0)}s`;
53 }
54
55 export function formatPercent(value?: number | null) {
56   if (value == null || Number.isNaN(value)) return '-';
57   return `${(value * 100).toFixed(1)}%`;
58 }
59
60 export function stageLabel(stage?: string | null) {
61   if (!stage) return '未开始';
62   return PIPELINE_STAGES.find((item) => item.key === stage)?.label ?? stage;
63 }
64
65 export function getTaskRunProgress(task: AnalysisTask) {
66   const configuredTotal = Number(task.totalRuns ?? task.params.runs ?? 1);
67   const total = Number.isFinite(configuredTotal) ? Math.max(1, Math.trunc(configuredTotal)) : 1;
68   const reportedCurrent = Number(task.currentRun ?? 0);
69   let current = Number.isFinite(reportedCurrent) ? Math.trunc(reportedCurrent) : 0;
70
71   // running 状态刚建立时, 首个算法事件可能还未写回 currentRun, 此时已经处于第 1 轮。
72   if (task.status === 'running' && current < 1) current = 1;
73   if (task.status === 'succeeded') current = total;
74
75   return { current: Math.max(0, Math.min(total, current)), total };
76 }
77
78 export function failureReasonLabel(reason?: string | null) {
79   switch (reason) {
80     case 'param_error':
81       return '参数错误';
82     case 'quality_error':
83       return '数据质量不通过';
84     case 'execution_error':
85       return '执行异常';
86     case 'timeout':
87       return '执行超时';
88     default:
89       return reason || '未知原因';
90   }
91 }
92
93 export function buildPipelineState(task: AnalysisTask) {
94   const currentKey = task.currentStage;
```



```
front/src/workbench/tasks/taskStatus.ts
```

```
95   const currentIndex = PIPELINE_STAGES.findIndex((item) => item.key === currentKey);
96   return PIPELINE_STAGES.map((stage, index) => {
97     let state: 'done' | 'running' | 'pending' | 'failed' = 'pending';
98     if (task.status === 'succeeded') {
99       state = 'done';
100    } else if (task.status === 'failed') {
101      if (currentIndex < 0) state = index === 0 ? 'failed' : 'pending';
102      else if (index < currentIndex) state = 'done';
103      else if (index === currentIndex) state = 'failed';
104      else state = 'pending';
105    } else if (task.status === 'running' || task.status === 'queued') {
106      if (currentIndex < 0) state = index === 0 ? 'running' : 'pending';
107      else if (index < currentIndex) state = 'done';
108      else if (index === currentIndex) state = 'running';
109      else state = 'pending';
110    } else if (task.status === 'cancelled') {
111      if (currentIndex >= 0 && index <= currentIndex) state = index === currentIndex ? 'failed' : 'done';
112    }
113    return { ...stage, state };
114  });
115 }
116
```

```
// ===== 文件: front/src/workbench/tasks/types.ts =====
```

```
1  export type TaskStatus =
2    | 'draft'
3    | 'queued'
4    | 'running'
5    | 'succeeded'
6    | 'failed'
7    | 'cancelled';
8
9  export type TaskMode = 'OMELET' | 'OMELET-SV';
10
11 export type TaskParams = {
12   nBase: number;
13   sigma: number;
14   lambda: number;
15   gamma: number;
16   anchor: number;
17   runs: number;
18   maxIter: number;
19   randomSeed: number;
20 };
21
22 export type TaskMetricsSummary = {
23   acc?: number | null;
24   nmi?: number | null;
25   ari?: number | null;
26   fl?: number | null;
```

```
front/src/workbench/tasks/types.ts
```

```
27  };
28
29  export type AnalysisTask = {
30    id: number;
31    name: string;
32    mode: TaskMode | string;
33    status: TaskStatus | string;
34    progress: number;
35    currentRun?: number;
36    totalRuns?: number;
37    queuePosition?: number | null;
38    currentIter: number;
39    maxIter: number;
40    currentStage?: string | null;
41    datasetId: number;
42    datasetName: string;
43    params: Partial<TaskParams> & Record<string, unknown>;
44    errorMessage?: string | null;
45    failureReason?: string | null;
46    metricsSummary?: TaskMetricsSummary | null;
47    runtimeSeconds?: number | null;
48    createdAt: string;
49    startedAt?: string | null;
50    finishedAt?: string | null;
51    updatedAt: string;
52  };
53
54  export type AnalysisTaskPage = {
55    items: AnalysisTask[];
56    total: number;
57    page: number;
58    pageSize: number;
59    totalPages: number;
60  };
61
62  export type AnalysisTaskStats = {
63    total: number;
64    draft: number;
65    queued: number;
66    running: number;
67    succeeded: number;
68    failed: number;
69    cancelled: number;
70    todayCompleted: number;
71    averageRuntimeSeconds?: number | null;
72    failureRate: number;
73  };
74
75  export type AnalysisTaskLog = {
76    id: number;
```

```
front/src/workbench/tasks/types.ts
```

```
77   action: string;
78   level: string;
79   message: string;
80   createdAt: string;
81   detail?: Record<string, unknown> | null;
82 };
83
84 export type TaskTemplate = {
85   id: number;
86   name: string;
87   mode: TaskMode | string;
88   params: Partial<TaskParams> & Record<string, unknown>;
89   createdAt: string;
90 };
91
92 export type TaskListQuery = {
93   page?: number;
94   pageSize?: number;
95   status?: string;
96   mode?: string;
97   datasetId?: number;
98   keyword?: string;
99   createdFrom?: string;
100  createdTo?: string;
101 };
102
103 export type CreateTaskPayload = {
104   datasetId: number;
105   name?: string;
106   mode?: TaskMode | string;
107   params?: Partial<TaskParams> & { lambda?: number };
108   startImmediately?: boolean;
109   templateId?: number;
110 };
111
112 export type DatasetOption = {
113   id: number;
114   name: string;
115   baseCount: number;
116   sampleCount: number;
117   qualityStatus: string;
118   qualityIssues?: string[];
119   hasLabels?: boolean;
120   classCount?: number;
121 };
122
123
124 // ===== 文件: front/src/workbench/workbench-polish.css =====
125 1 /* 工作台视觉层只覆盖已登录区域，登录页与首页继续使用各自的设计体系。 */
126 2 :root {
```

```
front/src/workbench/workbench-polish.css
```

```
3  --app-bg: #f3f5f7;
4  --app-bg-gradient: none;
5  --app-surface: #ffffff;
6  --app-surface-strong: #ffffff;
7  --app-surface-muted: #f8fafb;
8  --app-border: #e6e6eb;
9  --app-border-strong: #c8c8dc;
10 --app-text-primary: #1f2d3d;
11 --app-text-regular: #4b5c6f;
12 --app-text-muted: #738397;
13 --app-accent: #2f7del;
14 --app-accent-dark: #1f64bd;
15 --app-accent-soft: #eaf3ff;
16 --success: #23865f;
17 --warning: #c97a17;
18 --danger: #d64545;
19 --sidebar-width: 226px;
20 --sidebar-collapsed-width: 64px;
21 --header-height: 58px;
22 --app-gap: 18px;
23 --panel-radius: 8px;
24 --motion-fast: 160ms;
25 --motion-normal: 220ms;
26 --ease-out-quart: cubic-bezier(0.25, 1, 0.5, 1);
27 --z-sticky: 40;
28 --z-modal-backdrop: 80;
29 --z-modal: 90;
30 --z-toast: 100;
31 --z-tooltip: 110;
32 }
33
34 .app-shell {
35   color: var(--app-text-primary);
36   background: var(--app-bg);
37   font-weight: 500;
38 }
39
40 .app-background {
41   background: var(--app-bg);
42 }
43
44 .app-background__shape {
45   display: none;
46 }
47
48 .sidebar {
49   border-right-color: var(--app-border);
50   background: #fbfcfd;
51   box-shadow: none;
52   backdrop-filter: none;
```

front/src/workbench/workbench-polish.css

```
53 }
54
55 .sidebar-brand {
56   gap: 11px;
57   padding-inline: 15px;
58   border-bottom: 1px solid var(--app-border);
59   background: #ffffff;
60 }
61
62 .brand-logo {
63   width: 32px;
64   height: 32px;
65   flex-basis: 32px;
66   border-radius: 8px;
67 }
68
69 .brand-copy strong {
70   color: #203247;
71   font-size: 14px;
72   font-weight: 750;
73 }
74
75 .brand-copy small {
76   color: #7b8998;
77   font-size: 10px;
78   font-weight: 600;
79 }
80
81 .nav-list {
82   padding-block: 12px 18px;
83 }
84
85 .nav-group {
86   gap: 3px;
87   padding-block: 7px;
88 }
89
90 .nav-group + .nav-group {
91   border-top-color: #edf0f3;
92 }
93
94 .nav-group-label {
95   padding: 8px 16px 5px;
96   color: #8b98a6;
97   font-size: 10px;
98   font-weight: 700;
99 }
100
101 .nav-item {
102   width: calc(100% - 20px);
```

```
front/src/workbench/workbench-polish.css
```

```
103     height: 38px;
104     margin: 1px 10px;
105     padding-inline: 10px;
106     gap: 9px;
107     border: 0;
108     border-radius: 7px;
109     color: #536579;
110     transition:
111         color var(--motion-fast) ease,
112         background-color var(--motion-fast) ease,
113         transform var(--motion-fast) var(--ease-out-quart);
114 }
115
116 .nav-item:hover {
117     color: #26384d;
118     background: #f0f3f6;
119     transform: translateX(2px);
120 }
121
122 .nav-item.active {
123     color: #1f64bd;
124     background: #e7f1fd;
125     transform: none;
126 }
127
128 .nav-item.active .nav-icon {
129     color: #2f7del;
130 }
131
132 .nav-icon {
133     width: 22px;
134     min-width: 22px;
135     color: #718196;
136 }
137
138 .nav-label {
139     font-size: 13px;
140     font-weight: 650;
141 }
142
143 .sidebar.collapsed .nav-item {
144     width: 42px;
145     margin-inline: 11px;
146 }
147
148 .main-header {
149     z-index: var(--z-sticky);
150     height: var(--header-height);
151     padding-inline: 16px 14px;
152     border-bottom-color: var(--app-border);
```

```
front/src/workbench/workbench-polish.css
```

```
153   background: rgba(255, 255, 255, 0.96);
154   backdrop-filter: blur(10px);
155   transition: left var(--motion-normal) var(--ease-out-quart);
156 }
157
158 .header-section-title {
159   margin-left: 0;
160   color: #203247;
161   font-size: 14px;
162   font-weight: 750;
163 }
164
165 .header-context {
166   min-width: 0;
167   display: flex;
168   align-items: center;
169   gap: 8px;
170 }
171
172 .header-context > span {
173   color: #8794a3;
174   font-size: 12px;
175   font-weight: 600;
176 }
177
178 .header-context > svg {
179   flex: 0 0 auto;
180   color: #a2adb8;
181 }
182
183 .header-user {
184   width: 32px;
185   min-width: 32px;
186   height: 32px;
187   display: grid;
188   place-items: center;
189   padding: 0;
190   border: 0;
191   border-radius: 7px;
192   background: transparent;
193 }
194
195 .header-user-avatar {
196   width: 28px;
197   height: 28px;
198   display: grid;
199   place-items: center;
200   border-radius: 7px;
201   color: #ffffff;
202   background: #285f9f;
```

```
front/src/workbench/workbench-polish.css
```

```
203     font-size: 11px;
204     font-weight: 750;
205 }
206
207 .header-icon-button {
208     width: 32px;
209     min-width: 32px;
210     height: 32px;
211     border-radius: 7px;
212     color: #647589;
213 }
214
215 .header-icon-button:hover {
216     color: #1f64bd;
217     background: #eaf3ff;
218 }
219
220 .sidebar-toggle-floating {
221     left: calc(var(--sidebar-width) - 11px);
222     width: 22px;
223     height: 44px;
224     border-color: var(--app-border);
225     border-radius: 8px;
226     color: #69798a;
227     background: #ffffff;
228     box-shadow: 0 2px 6px rgba(31, 45, 61, 0.1);
229     backdrop-filter: none;
230     transition:
231         color var(--motion-fast) ease,
232         transform var(--motion-fast) var(--ease-out-quart);
233 }
234
235 .sidebar-collapsed .sidebar-toggle-floating {
236     left: calc(var(--sidebar-collapsed-width) - 11px);
237 }
238
239 .dashboard-content {
240     padding: calc(var(--header-height) + var(--app-gap)) var(--app-gap) 28px;
241     gap: 16px;
242 }
243
244 /* 日志页固定在当前视口内，页面级滚动交给事件列表内部处理。 */
245 .dashboard-content.is-log-layout {
246     height: 100vh;
247     height: 100dvh;
248     min-height: 0;
249     overflow-y: hidden;
250 }
251
252 .dashboard-content.is-log-layout > .task-logs-page {
```



```
front/src/workbench/workbench-polish.css
```

```
253     min-height: 0;
254     flex: 1;
255 }
256
257 .dashboard-content > * {
258     animation: workbench-content-in var(--motion-normal) var(--ease-out-quart) both;
259 }
260
261 .status-header,
262 .panel,
263 .summary-card,
264 .metric-card {
265     border-color: var(--app-border);
266     background: #ffffff;
267     backdrop-filter: none;
268 }
269
270 .panel {
271     border-radius: 8px;
272 }
273
274 .panel-header h2 {
275     color: #25374b;
276     font-size: 15px;
277     font-weight: 700;
278 }
279
280 .panel-header span {
281     color: #78889b;
282     font-size: 12px;
283     font-weight: 500;
284 }
285
286 .btn {
287     min-height: 34px;
288     border-radius: 7px;
289     font-size: 12px;
290     font-weight: 700;
291     transition:
292         color var(--motion-fast) ease,
293         border-color var(--motion-fast) ease,
294         background-color var(--motion-fast) ease,
295         transform var(--motion-fast) var(--ease-out-quart);
296 }
297
298 .btn-primary {
299     border-color: #2f7de1;
300     background: #2f7de1;
301 }
302
```

```
front/src/workbench/workbench-polish.css
```

```
303 .btn-primary:hover {
304   border-color: #256fc9;
305   background: #256fc9;
306 }
307
308 .btn-secondary {
309   border-color: var(--app-border);
310   background: #ffffff;
311 }
312
313 .btn-secondary:hover {
314   color: #1f64bd;
315   border-color: #9fc4ef;
316   background: #f2f7fd;
317 }
318
319 .btn-danger {
320   color: #ffffff;
321   border-color: #d64545;
322   background: #d64545;
323 }
324
325 .btn-disabled {
326   color: #9aa6b2;
327   border-color: #e3e7eb;
328   background: #f3f5f7;
329   cursor: not-allowed;
330   transform: none;
331 }
332
333 /* 页面头部不使用容器卡片，标题、上下文与操作保持在同一条视觉基线上。 */
334 .workbench-page,
335 .soft-page {
336   min-width: 0;
337   display: grid;
338   gap: 14px;
339 }
340
341 .workbench-page-header {
342   min-width: 0;
343   min-height: 48px;
344   display: flex;
345   align-items: center;
346   justify-content: space-between;
347   gap: 20px;
348   padding: 0 2px 2px;
349 }
350
351 .workbench-page-title-block {
352   min-width: 0;
```

```
front/src/workbench/workbench-polish.css
```

```
353     display: flex;
354     align-items: center;
355     gap: 11px;
356 }
357
358 .workbench-page-back {
359     flex: 0 0 auto;
360 }
361
362 .workbench-page-back-button {
363     min-width: 68px;
364     height: 34px;
365     display: inline-flex;
366     align-items: center;
367     justify-content: center;
368     gap: 6px;
369     padding-inline: 10px;
370     border: 1px solid var(--app-border-strong);
371     border-radius: 7px;
372     color: #536579;
373     background: #ffffff;
374     font: inherit;
375     font-size: 11px;
376     font-weight: 700;
377     cursor: pointer;
378     transition: color var(--motion-fast) ease, border-color var(--motion-fast) ease, background-color var(--motion-fast)
379         ease;
380
381 .workbench-page-back-button:hover {
382     color: #1f64bd;
383     border-color: #b8d2ed;
384     background: #f2f7fc;
385 }
386
387 .workbench-page-back-button:focus-visible {
388     outline: 2px solid rgba(47, 125, 225, 0.38);
389     outline-offset: 2px;
390 }
391
392 .workbench-page-icon {
393     width: 38px;
394     height: 38px;
395     flex: 0 0 38px;
396     display: grid;
397     place-items: center;
398     border: 1px solid #cfe2f8;
399     border-radius: 8px;
400     color: #1f64bd;
401     background: #edf5fe;
```

```
front/src/workbench/workbench-polish.css
```

```
402 }
403
404 .workbench-page-title-block > div {
405     min-width: 0;
406 }
407
408 .workbench-page-title-line {
409     min-width: 0;
410     display: flex;
411     align-items: center;
412     gap: 10px;
413 }
414
415 .workbench-page-title-line h1 {
416     margin: 0;
417     color: #203247;
418     font-size: 20px;
419     font-weight: 750;
420     line-height: 1.25;
421     letter-spacing: 0;
422     text-wrap: balance;
423 }
424
425 .workbench-page-title-block p {
426     max-width: 72ch;
427     margin: 3px 0 0;
428     overflow: hidden;
429     color: #738397;
430     font-size: 12px;
431     font-weight: 500;
432     line-height: 1.45;
433     text-overflow: ellipsis;
434     white-space: nowrap;
435 }
436
437 .workbench-page-actions,
438 .workbench-section-actions {
439     flex: 0 0 auto;
440     display: flex;
441     align-items: center;
442     gap: 8px;
443 }
444
445 .workbench-status {
446     min-height: 24px;
447     display: inline-flex;
448     align-items: center;
449     gap: 6px;
450     padding-inline: 8px;
451     border: 1px solid var(--app-border);
```

```
front/src/workbench/workbench-polish.css
```

```
452   border-radius: 999px;
453   color: #607184;
454   background: #f7f9fa;
455   font-size: 11px;
456   font-weight: 700;
457   white-space: nowrap;
458 }
459
460 .workbench-status i {
461   width: 7px;
462   height: 7px;
463   border-radius: 50%;
464   background: currentColor;
465 }
466
467 .workbench-status.success {
468   color: #1f7655;
469   border-color: #b9ddce;
470   background: #edf8f3;
471 }
472
473 .workbench-status.warning {
474   color: #aa6511;
475   border-color: #ecd2aa;
476   background: #fff7e9;
477 }
478
479 .workbench-status.error {
480   color: #bd3838;
481   border-color: #edc0c0;
482   background: #fff1f1;
483 }
484
485 .workbench-status.info {
486   color: #1f64bd;
487   border-color: #c8ddf5;
488   background: #edf5fe;
489 }
490
491 .workbench-status.is-pulsing i {
492   animation: status-pulse 1.8s ease-in-out infinite;
493 }
494
495 /* 指标带是一个整体信息区，分隔线承担层级，避免重复卡片造成视觉噪声。 */
496 .workbench-metric-strip {
497   min-width: 0;
498   display: grid;
499   grid-template-columns: repeat(4, minmax(0, 1fr));
500   border: 1px solid var(--app-border);
501   border-radius: 8px;
```

```
front/src/workbench/workbench-polish.css
```

```
502   background: #ffffff;
503   overflow: hidden;
504 }
505
506 .workbench-metric {
507   position: relative;
508   min-width: 0;
509   min-height: 86px;
510   padding: 13px 16px;
511   display: grid;
512   align-content: center;
513   gap: 4px;
514 }
515
516 .workbench-metric + .workbench-metric {
517   border-left: 1px solid var(--app-border);
518 }
519
520 .workbench-metric-label {
521   min-width: 0;
522   display: flex;
523   align-items: center;
524   gap: 6px;
525   color: #738397;
526 }
527
528 .workbench-metric-label span {
529   overflow: hidden;
530   font-size: 11px;
531   font-weight: 650;
532   text-overflow: ellipsis;
533   white-space: nowrap;
534 }
535
536 .workbench-metric strong {
537   min-width: 0;
538   overflow: hidden;
539   color: #25374b;
540   font-size: 22px;
541   font-weight: 760;
542   font-variant-numeric: tabular-nums;
543   line-height: 1.15;
544   text-overflow: ellipsis;
545   white-space: nowrap;
546 }
547
548 .workbench-metric small {
549   min-width: 0;
550   overflow: hidden;
551   color: #8a97a6;
```

```
front/src/workbench/workbench-polish.css
```

```
552     font-size: 11px;
553     font-weight: 500;
554     text-overflow: ellipsis;
555     white-space: nowrap;
556 }
557
558 .workbench-metric.tone-blue .workbench-metric-label svg,
559 .workbench-metric.tone-blue strong {
560     color: #2f7del;
561 }
562
563 .workbench-metric.tone-teal .workbench-metric-label svg,
564 .workbench-metric.tone-teal strong {
565     color: #14796f;
566 }
567
568 .workbench-metric.tone-green .workbench-metric-label svg,
569 .workbench-metric.tone-green strong {
570     color: #23865f;
571 }
572
573 .workbench-metric.tone-amber .workbench-metric-label svg,
574 .workbench-metric.tone-amber strong {
575     color: #b96c12;
576 }
577
578 .workbench-metric.tone-red .workbench-metric-label svg,
579 .workbench-metric.tone-red strong {
580     color: #d64545;
581 }
582
583 .workbench-section-header {
584     min-width: 0;
585     min-height: 40px;
586     display: flex;
587     align-items: center;
588     justify-content: space-between;
589     gap: 16px;
590 }
591
592 .workbench-section-header > div:first-child {
593     min-width: 0;
594 }
595
596 .workbench-section-header h2 {
597     margin: 0;
598     color: #25374b;
599     font-size: 15px;
600     font-weight: 720;
601     letter-spacing: 0;
```

```
front/src/workbench/workbench-polish.css
```

```
602 }
603
604 .workbench-section-header span {
605     display: block;
606     margin-top: 3px;
607     overflow: hidden;
608     color: #7b8a9c;
609     font-size: 11px;
610     font-weight: 500;
611     text-overflow: ellipsis;
612     white-space: nowrap;
613 }
614
615 .workbench-progress {
616     min-width: 0;
617     display: grid;
618     gap: 7px;
619 }
620
621 .workbench-progress > div:first-child {
622     display: flex;
623     align-items: center;
624     justify-content: space-between;
625     gap: 12px;
626     color: #738397;
627     font-size: 11px;
628     font-weight: 600;
629 }
630
631 .workbench-progress strong {
632     color: #34475b;
633     font-variant-numeric: tabular-nums;
634 }
635
636 .workbench-progress-track {
637     height: 7px;
638     overflow: hidden;
639     border-radius: 999px;
640     background: #e8ecef;
641 }
642
643 .workbench-progress-track > span {
644     position: relative;
645     display: block;
646     width: 100%;
647     height: 100%;
648     border-radius: inherit;
649     background: #2f7del;
650     transform-origin: left center;
651     transition: transform 420ms var(--ease-out-quart);
```



```
front/src/workbench/workbench-polish.css
```

```
652 }
653
654 .workbench-progress.tone-teal .workbench-progress-track > span {
655     background: #158578;
656 }
657
658 .workbench-progress.tone-green .workbench-progress-track > span {
659     background: #23865f;
660 }
661
662 .workbench-progress.tone-amber .workbench-progress-track > span {
663     background: #c97a17;
664 }
665
666 .workbench-progress.is-animated .workbench-progress-track > span::after {
667     position: absolute;
668     inset: 0;
669     content: '';
670     background: rgba(255, 255, 255, 0.22);
671     transform: translateX(-100%);
672     animation: progress-scan 1.8s ease-in-out infinite;
673 }
674
675 .workbench-notice {
676     min-width: 0;
677     min-height: 48px;
678     display: grid;
679     grid-template-columns: auto minmax(0, 1fr) auto;
680     align-items: center;
681     gap: 10px;
682     padding: 9px 12px;
683     border: 1px solid #c9def5;
684     border-radius: 8px;
685     color: #1f64bd;
686     background: #f2f7fd;
687 }
688
689 .workbench-notice > div:nth-child(2) {
690     min-width: 0;
691     display: grid;
692     gap: 2px;
693 }
694
695 .workbench-notice strong,
696 .workbench-notice span {
697     min-width: 0;
698     overflow: hidden;
699     text-overflow: ellipsis;
700     white-space: nowrap;
701 }
```

```
front/src/workbench/workbench-polish.css
```

```
702
703 .workbench-notice strong {
704     color: #2d4258;
705     font-size: 12px;
706     font-weight: 700;
707 }
708
709 .workbench-notice span {
710     color: #6f8093;
711     font-size: 11px;
712     font-weight: 500;
713 }
714
715 .workbench-notice.success {
716     color: #23865f;
717     border-color: #bddfcf;
718     background: #f0f8f4;
719 }
720
721 .workbench-notice.warning {
722     color: #b96c12;
723     border-color: #ecd2aa;
724     background: #fff8ec;
725 }
726
727 .workbench-notice.error {
728     color: #d64545;
729     border-color: #ecc4c4;
730     background: #fff3f3;
731 }
732
733 @keyframes workbench-content-in {
734     from {
735         opacity: 0.72;
736         transform: translateY(6px);
737     }
738
739     to {
740         opacity: 1;
741         transform: translateY(0);
742     }
743 }
744
745 @keyframes status-pulse {
746     0%,
747     100% {
748         box-shadow: 0 0 0 0 currentColor;
749         opacity: 1;
750     }
751 }
```

```
front/src/workbench/workbench-polish.css
```

```
752     50% {
753         box-shadow: 0 0 0 4px transparent;
754         opacity: 0.68;
755     }
756 }
757
758 @keyframes progress-scan {
759     0% {
760         transform: translateX(-100%);
761     }
762
763     70%,
764     100% {
765         transform: translateX(100%);
766     }
767 }
768
769 @media (min-width: 1580px) {
770     :root {
771         --app-gap: 22px;
772     }
773
774     .dashboard-content {
775         padding-bottom: 34px;
776     }
777 }
778
779 @media (prefers-reduced-motion: reduce) {
780     .dashboard-content > *,
781     .workbench-status i,
782     .workbench-progress-track > span,
783     .workbench-progress-track > span::after {
784         animation: none !important;
785         transition: none !important;
786     }
787
788     .nav-item,
789     .btn,
790     .sidebar-toggle-floating {
791         transition-duration: 0.01ms !important;
792     }
793 }
794
795 /* 算法流程页采用主分析区加诊断区，固定桌面端信息密度。 */
796 .ca-primary-grid,
797 .mkl-primary-grid,
798 .mkl-secondary-grid,
799 .result-primary-grid,
800 .result-support-grid,
801 .export-builder-grid {
```

```
front/src/workbench/workbench-polish.css
```

```
802   min-width: 0;
803   display: grid;
804   gap: 14px;
805   align-items: stretch;
806 }
807
808 .ca-primary-grid {
809   grid-template-columns: minmax(0, 1.72fr) minmax(310px, 0.78fr);
810 }
811
812 .ca-primary-grid > .panel,
813 .mkl-primary-grid > .panel,
814 .mkl-secondary-grid > .panel,
815 .result-primary-grid > .panel,
816 .result-support-grid > .panel,
817 .export-builder-grid > .panel {
818   height: 100%;
819 }
820
821 .ca-primary-grid .ca-matrix-panel {
822   min-height: 420px;
823 }
824
825 .ca-primary-grid .feature-list-panel {
826   display: flex;
827   flex-direction: column;
828 }
829
830 .ca-primary-grid .feature-list {
831   flex: 1;
832   align-content: start;
833 }
834
835 .ca-primary-grid .feature-row {
836   padding: 13px 12px;
837   background: #fbfcfd;
838 }
839
840 .ca-diagnostics-panel,
841 .mkl-diagnostics-panel,
842 .result-runs-panel,
843 .recent-exports-panel {
844   padding: 15px 16px;
845 }
846
847 .ca-diagnostics-grid {
848   min-width: 0;
849   display: grid;
850   grid-template-columns: minmax(260px, 0.9fr) minmax(320px, 1.2fr) minmax(280px, 0.9fr);
851   gap: 22px;
```

```
front/src/workbench/workbench-polish.css
```

```
852   padding-top: 12px;
853   border-top: 1px solid var(--app-border);
854 }
855
856 .ca-threshold-control,
857 .ca-distribution,
858 .ca-diagnostic-summary {
859   min-width: 0;
860 }
861
862 .ca-threshold-control > div:first-child {
863   display: flex;
864   align-items: center;
865   justify-content: space-between;
866   gap: 12px;
867 }
868
869 .ca-threshold-control label,
870 .ca-threshold-control strong {
871   color: #34475b;
872   font-size: 12px;
873   font-weight: 700;
874 }
875
876 .ca-threshold-control strong {
877   color: #1f64bd;
878   font-variant-numeric: tabular-nums;
879 }
880
881 .ca-threshold-control input[type='range'] {
882   width: 100%;
883   margin: 16px 0 7px;
884   accent-color: #2f7del;
885 }
886
887 .ca-threshold-scale {
888   display: flex;
889   justify-content: space-between;
890   color: #8b98a6;
891   font-size: 10px;
892   font-variant-numeric: tabular-nums;
893 }
894
895 .ca-distribution {
896   display: grid;
897   gap: 9px;
898 }
899
900 .ca-distribution-row {
901   min-width: 0;
```

```
front/src/workbench/workbench-polish.css
```

```
902   display: grid;
903   grid-template-columns: 78px minmax(0, 1fr) 34px;
904   align-items: center;
905   gap: 9px;
906   color: #69798b;
907   font-size: 11px;
908   font-variant-numeric: tabular-nums;
909 }
910
911 .ca-distribution-row > div {
912   height: 7px;
913   overflow: hidden;
914   border-radius: 999px;
915   background: #edf0f3;
916 }
917
918 .ca-distribution-row i {
919   display: block;
920   height: 100%;
921   border-radius: inherit;
922   background: #9ebbd7;
923 }
924
925 .ca-distribution-row i.mid {
926   background: #5fa7d1;
927 }
928
929 .ca-distribution-row i.high {
930   background: #2f7de1;
931 }
932
933 .ca-distribution-row i.peak {
934   background: #223c5a;
935 }
936
937 .ca-distribution-row strong {
938   color: #42556a;
939   text-align: right;
940 }
941
942 .ca-diagnostic-summary {
943   display: grid;
944   gap: 8px;
945 }
946
947 .ca-diagnostic-summary > div {
948   min-width: 0;
949   display: flex;
950   align-items: flex-start;
951   gap: 9px;
```

```
front/src/workbench/workbench-polish.css
```

```
952 }
953
954 .ca-diagnostic-summary svg {
955     flex: 0 0 auto;
956     margin-top: 1px;
957     color: #23865f;
958 }
959
960 .ca-diagnostic-summary > div:last-child svg {
961     color: #2f7de1;
962 }
963
964 .ca-diagnostic-summary span {
965     min-width: 0;
966     display: grid;
967     gap: 2px;
968 }
969
970 .ca-diagnostic-summary strong,
971 .ca-diagnostic-summary small {
972     overflow: hidden;
973     text-overflow: ellipsis;
974     white-space: nowrap;
975 }
976
977 .ca-diagnostic-summary strong {
978     color: #34475b;
979     font-size: 11px;
980 }
981
982 .ca-diagnostic-summary small {
983     color: #8290a0;
984     font-size: 10px;
985 }
986
987 .mkl-primary-grid {
988     grid-template-columns: minmax(0, 1.65fr) minmax(310px, 0.75fr);
989 }
990
991 .mkl-primary-grid .selectable-chart-panel .chart-surface {
992     height: 320px;
993 }
994
995 .mkl-primary-grid .task-panel {
996     display: flex;
997     flex-direction: column;
998 }
999
1000 .mkl-primary-grid .task-grid {
1001     grid-template-columns: repeat(2, minmax(0, 1fr));
```

```
front/src/workbench/workbench-polish.css
```

```
1002 }
1003
1004 .mkl-primary-grid .progress-block {
1005     margin-top: auto;
1006     padding-top: 16px;
1007 }
1008
1009 .mkl-secondary-grid {
1010     grid-template-columns: minmax(360px, 0.9fr) minmax(0, 1.3fr);
1011 }
1012
1013 .mkl-secondary-grid .pipeline-list {
1014     gap: 7px;
1015 }
1016
1017 .mkl-secondary-grid .pipeline-step {
1018     padding: 8px 10px;
1019 }
1020
1021 .mkl-secondary-grid .chart-surface {
1022     height: 282px;
1023 }
1024
1025 .mkl-kernel-rows {
1026     min-width: 0;
1027     display: grid;
1028     grid-template-columns: repeat(2, minmax(0, 1fr));
1029     gap: 0 24px;
1030     padding-top: 8px;
1031     border-top: 1px solid var(--app-border);
1032 }
1033
1034 .mkl-kernel-row {
1035     min-width: 0;
1036     display: grid;
1037     grid-template-columns: minmax(160px, 0.72fr) minmax(190px, 1fr);
1038     align-items: center;
1039     gap: 18px;
1040     padding: 11px 0;
1041     border-bottom: 1px solid #edf0f2;
1042 }
1043
1044 .mkl-kernel-row > div:first-child {
1045     min-width: 0;
1046     display: grid;
1047     gap: 3px;
1048 }
1049
1050 .mkl-kernel-row strong,
1051 .mkl-kernel-row span {
```



```
front/src/workbench/workbench-polish.css
```

```
1052     overflow: hidden;
1053     text-overflow: ellipsis;
1054     white-space: nowrap;
1055 }
1056
1057 .mkl-kernel-row strong {
1058     color: #34475b;
1059     font-size: 12px;
1060 }
1061
1062 .mkl-kernel-row span {
1063     color: #8491a1;
1064     font-size: 10px;
1065 }
1066
1067 .result-primary-grid {
1068     grid-template-columns: minmax(0, 1.65fr) minmax(320px, 0.75fr);
1069 }
1070
1071 .result-featured-chart .chart-surface {
1072     height: 340px;
1073 }
1074
1075 .result-quality-panel {
1076     display: flex;
1077     flex-direction: column;
1078     gap: 13px;
1079 }
1080
1081 .result-quality-score {
1082     display: flex;
1083     align-items: flex-start;
1084     justify-content: space-between;
1085     gap: 14px;
1086     padding: 12px 0;
1087     border-block: 1px solid var(--app-border);
1088 }
1089
1090 .result-quality-score > div {
1091     display: flex;
1092     align-items: baseline;
1093     gap: 5px;
1094 }
1095
1096 .result-quality-score span,
1097 .result-quality-score small {
1098     color: #78889b;
1099     font-size: 11px;
1100 }
1101
```

```
front/src/workbench/workbench-polish.css
```

```
1102 .result-quality-score strong {
1103     margin-left: 8px;
1104     color: #23865f;
1105     font-size: 28px;
1106     font-weight: 760;
1107     font-variant-numeric: tabular-nums;
1108 }
1109
1110 .result-quality-bars {
1111     display: grid;
1112     gap: 11px;
1113 }
1114
1115 .result-quality-panel .workbench-notice {
1116     margin-top: auto;
1117 }
1118
1119 .result-support-grid {
1120     grid-template-columns: repeat(2, minmax(0, 1fr));
1121 }
1122
1123 .result-support-grid .chart-surface,
1124 .result-support-grid .selectable-chart-panel .chart-surface {
1125     height: 270px;
1126 }
1127
1128 .result-runs-table {
1129     overflow: hidden;
1130     border: 1px solid var(--app-border);
1131     border-radius: 7px;
1132 }
1133
1134 .result-runs-row {
1135     min-width: 0;
1136     min-height: 42px;
1137     display: grid;
1138     grid-template-columns: 0.65fr repeat(5, minmax(72px, 1fr)) 0.8fr;
1139     align-items: center;
1140     gap: 10px;
1141     padding-inline: 12px;
1142     border-top: 1px solid #edf0f2;
1143     background: #ffffff;
1144 }
1145
1146 .result-runs-row:first-child {
1147     border-top: 0;
1148 }
1149
1150 .result-runs-row.header {
1151     min-height: 36px;
```

```
front/src/workbench/workbench-polish.css
```

```
1152     color: #778699;
1153     background: #f7f9fa;
1154     font-size: 10px;
1155     font-weight: 700;
1156 }
1157
1158 .result-runs-row > span {
1159     min-width: 0;
1160     overflow: hidden;
1161     color: #46596e;
1162     font-size: 11px;
1163     font-variant-numeric: tabular-nums;
1164     text-overflow: ellipsis;
1165     white-space: nowrap;
1166 }
1167
1168 .result-runs-row > span:first-child {
1169     color: #24384e;
1170     font-weight: 700;
1171 }
1172
1173 .export-builder-grid {
1174     grid-template-columns: minmax(0, 1.45fr) minmax(340px, 0.75fr);
1175 }
1176
1177 .export-selection-panel,
1178 .export-settings-panel {
1179     padding: 15px 16px;
1180 }
1181
1182 .text-action,
1183 .icon-action {
1184     border: 0;
1185     color: #1f64bd;
1186     background: transparent;
1187     cursor: pointer;
1188 }
1189
1190 .text-action {
1191     padding: 5px 2px;
1192     font-size: 11px;
1193     font-weight: 700;
1194 }
1195
1196 .text-action:hover {
1197     color: #164d93;
1198     text-decoration: underline;
1199 }
1200
1201 .icon-action {
```

```
front/src/workbench/workbench-polish.css
```

```
1202     width: 30px;
1203     height: 30px;
1204     display: grid;
1205     place-items: center;
1206     border-radius: 6px;
1207 }
1208
1209 .icon-action:hover {
1210     background: #eaf3ff;
1211 }
1212
1213 .export-item-list {
1214     overflow: hidden;
1215     border: 1px solid var(--app-border);
1216     border-radius: 7px;
1217 }
1218
1219 .export-item-row {
1220     min-width: 0;
1221     min-height: 58px;
1222     display: grid;
1223     grid-template-columns: auto auto minmax(0, 1fr) auto;
1224     align-items: center;
1225     gap: 11px;
1226     padding: 8px 12px;
1227     border-top: 1px solid #edf0f2;
1228     background: #ffffff;
1229     cursor: pointer;
1230     transition: background-color var(--motion-fast) ease;
1231 }
1232
1233 .export-item-row:first-child {
1234     border-top: 0;
1235 }
1236
1237 .export-item-row:hover,
1238 .export-item-row.selected {
1239     background: #f6f9fc;
1240 }
1241
1242 .export-item-row input {
1243     width: 16px;
1244     height: 16px;
1245     margin: 0;
1246     accent-color: #2f7del;
1247 }
1248
1249 .export-item-icon {
1250     width: 32px;
1251     height: 32px;
```

```
front/src/workbench/workbench-polish.css
```

```
1252     display: grid;
1253     place-items: center;
1254     border-radius: 7px;
1255     color: #2f7del;
1256     background: #eaf3ff;
1257 }
1258
1259 .export-item-copy {
1260     min-width: 0;
1261     display: grid;
1262     gap: 3px;
1263 }
1264
1265 .export-item-copy strong,
1266 .export-item-copy small {
1267     overflow: hidden;
1268     text-overflow: ellipsis;
1269     white-space: nowrap;
1270 }
1271
1272 .export-item-copy strong {
1273     color: #34475b;
1274     font-size: 12px;
1275 }
1276
1277 .export-item-copy small,
1278 .export-item-size {
1279     color: #8190a1;
1280     font-size: 10px;
1281 }
1282
1283 .export-settings-panel {
1284     display: flex;
1285     flex-direction: column;
1286     gap: 13px;
1287 }
1288
1289 .export-field {
1290     display: grid;
1291     gap: 6px;
1292 }
1293
1294 .export-field > span,
1295 .export-format-field legend {
1296     color: #6f8093;
1297     font-size: 11px;
1298     font-weight: 650;
1299 }
1300
1301 .export-field input {
```

```
front/src/workbench/workbench-polish.css
```

```
1302     width: 100%;
1303     height: 36px;
1304     border: 1px solid var(--app-border);
1305     border-radius: 7px;
1306     padding-inline: 10px;
1307     color: #34475b;
1308     background: #ffffff;
1309     font-size: 12px;
1310     font-weight: 600;
1311 }
1312
1313 .export-field input:focus-visible {
1314     outline: 2px solid #c8ddf5;
1315     border-color: #2f7del;
1316 }
1317
1318 .export-format-field {
1319     min-width: 0;
1320     margin: 0;
1321     padding: 0;
1322     border: 0;
1323 }
1324
1325 .export-format-field legend {
1326     margin-bottom: 6px;
1327 }
1328
1329 .export-format-options {
1330     display: grid;
1331     grid-template-columns: repeat(3, minmax(0, 1fr));
1332     gap: 7px;
1333 }
1334
1335 .export-format-options button {
1336     height: 40px;
1337     display: inline-flex;
1338     align-items: center;
1339     justify-content: center;
1340     gap: 7px;
1341     border: 1px solid var(--app-border);
1342     border-radius: 7px;
1343     color: #637488;
1344     background: #ffffff;
1345     cursor: pointer;
1346     font-size: 11px;
1347     font-weight: 650;
1348 }
1349
1350 .export-format-options button:hover {
1351     border-color: #abc8e8;
```

front/src/workbench/workbench-polish.css

```
1352     background: #f5f9fd;
1353 }
1354
1355 .export-format-options button.active {
1356     color: #1f64bd;
1357     border-color: #89b7eb;
1358     background: #eaf3ff;
1359 }
1360
1361 .export-manifest {
1362     display: grid;
1363     grid-template-columns: repeat(2, minmax(0, 1fr));
1364     gap: 0;
1365     border: 1px solid var(--app-border);
1366     border-radius: 7px;
1367     overflow: hidden;
1368 }
1369
1370 .export-manifest > div {
1371     min-width: 0;
1372     padding: 9px 10px;
1373     display: grid;
1374     gap: 3px;
1375     border-top: 1px solid #edf0f2;
1376     border-left: 1px solid #edf0f2;
1377     background: #f9fafb;
1378 }
1379
1380 .export-manifest > div:nth-child(-n + 2) {
1381     border-top: 0;
1382 }
1383
1384 .export-manifest > div:nth-child(odd) {
1385     border-left: 0;
1386 }
1387
1388 .export-manifest span,
1389 .export-manifest strong {
1390     overflow: hidden;
1391     text-overflow: ellipsis;
1392     white-space: nowrap;
1393 }
1394
1395 .export-manifest span {
1396     color: #8290a0;
1397     font-size: 10px;
1398 }
1399
1400 .export-manifest strong {
1401     color: #3c5065;
```

```
front/src/workbench/workbench-polish.css
```

```
1402     font-size: 11px;
1403 }
1404
1405 .export-submit {
1406     width: 100%;
1407     margin-top: auto;
1408 }
1409
1410 .recent-export-table {
1411     border: 1px solid var(--app-border);
1412     border-radius: 7px;
1413     overflow: hidden;
1414 }
1415
1416 .recent-export-row {
1417     min-width: 0;
1418     min-height: 46px;
1419     display: grid;
1420     grid-template-columns: auto minmax(220px, 1.3fr) minmax(170px, 0.8fr) 86px 110px auto;
1421     align-items: center;
1422     gap: 11px;
1423     padding-inline: 12px;
1424     border-top: 1px solid #edf0f2;
1425 }
1426
1427 .recent-export-row:first-child {
1428     border-top: 0;
1429 }
1430
1431 .recent-export-row > svg {
1432     color: #2f7del;
1433 }
1434
1435 .recent-export-row strong,
1436 .recent-export-row span,
1437 .recent-export-row time {
1438     min-width: 0;
1439     overflow: hidden;
1440     text-overflow: ellipsis;
1441     white-space: nowrap;
1442 }
1443
1444 .recent-export-row strong {
1445     color: #34475b;
1446     font-size: 11px;
1447 }
1448
1449 .recent-export-row span,
1450 .recent-export-row time {
1451     color: #7c8b9c;
```



```
front/src/workbench/workbench-polish.css
```

```
1452     font-size: 10px;
1453 }
1454
1455 @media (max-width: 1360px) {
1456     .ca-primary-grid,
1457     .mkl-primary-grid,
1458     .result-primary-grid,
1459     .export-builder-grid {
1460         grid-template-columns: minmax(0, 1.45fr) minmax(300px, 0.75fr);
1461     }
1462
1463     .ca-diagnostics-grid {
1464         grid-template-columns: minmax(220px, 0.8fr) minmax(260px, 1fr) minmax(240px, 0.8fr);
1465         gap: 16px;
1466     }
1467
1468     .mkl-kernel-row {
1469         grid-template-columns: minmax(140px, 0.8fr) minmax(170px, 1fr);
1470     }
1471 }
1472
1473 /* 核配置页控制矩阵预览的实际尺寸，防止正方形单元随列宽无限放大。 */
1474 .kernel-config-grid {
1475     grid-template-columns: minmax(360px, 0.78fr) minmax(480px, 1.22fr);
1476     gap: 14px;
1477 }
1478
1479 .kernel-config-grid > .panel {
1480     height: 100%;
1481 }
1482
1483 .soft-kernel-panel,
1484 .soft-matrix-panel {
1485     padding: 15px 16px;
1486 }
1487
1488 .soft-kernel-panel {
1489     display: flex;
1490     flex-direction: column;
1491 }
1492
1493 .soft-kernel-panel .soft-segmented {
1494     margin-top: 4px;
1495 }
1496
1497 .soft-formula-box {
1498     margin-top: 12px;
1499     border-color: #c8ddf5;
1500     background: #f1f7fd;
1501 }
```

```
front/src/workbench/workbench-polish.css
```

```
1502
1503 .soft-formula-box strong {
1504     color: #1f64bd;
1505 }
1506
1507 .soft-param-grid input {
1508     height: 36px;
1509     border-color: var(--app-border);
1510     border-radius: 7px;
1511     font-weight: 650;
1512 }
1513
1514 .soft-param-grid input:focus-visible {
1515     outline: 2px solid #c8ddf5;
1516     border-color: #2f7de1;
1517 }
1518
1519 .kernel-validation-list {
1520     margin-top: 13px;
1521     display: flex;
1522     flex-wrap: wrap;
1523     gap: 7px 12px;
1524 }
1525
1526 .kernel-validation-list span {
1527     display: inline-flex;
1528     align-items: center;
1529     gap: 5px;
1530     color: #617285;
1531     font-size: 10px;
1532     font-weight: 600;
1533 }
1534
1535 .kernel-validation-list svg {
1536     color: #23865f;
1537 }
1538
1539 .soft-kernel-grid {
1540     min-height: 286px;
1541     display: grid;
1542     grid-template-columns: repeat(6, 42px);
1543     grid-template-rows: repeat(6, 42px);
1544     place-content: center;
1545     gap: 5px;
1546     padding: 14px;
1547     background: #f7f9fa;
1548 }
1549
1550 .soft-kernel-grid span {
1551     width: 42px;
```

```
front/src/workbench/workbench-polish.css
```

```
1552     height: 42px;
1553     aspect-ratio: auto;
1554     border-radius: 5px;
1555     background: #2f7del;
1556     transition:
1557         transform var(--motion-fast) var(--ease-out-quart),
1558         opacity var(--motion-fast) ease;
1559 }
1560
1561 .soft-kernel-grid span:hover {
1562     transform: scale(1.08);
1563 }
1564
1565 .kernel-matrix-legend {
1566     margin-top: 9px;
1567     display: flex;
1568     align-items: center;
1569     justify-content: center;
1570     gap: 14px;
1571     color: #748498;
1572     font-size: 10px;
1573 }
1574
1575 .kernel-matrix-legend span {
1576     display: inline-flex;
1577     align-items: center;
1578     gap: 5px;
1579 }
1580
1581 .kernel-matrix-legend i {
1582     width: 11px;
1583     height: 11px;
1584     border-radius: 3px;
1585     background: rgba(47, 125, 225, 0.28);
1586 }
1587
1588 .kernel-matrix-legend i.mid {
1589     background: rgba(47, 125, 225, 0.62);
1590 }
1591
1592 .kernel-matrix-legend i.high {
1593     background: #2f7del;
1594 }
1595
1596 .kernel-source-list {
1597     margin-top: 11px;
1598     display: grid;
1599     grid-template-columns: repeat(2, minmax(0, 1fr));
1600     overflow: hidden;
1601     border: 1px solid var(--app-border);
```

```
front/src/workbench/workbench-polish.css
```

```
1602     border-radius: 7px;
1603 }
1604
1605 .kernel-source-list > div {
1606     min-width: 0;
1607     min-height: 44px;
1608     display: grid;
1609     grid-template-columns: auto auto minmax(0, 1fr) auto;
1610     align-items: center;
1611     gap: 8px;
1612     padding-inline: 10px;
1613     border-top: 1px solid #edf0f2;
1614     border-left: 1px solid #edf0f2;
1615 }
1616
1617 .kernel-source-list > div:nth-child(-n + 2) {
1618     border-top: 0;
1619 }
1620
1621 .kernel-source-list > div:nth-child(odd) {
1622     border-left: 0;
1623 }
1624
1625 .kernel-source-list svg {
1626     color: #2f7del;
1627 }
1628
1629 .kernel-source-list strong,
1630 .kernel-source-list span,
1631 .kernel-source-list b {
1632     overflow: hidden;
1633     text-overflow: ellipsis;
1634     white-space: nowrap;
1635 }
1636
1637 .kernel-source-list strong {
1638     color: #34475b;
1639     font-size: 11px;
1640 }
1641
1642 .kernel-source-list span {
1643     color: #8190a1;
1644     font-size: 10px;
1645 }
1646
1647 .kernel-source-list b {
1648     color: #1f64bd;
1649     font-size: 10px;
1650     font-variant-numeric: tabular-nums;
1651 }
```

```
front/src/workbench/workbench-polish.css
```

```
1652
1653 .performance-layout {
1654     min-width: 0;
1655     display: grid;
1656     grid-template-columns: minmax(0, 1.55fr) minmax(320px, 0.65fr);
1657     gap: 14px;
1658     align-items: stretch;
1659 }
1660
1661 .performance-layout > .panel {
1662     height: 100%;
1663 }
1664
1665 .performance-stability-panel {
1666     display: flex;
1667     flex-direction: column;
1668     gap: 14px;
1669 }
1670
1671 .performance-score {
1672     min-height: 76px;
1673     display: flex;
1674     align-items: center;
1675     justify-content: space-between;
1676     gap: 12px;
1677     padding: 12px 14px;
1678     border: 1px solid #d6e5dc;
1679     border-radius: 7px;
1680     background: #f2f8f5;
1681 }
1682
1683 .performance-score > div {
1684     display: flex;
1685     align-items: baseline;
1686     gap: 5px;
1687 }
1688
1689 .performance-score span,
1690 .performance-score small {
1691     color: #6e8076;
1692     font-size: 10px;
1693 }
1694
1695 .performance-score strong {
1696     margin-left: 8px;
1697     color: #23865f;
1698     font-size: 28px;
1699     font-variant-numeric: tabular-nums;
1700 }
1701
```

```
front/src/workbench/workbench-polish.css
```

```
1702 .performance-score svg {
1703     color: #23865f;
1704 }
1705
1706 .performance-progress-list {
1707     display: grid;
1708     gap: 12px;
1709 }
1710
1711 .performance-stability-panel .workbench-notice {
1712     margin-top: auto;
1713 }
1714
1715 .performance-comparison-panel {
1716     padding: 15px 16px;
1717 }
1718
1719 .performance-comparison-table {
1720     overflow: hidden;
1721     border: 1px solid var(--app-border);
1722     border-radius: 7px;
1723 }
1724
1725 .performance-comparison-row {
1726     min-height: 42px;
1727     display: grid;
1728     grid-template-columns: minmax(150px, 1.2fr) repeat(4, minmax(78px, 0.75fr)) minmax(96px, 0.75fr);
1729     align-items: center;
1730     gap: 12px;
1731     padding-inline: 12px;
1732     border-top: 1px solid #edf0f2;
1733     background: #ffffff;
1734 }
1735
1736 .performance-comparison-row:first-child {
1737     border-top: 0;
1738 }
1739
1740 .performance-comparison-row.header {
1741     min-height: 35px;
1742     color: #78889b;
1743     background: #f7f9fa;
1744     font-size: 10px;
1745     font-weight: 700;
1746 }
1747
1748 .performance-comparison-row strong,
1749 .performance-comparison-row span,
1750 .performance-comparison-row b {
1751     min-width: 0;
```

```
front/src/workbench/workbench-polish.css
```

```
1752     overflow: hidden;
1753     text-overflow: ellipsis;
1754     white-space: nowrap;
1755 }
1756
1757 .performance-comparison-row strong {
1758     color: #34475b;
1759     font-size: 11px;
1760 }
1761
1762 .performance-comparison-row span,
1763 .performance-comparison-row b {
1764     color: #66778b;
1765     font-size: 10px;
1766     font-variant-numeric: tabular-nums;
1767 }
1768
1769 .performance-comparison-row.best {
1770     background: #f2f8f5;
1771 }
1772
1773 .performance-comparison-row.best strong,
1774 .performance-comparison-row.best b {
1775     color: #23865f;
1776 }
1777
1778 .performance-comparison-row.good b {
1779     color: #1f64bd;
1780 }
1781
1782 .visualization-workspace {
1783     min-width: 0;
1784     display: grid;
1785     grid-template-columns: minmax(0, 1fr) 330px;
1786     gap: 14px;
1787     align-items: stretch;
1788 }
1789
1790 .visualization-featured,
1791 .visualization-index {
1792     height: 100%;
1793     padding: 15px 16px;
1794 }
1795
1796 .visualization-featured {
1797     display: flex;
1798     flex-direction: column;
1799 }
1800
1801 .visualization-chart {
```

```
front/src/workbench/workbench-polish.css
```

```
1802     width: 100%;
1803     height: 420px;
1804     min-height: 420px;
1805 }
1806
1807 .visualization-chart-footer {
1808     min-height: 32px;
1809     display: flex;
1810     align-items: center;
1811     justify-content: space-between;
1812     gap: 14px;
1813     padding-top: 9px;
1814     border-top: 1px solid var(--app-border);
1815 }
1816
1817 .visualization-chart-footer span {
1818     display: inline-flex;
1819     align-items: center;
1820     gap: 6px;
1821     color: #7c8b9c;
1822     font-size: 10px;
1823 }
1824
1825 .visualization-chart-footer svg {
1826     color: #23865f;
1827 }
1828
1829 .visualization-view-list {
1830     display: grid;
1831     gap: 7px;
1832 }
1833
1834 .visualization-view-list button {
1835     min-width: 0;
1836     min-height: 68px;
1837     display: grid;
1838     grid-template-columns: auto minmax(0, 1fr) auto;
1839     align-items: center;
1840     gap: 10px;
1841     padding: 9px 10px;
1842     border: 1px solid var(--app-border);
1843     border-radius: 7px;
1844     color: #56687b;
1845     background: #ffffff;
1846     cursor: pointer;
1847     text-align: left;
1848     transition:
1849         border-color var(--motion-fast) ease,
1850         background-color var(--motion-fast) ease,
1851         transform var(--motion-fast) var(--ease-out-quart);
```



```
front/src/workbench/workbench-polish.css
```

```
1852 }
1853
1854 .visualization-view-list button:hover {
1855     border-color: #b4cde7;
1856     background: #f6f9fc;
1857     transform: translateX(2px);
1858 }
1859
1860 .visualization-view-list button.active {
1861     color: #1f64bd;
1862     border-color: #9fc4ef;
1863     background: #edf5fe;
1864     transform: none;
1865 }
1866
1867 .visualization-view-icon {
1868     width: 34px;
1869     height: 34px;
1870     display: grid;
1871     place-items: center;
1872     border-radius: 7px;
1873     color: #2f7del;
1874     background: #eaf3ff;
1875 }
1876
1877 .visualization-view-list button > span:nth-child(2) {
1878     min-width: 0;
1879     display: grid;
1880     gap: 4px;
1881 }
1882
1883 .visualization-view-list strong,
1884 .visualization-view-list small {
1885     overflow: hidden;
1886     text-overflow: ellipsis;
1887     white-space: nowrap;
1888 }
1889
1890 .visualization-view-list strong {
1891     color: #34475b;
1892     font-size: 12px;
1893 }
1894
1895 .visualization-view-list small {
1896     color: #8190a1;
1897     font-size: 10px;
1898 }
1899
1900 .visualization-view-list i {
1901     color: #1f64bd;
```

```
front/src/workbench/workbench-polish.css
```

```
1902     font-size: 10px;
1903     font-style: normal;
1904     font-weight: 700;
1905 }
1906
1907 .report-workspace {
1908     min-width: 0;
1909     display: grid;
1910     grid-template-columns: minmax(360px, 0.78fr) minmax(0, 1.22fr);
1911     gap: 14px;
1912     align-items: stretch;
1913 }
1914
1915 .report-builder-panel,
1916 .report-preview-panel {
1917     height: 100%;
1918     padding: 15px 16px;
1919 }
1920
1921 .report-builder-panel {
1922     display: flex;
1923     flex-direction: column;
1924     gap: 13px;
1925 }
1926
1927 .report-fields {
1928     display: grid;
1929     grid-template-columns: repeat(2, minmax(0, 1fr));
1930     gap: 9px;
1931 }
1932
1933 .report-fields label {
1934     min-width: 0;
1935     display: grid;
1936     gap: 6px;
1937 }
1938
1939 .report-fields span,
1940 .report-format-field legend {
1941     color: #6f8093;
1942     font-size: 10px;
1943     font-weight: 650;
1944 }
1945
1946 .report-fields input {
1947     width: 100%;
1948     height: 36px;
1949     min-width: 0;
1950     padding-inline: 10px;
1951     border: 1px solid var(--app-border);
```

```
front/src/workbench/workbench-polish.css
```

```
1952   border-radius: 7px;
1953   color: #34475b;
1954   background: #ffffff;
1955   font-size: 11px;
1956   font-weight: 600;
1957 }
1958
1959 .report-fields input:focus-visible {
1960   outline: 2px solid #c8ddf5;
1961   border-color: #2f7de1;
1962 }
1963
1964 .report-format-field {
1965   margin: 0;
1966   padding: 0;
1967   border: 0;
1968 }
1969
1970 .report-format-field legend {
1971   margin-bottom: 6px;
1972 }
1973
1974 .report-format-field > div {
1975   display: grid;
1976   grid-template-columns: repeat(2, minmax(0, 1fr));
1977   gap: 7px;
1978 }
1979
1980 .report-format-field button {
1981   height: 36px;
1982   display: inline-flex;
1983   align-items: center;
1984   justify-content: center;
1985   gap: 7px;
1986   border: 1px solid var(--app-border);
1987   border-radius: 7px;
1988   color: #647589;
1989   background: #ffffff;
1990   cursor: pointer;
1991   font-size: 11px;
1992   font-weight: 650;
1993 }
1994
1995 .report-format-field button.active {
1996   color: #1f64bd;
1997   border-color: #9fc4ef;
1998   background: #edf5fe;
1999 }
2000
2001 .report-section-list {
```

```
front/src/workbench/workbench-polish.css
```

```
2002     overflow: hidden;
2003     border: 1px solid var(--app-border);
2004     border-radius: 7px;
2005 }
2006
2007 .report-section-list label {
2008     min-width: 0;
2009     min-height: 51px;
2010     display: grid;
2011     grid-template-columns: auto minmax(0, 1fr);
2012     align-items: center;
2013     gap: 10px;
2014     padding: 8px 10px;
2015     border-top: 1px solid #edf0f2;
2016     background: #ffffff;
2017     cursor: pointer;
2018 }
2019
2020 .report-section-list label:first-child {
2021     border-top: 0;
2022 }
2023
2024 .report-section-list label.selected {
2025     background: #f6f9fc;
2026 }
2027
2028 .report-section-list input {
2029     width: 15px;
2030     height: 15px;
2031     margin: 0;
2032     accent-color: #2f7del;
2033 }
2034
2035 .report-section-list label > span {
2036     min-width: 0;
2037     display: grid;
2038     gap: 3px;
2039 }
2040
2041 .report-section-list strong,
2042 .report-section-list small {
2043     overflow: hidden;
2044     text-overflow: ellipsis;
2045     white-space: nowrap;
2046 }
2047
2048 .report-section-list strong {
2049     color: #34475b;
2050     font-size: 11px;
2051 }
```

```
front/src/workbench/workbench-polish.css
```

```
2052
2053 .report-section-list small {
2054     color: #8290a0;
2055     font-size: 10px;
2056 }
2057
2058 .report-document {
2059     min-height: 525px;
2060     padding: 28px 32px 20px;
2061     display: flex;
2062     flex-direction: column;
2063     border: 1px solid #dfe4e9;
2064     border-radius: 4px;
2065     color: #33475b;
2066     background: #ffffff;
2067 }
2068
2069 .report-document > header {
2070     display: grid;
2071     gap: 7px;
2072     padding-bottom: 18px;
2073     border-bottom: 2px solid #274d78;
2074 }
2075
2076 .report-document > header span {
2077     color: #2f7del;
2078     font-size: 9px;
2079     font-weight: 750;
2080 }
2081
2082 .report-document > header strong {
2083     overflow-wrap: anywhere;
2084     color: #22384f;
2085     font-size: 22px;
2086     line-height: 1.25;
2087 }
2088
2089 .report-document > header small {
2090     color: #7c8b9c;
2091     font-size: 10px;
2092 }
2093
2094 .report-document-summary {
2095     margin-top: 18px;
2096     display: grid;
2097     grid-template-columns: repeat(4, minmax(0, 1fr));
2098     border-block: 1px solid #e3e7eb;
2099 }
2100
2101 .report-document-summary > div {
```

```
front/src/workbench/workbench-polish.css
```

```
2102   min-width: 0;
2103   padding: 11px;
2104   display: grid;
2105   gap: 4px;
2106   border-left: 1px solid #e3e7eb;
2107 }
2108
2109 .report-document-summary > div:first-child {
2110   border-left: 0;
2111 }
2112
2113 .report-document-summary span,
2114 .report-document-summary strong {
2115   overflow: hidden;
2116   text-overflow: ellipsis;
2117   white-space: nowrap;
2118 }
2119
2120 .report-document-summary span {
2121   color: #8290a0;
2122   font-size: 9px;
2123 }
2124
2125 .report-document-summary strong {
2126   color: #34475b;
2127   font-size: 11px;
2128 }
2129
2130 .report-document-metrics {
2131   margin-top: 15px;
2132   display: grid;
2133   grid-template-columns: repeat(4, minmax(0, 1fr));
2134   gap: 7px;
2135 }
2136
2137 .report-document-metrics span {
2138   padding: 8px;
2139   border-radius: 5px;
2140   color: #1f64bd;
2141   background: #edf5fe;
2142   font-size: 10px;
2143   font-weight: 700;
2144   text-align: center;
2145 }
2146
2147 .report-document-toc {
2148   margin-top: 24px;
2149 }
2150
2151 .report-document-toc h3 {
```

```
front/src/workbench/workbench-polish.css
```

```
2152     margin: 0 0 11px;
2153     color: #2d4258;
2154     font-size: 13px;
2155 }
2156
2157 .report-document-toc > div {
2158     min-height: 31px;
2159     display: flex;
2160     align-items: center;
2161     gap: 9px;
2162     color: #536579;
2163     font-size: 10px;
2164 }
2165
2166 .report-document-toc i {
2167     flex: 1;
2168     height: 1px;
2169     border-top: 1px dotted #c9d0d7;
2170 }
2171
2172 .report-document-toc p {
2173     color: #8a97a6;
2174     font-size: 10px;
2175 }
2176
2177 .report-document > footer {
2178     margin-top: auto;
2179     padding-top: 14px;
2180     display: flex;
2181     justify-content: space-between;
2182     border-top: 1px solid #e3e7eb;
2183     color: #8b98a6;
2184     font-size: 9px;
2185 }
2186
2187 .logs-workspace {
2188     padding: 15px 16px;
2189 }
2190
2191 .logs-toolbar {
2192     min-width: 0;
2193     display: flex;
2194     align-items: center;
2195     justify-content: space-between;
2196     gap: 12px;
2197     margin-bottom: 11px;
2198 }
2199
2200 .logs-search {
2201     width: min(420px, 42%);
```

```
front/src/workbench/workbench-polish.css
```

```
2202     height: 36px;
2203     display: flex;
2204     align-items: center;
2205     gap: 8px;
2206     padding-inline: 10px;
2207     border: 1px solid var(--app-border);
2208     border-radius: 7px;
2209     color: #8290a0;
2210     background: #ffffff;
2211 }
2212
2213 .logs-search:focus-within {
2214     border-color: #2f7de1;
2215     outline: 2px solid #c8ddf5;
2216 }
2217
2218 .logs-search input {
2219     flex: 1;
2220     min-width: 0;
2221     border: 0;
2222     outline: 0;
2223     color: #34475b;
2224     background: transparent;
2225     font-size: 11px;
2226 }
2227
2228 .logs-search input::placeholder {
2229     color: #8190a1;
2230     opacity: 1;
2231 }
2232
2233 .logs-level-filter {
2234     display: inline-flex;
2235     gap: 2px;
2236     padding: 3px;
2237     border: 1px solid var(--app-border);
2238     border-radius: 7px;
2239     background: #f7f9fa;
2240 }
2241
2242 .logs-level-filter button {
2243     height: 28px;
2244     padding-inline: 11px;
2245     border: 0;
2246     border-radius: 5px;
2247     color: #6f8093;
2248     background: transparent;
2249     cursor: pointer;
2250     font-size: 10px;
2251     font-weight: 650;
```



```
front/src/workbench/workbench-polish.css
```

```
2252 }
2253
2254 .logs-level-filter button.active {
2255     color: #1f64bd;
2256     background: #ffffff;
2257     box-shadow: 0 1px 3px rgba(31, 45, 61, 0.08);
2258 }
2259
2260 .logs-table {
2261     overflow: hidden;
2262     border: 1px solid var(--app-border);
2263     border-radius: 7px;
2264 }
2265
2266 .logs-row {
2267     min-width: 0;
2268     min-height: 52px;
2269     display: grid;
2270     grid-template-columns: 84px 110px minmax(280px, 1fr) 100px 152px;
2271     align-items: center;
2272     gap: 12px;
2273     padding-inline: 12px;
2274     border-top: 1px solid #edf0f2;
2275     background: #ffffff;
2276 }
2277
2278 .logs-row:first-child {
2279     border-top: 0;
2280 }
2281
2282 .logs-row.header {
2283     min-height: 36px;
2284     color: #79889a;
2285     background: #f7f9fa;
2286     font-size: 10px;
2287     font-weight: 700;
2288 }
2289
2290 .logs-row strong,
2291 .logs-row > span,
2292 .logs-row code,
2293 .logs-row time {
2294     min-width: 0;
2295     overflow: hidden;
2296     text-overflow: ellipsis;
2297     white-space: nowrap;
2298 }
2299
2300 .logs-row strong {
2301     color: #34475b;
```

```
front/src/workbench/workbench-polish.css
```

```
2302     font-size: 11px;
2303 }
2304
2305 .logs-row > span,
2306 .logs-row time {
2307     color: #6f8093;
2308     font-size: 10px;
2309 }
2310
2311 .logs-row code {
2312     width: max-content;
2313     max-width: 100%;
2314     padding: 3px 6px;
2315     border-radius: 4px;
2316     color: #52667c;
2317     background: #f0f3f6;
2318     font-family: ui-monospace, SFMono-Regular, Consolas, monospace;
2319     font-size: 9px;
2320 }
2321
2322 .logs-empty {
2323     min-height: 180px;
2324     display: grid;
2325     place-items: center;
2326     align-content: center;
2327     gap: 6px;
2328     color: #8190a1;
2329 }
2330
2331 .logs-empty strong {
2332     color: #42556a;
2333     font-size: 12px;
2334 }
2335
2336 .logs-empty span {
2337     font-size: 10px;
2338 }
2339
2340 @media (max-width: 1360px) {
2341     .kernel-config-grid {
2342         grid-template-columns: minmax(340px, 0.8fr) minmax(430px, 1.2fr);
2343     }
2344
2345     .visualization-workspace {
2346         grid-template-columns: minmax(0, 1fr) 300px;
2347     }
2348
2349     .report-workspace {
2350         grid-template-columns: minmax(340px, 0.82fr) minmax(0, 1.18fr);
2351     }
```

```
front/src/workbench/workbench-polish.css
```

```
2352 }
2353 /* 公式容器保留原有面板节奏，并在长公式时允许横向查看而不压缩数学符号。 */
2354 .math-formula {
2355     color: #172033;
2356     display: inline-flex;
2357     max-width: 100%;
2358     overflow-x: auto;
2359     overflow-y: hidden;
2360     padding: 2px 0;
2361     vertical-align: middle;
2362 }
2363
2364 .math-formula .katex {
2365     font-size: 1.06em;
2366 }
2367
2368 .math-formula-display {
2369     align-items: center;
2370     justify-content: center;
2371     width: 100%;
2372     min-height: 42px;
2373 }
2374
2375 .math-formula-fallback {
2376     color: #3c4a62;
2377     font-family: ui-monospace, SFMono-Regular, Menlo, Consolas, monospace;
2378     font-size: 0.875rem;
2379     line-height: 1.5;
2380 }
2381
2382 @media (prefers-reduced-motion: reduce) {
2383     .workbench-page *,
2384     .soft-page * {
2385         animation-duration: 0.01ms !important;
2386         transition-duration: 0.01ms !important;
2387         scroll-behavior: auto !important;
2388     }
2389 }
2390
2391 .result-state-panel,
2392 .result-summary-panel,
2393 .result-pairs-panel,
2394 .result-kernel-panel,
2395 .result-export-panel,
2396 .result-report-controls,
2397 .result-report-preview {
2398     min-width: 0;
2399 }
2400
2401 .result-state-panel {
```

```
front/src/workbench/workbench-polish.css
```

```
2402     display: grid;
2403     gap: 18px;
2404     max-width: 760px;
2405 }
2406
2407 .result-state-actions,
2408 .result-stat-line,
2409 .result-export-checks {
2410     align-items: center;
2411     display: flex;
2412     flex-wrap: wrap;
2413     gap: 10px;
2414 }
2415
2416 .result-two-column {
2417     align-items: start;
2418     display: grid;
2419     gap: 18px;
2420     grid-template-columns: minmax(0, 1.25fr) minmax(340px, 0.75fr);
2421     margin-top: 18px;
2422 }
2423
2424 .result-two-column.visualization {
2425     grid-template-columns: minmax(0, 1.35fr) minmax(280px, 0.45fr);
2426 }
2427
2428 .result-chart {
2429     height: 320px;
2430     min-height: 320px;
2431     width: 100%;
2432 }
2433
2434 .result-chart.large {
2435     height: 480px;
2436     min-height: 480px;
2437 }
2438
2439 /* 色阶独立于 ECharts 绘图区，避免与横轴标签及下方统计信息争用空间。 */
2440 .result-matrix-scale {
2441     min-height: 30px;
2442     display: grid;
2443     grid-template-columns: 64px minmax(140px, 220px) 64px;
2444     align-items: center;
2445     justify-content: center;
2446     gap: 8px;
2447     padding-top: 4px;
2448     color: #64758a;
2449     font-size: 10px;
2450     font-variant-numeric: tabular-nums;
2451 }
```

```
front/src/workbench/workbench-polish.css
```

```
2452
2453 .result-matrix-scale-bound.is-minimum {
2454     text-align: right;
2455 }
2456
2457 .result-matrix-scale-track {
2458     position: relative;
2459     display: block;
2460     height: 9px;
2461     border: 1px solid #d4e0ea;
2462     border-radius: 999px;
2463     background: linear-gradient(90deg, #edf4fa, #78a9d4, #1c5f91);
2464 }
2465
2466 .result-matrix-scale-indicator {
2467     position: absolute;
2468     top: 50%;
2469     width: 0;
2470     height: 0;
2471     opacity: 0;
2472     pointer-events: none;
2473     transform: translate(-50%, -50%);
2474     transition:
2475         left 200ms cubic-bezier(0.22, 1, 0.36, 1),
2476         opacity 140ms ease-out;
2477 }
2478
2479 .result-matrix-scale-indicator.is-active {
2480     opacity: 1;
2481 }
2482
2483 .result-matrix-scale-indicator i {
2484     position: absolute;
2485     left: -4px;
2486     top: -4px;
2487     width: 8px;
2488     height: 8px;
2489     border: 2px solid #ffffff;
2490     border-radius: 50%;
2491     background: #173d5e;
2492     box-shadow: 0 0 0 1px rgba(23, 61, 94, 0.3);
2493 }
2494
2495 .result-matrix-scale-indicator output {
2496     position: absolute;
2497     left: 0;
2498     bottom: 8px;
2499     min-width: 48px;
2500     padding: 2px 5px;
2501     border-radius: 4px;
```

```
front/src/workbench/workbench-polish.css
```

```
2502     background: #253f58;
2503     color: #ffffff;
2504     font-size: 10px;
2505     font-weight: 650;
2506     line-height: 16px;
2507     text-align: center;
2508     transform: translateX(-50%);
2509     white-space: nowrap;
2510 }
2511
2512 .result-matrix-scale-indicator output::after {
2513     position: absolute;
2514     left: 50%;
2515     top: 100%;
2516     width: 0;
2517     height: 0;
2518     border: 4px solid transparent;
2519     border-top-color: #253f58;
2520     content: '';
2521     transform: translateX(-50%);
2522 }
2523
2524 .result-matrix-scale-indicator.is-start output {
2525     left: -5px;
2526     transform: none;
2527 }
2528
2529 .result-matrix-scale-indicator.is-start output::after {
2530     left: 5px;
2531     transform: none;
2532 }
2533
2534 .result-matrix-scale-indicator.is-end output {
2535     right: -5px;
2536     left: auto;
2537     transform: none;
2538 }
2539
2540 .result-matrix-scale-indicator.is-end output::after {
2541     right: 5px;
2542     left: auto;
2543     transform: none;
2544 }
2545
2546 .result-matrix-scale.is-compact {
2547     min-height: 26px;
2548     padding-top: 2px;
2549 }
2550
2551 .result-stat-line {
```

```
front/src/workbench/workbench-polish.css
```

```
2552   border-top: 1px solid #e4eaf0;
2553   color: #58667b;
2554   font-size: 0.8125rem;
2555   justify-content: space-between;
2556   padding-top: 12px;
2557 }
2558
2559 .result-summary-list {
2560   display: grid;
2561   gap: 10px;
2562   grid-template-columns: repeat(2, minmax(0, 1fr));
2563   margin: 0 0 18px;
2564 }
2565
2566 .result-summary-list div {
2567   background: #f5f8fb;
2568   border-radius: 6px;
2569   min-width: 0;
2570   padding: 12px;
2571 }
2572
2573 .result-summary-list dt {
2574   color: #647289;
2575   font-size: 0.75rem;
2576   margin-bottom: 5px;
2577 }
2578
2579 .result-summary-list dd {
2580   color: #172033;
2581   font-size: 0.9375rem;
2582   font-weight: 700;
2583   margin: 0;
2584   overflow-wrap: anywhere;
2585 }
2586
2587 .result-summary-list.compact {
2588   margin-top: 16px;
2589 }
2590
2591 .result-formula-panel {
2592   align-items: center;
2593   display: grid;
2594   gap: 8px;
2595   margin-top: 18px;
2596 }
2597
2598 .result-formula-panel > span,
2599 .soft-formula-box > span {
2600   color: #627089;
2601   font-size: 0.8125rem;
```

front/src/workbench/workbench-polish.css

```
2602     font-weight: 700;
2603 }
2604
2605 .result-pair-list,
2606 .result-kernel-list,
2607 .result-export-history {
2608     display: grid;
2609     gap: 8px;
2610 }
2611
2612 .result-pair-list > div,
2613 .result-kernel-list > div,
2614 .result-artifact-row {
2615     align-items: center;
2616     border-bottom: 1px solid #e7edf3;
2617     display: flex;
2618     gap: 12px;
2619     justify-content: space-between;
2620     min-height: 42px;
2621 }
2622
2623 .result-pair-list > div:last-child,
2624 .result-kernel-list > div:last-child,
2625 .result-artifact-row:last-child {
2626     border-bottom: 0;
2627 }
2628
2629 .result-kernel-list small,
2630 .result-artifact-row small,
2631 .result-export-history small {
2632     color: #68768a;
2633 }
2634
2635 .result-select-field {
2636     flex: 0 0 auto;
2637     width: 108px;
2638 }
2639
2640 .result-top-pair-select-field {
2641     width: 92px;
2642 }
2643
2644 /* ===== 性能评估：结论、证据和复核明细 ===== */
2645 .result-evaluation-header-actions {
2646     display: flex;
2647     align-items: center;
2648     gap: 8px;
2649 }
2650
2651 .result-evaluation-page > .panel {
```



```
front/src/workbench/workbench-polish.css
```

```
2652 padding: 15px 16px;
2653 }
2654
2655 .result-evaluation-page > .panel > .workbench-section-header,
2656 .result-evaluation-analysis-grid .panel > .workbench-section-header {
2657   margin-bottom: 13px;
2658 }
2659
2660 .result-evaluation-conclusion-body {
2661   min-width: 0;
2662   display: grid;
2663   grid-template-columns: minmax(300px, 0.78fr) minmax(0, 1.22fr);
2664   border-top: 1px solid var(--app-border);
2665 }
2666
2667 .result-evaluation-conclusion-copy {
2668   min-width: 0;
2669   display: grid;
2670   align-content: center;
2671   gap: 8px;
2672   padding: 18px 26px 17px 2px;
2673   border-right: 1px solid var(--app-border);
2674 }
2675
2676 .result-evaluation-conclusion-state {
2677   width: max-content;
2678   max-width: 100%;
2679   display: inline-flex;
2680   align-items: center;
2681   gap: 7px;
2682   color: #237a59;
2683   font-size: 11px;
2684   font-weight: 700;
2685 }
2686
2687 .result-evaluation-conclusion-copy > strong {
2688   color: #263b50;
2689   font-size: 18px;
2690   font-weight: 750;
2691   line-height: 1.35;
2692 }
2693
2694 .result-evaluation-conclusion-copy p {
2695   max-width: 68ch;
2696   margin: 0;
2697   color: #6d7d8f;
2698   font-size: 11px;
2699   line-height: 1.7;
2700 }
2701
```

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```
2702 .result-evaluation-conclusion-facts {
2703     min-width: 0;
2704     display: grid;
2705     grid-template-columns: repeat(4, minmax(0, 1fr));
2706     margin: 0;
2707 }
2708
2709 .result-evaluation-conclusion-facts > div {
2710     min-width: 0;
2711     display: grid;
2712     align-content: center;
2713     gap: 5px;
2714     padding: 17px 16px;
2715     border-left: 1px solid #edf0f2;
2716 }
2717
2718 .result-evaluation-conclusion-facts > div:first-child {
2719     border-left: 0;
2720 }
2721
2722 .result-evaluation-conclusion-facts dt {
2723     min-width: 0;
2724     display: flex;
2725     align-items: center;
2726     gap: 6px;
2727     overflow: hidden;
2728     color: #748396;
2729     font-size: 10px;
2730     font-weight: 650;
2731     text-overflow: ellipsis;
2732     white-space: nowrap;
2733 }
2734
2735 .result-evaluation-conclusion-facts dt svg {
2736     flex: 0 0 auto;
2737     color: #2f7del;
2738 }
2739
2740 .result-evaluation-conclusion-facts dd {
2741     min-width: 0;
2742     margin: 0;
2743     overflow: hidden;
2744     color: #294158;
2745     font-size: 17px;
2746     font-weight: 750;
2747     font-variant-numeric: tabular-nums;
2748     text-overflow: ellipsis;
2749     white-space: nowrap;
2750 }
2751
```

front/src/workbench/workbench-polish.css

```
2752 .result-evaluation-conclusion-facts small {
2753     min-width: 0;
2754     overflow: hidden;
2755     color: #8995a3;
2756     font-size: 9px;
2757     text-overflow: ellipsis;
2758     white-space: nowrap;
2759 }
2760
2761 .result-evaluation-analysis-grid {
2762     min-width: 0;
2763     display: grid;
2764     grid-template-columns: minmax(0, 1.55fr) minmax(330px, 0.65fr);
2765     gap: 12px;
2766     align-items: stretch;
2767 }
2768
2769 .result-evaluation-analysis-grid > .panel {
2770     height: 100%;
2771     padding: 15px 16px;
2772 }
2773
2774 .result-evaluation-trend-panel,
2775 .result-evaluation-stability {
2776     min-width: 0;
2777     display: flex;
2778     flex-direction: column;
2779 }
2780
2781 .result-evaluation-metric-select {
2782     width: 118px;
2783 }
2784
2785 .result-evaluation-trend-controls {
2786     display: flex;
2787     align-items: center;
2788     gap: 8px;
2789 }
2790
2791 .result-evaluation-trend-mode {
2792     display: inline-flex;
2793     align-items: center;
2794     gap: 2px;
2795     padding: 3px;
2796     border: 1px solid var(--app-border);
2797     border-radius: 7px;
2798     background: #f6f8fa;
2799 }
2800
2801 .result-evaluation-trend-mode button {
```

```
front/src/workbench/workbench-polish.css
```

```
2802     height: 26px;
2803     padding-inline: 9px;
2804     border: 0;
2805     border-radius: 5px;
2806     color: #748397;
2807     background: transparent;
2808     font: inherit;
2809     font-size: 10px;
2810     font-weight: 700;
2811     cursor: pointer;
2812     transition: color var(--motion-fast) ease, background-color var(--motion-fast) ease;
2813 }
2814
2815 .result-evaluation-trend-mode button:hover {
2816     color: #2f5f91;
2817 }
2818
2819 .result-evaluation-trend-mode button[aria-pressed='true'] {
2820     color: #1f64bd;
2821     background: #ffffff;
2822     box-shadow: 0 1px 3px rgba(35, 55, 78, 0.12);
2823 }
2824
2825 .result-evaluation-trend-mode button:focus-visible {
2826     outline: 2px solid rgba(47, 125, 225, 0.35);
2827     outline-offset: 1px;
2828 }
2829
2830 .result-evaluation-trend-summary {
2831     min-width: 0;
2832     display: grid;
2833     grid-template-columns: repeat(3, minmax(0, 1fr));
2834     border-block: 1px solid var(--app-border);
2835 }
2836
2837 .result-evaluation-trend-summary > span {
2838     min-width: 0;
2839     display: flex;
2840     align-items: baseline;
2841     gap: 7px;
2842     padding: 9px 12px;
2843     border-left: 1px solid #edf0f2;
2844     color: #7c8a9b;
2845     font-size: 10px;
2846     white-space: nowrap;
2847 }
2848
2849 .result-evaluation-trend-summary > span:first-child {
2850     border-left: 0;
2851 }
```

```
front/src/workbench/workbench-polish.css
```

```
2852
2853 .result-evaluation-trend-summary strong {
2854   min-width: 0;
2855   overflow: hidden;
2856   color: #30465c;
2857   font-size: 12px;
2858   font-variant-numeric: tabular-nums;
2859   text-overflow: ellipsis;
2860 }
2861
2862 .result-evaluation-trend-chart {
2863   width: 100%;
2864   height: 290px;
2865   flex: 1 1 auto;
2866   min-height: 290px;
2867 }
2868
2869 .result-evaluation-stability-list {
2870   display: grid;
2871   border-block: 1px solid var(--app-border);
2872 }
2873
2874 .result-evaluation-stability-row {
2875   min-width: 0;
2876   min-height: 49px;
2877   display: grid;
2878   grid-template-columns: minmax(88px, 1fr) auto minmax(56px, auto);
2879   align-items: center;
2880   gap: 10px;
2881   padding-inline: 3px;
2882   border-top: 1px solid #edf0f2;
2883 }
2884
2885 .result-evaluation-stability-row:first-child {
2886   border-top: 0;
2887 }
2888
2889 .result-evaluation-stability-row > div {
2890   min-width: 0;
2891   display: grid;
2892   gap: 3px;
2893 }
2894
2895 .result-evaluation-stability-row strong,
2896 .result-evaluation-stability-row small {
2897   overflow: hidden;
2898   text-overflow: ellipsis;
2899   white-space: nowrap;
2900 }
2901
```

```
front/src/workbench/workbench-polish.css
```

```
2902 .result-evaluation-stability-row strong {
2903     color: #34495d;
2904     font-size: 11px;
2905 }
2906
2907 .result-evaluation-stability-row small {
2908     color: #8794a3;
2909     font-size: 9px;
2910 }
2911
2912 .result-evaluation-stability-row > b {
2913     min-width: 0;
2914     overflow: hidden;
2915     color: #475b70;
2916     font-size: 11px;
2917     font-variant-numeric: tabular-nums;
2918     text-align: right;
2919     text-overflow: ellipsis;
2920     white-space: nowrap;
2921 }
2922
2923 .result-evaluation-stability-status {
2924     width: max-content;
2925     max-width: 100%;
2926     min-height: 21px;
2927     display: inline-flex;
2928     align-items: center;
2929     justify-content: center;
2930     padding: 2px 7px;
2931     border: 1px solid #dce3e9;
2932     border-radius: 999px;
2933     color: #6c7c8d;
2934     background: #f7f9fa;
2935     font-size: 9px;
2936     font-weight: 700;
2937     white-space: nowrap;
2938 }
2939
2940 .result-evaluation-stability-status.is-success {
2941     color: #237a59;
2942     border-color: #cce1d6;
2943     background: #f0f7f3;
2944 }
2945
2946 .result-evaluation-stability-status.is-warning {
2947     color: #a86413;
2948     border-color: #ead8bd;
2949     background: #fbf6ee;
2950 }
2951
```

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```
2952 .result-evaluation-runtime-band {
2953     min-width: 0;
2954     display: grid;
2955     grid-template-columns: 30px minmax(0, 1fr) auto;
2956     align-items: center;
2957     gap: 10px;
2958     margin-top: 13px;
2959     padding: 10px 11px;
2960     border: 1px solid #dce5ec;
2961     border-radius: 7px;
2962     background: #f7f9fb;
2963 }
2964
2965 .result-evaluation-runtime-band > svg {
2966     color: #2f7del;
2967 }
2968
2969 .result-evaluation-runtime-band > div {
2970     min-width: 0;
2971     display: grid;
2972     gap: 2px;
2973 }
2974
2975 .result-evaluation-runtime-band span,
2976 .result-evaluation-runtime-band small {
2977     color: #7c8998;
2978     font-size: 9px;
2979 }
2980
2981 .result-evaluation-runtime-band strong {
2982     color: #34495d;
2983     font-size: 11px;
2984     font-variant-numeric: tabular-nums;
2985 }
2986
2987 .result-evaluation-method-note {
2988     min-width: 0;
2989     display: flex;
2990     align-items: flex-start;
2991     gap: 8px;
2992     margin-top: auto;
2993     padding-top: 13px;
2994     color: #7b8998;
2995     font-size: 9px;
2996     line-height: 1.55;
2997 }
2998
2999 .result-evaluation-method-note svg {
3000     flex: 0 0 auto;
3001     margin-top: 1px;
```

```
front/src/workbench/workbench-polish.css
```

```
3002     color: #6e8295;
3003 }
3004
3005 .result-evaluation-distribution-list {
3006     display: grid;
3007     border-block: 1px solid var(--app-border);
3008 }
3009
3010 .result-evaluation-distribution-row {
3011     min-width: 0;
3012     min-height: 68px;
3013     display: grid;
3014     grid-template-columns: minmax(130px, 0.55fr) minmax(310px, 1.1fr) minmax(270px, 1.35fr) minmax(126px, 0.55fr);
3015     align-items: center;
3016     gap: 18px;
3017     padding: 10px 3px;
3018     border-top: 1px solid #edf0f2;
3019 }
3020
3021 .result-evaluation-distribution-row:first-child {
3022     border-top: 0;
3023 }
3024
3025 .result-evaluation-distribution-name {
3026     min-width: 0;
3027     display: flex;
3028     align-items: center;
3029     gap: 9px;
3030 }
3031
3032 .result-evaluation-distribution-name > strong {
3033     min-width: 0;
3034     overflow: hidden;
3035     color: #2f4459;
3036     font-size: 12px;
3037     text-overflow: ellipsis;
3038     white-space: nowrap;
3039 }
3040
3041 .result-evaluation-distribution-values {
3042     min-width: 0;
3043     display: grid;
3044     grid-template-columns: repeat(3, minmax(0, 1fr));
3045 }
3046
3047 .result-evaluation-distribution-values > span {
3048     min-width: 0;
3049     display: grid;
3050     gap: 3px;
3051     padding-inline: 11px;
```



```
front/src/workbench/workbench-polish.css
```

```
3052   border-left: 1px solid #edf0f2;
3053   color: #8793a1;
3054   font-size: 9px;
3055 }
3056
3057 .result-evaluation-distribution-values > span:first-child {
3058   padding-left: 0;
3059   border-left: 0;
3060 }
3061
3062 .result-evaluation-distribution-values strong {
3063   min-width: 0;
3064   overflow: hidden;
3065   color: #42576c;
3066   font-size: 11px;
3067   font-variant-numeric: tabular-nums;
3068   text-overflow: ellipsis;
3069   white-space: nowrap;
3070 }
3071
3072 .result-evaluation-range {
3073   min-width: 0;
3074   display: grid;
3075   grid-template-columns: 54px minmax(80px, 1fr) 54px;
3076   align-items: center;
3077   gap: 8px;
3078 }
3079
3080 .result-evaluation-range > span {
3081   color: #718093;
3082   font-size: 9px;
3083   font-variant-numeric: tabular-nums;
3084   white-space: nowrap;
3085 }
3086
3087 .result-evaluation-range > span:last-child {
3088   text-align: right;
3089 }
3090
3091 .result-evaluation-range-track {
3092   position: relative;
3093   height: 6px;
3094   border-radius: 999px;
3095   background: #e3e9ef;
3096 }
3097
3098 .result-evaluation-range-track::before,
3099 .result-evaluation-range-track::after {
3100   position: absolute;
3101   top: -2px;
```

```
front/src/workbench/workbench-polish.css
```

```
3102     width: 1px;
3103     height: 10px;
3104     background: #9aabbba;
3105     content: '';
3106 }
3107
3108 .result-evaluation-range-track::before {
3109     left: 0;
3110 }
3111
3112 .result-evaluation-range-track::after {
3113     right: 0;
3114 }
3115
3116 .result-evaluation-range-track i {
3117     position: absolute;
3118     top: 50%;
3119     width: 10px;
3120     height: 10px;
3121     border: 2px solid #ffffff;
3122     border-radius: 50%;
3123     background: #2f7del;
3124     box-shadow: 0 0 0 1px #2f7del;
3125     transform: translate(-50%, -50%);
3126 }
3127
3128 .result-evaluation-distribution-row.is-acc .result-evaluation-range-track i {
3129     background: #23865f;
3130     box-shadow: 0 0 0 1px #23865f;
3131 }
3132
3133 .result-evaluation-distribution-row.is-ari .result-evaluation-range-track i {
3134     background: #14796f;
3135     box-shadow: 0 0 0 1px #14796f;
3136 }
3137
3138 .result-evaluation-distribution-row.is-fl .result-evaluation-range-track i {
3139     background: #59677d;
3140     box-shadow: 0 0 0 1px #59677d;
3141 }
3142
3143 .result-evaluation-best-run {
3144     min-width: 0;
3145     display: flex;
3146     align-items: center;
3147     gap: 8px;
3148     color: #ad6b17;
3149 }
3150
3151 .result-evaluation-best-run svg {
```

```
front/src/workbench/workbench-polish.css
```

```
3152     flex: 0 0 auto;
3153 }
3154
3155 .result-evaluation-best-run > span {
3156     min-width: 0;
3157     display: grid;
3158     gap: 2px;
3159     color: #8b96a3;
3160     font-size: 9px;
3161 }
3162
3163 .result-evaluation-best-run strong {
3164     overflow: hidden;
3165     color: #3c5065;
3166     font-size: 11px;
3167     text-overflow: ellipsis;
3168     white-space: nowrap;
3169 }
3170
3171 .result-evaluation-table {
3172     min-width: 0;
3173     overflow-x: auto;
3174     border: 1px solid var(--app-border);
3175     border-radius: 7px;
3176 }
3177
3178 .result-evaluation-table-row {
3179     min-width: 1040px;
3180     min-height: 47px;
3181     display: grid;
3182     grid-template-columns: minmax(92px, 0.75fr) minmax(86px, 0.7fr) repeat(4, minmax(92px, 1fr)) minmax(88px, 0.72fr)
3183         minmax(176px, 1.35fr);
3184     align-items: center;
3185     gap: 7px;
3186     padding-inline: 10px;
3187     border-top: 1px solid #e7edf3;
3188     background: #ffffff;
3189     transition: background-color var(--motion-fast) ease;
3190 }
3191 .result-evaluation-table-row:first-child {
3192     border-top: 0;
3193 }
3194
3195 .result-evaluation-table-row.header {
3196     min-height: 37px;
3197     color: #68798c;
3198     background: #f5f7f9;
3199     font-size: 10px;
3200     font-weight: 700;
```

```
front/src/workbench/workbench-polish.css
```

```
3201 }
3202
3203 .result-evaluation-table-row:not(.header):hover {
3204     background: #f8fafc;
3205 }
3206
3207 .result-evaluation-table-row.is-best {
3208     background: #f3f8f5;
3209 }
3210
3211 .result-evaluation-table-row.is-representative > strong:first-child {
3212     color: #1f64bd;
3213 }
3214
3215 .result-evaluation-table-row > strong,
3216 .result-evaluation-table-row > span {
3217     min-width: 0;
3218     overflow: hidden;
3219     color: #536579;
3220     font-size: 10px;
3221     font-variant-numeric: tabular-nums;
3222     text-overflow: ellipsis;
3223     white-space: nowrap;
3224 }
3225
3226 .result-evaluation-table-row > strong {
3227     color: #30465c;
3228     font-size: 11px;
3229 }
3230
3231 .result-evaluation-table-row > .is-focused {
3232     padding: 7px 8px;
3233     border-radius: 5px;
3234     color: #1f64bd;
3235     background: #edf5fe;
3236     font-weight: 750;
3237 }
3238
3239 .result-evaluation-row-tags {
3240     display: flex;
3241     align-items: center;
3242     gap: 5px;
3243 }
3244
3245 .result-evaluation-row-tags b,
3246 .result-evaluation-row-tags small {
3247     min-height: 21px;
3248     display: inline-flex;
3249     align-items: center;
3250     padding: 2px 7px;
```

```
front/src/workbench/workbench-polish.css
```

```
3251     border-radius: 999px;
3252     font-size: 9px;
3253     font-weight: 700;
3254 }
3255
3256 .result-evaluation-row-tags b.is-best {
3257     color: #237a59;
3258     background: #e9f5ef;
3259 }
3260
3261 .result-evaluation-row-tags b.is-representative {
3262     color: #1f64bd;
3263     background: #eaf3ff;
3264 }
3265
3266 .result-evaluation-row-tags small {
3267     color: #7d8997;
3268     background: #f3f5f7;
3269 }
3270
3271 .result-evaluation-context-grid {
3272     min-width: 0;
3273     display: grid;
3274     grid-template-columns: repeat(5, minmax(0, 1fr));
3275     margin: 0;
3276     border-block: 1px solid var(--app-border);
3277 }
3278
3279 .result-evaluation-context-grid > div {
3280     min-width: 0;
3281     display: grid;
3282     gap: 5px;
3283     padding: 12px 14px;
3284     border-left: 1px solid #edf0f2;
3285 }
3286
3287 .result-evaluation-context-grid > div:nth-child(5n + 1) {
3288     border-left: 0;
3289 }
3290
3291 .result-evaluation-context-grid > div:nth-child(n + 6) {
3292     border-top: 1px solid #edf0f2;
3293 }
3294
3295 .result-evaluation-context-grid dt {
3296     color: #8491a0;
3297     font-size: 9px;
3298 }
3299
3300 .result-evaluation-context-grid dd {
```

```
front/src/workbench/workbench-polish.css
```

```
3301     min-width: 0;
3302     margin: 0;
3303     overflow: hidden;
3304     color: #34495d;
3305     font-size: 11px;
3306     font-weight: 700;
3307     font-variant-numeric: tabular-nums;
3308     text-overflow: ellipsis;
3309     white-space: nowrap;
3310 }
3311
3312 .result-evaluation-empty {
3313     min-height: 124px;
3314     display: flex;
3315     align-items: center;
3316     justify-content: center;
3317     gap: 12px;
3318     color: #738397;
3319 }
3320
3321 .result-evaluation-empty > div {
3322     display: grid;
3323     gap: 4px;
3324 }
3325
3326 .result-evaluation-empty strong {
3327     color: #34495d;
3328     font-size: 13px;
3329 }
3330
3331 .result-evaluation-empty span {
3332     font-size: 10px;
3333 }
3334
3335 .result-table {
3336     display: grid;
3337     overflow: auto;
3338 }
3339
3340 .result-table-row {
3341     align-items: center;
3342     border-bottom: 1px solid #e7edf3;
3343     display: grid;
3344     gap: 12px;
3345     grid-template-columns: 0.8fr repeat(4, minmax(90px, 1fr)) minmax(95px, 1fr);
3346     min-height: 42px;
3347     min-width: 760px;
3348     padding: 0 8px;
3349 }
3350
```

```
front/src/workbench/workbench-polish.css
```

```
3351 .result-table-row.header {
3352     background: #f4f7fa;
3353     color: #516074;
3354     font-size: 0.75rem;
3355     font-weight: 700;
3356 }
3357
3358 .result-visualization-list {
3359     display: grid;
3360     gap: 6px;
3361 }
3362
3363 .result-visualization-list button,
3364 .result-export-history button {
3365     align-items: center;
3366     background: transparent;
3367     border: 1px solid transparent;
3368     border-radius: 6px;
3369     color: #263449;
3370     display: flex;
3371     font: inherit;
3372     gap: 10px;
3373     min-height: 42px;
3374     padding: 8px 10px;
3375     text-align: left;
3376 }
3377
3378 .result-visualization-list button:hover,
3379 .result-visualization-list button.active,
3380 .result-export-history button:hover {
3381     background: #eef5f8;
3382     border-color: #bad4df;
3383 }
3384
3385 .result-export-panel {
3386     display: grid;
3387     gap: 14px;
3388     align-content: start;
3389 }
3390
3391 .result-artifact-row > span {
3392     display: grid;
3393     gap: 3px;
3394 }
3395
3396 .result-export-checks label {
3397     align-items: center;
3398     display: inline-flex;
3399     gap: 5px;
3400     min-height: 30px;
```

```
front/src/workbench/workbench-polish.css
```

```
3401 }
3402
3403 .result-export-history button {
3404     align-items: flex-start;
3405     flex-direction: column;
3406     gap: 2px;
3407     width: 100%;
3408 }
3409
3410 /* 任务日志采用执行时间线和独立上下文栏，避免把不同长度的事件强行压进固定表格列。 */
3411 .task-logs-page {
3412     min-height: 0;
3413     grid-template-rows: auto auto minmax(0, 1fr);
3414     overflow: hidden;
3415 }
3416
3417 .task-log-workspace {
3418     min-width: 0;
3419     min-height: 0;
3420     height: 100%;
3421     display: grid;
3422     grid-template-columns: minmax(0, 1fr) 300px;
3423     align-items: stretch;
3424     gap: 14px;
3425 }
3426
3427 .task-log-stream-panel,
3428 .task-log-context-panel {
3429     min-width: 0;
3430     min-height: 0;
3431     height: 100%;
3432     padding: 16px;
3433 }
3434
3435 .task-log-stream-panel {
3436     display: grid;
3437     grid-template-rows: auto auto minmax(0, 1fr);
3438     align-content: stretch;
3439     gap: 13px;
3440     overflow: hidden;
3441 }
3442
3443 .task-log-toolbar {
3444     min-width: 0;
3445     display: grid;
3446     gap: 10px;
3447     padding-block: 12px;
3448     border-top: 1px solid #e7edf3;
3449     border-bottom: 1px solid #e7edf3;
3450 }
```



```
front/src/workbench/workbench-polish.css
```

```
3451
3452 .task-log-search-row,
3453 .task-log-filter-row {
3454     min-width: 0;
3455     display: flex;
3456     align-items: center;
3457     justify-content: space-between;
3458     gap: 12px;
3459 }
3460
3461 .task-log-search {
3462     width: min(430px, 70%);
3463     min-width: 220px;
3464     height: 36px;
3465     display: flex;
3466     align-items: center;
3467     gap: 8px;
3468     padding-left: 10px;
3469     border: 1px solid #c7d3df;
3470     border-radius: 7px;
3471     color: #687b8f;
3472     background: #ffffff;
3473     transition:
3474         border-color 180ms var(--ease-out-quart),
3475         box-shadow 180ms var(--ease-out-quart);
3476 }
3477
3478 .task-log-search:focus-within {
3479     border-color: #2f7del;
3480     box-shadow: 0 0 2px #dbeafd;
3481 }
3482
3483 .task-log-search input {
3484     min-width: 0;
3485     flex: 1;
3486     border: 0;
3487     outline: 0;
3488     color: #283b50;
3489     background: transparent;
3490     font: inherit;
3491     font-size: 12px;
3492 }
3493
3494 .task-log-search input::placeholder {
3495     color: #6f8093;
3496     opacity: 1;
3497 }
3498
3499 .task-log-search button {
3500     width: 31px;
```

```
front/src/workbench/workbench-polish.css
```

```
3501     height: 31px;
3502     flex: 0 0 31px;
3503     display: grid;
3504     place-items: center;
3505     border: 0;
3506     border-radius: 5px;
3507     color: #718298;
3508     background: transparent;
3509     cursor: pointer;
3510     transition:
3511         color 160ms var(--ease-out-quart),
3512         background-color 160ms var(--ease-out-quart);
3513 }
3514
3515 .task-log-search button:hover,
3516 .task-log-search button:focus-visible {
3517     color: #bd3838;
3518     background: #ffff1f;
3519     outline: 0;
3520 }
3521
3522 .task-log-result-count {
3523     flex: 0 0 auto;
3524     color: #68798d;
3525     font-size: 11px;
3526     font-variant-numeric: tabular-nums;
3527     white-space: nowrap;
3528 }
3529
3530 .task-log-segmented,
3531 .task-log-order {
3532     min-width: 0;
3533     display: inline-flex;
3534     align-items: center;
3535     gap: 2px;
3536     padding: 3px;
3537     border: 1px solid #d2dbe5;
3538     border-radius: 7px;
3539     background: #f5f7f9;
3540 }
3541
3542 .task-log-segmented button,
3543 .task-log-order button {
3544     min-height: 29px;
3545     display: inline-flex;
3546     align-items: center;
3547     justify-content: center;
3548     gap: 6px;
3549     border: 0;
3550     border-radius: 5px;
```

```
front/src/workbench/workbench-polish.css
```

```
3551     color: #5f7185;
3552     background: transparent;
3553     cursor: pointer;
3554     font: inherit;
3555     font-size: 11px;
3556     font-weight: 650;
3557     white-space: nowrap;
3558     transition:
3559         color 170ms var(--ease-out-quart),
3560         background-color 170ms var(--ease-out-quart),
3561         box-shadow 170ms var(--ease-out-quart);
3562 }
3563
3564 .task-log-segmented button {
3565     min-width: 66px;
3566     padding-inline: 9px;
3567 }
3568
3569 .task-log-order button {
3570     min-width: 84px;
3571     padding-inline: 8px;
3572 }
3573
3574 .task-log-segmented button:hover,
3575 .task-log-order button:hover {
3576     color: #275e9e;
3577     background: #edf4fb;
3578 }
3579
3580 .task-log-segmented button:focus-visible,
3581 .task-log-order button:focus-visible {
3582     outline: 2px solid #a9cdf5;
3583     outline-offset: 1px;
3584 }
3585
3586 .task-log-segmented button.is-active,
3587 .task-log-order button.is-active {
3588     color: #1f64bd;
3589     background: #ffffff;
3590     box-shadow: 0 1px 3px rgba(31, 45, 61, 0.1);
3591 }
3592
3593 .task-log-segmented small {
3594     min-width: 17px;
3595     height: 17px;
3596     display: inline-grid;
3597     place-items: center;
3598     padding-inline: 4px;
3599     border-radius: 999px;
3600     color: #718298;
```

```
front/src/workbench/workbench-polish.css
```

```
3601     background: #e8edf2;
3602     font-size: 9px;
3603     font-variant-numeric: tabular-nums;
3604     line-height: 1;
3605 }
3606
3607 .task-log-segmented button.is-active small {
3608     color: #1f64bd;
3609     background: #e5f0fd;
3610 }
3611
3612 .task-log-refresh svg {
3613     transform-origin: center;
3614 }
3615
3616 .task-log-refresh svg.is-spinning {
3617     animation: task-log-spin 800ms linear infinite;
3618 }
3619
3620 .task-log-list-region {
3621     min-width: 0;
3622     min-height: 0;
3623     display: grid;
3624     align-content: start;
3625     gap: 13px;
3626     overflow-x: hidden;
3627     overflow-y: auto;
3628     overscroll-behavior: contain;
3629     scrollbar-gutter: stable;
3630     scroll-padding-block: 8px;
3631 }
3632
3633 .task-log-list-region:focus-visible {
3634     border-radius: 5px;
3635     outline: 2px solid #a9cdf5;
3636     outline-offset: -2px;
3637 }
3638
3639 .task-log-timeline {
3640     min-width: 0;
3641     margin: 0;
3642     padding: 0;
3643     list-style: none;
3644 }
3645
3646 .task-log-event {
3647     --task-log-color: #2f7de1;
3648     --task-log-soft: #e7f1fd;
3649     position: relative;
3650     min-width: 0;
```

```
front/src/workbench/workbench-polish.css
```

```
3651   min-height: 78px;
3652   display: grid;
3653   grid-template-columns: 24px minmax(0, 1fr) 136px;
3654   gap: 11px;
3655   padding: 12px 8px 12px 0;
3656   border-bottom: 1px solid #e8edf2;
3657   transition: background-color 180ms var(--ease-out-quart);
3658 }
3659
3660 .task-log-event:last-child {
3661   border-bottom: 0;
3662 }
3663
3664 .task-log-event:hover,
3665 .task-log-event:focus-within {
3666   background: #f8fafc;
3667 }
3668
3669 .task-log-event.level-warning {
3670   --task-log-color: #b96c12;
3671   --task-log-soft: #fff0d8;
3672 }
3673
3674 .task-log-event.level-error {
3675   --task-log-color: #d64545;
3676   --task-log-soft: #fde5e5;
3677 }
3678
3679 .task-log-event.level-neutral {
3680   --task-log-color: #718298;
3681   --task-log-soft: #e9edf1;
3682 }
3683
3684 .task-log-event-marker {
3685   position: relative;
3686   align-self: stretch;
3687   display: flex;
3688   justify-content: center;
3689 }
3690
3691 .task-log-event-marker::after {
3692   position: absolute;
3693   top: 22px;
3694   bottom: -25px;
3695   width: 1px;
3696   background: #d8e1ea;
3697   content: '';
3698 }
3699
3700 .task-log-event:last-child .task-log-event-marker::after {
```

```
front/src/workbench/workbench-polish.css
```

```
3701     display: none;
3702 }
3703
3704 .task-log-event-marker i {
3705     position: relative;
3706     z-index: 1;
3707     width: 10px;
3708     height: 10px;
3709     margin-top: 5px;
3710     border: 2px solid #ffffff;
3711     border-radius: 50%;
3712     background: var(--task-log-color);
3713     box-shadow: 0 0 0 4px var(--task-log-soft);
3714     transition: box-shadow 180ms var(--ease-out-quart);
3715 }
3716
3717 .task-log-event:hover .task-log-event-marker i {
3718     box-shadow: 0 0 0 6px var(--task-log-soft);
3719 }
3720
3721 .task-log-event-content {
3722     min-width: 0;
3723     display: grid;
3724     align-content: start;
3725     gap: 6px;
3726 }
3727
3728 .task-log-event-heading {
3729     min-width: 0;
3730     display: flex;
3731     align-items: center;
3732     flex-wrap: wrap;
3733     gap: 7px;
3734 }
3735
3736 .task-log-event-heading strong {
3737     min-width: 0;
3738     color: #25374b;
3739     font-size: 13px;
3740     font-weight: 720;
3741     line-height: 1.35;
3742     overflow-wrap: anywhere;
3743 }
3744
3745 .task-log-event-heading code,
3746 .task-log-context-list code {
3747     max-width: 100%;
3748     overflow: hidden;
3749     padding: 3px 6px;
3750     border-radius: 4px;
```

```
front/src/workbench/workbench-polish.css
```

```
3751     color: #50677f;
3752     background: #edf1f5;
3753     font-family: ui-monospace, SFMono-Regular, Menlo, Consolas, monospace;
3754     font-size: 9px;
3755     font-weight: 600;
3756     line-height: 1.2;
3757     text-overflow: ellipsis;
3758     white-space: nowrap;
3759 }
3760
3761 .task-log-event-content p {
3762     max-width: 72ch;
3763     margin: 0;
3764     color: #52667b;
3765     font-size: 12px;
3766     line-height: 1.55;
3767     overflow-wrap: anywhere;
3768 }
3769
3770 .task-log-event-meta {
3771     min-width: 0;
3772     display: flex;
3773     align-items: center;
3774     flex-wrap: wrap;
3775     gap: 7px;
3776 }
3777
3778 .task-log-event-meta .workbench-status {
3779     min-height: 20px;
3780     padding-inline: 6px;
3781     font-size: 9px;
3782 }
3783
3784 .task-log-event-meta .workbench-status i {
3785     width: 5px;
3786     height: 5px;
3787 }
3788
3789 .task-log-run {
3790     min-height: 20px;
3791     display: inline-flex;
3792     align-items: center;
3793     padding-inline: 7px;
3794     border: 1px solid #d7e2ec;
3795     border-radius: 999px;
3796     color: #516b84;
3797     background: #f5f8fa;
3798     font-size: 9px;
3799     font-weight: 700;
3800     font-variant-numeric: tabular-nums;
```

```
front/src/workbench/workbench-polish.css
```

```
3801     white-space: nowrap;
3802 }
3803
3804 .task-log-event > time {
3805     align-self: start;
3806     display: grid;
3807     justify-items: end;
3808     gap: 2px;
3809     padding-top: 1px;
3810     color: #7a899a;
3811     font-variant-numeric: tabular-nums;
3812     white-space: nowrap;
3813 }
3814
3815 .task-log-event > time span {
3816     font-size: 10px;
3817 }
3818
3819 .task-log-event > time strong {
3820     color: #506277;
3821     font-size: 12px;
3822     font-weight: 700;
3823 }
3824
3825 .task-log-empty {
3826     min-height: 240px;
3827     display: grid;
3828     place-items: center;
3829     align-content: center;
3830     gap: 7px;
3831     padding: 24px;
3832     color: #708196;
3833     text-align: center;
3834 }
3835
3836 .task-log-list-region > .task-log-empty {
3837     min-height: 100%;
3838 }
3839
3840 .task-log-empty strong {
3841     color: #30455b;
3842     font-size: 13px;
3843 }
3844
3845 .task-log-empty span {
3846     max-width: 52ch;
3847     font-size: 11px;
3848     line-height: 1.6;
3849 }
3850
```



```
front/src/workbench/workbench-polish.css
```

```
3851 .task-log-empty .btn {
3852     margin-top: 5px;
3853 }
3854
3855 .task-log-skeleton {
3856     display: grid;
3857 }
3858
3859 .task-log-skeleton > div {
3860     min-height: 78px;
3861     display: grid;
3862     grid-template-columns: 24px minmax(0, 1fr) 136px;
3863     align-items: center;
3864     gap: 11px;
3865     border-bottom: 1px solid #e8edf2;
3866 }
3867
3868 .task-log-skeleton i,
3869 .task-log-skeleton span {
3870     display: block;
3871     border-radius: 5px;
3872     background: #e7ecf1;
3873     animation: task-log-skeleton-pulse 1.2s ease-in-out infinite alternate;
3874 }
3875
3876 .task-log-skeleton i {
3877     width: 10px;
3878     height: 10px;
3879     margin-inline: auto;
3880     border-radius: 50%;
3881 }
3882
3883 .task-log-skeleton span {
3884     width: min(420px, 80%);
3885     height: 12px;
3886 }
3887
3888 .task-log-skeleton span:last-child {
3889     width: 108px;
3890     justify-self: end;
3891 }
3892
3893 .task-log-context-panel {
3894     display: grid;
3895     align-content: start;
3896     gap: 13px;
3897     overflow-x: hidden;
3898     overflow-y: auto;
3899     overscroll-behavior: contain;
3900 }
```

front/src/workbench/workbench-polish.css

```
3901
3902 .task-log-context-list {
3903     margin: 0;
3904 }
3905
3906 .task-log-context-list > div {
3907     min-width: 0;
3908     min-height: 45px;
3909     display: grid;
3910     grid-template-columns: 82px minmax(0, 1fr);
3911     align-items: center;
3912     gap: 10px;
3913     border-bottom: 1px solid #e8edf2;
3914 }
3915
3916 .task-log-context-list dt {
3917     color: #748498;
3918     font-size: 10px;
3919 }
3920
3921 .task-log-context-list dd {
3922     min-width: 0;
3923     margin: 0;
3924     overflow: hidden;
3925     color: #31465c;
3926     font-size: 11px;
3927     font-weight: 700;
3928     overflow-wrap: anywhere;
3929 }
3930
3931 .task-log-context-list dd > code {
3932     display: inline-block;
3933     vertical-align: middle;
3934 }
3935
3936 .task-log-context-list .workbench-status {
3937     min-height: 21px;
3938     padding-inline: 6px;
3939     font-size: 9px;
3940 }
3941
3942 .task-log-time-range {
3943     min-width: 0;
3944     display: grid;
3945     gap: 9px;
3946     padding-top: 2px;
3947 }
3948
3949 .task-log-time-range h3 {
3950     margin: 0;
```

```
front/src/workbench/workbench-polish.css
```

```
3951   color: #344a60;
3952   font-size: 12px;
3953   font-weight: 720;
3954   letter-spacing: 0;
3955 }
3956
3957 .task-log-time-range > div {
3958   min-width: 0;
3959   min-height: 34px;
3960   display: grid;
3961   grid-template-columns: 66px minmax(0, 1fr);
3962   align-items: center;
3963   gap: 8px;
3964   padding-block: 6px;
3965   border-top: 1px solid #e8edf2;
3966 }
3967
3968 .task-log-time-range span {
3969   color: #78889a;
3970   font-size: 9px;
3971 }
3972
3973 .task-log-time-range time {
3974   min-width: 0;
3975   overflow: hidden;
3976   color: #40566d;
3977   font-size: 10px;
3978   font-variant-numeric: tabular-nums;
3979   text-align: right;
3980   text-overflow: ellipsis;
3981   white-space: nowrap;
3982 }
3983
3984 @keyframes task-log-spin {
3985   to { transform: rotate(360deg); }
3986 }
3987
3988 @keyframes task-log-skeleton-pulse {
3989   from { opacity: 0.55; }
3990   to { opacity: 1; }
3991 }
3992
3993 @media (max-width: 1380px) {
3994   .task-log-workspace {
3995     grid-template-columns: minmax(0, 1fr) 270px;
3996   }
3997
3998   .task-log-filter-row {
3999     align-items: flex-start;
4000     flex-direction: column;
```

```
front/src/workbench/workbench-polish.css
```

```
4001   }
4002 }
4003
4004 @media (max-width: 1120px) {
4005   .dashboard-content.is-log-layout {
4006     height: auto;
4007     min-height: 100vh;
4008     overflow-y: visible;
4009   }
4010
4011   .dashboard-content.is-log-layout > .task-logs-page {
4012     flex: none;
4013   }
4014
4015   .task-logs-page {
4016     grid-template-rows: auto;
4017     overflow: visible;
4018   }
4019
4020   .task-log-workspace {
4021     height: auto;
4022     grid-template-columns: minmax(0, 1fr);
4023   }
4024
4025   .task-log-stream-panel,
4026   .task-log-context-panel {
4027     height: auto;
4028     overflow: visible;
4029   }
4030
4031   .task-log-stream-panel {
4032     grid-template-rows: auto auto auto;
4033   }
4034
4035   .task-log-list-region {
4036     overflow: visible;
4037   }
4038
4039   .task-log-context-panel {
4040     grid-template-columns: minmax(0, 1fr) minmax(320px, 0.8fr);
4041   }
4042
4043   .task-log-context-panel .workbench-section-header {
4044     grid-column: 1 / -1;
4045   }
4046 }
4047
4048 /* 分析总览沿用结果页的图表组件，通过紧凑模式控制信息密度。 */
4049 .result-panel-actions,
4050 .text-action {
```

```
front/src/workbench/workbench-polish.css
```

```
4051     display: inline-flex;
4052     align-items: center;
4053     gap: 6px;
4054 }
4055
4056 .result-panel-actions {
4057     justify-content: flex-end;
4058     flex-wrap: wrap;
4059     gap: 8px;
4060 }
4061
4062 .result-chart-panel.is-compact .result-chart {
4063     height: 280px;
4064     min-height: 280px;
4065 }
4066
4067 .result-panel-empty {
4068     min-height: 220px;
4069     display: grid;
4070     place-items: center;
4071     align-content: center;
4072     gap: 7px;
4073     color: #738397;
4074     text-align: center;
4075 }
4076
4077 .result-panel-empty svg {
4078     color: #7aa3c8;
4079 }
4080
4081 .result-panel-empty strong {
4082     color: #34475b;
4083     font-size: 13px;
4084 }
4085
4086 .result-panel-empty span {
4087     max-width: 42ch;
4088     font-size: 11px;
4089     line-height: 1.5;
4090 }
4091
4092 .analysis-overview-page {
4093     gap: 16px;
4094 }
4095
4096 .analysis-overview-featured,
4097 .analysis-overview-behavior,
4098 .analysis-overview-details {
4099     min-width: 0;
4100     display: grid;
```

```
front/src/workbench/workbench-polish.css
```

```
4101     gap: 14px;
4102     align-items: stretch;
4103 }
4104
4105 .analysis-overview-featured {
4106     grid-template-columns: minmax(0, 1.62fr) minmax(330px, 0.78fr);
4107 }
4108
4109 .analysis-overview-behavior {
4110     grid-template-columns: minmax(0, 1.15fr) minmax(360px, 0.85fr);
4111 }
4112
4113 .analysis-overview-details {
4114     grid-template-columns: minmax(390px, 0.82fr) minmax(0, 1.18fr);
4115     align-items: start;
4116 }
4117
4118 .analysis-overview-featured > .panel,
4119 .analysis-overview-behavior > .panel {
4120     height: 100%;
4121 }
4122
4123 .analysis-overview-summary {
4124     display: flex;
4125     flex-direction: column;
4126 }
4127
4128 .analysis-overview-summary-list {
4129     min-width: 0;
4130     display: grid;
4131     grid-template-columns: repeat(2, minmax(0, 1fr));
4132     margin: 6px 0 14px;
4133 }
4134
4135 .analysis-overview-summary-list > div {
4136     min-width: 0;
4137     padding: 10px 3px;
4138     border-bottom: 1px solid #e7edf3;
4139 }
4140
4141 .analysis-overview-summary-list > div:nth-child(even):not(.is-wide) {
4142     padding-right: 12px;
4143 }
4144
4145 .analysis-overview-summary-list > div:nth-child(odd):not(.is-wide) {
4146     padding-left: 12px;
4147     border-left: 1px solid #e7edf3;
4148 }
4149
4150 .analysis-overview-summary-list .is-wide {
```

```
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```

```
4151     grid-column: 1 / -1;
4152     padding-top: 8px;
4153 }
4154
4155 .analysis-overview-summary-list dt {
4156     margin-bottom: 4px;
4157     color: #738397;
4158     font-size: 11px;
4159     font-weight: 600;
4160 }
4161
4162 .analysis-overview-summary-list dd {
4163     min-width: 0;
4164     margin: 0;
4165     overflow: hidden;
4166     color: #25374b;
4167     font-size: 13px;
4168     font-weight: 720;
4169     text-overflow: ellipsis;
4170     white-space: nowrap;
4171 }
4172
4173 .analysis-overview-summary .workbench-notice {
4174     margin-top: auto;
4175 }
4176
4177 .analysis-overview-flow {
4178     overflow: hidden;
4179     padding-bottom: 0;
4180 }
4181
4182 .analysis-overview-stage-list {
4183     min-width: 0;
4184     display: grid;
4185     grid-template-columns: repeat(5, minmax(0, 1fr));
4186     margin: 8px -16px 0;
4187     border-top: 1px solid #e3e8ed;
4188 }
4189
4190 .analysis-overview-stage {
4191     min-width: 0;
4192     min-height: 86px;
4193     display: grid;
4194     grid-template-columns: auto minmax(0, 1fr) auto;
4195     grid-template-rows: auto auto;
4196     align-content: center;
4197     align-items: center;
4198     column-gap: 9px;
4199     row-gap: 5px;
4200     padding: 13px 12px;
```

```
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```

```
4201     border: 0;
4202     color: #34475b;
4203     background: #ffffff;
4204     cursor: pointer;
4205     text-align: left;
4206     transition:
4207         color var(--motion-fast) ease,
4208         background-color var(--motion-fast) ease;
4209 }
4210
4211 .analysis-overview-stage + .analysis-overview-stage {
4212     border-left: 1px solid #e3e8ed;
4213 }
4214
4215 .analysis-overview-stage:hover {
4216     color: #1f64bd;
4217     background: #f3f7fb;
4218 }
4219
4220 .analysis-overview-stage-icon {
4221     grid-row: 1 / 3;
4222     width: 32px;
4223     height: 32px;
4224     display: grid;
4225     place-items: center;
4226     border-radius: 7px;
4227     color: #1f64bd;
4228     background: #eaf3ff;
4229 }
4230
4231 .analysis-overview-stage-copy {
4232     min-width: 0;
4233     display: grid;
4234     gap: 3px;
4235 }
4236
4237 .analysis-overview-stage-copy strong,
4238 .analysis-overview-stage-copy small {
4239     min-width: 0;
4240     overflow: hidden;
4241     text-overflow: ellipsis;
4242     white-space: nowrap;
4243 }
4244
4245 .analysis-overview-stage-copy strong {
4246     color: #2d4055;
4247     font-size: 12px;
4248     font-weight: 720;
4249 }
4250
```



```
front/src/workbench/workbench-polish.css
```

```
4251 .analysis-overview-stage-copy small {
4252     color: #7a899a;
4253     font-size: 10px;
4254 }
4255
4256 .analysis-overview-stage-status {
4257     min-width: 0;
4258     display: inline-flex;
4259     align-items: center;
4260     gap: 4px;
4261     color: #23865f;
4262     font-size: 10px;
4263     font-weight: 700;
4264     white-space: nowrap;
4265 }
4266
4267 .analysis-overview-stage.is-incomplete .analysis-overview-stage-icon {
4268     color: #8a6b2c;
4269     background: #fff7e9;
4270 }
4271
4272 .analysis-overview-stage.is-incomplete .analysis-overview-stage-status {
4273     color: #9a6b21;
4274 }
4275
4276 .analysis-overview-stage-arrow {
4277     grid-column: 3;
4278     grid-row: 1 / 3;
4279     color: #9aa8b6;
4280     transition: transform var(--motion-fast) var(--ease-out-quart);
4281 }
4282
4283 .analysis-overview-stage:hover .analysis-overview-stage-arrow {
4284     color: #1f64bd;
4285     transform: translateX(2px);
4286 }
4287
4288 .analysis-overview-details .result-chart-panel.is-compact .result-chart {
4289     height: 190px;
4290     min-height: 190px;
4291 }
4292
4293 .analysis-overview-details .result-kernel-list {
4294     grid-template-columns: repeat(2, minmax(0, 1fr));
4295     column-gap: 16px;
4296     row-gap: 0;
4297 }
4298
4299 .analysis-overview-details .result-kernel-list > div {
4300     min-width: 0;
```

```
front/src/workbench/workbench-polish.css
```

```
4301 }
4302
4303 .analysis-overview-details .result-kernel-list > div span {
4304     min-width: 0;
4305     overflow: hidden;
4306     font-size: 11px;
4307     text-overflow: ellipsis;
4308     white-space: nowrap;
4309 }
4310
4311 .analysis-overview-details .result-kernel-list > div strong,
4312 .analysis-overview-details .result-kernel-list > div small {
4313     flex: 0 0 auto;
4314     font-size: 10px;
4315 }
4316
4317 .analysis-overview-runs {
4318     min-width: 0;
4319 }
4320
4321 .analysis-overview-run-table {
4322     min-width: 0;
4323     margin-top: 6px;
4324     overflow: hidden;
4325     border: 1px solid #e4e9ee;
4326     border-radius: 6px;
4327 }
4328
4329 .analysis-overview-run-row {
4330     min-width: 620px;
4331     min-height: 45px;
4332     display: grid;
4333     grid-template-columns: minmax(84px, 1.1fr) repeat(4, minmax(66px, 1fr)) minmax(72px, 0.85fr);
4334     align-items: center;
4335     gap: 10px;
4336     padding-inline: 12px;
4337     border-top: 1px solid #e7edf3;
4338     color: #536579;
4339     font-size: 11px;
4340     font-variant-numeric: tabular-nums;
4341 }
4342
4343 .analysis-overview-run-row:first-child {
4344     border-top: 0;
4345 }
4346
4347 .analysis-overview-run-row.header {
4348     min-height: 38px;
4349     color: #65768a;
4350     background: #f5f7f9;
```

```
front/src/workbench/workbench-polish.css
```

```
4351     font-size: 10px;
4352     font-weight: 700;
4353 }
4354
4355 .analysis-overview-run-row strong {
4356     color: #2f4257;
4357     font-weight: 720;
4358 }
4359
4360 .analysis-overview-inline-empty {
4361     min-height: 250px;
4362     display: grid;
4363     place-items: center;
4364     color: #738397;
4365     font-size: 12px;
4366 }
4367
4368 .analysis-overview-follow-up {
4369     min-width: 0;
4370     display: grid;
4371     gap: 12px;
4372     padding: 16px;
4373 }
4374
4375 .analysis-overview-follow-up-heading {
4376     min-width: 0;
4377     display: flex;
4378     align-items: center;
4379     gap: 10px;
4380     color: #52667b;
4381 }
4382
4383 .analysis-overview-follow-up-heading > div {
4384     min-width: 0;
4385     display: grid;
4386     gap: 3px;
4387 }
4388
4389 .analysis-overview-follow-up-heading strong,
4390 .analysis-overview-follow-up-heading span {
4391     min-width: 0;
4392     overflow: hidden;
4393     text-overflow: ellipsis;
4394     white-space: nowrap;
4395 }
4396
4397 .analysis-overview-follow-up-heading strong {
4398     color: #2b4056;
4399     font-size: 13px;
4400 }
```

```
front/src/workbench/workbench-polish.css
```

```
4401
4402 .analysis-overview-follow-up-heading span {
4403     color: #738397;
4404     font-size: 11px;
4405 }
4406
4407 .analysis-overview-follow-up-actions {
4408     min-width: 0;
4409     display: grid;
4410     grid-template-columns: repeat(4, minmax(0, 1fr));
4411     gap: 10px;
4412 }
4413
4414 .analysis-overview-follow-up-actions button {
4415     min-width: 0;
4416     min-height: 76px;
4417     display: grid;
4418     grid-template-columns: auto minmax(0, 1fr) auto;
4419     align-items: center;
4420     gap: 9px;
4421     padding: 12px 14px;
4422     border: 1px solid #e3e8ed;
4423     border-radius: 12px;
4424     color: #607184;
4425     background: #ffffff;
4426     cursor: pointer;
4427     text-align: left;
4428     transition:
4429         border-color var(--motion-fast) ease,
4430         color var(--motion-fast) ease,
4431         background-color var(--motion-fast) ease,
4432         transform var(--motion-fast) ease;
4433 }
4434
4435 .analysis-overview-follow-up-actions button:hover {
4436     color: #1f64bd;
4437     background: #f5f8fb;
4438     border-color: #c8d8ec;
4439     transform: translateY(-1px);
4440 }
4441
4442 .analysis-overview-follow-up-actions button > span {
4443     min-width: 0;
4444     display: grid;
4445     gap: 3px;
4446 }
4447
4448 .analysis-overview-follow-up-actions strong,
4449 .analysis-overview-follow-up-actions small {
4450     min-width: 0;
```

```
front/src/workbench/workbench-polish.css
```

```
4451     overflow: hidden;
4452     text-overflow: ellipsis;
4453     white-space: nowrap;
4454 }
4455
4456 .analysis-overview-follow-up-actions strong {
4457     color: #34475b;
4458     font-size: 11px;
4459 }
4460
4461 .analysis-overview-follow-up-actions small {
4462     color: #7a899a;
4463     font-size: 10px;
4464 }
4465
4466 .analysis-overview-stage:focus-visible,
4467 .analysis-overview-follow-up-actions button:focus-visible,
4468 .text-action:focus-visible {
4469     outline: 2px solid rgba(47, 125, 225, 0.42);
4470     outline-offset: -2px;
4471 }
4472
4473 @media (max-width: 1360px) {
4474     .analysis-overview-featured {
4475         grid-template-columns: minmax(0, 1.48fr) minmax(310px, 0.72fr);
4476     }
4477
4478     .analysis-overview-behavior {
4479         grid-template-columns: minmax(0, 1.08fr) minmax(340px, 0.92fr);
4480     }
4481
4482     .analysis-overview-details {
4483         grid-template-columns: minmax(360px, 0.72fr) minmax(0, 1.28fr);
4484     }
4485
4486     .analysis-overview-stage {
4487         padding-inline: 9px;
4488     }
4489
4490     .analysis-overview-stage-icon {
4491         width: 30px;
4492         height: 30px;
4493     }
4494
4495     .analysis-overview-follow-up {
4496         padding: 14px;
4497     }
4498
4499     .analysis-overview-follow-up-actions button {
4500         padding-inline: 10px;
```

```
front/src/workbench/workbench-polish.css
```

```
4501     }
4502   }
4503
4504   /* 结果详情页共享两级工作区：首屏承担主要判断，次级区域补充诊断与交付信息。 */
4505   .result-detail-page {
4506     gap: 16px;
4507   }
4508
4509   .result-detail-page .panel {
4510     min-width: 0;
4511   }
4512
4513   .result-detail-primary,
4514   .result-detail-secondary,
4515   .result-export-workspace,
4516   .result-report-workspace,
4517   .result-visual-workspace {
4518     min-width: 0;
4519     display: grid;
4520     align-items: start;
4521     gap: 16px;
4522   }
4523
4524   .result-detail-primary {
4525     grid-template-columns: minmax(0, 1.35fr) minmax(360px, 0.65fr);
4526   }
4527
4528   .result-detail-secondary {
4529     grid-template-columns: repeat(2, minmax(0, 1fr));
4530   }
4531
4532   .result-detail-page .result-chart-panel,
4533   .result-diagnostic-panel,
4534   .result-ranked-table-panel,
4535   .result-kernel-definition-panel,
4536   .result-kernel-comparison-panel,
4537   .result-visual-toolbox,
4538   .result-view-catalog,
4539   .result-analysis-summary-panel,
4540   .result-distribution-panel,
4541   .result-artifact-panel,
4542   .result-export-builder,
4543   .result-export-history-panel,
4544   .result-report-controls,
4545   .result-report-preview {
4546     overflow: hidden;
4547   }
4548
4549   .result-panel-actions {
4550     display: flex;
```

```
front/src/workbench/workbench-polish.css
```

```
4551     align-items: center;
4552     gap: 8px;
4553 }
4554
4555 .result-panel-empty {
4556     min-height: 260px;
4557     display: grid;
4558     place-items: center;
4559     align-content: center;
4560     gap: 7px;
4561     color: #8090a2;
4562     text-align: center;
4563 }
4564
4565 .result-panel-empty strong {
4566     color: #3a4d62;
4567     font-size: 13px;
4568 }
4569
4570 .result-panel-empty span {
4571     max-width: 52ch;
4572     font-size: 11px;
4573     line-height: 1.6;
4574 }
4575
4576 /* CA 详情 */
4577 .result-ca-primary {
4578     grid-template-columns: minmax(0, 1.42fr) minmax(340px, 0.58fr);
4579 }
4580
4581 .result-diagnostic-panel {
4582     display: grid;
4583     align-content: start;
4584     gap: 14px;
4585 }
4586
4587 .result-diagnostic-list,
4588 .result-analysis-facts,
4589 .result-view-summary,
4590 .result-export-manifest {
4591     margin: 0;
4592 }
4593
4594 .result-diagnostic-list {
4595     display: grid;
4596     grid-template-columns: repeat(2, minmax(0, 1fr));
4597     border-block: 1px solid #e5eaf0;
4598 }
4599
4600 .result-diagnostic-list div {
```

```
front/src/workbench/workbench-polish.css
```

```
4601     min-width: 0;
4602     display: grid;
4603     gap: 4px;
4604     padding: 11px 4px;
4605     border-bottom: 1px solid #edf0f3;
4606 }
4607
4608 .result-diagnostic-list div:nth-child(odd) {
4609     padding-right: 14px;
4610 }
4611
4612 .result-diagnostic-list div:nth-child(even) {
4613     padding-left: 14px;
4614     border-left: 1px solid #edf0f3;
4615 }
4616
4617 .result-diagnostic-list div:nth-last-child(-n + 2) {
4618     border-bottom: 0;
4619 }
4620
4621 .result-diagnostic-list dt,
4622 .result-analysis-facts dt,
4623 .result-view-summary dt,
4624 .result-export-manifest dt {
4625     color: #76869a;
4626     font-size: 10px;
4627     font-weight: 600;
4628 }
4629
4630 .result-diagnostic-list dd,
4631 .result-analysis-facts dd,
4632 .result-view-summary dd,
4633 .result-export-manifest dd {
4634     min-width: 0;
4635     margin: 0;
4636     overflow: hidden;
4637     color: #2b3e53;
4638     font-size: 12px;
4639     font-weight: 720;
4640     font-variant-numeric: tabular-nums;
4641     text-overflow: ellipsis;
4642     white-space: nowrap;
4643 }
4644
4645 .result-formula-inline {
4646     min-height: 104px;
4647     display: grid;
4648     align-content: center;
4649     gap: 6px;
4650     padding: 14px 16px;
```



```
front/src/workbench/workbench-polish.css
```

```
4651     border: 1px solid #dce5ee;
4652     border-radius: 7px;
4653     background: #f6f8fa;
4654 }
4655
4656 .result-formula-inline > span {
4657     color: #6f8094;
4658     font-size: 11px;
4659     font-weight: 700;
4660 }
4661
4662 .result-formula-inline .result-ca-formula .katex {
4663     color: #203247;
4664     font-size: 1.25rem;
4665 }
4666
4667 .result-formula-inline .result-ca-formula.math-formula-fallback {
4668     color: #203247;
4669     font-size: 1rem;
4670     font-weight: 700;
4671 }
4672
4673 .result-ranked-table {
4674     margin-top: 8px;
4675     overflow-x: auto;
4676     border: 1px solid #e2e8ee;
4677     border-radius: 6px;
4678 }
4679
4680 .result-ranked-row {
4681     min-width: 720px;
4682     min-height: 43px;
4683     display: grid;
4684     grid-template-columns: 72px repeat(2, minmax(110px, 0.8fr)) minmax(110px, 0.9fr) minmax(160px, 1.2fr);
4685     align-items: center;
4686     gap: 12px;
4687     padding-inline: 14px;
4688     border-top: 1px solid #e8edf2;
4689     color: #586a7e;
4690     font-size: 11px;
4691     font-variant-numeric: tabular-nums;
4692 }
4693
4694 .result-ranked-row:first-child {
4695     border-top: 0;
4696 }
4697
4698 .result-ranked-row.header {
4699     min-height: 36px;
4700     color: #6f8093;
```

```
front/src/workbench/workbench-polish.css
```

```
4701     background: #f5f7f9;
4702     font-size: 10px;
4703     font-weight: 700;
4704 }
4705
4706 .result-ranked-row strong {
4707     color: #294158;
4708     font-weight: 730;
4709 }
4710
4711 .result-top-pair-control {
4712     display: flex;
4713     align-items: center;
4714     gap: 7px;
4715     color: #718196;
4716     font-size: 10px;
4717     font-weight: 650;
4718 }
4719
4720 /* 核函数配置采用单一工作区，目录、定义与权重按操作顺序组织，不再复制结果看板布局。 */
4721 .result-kernel-workbench,
4722 .result-kernel-stability-panel,
4723 .result-mkl-solver-panel,
4724 .result-mkl-iteration-panel,
4725 .result-mkl-mixture-panel {
4726     min-width: 0;
4727     display: grid;
4728     align-content: start;
4729     gap: 16px;
4730     overflow: hidden;
4731 }
4732
4733 .result-kernel-workspace-body {
4734     min-width: 0;
4735     display: grid;
4736     grid-template-columns: 220px minmax(0, 1fr) 270px;
4737     overflow: hidden;
4738     border: 1px solid #dfe6ed;
4739     border-radius: 7px;
4740 }
4741
4742 .result-kernel-catalog {
4743     min-width: 0;
4744     display: grid;
4745     align-content: start;
4746     background: #f6f8fa;
4747 }
4748
4749 .result-kernel-catalog > span {
4750     min-height: 38px;
```

```
front/src/workbench/workbench-polish.css
```

```
4751     display: flex;
4752     align-items: center;
4753     padding-inline: 14px;
4754     color: #718196;
4755     font-size: 10px;
4756     font-weight: 700;
4757 }
4758
4759 .result-kernel-catalog button {
4760     min-width: 0;
4761     min-height: 62px;
4762     display: flex;
4763     align-items: center;
4764     justify-content: space-between;
4765     gap: 10px;
4766     padding: 9px 12px;
4767     border: 0;
4768     border-top: 1px solid #e2e8ee;
4769     color: #607184;
4770     background: transparent;
4771     cursor: pointer;
4772     font: inherit;
4773     text-align: left;
4774     transition: color var(--motion-fast) ease, background-color var(--motion-fast) ease;
4775 }
4776
4777 .result-kernel-catalog button:hover {
4778     color: #1f64bd;
4779     background: #eef4fa;
4780 }
4781
4782 .result-kernel-catalog button.active {
4783     color: #1f64bd;
4784     background: #e7f1fd;
4785 }
4786
4787 .result-kernel-catalog button > span {
4788     min-width: 0;
4789     display: grid;
4790     gap: 3px;
4791 }
4792
4793 .result-kernel-catalog strong,
4794 .result-kernel-catalog small {
4795     overflow: hidden;
4796     text-overflow: ellipsis;
4797     white-space: nowrap;
4798 }
4799
4800 .result-kernel-catalog strong {
```

```
front/src/workbench/workbench-polish.css
```

```
4801     color: #30465c;
4802     font-size: 11px;
4803 }
4804
4805 .result-kernel-catalog small {
4806     color: #7d8b9c;
4807     font-size: 9px;
4808 }
4809
4810 .result-kernel-catalog em {
4811     flex: 0 0 auto;
4812     color: #4f6479;
4813     font-size: 10px;
4814     font-style: normal;
4815     font-variant-numeric: tabular-nums;
4816 }
4817
4818 .result-kernel-definition {
4819     min-width: 0;
4820     min-height: 286px;
4821     display: grid;
4822     grid-template-rows: auto minmax(120px, 1fr) auto;
4823     align-content: stretch;
4824     gap: 18px;
4825     padding: 24px 28px;
4826     border-left: 1px solid #dfe6ed;
4827     background: #ffffff;
4828 }
4829
4830 .result-kernel-definition > header,
4831 .result-kernel-selected-weight > header,
4832 .result-mkl-diagnostics > header {
4833     min-width: 0;
4834     display: flex;
4835     align-items: center;
4836     justify-content: space-between;
4837     gap: 12px;
4838 }
4839
4840 .result-kernel-definition > header > div,
4841 .result-mkl-diagnostics > header > div {
4842     min-width: 0;
4843     display: grid;
4844     gap: 3px;
4845 }
4846
4847 .result-kernel-definition > header span,
4848 .result-kernel-selected-weight > header span,
4849 .result-mkl-diagnostics > header span {
4850     color: #7b8a9b;
```

```
front/src/workbench/workbench-polish.css
```

```
4851     font-size: 10px;
4852     font-weight: 650;
4853 }
4854
4855 .result-kernel-definition > header strong,
4856 .result-mkl-diagnostics > header strong {
4857     overflow: hidden;
4858     color: #2a4056;
4859     font-size: 14px;
4860     text-overflow: ellipsis;
4861     white-space: nowrap;
4862 }
4863
4864 .result-kernel-definition .result-kernel-formula {
4865     min-height: 120px;
4866     align-self: center;
4867     justify-self: stretch;
4868     padding-block: 12px;
4869 }
4870
4871 .result-kernel-definition .result-kernel-formula .katex {
4872     color: #203247;
4873     font-size: 1.25rem;
4874 }
4875
4876 .result-kernel-definition .result-kernel-formula.math-formula-fallback {
4877     color: #203247;
4878     font-size: 1rem;
4879     font-weight: 700;
4880 }
4881
4882 .result-kernel-definition > p {
4883     max-width: 64ch;
4884     margin: 0;
4885     color: #68798d;
4886     font-size: 10px;
4887     line-height: 1.6;
4888 }
4889
4890 .result-kernel-selected-weight {
4891     min-width: 0;
4892     display: grid;
4893     align-content: center;
4894     gap: 15px;
4895     padding: 20px;
4896     border-left: 1px solid #dfe6ed;
4897     background: #f7f9fb;
4898 }
4899
4900 .result-kernel-selected-weight > header strong {
```

```
front/src/workbench/workbench-polish.css
```

```
4901     color: #1f64bd;
4902     font-size: 22px;
4903     font-weight: 760;
4904     font-variant-numeric: tabular-nums;
4905 }
4906
4907 .result-kernel-selected-weight dl {
4908     display: grid;
4909     margin: 0;
4910     border-block: 1px solid #dfe6ed;
4911 }
4912
4913 .result-kernel-selected-weight dl div {
4914     min-width: 0;
4915     display: flex;
4916     align-items: center;
4917     justify-content: space-between;
4918     gap: 12px;
4919     min-height: 38px;
4920     border-top: 1px solid #e5eaf0;
4921 }
4922
4923 .result-kernel-selected-weight dl div:first-child {
4924     border-top: 0;
4925 }
4926
4927 .result-kernel-selected-weight dt {
4928     color: #718196;
4929     font-size: 10px;
4930 }
4931
4932 .result-kernel-selected-weight dd {
4933     margin: 0;
4934     color: #30465c;
4935     font-size: 11px;
4936     font-weight: 730;
4937     font-variant-numeric: tabular-nums;
4938 }
4939
4940 .result-kernel-selected-weight > small {
4941     color: #78889a;
4942     font-size: 9px;
4943     line-height: 1.55;
4944 }
4945
4946 .result-parameter-strip {
4947     display: grid;
4948     grid-template-columns: repeat(6, minmax(0, 1fr));
4949     overflow: hidden;
4950     border: 1px solid #e1e7ed;
```

```
front/src/workbench/workbench-polish.css
```

```
4951     border-radius: 6px;
4952 }
4953
4954 .result-parameter-strip > div {
4955     min-width: 0;
4956     display: grid;
4957     gap: 3px;
4958     padding: 9px 10px;
4959     border-left: 1px solid #e5eaf0;
4960 }
4961
4962 .result-parameter-strip > div:first-child {
4963     border-left: 0;
4964 }
4965
4966 .result-parameter-strip span {
4967     color: #7a899b;
4968     font-size: 9px;
4969 }
4970
4971 .result-parameter-strip strong {
4972     overflow: hidden;
4973     color: #34495e;
4974     font-size: 11px;
4975     text-overflow: ellipsis;
4976     white-space: nowrap;
4977 }
4978
4979 .result-kernel-stability-chart {
4980     height: 300px;
4981     min-height: 300px;
4982 }
4983
4984 .result-kernel-stability-actions {
4985     display: flex;
4986     align-items: center;
4987     gap: 9px;
4988 }
4989
4990 .result-kernel-trend-toggle {
4991     display: inline-flex;
4992     align-items: center;
4993     overflow: hidden;
4994     padding: 2px;
4995     border: 1px solid #d6e0e9;
4996     border-radius: 6px;
4997     background: #f4f7f9;
4998 }
4999
5000 .result-kernel-trend-toggle button {
```

```
front/src/workbench/workbench-polish.css
```

```
5001   min-height: 27px;
5002   padding: 4px 10px;
5003   border: 0;
5004   border-radius: 4px;
5005   color: #68798c;
5006   background: transparent;
5007   cursor: pointer;
5008   font: inherit;
5009   font-size: 10px;
5010   font-weight: 650;
5011   transition: color var(--motion-fast) ease, background-color var(--motion-fast) ease;
5012 }
5013
5014 .result-kernel-trend-toggle button:hover:not(:disabled) {
5015   color: #245f9f;
5016   background: #eaf1f7;
5017 }
5018
5019 .result-kernel-trend-toggle button.active {
5020   color: #174f8c;
5021   background: #ffffff;
5022 }
5023
5024 .result-kernel-trend-toggle button.disabled {
5025   color: #9aa6b4;
5026   cursor: not-allowed;
5027 }
5028
5029 .result-kernel-trend-toggle button:focus-visible {
5030   outline: 2px solid rgba(47, 125, 225, 0.42);
5031   outline-offset: -2px;
5032 }
5033
5034 .result-kernel-comparison {
5035   overflow-x: auto;
5036   border: 1px solid #e2e8ee;
5037   border-radius: 6px;
5038 }
5039
5040 .result-kernel-comparison-row {
5041   min-width: 760px;
5042   min-height: 44px;
5043   display: grid;
5044   grid-template-columns: minmax(170px, 1.25fr) minmax(150px, 1fr) repeat(3, minmax(105px, 0.72fr));
5045   align-items: center;
5046   gap: 10px;
5047   padding-inline: 14px;
5048   border-top: 1px solid #e8edf2;
5049   color: #5d6e81;
5050   font-size: 10px;
```



```
front/src/workbench/workbench-polish.css
```

```
5051     font-variant-numeric: tabular-nums;
5052 }
5053
5054 .result-kernel-comparison-row:first-child {
5055     border-top: 0;
5056 }
5057
5058 .result-kernel-comparison-row.header {
5059     min-height: 36px;
5060     color: #718095;
5061     background: #f5f7f9;
5062     font-weight: 700;
5063 }
5064
5065 .result-kernel-comparison-row strong {
5066     color: #2d4257;
5067     font-size: 11px;
5068 }
5069
5070 /* 多核学习页面围绕求解过程组织，与核定义页面保持明确的任务分工。 */
5071 .result-mkl-summary {
5072     min-width: 0;
5073     display: grid;
5074     grid-template-columns: minmax(265px, 1.2fr) repeat(6, minmax(108px, 1fr));
5075     align-items: stretch;
5076     overflow: hidden;
5077     border: 1px solid #dce5ed;
5078     border-radius: 8px;
5079     background: #ffffff;
5080 }
5081
5082 .result-mkl-summary-lead {
5083     min-width: 0;
5084     display: flex;
5085     align-items: center;
5086     gap: 11px;
5087     padding: 15px 17px;
5088     background: #f4f8fb;
5089 }
5090
5091 .result-mkl-summary-icon {
5092     flex: 0 0 auto;
5093     display: grid;
5094     place-items: center;
5095     width: 38px;
5096     height: 38px;
5097     border: 1px solid #c8deef;
5098     border-radius: 8px;
5099     color: #1f64bd;
5100     background: #eaf3ff;
```

```
front/src/workbench/workbench-polish.css
```

```
5101 }
5102
5103 .result-mkl-summary-lead > div {
5104     min-width: 0;
5105     display: grid;
5106     gap: 2px;
5107 }
5108
5109 .result-mkl-summary-lead > div > span {
5110     color: #718196;
5111     font-size: 10px;
5112     font-weight: 700;
5113 }
5114
5115 .result-mkl-summary-lead strong {
5116     overflow: hidden;
5117     color: #243b52;
5118     font-size: 14px;
5119     text-overflow: ellipsis;
5120     white-space: nowrap;
5121 }
5122
5123 .result-mkl-summary-lead small {
5124     overflow: hidden;
5125     color: #708298;
5126     font-size: 9px;
5127     text-overflow: ellipsis;
5128     white-space: nowrap;
5129 }
5130
5131 .result-mkl-summary-lead > .workbench-status {
5132     margin-left: auto;
5133     flex: 0 0 auto;
5134 }
5135
5136 .result-mkl-summary-metrics {
5137     min-width: 0;
5138     display: contents;
5139     margin: 0;
5140 }
5141
5142 .result-mkl-summary-metrics > div {
5143     min-width: 0;
5144     display: grid;
5145     align-content: center;
5146     gap: 4px;
5147     padding: 11px 13px;
5148     border-left: 1px solid #e3eaf0;
5149 }
5150
```

```
front/src/workbench/workbench-polish.css
```

```
5151 .result-mkl-summary-metrics dt {
5152     min-width: 0;
5153     display: flex;
5154     align-items: center;
5155     gap: 5px;
5156     overflow: hidden;
5157     color: #78899a;
5158     font-size: 9px;
5159     white-space: nowrap;
5160 }
5161
5162 .result-mkl-summary-metrics dt svg {
5163     flex: 0 0 auto;
5164     color: #4c83bb;
5165 }
5166
5167 .result-mkl-summary-metrics dd {
5168     min-width: 0;
5169     margin: 0;
5170     overflow: hidden;
5171     color: #2d455b;
5172     font-size: 13px;
5173     font-weight: 760;
5174     font-variant-numeric: tabular-nums;
5175     text-overflow: ellipsis;
5176     white-space: nowrap;
5177 }
5178
5179 .result-mkl-summary-metrics small {
5180     min-width: 0;
5181     overflow: hidden;
5182     color: #7a8b9e;
5183     font-size: 9px;
5184     text-overflow: ellipsis;
5185     white-space: nowrap;
5186 }
5187
5188 .result-mkl-solver-workspace {
5189     min-width: 0;
5190     display: grid;
5191     grid-template-columns: minmax(0, 1fr) 310px;
5192     align-items: stretch;
5193     overflow: hidden;
5194     border: 1px solid #dfe6ed;
5195     border-radius: 7px;
5196 }
5197
5198 .result-mkl-chart-area {
5199     min-width: 0;
5200     padding: 8px 10px 0;
```

```
front/src/workbench/workbench-polish.css
```

```
5201 }
5202
5203 .result-mkl-convergence-chart {
5204     height: 390px;
5205     min-height: 390px;
5206 }
5207
5208 .result-mkl-diagnostics {
5209     min-width: 0;
5210     display: grid;
5211     align-content: start;
5212     gap: 16px;
5213     padding: 20px;
5214     border-left: 1px solid #dfe6ed;
5215     background: #f7f9fb;
5216 }
5217
5218 .result-mkl-toolbar {
5219     display: flex;
5220     align-items: center;
5221     justify-content: flex-end;
5222     flex-wrap: wrap;
5223     gap: 8px;
5224 }
5225
5226 .result-mkl-view-toggle {
5227     display: inline-flex;
5228     align-items: center;
5229     overflow: hidden;
5230     padding: 2px;
5231     border: 1px solid #d6e0e9;
5232     border-radius: 6px;
5233     background: #f4f7f9;
5234 }
5235
5236 .result-mkl-view-toggle button {
5237     min-height: 27px;
5238     padding: 4px 9px;
5239     border: 0;
5240     border-radius: 4px;
5241     color: #68798c;
5242     background: transparent;
5243     cursor: pointer;
5244     font: inherit;
5245     font-size: 10px;
5246     font-weight: 650;
5247     transition: color var(--motion-fast) ease, background-color var(--motion-fast) ease;
5248 }
5249
5250 .result-mkl-view-toggle button:hover:not(:disabled) {
```

```
front/src/workbench/workbench-polish.css
```

```
5251     color: #245f9f;
5252     background: #eaf1f7;
5253 }
5254
5255 .result-mkl-view-toggle button.active {
5256     color: #174f8c;
5257     background: #ffffff;
5258 }
5259
5260 .result-mkl-view-toggle button.disabled {
5261     color: #a2adba;
5262     cursor: not-allowed;
5263 }
5264
5265 .result-mkl-view-toggle button:focus-visible {
5266     outline: 2px solid rgba(47, 125, 225, 0.42);
5267     outline-offset: -2px;
5268 }
5269
5270 .result-mkl-run-control {
5271     display: inline-flex;
5272     align-items: center;
5273     gap: 6px;
5274     color: #718196;
5275     font-size: 10px;
5276     font-weight: 650;
5277 }
5278
5279 .result-mkl-diagnostic-note {
5280     display: flex;
5281     align-items: flex-start;
5282     gap: 7px;
5283     color: #3c6f63;
5284     font-size: 10px;
5285     line-height: 1.5;
5286 }
5287
5288 .result-mkl-diagnostic-note svg {
5289     flex: 0 0 auto;
5290     margin-top: 1px;
5291 }
5292
5293 .result-mkl-diagnostics dl {
5294     display: grid;
5295     grid-template-columns: repeat(2, minmax(0, 1fr));
5296     margin: 0;
5297     border-block: 1px solid #dfe6ed;
5298 }
5299
5300 .result-mkl-diagnostics dl div {
```

```
front/src/workbench/workbench-polish.css
```

```
5301   min-width: 0;
5302   display: grid;
5303   gap: 4px;
5304   padding: 10px 2px;
5305   border-top: 1px solid #e5eaf0;
5306 }
5307
5308 .result-mkl-diagnostics dl div:nth-child(-n + 2) {
5309   border-top: 0;
5310 }
5311
5312 .result-mkl-diagnostics dl div:nth-child(even) {
5313   padding-left: 12px;
5314   border-left: 1px solid #e5eaf0;
5315 }
5316
5317 .result-mkl-diagnostics dt {
5318   color: #78889a;
5319   font-size: 9px;
5320 }
5321
5322 .result-mkl-diagnostics dd {
5323   min-width: 0;
5324   margin: 0;
5325   overflow: hidden;
5326   color: #2d4359;
5327   font-size: 11px;
5328   font-weight: 730;
5329   font-variant-numeric: tabular-nums;
5330   text-overflow: ellipsis;
5331   white-space: nowrap;
5332 }
5333
5334 .result-mkl-diagnostics > p,
5335 .result-mkl-mixture-note {
5336   margin: 0;
5337   color: #758599;
5338   font-size: 9px;
5339   line-height: 1.55;
5340 }
5341
5342 .result-mkl-detail-grid {
5343   min-width: 0;
5344   display: grid;
5345   grid-template-columns: minmax(0, 1.18fr) minmax(360px, 0.82fr);
5346   align-items: stretch;
5347   gap: 16px;
5348 }
5349
5350 .result-mkl-iteration-panel,
```

```
front/src/workbench/workbench-polish.css
```

```
5351 .result-mkl-mixture-panel {
5352     min-height: 100%;
5353 }
5354
5355 .result-mkl-iteration-table {
5356     max-height: 520px;
5357     overflow: auto;
5358     border: 1px solid #e2e8ee;
5359     border-radius: 6px;
5360 }
5361
5362 .result-mkl-iteration-row {
5363     min-width: 740px;
5364     min-height: 42px;
5365     display: grid;
5366     grid-template-columns: 92px minmax(145px, 1.15fr) minmax(110px, 0.85fr) minmax(110px, 0.8fr) minmax(88px, 0.65fr);
5367     align-items: center;
5368     gap: 12px;
5369     padding-inline: 13px;
5370     border-top: 1px solid #e7ecf1;
5371     color: #5f7184;
5372     font-size: 10px;
5373     font-variant-numeric: tabular-nums;
5374 }
5375
5376 .result-mkl-iteration-row:first-child {
5377     border-top: 0;
5378 }
5379
5380 .result-mkl-iteration-row.header {
5381     min-height: 36px;
5382     color: #6f8093;
5383     background: #f5f7f9;
5384     font-weight: 700;
5385 }
5386
5387 .result-mkl-iteration-row strong {
5388     color: #30465c;
5389 }
5390
5391 .result-mkl-iteration-row.is-terminal {
5392     color: #285d52;
5393     background: #f1f8f5;
5394 }
5395
5396 .result-mkl-iteration-row.is-terminal strong {
5397     color: #1e705f;
5398 }
5399
5400 .result-mkl-mixture-list {
```

```
front/src/workbench/workbench-polish.css
```

```
5401     display: grid;
5402     gap: 0;
5403     align-content: start;
5404 }
5405
5406 .result-mkl-mixture-row {
5407     min-width: 0;
5408     display: grid;
5409     grid-template-columns: 30px minmax(0, 1fr);
5410     gap: 10px;
5411     padding: 12px 0;
5412     border-top: 1px solid #e5ebf0;
5413 }
5414
5415 .result-mkl-mixture-row:first-child {
5416     padding-top: 2px;
5417     border-top: 0;
5418 }
5419
5420 .result-mkl-mixture-rank {
5421     display: grid;
5422     place-items: start center;
5423     padding-top: 2px;
5424     color: #8b9aaa;
5425     font-size: 10px;
5426     font-weight: 760;
5427     font-variant-numeric: tabular-nums;
5428 }
5429
5430 .result-mkl-mixture-row.is-dominant .result-mkl-mixture-rank {
5431     color: #1f64bd;
5432 }
5433
5434 .result-mkl-mixture-body {
5435     min-width: 0;
5436     display: grid;
5437     gap: 7px;
5438 }
5439
5440 .result-mkl-mixture-body > header {
5441     min-width: 0;
5442     display: flex;
5443     align-items: center;
5444     justify-content: space-between;
5445     gap: 12px;
5446 }
5447
5448 .result-mkl-mixture-body > header > div {
5449     min-width: 0;
5450     display: grid;
```



```
front/src/workbench/workbench-polish.css
```

```
5451     gap: 2px;
5452 }
5453
5454 .result-mkl-mixture-body strong {
5455     min-width: 0;
5456     overflow: hidden;
5457     color: #34495d;
5458     font-size: 10px;
5459     text-overflow: ellipsis;
5460     white-space: nowrap;
5461 }
5462
5463 .result-mkl-mixture-body em {
5464     color: #2f4358;
5465     font-size: 11px;
5466     font-style: normal;
5467     font-weight: 730;
5468     font-variant-numeric: tabular-nums;
5469 }
5470
5471 .result-mkl-mixture-body header small {
5472     color: #8291a1;
5473     font-size: 9px;
5474 }
5475
5476 .result-mkl-mixture-body i {
5477     height: 8px;
5478     overflow: hidden;
5479     border-radius: 999px;
5480     background: #e8edf2;
5481 }
5482
5483 .result-mkl-mixture-body i b {
5484     display: block;
5485     height: 100%;
5486     border-radius: inherit;
5487     background: #2f7de1;
5488 }
5489
5490 .result-mkl-mixture-row.tone-2 i b {
5491     background: #23865f;
5492 }
5493
5494 .result-mkl-mixture-row.tone-3 i b {
5495     background: #c97a17;
5496 }
5497
5498 .result-mkl-mixture-row.tone-4 i b {
5499     background: #7967a8;
5500 }
```

front/src/workbench/workbench-polish.css

```
5501
5502 .result-mkl-mixture-meta {
5503     display: flex;
5504     flex-wrap: wrap;
5505     gap: 5px 12px;
5506     color: #7a899a;
5507     font-size: 9px;
5508 }
5509
5510 .result-mkl-mixture-note {
5511     padding-top: 11px;
5512     border-top: 1px solid #e5ebf0;
5513 }
5514
5515 .result-mkl-run-panel {
5516     min-width: 0;
5517 }
5518
5519 .result-mkl-run-table {
5520     overflow-x: auto;
5521     border: 1px solid #e2e8ee;
5522     border-radius: 6px;
5523 }
5524
5525 .result-mkl-run-row {
5526     min-width: 860px;
5527     min-height: 43px;
5528     display: grid;
5529     grid-template-columns: 92px minmax(120px, 1fr) 78px minmax(150px, 1fr) minmax(150px, 1fr) 110px 92px;
5530     align-items: center;
5531     gap: 10px;
5532     padding-inline: 14px;
5533     border-top: 1px solid #e7ecf1;
5534     color: #5f7184;
5535     font-size: 10px;
5536     font-variant-numeric: tabular-nums;
5537 }
5538
5539 .result-mkl-run-row:first-child {
5540     border-top: 0;
5541 }
5542
5543 .result-mkl-run-row.header {
5544     min-height: 36px;
5545     color: #6f8093;
5546     background: #f5f7f9;
5547     font-weight: 700;
5548 }
5549
5550 .result-mkl-run-row strong {
```

```
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```

```
5551     color: #30465c;
5552 }
5553
5554 .result-mkl-run-row.is-representative {
5555     color: #275d52;
5556     background: #f1f8f5;
5557 }
5558
5559 .result-mkl-run-row.is-representative strong {
5560     color: #1e705f;
5561 }
5562
5563 /* 可视化工作区 */
5564 .result-visual-workspace {
5565     grid-template-columns: minmax(0, 1fr) 320px;
5566     align-items: stretch;
5567 }
5568
5569 .result-visual-canvas .result-chart.large {
5570     height: 520px;
5571     min-height: 520px;
5572 }
5573
5574 .result-visual-toolbox {
5575     height: 100%;
5576     display: grid;
5577     align-content: start;
5578     gap: 15px;
5579     box-sizing: border-box;
5580 }
5581
5582 .result-visual-toolbox .result-visualization-list button {
5583     min-height: 56px;
5584     padding: 8px 10px;
5585     cursor: pointer;
5586 }
5587
5588 .result-visual-toolbox .result-visualization-list button > span,
5589 .result-view-catalog button > span {
5590     min-width: 0;
5591     display: grid;
5592     gap: 3px;
5593 }
5594
5595 .result-visual-toolbox .result-visualization-list strong,
5596 .result-visual-toolbox .result-visualization-list small,
5597 .result-view-catalog strong,
5598 .result-view-catalog small {
5599     overflow: hidden;
5600     text-overflow: ellipsis;
```

```
front/src/workbench/workbench-polish.css
```

```
5601   white-space: nowrap;
5602 }
5603
5604 .result-visual-toolbox .result-visualization-list strong,
5605 .result-view-catalog strong {
5606   color: #34495d;
5607   font-size: 11px;
5608 }
5609
5610 .result-visual-toolbox .result-visualization-list small,
5611 .result-view-catalog small {
5612   color: #7c8b9c;
5613   font-size: 9px;
5614 }
5615
5616 .result-view-summary {
5617   display: grid;
5618   grid-template-columns: repeat(2, minmax(0, 1fr));
5619   border-block: 1px solid #e4e9ee;
5620 }
5621
5622 .result-view-summary div {
5623   min-width: 0;
5624   display: grid;
5625   gap: 3px;
5626   padding: 10px 2px;
5627 }
5628
5629 .result-view-detail-button,
5630 .result-export-submit,
5631 .result-report-submit {
5632   width: 100%;
5633   justify-content: center;
5634 }
5635
5636 .result-view-catalog > div {
5637   display: grid;
5638   grid-template-columns: repeat(4, minmax(0, 1fr));
5639   margin-top: 8px;
5640   overflow: hidden;
5641   border: 1px solid #e3e8ee;
5642   border-radius: 6px;
5643 }
5644
5645 .result-view-catalog button {
5646   min-width: 0;
5647   min-height: 62px;
5648   display: grid;
5649   grid-template-columns: auto minmax(0, 1fr) auto;
5650   align-items: center;
```

```
front/src/workbench/workbench-polish.css
```

```
5651     gap: 9px;
5652     padding: 9px 11px;
5653     border: 0;
5654     border-left: 1px solid #e3e8ee;
5655     color: #607184;
5656     background: #ffffff;
5657     cursor: pointer;
5658     text-align: left;
5659     transition: color var(--motion-fast) ease, background-color var(--motion-fast) ease;
5660 }
5661
5662 .result-view-catalog button:first-child {
5663     border-left: 0;
5664 }
5665
5666 .result-view-catalog button:hover {
5667     color: #1f64bd;
5668     background: #f3f7fb;
5669 }
5670
5671 /* 结果分析 */
5672 .result-analysis-summary-panel {
5673     display: grid;
5674     align-content: start;
5675     gap: 12px;
5676     padding-block: 16px;
5677 }
5678
5679 .result-analysis-summary-heading {
5680     min-width: 0;
5681     display: flex;
5682     align-items: flex-start;
5683     justify-content: space-between;
5684     gap: 16px;
5685 }
5686
5687 .result-analysis-summary-body {
5688     min-width: 0;
5689     display: grid;
5690     grid-template-columns: minmax(0, 1fr) minmax(300px, 0.34fr);
5691     align-items: stretch;
5692     gap: 16px;
5693 }
5694
5695 .result-analysis-facts {
5696     display: grid;
5697     grid-template-columns: repeat(6, minmax(0, 1fr));
5698     overflow: hidden;
5699     border: 1px solid #e2e8ee;
5700     border-radius: 10px;
```

```
front/src/workbench/workbench-polish.css
```

```
5701 }
5702
5703 .result-analysis-facts div {
5704     min-width: 0;
5705     display: grid;
5706     align-content: center;
5707     gap: 4px;
5708     padding: 10px 14px;
5709     border-left: 1px solid #e8edf2;
5710 }
5711
5712 .result-analysis-facts div:first-child {
5713     border-left: 0;
5714 }
5715
5716 .result-analysis-facts div.is-primary dd {
5717     color: #185f93;
5718 }
5719
5720 .result-analysis-facts dt {
5721     color: #718195;
5722     font-size: 10px;
5723 }
5724
5725 .result-analysis-facts dd {
5726     margin: 0;
5727     color: #33495e;
5728     font-size: 12px;
5729     font-weight: 700;
5730     font-variant-numeric: tabular-nums;
5731     line-height: 1.35;
5732 }
5733
5734 .result-analysis-summary-body > .workbench-notice {
5735     height: 100%;
5736     align-content: center;
5737 }
5738
5739 .result-analysis-evidence {
5740     min-width: 0;
5741     display: grid;
5742     grid-template-columns: minmax(0, 1.48fr) minmax(420px, 0.82fr);
5743     grid-template-areas:
5744         "scatter convergence"
5745         "scatter trend";
5746     align-items: stretch;
5747     gap: 16px;
5748 }
5749
5750 .result-analysis-scatter,
```

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```
5751 .result-analysis-convergence,
5752 .result-analysis-trend {
5753   min-width: 0;
5754   display: grid;
5755 }
5756
5757 .result-analysis-scatter {
5758   grid-area: scatter;
5759 }
5760
5761 .result-analysis-convergence {
5762   grid-area: convergence;
5763 }
5764
5765 .result-analysis-trend {
5766   grid-area: trend;
5767 }
5768
5769 .result-analysis-evidence .result-chart-panel {
5770   min-height: 0;
5771 }
5772
5773 .result-analysis-scatter .result-chart {
5774   height: 486px;
5775   min-height: 486px;
5776 }
5777
5778 .result-analysis-convergence .result-chart,
5779 .result-analysis-trend .result-chart {
5780   height: 190px;
5781   min-height: 190px;
5782 }
5783
5784 .result-distribution-columns {
5785   display: grid;
5786   grid-template-columns: repeat(2, minmax(0, 1fr));
5787   gap: 28px;
5788   margin-top: 10px;
5789 }
5790
5791 .result-distribution-column {
5792   min-width: 0;
5793   display: grid;
5794   align-content: start;
5795   gap: 8px;
5796 }
5797
5798 .result-distribution-column > strong {
5799   padding-bottom: 8px;
5800   border-bottom: 1px solid #e4e9ee;
```

```
front/src/workbench/workbench-polish.css
```

```
5801     color: #30465c;
5802     font-size: 11px;
5803 }
5804
5805 .result-distribution-row {
5806     min-width: 0;
5807     min-height: 28px;
5808     display: grid;
5809     grid-template-columns: minmax(72px, 0.55fr) minmax(120px, 1.45fr) 38px 48px;
5810     align-items: center;
5811     gap: 9px;
5812     color: #607184;
5813     font-size: 10px;
5814     font-variant-numeric: tabular-nums;
5815 }
5816
5817 .result-distribution-row i {
5818     height: 7px;
5819     overflow: hidden;
5820     border-radius: 999px;
5821     background: #e8edf2;
5822 }
5823
5824 .result-distribution-row i b {
5825     display: block;
5826     height: 100%;
5827     border-radius: inherit;
5828     background: #2f7de1;
5829 }
5830
5831 .result-distribution-column + .result-distribution-column i b {
5832     background: #23865f;
5833 }
5834
5835 .result-distribution-row strong {
5836     color: #33495e;
5837     text-align: right;
5838 }
5839
5840 .result-distribution-row small {
5841     color: #7b8a9c;
5842     text-align: right;
5843 }
5844
5845 /* 结果导出 */
5846 .result-artifact-panel,
5847 .result-export-builder {
5848     display: grid;
5849     align-content: start;
5850     gap: 14px;
```


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```
5851 }
5852
5853 .result-export-composer {
5854     gap: 16px;
5855 }
5856
5857 .result-export-compose-grid {
5858     display: grid;
5859     grid-template-columns: minmax(360px, 0.72fr) minmax(0, 1.28fr);
5860     gap: 24px;
5861 }
5862
5863 .result-export-setup,
5864 .result-export-selection {
5865     min-width: 0;
5866     display: grid;
5867     align-content: start;
5868     gap: 14px;
5869 }
5870
5871 .result-export-selection {
5872     padding-left: 24px;
5873     border-left: 1px solid #e3e9ef;
5874 }
5875
5876 .result-export-name-field {
5877     display: grid;
5878     gap: 6px;
5879 }
5880
5881 .result-export-name-field > span,
5882 .result-export-selection > header {
5883     color: #607185;
5884     font-size: 10px;
5885     font-weight: 700;
5886 }
5887
5888 .result-export-name-field input {
5889     width: 100%;
5890     height: 38px;
5891     padding-inline: 11px;
5892     border: 1px solid #d9e1e8;
5893     border-radius: 6px;
5894     color: #30465c;
5895     background: #f8fafc;
5896     font: inherit;
5897 }
5898
5899 .result-export-presets {
5900     display: grid;
```

```
front/src/workbench/workbench-polish.css
```

```
5901     grid-template-columns: repeat(3, minmax(0, 1fr));
5902     gap: 7px;
5903 }
5904
5905 .result-export-presets button {
5906     min-width: 0;
5907     min-height: 64px;
5908     display: grid;
5909     grid-template-columns: minmax(0, 1fr) auto;
5910     align-items: center;
5911     gap: 8px;
5912     padding: 9px 10px;
5913     border: 1px solid #dfe6ed;
5914     border-radius: 6px;
5915     color: #637488;
5916     background: #ffffff;
5917     cursor: pointer;
5918     text-align: left;
5919     transition: border-color var(--motion-fast) ease, background-color var(--motion-fast) ease;
5920 }
5921
5922 .result-export-presets button:hover {
5923     border-color: #a9c9ed;
5924     background: #f3f7fb;
5925 }
5926
5927 .result-export-presets button > span:first-child {
5928     min-width: 0;
5929     display: grid;
5930     gap: 3px;
5931 }
5932
5933 .result-export-presets strong,
5934 .result-export-presets small {
5935     overflow: hidden;
5936     text-overflow: ellipsis;
5937     white-space: nowrap;
5938 }
5939
5940 .result-export-presets strong {
5941     color: #33495e;
5942     font-size: 10px;
5943 }
5944
5945 .result-export-presets small,
5946 .result-export-presets button > span:last-child {
5947     font-size: 9px;
5948 }
5949
5950 .result-export-selection > header {
```

```
front/src/workbench/workbench-polish.css
```

```
5951     display: flex;
5952     align-items: center;
5953     justify-content: space-between;
5954     gap: 12px;
5955 }
5956
5957 .result-artifact-list {
5958     display: grid;
5959     border-block: 1px solid #e4e9ee;
5960 }
5961
5962 .result-artifact-item {
5963     min-width: 0;
5964     min-height: 58px;
5965     display: grid;
5966     grid-template-columns: 34px minmax(180px, 1fr) minmax(90px, 0.35fr) auto;
5967     align-items: center;
5968     gap: 11px;
5969     padding: 8px 2px;
5970     border-top: 1px solid #e8edf2;
5971 }
5972
5973 .result-export-library .result-artifact-item {
5974     grid-template-columns: 30px minmax(0, 1fr) 68px auto;
5975     gap: 8px;
5976 }
5977
5978 .result-export-library .result-artifact-item .btn {
5979     width: 30px;
5980     height: 30px;
5981     padding: 0;
5982 }
5983
5984 .result-artifact-item:first-child {
5985     border-top: 0;
5986 }
5987
5988 .result-artifact-icon {
5989     width: 30px;
5990     height: 30px;
5991     display: grid;
5992     place-items: center;
5993     border-radius: 6px;
5994     color: #2f6fa9;
5995     background: #edf5fc;
5996 }
5997
5998 .result-artifact-item > span:not(.result-artifact-icon) {
5999     min-width: 0;
6000     display: grid;
```

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```
6001     gap: 3px;
6002 }
6003
6004 .result-artifact-item strong,
6005 .result-artifact-item small {
6006     min-width: 0;
6007     overflow: hidden;
6008     text-overflow: ellipsis;
6009     white-space: nowrap;
6010 }
6011
6012 .result-artifact-item strong {
6013     color: #30465c;
6014     font-size: 11px;
6015 }
6016
6017 .result-artifact-item small {
6018     color: #7c8b9c;
6019     font-size: 9px;
6020 }
6021
6022 .result-export-option-list {
6023     display: grid;
6024     grid-template-columns: repeat(2, minmax(0, 1fr));
6025     gap: 7px;
6026 }
6027
6028 .result-export-option-list label,
6029 .result-report-sections label {
6030     min-width: 0;
6031     display: flex;
6032     align-items: center;
6033     gap: 8px;
6034     border: 1px solid #dfe5eb;
6035     border-radius: 6px;
6036     color: #607184;
6037     background: #ffffff;
6038     cursor: pointer;
6039     transition: border-color var(--motion-fast) ease, background-color var(--motion-fast) ease;
6040 }
6041
6042 .result-export-option-list label {
6043     min-height: 54px;
6044     padding: 8px 9px;
6045 }
6046
6047 .result-export-option-list label:hover,
6048 .result-export-option-list label.selected,
6049 .result-report-sections label:hover,
6050 .result-report-sections label.selected {
```

```
front/src/workbench/workbench-polish.css
```

```
6051     border-color: #a9c9ed;
6052     background: #f0f6fd;
6053 }
6054
6055 .result-export-option-list input,
6056 .result-report-sections input {
6057     flex: 0 0 auto;
6058     accent-color: #2f7del;
6059 }
6060
6061 .result-export-option-list label > span,
6062 .result-report-sections label > span {
6063     min-width: 0;
6064     display: grid;
6065     gap: 2px;
6066 }
6067
6068 .result-export-option-list strong,
6069 .result-export-option-list small,
6070 .result-report-sections strong,
6071 .result-report-sections small {
6072     overflow: hidden;
6073     text-overflow: ellipsis;
6074     white-space: nowrap;
6075 }
6076
6077 .result-export-option-list strong,
6078 .result-report-sections strong {
6079     color: #34495d;
6080     font-size: 10px;
6081 }
6082
6083 .result-export-option-list small,
6084 .result-report-sections small {
6085     color: #7d8b9c;
6086     font-size: 9px;
6087 }
6088
6089 .result-export-manifest {
6090     display: grid;
6091     grid-template-columns: repeat(2, minmax(0, 1fr));
6092     border-block: 1px solid #e4e9ee;
6093 }
6094
6095 .result-export-manifest div {
6096     min-width: 0;
6097     display: grid;
6098     gap: 3px;
6099     padding: 9px 2px;
6100 }
```

```
front/src/workbench/workbench-polish.css
```

```
6101
6102 .result-export-manifest dt,
6103 .result-export-manifest dd {
6104     min-width: 0;
6105     overflow: hidden;
6106     text-overflow: ellipsis;
6107     white-space: nowrap;
6108 }
6109
6110 .result-export-submit {
6111     min-height: 38px;
6112 }
6113
6114 .result-export-library {
6115     min-width: 0;
6116     display: grid;
6117     grid-template-columns: minmax(0, 1.35fr) minmax(390px, 0.65fr);
6118     align-items: stretch;
6119     gap: 16px;
6120 }
6121
6122 .result-export-history-panel,
6123 .result-artifact-panel {
6124     min-width: 0;
6125     height: 100%;
6126     display: grid;
6127     grid-template-rows: auto minmax(0, 1fr);
6128     align-content: stretch;
6129 }
6130
6131 .result-export-archive-list {
6132     display: grid;
6133     margin-top: 4px;
6134     border-block: 1px solid #e2e8ee;
6135 }
6136
6137 .result-export-archive {
6138     min-width: 0;
6139     min-height: 82px;
6140     display: grid;
6141     grid-template-columns: 36px minmax(220px, 1fr) 190px auto auto;
6142     align-items: center;
6143     gap: 12px;
6144     padding-block: 11px;
6145     border-top: 1px solid #e8edf2;
6146 }
6147
6148 .result-export-archive:first-child {
6149     border-top: 0;
6150 }
```

```
front/src/workbench/workbench-polish.css
```

```
6151
6152 .result-export-archive-icon {
6153     width: 34px;
6154     height: 34px;
6155     display: grid;
6156     place-items: center;
6157     border-radius: 6px;
6158     color: #2471aa;
6159     background: #edf5fc;
6160 }
6161
6162 .result-export-archive-main,
6163 .result-export-archive-meta {
6164     min-width: 0;
6165     display: grid;
6166     gap: 4px;
6167 }
6168
6169 .result-export-archive-main > strong,
6170 .result-export-archive-main > small,
6171 .result-export-archive-meta time {
6172     overflow: hidden;
6173     text-overflow: ellipsis;
6174     white-space: nowrap;
6175 }
6176
6177 .result-export-archive-main > strong {
6178     color: #30465c;
6179     font-size: 11px;
6180 }
6181
6182 .result-export-archive-main > small,
6183 .result-export-archive-meta {
6184     color: #77879a;
6185     font-size: 9px;
6186 }
6187
6188 .result-export-archive-main > div {
6189     display: flex;
6190     flex-wrap: wrap;
6191     gap: 4px;
6192 }
6193
6194 .result-export-archive-main > div span {
6195     padding: 2px 6px;
6196     border-radius: 999px;
6197     color: #45657f;
6198     background: #eef3f7;
6199     font-size: 8px;
6200 }
```

```
front/src/workbench/workbench-polish.css
```

```
6201
6202 .result-export-archive-meta strong {
6203     color: #43586d;
6204     font-size: 10px;
6205 }
6206
6207 .result-export-empty {
6208     min-height: 190px;
6209     height: 100%;
6210     display: grid;
6211     place-items: center;
6212     align-content: center;
6213     gap: 7px;
6214     color: #75879a;
6215     text-align: center;
6216 }
6217
6218 .result-export-empty svg {
6219     color: #2f7de1;
6220 }
6221
6222 .result-export-empty strong {
6223     color: #344b61;
6224     font-size: 12px;
6225 }
6226
6227 .result-export-empty span {
6228     max-width: 48ch;
6229     font-size: 10px;
6230     line-height: 1.55;
6231 }
6232
6233 .result-export-history-table {
6234     margin-top: 8px;
6235     overflow-x: auto;
6236     border: 1px solid #e2e8ee;
6237     border-radius: 6px;
6238 }
6239
6240 .result-export-history-row {
6241     min-width: 760px;
6242     min-height: 48px;
6243     display: grid;
6244     grid-template-columns: 24px minmax(260px, 1fr) 100px 190px auto;
6245     align-items: center;
6246     gap: 10px;
6247     padding-inline: 12px;
6248     border-top: 1px solid #e8edf2;
6249     color: #65768a;
6250     font-size: 10px;
```



```
front/src/workbench/workbench-polish.css
```

```
6251 }
6252
6253 .result-export-history-row:first-child {
6254     border-top: 0;
6255 }
6256
6257 .result-export-history-row strong {
6258     overflow: hidden;
6259     color: #32485e;
6260     font-size: 11px;
6261     text-overflow: ellipsis;
6262     white-space: nowrap;
6263 }
6264
6265 /* 报告页左侧固定编辑语义，右侧保持接近最终文档的阅读宽度。 */
6266 .result-report-workspace {
6267     grid-template-columns: 360px minmax(0, 1fr);
6268 }
6269
6270 .result-report-controls {
6271     display: grid;
6272     align-content: start;
6273     gap: 15px;
6274     position: sticky;
6275     top: 16px;
6276 }
6277
6278 .result-report-format,
6279 .result-report-sections {
6280     display: grid;
6281     gap: 8px;
6282     margin: 0;
6283     padding: 0;
6284     border: 0;
6285 }
6286
6287 .result-report-format legend,
6288 .result-report-sections legend {
6289     margin-bottom: 7px;
6290     color: #56687c;
6291     font-size: 10px;
6292     font-weight: 700;
6293 }
6294
6295 .result-report-format .soft-segmented {
6296     display: grid;
6297     grid-template-columns: repeat(2, minmax(0, 1fr));
6298 }
6299
6300 .result-report-sections label {
```

```
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```

```
6301     min-height: 46px;
6302     padding: 7px 9px;
6303 }
6304
6305 .result-report-preview {
6306     min-width: 0;
6307     background: #f3f5f7;
6308 }
6309
6310 .result-report-preview-actions {
6311     display: flex;
6312     align-items: center;
6313     gap: 6px;
6314 }
6315
6316 .result-report-action {
6317     width: 28px;
6318     height: 28px;
6319     display: inline-grid;
6320     place-items: center;
6321     padding: 0;
6322     border: 1px solid #d6e0e9;
6323     border-radius: 5px;
6324     color: #526a81;
6325     background: #ffffff;
6326     cursor: pointer;
6327     transition: color var(--motion-fast) ease, border-color var(--motion-fast) ease, background-color var(--motion-fast)
        ease;
6328 }
6329
6330 .result-report-action:hover:not(:disabled) {
6331     border-color: #a9c9ed;
6332     color: #1f64bd;
6333     background: #f0f6fd;
6334 }
6335
6336 .result-report-action:disabled {
6337     color: #a6b2bf;
6338     background: #f5f7f9;
6339     cursor: not-allowed;
6340 }
6341
6342 .result-report-document {
6343     width: min(100%, 880px);
6344     min-height: 620px;
6345     margin: 12px auto 0;
6346     padding: 28px 34px;
6347     border: 1px solid #dfe4e9;
6348     border-radius: 4px;
6349     background: #ffffff;
```

```
front/src/workbench/workbench-polish.css
```

```
6350 }
6351
6352 .result-report-document > header {
6353     display: grid;
6354     gap: 6px;
6355     padding-bottom: 16px;
6356     border-bottom: 2px solid #2b4055;
6357 }
6358
6359 .result-report-document > header span {
6360     color: #63758a;
6361     font-size: 9px;
6362     font-weight: 700;
6363 }
6364
6365 .result-report-document > header strong {
6366     color: #203247;
6367     font-size: 22px;
6368     font-weight: 760;
6369     line-height: 1.3;
6370     overflow-wrap: anywhere;
6371 }
6372
6373 .result-report-document > header small {
6374     color: #7d8b9c;
6375     font-size: 10px;
6376 }
6377
6378 .report-preview-section {
6379     margin-top: 8px;
6380     border-top: 1px solid #e4e9ee;
6381 }
6382
6383 .report-preview-section > summary {
6384     min-height: 48px;
6385     display: flex;
6386     align-items: center;
6387     justify-content: space-between;
6388     gap: 12px;
6389     padding-block: 7px;
6390     color: #2d4258;
6391     cursor: pointer;
6392     list-style: none;
6393 }
6394
6395 .report-preview-section > summary::-webkit-details-marker {
6396     display: none;
6397 }
6398
6399 .report-preview-section > summary > span {
```

```
front/src/workbench/workbench-polish.css
```

```
6400     min-width: 0;
6401     display: grid;
6402     gap: 2px;
6403 }
6404
6405 .report-preview-section > summary h3 {
6406     margin: 0;
6407     font-size: 14px;
6408 }
6409
6410 .report-preview-section > summary small {
6411     color: #7b8998;
6412     font-size: 9px;
6413 }
6414
6415 .report-preview-section > summary > svg {
6416     flex: 0 0 auto;
6417     color: #73869a;
6418     transition: transform var(--motion-fast) ease, color var(--motion-fast) ease;
6419 }
6420
6421 .report-preview-section > summary:hover > svg {
6422     color: #1f64bd;
6423 }
6424
6425 .report-preview-section[open] > summary > svg {
6426     transform: rotate(180deg);
6427 }
6428
6429 .report-preview-section-body {
6430     padding-bottom: 12px;
6431     animation: report-preview-reveal 160ms ease-out;
6432 }
6433
6434 @keyframes report-preview-reveal {
6435     from { opacity: 0; transform: translateY(-4px); }
6436     to { opacity: 1; transform: translateY(0); }
6437 }
6438
6439 .report-preview-empty {
6440     margin: 0;
6441     padding: 10px 4px;
6442     color: #7d8b9c;
6443     font-size: 10px;
6444 }
6445
6446 .result-report-document h3 {
6447     margin: 0 0 10px;
6448     color: #2d4258;
6449     font-size: 14px;
```

```
front/src/workbench/workbench-polish.css
```

```
6450     font-weight: 730;
6451     letter-spacing: 0;
6452 }
6453
6454 .result-report-document .report-document-summary,
6455 .result-report-document .report-document-metrics {
6456     display: grid;
6457     grid-template-columns: repeat(3, minmax(0, 1fr));
6458     overflow: hidden;
6459     border: 1px solid #ele6eb;
6460     border-radius: 4px;
6461 }
6462
6463 .result-report-document .report-document-summary > div,
6464 .result-report-document .report-document-metrics > div {
6465     min-width: 0;
6466     display: grid;
6467     gap: 4px;
6468     padding: 11px;
6469     border-left: 1px solid #e6eae;
6470     border-top: 1px solid #e6eae;
6471 }
6472
6473 .result-report-document .report-document-summary > div:nth-child(-n + 3),
6474 .result-report-document .report-document-metrics > div {
6475     border-top: 0;
6476 }
6477
6478 .result-report-document .report-document-summary > div:nth-child(3n + 1),
6479 .result-report-document .report-document-metrics > div:nth-child(4n + 1) {
6480     border-left: 0;
6481 }
6482
6483 .result-report-document .report-document-metrics {
6484     grid-template-columns: repeat(4, minmax(0, 1fr));
6485 }
6486
6487 .result-report-document .report-document-summary span,
6488 .result-report-document .report-document-metrics span {
6489     color: #7b8998;
6490     font-size: 9px;
6491 }
6492
6493 .result-report-document .report-document-summary strong,
6494 .result-report-document .report-document-metrics strong {
6495     min-width: 0;
6496     overflow: hidden;
6497     color: #2a4055;
6498     font-size: 12px;
6499     text-overflow: ellipsis;
```

```
front/src/workbench/workbench-polish.css
```

```
6500   white-space: nowrap;
6501 }
6502
6503 .result-report-document .report-document-metrics small {
6504   color: #7d8b9b;
6505   font-size: 8px;
6506 }
6507
6508 .report-parameter-list,
6509 .report-artifact-list {
6510   display: grid;
6511   border-block: 1px solid #e3e8ed;
6512 }
6513
6514 .report-parameter-list > div,
6515 .report-artifact-list > div {
6516   min-width: 0;
6517   min-height: 34px;
6518   display: grid;
6519   align-items: center;
6520   gap: 9px;
6521   padding-inline: 4px;
6522   border-top: 1px solid #e8edf1;
6523   color: #647487;
6524   font-size: 10px;
6525 }
6526
6527 .report-parameter-list > div {
6528   grid-template-columns: minmax(96px, 0.65fr) minmax(0, 1.35fr);
6529 }
6530
6531 .report-parameter-list {
6532   grid-template-columns: repeat(2, minmax(0, 1fr));
6533 }
6534
6535 .report-parameter-list > div:nth-child(-n + 2) {
6536   border-top: 0;
6537 }
6538
6539 .report-parameter-list > div:nth-child(2n + 1) {
6540   border-left: 0;
6541 }
6542
6543 .report-artifact-list > div {
6544   grid-template-columns: 20px minmax(0, 1fr) auto;
6545 }
6546
6547 .report-parameter-list > div:first-child,
6548 .report-artifact-list > div:first-child {
6549   border-top: 0;
```

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```
6550 }
6551
6552 .report-parameter-list strong,
6553 .report-artifact-list strong {
6554     min-width: 0;
6555     overflow: hidden;
6556     color: #354a5f;
6557     text-overflow: ellipsis;
6558     white-space: nowrap;
6559 }
6560
6561 .report-run-table {
6562     overflow-x: auto;
6563     border: 1px solid #ele6eb;
6564     border-radius: 4px;
6565 }
6566
6567 .report-run-table > div {
6568     min-width: 560px;
6569     min-height: 36px;
6570     display: grid;
6571     grid-template-columns: minmax(100px, 0.9fr) repeat(4, minmax(76px, 1fr));
6572     align-items: center;
6573     gap: 8px;
6574     padding-inline: 10px;
6575     border-top: 1px solid #e7ecf0;
6576     color: #647487;
6577     font-size: 9px;
6578     font-variant-numeric: tabular-nums;
6579 }
6580
6581 .report-run-table > div:first-child {
6582     border-top: 0;
6583 }
6584
6585 .report-run-table > div.header {
6586     min-height: 32px;
6587     color: #607185;
6588     background: #f5f7f9;
6589     font-weight: 700;
6590 }
6591
6592 .report-run-table strong {
6593     color: #34495d;
6594 }
6595
6596 .report-preview-note {
6597     display: block;
6598     margin-top: 7px;
6599     color: #7d8b9c;
```

```
front/src/workbench/workbench-polish.css
```

```
6600     font-size: 9px;
6601 }
6602
6603 .result-view-catalog button:focus-visible,
6604 .result-visualization-list button:focus-visible,
6605 .result-kernel-catalog button:focus-visible,
6606 .result-report-action:focus-visible,
6607 .report-preview-section > summary:focus-visible,
6608 .result-export-option-list label:has(input:focus-visible),
6609 .result-report-sections label:has(input:focus-visible) {
6610     outline: 2px solid rgba(47, 125, 225, 0.42);
6611     outline-offset: 1px;
6612 }
6613
6614 @media (max-width: 1500px) {
6615     .result-mkl-summary {
6616         grid-template-columns: minmax(230px, 1.2fr) repeat(3, minmax(118px, 1fr));
6617     }
6618
6619     .result-detail-primary,
6620     .result-ca-primary {
6621         grid-template-columns: minmax(0, 1.22fr) minmax(330px, 0.78fr);
6622     }
6623
6624     .result-kernel-workspace-body {
6625         grid-template-columns: 200px minmax(0, 1fr) 240px;
6626     }
6627
6628     .result-kernel-definition {
6629         padding-inline: 22px;
6630     }
6631
6632     .result-mkl-solver-workspace {
6633         grid-template-columns: minmax(0, 1fr) 290px;
6634     }
6635
6636     .result-mkl-detail-grid {
6637         grid-template-columns: minmax(0, 1.12fr) minmax(340px, 0.88fr);
6638     }
6639
6640     .result-export-compose-grid {
6641         grid-template-columns: minmax(330px, 0.78fr) minmax(0, 1.22fr);
6642         gap: 18px;
6643     }
6644
6645     .result-export-library {
6646         grid-template-columns: minmax(0, 1.2fr) minmax(360px, 0.8fr);
6647     }
6648
6649     .result-export-archive {
```



```
front/src/workbench/workbench-polish.css
```

```
6650     grid-template-columns: 34px minmax(200px, 1fr) 150px auto;
6651   }
6652
6653   .result-export-archive > .workbench-status {
6654     display: none;
6655   }
6656
6657   .result-analysis-summary-body {
6658     grid-template-columns: minmax(0, 1fr) 300px;
6659   }
6660
6661   .result-analysis-facts {
6662     grid-template-columns: repeat(3, minmax(0, 1fr));
6663   }
6664
6665   .result-analysis-facts div:nth-child(4) {
6666     padding-left: 2px;
6667     border-left: 0;
6668   }
6669
6670   .result-analysis-facts div:nth-child(-n + 3) {
6671     border-bottom: 1px solid #e8edf2;
6672   }
6673
6674   .result-analysis-evidence {
6675     grid-template-columns: minmax(0, 1.32fr) minmax(380px, 0.88fr);
6676   }
6677 }
6678
6679 @media (max-width: 1360px) {
6680   .result-mkl-summary {
6681     grid-template-columns: minmax(220px, 1.2fr) repeat(2, minmax(116px, 1fr));
6682   }
6683
6684   .result-mkl-toolbar {
6685     justify-content: flex-start;
6686   }
6687
6688   .result-detail-primary,
6689   .result-detail-secondary,
6690   .result-ca-primary,
6691   .result-analysis-primary,
6692   .result-analysis-secondary {
6693     grid-template-columns: repeat(2, minmax(0, 1fr));
6694   }
6695
6696   .result-analysis-summary-body {
6697     grid-template-columns: 1fr;
6698   }
6699 }
```

```
front/src/workbench/workbench-polish.css
```

```
6700   .result-analysis-evidence {
6701     grid-template-columns: minmax(0, 1.2fr) minmax(340px, 0.8fr);
6702   }
6703
6704   .result-visual-workspace {
6705     grid-template-columns: minmax(0, 1fr) 292px;
6706   }
6707
6708   .result-export-presets {
6709     grid-template-columns: 1fr;
6710   }
6711
6712   .result-export-library {
6713     grid-template-columns: minmax(0, 1fr) 350px;
6714   }
6715
6716   .result-report-workspace {
6717     grid-template-columns: 330px minmax(0, 1fr);
6718   }
6719
6720   .result-report-document {
6721     padding-inline: 28px;
6722   }
6723
6724   .result-report-document .report-document-metrics {
6725     grid-template-columns: repeat(2, minmax(0, 1fr));
6726   }
6727
6728   .result-report-document .report-document-metrics > div:nth-child(2n + 1) {
6729     border-left: 0;
6730   }
6731
6732   .result-kernel-workspace-body {
6733     grid-template-columns: 190px minmax(0, 1fr) 220px;
6734   }
6735
6736   .result-kernel-definition,
6737   .result-kernel-selected-weight,
6738   .result-mkl-diagnostics {
6739     padding: 16px;
6740   }
6741
6742   .result-mkl-solver-workspace {
6743     grid-template-columns: minmax(0, 1fr) 270px;
6744   }
6745
6746   .result-mkl-detail-grid {
6747     grid-template-columns: minmax(0, 1fr) 330px;
6748   }
6749 }
```

```
front/src/workbench/workbench-polish.css
```

```
6750
6751 @media (max-width: 1080px) {
6752   .result-mkl-summary {
6753     grid-template-columns: repeat(3, minmax(0, 1fr));
6754   }
6755
6756   .result-mkl-summary-lead {
6757     grid-column: 1 / -1;
6758   }
6759
6760   .result-mkl-solver-workspace,
6761   .result-mkl-detail-grid {
6762     grid-template-columns: 1fr;
6763   }
6764
6765   .result-mkl-diagnostics {
6766     border-top: 1px solid #dfe6ed;
6767     border-left: 0;
6768   }
6769 }
6770
6771 @media (max-width: 1500px) {
6772   .result-evaluation-conclusion-body {
6773     grid-template-columns: minmax(280px, 0.72fr) minmax(0, 1.28fr);
6774   }
6775
6776   .result-evaluation-conclusion-facts {
6777     grid-template-columns: repeat(2, minmax(0, 1fr));
6778   }
6779
6780   .result-evaluation-conclusion-facts > div:nth-child(3) {
6781     border-left: 0;
6782     border-top: 1px solid #edf0f2;
6783   }
6784
6785   .result-evaluation-conclusion-facts > div:nth-child(4) {
6786     border-top: 1px solid #edf0f2;
6787   }
6788 }
6789
6790 @media (max-width: 1200px) {
6791   .result-evaluation-analysis-grid {
6792     grid-template-columns: 1fr;
6793   }
6794
6795   .result-evaluation-conclusion-body {
6796     grid-template-columns: 1fr;
6797   }
6798
6799   .result-evaluation-conclusion-copy {
```

```
front/src/workbench/workbench-polish.css
```

```
6800     padding-right: 2px;
6801     border-right: 0;
6802     border-bottom: 1px solid var(--app-border);
6803 }
6804
6805 .result-evaluation-context-grid {
6806     grid-template-columns: repeat(4, minmax(0, 1fr));
6807 }
6808
6809 .result-evaluation-context-grid > div:nth-child(5n + 1) {
6810     border-left: 1px solid #edf0f2;
6811 }
6812
6813 .result-evaluation-context-grid > div:nth-child(4n + 1) {
6814     border-left: 0;
6815 }
6816
6817 .result-evaluation-context-grid > div:nth-child(n + 5) {
6818     border-top: 1px solid #edf0f2;
6819 }
6820 }
6821
6822 @media (prefers-reduced-motion: reduce) {
6823     .result-evaluation-table-row {
6824         transition: none;
6825     }
6826 }
6827
```